

# RPR, NETN and Link 16

## Interface Control Document

An example, generated from the data model by Sen. The reference that follows — every package, every type, every table — is written by the tool and is rewritten whenever the model changes.

This page is not. It is an ordinary Typst file included at a point the generated document leaves open, which is how a project puts its own cover, approvals and appendices around a reference it never has to edit.

Placeholder for your own project or product assets

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# About this document

This is an example published with Sen's documentation. It was produced by running the model through `sen generate typst` and compiling the result, with no editing in between.

Everything from *Model overview* onwards is generated. It follows the model and only the model: if a type is added, renamed or given a new description, the next build says so, and nothing anyone wrote by hand is lost in the process.

This page, and the cover before it, are not generated. They are Typst files of your own that the document includes at points it leaves open for you:

|                                 |                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>--front-matter</code>     | A cover, before the contents.                                                                               |
| <code>--before-reference</code> | Anything between the contents and the model — a scope, a set of conventions, an approvals table. This page. |
| <code>--after-reference</code>  | Appendices, after the model.                                                                                |
| <code>--style</code>            | Every decision about how the document looks.                                                                |

The generator does not read any of those files. It writes an `#raw("#include")` and Typst resolves it when the document is compiled, so a cover page can be anything Typst can set — and changing it never means regenerating the model.

# Model overview

955 types in 4 packages. Every class descends from a single root.

## Packages

| Package | Types | Object classes | Fixed records | Enumerations | Variant records | Arrays | Quantities | Aliases | Optionals |
|---------|-------|----------------|---------------|--------------|-----------------|--------|------------|---------|-----------|
| hla     | 14    | 1              | —             | —            | —               | —      | —          | —       | 13        |
| link16  | 24    | —              | 16            | —            | —               | 4      | —          | 4       | —         |
| netn    | 464   | 59             | 197           | 61           | 10              | 56     | —          | 32      | 49        |
| rpr     | 453   | 63             | 160           | 109          | 9               | 58     | 22         | 22      | 10        |

## Class hierarchy across all packages

|                                      |     |
|--------------------------------------|-----|
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| — netn.COMRoot                       | 25  |
| — netn.CommunicationNetwork          | 26  |
| — netn.CommunicationNode             | 26  |
| — netn.Connection                    | 27  |
| — netn.DisruptionEffect              | 27  |
| — netn.LinkStates                    | 30  |
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| — netn.Weather                       | 48  |
| — netn.LandSurface                   | 29  |
| — netn.TroposphereLayer              | 46  |
| — netn.WaterSurface                  | 47  |
| — netn.Service                       | 45  |
| — netn.ORGRoot                       | 43  |
| — netn.EquipmentItem                 | 28  |
| — netn.AisEquipmentItem              | 23  |
| — netn.AidToNavigation               | 23  |
| — netn.Basestation                   | 23  |
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| — rpr.Human                          | 165 |
| — netn.NETNHuman                     | 37  |
| — netn.CBRNHuman                     | 24  |
| — rpr.NonHuman                       | 170 |



|                                    |     |
|------------------------------------|-----|
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| └─ rpr.Munition                    | 169 |
| └─ netn.NETNMunition               | 39  |
| └─ rpr.Platform                    | 172 |
| └─ rpr.Aircraft                    | 160 |
| └─ netn.NETNAircraft               | 32  |
| └─ netn.CBRNAircraft               | 23  |
| └─ rpr.AmphibiousVehicle           | 160 |
| └─ netn.NETNAmphibiousVehicle      | 34  |
| └─ netn.CBRNAmphibiousVehicle      | 24  |
| └─ rpr.GroundVehicle               | 165 |
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| └─ netn.CBRNGroundVehicle          | 24  |
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| └─ rpr.OtherLinearObject           | 170 |
| └─ rpr.PointObject                 | 172 |
| └─ rpr.BreachablePointObject       | 161 |
| └─ rpr.BurstPointObject            | 161 |
| └─ rpr.CraterObject                | 162 |
| └─ rpr.OtherPointObject            | 170 |



# Reading the tables

Every property carries three flags, in this order.

| Flag | Meaning                                                                              |
|------|--------------------------------------------------------------------------------------|
| R0   | Read-only: the value is published by whoever owns it and cannot be set from outside. |
| RW   | Read-write: the value can be set as well as read.                                    |
| D    | Dynamic: the value changes over the life of an instance.                             |
| S    | Static: the value is fixed once an instance exists.                                  |
| C    | Confirmed: delivery is acknowledged, and the update is retried until it arrives.     |
| BE   | Best effort: the update is sent once and may be lost.                                |

# hla

14 types.

Object classes

ObjectRoot

HLA Base Class that contains the RTI object ID

8

ObjectRoot

CLASS

HLA Base Class that contains the RTI object ID

| Name  | Type   | Flags  | Description       |
|-------|--------|--------|-------------------|
| rtiId | string | RW D C | HLA RTI object ID |

Optionals

|                |   |
|----------------|---|
| MaybeBool      | 8 |
| MaybeDuration  | 8 |
| MaybeF32       | 8 |
| MaybeF64       | 8 |
| MaybeI16       | 8 |
| MaybeI32       | 8 |
| MaybeI64       | 9 |
| MaybeString    | 9 |
| MaybeTimeStamp | 9 |
| MaybeU16       | 9 |
| MaybeU32       | 9 |
| MaybeU64       | 9 |
| MaybeU8        | 9 |

MaybeBool

OPTIONAL

```
optional<bool> MaybeBool;
```

MaybeDuration

OPTIONAL

```
optional<sen.Duration> MaybeDuration;
```

MaybeF32

OPTIONAL

```
optional<f32> MaybeF32;
```

MaybeF64

OPTIONAL

```
optional<f64> MaybeF64;
```

MaybeI16

OPTIONAL

```
optional<i16> MaybeI16;
```

MaybeI32

OPTIONAL

```
optional<i32> MaybeI32;
```

---

**MaybeI64** OPTIONAL

```
optional<i64> MaybeI64;
```

---

**MaybeString** OPTIONAL

```
optional<string> MaybeString;
```

---

**MaybeTimeStamp** OPTIONAL

```
optional<sen.TimeStamp> MaybeTimeStamp;
```

---

**MaybeU16** OPTIONAL

```
optional<u16> MaybeU16;
```

---

**MaybeU32** OPTIONAL

```
optional<u32> MaybeU32;
```

---

**MaybeU64** OPTIONAL

```
optional<u64> MaybeU64;
```

---

**MaybeU8** OPTIONAL

```
optional<u8> MaybeU8;
```

---

# link16

24 types.

## Fixed records

|                                            |                                                                                                                            |    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">JTIDSHeaderStruct</a>          | Link 16 Header Word (35 bits) for common messages, JTIDS voice, and VMF. Padded to 48 bits.                                | 10 |
| <a href="#">JTIDSLETRadioSignal</a>        | Link 16 radio signal containing Link 16 Enhanced Throughput (LET) encoded messages.                                        | 10 |
| <a href="#">JTIDSMessageRadioSignal</a>    | Link 16 radio signal containing Fixed Word Format (FWF) TADIL-J messages.                                                  | 11 |
| <a href="#">JTIDSVoiceCVSDRadioSignal</a>  | Link 16 radio signal containing a voice message encoded using CVSD.                                                        | 12 |
| <a href="#">JTIDSVoiceLPC10RadioSignal</a> | Link 16 radio signal containing a voice message encoded using LPC10.                                                       | 12 |
| <a href="#">JTIDSVoiceLPC12RadioSignal</a> | Link 16 radio signal containing a voice message encoded using LPC12.                                                       | 13 |
| <a href="#">LETHHeaderStruct</a>           | Link 16 Enhanced Throughput (LET) Header Word (40 bits). Padded to 48 bits.                                                | 14 |
| <a href="#">Link16RadioSignal</a>          | Link 16 radio signal common base class.                                                                                    | 14 |
| <a href="#">NTPTimestampStruct</a>         | 64-bit timestamp format according to RFC 5905. All bits set to one in both fields indicate a no statement/wildcard.        | 15 |
| <a href="#">RTTABRadioSignal</a>           | Link 16 radio signal containing an RTT/A or RTT/B message.                                                                 | 15 |
| <a href="#">RTTABStruct</a>                | JTIDS RTT A/B word (35 bits). Padded to 48 bits.                                                                           | 16 |
| <a href="#">RTTReplyRadioSignal</a>        | Link 16 radio signal containing an RTT Reply message.                                                                      | 16 |
| <a href="#">RTTReplyStruct</a>             | JTIDS RTT Reply word (35 bits). Padded to 48 bits.                                                                         | 17 |
| <a href="#">TADILJWordStruct</a>           | TADIL-J word (75 bits) used for Fixed Word Format (FWF) messages and the Variable Message Format (VMF). Padded to 80 bits. | 17 |
| <a href="#">TDLBinaryRadioSignal</a>       | Parent class for all TDL radio signals.                                                                                    | 17 |
| <a href="#">VMFRadioSignal</a>             | Link 16 radio signal containing Variable Message Format (VMF) messages.                                                    | 18 |

## JTIDSHeaderStruct STRUCTURE

Link 16 Header Word (35 bits) for common messages, JTIDS voice, and VMF. Padded to 48 bits.

| Name | Type                        | Description                                                                                                                                                                                                                      |
|------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data | <a href="#">OctetArray6</a> | In total 48 bits, with the following fields: Bits 0-2: Time Slot Type. Bit 3: Relay Transmission Indicator. Bits 4-18: Source Track Number of Sender. Bits 19-34: Secure Data Unit Serial Number. Bits 35-47: Padding, set to 0. |

Named by [JTIDSMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [VMFRadioSignal](#).

## JTIDSLETRadioSignal STRUCTURE

Link 16 radio signal containing Link 16 Enhanced Throughput (LET) encoded messages.

| Name                 | Type                                                     | Description                                                                                                                                 |
|----------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex       | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal.                                                                               |
| dataRate             | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                                                                                            |
| signalDataLength     | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                                                                                              |
| signalData           | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                                                                                                            |
| tacticalDataLinkType | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any).                                                                    |
| tDLMessageCount      | <a href="#">rpr.UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.                                                                         |
| nPGNumber            | <a href="#">NetworkParticipationGroupNumber</a>          | Network/Needline Participation Group (NPG) number. Not optional.                                                                            |
| netNumber            | <a href="#">NetworkNumber</a>                            | Network Number. Not optional.                                                                                                               |
| tsecCvll             | <a href="#">CryptoVariable</a>                           | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard). |

| Name                  | Type                                     | Description                                                                                                                                                                                                                                                                   |
|-----------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| msecCvll              | CryptoVariable                           | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | rpr.<br>SISOSTD002Version<br>Enum8       | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | rpr.Link16Version<br>Enum8               | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | TimeSlotIdentifier                       | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | NTPTimestampStruct                       | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| lETHeader             | LETHeaderStruct                          | LET Header Word. Not optional.                                                                                                                                                                                                                                                |
| tADILJMessage         | TADILJWordStruct<br>LengthlessArray1Plus | LET message data words. Not optional.                                                                                                                                                                                                                                         |

## JTIDSMessageradioSignal STRUCTURE

Link 16 radio signal containing Fixed Word Format (FWF) TADIL-J messages.

| Name                 | Type                                             | Description                                                                                                                                                                                                                                                                   |
|----------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex       | string                                           | The object instance ID of the radio transmitting this signal.                                                                                                                                                                                                                 |
| dataRate             | rpr.BitRateBitPer<br>SecondUnsigned<br>Integer32 | The rate at which the data is being transmitted.                                                                                                                                                                                                                              |
| signalDataLength     | rpr.BitsUnsigned<br>Integer16                    | The length of the signal data.                                                                                                                                                                                                                                                |
| signalData           | rpr.SignalData<br>LengthlessArray1Plus           | The signal data.                                                                                                                                                                                                                                                              |
| tacticalDataLinkType | rpr.TacticalDataLink<br>TypeEnum16               | The type of tactical data link used to transmitted this signal (if any).                                                                                                                                                                                                      |
| tDLMessageCount      | rpr.UnsignedInteger16                            | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber            | NetworkParticipation<br>GroupNumber              | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber            | NetworkNumber                                    | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll             | CryptoVariable                                   | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll             | CryptoVariable                                   | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version    | rpr.<br>SISOSTD002Version<br>Enum8               | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version        | rpr.Link16Version<br>Enum8                       | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID           | TimeSlotIdentifier                               | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |

| Name                  | Type                                                                     | Description                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>                                       | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard). |
| jTIDSHeader           | <a href="#">JTIDSHeaderStruct</a>                                        | Link 16 Header Word. Not optional.                                                                                                                                         |
| tADILJMessage         | <a href="#">TADILJWordStruct</a><br><a href="#">LengthlessArray1Plus</a> | Link 16 message data words. Not optional.                                                                                                                                  |

## JTIDSVoiceCVSDRadioSignal STRUCTURE

Link 16 radio signal containing a voice message encoded using CVSD.

| Name                  | Type                                                     | Description                                                                                                                                                                                                                                                                   |
|-----------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex        | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal.                                                                                                                                                                                                                 |
| dataRate              | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                                                                                                                                                                                                                              |
| signalDataLength      | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                                                                                                                                                                                                                                |
| signalData            | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                                                                                                                                                                                                                                              |
| tacticalDataLinkType  | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any).                                                                                                                                                                                                      |
| tDLMessageCount       | <a href="#">rpr.UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber             | <a href="#">NetworkParticipationGroupNumber</a>          | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | <a href="#">NetworkNumber</a>                            | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | <a href="#">CryptoVariable</a>                           | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>                           | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a>               | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>                   | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>                       | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>                       | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| dataLength            | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the encoded voice in the Data parameter, in bits. Not optional.                                                                                                                                                                                                 |
| jTIDSHeader           | <a href="#">JTIDSHeaderStruct</a>                        | Link 16 Header Word. Not optional.                                                                                                                                                                                                                                            |
| data                  | <a href="#">OctetLengthlessArray29Plus</a>               | JTIDS CVSD encoded voice data. Not optional.                                                                                                                                                                                                                                  |

## JTIDSVoiceLPC10RadioSignal STRUCTURE

Link 16 radio signal containing a voice message encoded using LPC10.



| Name                  | Type                                     | Description                                                                                                                                                                                                                                                                   |
|-----------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex        | string                                   | The object instance ID of the radio transmitting this signal.                                                                                                                                                                                                                 |
| dataRate              | rpr.BitRateBitPerSecondUnsignedInteger32 | The rate at which the data is being transmitted.                                                                                                                                                                                                                              |
| signalDataLength      | rpr.BitsUnsignedInteger16                | The length of the signal data.                                                                                                                                                                                                                                                |
| signalData            | rpr.SignalDataLengthlessArray1Plus       | The signal data.                                                                                                                                                                                                                                                              |
| tacticalDataLinkType  | rpr.TacticalDataLinkTypeEnum16           | The type of tactical data link used to transmitted this signal (if any).                                                                                                                                                                                                      |
| tDLMessageCount       | rpr.UnsignedInteger16                    | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber             | NetworkParticipationGroupNumber          | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | NetworkNumber                            | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | CryptoVariable                           | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | CryptoVariable                           | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | rpr.SISOSTD002VersionEnum8               | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | rpr.Link16VersionEnum8                   | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | TimeSlotIdentifier                       | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | NTPTimestampStruct                       | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| dataLength            | rpr.BitsUnsignedInteger16                | The length of the encoded voice in the Data parameter, in bits. Not optional.                                                                                                                                                                                                 |
| jTIDSHeader           | JTIDSHeaderStruct                        | Link 16 Header Word. Not optional.                                                                                                                                                                                                                                            |
| data                  | OctetLengthlessArray29Plus               | JTIDS LPC10 encoded voice data. Not optional.                                                                                                                                                                                                                                 |

## JTIDSVoiceLPC12RadioSignal STRUCTURE

Link 16 radio signal containing a voice message encoded using LPC12.

| Name                 | Type                                     | Description                                                              |
|----------------------|------------------------------------------|--------------------------------------------------------------------------|
| hostRadioIndex       | string                                   | The object instance ID of the radio transmitting this signal.            |
| dataRate             | rpr.BitRateBitPerSecondUnsignedInteger32 | The rate at which the data is being transmitted.                         |
| signalDataLength     | rpr.BitsUnsignedInteger16                | The length of the signal data.                                           |
| signalData           | rpr.SignalDataLengthlessArray1Plus       | The signal data.                                                         |
| tacticalDataLinkType | rpr.TacticalDataLinkTypeEnum16           | The type of tactical data link used to transmitted this signal (if any). |

| Name                  | Type                                            | Description                                                                                                                                                                                                                                                                   |
|-----------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tDLMessageCount       | <a href="#">rpr.UnsignedInteger16</a>           | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber             | <a href="#">NetworkParticipationGroupNumber</a> | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | <a href="#">NetworkNumber</a>                   | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | <a href="#">CryptoVariable</a>                  | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>                  | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a>      | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>          | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>              | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>              | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| dataLength            | <a href="#">rpr.BitsUnsignedInteger16</a>       | The length of the encoded voice in the Data parameter, in bits. Not optional.                                                                                                                                                                                                 |
| jTIDSHeader           | <a href="#">JTIDSHeaderStruct</a>               | Link 16 Header Word. Not optional.                                                                                                                                                                                                                                            |
| data                  | <a href="#">OctetLengthlessArray29Plus</a>      | JTIDS LPC12 encoded voice data. Not optional.                                                                                                                                                                                                                                 |

## LETHHeaderStruct STRUCTURE

Link 16 Enhanced Throughput (LET) Header Word (40 bits). Padded to 48 bits.

| Name | Type                        | Description                                                                                                                                                                                                                                                         |
|------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data | <a href="#">OctetArray6</a> | In total 48 bits, with the following fields: Bits 0-3: LET ID Symbol. Bit 4: Relay Transmission Indicator. Bits 5-8: LET Message Packing Type. Bits 9-23: Source Track Number of Sender. Bits 24-39: Secure Data Unit Serial Number. Bits 40-47: Padding, set to 0. |

Named by [JTIDSLETRadioSignal](#).

## Link16RadioSignal STRUCTURE

Link 16 radio signal common base class.

| Name             | Type                                                     | Description                                                   |
|------------------|----------------------------------------------------------|---------------------------------------------------------------|
| hostRadioIndex   | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal. |
| dataRate         | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.              |
| signalDataLength | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                |
| signalData       | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                              |

| Name                  | Type                                            | Description                                                                                                                                                                                                                                                                   |
|-----------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tacticalDataLinkType  | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>  | The type of tactical data link used to transmitted this signal (if any).                                                                                                                                                                                                      |
| tDLMessageCount       | <a href="#">rpr.UnsignedInteger16</a>           | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber             | <a href="#">NetworkParticipationGroupNumber</a> | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | <a href="#">NetworkNumber</a>                   | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | <a href="#">CryptoVariable</a>                  | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>                  | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a>      | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>          | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>              | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>              | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |

## NTPTimestampStruct STRUCTURE

64-bit timestamp format according to RFC 5905. All bits set to one in both fields indicate a no statement/wildcard.

| Name     | Type                                  | Description                                                         |
|----------|---------------------------------------|---------------------------------------------------------------------|
| seconds  | <a href="#">rpr.UnsignedInteger32</a> | Number of seconds since 0 h 1 January 1900 UTC.                     |
| fraction | <a href="#">rpr.UnsignedInteger32</a> | Fraction of a second, in $1/(2^{32})$ resolution (232 picoseconds). |

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [Link16RadioSignal](#), [RTTABRadioSignal](#), [RTTReplyRadioSignal](#), [VMFRadioSignal](#).

## RTTABRadioSignal STRUCTURE

Link 16 radio signal containing an RTT/A or RTT/B message.

| Name                 | Type                                                     | Description                                                              |
|----------------------|----------------------------------------------------------|--------------------------------------------------------------------------|
| hostRadioIndex       | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal.            |
| dataRate             | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                         |
| signalDataLength     | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                           |
| signalData           | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                                         |
| tacticalDataLinkType | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any). |
| tDLMessageCount      | <a href="#">rpr.UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.      |

| Name                  | Type                                            | Description                                                                                                                                                                                                                                                                   |
|-----------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nPGNumber             | <a href="#">NetworkParticipationGroupNumber</a> | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | <a href="#">NetworkNumber</a>                   | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | <a href="#">CryptoVariable</a>                  | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>                  | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a>      | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>          | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>              | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>              | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| rttab                 | <a href="#">RTTABStruct</a>                     | JTIDS RTT A/B message word. Not optional.                                                                                                                                                                                                                                     |

## RTTABStruct STRUCTURE

JTIDS RTT A/B word (35 bits). Padded to 48 bits.

| Name | Type                        | Description                                                                                                                                                                                                                                    |
|------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data | <a href="#">OctetArray6</a> | In total 48 bits, with the following fields: Bits 0-2: Time Slot Type. Bit 3: RTT Interrogation Type. Bits 4-18: variable content, see MIL-STD-6016 or STANAG 5516. Bits 19-34: Secure Data Unit Serial Number. Bits 35-47: Padding, set to 0. |

Named by [RTTABRadioSignal](#).

## RTTReplyRadioSignal STRUCTURE

Link 16 radio signal containing an RTT Reply message.

| Name                 | Type                                                     | Description                                                              |
|----------------------|----------------------------------------------------------|--------------------------------------------------------------------------|
| hostRadioIndex       | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal.            |
| dataRate             | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                         |
| signalDataLength     | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                           |
| signalData           | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                                         |
| tacticalDataLinkType | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any). |
| tDLMessageCount      | <a href="#">rpr.UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.      |
| nPGNumber            | <a href="#">NetworkParticipationGroupNumber</a>          | Network/Needline Participation Group (NPG) number. Not optional.         |
| netNumber            | <a href="#">NetworkNumber</a>                            | Network Number. Not optional.                                            |

| Name                  | Type                                       | Description                                                                                                                                                                                                                                                                   |
|-----------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tsecCvll              | <a href="#">CryptoVariable</a>             | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>             | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a> | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>     | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>         | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>         | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| rTTReply              | <a href="#">RTTReplyStruct</a>             | JTIDS RTT reply message word. Not optional.                                                                                                                                                                                                                                   |

## RTTReplyStruct STRUCTURE

JTIDS RTT Reply word (35 bits). Padded to 48 bits.

| Name | Type                        | Description                                                                                                                                         |
|------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| data | <a href="#">OctetArray6</a> | In total 48 bits, with the following fields: Bits 0-18: Time of Arrival. Bits 19-34: Secure Data Unit Serial Number. Bits 35-47: Padding, set to 0. |

Named by [RTTReplyRadioSignal](#).

## TADILJWordStruct STRUCTURE

TADIL-J word (75 bits) used for Fixed Word Format (FWF) messages and the Variable Message Format (VMF). Padded to 80 bits.

| Name | Type                         | Description                                                                                                                                                                                          |
|------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data | <a href="#">OctetArray10</a> | In total 80 bits, with the following fields: Bits 0-1: Word Format. Bits 2-69: variable content, see MIL-STD-6016 or STANAG 5516 for FWF J-Words. Bits 70-74: Parity. Bits 75-79: Padding, set to 0. |

Named by [TADILJWordStructLengthlessArray1Plus](#).

## TDLBinaryRadioSignal STRUCTURE

Parent class for all TDL radio signals.

| Name             | Type                                                     | Description                                                   |
|------------------|----------------------------------------------------------|---------------------------------------------------------------|
| hostRadioIndex   | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal. |
| dataRate         | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.              |
| signalDataLength | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                |
| signalData       | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                              |

| Name                 | Type                                           | Description                                                              |
|----------------------|------------------------------------------------|--------------------------------------------------------------------------|
| tacticalDataLinkType | <a href="#">rpr.TacticalDataLinkTypeEnum16</a> | The type of tactical data link used to transmitted this signal (if any). |
| tDLMessageCount      | <a href="#">rpr.UnsignedInteger16</a>          | The number of tactical data link messages contained in this signal.      |

## VMFRadioSignal STRUCTURE

Link 16 radio signal containing Variable Message Format (VMF) messages.

| Name                  | Type                                                     | Description                                                                                                                                                                                                                                                                   |
|-----------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex        | <a href="#">string</a>                                   | The object instance ID of the radio transmitting this signal.                                                                                                                                                                                                                 |
| dataRate              | <a href="#">rpr.BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                                                                                                                                                                                                                              |
| signalDataLength      | <a href="#">rpr.BitsUnsignedInteger16</a>                | The length of the signal data.                                                                                                                                                                                                                                                |
| signalData            | <a href="#">rpr.SignalDataLengthlessArray1Plus</a>       | The signal data.                                                                                                                                                                                                                                                              |
| tacticalDataLinkType  | <a href="#">rpr.TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any).                                                                                                                                                                                                      |
| tDLMessageCount       | <a href="#">rpr.UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.                                                                                                                                                                                                           |
| nPGNumber             | <a href="#">NetworkParticipationGroupNumber</a>          | Network/Needline Participation Group (NPG) number. Not optional.                                                                                                                                                                                                              |
| netNumber             | <a href="#">NetworkNumber</a>                            | Network Number. Not optional.                                                                                                                                                                                                                                                 |
| tsecCvll              | <a href="#">CryptoVariable</a>                           | Transmission Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                   |
| msecCvll              | <a href="#">CryptoVariable</a>                           | Message Security Crypto Variable Logic Label. Optional for TSA Levels 0-2; default value: all bits set to one (no statement/wildcard).                                                                                                                                        |
| sISOSTD002Version     | <a href="#">rpr.SISOSTD002VersionEnum8</a>               | Indicates which SISO-STD-002 version was used, i.e., whether byte swapping has or has not been used in the Message Data field (0 = SISO-STD-002-2006, legacy byte swapping employed; 1 = SISO-STD-002-2021, new method, bit stream, no byte swapping employed). Not optional. |
| link16Version         | <a href="#">rpr.Link16VersionEnum8</a>                   | Indicates which version of MIL-STD-6016 or STANAG 5516 is being used. Optional; default value: NoStatement.                                                                                                                                                                   |
| timeSlotID            | <a href="#">TimeSlotIdentifier</a>                       | Time slot identification. Optional for low-fidelity TSA Levels (0, 1); default value: all bits set to one (no statement/wildcard).                                                                                                                                            |
| perceivedTransmitTime | <a href="#">NTPTimestampStruct</a>                       | The time at which the radio signal is transmitted. Optional for low-fidelity synchronization (TSA Levels 0-3); default value: all bits set to one (no statement/wildcard).                                                                                                    |
| jTIDSHeader           | <a href="#">JTIDSHeaderStruct</a>                        | Link 16 Header Word. Not optional.                                                                                                                                                                                                                                            |
| messageData           | <a href="#">TADILJWordStructLengthlessArray1Plus</a>     | Link 16 message data words with VMF information fields. Not optional.                                                                                                                                                                                                         |

### Arrays

|                                                      |                                                                                     |    |
|------------------------------------------------------|-------------------------------------------------------------------------------------|----|
| <a href="#">OctetArray10</a>                         | Array of 10 octets.                                                                 | 18 |
| <a href="#">OctetArray6</a>                          | Array of 6 octets.                                                                  | 19 |
| <a href="#">OctetLengthlessArray29Plus</a>           | Array of at least 29 octets, encoded without the number of array elements.          | 19 |
| <a href="#">TADILJWordStructLengthlessArray1Plus</a> | Array of at least 1 TADILJWordStruct, encoded without the number of array elements. | 19 |

## OctetArray10 SEQUENCE

Array of 10 octets.

```
sequence<rpr.Octet, 10> OctetArray10;
```

Named by [TADILJWordStruct](#).

### OctetArray6 SEQUENCE

Array of 6 octets.

```
sequence<rpr.Octet, 6> OctetArray6;
```

Named by [JTIDSHeaderStruct](#), [LEHeaderStruct](#), [RTTABStruct](#), [RTTReplyStruct](#).

### OctetLengthlessArray29Plus SEQUENCE

Array of at least 29 octets, encoded without the number of array elements.

```
sequence<rpr.Octet> OctetLengthlessArray29Plus;
```

Named by [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#).

### TADILJWordStructLengthlessArray1Plus SEQUENCE

Array of at least 1 TADILJWordStruct, encoded without the number of array elements.

```
sequence<TADILJWordStruct> TADILJWordStructLengthlessArray1Plus;
```

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [VMFRadioSignal](#).

#### Aliases

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">CryptoVariable</a>                  | An integer identification of the crypto variable used for JTIDS transmission and message encryption. This variable allows for simulated crypto netting. Valid range: [0,127] and all bits set to one indicating no statement/wildcard.                                                                                                                                                                                                                                       | 19 |
| <a href="#">NetworkNumber</a>                   | Used to create virtual sub-circuits within NPG for stacked nets or between NPGs for multi-net operations. Valid range: [0,127].                                                                                                                                                                                                                                                                                                                                              | 19 |
| <a href="#">NetworkParticipationGroupNumber</a> | Used to segregate information within a JTIDS/MIDS network. Creates virtual networks of participants. Valid range: [0,511].                                                                                                                                                                                                                                                                                                                                                   | 20 |
| <a href="#">TimeSlotIdentifier</a>              | Bits 0-16 indicate the Time Slot Number; valid arrange [0,98303]. Time Slot 0 represents time slot A-1, and Time Slot 98303 represents C-32767. When the Epoch is 112, the last valid time slot is 45151 (end of the day). Bits 17-23 are padding and set to 0. Bits 24-31 indicate the Epoch number; valid range [0,112]. An epoch is 12.8 minutes long, 112.5 epochs in a 24 hour day. All bits set to one (including the padding field) indicate a no statement/wildcard. | 20 |

### CryptoVariable ALIAS

An integer identification of the crypto variable used for JTIDS transmission and message encryption. This variable allows for simulated crypto netting. Valid range: [0,127] and all bits set to one indicating no statement/wildcard.

```
alias CryptoVariable u8;
```

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [Link16RadioSignal](#), [RTTABRadioSignal](#), [RTTReplyRadioSignal](#), [VMFRadioSignal](#).

### NetworkNumber ALIAS

Used to create virtual sub-circuits within NPG for stacked nets or between NPGs for multi-net operations. Valid range: [0,127].

```
alias NetworkNumber u8;
```

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [Link16RadioSignal](#), [RTTABRadioSignal](#), [RTTReplyRadioSignal](#), [VMFRadioSignal](#).

---

## NetworkParticipationGroupNumber ALIAS

Used to segregate information within a JTIDS/MIDS network. Creates virtual networks of participants. Valid range: [0,511].

```
alias NetworkParticipationGroupNumber u16;
```

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [Link16RadioSignal](#), [RTTABRadioSignal](#), [RTTReplyRadioSignal](#), [VMFRadioSignal](#).

---

## TimeSlotIdentifier ALIAS

Bits 0-16 indicate the Time Slot Number; valid arrange [0,98303]. Time Slot 0 represents time slot A-1, and Time Slot 98303 represents C-32767. When the Epoch is 112, the last valid time slot is 45151 (end of the day). Bits 17-23 are padding and set to 0. Bits 24-31 indicate the Epoch number; valid range [0,112]. An epoch is 12.8 minutes long, 112.5 epochs in a 24 hour day. All bits set to one (including the padding field) indicate a no statement/wildcard.

```
alias TimeSlotIdentifier u32;
```

Named by [JTIDSLETRadioSignal](#), [JTIDSMMessageRadioSignal](#), [JTIDSVoiceCVSDRadioSignal](#), [JTIDSVoiceLPC10RadioSignal](#), [JTIDSVoiceLPC12RadioSignal](#), [Link16RadioSignal](#), [RTTABRadioSignal](#), [RTTReplyRadioSignal](#), [VMFRadioSignal](#).



# netn

464 types.

## Object classes

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">ATP45HazardArea</a>         | Represents the footprint generated by a CBRN warning and reporting simulation. This follows the NATO ATP-45 doctrine to generate a hazard area.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 23 |
| <a href="#">AidToNavigation</a>         | This class defines additional attributes for aid to navigation (ATON) equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 23 |
| <a href="#">AisEquipmentItem</a>        | This class defines additional attributes for AIS equipment. Depending on the kind of AIS equipment, further attributes are added in sub-classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 23 |
| <a href="#">Basestation</a>             | This class defines additional attributes for basestation equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 23 |
| <a href="#">CBRNAircraft</a>            | Extension of NETN_Aircraft to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 23 |
| <a href="#">CBRNAmphibiousVehicle</a>   | Extension of NETN_AmphibiousVehicle to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 24 |
| <a href="#">CBRNDetector</a>            | Represents a CBRN detector. This object is used to pass information to a CBRN federate, which will model the detector internally and publish a CBRN_DetectorAlarm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 24 |
| <a href="#">CBRNGroundVehicle</a>       | Extension of NETN_GroundVehicle to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 24 |
| <a href="#">CBRNHuman</a>               | Extension of NETN_Human to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 24 |
| <a href="#">CBRNMultidomainPlatform</a> | Extension of NETN_MultidomainPlatform to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 24 |
| <a href="#">CBRNSensor</a>              | Represents a CBRN sensor. This object is used to pass information to a CBRN federate, which will return sensor readings by publishing CBRN_SensorUpdate interactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 25 |
| <a href="#">CBRNSpacecraft</a>          | Extension of NETN_Spacecraft to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 25 |
| <a href="#">CBRNSubmersibleVessel</a>   | Extension of NETN_SubmersibleVehicle to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 25 |
| <a href="#">CBRNSurfaceVessel</a>       | Extension of NETN_SurfaceVessel to provide CBRN modelling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25 |
| <a href="#">COLPRO</a>                  | Represents a feature that provides Collective Protection (COLPRO) against a CBRN threat.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 25 |
| <a href="#">COMRoot</a>                 | Abstract root class for all NETN-COM objects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 25 |
| <a href="#">CheckPoint</a>              | A checkpoint defines a location where simulated aggregate and physical entities should stop and wait a specified time before continuing on their route. If the checkpoint is currently in operation the status attribute should be set to true. A system simulating movement of physical or aggregate entities can be either directly affected by the checkpoint by subscribing to its state or indirectly by receiving an NETN-ETR 'Wait' task.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 26 |
| <a href="#">CommunicationNetwork</a>    | The CommunicationNetwork class specifies details about the type of and service used by a logical communication network. This corresponds to the CommunicationNet-Elements of MSDL Units and Equipments. Instances of this object class should be considered as optional. No federate should rely on this data to work with communication networks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 26 |
| <a href="#">CommunicationNode</a>       | A CommunicationNode is the representation of the interface of a simulated entity to logical communication networks. The location of the CommunicationNode is derived from the referenced entity or specified explicitly (if the referenced entity is not registered in the federation). Connections to communication networks are established only if connectivity exists based on the status of available network devices and the simulation of links. Each potential connection is described in terms of requested connections (the equivalent to the information provided by MSDL).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 26 |
| <a href="#">Connection</a>              | Description of an outgoing connection from a sender to a set of receivers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 27 |
| <a href="#">DecontaminationStation</a>  | Represents a feature that provides treatment for CBRN exposure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 27 |
| <a href="#">DisruptionEffect</a>        | The 'DisruptionEffect' object class is used to represent the disruption of connections between communication nodes in an affected area. Communication models can subscribe to disruption-effect objects and use the information to simulate the effect of disruption on communication nodes, connections and link states.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 27 |
| <a href="#">EnvironmentCondition</a>    | Root class for all types of EnvironmentCondition objects. The region where the EnvironmentCondition applies is defined by the GeoReference attribute, as follows. If GeoReference is provided as a geographical area, and the sub class in question has surface semantics (water/land), the EnvironmentCondition applies only within that area of the surface. If GeoReference is provided as a geographical area, and the sub class in question defines a Layer attribute, the EnvironmentCondition applies only within the volume bounded both by the Layer and by the body described by projecting each point of the area along a line through the center of the earth. (This means, e.g., that any volumes described by two adjacent geodetic quadrangles will never overlap, and if they belong to the same layer, will also be adjacent.) If GeoReference is provided as a RPREntityReference, NETNEntityReference or GMLFeatureReference identifying a feature (as opposed to a geometry object), the EnvironmentCondition is considered to apply only to the immediate vicinity of the referenced entity/feature: any Layer attribute specified by a sub class is ignored in this case. If not provided, the EnvironmentCondition is considered global, only restricted by the semantics of its particular sub class. | 27 |
| <a href="#">EquipmentItem</a>           | Representation of an equipment item based on attributes defined in the NETN-ORG extensions of MSDL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 28 |
| <a href="#">FederateApplication</a>     | Representation of the allocation of initial modelling responsibility of units and equipment items to a federate application.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 28 |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Force                         | Representation of a specific force and its relationship with other forces. Units belonging to a force all have the same relationship with units belonging to another force. The relationships between forces can be asymmetrical, e.g. Force A can be hostile to B while B is neutral to A. The relation may change during a federation execution.                                                                                                                 | 29 |
| Installation                  | Representation of an installation/facility based on attributes defined in the NETN-ORG extensions of MSDL.                                                                                                                                                                                                                                                                                                                                                         | 29 |
| LandSurface                   | The environment conditions related to a land surface. Besides the common attributes such as Temperature, Wind and Precipitation etc. additional detail regarding snow, moisture, and ice conditions of the land surface can be represented. All attributes of LandSurfaceCondition objects are optional. An instance of this class without a GeoReference is considered to apply to all land surfaces.                                                             | 29 |
| LinkStates                    | Representation of the state of links in a physical network.                                                                                                                                                                                                                                                                                                                                                                                                        | 30 |
| Location                      | A location is a geographical point in latitude, longitude and altitude above mean sea level.                                                                                                                                                                                                                                                                                                                                                                       | 30 |
| METOCRoot                     | Superclass for Environment Conditions and Layers.                                                                                                                                                                                                                                                                                                                                                                                                                  | 30 |
| NETNAggregate                 | Aggregate extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30 |
| NETNAircraft                  | Extension of RPR-FOM Aircraft object class.                                                                                                                                                                                                                                                                                                                                                                                                                        | 32 |
| NETNAmphibiousVehicle         | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 34 |
| NETNCulturalFeature           | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 35 |
| NETNGroundVehicle             | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 36 |
| NETNHuman                     | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 37 |
| NETNMultiDomainPlatform       | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 37 |
| NETNMunition                  | Extension for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 39 |
| NETNNonHuman                  | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 39 |
| NETNSpacecraft                | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 40 |
| NETNSubmersibleVessel         | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 41 |
| NETNSurfaceVessel             | Extensions for NETN                                                                                                                                                                                                                                                                                                                                                                                                                                                | 42 |
| ORGRoot                       | Root object class for all NETN-ORG object classes. Includes unique identifier and unique name attributes inherited by all subclasses.                                                                                                                                                                                                                                                                                                                              | 43 |
| Path                          | A path is defined as a sequence of at least two locations. Each location in the path is defined geodetic coordinates and altitude above the mean sea level.                                                                                                                                                                                                                                                                                                        | 43 |
| PhysicalNetwork               | This object class represents type-specific parameters/bounds for a physical network. Physical networks are simulated by LinkState objects and do not require that an instance of PhysicalNetwork exist in the federation.                                                                                                                                                                                                                                          | 43 |
| ProbabilityHazardContourGroup | Represents the footprint covered by a CBRN hazard. This object covers the agent-specific effects which would be the output from a casualty model.                                                                                                                                                                                                                                                                                                                  | 44 |
| RawDataHazardContourGroup     | Represents the footprint covered by a CBRN hazard. This object covers the raw data which would be the output from a dispersion model.                                                                                                                                                                                                                                                                                                                              | 44 |
| Region                        | A region is an area on the earth's surface defined by a polygon or a circle.                                                                                                                                                                                                                                                                                                                                                                                       | 44 |
| SARaircraft                   | This class defines additional attributes for SAR aircraft equipment.                                                                                                                                                                                                                                                                                                                                                                                               | 44 |
| SEFacility                    | Base class for NETN Synthetic Environment object classes.                                                                                                                                                                                                                                                                                                                                                                                                          | 44 |
| SEGeoObject                   | Representation of geographically associated locations, paths, and regions with or without a defined name. E.g. a harbour may be represented as a Location with a specific name, a commonly used route can be represented as a named path and a region can be represented as an specific geographic area and reused as an object instance in the simulation.                                                                                                        | 45 |
| Service                       | All systems providing EnvironmentCondition information as either object instances or as responses to service requests must register a Service object instance.                                                                                                                                                                                                                                                                                                     | 45 |
| SubsurfaceLayer               | EnvironmentCondition of a subsurface water layer. If GeoReference is not provided, the condition is considered global within the specified Layer.                                                                                                                                                                                                                                                                                                                  | 45 |
| TroposphereLayer              | If overlapping Atmospheric Conditions exist the following merging rules apply: Humidity, AirTemperature and BarometricPressure are calculated as the average of the overlapping conditions. Visibility is calculated as the minimum visibility distance of the overlapping conditions. A TroposphereLayer instance without a GeoReference is considered global within its specified Layer.                                                                         | 46 |
| Unit                          | Representation of an organizational unit based on attributes defined in the NETN-ORG extensions of MSDL.                                                                                                                                                                                                                                                                                                                                                           | 46 |
| Vessel                        | This class defines additional attributes for vessel equipment.                                                                                                                                                                                                                                                                                                                                                                                                     | 47 |
| WaterSurface                  | Condition of sea surface in the specified region. An instance of this class without a GeoReference is considered to apply to all water surfaces.                                                                                                                                                                                                                                                                                                                   | 47 |
| Weather                       | Information about the general weather condition can be represented using WeatherCondition objects. Common data such as Temperature, Wind and Precipitation is represented but also details regarding Barometric Pressure and Humidity. Information about Visual Range and Haze conditions can also be represented. All attributes of WeatherCondition objects are optional. An instance of this class without a GeoReference is considered global above sea level. | 48 |

## ATP45HazardArea CLASS

Represents the footprint generated by a CBRN warning and reporting simulation. This follows the NATO ATP-45 doctrine to generate a hazard area.

| Name                | Type                                     | Flags   | Description                                                                              |
|---------------------|------------------------------------------|---------|------------------------------------------------------------------------------------------|
| time                | <a href="#">EpochTimeSecInt64</a>        | RO D BE | Simulation time at which this hazard area was generated.                                 |
| validityTime        | <a href="#">rpr.TimeSecondInteger32</a>  | RO D BE | Duration (in seconds) from the Time attribute that this hazard area prediction is valid. |
| locations           | <a href="#">GeocentricPolygon</a>        | RO D BE | Array of locations that define the hazard area                                           |
| aTP45HazardAreaType | <a href="#">ATP45HazardAreaTypeEnum8</a> | RO S BE | This hazard area's type                                                                  |
| agentClass          | <a href="#">AgentClassEnum8</a>          | RO S BE | The agent class                                                                          |
| uniqueID            | <a href="#">UuidArrayOfHLAbyte16</a>     | RO S BE | Unique representation of the area's ID.                                                  |

## AidToNavigation CLASS

This class defines additional attributes for aid to navigation (ATON) equipment.

## AisEquipmentItem CLASS

This class defines additional attributes for AIS equipment. Depending on the kind of AIS equipment, further attributes are added in sub-classes.

| Name              | Type                                                  | Flags  | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|-------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mmsi              | <a href="#">i32</a>                                   | RO S C | Optional. The MMSI number (Maritime Mobile Service Identity) of the AIS station. If the value is not provided then the subscribing federate that is responsible for the modelling of the AIS station shall generate a value. Note that the value of each individual AIS station must be unique across all AIS stations in the simulation. Recycling MMSI numbers is not advised since some systems may retain dead tracks for some time. |
| transmitterStatus | <a href="#">rpr.TransmitterOperationalStatusEnum8</a> | RO D C | Optional. The initial status of the AIS Transmitter(s) of the AIS station. If the value is not provided then the value shall assumed to be ON.                                                                                                                                                                                                                                                                                           |
| radioSystemType   | <a href="#">rpr.EntityTypeStruct</a>                  | RO S C | Optional. The type of transmitter. If the value is not provided then the subscribing federate that is responsible for the modelling of the AIS station shall determine the type of transmitter.                                                                                                                                                                                                                                          |

## Basestation CLASS

This class defines additional attributes for basestation equipment.

## CBRNAircraft CLASS

Extension of NETN\_Aircraft to provide CBRN modelling.

| Name          | Type                                   | Flags   | Description                                                |
|---------------|----------------------------------------|---------|------------------------------------------------------------|
| contamination | <a href="#">ArrayOfAgentMassStruct</a> | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

## CBRNAmphibiousVehicle CLASS

Extension of NETN\_AmphibiousVehicle to provide CBRN modelling.

| Name          | Type                                   | Flags   | Description                                                |
|---------------|----------------------------------------|---------|------------------------------------------------------------|
| contamination | <a href="#">ArrayOfAgentMassStruct</a> | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

## CBRNDetector CLASS

Represents a CBRN detector. This object is used to pass information to a CBRN federate, which will model the detector internally and publish a CBRN\_DetectorAlarm.

| Name             | Type                                            | Flags   | Description                                                                                                                                 |
|------------------|-------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueID         | <a href="#">UuidArrayOfHLAbyte16</a>            | RO S BE | Unique representation of the detector's ID.                                                                                                 |
| processingTime   | <a href="#">rpr.TimeSecondInteger32</a>         | RO S BE | Duration (in seconds) between the detector being exposed to a concentration of agent above its threshold and the detector raising an alarm. |
| averagingTime    | <a href="#">rpr.TimeSecondInteger32</a>         | RO S BE | Duration (in seconds) over which the detector will collect samples.                                                                         |
| detectableAgents | <a href="#">ArrayOfAgentConcentrationStruct</a> | RO S BE | Array of detectable agents and their thresholds.                                                                                            |
| alarm            | <a href="#">CBRNAlarmStruct</a>                 | RO D BE | Data representing the alarm of this detector. Defaults to no alarm if the attribute is not set.                                             |

## CBRNGroundVehicle CLASS

Extension of NETN\_GroundVehicle to provide CBRN modelling.

| Name          | Type                                   | Flags   | Description                                                |
|---------------|----------------------------------------|---------|------------------------------------------------------------|
| contamination | <a href="#">ArrayOfAgentMassStruct</a> | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

## CBRNHuman CLASS

Extension of NETN\_Human to provide CBRN modelling.

| Name        | Type                                      | Flags   | Description                                                                           |
|-------------|-------------------------------------------|---------|---------------------------------------------------------------------------------------|
| exposures   | <a href="#">ArrayOfCBRNExposureStruct</a> | RO D BE | Array of agents to which this entity has been exposed to. Defaults to an empty array. |
| treatments  | <a href="#">ArrayOfTreatmentStruct</a>    | RO D BE | The types of treatment that this entity has used. Defaults to an empty array.         |
| triageLevel | <a href="#">CBRNDamageEnum8</a>           | RO D BE | Triage level of this entity. Defaults to Uninjured.                                   |
| iPEType     | <a href="#">IPETypeEnum8</a>              | RO D BE | Type of IPE that the entity has donned. Defaults to None.                             |

## CBRNMultiDomainPlatform CLASS

Extension of NETN\_MultiDomainPlatform to provide CBRN modelling.

| Name          | Type                                   | Flags   | Description                                                |
|---------------|----------------------------------------|---------|------------------------------------------------------------|
| contamination | <a href="#">ArrayOfAgentMassStruct</a> | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

**CBRNSensor** CLASS

Represents a CBRN sensor. This object is used to pass information to a CBRN federate, which will return sensor readings by publishing CBRN\_SensorUpdate interactions.

| Name             | Type                           | Flags   | Description                                                                   |
|------------------|--------------------------------|---------|-------------------------------------------------------------------------------|
| uniqueID         | UuidArrayOfHLAByte16           | RO S BE | Unique representation of the sensor's ID.                                     |
| updateFrequency  | rpr.TimeSecondInteger32        | RO S BE | Duration (in seconds) that this sensor would like between updated readings.   |
| averagingTime    | rpr.TimeSecondInteger32        | RO S BE | Duration (in seconds) over which the sensor will collect samples.             |
| detectableAgents | ArrayOfAgentTypeEnum           | RO S BE | Array of agents that this sensor wishes to detect.                            |
| sensorReadings   | ArrayOfCBRNSensorReadingStruct | RO D BE | Latest sensor readings. Defaults to no readings if this attribute is not set. |

**CBRNSpacecraft** CLASS

Extension of NETN\_Spacecraft to provide CBRN modelling.

| Name          | Type                   | Flags   | Description                                                |
|---------------|------------------------|---------|------------------------------------------------------------|
| contamination | ArrayOfAgentMassStruct | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

**CBRNSubmersibleVessel** CLASS

Extension of NETN\_SubmersibleVehicle to provide CBRN modelling.

| Name          | Type                   | Flags   | Description                                                |
|---------------|------------------------|---------|------------------------------------------------------------|
| contamination | ArrayOfAgentMassStruct | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

**CBRNSurfaceVessel** CLASS

Extension of NETN\_SurfaceVessel to provide CBRN modelling.

| Name          | Type                   | Flags   | Description                                                |
|---------------|------------------------|---------|------------------------------------------------------------|
| contamination | ArrayOfAgentMassStruct | RO D BE | CBRN hazardous agent inside vehicle due to embedded units. |

**COLPRO** CLASS

Represents a feature that provides Collective Protection (COLPRO) against a CBRN threat.

| Name           | Type                                 | Flags   | Description                                               |
|----------------|--------------------------------------|---------|-----------------------------------------------------------|
| numberEntities | QuantityUInt32                       | RO S BE | The number of entities that this COLPRO can handle.       |
| protection     | ArrayOfProtectionEffectivenessStruct | RO S BE | The effectiveness that this COLPRO offers for each agent. |

**COMRoot** CLASS

Abstract root class for all NETN-COM objects.

| Name     | Type | Flags  | Description                                                  |
|----------|------|--------|--------------------------------------------------------------|
| uniqueId | UUID | RO D C | Required. Unique identifier for all NETN-COM object classes. |

## Checkpoint CLASS

A checkpoint defines a location where simulated aggregate and physical entities should stop and wait a specified time before continuing on their route. If the checkpoint is currently in operation the status attribute should be set to true. A system simulating movement of physical or aggregate entities can be either directly affected by the checkpoint by subscribing to its state or indirectly by receiving an NETN-ETR `Wait` task.

| Name      | Type                                    | Flags  | Description                                                                                                                                                                                                                                                         |
|-----------|-----------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delayTime | <a href="#">rpr.TimeSecondInteger32</a> | RO D C | Required. The time that an entity shall wait at the checkpoint before passing. The time is a nominal value; federates can use this for modifying delay time for different types of entities, e.g. add or subtract a value or multiply with a type depending factor. |

## CommunicationNetwork CLASS

The CommunicationNetwork class specifies details about the type of and service used by a logical communication network. This corresponds to the CommunicationNet-Elements of MSDL Units and Equipments. Instances of this object class should be considered as optional. No federate should rely on this data to work with communication networks.

| Name        | Type                                         | Flags  | Description                                                                                   |
|-------------|----------------------------------------------|--------|-----------------------------------------------------------------------------------------------|
| uniqueName  | <a href="#">Text64</a>                       | RO S C | Required. Unique communication network name. Unique in the context of communication networks. |
| networkType | <a href="#">CommunicationNetworkTypeEnum</a> | RO S C | Optional. The communication network type of use.                                              |
| serviceType | <a href="#">CommunicationServiceTypeEnum</a> | RO S C | Optional. The type of service used on the communication network.                              |

## CommunicationNode CLASS

A CommunicationNode is the representation of the interface of a simulated entity to logical communication networks. The location of the CommunicationNode is derived from the referenced entity or specified explicitly (if the referenced entity is not registered in the federation). Connections to communication networks are established only if connectivity exists based on the status of available network devices and the simulation of links. Each potential connection is described in terms of requested connections (the equivalent to the information provided by MSDL).

| Name                 | Type                                     | Flags  | Description                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| entityId             | <a href="#">UUID</a>                     | RO S C | Required. Reference by UUID to an object instance of one of the following classes: * NETN-MRM Aggregate Entity * NETN-Physical extensions of RPR Physical Entities. If the referenced entity exists in the federation, the location of the node is derived from the location of the entity. If the referenced entity does not exist in the federation, the location of the node is defined by the Location attribute. |
| location             | <a href="#">rpr.WorldLocationStruct</a>  | RO D C | Optional. Specifies the location of the CommunicationNode in case the entity referenced by EntityId is not registered in the federation. If the entity referenced by EntityId exists in the federation, the location of the communication node is derived from that entity and the value of the Location attribute shall be ignored.                                                                                  |
| requestedConnections | <a href="#">RequestedConnectionArray</a> | RO D C | Required. Possible (requested) connections for the communication node.                                                                                                                                                                                                                                                                                                                                                |

| Name                | Type                                    | Flags  | Description                                                                                                                                                                                                                                                                                                                                |
|---------------------|-----------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| networkDevices      | <a href="#">NetworkDeviceArray</a>      | RO D C | Required. Available network devices defining the association of a communication network (connection layer) with a physical network (link layer). Each network device can be associated with several communication networks but only to one physical network. Each network device also describes the transmitter and receiver capabilities. |
| incomingConnections | <a href="#">IncomingConnectionArray</a> | RO D C | Optional. Description of all incoming connections to the receiving entity.                                                                                                                                                                                                                                                                 |

## Connection CLASS

Description of an outgoing connection from a sender to a set of receivers.

| Name                     | Type                                    | Flags  | Description                                                                    |
|--------------------------|-----------------------------------------|--------|--------------------------------------------------------------------------------|
| communicationNetworkName | <a href="#">Text64</a>                  | RO S C | Required. A reference to the communication network this connection belongs to. |
| senderEntityId           | <a href="#">UUID</a>                    | RO S C | Required. A reference to the entity sending data using this connection.        |
| receivers                | <a href="#">ConnectionReceiverArray</a> | RO D C | Required. Characteristics of the connections to individual receiving entities. |

## DecontaminationStation CLASS

Represents a feature that provides treatment for CBRN exposure.

| Name                  | Type                                    | Flags   | Description                                                                |
|-----------------------|-----------------------------------------|---------|----------------------------------------------------------------------------|
| numberEntities        | <a href="#">QuantityUInt32</a>          | RO S BE | The number of entities that this decontamination station can handle.       |
| decontaminationPeriod | <a href="#">rpr.TimeSecondInteger32</a> | RO S BE | Duration in seconds it takes to decontaminate an entity.                   |
| treatments            | <a href="#">ArrayOfTreatmentStruct</a>  | RO D BE | The types of treatment that this decontamination station currently offers. |

## DisruptionEffect CLASS

The `DisruptionEffect` object class is used to represent the disruption of connections between communication nodes in an affected area. Communication models can subscribe to disruption-effect objects and use the information to simulate the effect of disruption on communication nodes, connections and link states.

| Name             | Type                               | Flags  | Description                                                                                                                                    |
|------------------|------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| affectedNetworks | <a href="#">ArrayOfText64</a>      | RO D C | Optional. Names of all affected communication networks. If not provided all networks in the specified area are affected.                       |
| affectedArea     | <a href="#">GeodeticQuadrangle</a> | RO D C | Optional. Area affected by disruption. If not provided the default is a global disruption and all connections are affected.                    |
| effect           | <a href="#">PercentFloat64</a>     | RO D C | Required. Level of disruption in percent. 100% equals No connectivity and 0% no disruption effect. The level of disruption can vary over time. |

## EnvironmentCondition CLASS

Root class for all types of EnvironmentCondition objects. The region where the EnvironmentCondition applies is defined by the GeoReference attribute, as follows. If GeoReference is provided as a geographical area, and the sub class in question has surface semantics (water/land), the EnvironmentCondition applies only within that area of the surface. If GeoReference



is provided as a geographical area, and the sub class in question defines a Layer attribute, the EnvironmentCondition applies only within the volume bounded both by the Layer and by the body described by projecting each point of the area along a line through the center of the earth. (This means, e.g., that any volumes described by two adjacent geodetic quadrangles will never overlap, and if they belong to the same layer, will also be adjacent.) If GeoReference is provided as a RPREntityReference, NETNEntityReference or GMLFeatureReference identifying a feature (as opposed to a geometry object), the EnvironmentCondition is considered to apply only to the immediate vicinity of the referenced entity/feature: any Layer attribute specified by a sub class is ignored in this case. If not provided, the EnvironmentCondition is considered global, only restricted by the semantics of its particular sub class.

| Name         | Type                | Flags  | Description                                                                            |
|--------------|---------------------|--------|----------------------------------------------------------------------------------------|
| serviceId    | UUID                | RO S C | Required. Identifies the METOC Service which produced the EnvironmentCondition object. |
| geoReference | GeoReferenceVariant | RO S C | Optional. A geographical location, region, feature or simulated object.                |

## EquipmentItem CLASS

Representation of an equipment item based on attributes defined in the NETN-ORG extensions of MSDL.

| Name                  | Type                               | Flags  | Description                                                                                                                                         |
|-----------------------|------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| nameNonSen            | Text255                            | RO S C | Required. A unique name of the equipment item.                                                                                                      |
| holdingUnit           | UUID                               | RO D C | Required. Identifies the lowest level unit in the ORBAT to which this equipment item belongs.                                                       |
| mountedOn             | UUID                               | RO D C | Optional. Reference to another equipment item on which this item is mounted or attached.                                                            |
| entityType            | rpr.EntityTypeStruct               | RO S C | Required. SISO-REF-010 code for entity type definitions. If unknown, use 0.0.0.0.0.0.0.                                                             |
| natoStockNumber       | NatoStockNumberArray13             | RO S C | Optional. NATO stock number code. 13 digits (0-9)                                                                                                   |
| symbolId              | string                             | RO D C | Required. A symbol identifier represented as a string.                                                                                              |
| symbolAmplification   | EquipmentSymbolAmplificationStruct | RO D C | Optional. Text amplifications for the equipment's symbol. Static or very low update rate.                                                           |
| disposition           | DispositionStruct                  | RO D C | Required. Detailed positional information on equipment item.                                                                                        |
| communicationNetworks | ArrayOfCommunicationNetworks       | RO D C | Optional. Set of Communication Networks that the equipment monitors and uses to communicate during a mission.                                       |
| resolution            | ModelResolutionTypeEnum32          | RO D C | Optional. Enumeration indicating the level of fidelity appropriate for instantiating the equipment in the simulation. Default value: NOT_SPECIFIED. |

## FederateApplication CLASS

Representation of the allocation of initial modelling responsibility of units and equipment items to a federate application.

| Name       | Type    | Flags  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|---------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nameNonSen | Text255 | RO S C | Required. A unique name in the organization to group units and equipment items. This group name is used to determine the initial allocation of modelling responsibility of units and equipment items to a federate. The federation agreements should specify how this name is to be interpreted and how the allocation will be performed. A common agreement and practice are to use the federate name as group name, i.e. the units and equipment items are allocated to the federate with that name. Typically one application will be |



| Name           | Type                        | Flags  | Description                                                                                                                                                                                                                                      |
|----------------|-----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| units          | <a href="#">ArrayOfUuid</a> | RO D C | responsible for performing the allocation. Alternatives for the group name are for example: federate type name, a functional name.<br>Optional. The units for which the federate has the responsibility of simulating. Default is an empty list. |
| equipmentItems | <a href="#">ArrayOfUuid</a> | RO D C | Optional. The equipment for which the federate has the responsibility of simulating. Default is an empty list.                                                                                                                                   |
| installations  | <a href="#">ArrayOfUuid</a> | RO D C | Optional. The installations for which the federate has the responsibility of simulating. Default is an empty list.                                                                                                                               |

## Force CLASS

Representation of a specific force and its relationship with other forces. Units belonging to a force all have the same relationship with units belonging to another force. The relationships between forces can be asymmetrical, e.g. Force A can be hostile to B while B is neutral to A. The relation may change during a federation execution.

| Name       | Type                             | Flags  | Description                                                                 |
|------------|----------------------------------|--------|-----------------------------------------------------------------------------|
| nameNonSen | <a href="#">Text255</a>          | RO S C | Required. Name of the force/side.                                           |
| relations  | <a href="#">ArrayOfRelations</a> | RO D C | Optional. Sequence of relations, force-relation. Default is an empty array. |

## Installation CLASS

Representation of an installation/facility based on attributes defined in the NETN-ORG extensions of MSDL.

| Name                | Type                                                  | Flags  | Description                                                                                  |
|---------------------|-------------------------------------------------------|--------|----------------------------------------------------------------------------------------------|
| nameNonSen          | <a href="#">Text255</a>                               | RO S C | Required. A unique name of the installation.                                                 |
| owner               | <a href="#">UUID</a>                                  | RO D C | Required. Reference to the force, unit or equipment item owning or hosting the installation. |
| entityType          | <a href="#">rpr.EntityTypeStruct</a>                  | RO S C | Required. SISO-REF-010 code for entity type definitions. If unknown, use 0.0.0.0.0.0.0.      |
| symbolId            | <a href="#">string</a>                                | RO D C | Required. A symbol identifier represented as a string.                                       |
| symbolAmplification | <a href="#">InstallationSymbolAmplificationStruct</a> | RO D C | Optional. Text amplifications for the installation's symbol. Static or very low update rate. |
| location            | <a href="#">GeodeticPoint</a>                         | RO D C | Required. Geographic location of the installation.                                           |

## LandSurface CLASS

The environment conditions related to a land surface. Besides the common attributes such as Temperature, Wind and Precipitation etc. additional detail regarding snow, moisture, and ice conditions of the land surface can be represented. All attributes of LandSurfaceCondition objects are optional. An instance of this class without a GeoReference is considered to apply to all land surfaces.

| Name | Type                       | Flags  | Description                                                                                                                                                                                                                                     |
|------|----------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| snow | <a href="#">SnowStruct</a> | RO D C | Optional. Current snow depth and density in the region/location specified by the EnvironmentCondition. Default is no snow. If EnvironmentConditions with overlapping regions/locations exist the average snow depth and density should be used. |

| Name         | Type                                   | Flags  | Description                                                                                                                                                                                                                                                                                                                    |
|--------------|----------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| moisture     | <a href="#">SurfaceMoistureEnum16</a>  | RO D C | Optional. Surface moisture in the region/location specified by the EnvironmentCondition. Default is 0 (Dry). If EnvironmentConditions with overlapping regions/locations exist, moisture is calculated as the highest enumerated value making the resulting value the one with most moisture.                                  |
| iceCondition | <a href="#">RoadIceConditionEnum16</a> | RO D C | Optional. Surface ice condition in the region/location specified by the EnvironmentCondition. Default is no ice. If EnvironmentConditions with overlapping regions/locations exist, surface ice condition is calculated as the highest enumerated value making the resulting value the one with the most severe ice condition. |

## LinkStates CLASS

Representation of the state of links in a physical network.

| Name                | Type                            | Flags  | Description                                                 |
|---------------------|---------------------------------|--------|-------------------------------------------------------------|
| physicalNetworkName | <a href="#">Text64</a>          | RO S C | Required. Reference to physical network by its unique name. |
| links               | <a href="#">LinkStatusArray</a> | RO D C | Required. Status of a set of physical network links.        |

## Location CLASS

A location is a geographical point in latitude, longitude and altitude above mean sea level.

| Name  | Type                          | Flags  | Description                                                                               |
|-------|-------------------------------|--------|-------------------------------------------------------------------------------------------|
| point | <a href="#">GeodeticPoint</a> | RO D C | Required. A geographical point in latitude, longitude, and altitude above mean sea level. |

## METOCRoot CLASS

Superclass for Environment Conditions and Layers.

| Name       | Type                   | Flags  | Description                                                                        |
|------------|------------------------|--------|------------------------------------------------------------------------------------|
| uniqueId   | <a href="#">UUID</a>   | RO S C | Required. Unique identifier for the Environment Condition object or METOC Service. |
| nameNonSen | <a href="#">string</a> | RO S C | Optional. Name of the Environment Condition or METOC Service.                      |

## NETNAggregate CLASS

Aggregate extensions for NETN

| Name     | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|--------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId | <a href="#">UuidArrayOfHLAbyte16</a> | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data. The unique identifier can serve dual purposes. It is a unique identification of the NETN_Aggregate object instance but can also be a reference to a NETN-ORG unit element with the same unique identifier. |

| Name               | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|--------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| status             | <a href="#">ActiveStatusEnum8</a>    | RO D BE | Required. Indicate if this aggregate unit currently is being simulated or not. E.g. units mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before setting the status to active. |
| subunitList        | <a href="#">ArrayOfUuid</a>          | RO D C  | Optional. Reference to disaggregated representations of subsets of the aggregate unit when registered in the federation. Each element should refer to an existing NETN_Aggregate object in the federation. If not published, disaggregation is not supported.                                                                                                                                                                                                                                                                                                                |
| parentUnit         | <a href="#">UuidArrayOfHLAbyte16</a> | RO D C  | Optional. Reference to parent aggregate entity. If not published, aggregation is not supported. The default value is 0000000000000000 (no parent unit).                                                                                                                                                                                                                                                                                                                                                                                                                      |
| dividedUnitList    | <a href="#">ArrayOfUuid</a>          | RO D C  | Optional. Reference to other aggregate or physical entities divided from the aggregate unit to represent specific subsets of holdings. If not published, a division is not supported.                                                                                                                                                                                                                                                                                                                                                                                        |
| sourceUnit         | <a href="#">UuidArrayOfHLAbyte16</a> | RO D BE | Optional. Reference to an active NETN_Aggregate instance, the source of a NETN-MRM division. If not published, merging is not supported. The default value is 0000000000000000 representing no source unit.                                                                                                                                                                                                                                                                                                                                                                  |
| embeddedUnitList   | <a href="#">ArrayOfUuid</a>          | RO D BE | Optional. Reference to units or platforms embarked on and transported by this unit. If not published, transport of embedded units not supported.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| higherHeadquarters | <a href="#">UuidArrayOfHLAbyte16</a> | RO D BE | Optional. A reference to an entity representing the aggregate unit's superior or headquarters from which orders are given and to which reports are sent. The highest level unit or headquarters will publish 0000000000000000 as its HigherHeadquarters value. The referenced entity may or may not be registered in the federation as a NETN_Aggregate and/or NETN-ORG unit. If not published, the aggregate does not have a superior unit or headquarter. The default value is 0000000000000000 (no higher headquarters).                                                  |
| mounted            | <a href="#">f64</a>                  | RO D BE | Optional. The percentage of aggregate personnel travelling on or in their organic transport. Default 100% - all personnel mounted.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| symbolId           | <a href="#">string</a>               | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| callsign           | <a href="#">string</a>               | RO S BE | Required. A callsign used to address the unit. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Name                | Type                                      | Flags   | Description                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|-------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| echelon             | <a href="#">EchelonEnum32</a>             | RO S BE | Optional. The size of the unit (level of command).                                                                                                                                                                                                                                                                                                             |
| entityList          | <a href="#">ArrayOfEntityStruct</a>       | RO D BE | Optional. This attribute provides data on all entities comprising the aggregate. Entities include equipment, e.g. platforms, weapons, sensors and lifeforms such as personnel. Each entity contains key status attributes and subunit allocation information. If not provided the status and allocation of entities is not modelled on an entity level.        |
| suppliesStatus      | <a href="#">NETNArrayOfSupplyStruct</a>   | RO D BE | Optional. The type and quantities of supplies available (on hand) to the unit. If not provided, the amount of available supplies is undefined.                                                                                                                                                                                                                 |
| equipmentStatus     | <a href="#">ArrayOfResourceStatus</a>     | RO D BE | Optional. This summarizes the health status of the equipment comprising the aggregate. If not provided, the status of equipment is undefined.                                                                                                                                                                                                                  |
| personnelStatus     | <a href="#">ArrayOfResourceStatus</a>     | RO D BE | Optional. This summarizes the health status of personnel comprising the aggregate. If not provided, the status of personnel is undefined.                                                                                                                                                                                                                      |
| visualSignature     | <a href="#">VisualSignatureStruct</a>     | RO D BE | Describes the unit's susceptibility to electro-optical detection.                                                                                                                                                                                                                                                                                              |
| hUMINTSignature     | <a href="#">HUMINTSignatureStruct</a>     | RO D BE | Describes the unit's susceptibility to human intelligence (HUMINT), i.e. information collected and provided by human sources.                                                                                                                                                                                                                                  |
| electronicSignature | <a href="#">ElectronicSignatureStruct</a> | RO D BE | Describes the aggregate's susceptibility to electronic detection both as a summary value and by identifying aggregate sensors together with their operational status.                                                                                                                                                                                          |
| combatValue         | <a href="#">f64</a>                       | RO D BE | Optional. A summary value (in per cent) of unit effectiveness based on the level of training, leadership, morale, personnel and equipment operational status, etc. The default value is 100%.                                                                                                                                                                  |
| coverStatus         | <a href="#">f64</a>                       | RO D BE | Optional. Describes the unit's protection from the effects of weapons fire. Default is 0% - Fully affected by weapon fire.                                                                                                                                                                                                                                     |
| captureStatus       | <a href="#">CaptureStatusEnum8</a>        | RO D BE | The status of an aggregate with respect to its control or influence over its own activities.                                                                                                                                                                                                                                                                   |
| mission             | <a href="#">MissionStruct</a>             | RO D BE | The operational task the aggregate has been ordered to perform.                                                                                                                                                                                                                                                                                                |
| activity            | <a href="#">AggregateMissionEnum16</a>    | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. Default is 0 (Other activity). |
| route               | <a href="#">PathVariantStruct</a>         | RO D C  | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                        |
| destination         | <a href="#">PointVariantStruct</a>        | RO D C  | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                 |
| weaponsControlOrder | <a href="#">WeaponControlOrderEnum8</a>   | RO D BE | Optional. Describes current Weapon Control Order Free, Tight, or Hold. Default is 0 - Other.                                                                                                                                                                                                                                                                   |

## NETNAircraft CLASS

Extension of RPR-FOM Aircraft object class.

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId         | <a href="#">UuidArrayOfHLAByte16</a>   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status           | <a href="#">ActiveStatusEnum8</a>      | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit       | <a href="#">UuidArrayOfHLAByte16</a>   | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list.                                                                                                                                                                              |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| symbolId         | <a href="#">string</a>                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## NETNAmphibiousVehicle CLASS

Extensions for NETN

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId         | <a href="#">UuidArrayOfHLAbyte16</a>   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status           | <a href="#">ActiveStatusEnum8</a>      | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit       | <a href="#">UuidArrayOfHLAbyte16</a>   | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list.                                                                                                                                                                              |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Name     | Type   | Flags   | Description                                            |
|----------|--------|---------|--------------------------------------------------------|
| symbolId | string | RO D BE | Optional. A symbol identifier represented as a string. |

## NETNCulturalFeature CLASS

Extensions for NETN

| Name             | Type                       | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId         | UuidArrayOf<br>HLAbyte16   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status           | ActiveStatusEnum8          | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| embeddedUnitList | ArrayOfUuid                | RO D BE | Optional. Reference to platforms or lifeforms embarked on this entity. If not published, embarkment/disembarkment of embedded units not supported. Entities are referenced by UUID. E.g. when soldiers are inside in an NETN_CulturalFeature entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier leaves he NETN_CulturalFeature the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list.                                                                                                                                                                   |
| callsign         | string                     | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity         | AggregateMission<br>Enum16 | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| symbolId         | string                     | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## NETNGroundVehicle CLASS

Extensions for NETN

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId         | <a href="#">UuidArrayOfHLAbyte16</a>   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status           | <a href="#">ActiveStatusEnum8</a>      | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit       | <a href="#">UuidArrayOfHLAbyte16</a>   | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list.                                                                                                                                                                              |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



| Name     | Type   | Flags   | Description                                            |
|----------|--------|---------|--------------------------------------------------------|
| symbolId | string | RO D BE | Optional. A symbol identifier represented as a string. |

## NETNHuman CLASS

Extensions for NETN

| Name        | Type                       | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId    | UuidArrayOf<br>HLAbyte16   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status      | ActiveStatusEnum8          | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit  | UuidArrayOf<br>HLAbyte16   | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 0000000000000000 representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| callsign    | string                     | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity    | AggregateMission<br>Enum16 | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| route       | PathVariantStruct          | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination | PointVariantStruct         | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| symbolId    | string                     | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## NETNMultiDomainPlatform CLASS

Extensions for NETN

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId         | <a href="#">UuidArrayOfHLAByte16</a>   | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status           | <a href="#">ActiveStatusEnum8</a>      | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit       | <a href="#">UuidArrayOfHLAByte16</a>   | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list.                                                                                                                                                                              |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                                                                 |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| symbolId         | <a href="#">string</a>                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**NETNMunition** CLASS

Extension for NETN

| Name        | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId    | <a href="#">UuidArrayOfHLAbyte16</a> | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status      | <a href="#">ActiveStatusEnum8</a>    | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| route       | <a href="#">PathVariantStruct</a>    | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| destination | <a href="#">PointVariantStruct</a>   | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| symbolId    | <a href="#">string</a>               | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**NETNNonHuman** CLASS

Extensions for NETN

| Name       | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId   | <a href="#">UuidArrayOfHLAbyte16</a> | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status     | <a href="#">ActiveStatusEnum8</a>    | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit | <a href="#">UuidArrayOfHLAbyte16</a> | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Name        | Type                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                              |
|-------------|------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                        |         | physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                    |
| callsign    | string                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                              |
| activity    | AggregateMissionEnum16 | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity). |
| route       | PathVariantStruct      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                  |
| destination | PointVariantStruct     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                           |
| symbolId    | string                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                   |

## NETNSpacecraft CLASS

Extensions for NETN

| Name       | Type                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId   | UuidArrayOfHLAbyte16 | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status     | ActiveStatusEnum8    | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit | UuidArrayOfHLAbyte16 | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|----------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list. |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                 |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                    |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| symbolId         | <a href="#">string</a>                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## NETNSubmersibleVessel CLASS

Extensions for NETN

| Name       | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId   | <a href="#">UuidArrayOfHLAbyte16</a> | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| status     | <a href="#">ActiveStatusEnum8</a>    | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the current status of the instance before status is set to active. |
| sourceUnit | <a href="#">UuidArrayOfHLAbyte16</a> | RO D BE | Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Name             | Type                                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|----------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| embeddedUnitList | <a href="#">ArrayOfUuid</a>            | RO D BE | merging is not supported. The default value is 00000000-0000-0000-0000-000000000000 (UUID with all zeros) representing no source unit.<br>Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list. |
| callsign         | <a href="#">string</a>                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| activity         | <a href="#">AggregateMissionEnum16</a> | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                                                                                                                                                              |
| route            | <a href="#">PathVariantStruct</a>      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| destination      | <a href="#">PointVariantStruct</a>     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| symbolId         | <a href="#">string</a>                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## NETNSurfaceVessel CLASS

Extensions for NETN

| Name     | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uniqueId | <a href="#">UuidArrayOfHLAbyte16</a> | RO S BE | Required. A unique identifier for the object. The Universally Unique Identifier (UUID) is either generated or defined as part of scenario initialization, e.g. using NETN-ORG MSDL data for defining specific platforms as equipment assigned to units.                                                                                                                                                                                                                                                                                                                                                                   |
| status   | <a href="#">ActiveStatusEnum8</a>    | RO D BE | Required. Indicate if this platform currently is being simulated or not. E.g. platforms mounted or embarked on transports can be set to inactive. During an inactive state, the attribute values may not reflect an accurate, current value. Therefore, any subscribing federate can ignore inactive units. All attributes must be updated to represent the current status of the instance before setting the state to Active. An inactive instance may have its instance attributes updated by a federate but reflected updates shall be ignored by receiving federates. All attributes must be updated to represent the |

| Name             | Type                   | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sourceUnit       | UuidArrayOfHLAbyte16   | RO D BE | current status of the instance before status is set to active.<br>Optional. Reference to an active NETN_Aggregate instance from which this physical entity was divided. If not published, merging is not supported. The default value is 0000000000000000 representing no source unit.                                                                                                                                                                                                                      |
| embeddedUnitList | ArrayOfUuid            | RO D BE | Optional. Reference to platforms or lifeforms embarked on this platform. If not published, transport of embedded units is not supported. Entities are referenced by UUID. E.g. when soldiers are transported in a NETN Physical entity, then references (UUIDs) to these soldier instances are added to the EmbeddedUnitList for the entity. When a soldier disembarks from the transporter, the reference (UUID) to that soldier is removed from the EmbeddedUnitList. The default value is an empty list. |
| callsign         | string                 | RO S BE | Required. A callsign used to address the platform. Callsigns should be unique in the context in which they are used but not required to be globally unique.                                                                                                                                                                                                                                                                                                                                                 |
| activity         | AggregateMissionEnum16 | RO D BE | Optional. The current activity of the platform. The value is based on the Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM) action-event-category-code. The JC3IEDM is a fully documented standard [NATO STANAG 5525] for an information exchange data model for the sharing of C2 information. The default value is 0 (Other activity).                                                                                                                                    |
| route            | PathVariantStruct      | RO D BE | Optional. The current path of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| destination      | PointVariantStruct     | RO D BE | Optional. The current destination of movement.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| symbolId         | string                 | RO D BE | Optional. A symbol identifier represented as a string.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## ORGRoot CLASS

Root object class for all NETN-ORG object classes. Includes unique identifier and unique name attributes inherited by all subclasses.

| Name     | Type | Flags  | Description                  |
|----------|------|--------|------------------------------|
| uniqueId | UUID | RO S C | Required. Unique Identifier. |

## Path CLASS

A path is defined as a sequence of at least two locations. Each location in the path is defined geodetic coordinates and altitude above the mean sea level.

| Name   | Type         | Flags  | Description                                                          |
|--------|--------------|--------|----------------------------------------------------------------------|
| points | GeodeticPath | RO D C | Required. A path with at least 2 points expressed as GeodeticPoints. |

## PhysicalNetwork CLASS

This object class represents type-specific parameters/bounds for a physical network. Physical networks are simulated by LinkState objects and do not require that an instance of PhysicalNetwork exist in the federation.

| Name        | Type                              | Flags  | Description                                                                             |
|-------------|-----------------------------------|--------|-----------------------------------------------------------------------------------------|
| uniqueName  | Text64                            | RO S C | Required. Unique physical network name. Uniqueness in the context of physical networks. |
| description | PhysicalNetworkDescriptionVariant | RO S C | Required. Characteristics of the physical network.                                      |

## ProbabilityHazardContourGroup CLASS

Represents the footprint covered by a CBRN hazard. This object covers the agent-specific effects which would be the output from a casualty model.

| Name         | Type                                  | Flags   | Description                                                                                    |
|--------------|---------------------------------------|---------|------------------------------------------------------------------------------------------------|
| time         | EpochTimeSecInt64                     | RO D BE | Simulation time at which this contour group was generated.                                     |
| agent        | AgentTypeEnum16                       | RO S BE | Agent reflected in this contour.                                                               |
| exposureType | ExposureTypeEnum8                     | RO S BE | Type of exposure.                                                                              |
| contours     | ArrayOfProbabilityHazardContourStruct | RO D BE | Array of contours. These should be ordered in a sequence of ascending PercentProbabilityLevel. |
| uniqueID     | UuidArrayOfHLAbyte16                  | RO S BE | Unique representation of the contour's ID.                                                     |

## RawDataHazardContourGroup CLASS

Represents the footprint covered by a CBRN hazard. This object covers the raw data which would be the output from a dispersion model.

| Name       | Type                              | Flags   | Description                                                                          |
|------------|-----------------------------------|---------|--------------------------------------------------------------------------------------|
| time       | EpochTimeSecInt64                 | RO D BE | Simulation time at which this contour group was generated.                           |
| agent      | AgentTypeEnum16                   | RO S BE | Agent reflected in this contour.                                                     |
| hazardType | HazardTypeEnum8                   | RO S BE | Type of hazard.                                                                      |
| contours   | ArrayOfRawDataHazardContourStruct | RO D BE | Array of contours. These should be ordered in a sequence of ascending ExposureLevel. |
| uniqueID   | UuidArrayOfHLAbyte16              | RO S BE | Unique representation of the contour's ID.                                           |

## Region CLASS

A region is an area on the earth's surface defined by a polygon or a circle.

| Name | Type              | Flags  | Description                                                                     |
|------|-------------------|--------|---------------------------------------------------------------------------------|
| area | AreaVariantStruct | RO D C | Required. Defines the area, either as a geodetic polygon, circle or quadrangle. |

## SARaircraft CLASS

This class defines additional attributes for SAR aircraft equipment.

## SEFacility CLASS

Base class for NETN Syntetic Environment object classes.

| Name     | Type                 | Flags  | Description                                                                  |
|----------|----------------------|--------|------------------------------------------------------------------------------|
| uniqueId | UuidArrayOfHLAbyte16 | RO S C | Required. Pre-defined or generated unique identifier of the object instance. |



| Name              | Type                                                 | Flags  | Description                                                                                                                          |
|-------------------|------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| locationReference | <a href="#">GeoLocationReferenceVariant</a>          | RO S C | Required. A reference that identifies a geographical location directly or indirectly by reference to a simulated object.             |
| nameNonSen        | <a href="#">string</a>                               | RO S C | Optional. Name of the facility.                                                                                                      |
| symbolId          | <a href="#">string</a>                               | RO S C | Optional. A symbol identifier represented as a string.                                                                               |
| status            | <a href="#">ActiveStatusEnum8</a>                    | RO D C | Optional. Specifies if the facility is active (operational) or not. Default value is 1 (Active).                                     |
| objectType        | <a href="#">rpr.EnvironmentObject<br/>TypeStruct</a> | RO S C | Optional. Specifying the domain, the kind and the specific identification of the environment object.                                 |
| forceId           | <a href="#">rpr.ForceIdentifier<br/>Enum8</a>        | RO D C | Optional. The force that operates the facility. Default is 0 (Other).                                                                |
| percentComplete   | <a href="#">f32</a>                                  | RO D C | Optional. Describes a level of completion when establishing/building of the facility. Expressed in percent [0-100]. Default is 100%. |
| damageState       | <a href="#">rpr.DamageStatus<br/>Enum32</a>          | RO D C | Optional. The damage state on the facility. Default value is 0 (NoDamage).                                                           |
| comment           | <a href="#">string</a>                               | RO D C | Optional. A descriptive text comment.                                                                                                |

## SEGeoObject CLASS

Representation of geographically associated locations, paths, and regions with or without a defined name. E.g. a harbour may be represented as a Location with a specific name, a commonly used route can be represented as a named path and a region can be represented as an specific geographic are and reused as an object instance in the simulation.

| Name       | Type                                      | Flags  | Description                                                         |
|------------|-------------------------------------------|--------|---------------------------------------------------------------------|
| uniqueId   | <a href="#">UuidArrayOf<br/>HLAbyte16</a> | RO S C | Required. Unique identifier for the GeoObject.                      |
| nameNonSen | <a href="#">string</a>                    | RO S C | Optional. Name of the GeoObject, e.g. Location name, Road name etc. |

## Service CLASS

All systems providing EnvironmentCondition information as either object instances or as responses to service requests must register a Service object instance.

| Name      | Type                                        | Flags  | Description                                                                                                                                                        |
|-----------|---------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| modelType | <a href="#">WeatherModelType<br/>Enum32</a> | RO S C | Required. Type of METOC model provided by the Service. Specifies whether the Service delivers Simulated, Real (Historical), Live (Current) or Standard model data. |

## SubsurfaceLayer CLASS

EnvironmentCondition of a subsurface water layer. If GeoReference is not provided, the condition is considered global within the specified Layer.

| Name        | Type                                                     | Flags  | Description                                                                                                                                                                                   |
|-------------|----------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| temperature | <a href="#">rpr.Temperature<br/>DegreeCelsiusFloat32</a> | RO D C | Optional. Temperature in the region/location specified by the EnvironmentCondition. If EnvironmentConditions with overlapping regions/locations exist the average temperature should be used. |
| layer       | <a href="#">LayerStruct</a>                              | RO D C | Optional. A body of water extending from a (negative) base altitude upwards (see                                                                                                              |

| Name       | Type               | Flags  | Description                                                                                                                                                                                                                                      |
|------------|--------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                    |        | LayerStruct). Any portion of the Layer extending above sea level is ignored. If Layer is not provided, it is considered to extend from the sea floor to the surface of the water.                                                                |
| current    | CurrentStruct      | RO D C | Optional. Describes current in the water layer. Default is no current. If EnvironmentConditions with overlapping regions/locations exist the average current direction and speed should be used.                                                 |
| salinity   | f32                | RO D C | Optional. Salinity of sea water on the practical salinity scale 1978 (PSS-78). Default value is 35 (equivalent to 35 parts per thousand). If EnvironmentConditions with overlapping regions/locations exist the average salinity should be used. |
| bottomType | SedimentTypeEnum32 | RO D C | Optional. Type of sediment on the sea floor. Default is NoSediment. If EnvironmentConditions with overlapping regions/locations exist, the sediment type in the overlapping region is determined by the latest updated value.                    |

## TroposphereLayer CLASS

If overlapping Atmospheric Conditions exist the following merging rules apply: Humidity, AirTemperature and Barometric-Pressure are calculated as the average of the overlapping conditions. Visibility is calculated as the minimum visibility distance of the overlapping conditions. A TroposphereLayer instance without a GeoReference is considered global within its specified Layer.

| Name  | Type        | Flags  | Description                                                                                                                                                                                                                                                     |
|-------|-------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| layer | LayerStruct | RO D C | Optional. Volume of atmosphere extending from a base altitude upwards (see LayerStruct). Any portion of the Layer extending below sea level is ignored. If Layer is not provided, it is considered to extend from the surface of the earth upward indefinitely. |
| cloud | CloudStruct | RO D C | Optional. Data about Clouds in the Atmospheric layer. Default is no clouds. If EnvironmentConditions with overlapping regions/locations exist, the cloud condition in the overlapping region is determined by the latest updated value.                         |

## Unit CLASS

Representation of an organizational unit based on attributes defined in the NETN-ORG extensions of MSDL.

| Name         | Type    | Flags  | Description                                                                                                                                                                                              |
|--------------|---------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nameNonSen   | Text255 | RO S C | Required. A unique name of the unit.                                                                                                                                                                     |
| superiorUnit | UUID    | RO D C | Required. Reference to the superior unit. If this unit has no superior unit (i.e. is the top-unit) then the value of this attribute shall be 00000000-0000-0000-0000-000000000000 (UUID with all zeros). |
| embarkedIn   | UUID    | RO D C | Optional. A reference to a unit or equipment item which is carrying/transporting the unit. Default value is 00000000-0000-0000-0000-000000000000                                                         |

| Name                  | Type                          | Flags  | Description                                                                                                                                                                                                                                          |
|-----------------------|-------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| force                 | UUID                          | RO D C | (UUID with all zeros) indicating that the unit is not embarked in or mounted on any other unit or equipment.<br>Optional. A reference to the Force that the unit belongs to. A missing value is interpreted as the force being unknown/undetermined. |
| holdings              | ArrayOfHoldings               | RO D C | Optional. A set of holdings of the unit. Default is an empty list.                                                                                                                                                                                   |
| entityType            | rpr.EntityTypeStruct          | RO S C | Required. The entity type to use for representation of the unit as an aggregate entity. Entity types should be based on SISO-REF-010 and/or defined by federation agreements.                                                                        |
| symbolId              | string                        | RO D C | Required. A symbol identifier represented as a string.                                                                                                                                                                                               |
| symbolAmplification   | UnitSymbolAmplificationStruct | RO D C | Optional. Text amplifications for the unit's symbol. Static or very low update rate.                                                                                                                                                                 |
| disposition           | DispositionStruct             | RO D C | Required. Detailed positional information on unit.                                                                                                                                                                                                   |
| formation             | FormationStruct               | RO D C | Optional. Formation of this unit.                                                                                                                                                                                                                    |
| isHq                  | bool                          | RO S C | Optional. Indicate whether the unit has a command function, e.g. if it is an HQ or not.                                                                                                                                                              |
| echelon               | EchelonEnum32                 | RO S C | Optional. Symbol modifier identifying the command level. Default NONE.                                                                                                                                                                               |
| communicationNetworks | ArrayOfCommunicationNetworks  | RO D C | Optional. Set of Communication Networks that the unit monitors and uses to communicate during a mission.                                                                                                                                             |

## Vessel CLASS

This class defines additional attributes for vessel equipment.

| Name | Type | Flags  | Description                                                                                                                                                                                                                   |
|------|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imo  | i32  | RO S C | Optional. The International Maritime Organization (IMO) number to identify the vessel. If the value is not provided then the subscribing federate that is responsible for the modelling of the vessel shall generate a value. |

## WaterSurface CLASS

Condition of sea surface in the specified region. An instance of this class without a GeoReference is considered to apply to all water surfaces.

| Name     | Type             | Flags  | Description                                                                                                                                                                                                                                      |
|----------|------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seaState | SeaStateEnum16   | RO D C | Optional. State of sea surface. Default is Calm_glassy. If EnvironmentConditions with overlapping regions/locations exist, the sea state in the overlapping region is determined by the latest updated value.                                    |
| salinity | f32              | RO D C | Optional. Salinity of sea water on the practical salinity scale 1978 (PSS-78). Default value is 35 (equivalent to 35 parts per thousand). If EnvironmentConditions with overlapping regions/locations exist the average salinity should be used. |
| tide     | rpr.MeterFloat32 | RO D C | Optional. The height of the current tide relative to MSL. Default is 0. If EnvironmentConditions                                                                                                                                                 |

| Name    | Type                          | Flags  | Description                                                                                                                                                                                                                                                                                                                   |
|---------|-------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ice     | <a href="#">IceStruct</a>     | RO D C | with overlapping regions/locations exist, the value for the overlapping region is the average Tide value.<br>Optional. Ice condition on surface. Default is no ice. If EnvironmentConditions with overlapping regions/locations exist, the ice condition in the overlapping region is determined by the latest updated value. |
| current | <a href="#">CurrentStruct</a> | RO D C | Optional. Current on water surface. Default is no current. Ignored if surface has 100% Ice coverage. If EnvironmentConditions with overlapping regions/locations exist, the current in the overlapping region is the average current direction and speed.                                                                     |
| wave    | <a href="#">WaveStruct</a>    | RO D C | Optional. Waves on the water surface. Default is no waves. Ignored if surface has 100% Ice coverage. If EnvironmentConditions with overlapping regions/locations exist, the wave condition in the overlapping region is determined by the latest updated value.                                                               |
| swell   | <a href="#">WaveStruct</a>    | RO D C | Optional. Swell of the body of water. Default is no Swell. Ignored if surface has 100% Ice coverage. If EnvironmentConditions with overlapping regions/locations exist, the swell condition in the overlapping region is determined by the latest updated value.                                                              |

## Weather CLASS

Information about the general weather condition can be represented using WeatherCondition objects. Common data such as Temperature, Wind and Precipitaion is represented but also details regarding Barometric Pressure and Humidity. Information about Visual Range and Haze conditions can also be represented. All attributes of WeatherCondition objects are optional. An instance of this class without a GeoReference is considered global above sea level.

| Name          | Type                                                | Flags  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| temperature   | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | RO D C | Optional. Temperature in the region/location specified by the EnvironmentCondition. If EnvironmentConditions with overlapping regions/locations exist the average temperature should be used.                                                                                                                                                                                                                                                                              |
| wind          | <a href="#">WindStruct</a>                          | RO D C | Optional. Wind speed and direction in the region/location specified by the EnvironmentCondition. If EnvironmentConditions with overlapping regions/locations exist the average wind speed and direction should be used.                                                                                                                                                                                                                                                    |
| precipitation | <a href="#">PrecipitationStruct</a>                 | RO D C | Optional. Current precipitation type and intensity in the region/location specified by the Environment Condition. Default is No Precipitation. If EnvironmentConditions with overlapping regions/locations exist the average intensity should be used. Conflicting precipitation types are resolved according to the following precedence: Snow, Hail, Rain, No Precipitation. E.g. If there is one overlapping Environment Condition with Snow the result is always Snow. |
| haze          | <a href="#">HazeStruct</a>                          | RO D C | Optional. Current Haze type and density in the region/location specified by the Environment Condition. Default is No Haze. If                                                                                                                                                                                                                                                                                                                                              |

| Name               | Type             | Flags  | Description                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| humidity           | f32              | RO D C | EnvironmentConditions with overlapping regions/locations exist having the same Haze type the average density should be used.<br>Optional. Humidity in percent in the region/location specified by the Environment Condition. Default is 75% (Normal value) If EnvironmentConditions with overlapping regions/locations exist the average humidity should be used. |
| barometricPressure | f32              | RO D C | Optional. Barometric pressure measured in millibar or hectopascal (1 mbar = 1hPa) in the region/location specified by the Environment Condition. If EnvironmentConditions with overlapping regions/locations exist the average barometric pressure should be used.                                                                                                |
| visibilityRange    | rpr.MeterFloat32 | RO D C | Optional. The distance at which an object or light can be clearly discerned by the human eye in the region/location specified by the Environment Condition. If EnvironmentConditions with overlapping regions/locations exist the minimum distance should be used.                                                                                                |

## Fixed records

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">AISRadioSignal</a>                             | This is the root interaction class for all AIS messages. The interaction class contains a reference to the embedded system that represents the AIS transmitter. The embedded system should be modelled as a RadioTransmitter, an RPR-FOM object class that is subclassed from Embedded System.                                                                                                                             | 56 |
| <a href="#">AcceptOffer</a>                                | Use the AcceptOffer interaction to accept an offer received in an OfferService interaction.                                                                                                                                                                                                                                                                                                                                | 56 |
| <a href="#">AcknowledgeData</a>                            | Acknowledge data for Message Types 7 and 13.                                                                                                                                                                                                                                                                                                                                                                               | 56 |
| <a href="#">Activate</a>                                   | Request federate to change of status of AggregateUnit to Active. If required the unit will be registered in the federation.                                                                                                                                                                                                                                                                                                | 57 |
| <a href="#">AddPassage</a>                                 | Tasking of an entity to lay/build a passage between the two given points. The passage can, for example, be a passage through an obstacle or a bridge over a river. The tasked entity should be within a certain distance tolerance (specified in the federation agreement) of one of the points of the passage. A passage object should be registered in the federation with the provided UUID.                            | 57 |
| <a href="#">AgentConcentrationStruct</a>                   | Concentration value associated with a specific agent.                                                                                                                                                                                                                                                                                                                                                                      | 57 |
| <a href="#">AgentMassStruct</a>                            | Mass of contaminant inside a vehicle brought in by an embedded unit.                                                                                                                                                                                                                                                                                                                                                       | 58 |
| <a href="#">Aggregate</a>                                  | Instruction to the AggregateFederate to perform aggregation of the specified AggregateUnit's parts.                                                                                                                                                                                                                                                                                                                        | 58 |
| <a href="#">AisBaseStationReportAndUTCDateResponseData</a> | Common reporting structure for Message Types 4 and 11 for reporting UTC time and date and, at the same time, position. It is also used by AIS stations for determining if the station is within 120 NM for response to Messages Types 20 and 23.                                                                                                                                                                           | 58 |
| <a href="#">AisBinaryMessageData</a>                       | Binary Payload for Message Types 6 and 8. The interpretation of the binary payload is controlled by: - The Designated Area Code (DAC), which is a jurisdiction code: 366 for the United States. It uses the same encoding as the area designator in MMMSIs; see [ITU-MID]. 1 designates international (ITU) messages. - The FID, which is the Functional ID for a message subtype. In some sources this is abbreviated FI. | 58 |
| <a href="#">AisMessage</a>                                 | Super class for all AIS message types.                                                                                                                                                                                                                                                                                                                                                                                     | 59 |
| <a href="#">AisMessage1</a>                                | Message Type 1: Position Report Class A. Message type for a scheduled position report; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor.                  | 59 |
| <a href="#">AisMessage10</a>                               | Message Type 10: UTC/Date Inquiry. Request for UTC/Date information from an AIS base station.                                                                                                                                                                                                                                                                                                                              | 59 |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">AisMessage11</a>           | Message Type 11: UTC/Date Response. Identical to Message Type 4, with the semantics of a response to inquiry. This message type is only transmitted from a mobile station as a result of a UTC request message (Message Type 10).                                                                                                                                                                                                  | 60 |
| <a href="#">AisMessage12</a>           | Type 12: Addressed Safety-Related Message. This is a point-to-point text message. The payload is interpreted as six-bit text.                                                                                                                                                                                                                                                                                                      | 60 |
| <a href="#">AisMessage13</a>           | Message Type 13: Safety-Related Acknowledgement. This message type is a receipt acknowledgement to senders of previous messages of Message Type 12.                                                                                                                                                                                                                                                                                | 60 |
| <a href="#">AisMessage14</a>           | Type 14: Safety-Related Broadcast Message. This is a broadcast text message. The payload is interpreted as six-bit text.                                                                                                                                                                                                                                                                                                           | 60 |
| <a href="#">AisMessage17</a>           | Message Type 17: DGNSS Broadcast Binary Message. This message type is used to broadcast differential corrections for GPS.                                                                                                                                                                                                                                                                                                          | 61 |
| <a href="#">AisMessage18</a>           | Message Type 18: Standard Class B CS Position Report. A less detailed report than message types 1-3 for vessels using Class B transmitters. Omits navigational status and rate of turn.                                                                                                                                                                                                                                            | 61 |
| <a href="#">AisMessage19</a>           | Message Type 19: Extended Class B CS Position Report. A slightly more detailed report than Message Type 18 for vessels using Class B transmitters. Omits navigational status and rate of turn. Fields are as in the common navigation block and the Message Type 5 message.                                                                                                                                                        | 61 |
| <a href="#">AisMessage2</a>            | Message Type 2: Position Report Class A. Message type for an assigned scheduled position report; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor.                | 62 |
| <a href="#">AisMessage21</a>           | Message Type 21: Aid-to-Navigation Report. Identification and location message to be emitted by aids to navigation such as buoys and lighthouses.                                                                                                                                                                                                                                                                                  | 62 |
| <a href="#">AisMessage24</a>           | Message Type 24: Static Data Report. Equivalent of Message Type 5 for ships using Class B equipment. Also used to associate an MMSI with a name on either class A or class B equipment. This message type may be in part A or part B format. According to the standard, parts A and B are expected to be broadcast in adjacent pairs.                                                                                              | 63 |
| <a href="#">AisMessage27</a>           | Message Type 27: Long Range AIS Broadcast message. This message type is primarily intended for long-range detection of AIS Class A equipped vessels (typically by satellite). This message has a similar content to Messages 1, 2 and 3, but the total number of bits has been compressed to allow for increased propagation delays associated with long-range detection.                                                          | 64 |
| <a href="#">AisMessage3</a>            | Message Type 3: Position Report Class A. Message type for a special position report, response to interrogation; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor. | 64 |
| <a href="#">AisMessage4</a>            | Message Type 4: Base Station Report. This message is to be used by fixed-location base stations to periodically report a position and time reference.                                                                                                                                                                                                                                                                              | 64 |
| <a href="#">AisMessage5</a>            | Message Type 5: Static and Voyage Related Data. This message type is transmitted every 6 minutes and should only be used by Class A shipborne and SAR aircraft AIS stations when reporting static or voyage related data.                                                                                                                                                                                                          | 65 |
| <a href="#">AisMessage6</a>            | Message Type 6: Binary Addressed Message. This message type is an addressed point-to-point message with a binary payload.                                                                                                                                                                                                                                                                                                          | 65 |
| <a href="#">AisMessage7</a>            | Message Type 7: Binary Acknowledge. This message type is a receipt acknowledgement to the senders of a previous messages of Message Type 6. Up to 4 destination MMSIs can be acknowledged in one message.                                                                                                                                                                                                                          | 66 |
| <a href="#">AisMessage8</a>            | Message Type 8: Binary Broadcast Message. This message type is a broadcast message with a binary payload.                                                                                                                                                                                                                                                                                                                          | 66 |
| <a href="#">AisMessage9</a>            | Message type 9: Standard SAR Aircraft Position Report. Tracking information for search-and-rescue aircraft.                                                                                                                                                                                                                                                                                                                        | 66 |
| <a href="#">AisNavigationData</a>      | Message Types 1, 2 and 3 share a common reporting structure for navigational data, captured in this class.                                                                                                                                                                                                                                                                                                                         | 67 |
| <a href="#">AisStaticAndVoyageData</a> | Common Static and Voyage Data for Message Types 5, 19 and 24.                                                                                                                                                                                                                                                                                                                                                                      | 67 |

|                              |                                                                                                                                                                                                                                                                                                                                                |    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| AppointmentStruct            | Date ; Time and Location of an appointment. When date and time is set to zero (0), implies no date-time specification                                                                                                                                                                                                                          | 67 |
| CBRNAlarmStruct              | Properties of a CBRN alarm.                                                                                                                                                                                                                                                                                                                    | 68 |
| CBRNCasualty                 | Represents a CBRN casualty caused by exposure. This is for use with federates that cannot support the TriageLevel attribute in CBRN_Human.                                                                                                                                                                                                     | 68 |
| CBRNDetectorAlarm            | Represents the alarm trigger of a previously registered CBRN detector.                                                                                                                                                                                                                                                                         | 68 |
| CBRNExposureStruct           | Dosage exposure value associated with a specific agent.                                                                                                                                                                                                                                                                                        | 68 |
| CBRNFacilityUpdate           | Represents an order for the specified entities to enter or leave the CBRN facility.                                                                                                                                                                                                                                                            | 68 |
| CBRNPlatformUpdate           | Represents an update to the contaminating mass inside a vehicle due to embedded entities.                                                                                                                                                                                                                                                      | 68 |
| CBRNRelease                  | Communicates information associated with the release of hazardous agent.                                                                                                                                                                                                                                                                       | 69 |
| CBRNSensorReadingStruct      | Timed CBRN sensor reading.                                                                                                                                                                                                                                                                                                                     | 69 |
| CBRNSensorUpdate             | Sends information about the current state of a previously registered CBRN sensor.                                                                                                                                                                                                                                                              | 69 |
| CBRNTreatmentCommand         | Represents an order for the specified entities to receive the list of treatments.                                                                                                                                                                                                                                                              | 69 |
| COLPROUpdate                 | Represents an order for the specified entities to enter or leave the COLPRO.                                                                                                                                                                                                                                                                   | 70 |
| CancelAllTasks               | Management task to an entity to cancel all queued and ongoing tasks.                                                                                                                                                                                                                                                                           | 70 |
| CancelOffer                  | Used by provider to cancel an already made offer before it has been accepted. Used if the OfferTimeOut has passed.                                                                                                                                                                                                                             | 70 |
| CancelRequestLOG             | Used by either a service consuming entity or a service providing entity to inform about early termination of the service delivery or in some cases termination of the service request before delivery has begun.                                                                                                                               | 70 |
| CancelRequestTMR             | A `CancelRequest` interaction can be sent by the **Request Federate** to cancel a request.                                                                                                                                                                                                                                                     | 70 |
| CancelSpecifiedTasks         | Management task to an entity to cancel all specified tasks.                                                                                                                                                                                                                                                                                    | 71 |
| CapabilitiesSupportedETR     | Provides the set of ETR_Task, ETR_SimCon and ETR_Report interactions that the federate modelling the entity supports. This interaction is in response to a QuerySupportedCapabilities.                                                                                                                                                         | 71 |
| CapabilitiesSupportedMRM     | An interaction sent in response to a QuerySupportedCapabilities request. The response includes a list of names of the supported capabilities for the Aggregate unit specified in the query. The names are one or more of 'Aggregate', 'Disaggregate', 'Divide', 'Merge', 'Activate' and 'Inactivate'.                                          | 71 |
| ClearObstacle                | Tasking of an entity to clear an obstacle or minefield with the given UUID. The tasked entity should be within a certain distance (tolerance specified in the federation agreement) of one of the points of the geometry of the obstacle.                                                                                                      | 72 |
| CloudStruct                  | Description of cloud layer type, coverage and density.                                                                                                                                                                                                                                                                                         | 72 |
| CommunicationNetworkStruct   | CommunicationNetInstance that equipment monitors or uses to communicate with a mission.                                                                                                                                                                                                                                                        | 72 |
| ConnectionReceiverStruct     | Characteristics of a connection to a receiver.                                                                                                                                                                                                                                                                                                 | 72 |
| CreateMinefield              | Tasking of an entity to create a minefield within the specified geometry. When the task is completed, a minefield object should be published in the federation (e.g. RPR-FOM Minefield). This tasking interaction is different from the RPR-FOM MinefieldObjectTransaction interaction, which asks a federate to create a minefield magically. | 73 |
| CreateObstacle               | Tasking of an entity to create an obstacle with the given geometry. The tasked entity should be within a certain distance (tolerance specified in the federation agreement) of one of the points of the geometry.                                                                                                                              | 73 |
| CurrentStruct                | Water current direction and speed.                                                                                                                                                                                                                                                                                                             | 74 |
| DamageStatusReport           | Report on a unit's damage status.                                                                                                                                                                                                                                                                                                              | 74 |
| Deactivate                   | Request change of status of AggregateUnit to Inactive and if indicated remove it from the federation.                                                                                                                                                                                                                                          | 74 |
| DecontaminationStationUpdate | Represents an order for the specified entities to enter or leave the decontamination station.                                                                                                                                                                                                                                                  | 74 |
| Disaggregate                 | Instruction to perform a full disaggregation of a AggregatedUnit. All subunits and platforms will be registered in the federation.                                                                                                                                                                                                             | 75 |
| Dismount                     | Tasking of an entity to dismount from a mounted position. When the task is completed, the tasked entity is no longer attached to or embarked in another entity.                                                                                                                                                                                | 75 |
| DispositionStruct            | Positioning of unit or equipment item.                                                                                                                                                                                                                                                                                                         | 75 |



|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| DisruptCommunication                  | Tasking of an entity to introduce a communication network disruption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 75 |
| Divide                                | Instruction to divide the simulated AggregateUnit into multiple simulated object. The resources are divided among the simulated entities. After successful division two simulated entities represent the entire Unit. 1) The original AggregateUnit and 2) the divided unit or platform. Both these entities are simulated until merged.                                                                                                                                                                                                                                                                                                                                                                                                | 76 |
| ETRReport                             | A base interaction class for more specialized report interaction classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 76 |
| ETRRoot                               | Root interaction class for the Entity Tasking and Reporting (ETR) interaction classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 77 |
| ETRSimCon                             | A base interaction class for EXCON Task interaction classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 77 |
| ETRTask                               | A base interaction class for more specialized task interaction classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 77 |
| ETRTaskManagement                     | A base interaction class for more specialized task management interaction classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 77 |
| ElectronicSignatureStruct             | A summary percentage of an aggregate's susceptibility to detection of its electronic emissions. Zero percent means that the aggregate has no electronic emissions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 77 |
| EntityStruct                          | An entity represented to the federation as part of the aggregate object which owns it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 78 |
| EquipmentSymbolAmplificationStruct    | Text amplifiers for Equipment symbols.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 78 |
| EstablishCheckPoint                   | Tasking of an entity to establish a checkpoint. The task defines a location where a checkpoint shall be established and then operated. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the checkpoint. If not, a separate move task should be issued first. During the execution of the task, a CheckPoint object (NETN-SE) should be registered in the federation. The completion and operational status of the CheckPoint object should be updated continuously during the execution of the task or at task completion. Immediately after the checkpoint has been established, it will be operated by the tasked entity for a specified duration of time. | 79 |
| FireAtEntity                          | Tasking of an entity to fire at another specified target entity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 79 |
| FireAtEntityWM                        | Tasking of an entity to fire at a specified target entity with the specified weapon and munition data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 80 |
| FireAtLocation                        | Tasking of an entity to fire at a location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 80 |
| FireAtLocationWM                      | Tasking of an entity to fire at a location with the specified weapon and munition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 81 |
| FollowEntity                          | Tasking of an entity to follow another entity at a specified distance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 81 |
| FormationPositionStruct               | The position of this unit/equipment in the higher unit formation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 81 |
| FormationStruct                       | The formation of this unit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 82 |
| GeodeticCircle                        | A geodetic point and radius specifying a circle on the surface of the earth WGS84 where the radius is a great circle distance on the surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 82 |
| GeodeticLocation                      | A geodetic point, specified by latitude and longitude, with unspecified altitude. WGS84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 82 |
| GeodeticPoint                         | A geodetic point, specified by latitude, longitude and altitude.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 82 |
| GeodeticQuadrangle                    | A latitude-longitude quadrangle is a region bounded by two meridians and two parallels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 83 |
| HUMINTSignatureStruct                 | Describes the unit's susceptibility to human intelligence (HUMINT), i.e. information collected and provided by human sources.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 83 |
| HazeStruct                            | Type and density of haze material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 83 |
| Holding                               | Static and dynamic data about the holding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 83 |
| IPECommand                            | Represents an order for the specified entities to don protective equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 83 |
| IceStruct                             | Ice type, thickness and coverage on water surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 84 |
| InWeaponRangeReport                   | Report on a unit's ability to reach specific targets with its weapon systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 84 |
| IncomingConnectionStruct              | Characteristics of a specific incoming connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 84 |
| InitiateTransfer                      | An `InitiateTransfer` interaction is sent by a <b>Trigger Federate</b> to initiate a TMR request. Parameters include information on the participating federates, which object instances and what attributes are included in the transfer. The <b>Trigger Federate</b> shall generate a unique `EventId` to be used in all subsequent TMR interactions related to the event.                                                                                                                                                                                                                                                                                                                                                             | 84 |
| InstallationSymbolAmplificationStruct | Text amplifiers for Installation symbols.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 85 |
| LOGInteraction                        | Base class for all NETN Logistics Pattern Service Transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 86 |
| LandSurfaceCondition                  | Response with condition for land surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 86 |



|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| LayerStruct                                          | A layer of the Earth's environment expressed as a base altitude and a thickness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 86 |
| LinkStatusStruct                                     | Status of a physical network link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 87 |
| METOCInteraction                                     | Root class for requesting environment data from a METOC Service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 87 |
| MRMInteraction                                       | Base class for all MRM interactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 87 |
| MagicMove                                            | A simulation control task that instructs the entity to immediately jump to the specified location and update the new position and heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 87 |
| MagicResource                                        | A simulation control task that instructs the entity to set amount of available resources immediately                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 87 |
| Merge                                                | Instruction to merge the simulated AggregateUnit with the selected divided parts. After successful merge the divided parts are removed from the federation and their resources are combined with the AggregatedUnit.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 88 |
| MissionStruct                                        | The operational task the aggregate has been ordered to perform, the time the mission was assigned, and the estimated completion time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 88 |
| Mount                                                | Tasking of entity to mount the specified entity. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the entity to mount.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 88 |
| Move                                                 | Tasking of an entity to move in a specified direction for a given duration of time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 89 |
| MoveIntoFormation                                    | Tasking of an entity to move into the given formation on the given location with the given heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 89 |
| MoveToEntity                                         | Tasking of an entity to move to another entity. If a path is provided, the entity should follow the path as its route to the entity. The entity should align with the path from its current position to the nearest position or waypoint on the path. The entity should leave the path at position or waypoint on the path closest to the destination entity. The entity moves directly towards the destination entity if no path (or a zero-length path) is provided.                                                                                                                                                                                                       | 89 |
| MoveToLocation                                       | Tasking of an entity to move to the specified destination location. If a path is provided, the entity should follow the path as its route to the destination. The entity should align with the path from its current position to the nearest position or waypoint on the path. The entity should leave the path at position or waypoint on the path closest to the destination. The entity moves directly towards the destination location if no path (or a zero-length path) is provided.                                                                                                                                                                                   | 90 |
| NETNSupplyStruct                                     | Description of supply. Same encoding as RPR2 SupplyStruct.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 90 |
| NetworkDeviceEmptyCharacteristicsStruct              | Empty placeholder for future network device characteristics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 91 |
| NetworkDeviceGenericTransmitterCharacteristicsStruct | Generic description of network device transmitter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 91 |
| NetworkDeviceStruct                                  | Describes the type and capability of a network device. Each device must have at least a transmitter (TX) or receiver (RX). Transceivers should define both.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 91 |
| Observe                                              | Tasking of an entity to observe an area.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 91 |
| OfferRepair                                          | Is sent by a federate simulating the service providing entity in response to a RequestRepair interaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 92 |
| OfferService                                         | The OfferService is a response to a RequestService. Subclasses of this interaction for specific types of offers contain a more detailed description of the offer. This information may include when, where, how the service can be delivered.                                                                                                                                                                                                                                                                                                                                                                                                                                | 92 |
| OfferSupply                                          | Used by a supply service provider to indicate which of the requested material (amount and type) can be offered. In this request the consumer decides whether the loading is done by the provider or by the consumer.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 93 |
| OfferTransfer                                        | An `OfferTransfer` interaction is sent by a <b>Response Federate</b> as a reply to a `RequestTransfer` to indicate if a transfer can be accomplished or not. <b>Special Case</b> : When the transfer type is Divesting, the delivery order of the interaction `OfferTransfer` and the HLA ownership callback service `requestAttributeOwnershipRelease` at the receiving federate is not determined. Due to this, receiving a `requestAttributeOwnershipRelease` callback should be treated in the same way as receiving a `OfferTransfer` with a positive offer. The user-supplied tag in the callback contains the `EventId` which identifies a specific transfer process. | 93 |
| OfferTransport                                       | An Offer for a Transport support. The OfferTransport interaction shall be sent by the service providing federate in response to a RequestTransport interaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 93 |
| OperateCheckPoint                                    | Tasking of an entity to operate a checkpoint. The tasked entity should be within a certain distance (tolerance specified in the federation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 94 |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|                                | agreements) of the location of the checkpoint. If not, a separate move task should be issued first. The tasked entity activates an inactive checkpoint and operates the checkpoint for the specified duration of time.                                                                                                                                                                                                                    |     |
| Patrol                         | Tasking of an entity to perform a patrol. The tasked entity moves from its current position to the start of the patrol route and then moves according to patrol route from its start point in the path, through all waypoints.                                                                                                                                                                                                            | 94  |
| PatrolRepeating                | Tasking of an entity to repeat a patrol for a given duration. When the duration time has passed, then the last cycle of the patrol is completed before the task ends. If the time of a cycle takes longer then the interval time, then the cycle starts directly (without delay). If the time of a cycle takes less then the interval time, then the entity waits at the first point of the patrol route until the next cycle is started. | 95  |
| PhysicalGenericNetworkStruct   | Characteristics of a generic physical network.                                                                                                                                                                                                                                                                                                                                                                                            | 95  |
| PhysicalUndefinedNetworkStruct | (Empty) Characteristics of an undefined physical network.                                                                                                                                                                                                                                                                                                                                                                                 | 96  |
| PositionStatusReport           | Report on a unit's position, speed, and heading.                                                                                                                                                                                                                                                                                                                                                                                          | 96  |
| PrecipitationStruct            | Type and intensity of precipitation.                                                                                                                                                                                                                                                                                                                                                                                                      | 96  |
| ProbabilityHazardContourStruct | Represents the footprint covered by a CBRN hazard. This object covers the agent-specific effects which would be the output from a casualty model.                                                                                                                                                                                                                                                                                         | 96  |
| ProtectionEffectivenessStruct  | Protection effectiveness associated with a specific agent.                                                                                                                                                                                                                                                                                                                                                                                | 97  |
| QuerySupportedCapabilitiesETR  | Query which tasks, reports and SimCon actions that an entity supports. The queried entity shall respond with a CapabilitiesSupported message.                                                                                                                                                                                                                                                                                             | 97  |
| QuerySupportedCapabilitiesMRM  | A request to query the capabilities of a specified federation to provide support for MRM events. The queried federate shall respond with a CapabilitiesSupported interaction.                                                                                                                                                                                                                                                             | 97  |
| RawDataHazardContourStruct     | Contour locations bounding a given exposure value.                                                                                                                                                                                                                                                                                                                                                                                        | 97  |
| ReadyToReceiveService          | Issued by a service consumer to indicate that service delivery can start. The time of service delivery may be significantly later than indicating ready for service delivery.                                                                                                                                                                                                                                                             | 97  |
| RejectOffer                    | Used to reject an offer made by a service providing entity as indicated in a OfferService interaction. By issuing a RejectOffer interaction the service consuming entity informs the providing entity that the offer has been rejected.                                                                                                                                                                                                   | 98  |
| Relation                       | The specific value that represents the perceived hostility status.                                                                                                                                                                                                                                                                                                                                                                        | 98  |
| ReleaseDistributionStruct      | Mean and variance of the distribution of the particles or droplets in a release.                                                                                                                                                                                                                                                                                                                                                          | 98  |
| ReleaseDynamicsStruct          | Defines the dynamic properties of a release.                                                                                                                                                                                                                                                                                                                                                                                              | 98  |
| ReleaseSizeStruct              | Defines the properties of the initial size of a release.                                                                                                                                                                                                                                                                                                                                                                                  | 98  |
| RemoveCheckPoint               | Tasking of an entity to remove a checkpoint. This task removes a previously established checkpoint. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the checkpoint. If not, a separate move task should be issued first. After completion of the task, the checkpoint object should be deleted from the federation.                                           | 98  |
| RemovePassage                  | Task entity to remove the passage with the given UUID. The tasked entity should be within a certain distance tolerance (specified in the federation agreement) of one of the points of the passage to make the task possible.                                                                                                                                                                                                             | 99  |
| RepairComplete                 | This interaction is sent by the provider when the repair service is delivered to the consumer                                                                                                                                                                                                                                                                                                                                             | 99  |
| RepairStruct                   | Repairs associated with a specific materiel                                                                                                                                                                                                                                                                                                                                                                                               | 99  |
| RequestLandSurfaceCondition    | Request for land surface condition data.                                                                                                                                                                                                                                                                                                                                                                                                  | 100 |
| RequestMETOC                   | A request to a specified METOC Service to provide METOC data for a specific geographical reference. The request can result in either a response interaction including the requested data or registration of an EnvironmentCondition object for continuous updates.                                                                                                                                                                        | 100 |
| RequestMRM                     | A base class for all MRM Request events.                                                                                                                                                                                                                                                                                                                                                                                                  | 100 |
| RequestRepair                  | Sent by the consumer when a repair is needed. Specifies entity and type of repair                                                                                                                                                                                                                                                                                                                                                         | 100 |
| RequestService                 | A consumer federate initiates service negotiation using `RequestService`. A unique ServiceId and a reference to a `ConsumerEntity` are required parameters. A reference to a specific `ProviderEntity` and a system wall-clock time for when offers are expected `RequestTimeOut` are optional. Requests for specific types of services are defined as subclasses to `RequestService` and                                                 | 101 |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|                                   | include parameters for detailing the requirements of the request. This may include information when, where and how the service should be delivered.                                                                                                                                                                                                                                                                                  |     |
| RequestSubsurfaceLayerCondition   | Request for sub-surface condition data.                                                                                                                                                                                                                                                                                                                                                                                              | 101 |
| RequestSupply                     | The consumer sends a `RequestSupply` interaction to request supplies. The amount and type of supplies are specified in the required `SuppliesData` parameter. The required `TransferDirection` parameter indicates whether supplies are transferred from Consumer to provider or from Provider to consumer.                                                                                                                          | 102 |
| RequestTransfer                   | A `RequestTransfer` interaction is sent by the requesting federate to the responding federate. If the request is the result of a `InitiateTransfer` interaction, the parameters from that interaction shall be copied and used in the request including `EventId`. If the request is not triggered the <b>Request Federate</b> shall generate a unique `EventId` to be used in all subsequent TMR interactions related to the event. | 102 |
| RequestTransport                  | A request for a Transport service. Use a RequestTransport interaction to initiate a transport, embarkment or disembarkment of a platform.                                                                                                                                                                                                                                                                                            | 103 |
| RequestTroposphereLayerCondition  | Request for tropospheric environment condition data.                                                                                                                                                                                                                                                                                                                                                                                 | 103 |
| RequestWaterSurfaceCondition      | Request for water surface condition data.                                                                                                                                                                                                                                                                                                                                                                                            | 103 |
| RequestWeatherCondition           | Request for general weather data.                                                                                                                                                                                                                                                                                                                                                                                                    | 104 |
| RequestedConnection               | Characterisation of a requested connection.                                                                                                                                                                                                                                                                                                                                                                                          | 104 |
| ResourceStatusNumberStruct        | The name of a resource and the number of instances of that resource by health status.                                                                                                                                                                                                                                                                                                                                                | 104 |
| ResourceStatusReport              | Report on a unit's remaining amount resources.                                                                                                                                                                                                                                                                                                                                                                                       | 104 |
| ResponseMETOC                     | Response to a RequestEnvironmentCondition. RequestId parameter should match the corresponding parameter in the request.                                                                                                                                                                                                                                                                                                              | 105 |
| ResponseMRM                       | A response from the receiving federate indicating ability to comply with request.                                                                                                                                                                                                                                                                                                                                                    | 105 |
| SendSafetyRelatedBroadcastMessage | Tasks entity (the source) to send a safety related broadcast message (AIS message type 14). The source must represent an AIS station (vessel, SAR aircraft, etc).                                                                                                                                                                                                                                                                    | 105 |
| SendSafetyRelatedMessage          | Tasks entity (the source) to send a safety related message (AIS message type 12) to another entity (the destination). Both source and destination must represent an AIS station (vessel, SAR aircraft, etc).                                                                                                                                                                                                                         | 106 |
| SensorReport                      | Report on a unit's sensor detection of entities.                                                                                                                                                                                                                                                                                                                                                                                     | 106 |
| SensorStruct                      | Defines a sensor, operational status, damage status, coverage and ID                                                                                                                                                                                                                                                                                                                                                                 | 106 |
| ServiceComplete                   | Used by a service providing entity to inform the service consuming entity that the service has been delivered.                                                                                                                                                                                                                                                                                                                       | 107 |
| ServiceReceived                   | Used by a service consuming entity to inform the service providing entity that the service has been delivered.                                                                                                                                                                                                                                                                                                                       | 107 |
| ServiceStarted                    | Issued by a service provider to inform about the start of service delivery. The time of service delivery start may be significantly later than receiving an indication from the consumer that the service delivery can start.                                                                                                                                                                                                        | 107 |
| SetOrderedAltitude                | Tasking of an entity to set move to specified altitude.                                                                                                                                                                                                                                                                                                                                                                              | 107 |
| SetOrderedSpeed                   | Tasking of an entity to change speed.                                                                                                                                                                                                                                                                                                                                                                                                | 107 |
| SetRulesOfEngagement              | Tasking of an entity to change its rules of engagement.                                                                                                                                                                                                                                                                                                                                                                              | 108 |
| SetTransmitterStatus              | Tasking of an entity to switch on/off all of its transmitters.                                                                                                                                                                                                                                                                                                                                                                       | 108 |
| SnowStruct                        | Depth and density of snow cover.                                                                                                                                                                                                                                                                                                                                                                                                     | 109 |
| SpotReport                        | Report on a unit's awareness of spotted entities.                                                                                                                                                                                                                                                                                                                                                                                    | 109 |
| SpottedEntity                     | Description of the observed entity. The symbol contains information about the spotted entity's relation to the spotter and details about the type and echelon at the spotted entity.                                                                                                                                                                                                                                                 | 109 |
| SpottedEquipment                  | Equipment at the spotted entity.                                                                                                                                                                                                                                                                                                                                                                                                     | 110 |
| StatusReport                      | Report on a unit's own (perceived) state. This report should be generated with a frequency specified in the federation agreements.                                                                                                                                                                                                                                                                                                   | 110 |
| StopAtSideOfRoad                  | Tasking of an entity to stop at the side of the road. This task is only relevant for an entity that is moving along a road to a destination. The executing move task is cancelled, and a new move is defined towards a position at the side of the road (the simulator has to calculate this location).                                                                                                                              | 110 |
| SubsurfaceLayerCondition          | Response with subsurface body of water condition data.                                                                                                                                                                                                                                                                                                                                                                               | 110 |
| SupplyComplete                    | This interaction is sent by the provider when the supply is delivered to the consumer                                                                                                                                                                                                                                                                                                                                                | 111 |
| TMRInteraction                    | The `TMR_Interaction` interaction class is the base for all Transfer of Modelling Responsibility interactions. It provides a single parameter                                                                                                                                                                                                                                                                                        | 111 |

|                               |                                                                                                                                           |     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----|
|                               | used in all TMR interactions to uniquely identify TMR transactions between participating federates.                                       |     |
| TaskStatusReport              | A management task report regarding the status of a specific task assigned to an entity.                                                   | 111 |
| TransferResult                | A `TransferResult` interaction is sent by the <b>Request Federate</b> to indicate successful or unsuccessful completion of the transfer.  | 112 |
| TransportDestroyedEntities    | Used by a service provider to update information on damage state of entities under transport.                                             | 112 |
| TransportDisembarkmentStatus  | Is sent by the service provider federate, to inform the service consumer of the disembarkment state, after the ServiceStarted interaction | 112 |
| TransportEmbarkmentStatus     | Is sent by the service provider federate, to inform the service consumer of the embarkment state, after the ServiceStarted interaction    | 113 |
| TreatmentStruct               | Defines the properties for a CBRN treatment.                                                                                              | 113 |
| TroposphereLayerCondition     | Response with environment condition in volume of air.                                                                                     | 113 |
| TurnToHeading                 | Tasking of an entity to turn to the specified heading.                                                                                    | 114 |
| TurnToOrientation             | Tasking of an entity to rotate to a specified orientation, including pitch and roll.                                                      | 114 |
| UnderAttackStatusReport       | Report from a unit that it is under attack.                                                                                               | 114 |
| UnitSymbolAmplificationStruct | Text amplifiers for Unit symbols.                                                                                                         | 115 |
| VisualSignatureStruct         | Specifies the visual structure                                                                                                            | 115 |
| Wait                          | Tasking of an entity to wait for a duration of time.                                                                                      | 116 |
| WaterSurfaceCondition         | Response with water surface condition data.                                                                                               | 116 |
| WaveStruct                    | Water surface wave conditions and direction.                                                                                              | 117 |
| WeatherCondition              | Response with general weather condition data.                                                                                             | 117 |
| WindStruct                    | Wind direction and speeds.                                                                                                                | 117 |

## AISRadioSignal STRUCTURE

This is the root interaction class for all AIS messages. The interaction class contains a reference to the embedded system that represents the AIS transmitter. The embedded system should be modelled as a RadioTransmitter, an RPR-FOM object class that is subclassed from Embedded System.

| Name           | Type   | Description                                                                                                                       |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |

## AcceptOffer STRUCTURE

Use the AcceptOffer interaction to accept an offer received in an OfferService interaction.

| Name      | Type          | Description                                                                       |
|-----------|---------------|-----------------------------------------------------------------------------------|
| requestId | TransactionId | Required: Unique identifier for the requested service related to the transaction. |
| offerId   | TransactionId | Required: Unique offer identifier.                                                |

## AcknowledgeData STRUCTURE

Acknowledge data for Message Types 7 and 13.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |
| userId         | MMSIType      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| destination1   | MMSIType      | Required. MMSI number 1.                                                                                                          |
| destination2   | i32           | Optional (Default: not available). MMSI number 2.                                                                                 |

| Name         | Type | Description                                       |
|--------------|------|---------------------------------------------------|
| destination3 | i32  | Optional (Default: not available). MMSI number 3. |
| destination4 | i32  | Optional (Default: not available). MMSI number 4. |

## Activate STRUCTURE

Request federate to change of status of AggregateUnit to Active. If required the unit will be registered in the federation.

| Name          | Type                | Description                                                                                                                                                                                                                                             |
|---------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | TransactionId       | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate      | string              | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | UuidArrayOfHLByte16 | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |

## AddPassage STRUCTURE

Tasking of an entity to lay/build a passage between the two given points. The passage can, for example, be a passage through an obstacle or a bridge over a river. The tasked entity should be within a certain distance tolerance (specified in the federation agreement) of one of the points of the passage. A passage object should be registered in the federation with the provided UUID.

| Name                    | Type                    | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId           | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                    | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                  | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18              | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                  | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8           | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64           | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| point1                  | rpr.WorldLocationStruct | Required. The line between point1 and point2 defines the geometry of the passage.                                                                                                                                             |
| point2                  | rpr.WorldLocationStruct | Required. The line between point1 and point2 defines the geometry of the passage.                                                                                                                                             |
| uniqueId                | UUID                    | Required. A unique UUID of the passage.                                                                                                                                                                                       |

## AgentConcentrationStruct STRUCTURE

Concentration value associated with a specific agent.

| Name              | Type                     | Description                              |
|-------------------|--------------------------|------------------------------------------|
| meanConcentration | MassConcentrationFloat32 | Concentration in kg m-3.                 |
| agent             | AgentTypeEnum16          | Enumeration representation of the agent. |

Named by [ArrayOfAgentConcentrationStruct](#), [CBRNSensorReadingStruct](#).

## AgentMassStruct STRUCTURE

Mass of contaminant inside a vehicle brought in by an embedded unit.

| Name             | Type                                    | Description                              |
|------------------|-----------------------------------------|------------------------------------------|
| contaminatedMass | <a href="#">rpr.MassKilogramFloat32</a> | Mass in kg of contaminant.               |
| agent            | <a href="#">AgentTypeEnum16</a>         | Enumeration representation of the agent. |

Named by [ArrayOfAgentMassStruct](#).

## Aggregate STRUCTURE

Instruction to the AggregateFederate to perform aggregation of the specified AggregateUnit's parts.

| Name           | Type                                 | Description                                                                                                                                                                                                                                                    |
|----------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                        |
| federate       | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned.        |
| aggregateUnit  | <a href="#">UuidArrayOfHLAbyte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                                    |
| removeSubunits | <a href="#">bool</a>                 | Optional. Indicates if the disaggregate subunits (represented as aggregates in the federation) should be deleted from the federation execution. Default is TRUE - all subunits should be deleted. If FALSE all disaggregate subunits shall be set to inactive. |

## AisBaseStationReportAndUTCDateResponseData STRUCTURE

Common reporting structure for Message Types 4 and 11 for reporting UTC time and date and, at the same time, position. It is also used by AIS stations for determining if the station is within 120 NM for response to Messages Types 20 and 23.

| Name           | Type                             | Description                                                                                                                       |
|----------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>           | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a>    | Required. Message type identifier.                                                                                                |
| userId         | <a href="#">MMSIType</a>         | Required. The message is from the vessel identified by this MMSI.                                                                 |
| uTctime        | <a href="#">i64</a>              | Optional (Default: not available). Time of report.                                                                                |
| position       | <a href="#">GeodeticLocation</a> | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |

## AisBinaryMessageData STRUCTURE

Binary Payload for Message Types 6 and 8. The interpretation of the binary payload is controlled by: - The Designated Area Code (DAC), which is a jurisdiction code: 366 for the United States. It uses the same encoding as the area designator in MMMSIs; see [ITU-MID]. 1 designates international (ITU) messages. - The FID, which is the Functional ID for a message subtype. In some sources this is abbreviated FI.

| Name           | Type                          | Description                                                                                                                       |
|----------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a> | Required. Message type identifier.                                                                                                |

| Name               | Type                                   | Description                                                       |
|--------------------|----------------------------------------|-------------------------------------------------------------------|
| userId             | <a href="#">MMSIType</a>               | Required. The message is from the vessel identified by this MMSI. |
| designatedAreaCode | <a href="#">DesignatedAreaCodeType</a> | Required. Designated area code (DAC).                             |
| functionId         | <a href="#">FunctionIdType</a>         | Required. Functional ID (FID).                                    |
| data               | <a href="#">BinArrayType</a>           | Required. Binary data up to 920 bits.                             |

## AisMessage STRUCTURE

Super class for all AIS message types.

| Name           | Type                          | Description                                                                                                                       |
|----------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a> | Required. Message type identifier.                                                                                                |
| userId         | <a href="#">MMSIType</a>      | Required. The message is from the vessel identified by this MMSI.                                                                 |

## AisMessage1 STRUCTURE

Message Type 1: Position Report Class A. Message type for a scheduled position report; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor.

| Name                     | Type                                              | Description                                                                                                                       |
|--------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex           | <a href="#">string</a>                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId                | <a href="#">MsgIdEnumType</a>                     | Required. Message type identifier.                                                                                                |
| userId                   | <a href="#">MMSIType</a>                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| navigationalStatus       | <a href="#">NavigationStatusEnumType</a>          | Optional (Default: not defined). Navigational status.                                                                             |
| rateOfTurn               | <a href="#">f32</a>                               | Optional (Default: not available). Rate of turn.                                                                                  |
| speedOverGround          | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Optional (Default: not available). Speed over ground.                                                                             |
| position                 | <a href="#">GeodeticLocation</a>                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround         | <a href="#">f32</a>                               | Optional (Default: not available). Course over ground.                                                                            |
| trueHeading              | <a href="#">f32</a>                               | Optional (Default: not available). True heading.                                                                                  |
| uTctime                  | <a href="#">i64</a>                               | Optional (Default: not available). Time of the report.                                                                            |
| specialManeuverIndicator | <a href="#">ManeuverIndicatorEnumType</a>         | Optional (Default: not available). Maneuver indication.                                                                           |

## AisMessage10 STRUCTURE

Message Type 10: UTC/Date Inquiry. Request for UTC/Date information from an AIS base station.

| Name           | Type                          | Description                                                                                                                       |
|----------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a> | Required. Message type identifier.                                                                                                |



| Name        | Type     | Description                                                       |
|-------------|----------|-------------------------------------------------------------------|
| userId      | MMSIType | Required. The message is from the vessel identified by this MMSI. |
| destination | MMSIType | Required. Destination MMSI.                                       |

### AisMessage11 STRUCTURE

Message Type 11: UTC/Date Response. Identical to Message Type 4, with the semantics of a response to inquiry. This message type is only transmitted from a mobile station as a result of a UTC request message (Message Type 10).

| Name           | Type             | Description                                                                                                                       |
|----------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string           | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType    | Required. Message type identifier.                                                                                                |
| userId         | MMSIType         | Required. The message is from the vessel identified by this MMSI.                                                                 |
| uTCtime        | i64              | Optional (Default: not available). Time of report.                                                                                |
| position       | GeodeticLocation | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |

### AisMessage12 STRUCTURE

Type 12: Addressed Safety-Related Message. This is a point-to-point text message. The payload is interpreted as six-bit text.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |
| userId         | MMSIType      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| destination    | MMSIType      | Required. Destination MMSI.                                                                                                       |
| message        | string        | Required. 1-156 chars of six-bit text.                                                                                            |

### AisMessage13 STRUCTURE

Message Type 13: Safety-Related Acknowledgement. This message type is a receipt acknowledgement to senders of previous messages of Message Type 12.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |
| userId         | MMSIType      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| destination1   | MMSIType      | Required. MMSI number 1.                                                                                                          |
| destination2   | i32           | Optional (Default: not available). MMSI number 2.                                                                                 |
| destination3   | i32           | Optional (Default: not available). MMSI number 3.                                                                                 |
| destination4   | i32           | Optional (Default: not available). MMSI number 4.                                                                                 |

### AisMessage14 STRUCTURE

Type 14: Safety-Related Broadcast Message. This is a broadcast text message. The payload is interpreted as six-bit text.



| Name           | Type                          | Description                                                                                                                       |
|----------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a> | Required. Message type identifier.                                                                                                |
| userId         | <a href="#">MMSIType</a>      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| message        | <a href="#">string</a>        | Required. 1-161 chars of six-bit text.                                                                                            |

## AisMessage17 STRUCTURE

Message Type 17: DGNSS Broadcast Binary Message. This message type is used to broadcast differential corrections for GPS.

| Name           | Type                             | Description                                                                                                                       |
|----------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>           | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a>    | Required. Message type identifier.                                                                                                |
| userId         | <a href="#">MMSIType</a>         | Required. The message is from the vessel identified by this MMSI.                                                                 |
| position       | <a href="#">GeodeticLocation</a> | Optional (Default: not available). Position data.                                                                                 |

## AisMessage18 STRUCTURE

Message Type 18: Standard Class B CS Position Report. A less detailed report than message types 1-3 for vessels using Class B transmitters. Omits navigational status and rate of turn.

| Name             | Type                                              | Description                                                                                                                       |
|------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex   | <a href="#">string</a>                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId        | <a href="#">MsgIdEnumType</a>                     | Required. Message type identifier.                                                                                                |
| userId           | <a href="#">MMSIType</a>                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| speedOverGround  | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Optional (Default: not available). Speed over ground.                                                                             |
| position         | <a href="#">GeodeticLocation</a>                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround | <a href="#">f32</a>                               | Optional (Default: not available). Course over ground.                                                                            |
| trueHeading      | <a href="#">f32</a>                               | Optional (Default: not available). True heading.                                                                                  |
| uTcTime          | <a href="#">i64</a>                               | Optional (Default: not available). Time of the report.                                                                            |

## AisMessage19 STRUCTURE

Message Type 19: Extended Class B CS Position Report. A slightly more detailed report than Message Type 18 for vessels using Class B transmitters. Omits navigational status and rate of turn. Fields are as in the common navigation block and the Message Type 5 message.

| Name           | Type                          | Description                                                                                                                       |
|----------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | <a href="#">string</a>        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | <a href="#">MsgIdEnumType</a> | Required. Message type identifier.                                                                                                |
| userId         | <a href="#">MMSIType</a>      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| callsign       | <a href="#">string</a>        | Required. 7 six-bit characters for the callsign.                                                                                  |

| Name               | Type                                  | Description                                                          |
|--------------------|---------------------------------------|----------------------------------------------------------------------|
| name               | string                                | Required. 20 six-bit characters for the name.                        |
| shipType           | u8                                    | Optional (Default: not available). Ship type.                        |
| dimensionBow       | rpr.LengthMeter<br>Float32            | Optional (Default: not available). GPS Ant. distance from bow.       |
| dimensionStern     | rpr.LengthMeter<br>Float32            | Optional (Default: not available). GPS Ant. distance from stern.     |
| dimensionPort      | rpr.LengthMeter<br>Float32            | Optional (Default: not available). GPS Ant. distance from port.      |
| dimensionStarboard | rpr.LengthMeter<br>Float32            | Optional (Default: not available). GPS Ant. distance from starboard. |
| speedOverGround    | rpr.VelocityMeterPer<br>SecondFloat32 | Optional (Default: not available). Speed over ground.                |
| position           | GeodeticLocation                      | Optional (Default: not available). AIS (Lat,Lon) position.           |
| courseOverGround   | f32                                   | Optional (Default: not available). Course over ground.               |
| trueHeading        | f32                                   | Optional (Default: not available). True heading.                     |
| uTctime            | i64                                   | Optional (Default: not available). Time of the report.               |

## AisMessage2 STRUCTURE

Message Type 2: Position Report Class A. Message type for an assigned scheduled position report; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor.

| Name                         | Type                                  | Description                                                                                                                       |
|------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex               | string                                | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId                    | MsgIdEnumType                         | Required. Message type identifier.                                                                                                |
| userId                       | MMSIType                              | Required. The message is from the vessel identified by this MMSI.                                                                 |
| navigationalStatus           | NavigationStatusEnum<br>Type          | Optional (Default: not defined). Navigational status.                                                                             |
| rateOfTurn                   | f32                                   | Optional (Default: not available). Rate of turn.                                                                                  |
| speedOverGround              | rpr.VelocityMeterPer<br>SecondFloat32 | Optional (Default: not available). Speed over ground.                                                                             |
| position                     | GeodeticLocation                      | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround             | f32                                   | Optional (Default: not available). Course over ground.                                                                            |
| trueHeading                  | f32                                   | Optional (Default: not available). True heading.                                                                                  |
| uTctime                      | i64                                   | Optional (Default: not available). Time of the report.                                                                            |
| specialManeuver<br>Indicator | ManeuverIndicator<br>EnumType         | Optional (Default: not available). Maneuver indication.                                                                           |

## AisMessage21 STRUCTURE

Message Type 21: Aid-to-Navigation Report. Identification and location message to be emitted by aids to navigation such as buoys and lighthouses.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |

| Name               | Type                       | Description                                                                                                                                                                                                          |
|--------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| userId             | MMSIType                   | Required. The message is from the vessel identified by this MMSI.                                                                                                                                                    |
| aidType            | AidTypeEnumType            | Optional (Default: not specified). Type of aid.                                                                                                                                                                      |
| name               | string                     | Optional (Default: 34 six-bit @ charachters). Up to max 34 six-bit charachters for the name of the aid.                                                                                                              |
| position           | GeodeticLocation           | Optional (Default: not available). AIS (Lat,Lon) position.                                                                                                                                                           |
| dimensionBow       | rpr.LengthMeter<br>Float32 | Optional (Default: not available). Distance from bow.                                                                                                                                                                |
| dimensionStern     | rpr.LengthMeter<br>Float32 | Optional (Default: not available). Distance from stern.                                                                                                                                                              |
| dimensionPort      | rpr.LengthMeter<br>Float32 | Optional (Default: not available). Distance from port.                                                                                                                                                               |
| dimensionStarboard | rpr.LengthMeter<br>Float32 | Optional (Default: not available). Distance from starboard.                                                                                                                                                          |
| uTctime            | i64                        | Optional (Default: not available). Time of the report.                                                                                                                                                               |
| offPositionFlag    | bool                       | Optional (Default: false). Off-Position Indicator. The Off-Position Indicator is for floating Aids-to-Navigation only: false means on position; true means off position. Only valid if UTCsecond is available.       |
| virtualFlag        | bool                       | Optional (Default: false). Virtual-aid flag. The Virtual Aid flag is interpreted as follows: false = real Aid to Navigation at indicated position; true = virtual Aid to Navigation simulated by nearby AIS station. |

## AisMessage24 STRUCTURE

Message Type 24: Static Data Report. Equivalent of Message Type 5 for ships using Class B equipment. Also used to associate an MMSI with a name on either class A or class B equipment. This message type may be in part A or part B format. According to the standard, parts A and B are expected to be broadcast in adjacent pairs.

| Name               | Type                       | Description                                                                                                                       |
|--------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex     | string                     | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId          | MsgIdEnumType              | Required. Message type identifier.                                                                                                |
| userId             | MMSIType                   | Required. The message is from the vessel identified by this MMSI.                                                                 |
| callsign           | string                     | Required. 7 six-bit characters for the callsign.                                                                                  |
| name               | string                     | Required. 20 six-bit characters for the name.                                                                                     |
| shipType           | u8                         | Optional (Default: not available). Ship type.                                                                                     |
| dimensionBow       | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from bow.                                                                    |
| dimensionStern     | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from stern.                                                                  |
| dimensionPort      | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from port.                                                                   |
| dimensionStarboard | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from starboard.                                                              |
| partNumber         | PartNumberEnumType         | Required. Part number.                                                                                                            |
| manufacturerId     | ManufacturerIdType         | Required. (Part B) 3 six-bit chars.                                                                                               |
| unitModel          | UnitModelType              | Required. (Part B) Unit Model Code.                                                                                               |
| serialNumber       | SerialNumberType           | Required. (Part B) Serial Number.                                                                                                 |

## AisMessage27 STRUCTURE

Message Type 27: Long Range AIS Broadcast message. This message type is primarily intended for long-range detection of AIS Class A equipped vessels (typically by satellite). This message has a similar content to Messages 1, 2 and 3, but the total number of bits has been compressed to allow for increased propagation delays associated with long-range detection.

| Name             | Type                              | Description                                                                                                                       |
|------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex   | string                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId        | MsgIdEnumType                     | Required. Message type identifier.                                                                                                |
| userId           | MMSIType                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| navigationStatus | NavigationStatusEnumType          | Optional (Default: not defined). Navigational status.                                                                             |
| position         | GeodeticLocation                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| speedOverGround  | rpr.VelocityMeterPerSecondFloat32 | Optional (Default: not available). Speed over ground.                                                                             |
| courseOverGround | f32                               | Optional (Default: not available). Course over ground.                                                                            |

## AisMessage3 STRUCTURE

Message Type 3: Position Report Class A. Message type for a special position report, response to interrogation; Class A shipborne mobile equipment. This message transmits information pertaining to a ships navigation: Longitude and latitude, time, heading, speed, ships navigation status (under power, at anchor...). This message is transmitted every 2 to 10 seconds while underway, and every 3 minutes while at anchor.

| Name                     | Type                              | Description                                                                                                                       |
|--------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex           | string                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId                | MsgIdEnumType                     | Required. Message type identifier.                                                                                                |
| userId                   | MMSIType                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| navigationalStatus       | NavigationStatusEnumType          | Optional (Default: not defined). Navigational status.                                                                             |
| rateOfTurn               | f32                               | Optional (Default: not available). Rate of turn.                                                                                  |
| speedOverGround          | rpr.VelocityMeterPerSecondFloat32 | Optional (Default: not available). Speed over ground.                                                                             |
| position                 | GeodeticLocation                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround         | f32                               | Optional (Default: not available). Course over ground.                                                                            |
| trueHeading              | f32                               | Optional (Default: not available). True heading.                                                                                  |
| uTcTime                  | i64                               | Optional (Default: not available). Time of the report.                                                                            |
| specialManeuverIndicator | ManeuverIndicatorEnumType         | Optional (Default: not available). Maneuver indication.                                                                           |

## AisMessage4 STRUCTURE

Message Type 4: Base Station Report. This message is to be used by fixed-location base stations to periodically report a position and time reference.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |

| Name     | Type             | Description                                                       |
|----------|------------------|-------------------------------------------------------------------|
| userId   | MMSIType         | Required. The message is from the vessel identified by this MMSI. |
| uTctime  | i64              | Optional (Default: not available). Time of report.                |
| position | GeodeticLocation | Optional (Default: not available). AIS (Lat,Lon) position.        |

## AisMessage5 STRUCTURE

Message Type 5: Static and Voyage Related Data. This message type is transmitted every 6 minutes and should only be used by Class A shipborne and SAR aircraft AIS stations when reporting static or voyage related data.

| Name               | Type                       | Description                                                                                                                                                                                   |
|--------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex     | string                     | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object.                                                             |
| messageId          | MsgIdEnumType              | Required. Message type identifier.                                                                                                                                                            |
| userId             | MMSIType                   | Required. The message is from the vessel identified by this MMSI.                                                                                                                             |
| callsign           | string                     | Required. 7 six-bit characters for the callsign.                                                                                                                                              |
| name               | string                     | Required. 20 six-bit characters for the name.                                                                                                                                                 |
| shipType           | u8                         | Optional (Default: not available). Ship type.                                                                                                                                                 |
| dimensionBow       | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from bow.                                                                                                                                |
| dimensionStern     | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from stern.                                                                                                                              |
| dimensionPort      | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from port.                                                                                                                               |
| dimensionStarboard | rpr.LengthMeter<br>Float32 | Optional (Default: not available). GPS Ant. distance from starboard.                                                                                                                          |
| imo                | i32                        | Optional (Default: zero). The IMO ship ID number. Not applicable to SAR aircraft.                                                                                                             |
| eTAtime            | i64                        | Optional (Default: not available). Estimated time of arrival. Not applicable to SAR aircraft. If a zero value is provided then this shall be interpreted as not available.                    |
| draught            | f32                        | Optional (Default: not available). Maximum present static draught.                                                                                                                            |
| destination        | string                     | Optional (Default: not available, 20 six-bit '@' value). 20 six-bit characters for the destination. For SAR aircraft, the use of this field may be decided by the responsible administration. |

## AisMessage6 STRUCTURE

Message Type 6: Binary Addressed Message. This message type is an addressed point-to-point message with a binary payload.

| Name               | Type                       | Description                                                                                                                       |
|--------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex     | string                     | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId          | MsgIdEnumType              | Required. Message type identifier.                                                                                                |
| userId             | MMSIType                   | Required. The message is from the vessel identified by this MMSI.                                                                 |
| designatedAreaCode | DesignatedAreaCode<br>Type | Required. Designated area code (DAC).                                                                                             |
| functionId         | FunctionIdType             | Required. Functional ID (FID).                                                                                                    |
| data               | BinArrayType               | Required. Binary data up to 920 bits.                                                                                             |

| Name        | Type     | Description                 |
|-------------|----------|-----------------------------|
| destination | MMSIType | Required. Destination MMSI. |

## AisMessage7 STRUCTURE

Message Type 7: Binary Acknowledge. This message type is a receipt acknowledgement to the senders of a previous messages of Message Type 6. Up to 4 destination MMSIs can be acknowledged in one message.

| Name           | Type          | Description                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex | string        | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId      | MsgIdEnumType | Required. Message type identifier.                                                                                                |
| userId         | MMSIType      | Required. The message is from the vessel identified by this MMSI.                                                                 |
| destination1   | MMSIType      | Required. MMSI number 1.                                                                                                          |
| destination2   | i32           | Optional (Default: not available). MMSI number 2.                                                                                 |
| destination3   | i32           | Optional (Default: not available). MMSI number 3.                                                                                 |
| destination4   | i32           | Optional (Default: not available). MMSI number 4.                                                                                 |

## AisMessage8 STRUCTURE

Message Type 8: Binary Broadcast Message. This message type is a broadcast message with a binary payload.

| Name               | Type                   | Description                                                                                                                       |
|--------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex     | string                 | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId          | MsgIdEnumType          | Required. Message type identifier.                                                                                                |
| userId             | MMSIType               | Required. The message is from the vessel identified by this MMSI.                                                                 |
| designatedAreaCode | DesignatedAreaCodeType | Required. Designated area code (DAC).                                                                                             |
| functionId         | FunctionIdType         | Required. Functional ID (FID).                                                                                                    |
| data               | BinArrayType           | Required. Binary data up to 920 bits.                                                                                             |

## AisMessage9 STRUCTURE

Message type 9: Standard SAR Aircraft Position Report. Tracking information for search-and-rescue aircraft.

| Name             | Type                              | Description                                                                                                                       |
|------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex   | string                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId        | MsgIdEnumType                     | Required. Message type identifier.                                                                                                |
| userId           | MMSIType                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| altitude         | f64                               | Optional (Default: not available). Altitude.                                                                                      |
| speedOverGround  | rpr.VelocityMeterPerSecondFloat32 | Optional (Default: not available). Speed over ground.                                                                             |
| position         | GeodeticLocation                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround | f32                               | Optional (Default: not available). Course over ground.                                                                            |
| uTctime          | i64                               | Optional (Default: not available). Time of the report.                                                                            |

## AisNavigationData STRUCTURE

Message Types 1, 2 and 3 share a common reporting structure for navigational data, captured in this class.

| Name                     | Type                                              | Description                                                                                                                       |
|--------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex           | <a href="#">string</a>                            | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId                | <a href="#">MsgIdEnumType</a>                     | Required. Message type identifier.                                                                                                |
| userId                   | <a href="#">MMSIType</a>                          | Required. The message is from the vessel identified by this MMSI.                                                                 |
| navigationalStatus       | <a href="#">NavigationStatusEnumType</a>          | Optional (Default: not defined). Navigational status.                                                                             |
| rateOfTurn               | <a href="#">f32</a>                               | Optional (Default: not available). Rate of turn.                                                                                  |
| speedOverGround          | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Optional (Default: not available). Speed over ground.                                                                             |
| position                 | <a href="#">GeodeticLocation</a>                  | Optional (Default: not available). AIS (Lat,Lon) position.                                                                        |
| courseOverGround         | <a href="#">f32</a>                               | Optional (Default: not available). Course over ground.                                                                            |
| trueHeading              | <a href="#">f32</a>                               | Optional (Default: not available). True heading.                                                                                  |
| uTcTime                  | <a href="#">i64</a>                               | Optional (Default: not available). Time of the report.                                                                            |
| specialManeuverIndicator | <a href="#">ManeuverIndicatorEnumType</a>         | Optional (Default: not available). Maneuver indication.                                                                           |

## AisStaticAndVoyageData STRUCTURE

Common Static and Voyage Data for Message Types 5, 19 and 24.

| Name               | Type                                   | Description                                                                                                                       |
|--------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| hostRadioIndex     | <a href="#">string</a>                 | Optional (Default: not available). The HostRadioIndex is a unique string that identifies the name of the RadioTransmitter object. |
| messageId          | <a href="#">MsgIdEnumType</a>          | Required. Message type identifier.                                                                                                |
| userId             | <a href="#">MMSIType</a>               | Required. The message is from the vessel identified by this MMSI.                                                                 |
| callsign           | <a href="#">string</a>                 | Required. 7 six-bit characters for the callsign.                                                                                  |
| name               | <a href="#">string</a>                 | Required. 20 six-bit characters for the name.                                                                                     |
| shipType           | <a href="#">u8</a>                     | Optional (Default: not available). Ship type.                                                                                     |
| dimensionBow       | <a href="#">rpr.LengthMeterFloat32</a> | Optional (Default: not available). GPS Ant. distance from bow.                                                                    |
| dimensionStern     | <a href="#">rpr.LengthMeterFloat32</a> | Optional (Default: not available). GPS Ant. distance from stern.                                                                  |
| dimensionPort      | <a href="#">rpr.LengthMeterFloat32</a> | Optional (Default: not available). GPS Ant. distance from port.                                                                   |
| dimensionStarboard | <a href="#">rpr.LengthMeterFloat32</a> | Optional (Default: not available). GPS Ant. distance from starboard.                                                              |

## AppointmentStruct STRUCTURE

Date ; Time and Location of an appointment. When date and time is set to zero (0), implies no date-time specification

| Name     | Type                                    | Description                                                  |
|----------|-----------------------------------------|--------------------------------------------------------------|
| dateTime | <a href="#">EpochTimeSecInt64</a>       | Date time in second since 1 january 1970 for the appointment |
| location | <a href="#">rpr.WorldLocationStruct</a> | Location of the appointment                                  |

Named by [OfferRepair](#), [OfferService](#), [OfferSupply](#), [OfferTransport](#), [RequestRepair](#), [RequestService](#), [RequestSupply](#), [RequestTransport](#).



## CBRNAlarmStruct STRUCTURE

Properties of a CBRN alarm.

| Name     | Type                                    | Description                                                    |
|----------|-----------------------------------------|----------------------------------------------------------------|
| time     | <a href="#">EpochTimeSecInt64</a>       | Simulation time of this detector alarm.                        |
| location | <a href="#">rpr.WorldLocationStruct</a> | Location of the detector alarm.                                |
| agent    | <a href="#">AgentTypeEnum16</a>         | Agent that the detector believes to have triggered this alarm. |

Named by [CBRNDetector](#), [CBRNDetectorAlarm](#).

## CBRNCasualty STRUCTURE

Represents a CBRN casualty caused by exposure. This is for use with federates that cannot support the TriageLevel attribute in CBRN\_Human.

| Name        | Type                                      | Description                                          |
|-------------|-------------------------------------------|------------------------------------------------------|
| uniqueID    | <a href="#">UuidArrayOfHLAbyte16</a>      | Unique representation of the entity's ID.            |
| triageLevel | <a href="#">CBRNDamageEnum8</a>           | Triage level of this entity.                         |
| exposures   | <a href="#">ArrayOfCBRNExposureStruct</a> | Array of agents to which this unit has been exposed. |

## CBRNDetectorAlarm STRUCTURE

Represents the alarm trigger of a previously registered CBRN detector.

| Name       | Type                                 | Description                                 |
|------------|--------------------------------------|---------------------------------------------|
| detectorID | <a href="#">UuidArrayOfHLAbyte16</a> | Unique representation of the detector's ID. |
| alarm      | <a href="#">CBRNAlarmStruct</a>      | Details of the alarm.                       |

## CBRNExposureStruct STRUCTURE

Dosage exposure value associated with a specific agent.

| Name     | Type                                               | Description                                                                 |
|----------|----------------------------------------------------|-----------------------------------------------------------------------------|
| exposure | <a href="#">DosageKgSecondPerMeterCubedFloat32</a> | Dosage value (in kg.s.m-3) which has caused this casualty. Defaults to 0.0. |
| agent    | <a href="#">AgentTypeEnum16</a>                    | Enumeration representation of the agent.                                    |

Named by [ArrayOfCBRNExposureStruct](#).

## CBRNFacilityUpdate STRUCTURE

Represents an order for the specified entities to enter or leave the CBRN facility.

| Name             | Type                                 | Description                                                                                                   |
|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| facilityID       | <a href="#">UuidArrayOfHLAbyte16</a> | The unique ID of the CBRN facility.                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>          | List of unique IDs of the entities who will enter or exit the CBRN facility.                                  |
| isEntry          | <a href="#">bool</a>                 | Determines whether the entities are entering (IsEntry = true) or exiting (IsEntry = false) the CBRN facility. |

## CBRNPlatformUpdate STRUCTURE

Represents an update to the contaminating mass inside a vehicle due to embedded entities.



| Name          | Type                                   | Description                                                             |
|---------------|----------------------------------------|-------------------------------------------------------------------------|
| platformID    | <a href="#">UuidArrayOfHLAbyte16</a>   | The unique ID of the platform.                                          |
| contamination | <a href="#">ArrayOfAgentMassStruct</a> | New state of CBRN hazardous agent inside vehicle due to embedded units. |

## CBRNRelease STRUCTURE

Communicates information associated with the release of hazardous agent.

| Name            | Type                                     | Description                                                                                                                       |
|-----------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| uniqueID        | <a href="#">UuidArrayOfHLAbyte16</a>     | Unique representation of the release's ID.                                                                                        |
| agent           | <a href="#">AgentTypeEnum16</a>          | The type of released CBRN hazardous agent.                                                                                        |
| location        | <a href="#">rpr.WorldLocationStruct</a>  | Initial location of the release in the geocentric location system.                                                                |
| mass            | <a href="#">rpr.MassKilogramFloat32</a>  | Total released agent mass in kg.                                                                                                  |
| duration        | <a href="#">rpr.TimeSecondInteger32</a>  | Duration in seconds over which the release takes place.                                                                           |
| releaseSize     | <a href="#">ReleaseSizeStruct</a>        | The initial size of the release including initial Gaussian sigmas of the released puff and mean & variance of released particles. |
| releaseDynamics | <a href="#">ReleaseDynamicsStruct</a>    | Temperature difference and density ratio of released material relative to the atmosphere.                                         |
| releaseVelocity | <a href="#">rpr.VelocityVectorStruct</a> | Velocity of the source term.                                                                                                      |

## CBRNSensorReadingStruct STRUCTURE

Timed CBRN sensor reading.

| Name    | Type                                     | Description                                    |
|---------|------------------------------------------|------------------------------------------------|
| time    | <a href="#">EpochTimeSecInt64</a>        | Simulation time of this sensor reading.        |
| reading | <a href="#">AgentConcentrationStruct</a> | Agent and concentration value in this reading. |

Named by [ArrayOfCBRNSensorReadingStruct](#).

## CBRNSensorUpdate STRUCTURE

Sends information about the current state of a previously registered CBRN sensor.

| Name     | Type                                            | Description                               |
|----------|-------------------------------------------------|-------------------------------------------|
| time     | <a href="#">EpochTimeSecInt64</a>               | Time of this sensor update                |
| sensorID | <a href="#">UuidArrayOfHLAbyte16</a>            | Unique representation of the sensor's ID. |
| readings | <a href="#">ArrayOfAgentConcentrationStruct</a> | Readings for this sensor                  |

## CBRNTreatmentCommand STRUCTURE

Represents an order for the specified entities to receive the list of treatments.

| Name     | Type                        | Description                                                        |
|----------|-----------------------------|--------------------------------------------------------------------|
| unitList | <a href="#">ArrayOfUuid</a> | List of unique IDs of the entities who will receive the treatment. |

| Name       | Type                            | Description                                      |
|------------|---------------------------------|--------------------------------------------------|
| treatments | <a href="#">TreatmentStruct</a> | Type of treatment to be applied to the entities. |

## COLPROUpdate STRUCTURE

Represents an order for the specified entities to enter or leave the COLPRO.

| Name             | Type                                 | Description                                                                                                   |
|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| facilityID       | <a href="#">UuidArrayOfHLAByte16</a> | The unique ID of the CBRN facility.                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>          | List of unique IDs of the entities who will enter or exit the CBRN facility.                                  |
| isEntry          | <a href="#">bool</a>                 | Determines whether the entities are entering (IsEntry = true) or exiting (IsEntry = false) the CBRN facility. |

## CancelAllTasks STRUCTURE

Management task to an entity to cancel all queued and ongoing tasks.

| Name                    | Type                          | Description                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskManagementId        | <a href="#">TransactionId</a> | Required. Unique identifier for the management task.                                                                                                                                                                                                |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that should execute the task.                                                                                                                                                                                     |
| tasker                  | <a href="#">string</a>        | Optional. Identifies the commander of the task.                                                                                                                                                                                                     |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer task management messages. If not provided, the message transmission should not be modelled and federates should receive and act on the task management messages directly. |

## CancelOffer STRUCTURE

Used by provider to cancel an already made offer before it has been accepted. Used if the OfferTimeOut has passed.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |
| offerId   | <a href="#">TransactionId</a> | Required: Reference to the cancelled offer.                                       |

## CancelRequestLOG STRUCTURE

Used by either a service consuming entity or a service providing entity to inform about early termination of the service delivery or in some cases termination of the service request before delivery has begun.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |
| reason    | <a href="#">string</a>        | Allows to inform about the reason of the cancel (free text)                       |

## CancelRequestTMR STRUCTURE

A `CancelRequest` interaction can be sent by the **Request Federate** to cancel a request.

| Name    | Type                          | Description                                                                                                                                                                     |
|---------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId | <a href="#">TransactionId</a> | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter |

| Name   | Type                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                              | is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |
| reason | CancellationReason<br>Enum32 | Optional. An enumerated value that describes the reason for the cancellation: `Other` = When a Requesting Federate for some reason except time out decides not to complete a transfer. `TimeOut` = When a model's time limit for receiving a TMR offer has passed.                                                                                                                                                                          |

## CancelSpecifiedTasks STRUCTURE

Management task to an entity to cancel all specified tasks.

| Name                        | Type           | Description                                                                                                                                                                                                                                         |
|-----------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskManagementId            | TransactionId  | Required. Unique identifier for the management task.                                                                                                                                                                                                |
| taskee                      | UUID           | Required. Reference to the entity that should execute the task.                                                                                                                                                                                     |
| tasker                      | string         | Optional. Identifies the commander of the task.                                                                                                                                                                                                     |
| communicationNetwork<br>Ids | ArrayOfText64  | Optional. Reference to communication networks (NETN-COM) used to transfer task management messages. If not provided, the message transmission should not be modelled and federates should receive and act on the task management messages directly. |
| tasks                       | ArrayOfTaskIds | Required. Tasks that should be cancelled.                                                                                                                                                                                                           |

## CapabilitiesSupportedETR STRUCTURE

Provides the set of ETR\_Task, ETR\_SimCon and ETR\_Report interactions that the federate modelling the entity supports. This interaction is in response to a QuerySupportedCapabilities.

| Name            | Type          | Description                                                                                                                                                     |
|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId          | TransactionId | Required. Unique identifier for the simulation control task.                                                                                                    |
| taskee          | UUID          | Required. Reference to the entity that should execute the task.                                                                                                 |
| tasker          | string        | Optional. Identifies the commander of the task.                                                                                                                 |
| capabilityNames | ArrayOfNames  | Required. Array of task and report names that the entity supports; the names match with the FOM class names of NETN-ETR Tasks, Reports and SimCon interactions. |

## CapabilitiesSupportedMRM STRUCTURE

An interaction sent in response to a QuerySupportedCapabilities request. The response includes a list of names of the supported capabilities for the Aggregate unit specified in the query. The names are one or more of 'Aggregate', 'Disaggregate', 'Divide', 'Merge', 'Activate' and 'Inactivate'.

| Name            | Type              | Description                                                                                                                                                                                                        |
|-----------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId         | TransactionId     | Unique identifier for all MRM interactions belonging to the same request/response event.                                                                                                                           |
| capabilityNames | ArrayOfStringType | Required. A list of names of the supported capabilities for the Aggregate entity specified in the query. The names are one or more of 'Aggregate', 'Disaggregate', 'Divide', 'Merge', 'Activate' and 'Inactivate'. |

## ClearObstacle STRUCTURE

Tasking of an entity to clear an obstacle or minefield with the given UUID. The tasked entity should be within a certain distance (tolerance specified in the federation agreement) of one of the points of the geometry of the obstacle.

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                | <a href="#">UUID</a>          | Required. Reference to the Obstacle to clear.                                                                                                                                                                                 |

## CloudStruct STRUCTURE

Description of cloud layer type, coverage and density.

| Name     | Type                               | Description                |
|----------|------------------------------------|----------------------------|
| type     | <a href="#">CloudTypeEnum32</a>    | Type of cloud              |
| coverage | <a href="#">rpr.PercentFloat32</a> | Cloud coverage as percent. |
| density  | <a href="#">MassDensityFloat32</a> | Average density of clouds. |

Named by [TroposphereLayer](#), [TroposphereLayerCondition](#).

## CommunicationNetworkStruct STRUCTURE

CommunicationNetInstance that equipment monitors or uses to communicate with a mission.

| Name        | Type                                           | Description                                                                                         |
|-------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| service     | <a href="#">CommunicationServiceTypeEnum32</a> | Required. The type of service being used for communication.                                         |
| owner       | <a href="#">UUID</a>                           | UUID reference to the Unit that owns a particular item of data as part of that unit's organization. |
| networkName | <a href="#">Text64</a>                         | Name of the Communication network.                                                                  |

Named by [ArrayOfCommunicationNetworks](#).

## ConnectionReceiverStruct STRUCTURE

Characteristics of a connection to a receiver.

| Name             | Type                                                 | Description                                                                               |
|------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------|
| receiverEntityId | <a href="#">UUID</a>                                 | Reference to a simulated entity receiving data through the connection.                    |
| latency          | <a href="#">rpr.TimeMillisecondUnsignedInteger32</a> | Latency in reaching the receiving entity.                                                 |
| bandwidth        | <a href="#">BitsPerSecond</a>                        | The bandwidth of the connection to the receiving entity,                                  |
| reliability      | <a href="#">PercentFloat64</a>                       | The reliability of the connection to the receiving entity.                                |
| hopCount         | <a href="#">rpr.Integer16</a>                        | Number of hops required to reach the receiver. If set to 0 = not specified or calculated. |

Named by [ConnectionReceiverArray](#).

## CreateMinefield STRUCTURE

Tasking of an entity to create a minefield within the specified geometry. When the task is completed, a minefield object should be published in the federation (e.g. RPR-FOM Minefield). This tasking interaction is different from the RPR-FOM MinefieldObjectTransaction interaction, which asks a federate to create a minefield magically.

| Name                        | Type                                   | Description                                                                                                                                                                                                                   |
|-----------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                      | <a href="#">TransactionId</a>          | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                      | <a href="#">UUID</a>                   | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                      | <a href="#">string</a>                 | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen                   | <a href="#">Datetime18</a>             | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                         | <a href="#">string</a>                 | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                    | <a href="#">TaskModeEnum8</a>          | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids | <a href="#">ArrayOfText64</a>          | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                    | <a href="#">UUID</a>                   | Required. A unique UUID of the Obstacle.                                                                                                                                                                                      |
| obstacleType                | <a href="#">rpr.EntityTypeStruct</a>   | Required. The type of the obstacle defined by the entity type of the obstacle (SISO REF-010 Land Culture Features) .                                                                                                          |
| geometry                    | <a href="#">AreaVariantStruct</a>      | Required. Area of the obstacle.                                                                                                                                                                                               |
| typeOfMines                 | <a href="#">rpr.EntityTypeStruct</a>   | Required. Type of mines used defined in SISO-REF-010.                                                                                                                                                                         |
| mineCountMineDensity        | <a href="#">MineCountVariantStruct</a> | Required. Count or density of mines.                                                                                                                                                                                          |

## CreateObstacle STRUCTURE

Tasking of an entity to create an obstacle with the given geometry. The tasked entity should be within a certain distance (tolerance specified in the federation agreement) of one of the points of the geometry.

| Name                        | Type                                 | Description                                                                                                                                                                                                                   |
|-----------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                      | <a href="#">TransactionId</a>        | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                      | <a href="#">UUID</a>                 | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                      | <a href="#">string</a>               | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen                   | <a href="#">Datetime18</a>           | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                         | <a href="#">string</a>               | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                    | <a href="#">TaskModeEnum8</a>        | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids | <a href="#">ArrayOfText64</a>        | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                    | <a href="#">UUID</a>                 | Required. A unique UUID of the Obstacle.                                                                                                                                                                                      |
| obstacleType                | <a href="#">rpr.EntityTypeStruct</a> | Required. The type of the obstacle defined by the entity type of the obstacle (SISO REF-010 Land Culture Features) .                                                                                                          |
| geometry                    | <a href="#">AreaVariantStruct</a>    | Required. Area of the obstacle.                                                                                                                                                                                               |

## CurrentStruct STRUCTURE

Water current direction and speed.

| Name      | Type                                              | Description           |
|-----------|---------------------------------------------------|-----------------------|
| direction | <a href="#">DirectionDegreesFloat32</a>           | Direction of current. |
| speed     | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Speed of current.     |

Named by [SubsurfaceLayer](#), [SubsurfaceLayerCondition](#), [WaterSurface](#), [WaterSurfaceCondition](#).

## DamageStatusReport STRUCTURE

Report on a unit's damage status.

| Name                    | Type                                       | Description                                                                                                                                                                                                                      |
|-------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                | <a href="#">TransactionId</a>              | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                    | <a href="#">DateTime18</a>                 | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>              | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | <a href="#">string</a>                     | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| entityId                | <a href="#">UUID</a>                       | Required. The entity that is reported about.                                                                                                                                                                                     |
| damageType              | <a href="#">DamageStatusEnhancedEnum32</a> | Required. Damage state of the reported entity.                                                                                                                                                                                   |

## Deactivate STRUCTURE

Request change of status of AggregateUnit to Inactive and if indicated remove it from the federation.

| Name          | Type                                 | Description                                                                                                                                                                                                                                             |
|---------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate      | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | <a href="#">UuidArrayOfHLAbyte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |
| removeUnit    | <a href="#">bool</a>                 | Optional. Indicates if the Aggregate Unit shall be removed as an object instance in the federation. Default = FALSE - keep object instance in federation.                                                                                               |

## DecontaminationStationUpdate STRUCTURE

Represents an order for the specified entities to enter or leave the decontamination station.

| Name             | Type                                 | Description                                                                                                   |
|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| facilityID       | <a href="#">UuidArrayOfHLAbyte16</a> | The unique ID of the CBRN facility.                                                                           |
| embeddedUnitList | <a href="#">ArrayOfUuid</a>          | List of unique IDs of the entities who will enter or exit the CBRN facility.                                  |
| isEntry          | <a href="#">bool</a>                 | Determines whether the entities are entering (IsEntry = true) or exiting (IsEntry = false) the CBRN facility. |

## Disaggregate STRUCTURE

Instruction to perform a full disaggregation of a AggregatedUnit. All subunits and platforms will be registered in the federation.

| Name          | Type                                 | Description                                                                                                                                                                                                                                             |
|---------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate      | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | <a href="#">UuidArrayOfHLAbyte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |

## Dismount STRUCTURE

Tasking of an entity to dismount from a mounted position. When the task is completed, the tasked entity is no longer attached to or embarked in another entity.

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |

## DispositionStruct STRUCTURE

Positioning of unit or equipment item.

| Name                | Type                                              | Description                                                       |
|---------------------|---------------------------------------------------|-------------------------------------------------------------------|
| location            | <a href="#">GeodeticPoint</a>                     | Geodetic location.                                                |
| directionOfMovement | <a href="#">DirectionDegreesFloat32</a>           | Direction of movement.                                            |
| speed               | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Speed of movement.                                                |
| formationPosition   | <a href="#">FormationPositionStruct</a>           | The position of this unit/equipment in the higher unit formation. |

Named by [EquipmentItem](#), [Unit](#).

## DisruptCommunication STRUCTURE

Tasking of an entity to introduce a communication network disruption.

| Name   | Type                          | Description                               |
|--------|-------------------------------|-------------------------------------------|
| taskId | <a href="#">TransactionId</a> | Required. Unique identifier for the task. |

| Name                             | Type               | Description                                                                                                                                                                                                                   |
|----------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskee                           | UUID               | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                           | string             | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen                        | Datetime18         | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                              | string             | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                         | TaskModeEnum8      | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids      | ArrayOfText64      | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| affectedCommunication<br>Network | Text64             | Required. Reference to the affected Communication Network. If not provided all communication networks are affected.                                                                                                           |
| affectedArea                     | GeodeticQuadrangle | Required. Area affected by disruption.                                                                                                                                                                                        |
| disruptionEnabled                | bool               | Required. Disruption enabled or disabled. Default is True = Enabled.                                                                                                                                                          |

## Divide STRUCTURE

Instruction to divide the simulated AggregateUnit into multiple simulated object. The resources are divided among the simulated entities. After successful division two simulated entities represent the entire Unit. 1) The original AggregateUnit and 2) the divided unit or platform. Both these entities are simulated until merged.

| Name                         | Type                        | Description                                                                                                                                                                                                                                             |
|------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId                      | TransactionId               | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate                     | string                      | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit                | UuidArrayOf<br>HLAbyte16    | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |
| equipment                    | ArrayOfResource<br>Status   | Optional. Amount of equipment of different type and health status to be divided.                                                                                                                                                                        |
| personnel                    | ArrayOfResource<br>Status   | Optional. Amount of personnel of different type and health status to be divided.                                                                                                                                                                        |
| supplies                     | NETNArrayOfSupply<br>Struct | Optional. Amount of supplies to divide.                                                                                                                                                                                                                 |
| registerPhysical<br>Entities | bool                        | Optional. If true all Equipment of type Platform and Lifeform are published as individual objects in the federation.                                                                                                                                    |

## ETRReport STRUCTURE

A base interaction class for more specialized report interaction classes.

| Name                        | Type          | Description                                                                                                                                                                                                                      |
|-----------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                    | TransactionId | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                        | Datetime18    | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetwork<br>Ids | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |



| Name     | Type   | Description                                                   |
|----------|--------|---------------------------------------------------------------|
| comments | string | Optional. Any additional comments associated with the report. |

## ETRoot STRUCTURE

Root interaction class for the Entity Tasking and Reporting (ETR) interaction classes.

## ETRSimCon STRUCTURE

A base interaction class for EXCON Task interaction classes.

| Name   | Type          | Description                                                     |
|--------|---------------|-----------------------------------------------------------------|
| taskId | TransactionId | Required. Unique identifier for the simulation control task.    |
| taskee | UUID          | Required. Reference to the entity that should execute the task. |
| tasker | string        | Optional. Identifies the commander of the task.                 |

## ETRTask STRUCTURE

A base interaction class for more specialized task interaction classes.

| Name                        | Type          | Description                                                                                                                                                                                                                   |
|-----------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                      | TransactionId | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                      | UUID          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                      | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen                   | Datetime18    | Required. Time when the task execution should start. Use the value 000000000000000.000 to indicate that the time is undefined.                                                                                                |
| why                         | string        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                    | TaskModeEnum8 | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |

## ETRTaskManagement STRUCTURE

A base interaction class for more specialized task management interaction classes.

| Name                        | Type          | Description                                                                                                                                                                                                                                         |
|-----------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskManagementId            | TransactionId | Required. Unique identifier for the management task.                                                                                                                                                                                                |
| taskee                      | UUID          | Required. Reference to the entity that should execute the task.                                                                                                                                                                                     |
| tasker                      | string        | Optional. Identifies the commander of the task.                                                                                                                                                                                                     |
| communicationNetwork<br>Ids | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer task management messages. If not provided, the message transmission should not be modelled and federates should receive and act on the task management messages directly. |

## ElectronicSignatureStruct STRUCTURE

A summary percentage of an aggregates susceptibility to detection of its electronic emissions. Zero percent means that the aggregate has no electronic emissions.

| Name                       | Type                                         | Description                                                                                                                                                       |
|----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| electronicSignaturePercent | <a href="#">rpr.PercentUnsignedInteger32</a> | A summary percentage of an aggregates susceptibility to detection of its electronic emissions. Zero percent means that the aggregate has no electronic emissions. |
| sensorArray                | <a href="#">ArrayOfSensorStruct</a>          | A list of sensors owned by the aggregate together with their respective operational status and range                                                              |

Named by [NETNAggregate](#).

## EntityStruct STRUCTURE

An entity represented to the federation as part of the aggregate object which owns it.

| Name             | Type                                       | Description                                                                                                                                                                      |
|------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| callsign         | <a href="#">string</a>                     | The unique identifier of the object.                                                                                                                                             |
| entityCategory   | <a href="#">EntityCategoryEnum32</a>       | Indicates whether the entity is equipment, person, emitter, etc.                                                                                                                 |
| entityStatus     | <a href="#">DamageStatusEnhancedEnum32</a> | The damage state of the entity.                                                                                                                                                  |
| isDistinctObject | <a href="#">bool</a>                       | A BaseEntity object has been created to represent this entity (true) or not (false). Default is false.                                                                           |
| isUnavailable    | <a href="#">bool</a>                       | This entity is in use by another object (true) or not (false). Default is false.                                                                                                 |
| facing           | <a href="#">DirectionDegreesFloat32</a>    | Direction is measured in degrees clockwise from orientation of unit. Default is 0.                                                                                               |
| concealment      | <a href="#">ConcealmentEnum32</a>          | Indicates whether the entity is concealed and, if so, how                                                                                                                        |
| offsetLocation   | <a href="#">rpr.RelativePositionStruct</a> | The entity location given as an offset from the location of the aggregate unit in meters.                                                                                        |
| unitAllocation   | <a href="#">UuidArrayOfHLAbyte16</a>       | Reference to unit by UUID to which this entity is allocated as a resource. Unit may or may not be represented in the federation as a NETN_Aggregate and/or NETN-ORG Unit object. |

Named by [ArrayOfEntityStruct](#).

## EquipmentSymbolAmplificationStruct STRUCTURE

Text amplifiers for Equipment symbols.

| Name                  | Type                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| quantity              | <a href="#">Text9</a>  | A text amplifier in an equipment symbol that identifies the number of items present.                                                                                                                                                                                                                                                                                                                                                        |
| staffComments         | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| additionalInformation | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| evaluationRating      | <a href="#">Text2</a>  | A text amplifier for units, equipment and installations that consists of a one-letter reliability rating and a one-number credibility rating. Reliability Ratings: A-completely reliable B-usually reliable C-fairly reliable D-not usually reliable E-unreliable F-reliability cannot be judged Credibility Ratings: 1-confirmed by other sources 2-probably true 3-possibly true 4-doubtfully true 5-improbable 6-truth cannot be judged. |
| signatureEquipment    | <a href="#">Text1</a>  | A text amplifier for hostile equipment; "!" indicates detectable electronic signatures.                                                                                                                                                                                                                                                                                                                                                     |
| hostile               | <a href="#">Text3</a>  | A text amplifier for equipment; letters 'ENY' denote hostile symbols.                                                                                                                                                                                                                                                                                                                                                                       |
| iffSifAis             | <a href="#">Text15</a> | A text amplifier displaying IFF/SIF/AIS identification modes and codes.                                                                                                                                                                                                                                                                                                                                                                     |
| uniqueDesignation     | <a href="#">Text21</a> | A text amplifier for units, equipment and installations that uniquely identifies a particular symbol or track number.                                                                                                                                                                                                                                                                                                                       |

| Name              | Type   | Description                                                                                                                                              |
|-------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| equipmentType     | Text24 | Identifies acquisitions number when used with SIGINT symbology.<br>A text amplifier for equipment that indicates types of equipment.                     |
| engagementBarText | Text8  | A text amplifier placed immediately atop the symbol in the engagement bar. May denote, 1) local/remote status; 2) engagement status; and 3) weapon type. |

Named by [EquipmentItem](#).

## EstablishCheckPoint STRUCTURE

Tasking of an entity to establish a checkpoint. The task defines a location where a checkpoint shall be established and then operated. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the checkpoint. If not, a separate move task should be issued first. During the execution of the task, a CheckPoint object (NETN-SE) should be registered in the federation. The completion and operational status of the CheckPoint object should be updated continuously during the execution of the task or at task completion. Immediately after the checkpoint has been established, it will be operated by the tasked entity for a specified duration of time.

| Name                        | Type                        | Description                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                      | TransactionId               | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                      | UUID                        | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                      | string                      | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen                   | Datetime18                  | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                         | string                      | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                    | TaskModeEnum8               | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids | ArrayOfText64               | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| name                        | string                      | Required. The name of the Check Point.                                                                                                                                                                                        |
| location                    | rpr.WorldLocation<br>Struct | Required. The location of the Check Point.                                                                                                                                                                                    |
| uniqueId                    | UUID                        | Required. Unique identifier for the SE_CheckPoint object to be created as a result of this task.                                                                                                                              |
| duration                    | TimeSecInt32                | Required. The duration of the initial operation of the Check Point after it has been established.                                                                                                                             |
| locationUuid                | UUID                        | Optional. The location for the checkpoint by reference to NETN-SE SE_GeoObject.                                                                                                                                               |

## FireAtEntity STRUCTURE

Tasking of an entity to fire at another specified target entity.

| Name      | Type          | Description                                                                                                                   |
|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| taskId    | TransactionId | Required. Unique identifier for the task.                                                                                     |
| taskee    | UUID          | Required. Reference to the entity that will execute the task.                                                                 |
| tasker    | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                      |
| startWhen | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined. |
| why       | string        | Optional. A text describing the reason for this task.                                                                         |

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| target                  | <a href="#">UUID</a>          | Required. Reference to the target entity.                                                                                                                                                                                     |

## FireAtEntityWM STRUCTURE

Tasking of an entity to fire at a specified target entity with the specified weapon and munition data.

| Name                    | Type                                 | Description                                                                                                                                                                                                                   |
|-------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a>        | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>                 | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>               | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>           | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>               | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a>        | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>        | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| target                  | <a href="#">UUID</a>                 | Required. Reference to the target entity.                                                                                                                                                                                     |
| weaponTypeToUse         | <a href="#">rpr.EntityTypeStruct</a> | Required. Preferred weapon type.                                                                                                                                                                                              |
| munitionTypeToUse       | <a href="#">rpr.EntityTypeStruct</a> | Required. Preferred munition type.                                                                                                                                                                                            |
| quantityFired           | <a href="#">rpr.Integer16</a>        | Required. Number of rounds.                                                                                                                                                                                                   |
| rateOfFire              | <a href="#">rpr.Float32</a>          | Required. Number of rounds per second.                                                                                                                                                                                        |

## FireAtLocation STRUCTURE

Tasking of an entity to fire at a location.

| Name                    | Type                                    | Description                                                                                                                                                                                                                   |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a>           | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>                    | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>                  | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>              | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>                  | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a>           | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>           | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| location                | <a href="#">rpr.WorldLocationStruct</a> | Required. The location to fire at.                                                                                                                                                                                            |
| duration                | <a href="#">TimeSecInt32</a>            | Required. The duration of the fire (seconds).                                                                                                                                                                                 |
| locationUuid            | <a href="#">UUID</a>                    | Optional. The location to fire at by reference to a NETN SE, SE_GeoObject.                                                                                                                                                    |

## FireAtLocationWM STRUCTURE

Tasking of an entity to fire at a location with the specified weapon and munition.

| Name                    | Type                    | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId           | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                    | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                  | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18              | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                  | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8           | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64           | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| location                | rpr.WorldLocationStruct | Required. The location to fire at.                                                                                                                                                                                            |
| duration                | TimeSecInt32            | Required. The duration of the fire (seconds).                                                                                                                                                                                 |
| locationUuid            | UUID                    | Optional. The location to fire at by reference to a NETN SE, SE_GeoObject.                                                                                                                                                    |
| weaponTypeToUse         | rpr.EntityTypeStruct    | Required. Preferred weapon type.                                                                                                                                                                                              |
| munitionTypeToUse       | rpr.EntityTypeStruct    | Required. Preferred munition type.                                                                                                                                                                                            |
| quantityFired           | rpr.Integer16           | Required. Number of rounds to fire                                                                                                                                                                                            |
| rateOfFire              | rpr.Float32             | Required. Number of rounds per second.                                                                                                                                                                                        |

## FollowEntity STRUCTURE

Tasking of an entity to follow another entity at a specified distance.

| Name                    | Type             | Description                                                                                                                                                                                                                   |
|-------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId    | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID             | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string           | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18       | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string           | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8    | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64    | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| entity                  | UUID             | Required. Reference to the entity to follow.                                                                                                                                                                                  |
| behind                  | rpr.MeterFloat32 | Required. The distance in meters behind the entity to follow.                                                                                                                                                                 |
| right                   | rpr.MeterFloat32 | Required. The distance in meters to the right of the entity to follow.                                                                                                                                                        |
| above                   | rpr.MeterFloat32 | Required. The distance in meters above the entity to follow.                                                                                                                                                                  |
| duration                | TimeSecInt32     | Required. The duration of the follow action (seconds).                                                                                                                                                                        |

## FormationPositionStruct STRUCTURE

The position of this unit/equipment in the higher unit formation.

| Name              | Type                                                        | Description                                                                                                                |
|-------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| outOfFormation    | <a href="#">bool</a>                                        | Identifies if the element is maneuvering independently from the higher unit's formation.                                   |
| formationOrder    | <a href="#">FormationInt32</a>                              | The placement, 1 to N, of a subordinate in the superior unit formation.                                                    |
| sensorOrientation | <a href="#">DirectionDegrees</a><br><a href="#">Float32</a> | Orientation of the main sensor, sights, and weapon in compass degrees relative to orientation of the platform (equipment). |

Named by [DispositionStruct](#).

## FormationStruct STRUCTURE

The formation of this unit.

| Name                  | Type                                                            | Description                                                                           |
|-----------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| formationLocationType | <a href="#">FormationLocation</a><br><a href="#">TypeEnum32</a> | The method used to correlate formation to location as center of mass or lead element. |
| formationSpacing      | <a href="#">rpr.MeterFloat32</a>                                | The default spacing in metres between subordinate elements one echelon below.         |
| formationOrientation  | <a href="#">DirectionDegrees</a><br><a href="#">Float32</a>     | Orientation in compass degrees for unit formations and equipment.                     |
| formationData         | <a href="#">FormationDataStruct</a>                             | The formation data specific to the specified formation type.                          |

Named by [Unit](#).

## GeodeticCircle STRUCTURE

A geodetic point and radius specifying a circle on the surface of the earth WGS84 where the radius is a great circle distance on the surface.

| Name        | Type                             | Description                                          |
|-------------|----------------------------------|------------------------------------------------------|
| centerPoint | <a href="#">GeodeticLocation</a> | The center of the circular area. Lat, Long on WGS84. |
| radius      | <a href="#">rpr.MeterFloat32</a> | The radius of the circular area.                     |

Named by [AreaVariantStruct](#), [GeoLocationReferenceVariant](#), [GeoReferenceVariant](#).

## GeodeticLocation STRUCTURE

A geodetic point, specified by latitude and longitude, with unspecified altitude. WGS84

| Name      | Type                                                      | Description               |
|-----------|-----------------------------------------------------------|---------------------------|
| latitude  | <a href="#">LatLongDegrees</a><br><a href="#">Float64</a> | The latitude in degrees.  |
| longitude | <a href="#">LatLongDegrees</a><br><a href="#">Float64</a> | The longitude in degrees. |

Named by [AisBaseStationReportAndUTCDateResponseData](#), [AisMessage1](#), [AisMessage11](#), [AisMessage17](#), [AisMessage18](#), [AisMessage19](#), [AisMessage2](#), [AisMessage21](#), [AisMessage27](#), [AisMessage3](#), [AisMessage4](#), [AisMessage9](#), [AisNavigationData](#), [GeoLocationReferenceVariant](#), [GeoReferenceVariant](#), [GeodeticCircle](#), [GeodeticPath](#), [GeodeticPolygon](#), [GeodeticQuadrangle](#).

## GeodeticPoint STRUCTURE

A geodetic point, specified by latitude, longitude and altitude.

| Name      | Type                                                      | Description               |
|-----------|-----------------------------------------------------------|---------------------------|
| latitude  | <a href="#">LatLongDegrees</a><br><a href="#">Float64</a> | The latitude in degrees.  |
| longitude | <a href="#">LatLongDegrees</a><br><a href="#">Float64</a> | The longitude in degrees. |

| Name     | Type                                 | Description                 |
|----------|--------------------------------------|-----------------------------|
| altitude | <a href="#">AltitudeMeterFloat64</a> | Height Above Mean Sea Level |

Named by [DispositionStruct](#), [Installation](#), [Location](#).

## GeodeticQuadrangle STRUCTURE

A latitude-longitude quadrangle is a region bounded by two meridians and two parallels.

| Name   | Type                             | Description        |
|--------|----------------------------------|--------------------|
| point1 | <a href="#">GeodeticLocation</a> | Lat, Long on WGS84 |
| point2 | <a href="#">GeodeticLocation</a> | Lat, Long on WGS84 |

Named by [AreaVariantStruct](#), [DisruptCommunication](#), [DisruptionEffect](#), [GeoLocationReferenceVariant](#), [GeoReferenceVariant](#).

## HUMINTSignatureStruct STRUCTURE

Describes the unit's susceptibility to human intelligence (HUMINT), i.e. information collected and provided by human sources.

| Name                       | Type                                              | Description                                                                                                                                                            |
|----------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hUMINTSignature<br>Percent | <a href="#">rpr.PercentUnsigned<br/>Integer32</a> | A summary percentage of an aggregates susceptibility to detection by human intelligence collectors. Zero percent signature means an aggregate is impervious to HUMINT. |

Named by [NETNAggregate](#).

## HazeStruct STRUCTURE

Type and density of haze material.

| Name    | Type                               | Description                                      |
|---------|------------------------------------|--------------------------------------------------|
| type    | <a href="#">HazeTypeEnum32</a>     | Type of Haze.                                    |
| density | <a href="#">MassDensityFloat32</a> | Concentration/Density of haze measured in Kg/m3. |

Named by [LandSurfaceCondition](#), [TroposphereLayerCondition](#), [WaterSurfaceCondition](#), [Weather](#), [WeatherCondition](#).

## Holding STRUCTURE

Static and dynamic data about the holding.

| Name                       | Type                                        | Description                                                                             |
|----------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------|
| operationalCount           | <a href="#">QuantityFloat64</a>             | The number of specific OBJECT-TYPEs a specific OBJECT-ITEM has available for operations |
| onHandQuantity             | <a href="#">QuantityFloat64</a>             | Current number of the holding.                                                          |
| requiredOnHand<br>Quantity | <a href="#">QuantityFloat64</a>             | The required number of the holding.                                                     |
| isEquipment                | <a href="#">bool</a>                        | True if is equipment, false for expendables.                                            |
| entityType                 | <a href="#">rpr.EntityTypeStruct</a>        | DIS enumeration. Use 0.0.0.0.0.0 when not defined.                                      |
| nSNcode                    | <a href="#">NatoStockNumber<br/>Array13</a> | 13 digits (0-9)                                                                         |
| nSNname                    | <a href="#">string</a>                      | Nato Stock Number                                                                       |

Named by [ArrayOfHoldings](#).

## IPECommand STRUCTURE

Represents an order for the specified entities to don protective equipment.

| Name     | Type                        | Description                                          |
|----------|-----------------------------|------------------------------------------------------|
| unitList | <a href="#">ArrayOfUuid</a> | List of unique IDs of the entities who will don IPE. |

| Name    | Type                        | Description                             |
|---------|-----------------------------|-----------------------------------------|
| iPEType | <a href="#">IPTypeEnum8</a> | Type of IPE that the entities will don. |

## IceStruct STRUCTURE

Ice type, thickness and coverage on water surface.

| Name      | Type                               | Description                        |
|-----------|------------------------------------|------------------------------------|
| type      | <a href="#">IceTypeEnum16</a>      | Type of ice on water surface.      |
| thickness | <a href="#">rpr.MeterFloat32</a>   | Thickness of ice on water surface. |
| coverage  | <a href="#">rpr.PercentFloat32</a> | Ice coverage on water surface.     |

Named by [WaterSurface](#), [WaterSurfaceCondition](#).

## InWeaponRangeReport STRUCTURE

Report on a unit's ability to reach specific targets with its weapon systems.

| Name                        | Type                                 | Description                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                    | <a href="#">TransactionId</a>        | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                        | <a href="#">Datetime18</a>           | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetwork<br>Ids | <a href="#">ArrayOfText64</a>        | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                    | <a href="#">string</a>               | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| observer                    | <a href="#">UUID</a>                 | Required. Reference to the entity that observed another entity in range of its weapons.                                                                                                                                          |
| weaponType                  | <a href="#">rpr.EntityTypeStruct</a> | Required. The type of the weapon that is in range.                                                                                                                                                                               |
| entitiesInWeaponRange       | <a href="#">ArrayOfUuid</a>          | Required. Reference to entities in range of the weapon.                                                                                                                                                                          |

## IncomingConnectionStruct STRUCTURE

Characteristics of a specific incoming connection.

| Name                         | Type                                                      | Description                                                                                         |
|------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| communicationNetwork<br>Name | <a href="#">Text64</a>                                    | Reference to the communication network associated with the incoming connection.                     |
| senderEntityId               | <a href="#">UUID</a>                                      | Reference by UUID to the simulated entity sending on the connection.                                |
| latency                      | <a href="#">rpr.TimeMillisecond<br/>UnsignedInteger32</a> | Delay in data delivery.                                                                             |
| bandwidth                    | <a href="#">BitsPerSecond</a>                             | Amount of data per time unit.                                                                       |
| reliability                  | <a href="#">PercentFloat64</a>                            | Reliability of connection in terms of availability. 100% = always available and no connection loss. |
| hopCount                     | <a href="#">rpr.Integer16</a>                             | The number of hops in physical network required for sender entity to reach receiver entity.         |

Named by [IncomingConnectionArray](#).

## InitiateTransfer STRUCTURE

An `InitiateTransfer` interaction is sent by a **Trigger Federate** to initiate a TMR request. Parameters include information on the participating federates, which object instances and what attributes are included in the transfer. The **Trigger Federate** shall generate a unique `EventId` to be used in all subsequent TMR interactions related to the event.



| Name             | Type                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId          | <a href="#">TransactionId</a>      | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |
| requestFederate  | <a href="#">string</a>             | Required. A name specifying the federate requesting the transfer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| responseFederate | <a href="#">string</a>             | Required unless TransferType is AcquireWithoutNegotiation: A Federate Name specifying the federate that is the responding federate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| instances        | <a href="#">ArrayOfUuid</a>        | Required. References (UUID) to NETN Platform, Lifeform or Aggregate instances included in the transfer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| attributes       | <a href="#">AttributeNamesType</a> | Required. The common set of attributes for all referenced instances to be included in the transfer. Reference to attributes is by attribute name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| transferType     | <a href="#">TransferEnumType</a>   | Required. An enumerated value that indicates the direction of the transfer of the ownership and whether there shall be any negotiation: `Acquire` = Requesting Federate acquires instance attribute ownership from the Responding Federate (current owner), `Divest` = Requesting Federate (current owner) releases instance attribute ownership to Responding Federate and `AcquireWithoutNegotiating` = Requesting Federate acquires unowned instance attributes.                                                                                                                                                         |

## InstallationSymbolAmplificationStruct STRUCTURE

Text amplifiers for Installation symbols.

| Name                  | Type                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| staffComments         | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| additionalInformation | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| evaluationRating      | <a href="#">Text2</a>  | A text amplifier for units, equipment and installations that consists of a one-letter reliability rating and a one-number credibility rating. Reliability Ratings: A-completely reliable B-usually reliable C-fairly reliable D-not usually reliable E-unreliable F-reliability cannot be judged Credibility Ratings: 1-confirmed by other sources 2-probably true 3-possibly true 4-doubtfully true 5-improbable 6-truth cannot be judged. |
| combatEffectiveness   | <a href="#">Text5</a>  | A text amplifier for units and installations that indicates unit effectiveness or installation capability.                                                                                                                                                                                                                                                                                                                                  |
| iffSifAis             | <a href="#">Text15</a> | A text amplifier displaying IFF/SIF/AIS identification modes and codes.                                                                                                                                                                                                                                                                                                                                                                     |
| uniqueDesignation     | <a href="#">Text21</a> | A text amplifier for units, equipment and installations that uniquely identifies a particular symbol or track number. Identifies acquisitions number when used with SIGINT symbology.                                                                                                                                                                                                                                                       |
| engagementBarText     | <a href="#">Text8</a>  | A text amplifier placed immediately atop the symbol in the engagement bar. May denote, 1) local/remote status; 2) engagement status; and 3) weapon type.                                                                                                                                                                                                                                                                                    |

Named by [Installation](#).

## LOGInteraction STRUCTURE

Base class for all NETN Logistics Pattern Service Transactions.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |

## LandSurfaceCondition STRUCTURE

Response with condition for land surface.

| Name                | Type                                                | Description                                                                                                                                                                      |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId             | <a href="#">TransactionId</a>                       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                              |
| environmentObjectId | <a href="#">UUID</a>                                | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                |
| geoReference        | <a href="#">GeoReferenceVariant</a>                 | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global. |
| status              | <a href="#">bool</a>                                | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                    |
| temperature         | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | Optional. Average temperature in the region/location.                                                                                                                            |
| wind                | <a href="#">WindStruct</a>                          | Optional. Average wind speed and direction in the region/location.                                                                                                               |
| precipitation       | <a href="#">PrecipitationStruct</a>                 | Optional. Average precipitation intensity region/location for a precipitation type in the following order of precedence Snow, Hail, Rain, NoPrecipitation.                       |
| haze                | <a href="#">HazeStruct</a>                          | Optional. Average haze density in the region/location.                                                                                                                           |
| humidity            | <a href="#">f32</a>                                 | Optional. Average humidity in percent in the region/location.                                                                                                                    |
| barometricPressure  | <a href="#">f32</a>                                 | Optional. Average barometric pressure measured in millibar or hectopascal (1 mbar = 1hPa) in the region/location.                                                                |
| visibilityRange     | <a href="#">rpr.MeterFloat32</a>                    | Optional. Average distance at which an object or light can be clearly discerned by the human eye in the region/location.                                                         |
| snow                | <a href="#">SnowStruct</a>                          | Optional. Average snow depth and density in the region/location.                                                                                                                 |
| moisture            | <a href="#">SurfaceMoistureEnum16</a>               | Optional. Maximum surface moisture in the region/location.                                                                                                                       |
| iceCondition        | <a href="#">RoadIceConditionEnum16</a>              | Optional. Most severe ice condition in the region/location.                                                                                                                      |

## LayerStruct STRUCTURE

A layer of the Earth's environment expressed as a base altitude and a thickness.

| Name      | Type                             | Description                                                                                                                                |
|-----------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| base      | <a href="#">rpr.MeterFloat32</a> | The altitude of the bottom end of the layer, in meters above mean sea level (MSL). Negative values means the base is below mean sea level. |
| thickness | <a href="#">rpr.MeterFloat32</a> | Thickness of the layer in meters above the base altitude. Always a positive value.                                                         |

Named by [RequestSubsurfaceLayerCondition](#), [RequestTroposphereLayerCondition](#), [SubsurfaceLayer](#), [TroposphereLayer](#).

## LinkStatusStruct STRUCTURE

Status of a physical network link.

| Name                | Type                                     | Description                                                                                         |
|---------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| transmitterDeviceId | UUID                                     | Reference to Transmitting network device.                                                           |
| receiverDeviceId    | UUID                                     | Reference to Receiving network device.                                                              |
| latency             | rpr.TimeMillisecond<br>UnsignedInteger32 | Delay in data delivery.                                                                             |
| bandwidth           | BitsPerSecond                            | Amount of data per time unit.                                                                       |
| reliability         | PercentFloat64                           | Reliability of connection in terms of availability. 100% = always available and no connection loss. |
| isBidirectional     | bool                                     | True is link is bi-directional. By default this value should be set to false.                       |

Named by [LinkStatusArray](#).

## METOCInteraction STRUCTURE

Root class for requesting environment data from a METOC Service.

| Name    | Type          | Description                                                                                                         |
|---------|---------------|---------------------------------------------------------------------------------------------------------------------|
| eventId | TransactionId | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction. |

## MRMInteraction STRUCTURE

Base class for all MRM interactions.

| Name    | Type          | Description                                                                             |
|---------|---------------|-----------------------------------------------------------------------------------------|
| eventId | TransactionId | Unique identifier for all MRM interactions belonging to the same request/respons event. |

## MagicMove STRUCTURE

A simulation control task that instructs the entity to immediately jump to the specified location and update the new position and heading.

| Name         | Type                        | Description                                                     |
|--------------|-----------------------------|-----------------------------------------------------------------|
| taskId       | TransactionId               | Required. Unique identifier for the simulation control task.    |
| taskee       | UUID                        | Required. Reference to the entity that should execute the task. |
| tasker       | string                      | Optional. Identifies the commander of the task.                 |
| location     | rpr.WorldLocation<br>Struct | Required. Location of the entity.                               |
| heading      | rpr.AngleRadian<br>Float32  | Required. Heading of the entity.                                |
| locationUuid | UUID                        | Optional. The new location for the entity. Referred by a UUID.  |

## MagicResource STRUCTURE

A simulation control task that instructs the entity to set amount of available resources immediately

| Name   | Type          | Description                                                     |
|--------|---------------|-----------------------------------------------------------------|
| taskId | TransactionId | Required. Unique identifier for the simulation control task.    |
| taskee | UUID          | Required. Reference to the entity that should execute the task. |
| tasker | string        | Optional. Identifies the commander of the task.                 |

| Name     | Type                             | Description                                                             |
|----------|----------------------------------|-------------------------------------------------------------------------|
| resource | <a href="#">NETNSupplyStruct</a> | Required. Defines the resource type that will get a new quantity value. |

## Merge STRUCTURE

Instruction to merge the simulated AggregateUnit with the selected divided parts. After successful merge the divided parts are removed from the federation and their resources are combined with the AggregatedUnit.

| Name          | Type                                 | Description                                                                                                                                                                                                                                             |
|---------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/response event.                                                                                                                                                                |
| federate      | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | <a href="#">UuidArrayOfHLAByte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |
| subunits      | <a href="#">ArrayOfUuid</a>          | Required. A set of unique identifiers of subelements of the AggregateUnit. These can be any subunit and/or equipment defined at a subunit on any level.                                                                                                 |

## MissionStruct STRUCTURE

The operational task the aggregate has been ordered to perform, the time the mission was assigned, and the estimated completion time.

| Name        | Type                                   | Description                                                |
|-------------|----------------------------------------|------------------------------------------------------------|
| startTime   | <a href="#">EpochTimeSecInt64</a>      | An optional field providing the mission start time         |
| endTime     | <a href="#">EpochTimeSecInt64</a>      | An optional field providing the mission estimated end time |
| missionEnum | <a href="#">AggregateMissionEnum16</a> | The mission assigned to the aggregate                      |

Named by [NETNAggregate](#).

## Mount STRUCTURE

Tasking of entity to mount the specified entity. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the entity to mount.

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| entityId                | <a href="#">UUID</a>          | Required. Reference to the entity to mount.                                                                                                                                                                                   |

## Move STRUCTURE

Tasking of an entity to move in a specified direction for a given duration of time.

| Name                    | Type                       | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId              | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                       | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                     | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                 | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                     | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8              | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64              | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| direction               | rpr.AngleRadian<br>Float32 | Required. Direction of movement (azimuth direction).                                                                                                                                                                          |
| duration                | TimeSecInt32               | Required. The duration of the move.                                                                                                                                                                                           |

## MoveIntoFormation STRUCTURE

Tasking of an entity to move into the given formation on the given location with the given heading.

| Name                    | Type                        | Description                                                                                                                                                                                                                   |
|-------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId               | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                        | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                      | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                  | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                      | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8               | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64               | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| formation               | rpr.FormationEnum32         | Required. The type of positional arrangement the entity.                                                                                                                                                                      |
| location                | rpr.WorldLocation<br>Struct | Required. The location to move to.                                                                                                                                                                                            |
| distance                | rpr.MeterFloat32            | Required. The minimum distance between entities in the formation.                                                                                                                                                             |
| heading                 | rpr.AngleRadian<br>Float32  | Required. The heading of the formation.                                                                                                                                                                                       |
| locationUuid            | UUID                        | Optional. The location to move to by reference to a NETN SE, SE_GeoObject.                                                                                                                                                    |

## MoveToEntity STRUCTURE

Tasking of an entity to move to another entity. If a path is provided, the entity should follow the path as its route to the entity. The entity should align with the path from its current position to the nearest position or waypoint on the path. The entity should leave the path at position or waypoint on the path closest to the destination entity. The entity moves directly towards the destination entity if no path (or a zero-length path) is provided.

| Name                    | Type                       | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId              | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                       | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                     | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                 | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                     | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8              | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64              | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| entity                  | UUID                       | Required. Reference to the entity to move to.                                                                                                                                                                                 |
| path                    | ArrayOfWorldLocationStruct | Optional. Route to the specified entity.                                                                                                                                                                                      |
| moveType                | MoveTypeEnum32             | Required. Indicates if roads have to be followed.                                                                                                                                                                             |
| pathUuid                | UUID                       | Optional. Path to use to get to the Location by reference to a NETN_SE, SE_GeoObject.                                                                                                                                         |

## MoveToLocation STRUCTURE

Tasking of an entity to move to the specified destination location. If a path is provided, the entity should follow the path as its route to the destination. The entity should align with the path from its current position to the nearest position or waypoint on the path. The entity should leave the path at position or waypoint on the path closest to the destination. The entity moves directly towards the destination location if no path (or a zero-length path) is provided.

| Name                    | Type                       | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId              | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                       | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                     | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                 | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                     | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8              | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64              | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| location                | rpr.WorldLocationStruct    | Required. The location to move to.                                                                                                                                                                                            |
| path                    | ArrayOfWorldLocationStruct | Optional. A route to use in order to move to the Location.                                                                                                                                                                    |
| moveType                | MoveTypeEnum32             | Required. Indicates if roads have to be followed.                                                                                                                                                                             |
| pathUuid                | UUID                       | Optional. Path to use to get to the Location by reference to a NETN_SE, SE_GeoObject.                                                                                                                                         |
| locationUuid            | UUID                       | Optional. The location to move to, by reference to a NETN_SE, SE_GeoObject.                                                                                                                                                   |

## NETNSupplyStruct STRUCTURE

Description of supply. Same encoding as RPR2 SupplyStruct.

| Name       | Type                                 | Description                                                                                                                                           |
|------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| supplyType | <a href="#">rpr.EntityTypeStruct</a> | The type of supply (as described in the Bit Encoded Values for Use with Protocols for Distributed Interactive Simulation Applications)                |
| quantity   | <a href="#">QuantityFloat32</a>      | The number of units of the supply type. The unit measure depends on the supply type and shall use the SI units of measurement used for such supplies. |

Named by [MagicResource](#), [NETNArrayOfSupplyStruct](#), [ResourceStatusReport](#).

## NetworkDeviceEmptyCharactersticsStruct STRUCTURE

Empty placeholder for future network device characteristics.

Named by [NetworkDeviceReceiverCharacteristicsVariant](#), [NetworkDeviceTransmitterCharacteristicsVariant](#).

## NetworkDeviceGenericTransmitterCharacteristicsStruct STRUCTURE

Generic description of network device transmitter.

| Name         | Type                             | Description                                                                                                                                                                     |
|--------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| maxLinkRange | <a href="#">rpr.MeterFloat32</a> | Range limit for the link in meters. By default this value should be set to zero. If zero the max link range is based on the default value from the associated physical network. |

Named by [NetworkDeviceTransmitterCharacteristicsVariant](#).

## NetworkDeviceStruct STRUCTURE

Describes the type and capability of a network device. Each device must have at least a transmitter (TX) or receiver (RX). Transceivers should define both.

| Name                  | Type                                                           | Description                                                                               |
|-----------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| networkDeviceId       | <a href="#">UUID</a>                                           | Unique identifier for network device.                                                     |
| networkDeviceModel    | <a href="#">Text64</a>                                         | Optional name of the model of network device (e.g. a radio model name).                   |
| communicationNetworks | <a href="#">CommunicationNetworkArray</a>                      | Reference to all communication networks this device is used for.                          |
| physicalNetworkName   | <a href="#">Text64</a>                                         | Reference to physical network this device is connected to.                                |
| tx                    | <a href="#">NetworkDeviceTransmitterCharacteristicsVariant</a> | Transmitter characteristics of the network device.                                        |
| rx                    | <a href="#">NetworkDeviceReceiverCharacteristicsVariant</a>    | Receiver characteristics of the network device.                                           |
| isRelay               | <a href="#">bool</a>                                           | True if this network device acts as a relay. By default this value should be set to true. |

Named by [NetworkDeviceArray](#).

## Observe STRUCTURE

Tasking of an entity to observe an area.

| Name   | Type                          | Description                                                                              |
|--------|-------------------------------|------------------------------------------------------------------------------------------|
| taskId | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                |
| taskee | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                            |
| tasker | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined. |

| Name                        | Type                              | Description                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| startWhen                   | <a href="#">Datetime18</a>        | Required. Time when the task execution should start. Use the value 000000000000000.000 to indicate that the time is undefined.                                                                                                |
| why                         | <a href="#">string</a>            | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                    | <a href="#">TaskModeEnum8</a>     | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetwork<br>Ids | <a href="#">ArrayOfText64</a>     | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| observationArea             | <a href="#">AreaVariantStruct</a> | Required. Area to observe.                                                                                                                                                                                                    |

## OfferRepair STRUCTURE

Is sent by a federate simulating the service providing entity in response to a RequestRepair interaction.

| Name             | Type                                 | Description                                                                                                                                                              |
|------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction.                                                                                        |
| offerId          | <a href="#">TransactionId</a>        | Required: Unique offer identifier.                                                                                                                                       |
| providerEntity   | <a href="#">UuidArrayOfHLAbyte16</a> | Required: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown.             |
| offerType        | <a href="#">OfferTypeEnum32</a>      | Optional: Specifies if the offer is Negative (no Offer), Positive (Complete as requested) or Modified (not the same as the requested).                                   |
| offerTimeOut     | <a href="#">i64</a>                  | Optional: The system wallclock time (in seconds after 1 January 1970) when the offer stops being valid. If not provided, the offer valid until otherwise specified.      |
| startAppointment | <a href="#">AppointmentStruct</a>    | Optional: Time and location of the start of service delivery.                                                                                                            |
| repairData       | <a href="#">ArrayOfRepairStruct</a>  | Required: List of the type of repairs offered. May be different from the list of requested repairs. List of all offered repairs if isOffering = true otherwise Undefined |

## OfferService STRUCTURE

The OfferService is a response to a RequestService. Subclasses of this interaction for specific types of offers contain a more detailed description of the offer. This information may include when, where, how the service can be delivered.

| Name             | Type                                 | Description                                                                                                                                                         |
|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction.                                                                                   |
| offerId          | <a href="#">TransactionId</a>        | Required: Unique offer identifier.                                                                                                                                  |
| providerEntity   | <a href="#">UuidArrayOfHLAbyte16</a> | Required: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown.        |
| offerType        | <a href="#">OfferTypeEnum32</a>      | Optional: Specifies if the offer is Negative (no Offer), Positive (Complete as requested) or Modified (not the same as the requested).                              |
| offerTimeOut     | <a href="#">i64</a>                  | Optional: The system wallclock time (in seconds after 1 January 1970) when the offer stops being valid. If not provided, the offer valid until otherwise specified. |
| startAppointment | <a href="#">AppointmentStruct</a>    | Optional: Time and location of the start of service delivery.                                                                                                       |



## OfferSupply STRUCTURE

Used by a supply service provider to indicate which of the requested materiel (amount and type) can be offered. In this request the consumer decides whether the loading is done by the provider or by the consumer.

| Name             | Type                    | Description                                                                                                                                                                  |
|------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | TransactionId           | Required: Unique identifier for the requested service related to the transaction.                                                                                            |
| offerId          | TransactionId           | Required: Unique offer identifier.                                                                                                                                           |
| providerEntity   | UuidArrayOfHLAbyte16    | Required: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown.                 |
| offerType        | OfferTypeEnum32         | Optional: Specifies if the offer is Negative (no Offer), Positive (Complete as requested) or Modified (not the same as the requested).                                       |
| offerTimeOut     | i64                     | Optional: The system wallclock time (in seconds after 1 January 1970) when the offer stops being valid. If not provided, the offer valid until otherwise specified.          |
| startAppointment | AppointmentStruct       | Optional: Time and location of the start of service delivery.                                                                                                                |
| suppliesData     | NETNArrayOfSupplyStruct | Required: List of type and quantity of supplies offered. May be different from the list of requested supplies. All offered supplies if isOffering = true otherwise Undefined |

## OfferTransfer STRUCTURE

An `OfferTransfer` interaction is sent by a **Response Federate** as a reply to a `RequestTransfer` to indicate if a transfer can be accomplished or not. **Special Case**: When the transfer type is Divesting, the delivery order of the interaction `OfferTransfer` and the HLA ownership callback service `requestAttributeOwnershipRelease` at the receiving federate is not determined. Due to this, receiving a `requestAttributeOwnershipRelease` callback should be treated in the same way as receiving a `OfferTransfer` with a positive offer. The user-supplied tag in the callback contains the `EventId` which identifies a specific transfer process.

| Name       | Type                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId    | TransactionId         | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |
| isOffering | bool                  | Required. True if and only if for all instances all attributes in the request are offered. False if for any instance any attribute in the request is not offered.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| reason     | NoOfferReasonEnumType | Optional. An enumerated value that shall specify the reason for a negative offer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## OfferTransport STRUCTURE

An Offer for a Transport support. The OfferTransport interaction shall be sent by the service providing federate in response to a RequestTransport interaction.

| Name      | Type          | Description                                                                       |
|-----------|---------------|-----------------------------------------------------------------------------------|
| requestId | TransactionId | Required: Unique identifier for the requested service related to the transaction. |
| offerId   | TransactionId | Required: Unique offer identifier.                                                |

| Name             | Type                 | Description                                                                                                                                                         |
|------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| providerEntity   | UuidArrayOfHLAByte16 | Required: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown.        |
| offerType        | OfferTypeEnum32      | Optional: Specifies if the offer is Negative (no Offer), Positive (Complete as requested) or Modified (not the same as the requested).                              |
| offerTimeOut     | i64                  | Optional: The system wallclock time (in seconds after 1 January 1970) when the offer stops being valid. If not provided, the offer valid until otherwise specified. |
| startAppointment | AppointmentStruct    | Optional: Time and location of the start of service delivery.                                                                                                       |
| transportData    | ArrayOfUuid          | Required: Entities to be transported.                                                                                                                               |
| endAppointment   | AppointmentStruct    | Optional: Location and time for disembarkment.                                                                                                                      |
| transporters     | ArrayOfUuid          | Optional: Platform list with transporters                                                                                                                           |

## OperateCheckpoint STRUCTURE

Tasking of an entity to operate a checkpoint. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the checkpoint. If not, a separate move task should be issued first. The tasked entity activates an inactive checkpoint and operates the checkpoint for the specified duration of time.

| Name                    | Type          | Description                                                                                                                                                                                                                   |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8 | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                | UUID          | Required. Reference to the checkpoint to operate.                                                                                                                                                                             |
| duration                | TimeSecInt32  | Required. The duration of the operation of the checkpoint.                                                                                                                                                                    |

## Patrol STRUCTURE

Tasking of an entity to perform a patrol. The tasked entity moves from its current position to the start of the patrol route and then moves according to patrol route from its start point in the path, through all waypoints.

| Name                    | Type          | Description                                                                                                                   |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId | Required. Unique identifier for the task.                                                                                     |
| taskee                  | UUID          | Required. Reference to the entity that will execute the task.                                                                 |
| tasker                  | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                      |
| startWhen               | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined. |
| why                     | string        | Optional. A text describing the reason for this task.                                                                         |
| taskMode                | TaskModeEnum8 | Required. Determines the task mode.                                                                                           |
| communicationNetworkIds | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task         |

| Name       | Type                                        | Description                                                                                                                                                       |
|------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| path       | <a href="#">ArrayOfWorldLocation Struct</a> | transmission should not be modelled and federates should receive and act on the task messages directly.<br>Required. Specifies the route to patrol.               |
| patrolType | <a href="#">PatrolTypeEnum32</a>            | Required. Defines how the patrol should be excuted. The full movement from start point, through the patrol route and back to the start point is called one cycle. |
| moveType   | <a href="#">PatrolMoveTypeEnum32</a>        | Required. Defines how the entity shall move during the patrol route.                                                                                              |
| pathUuid   | <a href="#">UUID</a>                        | Optional. Path for the patrol by reference to a NETN_SE, SE_GeoObject.                                                                                            |

## PatrolRepeating STRUCTURE

Tasking of an entity to repeat a patrol for a given duration. When the duration time has passed, then the last cycle of the patrol is completed before the task ends. If the time of a cycle takes longer then the interval time, then the cycle starts directly (without delay). If the time of a cycle takes less then the interval time, then the entity waits at the first point of the patrol route until the next cycle is started.

| Name                    | Type                                        | Description                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a>               | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>                        | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>                      | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>                  | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>                      | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a>               | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>               | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| path                    | <a href="#">ArrayOfWorldLocation Struct</a> | Required. Specifies the route to patrol.                                                                                                                                                                                      |
| patrolType              | <a href="#">PatrolTypeEnum32</a>            | Required. Defines how the patrol should be excuted. The full movement from start point, through the patrol route and back to the start point is called one cycle.                                                             |
| moveType                | <a href="#">PatrolMoveTypeEnum32</a>        | Required. Defines how the entity shall move during the patrol route.                                                                                                                                                          |
| pathUuid                | <a href="#">UUID</a>                        | Optional. Path for the patrol by reference to a NETN_SE, SE_GeoObject.                                                                                                                                                        |
| duration                | <a href="#">TimeSecInt32</a>                | Required. The duration of the total patrol task.                                                                                                                                                                              |
| intervalTime            | <a href="#">TimeSecInt32</a>                | Required. The time between the start of two cycles, i.e. the time to complete a cycle and any waiting time before starting the next cycle.                                                                                    |

## PhysicalGenericNetworkStruct STRUCTURE

Characteristics of a generic physical network.

| Name                | Type                             | Description                                                                             |
|---------------------|----------------------------------|-----------------------------------------------------------------------------------------|
| maxHopCount         | <a href="#">rpr.Integer32</a>    | The maximum nuber of hops allowed in the physical network.                              |
| defaultMaxLinkRange | <a href="#">rpr.MeterFloat32</a> | The maximum distance between sending and receiving entities in a physical network link. |

| Name        | Type                                                 | Description                                                                                         |
|-------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| latency     | <a href="#">rpr.TimeMillisecondUnsignedInteger32</a> | Delay in data delivery.                                                                             |
| bandwidth   | <a href="#">BitsPerSecond</a>                        | Amount of data per time unit.                                                                       |
| reliability | <a href="#">PercentFloat64</a>                       | Reliability of connection in terms of availability. 100% = always available and no connection loss. |

Named by [PhysicalNetworkDescriptionVariant](#).

## PhysicalUndefinedNetworkStruct STRUCTURE

(Empty) Characteristics of an undefined physical network.

Named by [PhysicalNetworkDescriptionVariant](#).

## PositionStatusReport STRUCTURE

Report on a unit's position, speed, and heading.

| Name                        | Type                                              | Description                                                                                                                                                                                                                      |
|-----------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                    | <a href="#">TransactionId</a>                     | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                        | <a href="#">Datetime18</a>                        | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetwork<br>Ids | <a href="#">ArrayOfText64</a>                     | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                    | <a href="#">string</a>                            | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| entityId                    | <a href="#">UUID</a>                              | Required. The entity that is reported about.                                                                                                                                                                                     |
| position                    | <a href="#">rpr.WorldLocationStruct</a>           | Required. Position of the entity at the specified time.                                                                                                                                                                          |
| heading                     | <a href="#">rpr.AngleRadianFloat32</a>            | Required. Heading of the entity.                                                                                                                                                                                                 |
| speed                       | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Required. Speed of the entity.                                                                                                                                                                                                   |

## PrecipitationStruct STRUCTURE

Type and intensity of precipitation.

| Name      | Type                                          | Description                             |
|-----------|-----------------------------------------------|-----------------------------------------|
| type      | <a href="#">PrecipitationTypeEnum32</a>       | Type of precipitation.                  |
| intensity | <a href="#">PrecipitationIntensityFloat32</a> | Amount of precipitation in mm/per hour. |

Named by [LandSurfaceCondition](#), [TroposphereLayerCondition](#), [WaterSurfaceCondition](#), [Weather](#), [WeatherCondition](#).

## ProbabilityHazardContourStruct STRUCTURE

Represents the footprint covered by a CBRN hazard. This object covers the agent-specific effects which would be the output from a casualty model.

| Name                        | Type                               | Description                                                                             |
|-----------------------------|------------------------------------|-----------------------------------------------------------------------------------------|
| percentProbability<br>Level | <a href="#">rpr.PercentFloat32</a> | Percentage of entities within this contour being affected to the define exposure level. |
| locations                   | <a href="#">GeocentricPolygon</a>  | Array of locations that define the probability contour.                                 |

Named by [ArrayOfProbabilityHazardContourStruct](#).

## ProtectionEffectivenessStruct STRUCTURE

Protection effectiveness associated with a specific agent.

| Name          | Type                               | Description                                          |
|---------------|------------------------------------|------------------------------------------------------|
| effectiveness | <a href="#">rpr.PercentFloat32</a> | Percentage by which this protection reduces effects. |
| agent         | <a href="#">AgentTypeEnum16</a>    | Enumeration representation of the agent.             |

Named by [ArrayOfProtectionEffectivenessStruct](#).

## QuerySupportedCapabilitiesETR STRUCTURE

Query which tasks, reports and SimCon actions that an entity supports. The queried entity shall respond with a CapabilitiesSupported message.

| Name   | Type                          | Description                                                     |
|--------|-------------------------------|-----------------------------------------------------------------|
| taskId | <a href="#">TransactionId</a> | Required. Unique identifier for the simulation control task.    |
| taskee | <a href="#">UUID</a>          | Required. Reference to the entity that should execute the task. |
| tasker | <a href="#">string</a>        | Optional. Identifies the commander of the task.                 |

## QuerySupportedCapabilitiesMRM STRUCTURE

A request to query the capabilities of a specified federation to provide support for MRM events. The queried federate shall respond with a CapabilitiesSupported interaction.

| Name          | Type                                 | Description                                                                                                                                                                                                                                             |
|---------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate      | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | <a href="#">UuidArrayOfHLAbyte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |

## RawDataHazardContourStruct STRUCTURE

Contour locations bounding a given exposure value.

| Name          | Type                              | Description                                        |
|---------------|-----------------------------------|----------------------------------------------------|
| exposureLevel | <a href="#">ExposureFloat32</a>   | Quantity of exposure, measured in SI units.        |
| locations     | <a href="#">GeocentricPolygon</a> | Array of locations that define the hazard contour. |

Named by [ArrayOfRawDataHazardContourStruct](#).

## ReadyToReceiveService STRUCTURE

Issued by a service consumer to indicate that service delivery can start. The time of service delivery may be significantly later than indicating ready for service delivery.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |

## RejectOffer STRUCTURE

Used to reject an offer made by a service providing entity as indicated in a OfferService interaction. By issuing a RejectOffer interaction the service consuming entity informs the providing entity that the offer has been rejected.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |
| reason    | <a href="#">string</a>        | Optional: Allows to inform about the reason of the cancel (free text)             |

## Relation STRUCTURE

The specific value that represents the perceived hostility status.

| Name         | Type                                      | Description                                        |
|--------------|-------------------------------------------|----------------------------------------------------|
| affiliate    | <a href="#">UUID</a>                      | The force that the current force has a relation to |
| relationship | <a href="#">HostilityStatusCodeEnum32</a> | The level of the relation                          |

Named by [ArrayOfRelations](#).

## ReleaseDistributionStruct STRUCTURE

Mean and variance of the distribution of the particles or droplets in a release.

| Name     | Type                                         | Description                                                     |
|----------|----------------------------------------------|-----------------------------------------------------------------|
| mean     | <a href="#">MeanMetersFloat32</a>            | Mean size of the released material in meters.                   |
| variance | <a href="#">VarianceMetersSquaredFloat32</a> | Variance of the distribution of released particulates/droplets. |

Named by [ReleaseSizeStruct](#).

## ReleaseDynamicsStruct STRUCTURE

Defines the dynamic properties of a release.

| Name                  | Type                                                | Description                                                                            |
|-----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------|
| densityRatio          | <a href="#">DensityRatioFloat32</a>                 | Ratio of material density relative to the atmosphere.                                  |
| temperatureDifference | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | Difference in temperature between the released material and the atmosphere in Celsius. |

Named by [CBRNRelease](#).

## ReleaseSizeStruct STRUCTURE

Defines the properties of the initial size of a release.

| Name                | Type                                      | Description                                                                              |
|---------------------|-------------------------------------------|------------------------------------------------------------------------------------------|
| sigmaArray          | <a href="#">ArrayOfSigmas6</a>            | Initial Gaussian sigmas of the released puff.                                            |
| releaseDistribution | <a href="#">ReleaseDistributionStruct</a> | Details regarding the size of the released particles/droplets for non-gaseous materials. |

Named by [CBRNRelease](#).

## RemoveCheckPoint STRUCTURE

Tasking of an entity to remove a checkpoint. This task removes a previously established checkpoint. The tasked entity should be within a certain distance (tolerance specified in the federation agreements) of the location of the checkpoint. If not, a separate move task should be issued first. After completion of the task, the checkpoint object should be deleted from the federation.

| Name                    | Type          | Description                                                                                                                                                                                                                   |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8 | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                | UUID          | Required. Reference to the CheckPoint to remove.                                                                                                                                                                              |

## RemovePassage STRUCTURE

Task entity to remove the passage with the given UUID. The tasked entity should be within a certain distance tolerance (specified in the federation agreement) of one of the points of the passage to make the task possible.

| Name                    | Type          | Description                                                                                                                                                                                                                   |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8 | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| uniqueId                | UUID          | Required. Reference to a A unique UUID of the passage.                                                                                                                                                                        |

## RepairComplete STRUCTURE

This interaction is sent by the provider when the repair service is delivered to the consumer

| Name       | Type                | Description                                                                                      |
|------------|---------------------|--------------------------------------------------------------------------------------------------|
| requestId  | TransactionId       | Required: Unique identifier for the requested service related to the transaction.                |
| repairData | ArrayOfRepairStruct | Required: List of the type of repairs done. May be different from the list of requested repairs. |

## RepairStruct STRUCTURE

Repairs associated with a specific materiel

| Name       | Type                  | Description                                               |
|------------|-----------------------|-----------------------------------------------------------|
| materielId | UuidArrayOfHLAbyte16  | UUID for the materiel.                                    |
| repairs    | ArrayOfRepairTypeEnum | List of the types of repair associated with the materiel. |

Named by [ArrayOfRepairStruct](#).

## RequestLandSurfaceCondition STRUCTURE

Request for land surface condition data.

| Name           | Type                                | Description                                                                                                                                                                        |
|----------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | <a href="#">TransactionId</a>       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | <a href="#">GeoReferenceVariant</a> | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | <a href="#">bool</a>                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | <a href="#">UUID</a>                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |

## RequestMETOC STRUCTURE

A request to a specified METOC Service to provide METOC data for a specific geographical reference. The request can result in either a response interaction including the requested data or registration of an EnvironmentCondition object for continuous updates.

| Name           | Type                                | Description                                                                                                                                                                        |
|----------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | <a href="#">TransactionId</a>       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | <a href="#">GeoReferenceVariant</a> | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | <a href="#">bool</a>                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | <a href="#">UUID</a>                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |

## RequestMRM STRUCTURE

A base class for all MRM Request events.

| Name          | Type                                 | Description                                                                                                                                                                                                                                             |
|---------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId       | <a href="#">TransactionId</a>        | Unique identifier for all MRM interactions belonging to the same request/respons event.                                                                                                                                                                 |
| federate      | <a href="#">string</a>               | Required. Intended federate responsible for performing the requested action. Sending federate should ensure that receiving federate can perform requested action. If not able to perform, a response interaction indicating failure should be returned. |
| aggregateUnit | <a href="#">UuidArrayOfHLAbyte16</a> | Required for all requests except QuerySupportedCapabilities. Unique identifier for the NETN_Aggregate for which this request is related to.                                                                                                             |

## RequestRepair STRUCTURE

Sent by the consumer when a repair is needed. Specifies entity and type of repair



| Name             | Type                                 | Description                                                                                                                                                  |
|------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction.                                                                            |
| consumerEntity   | <a href="#">UuidArrayOfHLAByte16</a> | Required: Unique identifier for a simulated entity that is the intended receiver of service                                                                  |
| providerEntity   | <a href="#">UuidArrayOfHLAByte16</a> | Optional: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown. |
| requestTimeOut   | <a href="#">i64</a>                  | Optional: Wallclock time. The timeout time in seconds after 1 January 1970.                                                                                  |
| startAppointment | <a href="#">AppointmentStruct</a>    | Optional: The time and location of the start of the service delivery.                                                                                        |
| repairData       | <a href="#">ArrayOfRepairStruct</a>  | Required: List of all requested repairs.                                                                                                                     |

## RequestService STRUCTURE

A consumer federate initiates service negotiation using `RequestService`. A unique ServiceId and a reference to a `ConsumerEntity` are required parameters. A reference to a specific `ProviderEntity` and a system wall-clock time for when offers are expected `RequestTimeOut` are optional. Requests for specific types of services are defined as subclasses to `RequestService` and include parameters for detailing the requirements of the request. This may include information when, where and how the service should be delivered.

| Name             | Type                                 | Description                                                                                                                                                  |
|------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction.                                                                            |
| consumerEntity   | <a href="#">UuidArrayOfHLAByte16</a> | Required: Unique identifier for a simulated entity that is the intended receiver of service                                                                  |
| providerEntity   | <a href="#">UuidArrayOfHLAByte16</a> | Optional: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown. |
| requestTimeOut   | <a href="#">i64</a>                  | Optional: Wallclock time. The timeout time in seconds after 1 January 1970.                                                                                  |
| startAppointment | <a href="#">AppointmentStruct</a>    | Optional: The time and location of the start of the service delivery.                                                                                        |

## RequestSubsurfaceLayerCondition STRUCTURE

Request for sub-surface condition data.

| Name           | Type                                | Description                                                                                                                                                                        |
|----------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | <a href="#">TransactionId</a>       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | <a href="#">GeoReferenceVariant</a> | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | <a href="#">bool</a>                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | <a href="#">UUID</a>                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |
| layer          | <a href="#">LayerStruct</a>         | Optional. A description of layer for request of layered subsurface environment conditions. Default is the entire body of water in the identified area.                             |

## RequestSupply STRUCTURE

The consumer sends a `RequestSupply` interaction to request supplies. The amount and type of supplies are specified in the required `SuppliesData` parameter. The required `TransferDirection` parameter indicates whether supplies are transferred from Consumer to provider or from Provider to consumer.

| Name              | Type                    | Description                                                                                                                                                  |
|-------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId         | TransactionId           | Required: Unique identifier for the requested service related to the transaction.                                                                            |
| consumerEntity    | UuidArrayOfHLAByte16    | Required: Unique identifier for a simulated entity that is the intended receiver of service                                                                  |
| providerEntity    | UuidArrayOfHLAByte16    | Optional: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown. |
| requestTimeout    | i64                     | Optional: Wallclock time. The timeout time in seconds after 1 January 1970.                                                                                  |
| startAppointment  | AppointmentStruct       | Optional: The time and location of the start of the service delivery.                                                                                        |
| suppliesData      | NETNArrayOfSupplyStruct | Required: List of type and quantity of supplies requested.                                                                                                   |
| transferDirection | TransferDirectionEnum32 | Required: Indicates if the transfer of supplies are from Consumer to Provider or from Provider to Consumer.                                                  |

## RequestTransfer STRUCTURE

A `RequestTransfer` interaction is sent by the requesting federate to the responding federate. If the request is the result of a `InitiateTransfer` interaction, the parameters from that interaction shall be copied and used in the request including `EventId`. If the request is not triggered the **Request Federate** shall generate a unique `EventId` to be used in all subsequent TMR interactions related to the event.

| Name             | Type              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId          | TransactionId     | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |
| responseFederate | string            | Required unless TransferType is AcquireWithoutNegotiation: A Federate Name specifying the federate that is the responding federate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| instances        | ArrayOfUuid       | Required. References (UUID) to NETN Platform, Lifeform or Aggregate instances included in the transfer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| attributes       | AttributeNameType | Required. The common set of attributes for all referenced instances to be included in the transfer. Reference to attributes is by attribute name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| transferType     | TransferEnumType  | Required. An enumerated value that indicates the direction of the transfer of the ownership and whether there shall be any negotiation: `Acquire` = Requesting Federate acquires instance attribute ownership from the Responding Federate (current owner), `Divest` = Requesting Federate (current owner) releases instance attribute ownership to Responding Federate and `AcquireWithoutNegotiating` = Requesting Federate acquires unowned instance attributes.                                                                                                                                                         |

## RequestTransport STRUCTURE

A request for a Transport service. Use a RequestTransport interaction to initiate a transport, embarkment or disembarkment of a platform.

| Name             | Type                 | Description                                                                                                                                                  |
|------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestId        | TransactionId        | Required: Unique identifier for the requested service related to the transaction.                                                                            |
| consumerEntity   | UuidArrayOfHLAByte16 | Required: Unique identifier for a simulated entity that is the intended receiver of service                                                                  |
| providerEntity   | UuidArrayOfHLAByte16 | Optional: Unique identifier of a simulated entity intended to perform the service. If not provided, the entity performing the service is considered unknown. |
| requestTimeout   | i64                  | Optional: Wallclock time. The timeout time in seconds after 1 January 1970.                                                                                  |
| startAppointment | AppointmentStruct    | Optional: The time and location of the start of the service delivery.                                                                                        |
| transportData    | ArrayOfUuid          | Required: Entities to be transported.                                                                                                                        |
| endAppointment   | AppointmentStruct    | Optional: Location and time for disembarkment.                                                                                                               |

## RequestTroposphereLayerCondition STRUCTURE

Request for tropospheric environment condition data.

| Name           | Type                | Description                                                                                                                                                                        |
|----------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | TransactionId       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | GeoReferenceVariant | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | bool                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | UUID                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |
| layer          | LayerStruct         | Optional. A description of layer for request of layered conditions troposphere environment condition. Default is the entire volume of air in the identified area.                  |

## RequestWaterSurfaceCondition STRUCTURE

Request for water surface condition data.

| Name           | Type                | Description                                                                                                                                                                        |
|----------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | TransactionId       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | GeoReferenceVariant | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | bool                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | UUID                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |

## RequestWeatherCondition STRUCTURE

Request for general weather data.

| Name           | Type                                | Description                                                                                                                                                                        |
|----------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId        | <a href="#">TransactionId</a>       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                                |
| geoReference   | <a href="#">GeoReferenceVariant</a> | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. If not provided, the request is for a global environment condition. |
| updateAsObject | <a href="#">bool</a>                | Optional. Indicates if the service is requested to represent the environment condition as an EnvironmentCondition object instance. Default is False.                               |
| serviceId      | <a href="#">UUID</a>                | Required: Unique identifier of the model used for providing METOC information. Multiple Models may exist.                                                                          |

## RequestedConnection STRUCTURE

Characterisation of a requested connection.

| Name                      | Type                               | Description                                                                                                                                           |
|---------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| communicationNetwork Name | <a href="#">Text64</a>             | Reference to the communication network which the requested connection should belong to.                                                               |
| requestedConnection Type  | <a href="#">ConnectionTypeEnum</a> | The type of connection requested.                                                                                                                     |
| connectionId              | <a href="#">UUID</a>               | Requested Id for the connection. If no Id is requested the value of this field should be all zeros.                                                   |
| destinationEntityIds      | <a href="#">ArrayOfUuid</a>        | List of entities intended as destinations for this requested connection. Leave empty in case of broadcasting or multicasting.                         |
| networkDeviceId           | <a href="#">UUID</a>               | Reference to the Network Device for the requested connection. If no specific network device is requested the value of this field should be all zeros. |

Named by [RequestedConnectionArray](#).

## ResourceStatusNumberStruct STRUCTURE

The name of a resource and the number of instances of that resource by health status.

| Name                        | Type                                 | Description                                                                                                                                  |
|-----------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| numberHealthyOrIntact       | <a href="#">QuantityFloat64</a>      | The number of healthy or intact resources                                                                                                    |
| numberSlightlyDamaged       | <a href="#">QuantityFloat64</a>      | The number of slightly damaged resources                                                                                                     |
| numberModerately Damaged    | <a href="#">QuantityFloat64</a>      | The number of moderately damaged resources                                                                                                   |
| numberSignificantly Damaged | <a href="#">QuantityFloat64</a>      | The number of significantly damaged resources                                                                                                |
| numberDestroyed             | <a href="#">QuantityFloat64</a>      | The number of destroyed or consumed resources                                                                                                |
| resourceName                | <a href="#">string</a>               | The name of the resource                                                                                                                     |
| resourceType                | <a href="#">rpr.EntityTypeStruct</a> | The type of the resource (as described in the Bit Encoded Values for Use with Protocols for Distributed Interactive Simulation Applications) |

Named by [ArrayOfResourceStatus](#).

## ResourceStatusReport STRUCTURE

Report on a unit's remaining amount resources.

| Name                    | Type                             | Description                                                                                                                                                                                                                      |
|-------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                | <a href="#">TransactionId</a>    | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                    | <a href="#">Datetime18</a>       | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>    | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | <a href="#">string</a>           | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| entityId                | <a href="#">UUID</a>             | Required. The entity that is reported about.                                                                                                                                                                                     |
| resource                | <a href="#">NETNSupplyStruct</a> | Required. The type of resource and remaining quantity.                                                                                                                                                                           |

## ResponseMETOC STRUCTURE

Response to a RequestEnvironmentCondition. RequestId parameter should match the corresponding parameter in the request.

| Name                | Type                                | Description                                                                                                                                                                      |
|---------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId             | <a href="#">TransactionId</a>       | Required: Unique identifier of the request. Will be referenced in a resulting METOC_Response interaction.                                                                        |
| environmentObjectId | <a href="#">UUID</a>                | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                |
| geoReference        | <a href="#">GeoReferenceVariant</a> | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global. |
| status              | <a href="#">bool</a>                | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                    |

## ResponseMRM STRUCTURE

A response from the receiving federate indicating ability to comply with request.

| Name    | Type                          | Description                                                                              |
|---------|-------------------------------|------------------------------------------------------------------------------------------|
| eventId | <a href="#">TransactionId</a> | Unique identifier for all MRM interactions belonging to the same request/response event. |
| status  | <a href="#">bool</a>          | Required. Specifies the result of the request action. TRUE indicates success.            |

## SendSafetyRelatedBroadcastMessage STRUCTURE

Tasks entity (the source) to send a safety related broadcast message (AIS message type 14). The source must represent an AIS station (vessel, SAR aircraft, etc).

| Name                    | Type                          | Description                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined. |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task         |

| Name    | Type   | Description                                                                                                                                       |
|---------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| message | string | transmission should not be modelled and federates should receive and act on the task messages directly.<br>Required. 1-161 chars of six-bit text. |

## SendSafetyRelatedMessage STRUCTURE

Tasks entity (the source) to send a safety related message (AIS message type 12) to another entity (the destination). Both source and destination must represent an AIS station (vessel, SAR aircraft, etc).

| Name                    | Type                 | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId        | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                 | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string               | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18           | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string               | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8        | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64        | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| entity                  | UuidArrayOfHLAbyte16 | Required. The entity that represents the destination. This must be an AIS station.                                                                                                                                            |
| message                 | string               | Required. 1-156 chars of six-bit text.                                                                                                                                                                                        |

## SensorReport STRUCTURE

Report on a unit's sensor detection of entities.

| Name                    | Type                     | Description                                                                                                                                                                                                                      |
|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                | TransactionId            | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                    | Datetime18               | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetworkIds | ArrayOfText64            | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | string                   | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| observer                | UUID                     | Required. Entity that observed the enemy (or neutral/unknown entity) and sends the spot report.                                                                                                                                  |
| identificationLevel     | IdentificationLevelEnum8 | Required. The identification level of the entit(y)(ies) spotted.                                                                                                                                                                 |
| spottedEntities         | ArrayOfSpottedEntities   | Required. Spotted entities at the time specified in the parameter 'When'.                                                                                                                                                        |
| sensorType              | rpr.EntityTypeStruct     | Required. The type of the sensor that detected the entities.                                                                                                                                                                     |

## SensorStruct STRUCTURE

Defines a sensor, operational status, damage status, coverage and ID

| Name              | Type                   | Description                          |
|-------------------|------------------------|--------------------------------------|
| sensorStateEnum   | SensorStateEnum32      | The operational status of the sensor |
| sensorDamageState | rpr.DamageStatusEnum32 | The damage status of the sensor      |

| Name           | Type                         | Description                     |
|----------------|------------------------------|---------------------------------|
| sensorCoverage | <a href="#">RangeFloat32</a> | The maximum range of the sensor |
| sensorID       | <a href="#">string</a>       | A sensor owned by the aggregate |

Named by [ArrayOfSensorStruct](#).

## ServiceComplete STRUCTURE

Used by a service providing entity to inform the service consuming entity that the service has been delivered.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |

## ServiceReceived STRUCTURE

Used by a service consuming entity to inform the service providing entity that the service has been delivered.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |

## ServiceStarted STRUCTURE

Issued by a service provider to inform about the start of service delivery. The time of service delivery start may be significantly later than receiving an indication from the consumer that the service delivery can start.

| Name      | Type                          | Description                                                                       |
|-----------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |

## SetOrderedAltitude STRUCTURE

Tasking of an entity to set move to specified altitude.

| Name                    | Type                                 | Description                                                                                                                                                                                                                   |
|-------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a>        | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>                 | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>               | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>           | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>               | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a>        | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>        | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| altitude                | <a href="#">AltitudeMeterFloat64</a> | Required. Defines the altitude.                                                                                                                                                                                               |
| altitudeType            | <a href="#">AltitudeTypeEnum8</a>    | Required. Above the ground or sea.                                                                                                                                                                                            |

## SetOrderedSpeed STRUCTURE

Tasking of an entity to change speed.

| Name   | Type                          | Description                               |
|--------|-------------------------------|-------------------------------------------|
| taskId | <a href="#">TransactionId</a> | Required. Unique identifier for the task. |

| Name                    | Type                              | Description                                                                                                                                                                                                                   |
|-------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskee                  | UUID                              | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                            | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                        | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                            | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8                     | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64                     | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| speed                   | rpr.VelocityMeterPerSecondFloat32 | Required. Determines the ordered speed to set for an entity, in m/s.                                                                                                                                                          |

## SetRulesOfEngagement STRUCTURE

Tasking of an entity to change its rules of engagement.

| Name                    | Type                   | Description                                                                                                                                                                                                                   |
|-------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId          | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                   | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                 | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18             | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                 | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8          | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64          | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| rulesOfEngagement       | RulesOfEngagementEnum8 | Required. Determines the rules of engagement for an entity.                                                                                                                                                                   |

## SetTransmitterStatus STRUCTURE

Tasking of an entity to switch on/off all of its transmitters.

| Name                    | Type          | Description                                                                                                                                                                                                                   |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8 | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64 | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |



| Name   | Type                                                  | Description                          |
|--------|-------------------------------------------------------|--------------------------------------|
| status | <a href="#">rpr.TransmitterOperationalStatusEnum8</a> | Required. Status on the transmitter. |

## SnowStruct STRUCTURE

Depth and density of snow cover.

| Name    | Type                               | Description            |
|---------|------------------------------------|------------------------|
| depth   | <a href="#">rpr.MeterFloat32</a>   | Snow depth on terrain. |
| density | <a href="#">MassDensityFloat32</a> | kg/m3                  |

Named by [LandSurface](#), [LandSurfaceCondition](#).

## SpotReport STRUCTURE

Report on a unit's awareness of spotted entities.

| Name                    | Type                                     | Description                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                | <a href="#">TransactionId</a>            | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                    | <a href="#">Datetime18</a>               | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetworkIds | <a href="#">ArrayOfText64</a>            | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | <a href="#">string</a>                   | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| observer                | <a href="#">UUID</a>                     | Required. Entity that observed the enemy (or neutral/unknown entity) and sends the spot report.                                                                                                                                  |
| identificationLevel     | <a href="#">IdentificationLevelEnum8</a> | Required. The identification level of the entit(y)(ies) spotted.                                                                                                                                                                 |
| spottedEntities         | <a href="#">ArrayOfSpottedEntities</a>   | Required. Spotted entities at the time specified in the parameter 'When'.                                                                                                                                                        |

## SpottedEntity STRUCTURE

Description of the observed entity. The symbol contains information about the spotted entity's relation to the spotter and details about the type and echelon at the spotted entity.

| Name          | Type                                              | Description                                                                                                                                                                                                                                                                                                                              |
|---------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spottedEntity | <a href="#">UUID</a>                              | The unique identifier for the entity                                                                                                                                                                                                                                                                                                     |
| location      | <a href="#">rpr.WorldLocationStruct</a>           | The location where the entity were spotted                                                                                                                                                                                                                                                                                               |
| orientation   | <a href="#">rpr.OrientationStruct</a>             | The orientation in Psi, Theta, Phi [radians]                                                                                                                                                                                                                                                                                             |
| speed         | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | The speed of the entity (m/s). Default value: 0 m/s                                                                                                                                                                                                                                                                                      |
| equipment     | <a href="#">ArrayOfSpottedEquipment</a>           | Spotted equipment list, type and number. Default value: Empty array.                                                                                                                                                                                                                                                                     |
| activity      | <a href="#">AggregateMissionEnum16</a>            | The current activity of the entity. (From NETN-Base) Default value: Moving (213)                                                                                                                                                                                                                                                         |
| symbol        | <a href="#">string</a>                            | Symbol identifier for the entity. The detailed level may vary and the publishing federate decides the level of details in the symbol description. E.g. Default Ground value: app6b:SUGP-----***** Default Air value: app6b:SUAP-----***** Default Sea Surface value: app6b:SUSP-----***** Default Subsurface value: app6b:SUUP-----***** |

Named by [ArrayOfSpottedEntities](#).

## SpottedEquipment STRUCTURE

Equipment at the spotted entity.

| Name              | Type                                 | Description                                                                                                                                                                                             |
|-------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type              | <a href="#">rpr.EntityTypeStruct</a> | Use entity types from Supply Types, Expendibles or Sensors/Emitters tables (SISO-REF-010-2010, 4.3), for aggregate entities, also use entity types from platform tables to describe the equipment type. |
| numberOfEquipment | <a href="#">QuantityInt32</a>        | The number of the equipment.                                                                                                                                                                            |

Named by [ArrayOfSpottedEquipment](#).

## StatusReport STRUCTURE

Report on a unit's own (perceived) state. This report should be generated with a frequency specified in the federation agreements.

| Name                    | Type                          | Description                                                                                                                                                                                                                      |
|-------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportId                | <a href="#">TransactionId</a> | Required. Unique identifier for the report.                                                                                                                                                                                      |
| when                    | <a href="#">Datetime18</a>    | Required. Date and time when the reported status was valid.                                                                                                                                                                      |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | <a href="#">string</a>        | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| entityId                | <a href="#">UUID</a>          | Required. The entity that is reported about.                                                                                                                                                                                     |

## StopAtSideOfRoad STRUCTURE

Tasking of an entity to stop at the side of the road. This task is only relevant for an entity that is moving along a road to a destination. The executing move task is cancelled, and a new move is defined towards a position at the side of the road (the simulator has to calculate this location).

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |

## SubsurfaceLayerCondition STRUCTURE

Response with subsurface body of water condition data.

| Name    | Type                          | Description                                                                                                         |
|---------|-------------------------------|---------------------------------------------------------------------------------------------------------------------|
| eventId | <a href="#">TransactionId</a> | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction. |

| Name                | Type                                    | Description                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| environmentObjectId | UUID                                    | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                                                                 |
| geoReference        | GeoReferenceVariant                     | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global.                                                  |
| status              | bool                                    | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                                                                     |
| temperature         | rpr.Temperature<br>DegreeCelsiusFloat32 | Optional. Average temperature in the body of water.                                                                                                                                                                               |
| current             | CurrentStruct                           | Optional. Average current direction and speed in the body of water.                                                                                                                                                               |
| salinity            | f32                                     | Optional. Average salinity in the body of water.                                                                                                                                                                                  |
| bottomType          | SedimentTypeEnum32                      | Optional. Type of sediment on the sea floor. Default is 0 (NoSediment). If EnvironmentConditions with overlapping regions/locations exist, the sediment type in the overlapping region is determined by the latest updated value. |

## SupplyComplete STRUCTURE

This interaction is sent by the provider when the supply is delivered to the consumer

| Name         | Type                        | Description                                                                                 |
|--------------|-----------------------------|---------------------------------------------------------------------------------------------|
| requestId    | TransactionId               | Required: Unique identifier for the requested service related to the transaction.           |
| suppliesData | NETNArrayOfSupply<br>Struct | Required: The amount of supplies, by type, that were transferred from provider to consumer. |

## TMRInteraction STRUCTURE

The `TMR\_Interaction` interaction class is the base for all Transfer of Modelling Responsibility interactions. It provides a single parameter used in all TMR interactions to uniquely identify TMR transactions between participating federates.

| Name    | Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId | TransactionId | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |

## TaskStatusReport STRUCTURE

A management task report regarding the status of a specific task assigned to an entity.

| Name             | Type          | Description                                                     |
|------------------|---------------|-----------------------------------------------------------------|
| taskManagementId | TransactionId | Required. Unique identifier for the management task.            |
| taskee           | UUID          | Required. Reference to the entity that should execute the task. |
| tasker           | string        | Optional. Identifies the commander of the task.                 |

| Name                    | Type                             | Description                                                                                                                                                                                                                                         |
|-------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| communicationNetworkIds | <a href="#">ArrayOfText64</a>    | Optional. Reference to communication networks (NETN-COM) used to transfer task management messages. If not provided, the message transmission should not be modelled and federates should receive and act on the task management messages directly. |
| taskId                  | <a href="#">TransactionId</a>    | Required. Identifies the task associated with the task report.                                                                                                                                                                                      |
| taskStatus              | <a href="#">TaskStatusEnum32</a> | Required. Indicates the status of the task.                                                                                                                                                                                                         |
| when                    | <a href="#">Datetime18</a>       | Required. Creation date and time of the report.                                                                                                                                                                                                     |
| comments                | <a href="#">string</a>           | Optional. Any additional comments. For example reason for cancelling, errors, etc.                                                                                                                                                                  |

## TransferResult STRUCTURE

A `TransferResult` interaction is sent by the **Request Federate** to indicate successful or unsuccessful completion of the transfer.

| Name        | Type                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId     | <a href="#">TransactionId</a> | Required. The `EventId` is a unique reference to a specific execution of a TMR pattern. It consists of two values, a Counter and a Federate Handle. The value of this parameter is also encoded and used as the User Supplied Tag in the HLA Ownership services to allow a correlation between the TMR interactions and HLA Ownership services. The `EventId` is generated by a trigger federate or, if no trigger federate is used, directly by the requesting federate. The value of this parameter should match all related TMR pattern interactions belonging to the same Transfer of Modelling Responsibilities event. |
| isCompleted | <a href="#">bool</a>          | Required. A Boolean value indicating the result of the transfer. TRUE = Transfer Completed Successfully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## TransportDestroyedEntities STRUCTURE

Used by a service provider to update information on damage state of entities under transport.

| Name             | Type                          | Description                                                                       |
|------------------|-------------------------------|-----------------------------------------------------------------------------------|
| requestId        | <a href="#">TransactionId</a> | Required: Unique identifier for the requested service related to the transaction. |
| destroyedObjects | <a href="#">ArrayOfUuid</a>   | Required: Identifies the objects that have been destroyed during transport.       |

## TransportDisembarkmentStatus STRUCTURE

Is sent by the service provider federate, to inform the service consumer of the disembarkment state, after the ServiceStarted interaction

| Name                    | Type                                 | Description                                                                       |
|-------------------------|--------------------------------------|-----------------------------------------------------------------------------------|
| requestId               | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction. |
| disembarkedObjects      | <a href="#">ArrayOfUuid</a>          | Required: References to disembarked entities. Updated during disembarkment.       |
| transportUnitIdentifier | <a href="#">UuidArrayOfHLAbyte16</a> | Required: UUID of entity performing the transport.                                |

## TransportEmbarkmentStatus STRUCTURE

Is sent by the service provider federate, to inform the service consumer of the embarkment state, after the ServiceStarted interaction

| Name                     | Type                                 | Description                                                                       |
|--------------------------|--------------------------------------|-----------------------------------------------------------------------------------|
| requestId                | <a href="#">TransactionId</a>        | Required: Unique identifier for the requested service related to the transaction. |
| embarkedObjects          | <a href="#">ArrayOfUuid</a>          | Required: List of entities currently embarked. Updated during embarkment.         |
| transportUnit Identifier | <a href="#">UuidArrayOfHLAbyte16</a> | Required: Referens to entity providing the transport.                             |

## TreatmentStruct STRUCTURE

Defines the properties for a CBRN treatment.

| Name            | Type                                    | Description                                                                                      |
|-----------------|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| uniqueID        | <a href="#">UuidArrayOfHLAbyte16</a>    | The ID of this treatment.                                                                        |
| effectiveness   | <a href="#">rpr.PercentFloat32</a>      | Percentage by which this treatment reduces effects.                                              |
| duration        | <a href="#">rpr.TimeSecondInteger32</a> | Duration in seconds for which this treatment will remain effective.                              |
| treatmentWindow | <a href="#">rpr.TimeSecondInteger32</a> | The time window in seconds between exposure and treatment for which this treatment is effective. |
| treatableAgents | <a href="#">ArrayOfAgentTypeEnum</a>    | Array of agents that can be treated using this treatment.                                        |

Named by [ArrayOfTreatmentStruct](#), [CBRNTreatmentCommand](#).

## TroposphereLayerCondition STRUCTURE

Response with environment condition in volume of air.

| Name                | Type                                                | Description                                                                                                                                                                      |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId             | <a href="#">TransactionId</a>                       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                              |
| environmentObjectId | <a href="#">UUID</a>                                | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                |
| geoReference        | <a href="#">GeoReferenceVariant</a>                 | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global. |
| status              | <a href="#">bool</a>                                | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                    |
| temperature         | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | Optional. Average temperature in the region/location.                                                                                                                            |
| wind                | <a href="#">WindStruct</a>                          | Optional. Average wind speed and direction in the region/ location.                                                                                                              |
| precipitation       | <a href="#">PrecipitationStruct</a>                 | Optional. Average precipitation intensisty region/location for a precipitation type in the following order of precedence Snow, Hail, Rain, NoPrecipitation.                      |
| haze                | <a href="#">HazeStruct</a>                          | Optional. Average haze density in the region/location,                                                                                                                           |
| humidity            | <a href="#">f32</a>                                 | Optional. Average humidity in percent in the region/location.                                                                                                                    |
| barometricPressure  | <a href="#">f32</a>                                 | Optional. Average barometric pressure measured in millibar or hectopascal (1 mbar = 1hPa) in the region/location.                                                                |
| visibilityRange     | <a href="#">rpr.MeterFloat32</a>                    | Optional. Average distance at which an object or light can be clearly discerned by the human eye in the region/location.                                                         |

| Name  | Type        | Description            |
|-------|-------------|------------------------|
| cloud | CloudStruct | Optional. Cloud cover. |

## TurnToHeading STRUCTURE

Tasking of an entity to turn to the specified heading.

| Name                    | Type                       | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId              | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                       | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                     | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                 | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                     | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8              | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64              | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| heading                 | rpr.AngleRadian<br>Float32 | Required. The heading an entity has to turn to.                                                                                                                                                                               |

## TurnToOrientation STRUCTURE

Tasking of an entity to rotate to a specified orientation, including pitch and roll.

| Name                    | Type                       | Description                                                                                                                                                                                                                   |
|-------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | TransactionId              | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | UUID                       | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | string                     | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | Datetime18                 | Required. Time when the task execution should start. Use the value 00000000000000.000 to indicate that the time is undefined.                                                                                                 |
| why                     | string                     | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | TaskModeEnum8              | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | ArrayOfText64              | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| heading                 | rpr.AngleRadian<br>Float32 | Required. The heading an entity has to turn to.                                                                                                                                                                               |
| pitch                   | rpr.AngleRadian<br>Float32 | Required. Defines the angle of the pitch.                                                                                                                                                                                     |
| roll                    | rpr.AngleRadian<br>Float32 | Required. Defines the angle of the roll.                                                                                                                                                                                      |

## UnderAttackStatusReport STRUCTURE

Report from a unit that it is under attack.

| Name     | Type          | Description                                                 |
|----------|---------------|-------------------------------------------------------------|
| reportId | TransactionId | Required. Unique identifier for the report.                 |
| when     | Datetime18    | Required. Date and time when the reported status was valid. |

| Name                    | Type                                                       | Description                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| communicationNetworkIds | <a href="#">ArrayOfText64</a>                              | Optional. Reference to communication networks (NETN-COM) used to transfer report messages. If not provided, the report transmission should not be modelled and federates should receive and act on the report messages directly. |
| comments                | <a href="#">string</a>                                     | Optional. Any additional comments associated with the report.                                                                                                                                                                    |
| entityId                | <a href="#">UUID</a>                                       | Required. The entity that is reported about.                                                                                                                                                                                     |
| fromDirection           | <a href="#">rpr.AngleRadian</a><br><a href="#">Float32</a> | Required. Direction from which the entity is attacked, between $0..2\pi$ or -1 when not specified.                                                                                                                               |
| severeness              | <a href="#">AttackTypeEnum32</a>                           | Required. Severeness of the attack upon the reporting entity.                                                                                                                                                                    |

## UnitSymbolAmplificationStruct STRUCTURE

Text amplifiers for Unit symbols.

| Name                  | Type                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reinforcedOrReduced   | <a href="#">Text3</a>  | A text amplifier in a unit symbol that displays (+) for reinforced, (-) for reduced, ( $\pm$ ) reinforced and reduced.                                                                                                                                                                                                                                                                                                                      |
| staffComments         | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| additionalInformation | <a href="#">Text20</a> | A text amplifier for units, equipment and installations.                                                                                                                                                                                                                                                                                                                                                                                    |
| evaluationRating      | <a href="#">Text2</a>  | A text amplifier for units, equipment and installations that consists of a one-letter reliability rating and a one-number credibility rating. Reliability Ratings: A-completely reliable B-usually reliable C-fairly reliable D-not usually reliable E-unreliable F-reliability cannot be judged Credibility Ratings: 1-confirmed by other sources 2-probably true 3-possibly true 4-doubtfully true 5-improbable 6-truth cannot be judged. |
| combatEffectiveness   | <a href="#">Text5</a>  | A text amplifier for units and installations that indicates unit effectiveness or installation capability.                                                                                                                                                                                                                                                                                                                                  |
| higherFormation       | <a href="#">Text21</a> | A text amplifier for units that indicates number or title of higher echelon command (corps are designated by Roman numerals).                                                                                                                                                                                                                                                                                                               |
| iffSifAis             | <a href="#">Text15</a> | A text amplifier displaying IFF/SIF/AIS identification modes and codes.                                                                                                                                                                                                                                                                                                                                                                     |
| uniqueDesignation     | <a href="#">Text21</a> | A text amplifier for units, equipment and installations that uniquely identifies a particular symbol or track number. Identifies acquisitions number when used with SIGINT symbology.                                                                                                                                                                                                                                                       |
| specialHeadquarters   | <a href="#">Text9</a>  | A text modifier for a unit's special C2 Headquarters.                                                                                                                                                                                                                                                                                                                                                                                       |
| engagementBarText     | <a href="#">Text8</a>  | A text amplifier placed immediately atop the symbol in the engagement bar. May denote, 1) local/remote status; 2) engagement status; and 3) weapon type.                                                                                                                                                                                                                                                                                    |

Named by [Unit](#).

## VisualSignatureStruct STRUCTURE

Specifies the visual structure

| Name                | Type                                                             | Description                                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dV0SignaturePercent | <a href="#">rpr.PercentUnsigned</a><br><a href="#">Integer32</a> | A summary percentage of an aggregates susceptibility to detection by direct view optics, i.e. the human eye, binoculars, or telescopes. A unit with zero percent signature would be concealed from DVO detection. |
| i2SignaturePercent  | <a href="#">rpr.PercentUnsigned</a><br><a href="#">Integer32</a> | A summary percentage of an aggregates susceptibility to detection by Image Intensifying sensors. A unit with zero percent signature would be invisible to image intensifiers (I2).                                |

| Name                    | Type                                         | Description                                                                                                                                                     |
|-------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| thermalSignaturePercent | <a href="#">rpr.PercentUnsignedInteger32</a> | A summary percentage of an aggregates susceptibility to detection by thermal sensors. A unit with zero percent signature would be invisible to thermal sensors. |

Named by [NETNAggregate](#).

## Wait STRUCTURE

Tasking of an entity to wait for a duration of time.

| Name                    | Type                          | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| taskId                  | <a href="#">TransactionId</a> | Required. Unique identifier for the task.                                                                                                                                                                                     |
| taskee                  | <a href="#">UUID</a>          | Required. Reference to the entity that will execute the task.                                                                                                                                                                 |
| tasker                  | <a href="#">string</a>        | Optional. Callsign of the commander of the task. If missing, the commander is undefined.                                                                                                                                      |
| startWhen               | <a href="#">Datetime18</a>    | Required. Time when the task execution should start. Use the value 000000000000000.000 to indicate that the time is undefined.                                                                                                |
| why                     | <a href="#">string</a>        | Optional. A text describing the reason for this task.                                                                                                                                                                         |
| taskMode                | <a href="#">TaskModeEnum8</a> | Required. Determines the task mode.                                                                                                                                                                                           |
| communicationNetworkIds | <a href="#">ArrayOfText64</a> | Optional. Reference to communication networks (NETN-COM) used to transfer tasking messages. If not provided, the task transmission should not be modelled and federates should receive and act on the task messages directly. |
| duration                | <a href="#">TimeSecInt32</a>  | Required. Specifies the wait time (seconds).                                                                                                                                                                                  |

## WaterSurfaceCondition STRUCTURE

Response with water surface condition data.

| Name                | Type                                                | Description                                                                                                                                                                      |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId             | <a href="#">TransactionId</a>                       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                              |
| environmentObjectId | <a href="#">UUID</a>                                | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                |
| geoReference        | <a href="#">GeoReferenceVariant</a>                 | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global. |
| status              | <a href="#">bool</a>                                | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                    |
| temperature         | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | Optional. Average temperature in the region/location.                                                                                                                            |
| wind                | <a href="#">WindStruct</a>                          | Optional. Average wind speed and direction in the region/location.                                                                                                               |
| precipitation       | <a href="#">PrecipitationStruct</a>                 | Optional. Average precipitation intensity region/location for a precipitation type in the following order of precedence Snow, Hail, Rain, NoPrecipitation.                       |
| haze                | <a href="#">HazeStruct</a>                          | Optional. Average haze density in the region/location.                                                                                                                           |
| humidity            | <a href="#">f32</a>                                 | Optional. Average humidity in percent in the region/location.                                                                                                                    |
| barometricPressure  | <a href="#">f32</a>                                 | Optional. Average barometric pressure measured in millibar or hectopascal (1 mbar = 1hPa) in the region/location.                                                                |
| visibilityRange     | <a href="#">rpr.MeterFloat32</a>                    | Optional. Average distance at which an object or light can be clearly discerned by the human eye in the region/location.                                                         |
| seaState            | <a href="#">SeaStateEnum16</a>                      | Optional. Sea state data.                                                                                                                                                        |
| salinity            | <a href="#">f32</a>                                 | Optional. Salinity in the surface water.                                                                                                                                         |



| Name    | Type                             | Description                                                     |
|---------|----------------------------------|-----------------------------------------------------------------|
| tide    | <a href="#">rpr.MeterFloat32</a> | Optional. The height relative to the MSL.                       |
| ice     | <a href="#">IceStruct</a>        | Optional. Ice conditions on the water surface.                  |
| current | <a href="#">CurrentStruct</a>    | Optional. Surface water Current direction and speed. N/A if Ice |
| wave    | <a href="#">WaveStruct</a>       | Optional. Surface Wave data. N/A if Ice                         |
| swell   | <a href="#">WaveStruct</a>       | Optional. Surface water Swell data. N/A if Ice                  |

## WaveStruct STRUCTURE

Water surface wave conditions and direction.

| Name      | Type                                    | Description                                                                                            |
|-----------|-----------------------------------------|--------------------------------------------------------------------------------------------------------|
| length    | <a href="#">rpr.MeterFloat32</a>        | Distance from a particular phase on a wave to the same phase on an adjacent wave.                      |
| height    | <a href="#">rpr.MeterFloat32</a>        | Average vertical distance between trough and crest produced by the wave.                               |
| period    | <a href="#">rpr.TimeSecondInteger32</a> | Wave period is the distance between two waves passing through a stationary point, measured in seconds. |
| direction | <a href="#">DirectionDegreesFloat32</a> | The direction in which the wave propagates.                                                            |

Named by [WaterSurface](#), [WaterSurfaceCondition](#).

## WeatherCondition STRUCTURE

Response with general weather condition data.

| Name                | Type                                                | Description                                                                                                                                                                      |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventId             | <a href="#">TransactionId</a>                       | Required: Unique identifier of the request. Will be referenced in a resulting resulting METOC_Response interaction.                                                              |
| environmentObjectId | <a href="#">UUID</a>                                | Optional. Reference to an existing environment condition object if the corresponding request includes UpdateAsObject set to true.                                                |
| geoReference        | <a href="#">GeoReferenceVariant</a>                 | Optional. Geographical reference to indicate for which point, area, path or object this request is related to. Default if not provided the environment condition data is global. |
| status              | <a href="#">bool</a>                                | Required: Specifies the result of the request action. TRUE indicates success.                                                                                                    |
| temperature         | <a href="#">rpr.TemperatureDegreeCelsiusFloat32</a> | Optional. Average temperature in the region/location.                                                                                                                            |
| wind                | <a href="#">WindStruct</a>                          | Optional. Average wind speed and direction in the region/ location.                                                                                                              |
| precipitation       | <a href="#">PrecipitationStruct</a>                 | Optional. Average precipitation intensisty region/location for a precipitation type in the following order of precedence Snow, Hail, Rain, NoPrecipitation.                      |
| haze                | <a href="#">HazeStruct</a>                          | Optional. Average haze density in the region/location,                                                                                                                           |
| humidity            | <a href="#">f32</a>                                 | Optional. Average humidity in percent in the region/location.                                                                                                                    |
| barometricPressure  | <a href="#">f32</a>                                 | Optional. Average barometric pressure measured in millibar or hectopascal (1 mbar = 1hPa) in the region/location.                                                                |
| visibilityRange     | <a href="#">rpr.MeterFloat32</a>                    | Optional. Average distance at which an object or light can be clearly discerned by the human eye in the region/location.                                                         |

## WindStruct STRUCTURE

Wind direction and speeds.

| Name            | Type                                              | Description                                                                |
|-----------------|---------------------------------------------------|----------------------------------------------------------------------------|
| direction       | <a href="#">DirectionDegreesFloat32</a>           | Required: Direction in degrees clockwise from due north.                   |
| horizontalSpeed | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Required: Wind speed (m/s) in the specified WindDirection.                 |
| verticalSpeed   | <a href="#">rpr.VelocityMeterPerSecondFloat32</a> | Optional. Wind speed in m/s. Negative number is a downdraft. Default is 0. |

Named by [LandSurfaceCondition](#), [TroposphereLayerCondition](#), [WaterSurfaceCondition](#), [Weather](#), [WeatherCondition](#).

## Enumerations

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">ATP45HazardAreaTypeEnum8</a>       | Type of ATP-45 Hazard Area (simple or detailed).                                                                                                                                                                                                                                                                                                                                                                    | 119 |
| <a href="#">ActiveStatusEnum8</a>              | A state which indicates the status of an object concerning its participation in the simulation. An object in an inactive state is not simulated and does not interact with other objects.                                                                                                                                                                                                                           | 119 |
| <a href="#">AgentClassEnum8</a>                | Class of Agent for an ATP-45 Hazard Area.                                                                                                                                                                                                                                                                                                                                                                           | 119 |
| <a href="#">AgentTypeEnum16</a>                | Type of CBRN hazardous agent.                                                                                                                                                                                                                                                                                                                                                                                       | 120 |
| <a href="#">AggregateMissionEnum16</a>         | Representation of the general class or nature of activity related to a unit's mission. Enumerations are based on JC3IEDM action-event-category-code.                                                                                                                                                                                                                                                                | 120 |
| <a href="#">AidTypeEnumType</a>                | Enumeration value for the aid type. According to [IALA], the aid type field has values 1-15 for fixed and 16-31 for floating aids to navigation.                                                                                                                                                                                                                                                                    | 123 |
| <a href="#">AirFormationTypeEnum32</a>         | Air formation                                                                                                                                                                                                                                                                                                                                                                                                       | 123 |
| <a href="#">AltitudeTypeEnum8</a>              | The reference for altitude. AMSL = Above Mean Sea Level or AGL = Above Ground Level.                                                                                                                                                                                                                                                                                                                                | 123 |
| <a href="#">AttackTypeEnum32</a>               | The kind of attack by the enemy.                                                                                                                                                                                                                                                                                                                                                                                    | 124 |
| <a href="#">CBRNDamageEnum8</a>                | Level of damage due to CBRN exposure.                                                                                                                                                                                                                                                                                                                                                                               | 124 |
| <a href="#">CancellationReasonEnum32</a>       | Describes the reason for a cancellation.                                                                                                                                                                                                                                                                                                                                                                            | 124 |
| <a href="#">CaptureStatusEnum8</a>             | The status of a person or unit with respect to their control or influence over their own activities. Default: 1 - Not Captured.                                                                                                                                                                                                                                                                                     | 124 |
| <a href="#">CloudTypeEnum32</a>                | Classification of different types of clouds.                                                                                                                                                                                                                                                                                                                                                                        | 124 |
| <a href="#">CommunicationNetworkTypeEnum</a>   | The type of a communication network.                                                                                                                                                                                                                                                                                                                                                                                | 125 |
| <a href="#">CommunicationServiceTypeEnum</a>   | The type of service used on a communication network. Service types as defined in MSDL. DATTRF: Data transfer FAX: Facsimile IFF: Identify Friend or Foe IMAGE: Image MCI: Multilateral Interoperability Programme (MIP) Common Interface Service MHS: Message Handling Service TDL: Tactical Data Link VIDSVC: Video Service VOCSVC:Voice Service NOS: Not Otherwise Specified                                      | 125 |
| <a href="#">CommunicationServiceTypeEnum32</a> | List of Communications Net Types for Army Units.                                                                                                                                                                                                                                                                                                                                                                    | 125 |
| <a href="#">ConcealmentEnum32</a>              | The reason for the objects concealment                                                                                                                                                                                                                                                                                                                                                                              | 125 |
| <a href="#">ConnectionTypeEnum</a>             | The type of a connection.                                                                                                                                                                                                                                                                                                                                                                                           | 126 |
| <a href="#">DamageStatusEnhancedEnum32</a>     | The damage status of an object.                                                                                                                                                                                                                                                                                                                                                                                     | 126 |
| <a href="#">EchelonEnum32</a>                  | The echelon level of a unit.                                                                                                                                                                                                                                                                                                                                                                                        | 126 |
| <a href="#">EntityCategoryEnum32</a>           | Category of entity                                                                                                                                                                                                                                                                                                                                                                                                  | 126 |
| <a href="#">ExposureTypeEnum8</a>              | Type of exposure represented in a contour group.                                                                                                                                                                                                                                                                                                                                                                    | 127 |
| <a href="#">FormationLocationTypeEnum32</a>    | Enumerated choice for the method used to correlate formation to location as center of mass or lead element.                                                                                                                                                                                                                                                                                                         | 127 |
| <a href="#">FormationTypeEnum32</a>            | Enumerated choice for the type of formation being Ground, Air, Surface or Subsurface.                                                                                                                                                                                                                                                                                                                               | 127 |
| <a href="#">GeoLocationTypeEnum32</a>          | Specifies different ways to reference geographical locations.                                                                                                                                                                                                                                                                                                                                                       | 127 |
| <a href="#">GroundFormationTypeEnum32</a>      | Specifies the formations for groundunits.                                                                                                                                                                                                                                                                                                                                                                           | 127 |
| <a href="#">HazardTypeEnum8</a>                | Type of dispersion output represented in a contour group.                                                                                                                                                                                                                                                                                                                                                           | 128 |
| <a href="#">HazeTypeEnum32</a>                 | Type of visibility obstruction.                                                                                                                                                                                                                                                                                                                                                                                     | 128 |
| <a href="#">HostilityStatusCodeEnum32</a>      | The specific value that represents the perceived hostility status.                                                                                                                                                                                                                                                                                                                                                  | 128 |
| <a href="#">IPETTypeEnum8</a>                  | Types of Individual Protective Equipment.                                                                                                                                                                                                                                                                                                                                                                           | 128 |
| <a href="#">IceTypeEnum16</a>                  | Type of Ice.                                                                                                                                                                                                                                                                                                                                                                                                        | 129 |
| <a href="#">IdentificationLevelEnum8</a>       | The identification level of an object.                                                                                                                                                                                                                                                                                                                                                                              | 129 |
| <a href="#">ManeuverIndicatorEnumType</a>      | Enumeration value to indicate a maneuver.                                                                                                                                                                                                                                                                                                                                                                           | 129 |
| <a href="#">MineCountEnum32</a>                | Type of mine count.                                                                                                                                                                                                                                                                                                                                                                                                 | 129 |
| <a href="#">ModelResolutionTypeEnum32</a>      | Enumeration indicating the level of fidelity appropriate for instanting the unit or equipment in the simulation.                                                                                                                                                                                                                                                                                                    | 129 |
| <a href="#">MoveTypeEnum32</a>                 | CrossCountry: move directly to the destination without taking into account the roads. OnlyRoads: stay on the roads to get to the closest point to the destination that is still on a road; if there is no road very near to the start (within around 10 meters) there will be no movement at all. RoadsAndCrossCountry: move to the destination by taking into account the roads; it is allowed to go off the road. | 129 |
| <a href="#">MsgIdEnumType</a>                  | Enumeration value for the message type.                                                                                                                                                                                                                                                                                                                                                                             | 130 |
| <a href="#">NavigationStatusEnumType</a>       | Enumeration value to indicate the navigational status.                                                                                                                                                                                                                                                                                                                                                              | 130 |
| <a href="#">NoOfferReasonEnumType</a>          | Describes the reason why not accepting a TMR request                                                                                                                                                                                                                                                                                                                                                                | 130 |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">OfferTypeEnum32</a>               | Type of the offer (With restriction, positive, negative)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 131 |
| <a href="#">PartNumberEnumType</a>            | Identifier for the message part number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 131 |
| <a href="#">PathTypeEnum32</a>                | Specifies if a path is defined by waypoints or by reference to a path object in the federation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 131 |
| <a href="#">PatrolMoveTypeEnum32</a>          | Defines how the movement shall be done during the execution of the patrol. Other - Federate application specific.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 131 |
| <a href="#">PatrolTypeEnum32</a>              | Other - The behaviour is federate application specific (e.g. wait for next command, etc.) Circle - At the end of route, move to start point of patrol route (federate application determines the route) Reverse - Follow the patrol route in reverse order.                                                                                                                                                                                                                                                                                                                  | 131 |
| <a href="#">PhysicalNetworkTypeEnum</a>       | The type of a physical network.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 131 |
| <a href="#">PointTypeEnum32</a>               | Specifies if a point is defined by a location or by reference to a point object in the federation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 132 |
| <a href="#">PrecipitationTypeEnum32</a>       | Type of precipitation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 132 |
| <a href="#">RoadIceConditionEnum16</a>        | Ice condition for roads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 132 |
| <a href="#">RulesOfEngagementEnum8</a>        | Rules of engagement for a simulated entity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 132 |
| <a href="#">SeaStateEnum16</a>                | State of the sea measured in Douglas Sea Scale.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 132 |
| <a href="#">SedimentTypeEnum32</a>            | The type of sediment on the sea floor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 133 |
| <a href="#">SensorStateEnum32</a>             | The emission states of aggregate sensors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 133 |
| <a href="#">SubsurfaceFormationTypeEnum32</a> | Subsurface formation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 133 |
| <a href="#">SurfaceFormationTypeEnum32</a>    | Surface formation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 133 |
| <a href="#">SurfaceMoistureEnum16</a>         | Road surface wetness or soil moisture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 133 |
| <a href="#">TaskModeEnum8</a>                 | Execute task in non-concurrent mode or concurrent mode. In the non-concurrent mode the task is placed on the task list for the entity, which serves as a waiting list. Once the task is at the head of the task list, and the currently executing task completes, it is removed from the task list and started. Using this task mode, tasks are executed one after the other. In the concurrent mode, the task is executed concurrently with other tasks. As soon as the task is accepted for an entity, it is started. With this task mode, there is no task list involved. | 133 |
| <a href="#">TaskStatusEnum32</a>              | The status of a task.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 134 |
| <a href="#">TransferDirectionEnum32</a>       | Indicates the direction of flow of material or supplies during service delivery.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 134 |
| <a href="#">TransferEnumType</a>              | Transfer type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 134 |
| <a href="#">WeaponControlOrderEnum8</a>       | The enumerations for weapon control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 134 |
| <a href="#">WeatherModelTypeEnum32</a>        | Type of weather model used by a METOC service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 134 |

## ATP45HazardAreaTypeEnum8 ENUMERATION

Type of ATP-45 Hazard Area (simple or detailed).

Held as u8.

|          |   |
|----------|---|
| simple   | 0 |
| detailed | 1 |

Named by [ATP45HazardArea](#).

## ActiveStatusEnum8 ENUMERATION

A state which indicates the status of an object concerning its participation in the simulation. An object in an inactive state is not simulated and does not interact with other objects.

Held as u8.

|          |   |
|----------|---|
| other    | 0 |
| active   | 1 |
| inactive | 2 |

Named by [NETNAggregate](#), [NETNAircraft](#), [NETNAmphibiousVehicle](#), [NETNCulturalFeature](#), [NETNGroundVehicle](#), [NETNHuman](#), [NETNMultiDomainPlatform](#), [NETNMunition](#), [NETNNonHuman](#), [NETNSpacecraft](#), [NETNSubmersibleVessel](#), [NETNSurfaceVessel](#), [SEFacility](#).

## AgentClassEnum8 ENUMERATION

Class of Agent for an ATP-45 Hazard Area.

Held as u8.

|              |   |
|--------------|---|
| chemical     | 0 |
| biological   | 1 |
| radiological | 2 |
| nuclear      | 3 |

Named by [ATP45HazardArea](#).

## AgentTypeEnum16 ENUMERATION

Type of CBRN hazardous agent.

Held as i16.

|                         |      |
|-------------------------|------|
| chemical                | 1000 |
| choking                 | 1100 |
| cg                      | 1102 |
| persistentNerve         | 1200 |
| ga                      | 1201 |
| gb                      | 1202 |
| gd                      | 1203 |
| gf                      | 1204 |
| vx                      | 1205 |
| nonPersistentNerve      | 1300 |
| vesicant                | 1400 |
| hd                      | 1401 |
| bloodAgent              | 1500 |
| ac                      | 1501 |
| ck                      | 1502 |
| riotControl             | 1700 |
| toxicIndustrialChemical | 1800 |
| tic1                    | 1801 |
| tic2                    | 1802 |
| biological              | 2000 |
| bacterial               | 2100 |
| anthrax                 | 2101 |
| plague                  | 2102 |
| toxin                   | 2200 |
| virus                   | 2300 |
| radiological            | 3000 |
| alpha                   | 3100 |
| beta                    | 3200 |
| gamma                   | 3300 |
| delta                   | 3400 |
| nuclear                 | 4000 |

Named by [AgentConcentrationStruct](#), [AgentMassStruct](#), [ArrayOfAgentTypeEnum](#), [CBRNAlarmStruct](#), [CBRNExposureStruct](#), [CBRNRelease](#), [ProbabilityHazardContourGroup](#), [ProtectionEffectivenessStruct](#), [RawDataHazardContourGroup](#).

## AggregateMissionEnum16 ENUMERATION

Representation of the general class or nature of activity related to a unit's mission. Enumerations are based on JC3IEDM action-event-category-code.

Held as i16.

|                        |    |                             |    |                          |    |
|------------------------|----|-----------------------------|----|--------------------------|----|
| abdication             | 1  | armsTrade                   | 22 | bellyLanding             | 43 |
| accident               | 2  | arrestingLegal              | 23 | blocking                 | 44 |
| accidentAircraftGround | 3  | arrestingOrObstructing      | 24 | bombing                  | 45 |
| accidentMine           | 4  | arson                       | 25 | bombingAccidental        | 46 |
| accidentTraffic        | 5  | artilleryFire               | 26 | bombingDeliberate        | 47 |
| accidentWeapon         | 6  | assassination               | 27 | boobyTrapDiscovery       | 48 |
| accidentWorkplace      | 7  | assembling                  | 28 | borderCrossingEscorted   | 49 |
| advancing              | 8  | assistingACriminal          | 29 | borderCrossingForced     | 50 |
| aerialEngagement       | 9  | atmosphericPollution        | 30 | borderCrossingIllegal    | 51 |
| aerialShootDown        | 10 | attackDeliberate            | 31 | borderCrossingNotPlanned | 52 |
| airAssault             | 11 | attackDiversion             | 32 | borderCrossingPlanned    | 53 |
| airborneAssault        | 12 | attackElectronic            | 33 | borderCrossingSurveilled | 54 |
| aircraftCrash          | 13 | attackHasty                 | 34 | borderIncursion          | 55 |
| aircraftLanding        | 14 | attackMain                  | 35 | borderRaid               | 56 |
| aircraftLaunchActivity | 15 | attackNotOtherwiseSpecified | 36 | breaching                | 57 |
| aircraftLoss           | 16 | attackSupporting            | 37 | buildUp                  | 58 |
| airspaceViolation      | 17 | attemptedMurder             | 38 | burnedOutObject          | 59 |
| alertCancellation      | 18 | attemptedRape               | 39 | bypass                   | 60 |
| ambush                 | 19 | attemptedRobbery            | 40 | canalise                 | 61 |
| amphibiousOperation    | 20 | attemptedSuicide            | 41 | capture                  | 62 |
| armsProduction         | 21 | avoiding                    | 42 | carrierLaunch            | 63 |

|                            |     |                             |     |                              |     |
|----------------------------|-----|-----------------------------|-----|------------------------------|-----|
| carrierRecovery            | 64  | drugManufacturingIllegal    | 131 | locating                     | 200 |
| cbrnEvent                  | 65  | drugOperation               | 132 | looting                      | 201 |
| ceremonyOrParade           | 66  | drugStorage                 | 133 | maintaining                  | 202 |
| civilDemonstrationIllegal  | 67  | drugTransportation          | 134 | marking                      | 203 |
| civilDemonstrationLegal    | 68  | earlyWarningAlert           | 135 | martialLawImplementation     | 204 |
| civilDisobedience          | 69  | earthquake                  | 136 | massingOfForces              | 205 |
| civilUnrest                | 70  | electionAssociatedViolence  | 137 | massiveDeportationOr         | 206 |
| civilWar                   | 71  | electronicEmission          | 138 | Banishment                   |     |
| clearingAir                | 72  | electronicWarfare           | 139 | medicalEvacuation            | 207 |
| clearingLandCombat         | 73  | enemyContact                | 140 | militaryMobilisation         | 208 |
| clearingObstacle           | 74  | engaging                    | 141 | mineLaying                   | 209 |
| clearingRadioNet           | 75  | enveloping                  | 142 | missingIndividual            | 210 |
| codewordExecution          | 76  | epidemic                    | 143 | missionStaging               | 211 |
| collisionMidAir            | 77  | equipmentFailure            | 144 | mortarFire                   | 212 |
| collisionObstacle          | 78  | escaping                    | 145 | moving                       | 213 |
| communicationsActivation   | 79  | escorting                   | 146 | murder                       | 214 |
| communicationsDeactivation | 80  | evacuating                  | 147 | mutualAssistancePact         | 215 |
| communicationsDisruption   | 81  | execution                   | 148 | Agreement                    |     |
| communicationsInterception | 82  | exploitation                | 149 | nationalElection             | 216 |
| communicationsOutage       | 83  | explosion                   | 150 | nationalHoliday              | 217 |
| communicationsRestoration  | 84  | famine                      | 151 | nationalStateOfEmergency     | 218 |
| conductingConference       | 85  | fire                        | 152 | naturalDisaster              | 219 |
| conductingForwardPassage   | 86  | firefighting                | 153 | navalGunFire                 | 220 |
| OfLines                    |     | fix                         | 154 | navalPlatformFlight          | 221 |
| conductingMediaInterview   | 87  | fixAcoustic                 | 155 | Operations                   |     |
| conductingPreparatoryFire  | 88  | fixElectromagnetic          | 156 | networkSeizure               | 222 |
| conductingRearwardPassage  | 89  | fixElectroOptical           | 157 | neutralizeChemical           | 223 |
| OfLines                    |     | flood                       | 158 | neutralizeCombat             | 224 |
| conductingRecreational     | 90  | followingAndAssuming        | 159 | neutralizeExplosive          | 225 |
| Activities                 |     | followingAndSupporting      | 160 | obscure                      | 226 |
| conductingRoadService      | 91  | forcedLanding               | 161 | observing                    | 227 |
| conductingSocialEvents     | 92  | friendlyFire                | 162 | occupying                    | 228 |
| conductingSportingEvents   | 93  | generatingChemicalSmoke     | 163 | oceansSeasOrWaterPollution   | 229 |
| confiscation               | 94  | genocide                    | 164 | offensiveOrCounteroffensive  | 230 |
| consolidatingOfAPosition   | 95  | governmentalCollapse        | 165 | organisedCrime               | 231 |
| constructing               | 96  | guarding                    | 166 | outbreakOfRacialOrTribalOr   | 232 |
| containing                 | 97  | gunneryAirToAir             | 167 | EthnicWarfare                |     |
| cooperating                | 98  | harassing                   | 168 | patrolling                   | 233 |
| counterAttack              | 99  | hiding                      | 169 | peaceConference              | 234 |
| counterAttackByFire        | 100 | hijackingBoat               | 170 | peaceTreatyAgreement         | 235 |
| counterBatteryFire         | 101 | hijackingLandVehicle        | 171 | penetrating                  | 236 |
| coupDetat                  | 102 | hijackingNotOtherwise       | 172 | pestilence                   | 237 |
| covering                   | 103 | Specified                   |     | petroleumProductSpills       | 238 |
| crimeAgainstHumanity       | 104 | hijackingPlane              | 173 | picketing                    | 239 |
| criminalIncident           | 105 | holdDefensive               | 174 | poisoning                    | 240 |
| crossing                   | 106 | holdOffensive               | 175 | politicalDemonstration       | 241 |
| dazzle                     | 107 | hostageTaking               | 176 | politicalExecution           | 242 |
| deathNaturalCauses         | 108 | humanRightsViolation        | 177 | pOWReturn                    | 243 |
| deathOfChiefOfState        | 109 | hunting                     | 178 | prisonerExchange             | 244 |
| deathOfSpiritualLeader     | 110 | identifying                 | 179 | procuring                    | 245 |
| deception                  | 111 | illumination                | 180 | protectionElectronic         | 246 |
| deceptionElectronic        | 112 | indirectFire                | 181 | providingAccommodation       | 247 |
| defeat                     | 113 | indiscriminateShooting      | 182 | providingAgriculturalSupport | 248 |
| defending                  | 114 | industrialEspionageIncident | 183 | providingBedding             | 249 |
| deflecting                 | 115 | infiltration                | 184 | providingCamps               | 250 |
| delaying                   | 116 | interception                | 185 | providingConstruction        | 251 |
| demolition                 | 117 | interdiction                | 186 | Services                     |     |
| demonstration              | 118 | intimidation                | 187 | providingDecontamination     | 252 |
| denying                    | 119 | invasion                    | 188 | Services                     |     |
| deploying                  | 120 | isolation                   | 189 | providingEducationServices   | 253 |
| destroying                 | 121 | issuingMediaArticle         | 190 | providingHealthcareServices  | 254 |
| disease                    | 122 | issuingMediaDocumentary     | 191 | providingHostNationSupport   | 255 |
| disengaging                | 123 | issuingPressRelease         | 192 | providingInfrastructure      | 256 |
| disrupting                 | 124 | jamming                     | 193 | providingLaundryServices     | 257 |
| distributing               | 125 | kidnapping                  | 194 | providingRepairServices      | 258 |
| diversion                  | 126 | labourStrike                | 195 | providingSecurityServices    | 259 |
| driveByShooting            | 127 | leaguer                     | 196 | providingShelter             | 260 |
| drought                    | 128 | letterBombExplosion         | 197 | providingStorageServices     | 261 |
| drugConsumptionIllegal     | 129 | letterBombIncident          | 198 |                              |     |
| drugDistributionIllegal    | 130 | localElection               | 199 |                              |     |

|                             |     |                           |     |
|-----------------------------|-----|---------------------------|-----|
| providingTransshipment      | 262 | troublemakingHooliganism  | 331 |
| Services                    |     | troublemakingInciting     | 332 |
| proxyBombing                | 263 | troublemakingIntimidating | 333 |
| psychologicalOperation      | 264 | turning                   | 334 |
| publishingMediaArticle      | 265 | unexplodedOrdnance        | 335 |
| publishingMediaDocumentary  | 266 | Discovery                 |     |
| publishingPressRelease      | 267 | vandalismOrRapeOrLootOr   | 336 |
| pursuing                    | 268 | RansackOrPlunderOrSack    |     |
| rape                        | 269 | verifying                 | 337 |
| reconnaissance              | 270 | vesselSinking             | 338 |
| reconnaissanceInForce       | 271 | volcanicEruption          | 339 |
| reconstituting              | 272 | warOrCrisisAlert          | 340 |
| recovering                  | 273 | warOrMilitaryConference   | 341 |
| recuperating                | 274 | warCrime                  | 342 |
| redeployment                | 275 | weaponFiring              | 343 |
| refugeeMovement             | 276 | withdrawal                | 344 |
| reinforcing                 | 277 | withdrawalUnderPressure   | 345 |
| reliefInPlace               | 278 | witnessing                | 346 |
| religiousDemonstration      | 279 | notOtherwiseSpecified     | 347 |
| religiousViolence           | 280 | other                     | 0   |
| religiousWarfare            | 281 |                           |     |
| rendezvous                  | 282 |                           |     |
| reorganising                | 283 |                           |     |
| repairing                   | 284 |                           |     |
| resting                     | 285 |                           |     |
| resupplying                 | 286 |                           |     |
| retain                      | 287 |                           |     |
| retire                      | 288 |                           |     |
| revolution                  | 289 |                           |     |
| riot                        | 290 |                           |     |
| robbery                     | 291 |                           |     |
| rocketFire                  | 292 |                           |     |
| sabotage                    | 293 |                           |     |
| screening                   | 294 |                           |     |
| secessionOfPortionOfCountry | 295 |                           |     |
| securing                    | 296 |                           |     |
| securityCompromise          | 297 |                           |     |
| securityViolation           | 298 |                           |     |
| seizing                     | 299 |                           |     |
| servingAsABreakoutForce     | 300 |                           |     |
| servingAsABridgeheadForce   | 301 |                           |     |
| servingAsAFlankGuard        | 302 |                           |     |
| servingAsAMainBody          | 303 |                           |     |
| servingAsAnAdvanceGuard     | 304 |                           |     |
| servingAsAnInPlaceForce     | 305 |                           |     |
| servingAsARearGuard         | 306 |                           |     |
| servingAsAReserve           | 307 |                           |     |
| settingUp                   | 308 |                           |     |
| shooting                    | 309 |                           |     |
| sniperAttack                | 310 |                           |     |
| spaceAccident               | 311 |                           |     |
| spying                      | 312 |                           |     |
| stateOfWar                  | 313 |                           |     |
| strafingAerial              | 314 |                           |     |
| strike                      | 315 |                           |     |
| suicide                     | 316 |                           |     |
| supporting                  | 317 |                           |     |
| suppressing                 | 318 |                           |     |
| surrender                   | 319 |                           |     |
| surveillanceElectronic      | 320 |                           |     |
| suspensionOfHostilities     | 321 |                           |     |
| terrorism                   | 322 |                           |     |
| threaten                    | 323 |                           |     |
| torture                     | 324 |                           |     |
| transporting                | 325 |                           |     |
| traversing                  | 326 |                           |     |
| treatyViolation             | 327 |                           |     |
| troublemakingAgitating      | 328 |                           |     |
| troublemakingBullying       | 329 |                           |     |
| troublemakingHarassing      | 330 |                           |     |

Named by [MissionStruct](#), [NETNAggregate](#), [NETNAircraft](#), [NETNAmphibiousVehicle](#), [NETNCulturalFeature](#), [NETNGroundVehicle](#), [NETNHuman](#), [NETNMultiDomainPlatform](#), [NETNNonHuman](#), [NETNSpacecraft](#), [NETNSubmersibleVessel](#), [NETNSurfaceVessel](#), [SpottedEntity](#).

## AidTypeEnumType ENUMERATION

Enumeration value for the aid type. According to [IALA], the aid type field has values 1-15 for fixed and 16-31 for floating aids to navigation.

Held as u8.

|                                         |    |
|-----------------------------------------|----|
| notSpecified                            | 0  |
| referencePoint                          | 1  |
| racon                                   | 2  |
| fixedStructureOffShore                  | 3  |
| spare1                                  | 4  |
| lightWithoutSectors                     | 5  |
| lightWithSectors                        | 6  |
| leadingLightFront                       | 7  |
| leadingLightRear                        | 8  |
| beaconCardinalN                         | 9  |
| beaconCardinalE                         | 10 |
| beaconCardinalS                         | 11 |
| beaconCardinalW                         | 12 |
| beaconPortHand                          | 13 |
| beaconStarboardHand                     | 14 |
| beaconPreferredChannelPort<br>Hand      | 15 |
| beaconPreferredChannel<br>StarboardHand | 16 |
| beaconIsolatedDanger                    | 17 |
| beaconSafeWater                         | 18 |
| beaconSpecialMark                       | 19 |
| cardinalMarkN                           | 20 |
| cardinalMarkE                           | 21 |
| cardinalMarkS                           | 22 |
| cardinalMarkW                           | 23 |
| portHandMark                            | 24 |
| starboardHandMark                       | 25 |
| preferredChannelPortHand                | 26 |
| preferredChannelStarboard<br>Hand       | 27 |
| isolatedDanger                          | 28 |
| safeWater                               | 29 |
| specialMark                             | 30 |
| lightVessel                             | 31 |

Named by [AisMessage21](#).

## AirFormationTypeEnum32 ENUMERATION

Air formation

Held as i32.

|              |   |
|--------------|---|
| notSpecified | 0 |
|--------------|---|

Named by [FormationDataStruct](#).

## AltitudeTypeEnum8 ENUMERATION

The reference for altitude. AMSL = Above Mean Sea Level or AGL = Above Ground Level.

Held as u8.

|      |   |
|------|---|
| amsl | 1 |
| agl  | 2 |

Named by [SetOrderedAltitude](#).

## AttackTypeEnum32 ENUMERATION

The kind of attack by the enemy.

Held as i32.

|                |   |
|----------------|---|
| none           | 0 |
| artillery      | 1 |
| heavyArtillery | 2 |
| other          | 3 |

Named by [UnderAttackStatusReport](#).

## CBRNDamageEnum8 ENUMERATION

Level of damage due to CBRN exposure.

Held as u8.

|           |   |
|-----------|---|
| uninjured | 0 |
| t3        | 1 |
| t2        | 2 |
| t1        | 3 |
| t4        | 4 |
| dead      | 5 |

Named by [CBRNCasualty](#), [CBRNHuman](#).

## CancellationReasonEnum32 ENUMERATION

Describes the reason for a cancellation.

Held as i32.

|         |   |
|---------|---|
| other   | 0 |
| timeOut | 1 |

Named by [CancelRequestTMR](#).

## CaptureStatusEnum8 ENUMERATION

The status of a person or unit with respect to their control or influence over their own activities. Default: 1 - Not Captured.

Held as u8.

|                     |   |
|---------------------|---|
| notCaptured         | 1 |
| captured            | 2 |
| attemptingSurrender | 3 |
| other               | 0 |

Named by [NETNAggregate](#).

## CloudTypeEnum32 ENUMERATION

Classification of different types of clouds.

Held as i32.

|                            |    |
|----------------------------|----|
| noCloud                    | 1  |
| altocumulus                | 5  |
| altostratus                | 6  |
| cirrocumulus               | 3  |
| cirrostratus               | 4  |
| cirrus                     | 2  |
| cumulonimbus               | 12 |
| cumulus                    | 9  |
| nimbostratus               | 8  |
| stratocumulus              | 7  |
| stratus                    | 10 |
| unknown                    | 13 |
| cumulusCongestus           | 11 |
| invalidCloudClassification | 0  |



Named by [CloudStruct](#).

---

## CommunicationNetworkTypeEnum ENUMERATION

The type of a communication network.

Held as u8.

|                           |   |
|---------------------------|---|
| other                     | 0 |
| commandNet                | 1 |
| operationsIntelligenceNet | 2 |
| adminLogisticsNet         | 3 |
| fireSupportNet            | 4 |

Named by [CommunicationNetwork](#).

---

## CommunicationServiceTypeEnum ENUMERATION

The type of service used on a communication network. Service types as defined in MSDL. DATTRF: Data transfer FAX: Facsimile IFF: Identify Friend or Foe IMAGE: Image MCI: Multilateral Interoperability Programme (MIP) Common Interface Service MHS: Message Handling Service TDL: Tactical Data Link VIDSVC: Video Service VOCSVC:Voice Service NOS: Not Otherwise Specified

Held as u8.

|        |   |
|--------|---|
| dattrf | 0 |
| fax    | 1 |
| iif    | 2 |
| image  | 3 |
| mci    | 4 |
| mhs    | 5 |
| tdl    | 6 |
| vidsvc | 7 |
| vocsvc | 8 |
| nos    | 9 |

Named by [CommunicationNetwork](#).

---

## CommunicationServiceTypeEnum32 ENUMERATION

List of Communications Net Types for Army Units.

Held as i32.

|        |   |
|--------|---|
| dattrf | 0 |
| fax    | 1 |
| iif    | 2 |
| image  | 3 |
| mci    | 4 |
| mhs    | 5 |
| tdl    | 6 |
| vidsvc | 7 |
| vocsvc | 8 |
| nos    | 9 |

Named by [CommunicationNetworkStruct](#).

---

## ConcealmentEnum32 ENUMERATION

The reason for the objects concealment

Held as u32.

|                           |   |
|---------------------------|---|
| invalid                   | 0 |
| inOpen                    | 1 |
| mountedInternally         | 2 |
| mountedExternally         | 3 |
| underNet                  | 4 |
| underGround               | 5 |
| insideStructure           | 6 |
| fightingPositionCovered   | 7 |
| fightingPositionUncovered | 8 |

Named by [EntityStruct](#).

---

## ConnectionTypeEnum ENUMERATION

The type of a connection.

Held as u8.

|                      |   |
|----------------------|---|
| broadcastTransceiver | 0 |
| broadcastTransmitter | 1 |
| broadcastReceiver    | 2 |
| peerToPeer           | 3 |
| unidirectional       | 4 |
| multicast            | 5 |
| intercept            | 6 |

Named by [RequestedConnection](#).

---

## DamageStatusEnhancedEnum32 ENUMERATION

The damage status of an object.

Held as i32.

|                   |   |
|-------------------|---|
| noDamage          | 0 |
| slightDamage      | 1 |
| moderateDamage    | 2 |
| significantDamage | 3 |
| destroyed         | 4 |

Named by [DamageStatusReport](#), [EntityStruct](#).

---

## EchelonEnum32 ENUMERATION

The echelon level of a unit.

Held as i32.

|            |    |
|------------|----|
| none       | 0  |
| team       | 1  |
| crew       | 2  |
| squad      | 3  |
| section    | 4  |
| platoon    | 5  |
| detachment | 6  |
| company    | 7  |
| battery    | 8  |
| troop      | 9  |
| battalion  | 10 |
| squadron   | 11 |
| regiment   | 12 |
| group      | 13 |
| brigade    | 14 |
| division   | 15 |
| corps      | 16 |
| army       | 17 |
| armygroup  | 18 |
| front      | 19 |
| region     | 20 |

Named by [NETNAggregate](#), [Unit](#).

---

## EntityCategoryEnum32 ENUMERATION

Category of entity

Held as u32.

|                 |   |             |   |
|-----------------|---|-------------|---|
| invalid         | 0 | radioEntity | 4 |
| equipmentEntity | 1 |             |   |
| personnelEntity | 2 |             |   |
| emitterEntity   | 3 |             |   |

Named by [EntityStruct](#).

---

## ExposureTypeEnum8 ENUMERATION

Type of exposure represented in a contour group.

Held as u8.

|                |   |
|----------------|---|
| exposure       | 0 |
| incapacitation | 1 |
| lethal         | 2 |

Named by [ProbabilityHazardContourGroup](#).

---

## FormationLocationTypeEnum32 ENUMERATION

Enumerated choice for the method used to correlate formation to location as center of mass or lead element.

Held as i32.

|              |   |
|--------------|---|
| leadElement  | 0 |
| centerOfMass | 1 |

Named by [FormationStruct](#).

---

## FormationTypeEnum32 ENUMERATION

Enumerated choice for the type of formation being Ground, Air, Surface or Subsurface.

Held as i32.

|            |   |
|------------|---|
| ground     | 0 |
| air        | 1 |
| surface    | 2 |
| subsurface | 3 |

---

## GeoLocationTypeEnum32 ENUMERATION

Specifies different ways to reference geographical locations.

Held as i32.

|               |   |
|---------------|---|
| point         | 1 |
| circle        | 2 |
| polygon       | 3 |
| quadrangle    | 4 |
| path          | 5 |
| netnUuid      | 6 |
| rprEntity     | 7 |
| nameReference | 8 |

---

## GroundFormationTypeEnum32 ENUMERATION

Specifies the formations for groundunits.

Held as i32.

|                 |    |
|-----------------|----|
| column          | 0  |
| staggeredColumn | 1  |
| echelonLeft     | 2  |
| echelonRight    | 3  |
| line            | 4  |
| wedge           | 5  |
| vee             | 6  |
| assaultVee      | 7  |
| fseColumn       | 8  |
| stack           | 9  |
| none            | 10 |

Named by [FormationDataStruct](#).

**HazardTypeEnum8** ENUMERATION

Type of dispersion output represented in a contour group.

Held as u8.

|                      |   |
|----------------------|---|
| concentration        | 0 |
| deposition           | 1 |
| dosage               | 2 |
| radiologicalDose     | 3 |
| radiologicalDoseRate | 4 |

Named by [RawDataHazardContourGroup](#).

**HazeTypeEnum32** ENUMERATION

Type of visibility obstruction.

Held as i32.

|             |    |
|-------------|----|
| none        | 0  |
| fog         | 1  |
| iceFog      | 2  |
| steamFog    | 3  |
| mist        | 4  |
| smoke       | 5  |
| volcanicAsh | 6  |
| dust        | 7  |
| sand        | 8  |
| snow        | 9  |
| otherBlue   | 10 |
| otherRed    | 11 |
| otherGreen  | 12 |

Named by [HazeStruct](#).

**HostilityStatusCodeEnum32** ENUMERATION

The specific value that represents the perceived hostility status.

Held as i32.

|        |    |
|--------|----|
| afr    | 0  |
| aho    | 1  |
| aiv    | 2  |
| ant    | 3  |
| faker  | 4  |
| fr     | 5  |
| ho     | 6  |
| iv     | 7  |
| joker  | 8  |
| neutrl | 9  |
| pendng | 10 |
| suspct | 11 |
| unk    | 12 |

Named by [Relation](#).

**IPTypeEnum8** ENUMERATION

Types of Individual Protective Equipment.

Held as u8.

|        |   |       |   |
|--------|---|-------|---|
| none   | 0 | fourR | 8 |
| one    | 1 |       |   |
| oneR   | 2 |       |   |
| two    | 3 |       |   |
| twoR   | 4 |       |   |
| three  | 5 |       |   |
| threeR | 6 |       |   |
| four   | 7 |       |   |

Named by [CBRNHuman](#), [IPECommand](#).

## IceTypeEnum16 ENUMERATION

Type of Ice.

Held as i16.

|      |   |
|------|---|
| none | 0 |
| ice  | 1 |

Named by [IceStruct](#).

## IdentificationLevelEnum8 ENUMERATION

The identification level of an object.

Held as u8.

|               |   |
|---------------|---|
| lost          | 0 |
| detected      | 1 |
| classified    | 2 |
| identified    | 3 |
| fullKnowledge | 4 |

Named by [SensorReport](#), [SpotReport](#).

## ManeuverIndicatorEnumType ENUMERATION

Enumeration value to indicate a maneuver.

Held as u8.

|                   |   |
|-------------------|---|
| notAvailable      | 0 |
| noSpecialManeuver | 1 |
| specialManeuver   | 2 |

Named by [AisMessage1](#), [AisMessage2](#), [AisMessage3](#), [AisNavigationData](#).

## MineCountEnum32 ENUMERATION

Type of mine count.

Held as i32.

|             |   |
|-------------|---|
| mineNumber  | 0 |
| mineDensity | 1 |

## ModelResolutionTypeEnum32 ENUMERATION

Enumeration indicating the level of fidelity appropriate for instancing the unit or equipment in the simulation.

Held as i32.

|              |   |
|--------------|---|
| none         | 0 |
| minimal      | 1 |
| standard     | 2 |
| enhanced     | 3 |
| high         | 4 |
| notSpecified | 5 |

Named by [EquipmentItem](#).

## MoveTypeEnum32 ENUMERATION

CrossCountry: move directly to the destination without taking into account the roads. OnlyRoads: stay on the roads to get to the closest point to the destination that is still on a road; if there is no road very near to the start (within around 10 meters) there will be no movement at all. RoadsAndCrossCountry: move to the destination by taking into account the roads; it is allowed to go off the road.

Held as i32.

|                      |   |
|----------------------|---|
| crossCountry         | 0 |
| onlyRoads            | 1 |
| roadsAndCrossCountry | 2 |

Named by [MoveToEntity](#), [MoveToLocation](#).

## MsgIdEnumType ENUMERATION

Enumeration value for the message type.

Held as u8.

|       |    |
|-------|----|
| msg1  | 1  |
| msg2  | 2  |
| msg3  | 3  |
| msg4  | 4  |
| msg5  | 5  |
| msg6  | 6  |
| msg7  | 7  |
| msg8  | 8  |
| msg9  | 9  |
| msg10 | 10 |
| msg11 | 11 |
| msg12 | 12 |
| msg13 | 13 |
| msg14 | 14 |
| msg17 | 17 |
| msg18 | 18 |
| msg19 | 19 |
| msg21 | 21 |
| msg24 | 24 |
| msg27 | 27 |

Named by [AcknowledgeData](#), [AisBaseStationReportAndUTCDateResponseData](#), [AisBinaryMessageData](#), [AisMessage](#), [AisMessage1](#), [AisMessage10](#), [AisMessage11](#), [AisMessage12](#), [AisMessage13](#), [AisMessage14](#), [AisMessage17](#), [AisMessage18](#), [AisMessage19](#), [AisMessage2](#), [AisMessage21](#), [AisMessage24](#), [AisMessage27](#), [AisMessage3](#), [AisMessage4](#), [AisMessage5](#), [AisMessage6](#), [AisMessage7](#), [AisMessage8](#), [AisMessage9](#), [AisNavigationData](#), [AisStaticAndVoyageData](#).

## NavigationStatusEnumType ENUMERATION

Enumeration value to indicate the navigational status.

Held as u8.

|                            |    |
|----------------------------|----|
| underWayUsingEngine        | 0  |
| atAnchor                   | 1  |
| notUnderCommand            | 2  |
| restrictedManoeuverability | 3  |
| constrainedByHerDraught    | 4  |
| moored                     | 5  |
| aground                    | 6  |
| engagedInFishing           | 7  |
| underWaySailing            | 8  |
| reserved1                  | 9  |
| reserved2                  | 10 |
| reserved3                  | 11 |
| reserved4                  | 12 |
| reserved5                  | 13 |
| aisSartIsActive            | 14 |
| notDefined                 | 15 |

Named by [AisMessage1](#), [AisMessage2](#), [AisMessage27](#), [AisMessage3](#), [AisNavigationData](#).

## NoOfferReasonEnumType ENUMERATION

Describes the reason why not accepting a TMR request

Held as i32.

|                           |   |
|---------------------------|---|
| other                     | 0 |
| capabilityDoesNotMatch    | 1 |
| attributeSetTooRestricted | 2 |

|                                          |   |
|------------------------------------------|---|
| attributeSetTooExtensive                 | 3 |
| federateTooBusy                          | 4 |
| attributeSetNotCompatibleWithPublication | 5 |
| ownershipStateNotCompatibleWithRequest   | 6 |
| entityNotKnown                           | 7 |

Named by [OfferTransfer](#).

## OfferTypeEnum32 ENUMERATION

Type of the offer (With restriction, positive, negative)

Held as i32.

|                                   |   |
|-----------------------------------|---|
| requestAcknowledgeWithRestriction | 0 |
| requestAcknowledgePositive        | 1 |
| requestAcknowledgeNegative        | 2 |

Named by [OfferRepair](#), [OfferService](#), [OfferSupply](#), [OfferTransport](#).

## PartNumberEnumType ENUMERATION

Identifier for the message part number.

Held as u8.

|       |   |
|-------|---|
| parta | 0 |
| partb | 1 |

Named by [AisMessage24](#).

## PathTypeEnum32 ENUMERATION

Specifies if a path is defined by waypoints or by reference to a path object in the federation.

Held as i32.

|               |   |
|---------------|---|
| waypoints     | 0 |
| uuidReference | 1 |

## PatrolMoveTypeEnum32 ENUMERATION

Defines how the movement shall be done during the execution of the patrol. Other - Federate application specific.

Held as i32.

|         |   |
|---------|---|
| other   | 0 |
| stealth | 1 |
| caution | 2 |
| open    | 3 |

Named by [Patrol](#), [PatrolRepeating](#).

## PatrolTypeEnum32 ENUMERATION

Other - The behaviour is federate application specific (e.g. wait for next command, etc.) Circle - At the end of route, move to start point of patrol route (federate application determines the route) Reverse - Follow the patrol route in reverse order.

Held as i32.

|         |   |
|---------|---|
| other   | 0 |
| circle  | 1 |
| reverse | 2 |

Named by [Patrol](#), [PatrolRepeating](#).

## PhysicalNetworkTypeEnum ENUMERATION

The type of a physical network.

Held as u8.

|           |   |           |   |         |   |
|-----------|---|-----------|---|---------|---|
| noNetwork | 0 | undefined | 1 | generic | 2 |
|-----------|---|-----------|---|---------|---|

|       |   |
|-------|---|
| radio | 3 |
| cable | 4 |
| laser | 5 |

---

## PointTypeEnum32 ENUMERATION

Specifies if a point is defined by a location or by reference to a point object in the federation.

Held as i32.

|               |   |
|---------------|---|
| location      | 0 |
| uuidReference | 1 |

---

## PrecipitationTypeEnum32 ENUMERATION

Type of precipitation.

Held as i32.

|                 |   |
|-----------------|---|
| noPrecipitation | 0 |
| rain            | 1 |
| freezingRain    | 2 |
| snow            | 3 |
| hail            | 4 |

Named by [PrecipitationStruct](#).

---

## RoadIceConditionEnum16 ENUMERATION

Ice condition for roads.

Held as i16.

|                 |   |
|-----------------|---|
| none            | 0 |
| patches         | 1 |
| blackIce        | 2 |
| slush           | 3 |
| sheetIce        | 4 |
| extremeSheetIce | 5 |

Named by [LandSurface](#), [LandSurfaceCondition](#).

---

## RulesOfEngagementEnum8 ENUMERATION

Rules of engagement for a simulated entity.

Held as u8.

|                   |   |
|-------------------|---|
| holdFire          | 0 |
| fireAtWill        | 1 |
| fireWhenFiredUpon | 2 |

Named by [SetRulesOfEngagement](#).

---

## SeaStateEnum16 ENUMERATION

State of the sea measured in Douglas Sea Scale.

Held as i16.

|                |   |
|----------------|---|
| calmGlassy     | 0 |
| calmRippled    | 1 |
| smoothWavelets | 2 |
| slight         | 3 |
| moderate       | 4 |
| rough          | 5 |
| veryRough      | 6 |
| high           | 7 |
| veryHigh       | 8 |
| phenomenal     | 9 |

Named by [WaterSurface](#), [WaterSurfaceCondition](#).



## SedimentTypeEnum32 ENUMERATION

The type of sediment on the sea floor.

Held as i32.

|            |   |
|------------|---|
| noSediment | 0 |
| sand       | 1 |
| mud        | 2 |
| gravel     | 3 |
| rock       | 4 |

Named by [SubsurfaceLayer](#), [SubsurfaceLayerCondition](#).

## SensorStateEnum32 ENUMERATION

The emission states of aggregate sensors

Held as u32.

|                  |   |
|------------------|---|
| onButNotEmitting | 2 |
| off              | 1 |
| other            | 0 |
| onAndEmitting    | 3 |

Named by [SensorStruct](#).

## SubsurfaceFormationTypeEnum32 ENUMERATION

Subsurface formation

Held as i32.

|              |   |
|--------------|---|
| notSpecified | 0 |
|--------------|---|

Named by [FormationDataStruct](#).

## SurfaceFormationTypeEnum32 ENUMERATION

Surface formation

Held as i32.

|              |   |
|--------------|---|
| notSpecified | 0 |
|--------------|---|

Named by [FormationDataStruct](#).

## SurfaceMoistureEnum16 ENUMERATION

Road surface wetness or soil moisture.

Held as i16.

|                |   |
|----------------|---|
| dry            | 0 |
| wet            | 1 |
| flooded        | 2 |
| heavilyFlooded | 3 |

Named by [LandSurface](#), [LandSurfaceCondition](#).

## TaskModeEnum8 ENUMERATION

Execute task in non-concurrent mode or concurrent mode. In the non-concurrent mode the task is placed on the task list for the entity, which serves as a waiting list. Once the task is at the head of the task list, and the currently executing task completes, it is removed from the task list and started. Using this task mode, tasks are executed one after the other. In the concurrent mode, the task is executed concurrently with other tasks. As soon as the task is accepted for an entity, it is started. With this task mode, there is no task list involved.

Held as u8.

|                   |   |
|-------------------|---|
| nonConcurrentMode | 0 |
| concurrentMode    | 1 |

Named by [AddPassage](#), [ClearObstacle](#), [CreateMinefield](#), [CreateObstacle](#), [Dismount](#), [DisruptCommunication](#), [ETRTask](#), [EstablishCheckPoint](#), [FireAtEntity](#), [FireAtEntityWM](#), [FireAtLocation](#), [FireAtLocationWM](#), [FollowEntity](#), [Mount](#), [Move](#), [MoveIntoFormation](#), [MoveToEntity](#), [MoveToLocation](#), [Observe](#), [OperateCheckPoint](#), [Patrol](#), [PatrolRepeating](#), [RemoveCheckPoint](#), [RemovePassage](#), [SendSafetyRelatedBroadcastMessage](#), [SendSafetyRelatedMessage](#), [SetOrderedAltitude](#), [SetOrderedSpeed](#), [SetRulesOfEngagement](#), [SetTransmitterStatus](#), [StopAtSideOfRoad](#), [TurnToHeading](#), [TurnToOrientation](#), [Wait](#).

---

## TaskStatusEnum32 ENUMERATION

The status of a task.

Held as i32.

|           |   |
|-----------|---|
| accepted  | 0 |
| refused   | 1 |
| cancelled | 2 |
| executing | 3 |
| completed | 4 |
| error     | 5 |

Named by [TaskStatusReport](#).

---

## TransferDirectionEnum32 ENUMERATION

Indicates the direction of flow of material or supplies during service delivery.

Held as i32.

|              |   |
|--------------|---|
| fromConsumer | 0 |
| fromProvider | 1 |

Named by [RequestSupply](#).

---

## TransferEnumType ENUMERATION

Transfer type

Held as i32.

|                           |   |
|---------------------------|---|
| other                     | 0 |
| acquire                   | 1 |
| divest                    | 2 |
| acquireWithoutNegotiating | 3 |

Named by [InitiateTransfer](#), [RequestTransfer](#).

---

## WeaponControlOrderEnum8 ENUMERATION

The enumerations for weapon control

Held as u8.

|              |   |
|--------------|---|
| weaponsFree  | 1 |
| weaponsHold  | 3 |
| other        | 0 |
| weaponsTight | 2 |

Named by [NETNAggregate](#).

---

## WeatherModelTypeEnum32 ENUMERATION

Type of weather model used by a METOC service.

Held as i32.

|                |   |
|----------------|---|
| none           | 0 |
| simulated      | 1 |
| realHistorical | 2 |
| live           | 3 |
| standardModel  | 4 |

Named by [Service](#).

**Variant records**

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                    |     |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AreaVariantStruct</a>                              | Description of an area relative to the earth's surface.                                                                                                                                                                                                                                                                                                                            | 135 |
| <a href="#">FormationDataStruct</a>                            | Struct for enumerated choice for the type of formation being Ground, Air, Surface or Subsurface.                                                                                                                                                                                                                                                                                   | 135 |
| <a href="#">GeoLocationReferenceVariant</a>                    | The area affected by an Environment Condition can be expressed as: - a location on the earth's surface represented by a Point, - an area on the earth's surface, represented by a Quadrangle, GeodeticPolygon, or GeodeticCircle, or - a reference to some other object/data. Objects that can be referenced are RPR entities, NETN entities, NETN-SE GeoObjects and GML Features. | 135 |
| <a href="#">GeoReferenceVariant</a>                            | The area affected by an Environment Condition can be expressed as: - a location on the earth's surface represented by a Point, - an area on the earth's surface, represented by a Quadrangle, GeodeticPolygon, or GeodeticCircle, or - a reference to some other object/data. Objects that can be referenced are RPR entities, NETN entities, NETN-SE GeoObjects and GML Features. | 136 |
| <a href="#">MineCountVariantStruct</a>                         | The number or density of mines.                                                                                                                                                                                                                                                                                                                                                    | 136 |
| <a href="#">NetworkDeviceReceiverCharacteristicsVariant</a>    | Characteristics of a network device receiver.                                                                                                                                                                                                                                                                                                                                      | 136 |
| <a href="#">NetworkDeviceTransmitterCharacteristicsVariant</a> | Characteristics of network device transmitter.                                                                                                                                                                                                                                                                                                                                     | 137 |
| <a href="#">PathVariantStruct</a>                              | Defines a path, either as a polygonal chain of waypoints or a UUID that refers to a path object in the federation.                                                                                                                                                                                                                                                                 | 137 |
| <a href="#">PhysicalNetworkDescriptionVariant</a>              | Characteristics of physical networks depending on their type.                                                                                                                                                                                                                                                                                                                      | 137 |
| <a href="#">PointVariantStruct</a>                             | Defines the point, either a Location or a UUID reference to a point object in the federation.                                                                                                                                                                                                                                                                                      | 137 |

**AreaVariantStruct** VARIANT

Description of an area relative to the earth's surface.

| Type                               | Description                                                                                                                                                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">GeodeticPolygon</a>    | A sequence of geodetic locations defining a geographical area bounded by a closed path where the first and last locations in the sequence are connected. Each point is a geodetic coordinate in WGS84 on the earth surface, and each segment is a great circle between locations. |
| <a href="#">GeodeticCircle</a>     | A geodetic point and radius specifying a circle on the surface of the earth WGS84 where the radius is a great circle distance on the surface.                                                                                                                                     |
| <a href="#">GeodeticQuadrangle</a> | A latitude-longitude quadrangle is a region bounded by two meridians and two parallels.                                                                                                                                                                                           |

Named by [CreateMinefield](#), [CreateObstacle](#), [Observe](#), [Region](#).

**FormationDataStruct** VARIANT

Struct for enumerated choice for the type of formation being Ground, Air, Surface or Subsurface.

| Type                                          | Description                                     |
|-----------------------------------------------|-------------------------------------------------|
| <a href="#">GroundFormationTypeEnum32</a>     | Ground formations.                              |
| <a href="#">AirFormationTypeEnum32</a>        | Air formations, currently not specified         |
| <a href="#">SurfaceFormationTypeEnum32</a>    | Surface formations, currently not specified.    |
| <a href="#">SubsurfaceFormationTypeEnum32</a> | Subsurface formations, currently not specified. |

Named by [FormationStruct](#).

**GeoLocationReferenceVariant** VARIANT

The area affected by an Environment Condition can be expressed as: - a location on the earth's surface represented by a Point, - an area on the earth's surface, represented by a Quadrangle, GeodeticPolygon, or GeodeticCircle, or - a reference to some other object/data. Objects that can be referenced are RPR entities, NETN entities, NETN-SE GeoObjects and GML Features.

| Type                                       | Description                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">GeodeticLocation</a>           | A location on the surface of the earth defined by lat, lon WGS84.                                                                                  |
| <a href="#">GeodeticCircle</a>             | A geodetic point and radius specifying a circle on the surface of the earth WGS84 where the radius is a great circle distance on the surface.      |
| <a href="#">GeodeticQuadrangle</a>         | A latitude-longitude quadrangle is a region bounded by two meridians and two parallels.                                                            |
| <a href="#">GeodeticPolygon</a>            | A polygon on the surface of the earth defined by a number of great-circle arcs represented as lat, lon Locations. WGS84.                           |
| <a href="#">UUID</a>                       | Reference to an object instance of class Location, Path or Region, defined in NETN-SE FOM Module or a NETN Physical or Aggregate entity with UUID. |
| <a href="#">rpr.EntityIdentifierStruct</a> | Reference to an instance of a leaf node subclass of RPR-FOM BaseEntity by its EntityIdentifier. E.g. an Aircraft, SurfaceVessel etc.               |

Named by [SEFacility](#).

## GeoReferenceVariant VARIANT

The area affected by an Environment Condition can be expressed as: - a location on the earth's surface represented by a Point, - an area on the earth's surface, represented by a Quadrangle, GeodeticPolygon, or GeodeticCircle, or - a reference to some other object/data. Objects that can be referenced are RPR entities, NETN entities, NETN-SE GeoObjects and GML Features.

| Type                                       | Description                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">GeodeticLocation</a>           | A location on the surface of the earth defined by lat, lon WGS84.                                                                                  |
| <a href="#">GeodeticCircle</a>             | A geodetic point and radius specifying a circle on the surface of the earth WGS84 where the radius is a great circle distance on the surface.      |
| <a href="#">GeodeticQuadrangle</a>         | A latitude-longitude quadrangle is a region bounded by two meridians and two parallels.                                                            |
| <a href="#">GeodeticPolygon</a>            | A polygon on the surface of the earth defined by a number of great-circle arcs represented as lat, lon Locations. WGS84.                           |
| <a href="#">UUID</a>                       | Reference to an object instance of class Location, Path or Region, defined in NETN-SE FOM Module or a NETN Physical or Aggregate entity with UUID. |
| <a href="#">rpr.EntityIdentifierStruct</a> | Reference to an instance of a leaf node subclass of RPR-FOM BaseEntity by its EntityIdentifier. E.g. an Aircraft, SurfaceVessel etc.               |
| <a href="#">string</a>                     | Reference to a GML feature by its GML Identifier.                                                                                                  |

Named by [EnvironmentCondition](#), [LandSurfaceCondition](#), [RequestLandSurfaceCondition](#), [RequestMETOC](#), [RequestSubsurfaceLayerCondition](#), [RequestTroposphereLayerCondition](#), [RequestWaterSurfaceCondition](#), [RequestWeatherCondition](#), [ResponseMETOC](#), [SubsurfaceLayerCondition](#), [TroposphereLayerCondition](#), [WaterSurfaceCondition](#), [WeatherCondition](#).

## MineCountVariantStruct VARIANT

The number or density of mines.

| Type                               | Description          |
|------------------------------------|----------------------|
| <a href="#">rpr.Integer32</a>      | Number of mines.     |
| <a href="#">rpr.PercentFloat32</a> | Percentage of mines. |

Named by [CreateMinefield](#).

## NetworkDeviceReceiverCharacteristicsVariant VARIANT

Characteristics of a network device receiver.

| Type                                                    | Description |
|---------------------------------------------------------|-------------|
| <a href="#">NetworkDeviceEmptyCharacteristicsStruct</a> | No network. |
| <a href="#">NetworkDeviceEmptyCharacteristicsStruct</a> |             |
| <a href="#">NetworkDeviceEmptyCharacteristicsStruct</a> | NA          |

Named by [NetworkDeviceStruct](#).

## NetworkDeviceTransmitterCharacteristicsVariant VARIANT

Characteristics of network device transmitter.

| Type                                                                 | Description                                           |
|----------------------------------------------------------------------|-------------------------------------------------------|
| <a href="#">NetworkDeviceEmptyCharacteristicsStruct</a>              | No Network                                            |
| <a href="#">NetworkDeviceEmptyCharacteristicsStruct</a>              |                                                       |
| <a href="#">NetworkDeviceGenericTransmitterCharacteristicsStruct</a> | A generic description of transmitter characteristics. |

Named by [NetworkDeviceStruct](#).

## PathVariantStruct VARIANT

Defines a path, either as a polygonal chain of waypoints or a UUID that refers to a path object in the federation.

| Type                                       | Description                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">ArrayOfWorldLocationStruct</a> | The path defined by waypoints, not necessarily registered in the federation execution as a NETN_GeoObject.Path. The array can be empty (size=0). |
| <a href="#">UuidArrayOfHLAbyte16</a>       | A UUID that refers to a NETN_GeoObject.Path that is registered in the federation execution.                                                      |

Named by [NETNAggregate](#), [NETNAircraft](#), [NETNAmphibiousVehicle](#), [NETNGroundVehicle](#), [NETNHuman](#), [NETNMultiDomainPlatform](#), [NETNMunition](#), [NETNNonHuman](#), [NETNSpacecraft](#), [NETNSubmersibleVessel](#), [NETNSurfaceVessel](#).

## PhysicalNetworkDescriptionVariant VARIANT

Characteristics of physical networks depending on their type.

| Type                                           | Description                                               |
|------------------------------------------------|-----------------------------------------------------------|
| <a href="#">PhysicalUndefinedNetworkStruct</a> | (Empty) Characteristics of an undefined physical network. |
| <a href="#">PhysicalGenericNetworkStruct</a>   | Characteristics of a generic physical network.            |

Named by [PhysicalNetwork](#).

## PointVariantStruct VARIANT

Defines the point, either a Location or a UUID reference to a point object in the federation.

| Type                                    | Description                                   |
|-----------------------------------------|-----------------------------------------------|
| <a href="#">rpr.WorldLocationStruct</a> | The geocentric location.                      |
| <a href="#">UuidArrayOfHLAbyte16</a>    | A UUID that refers to a NETN_GeoObject.Point. |

Named by [NETNAggregate](#), [NETNAircraft](#), [NETNAmphibiousVehicle](#), [NETNGroundVehicle](#), [NETNHuman](#), [NETNMultiDomainPlatform](#), [NETNMunition](#), [NETNNonHuman](#), [NETNSpacecraft](#), [NETNSubmersibleVessel](#), [NETNSurfaceVessel](#).

### Arrays

|                                                 |                                                             |     |
|-------------------------------------------------|-------------------------------------------------------------|-----|
| <a href="#">ArrayOfAgentConcentrationStruct</a> | Array of agents and their concentrations.                   | 139 |
| <a href="#">ArrayOfAgentMassStruct</a>          | Array of agents and their masses.                           | 139 |
| <a href="#">ArrayOfAgentTypeEnum</a>            | Array of agents.                                            | 139 |
| <a href="#">ArrayOfCBRNExposureStruct</a>       | Array of agents and their dosages.                          | 139 |
| <a href="#">ArrayOfCBRNSensorReadingStruct</a>  | Array of sensor readings.                                   | 139 |
| <a href="#">ArrayOfCommunicationNetworks</a>    | An array of Communication Networks.                         | 139 |
| <a href="#">ArrayOfEntityStruct</a>             | Data for one or more entities that comprise an entity list. | 139 |
| <a href="#">ArrayOfHoldings</a>                 | Sequence of holdings.                                       | 139 |
| <a href="#">ArrayOfNames</a>                    | Array of Names.                                             | 140 |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ArrayOfProbabilityHazardContourStruct | Array of Probability Hazard Contours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 140 |
| ArrayOfProtectionEffectivenessStruct  | Array of the protection's effectiveness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 140 |
| ArrayOfRawDataHazardContourStruct     | Array of Raw Data Hazard Contours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 140 |
| ArrayOfRelations                      | Sequence of relations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 140 |
| ArrayOfRepairStruct                   | List of repair descriptions (equipment and type of repairs).                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 140 |
| ArrayOfRepairTypeEnum                 | List of repair types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 140 |
| ArrayOfResourceStatus                 | The array of health states for a named resource.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 140 |
| ArrayOfSensorStruct                   | Array with definitions of sensors, 1+ cardinality                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 141 |
| ArrayOfSigmas6                        | Size of the initial Gaussian puff sigma values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 141 |
| ArrayOfSpottedEntities                | The spotted entities at a specific time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 141 |
| ArrayOfSpottedEquipment               | An array with spotted equipment at the spotted entity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 141 |
| ArrayOfStringType                     | A generic representation of a set of strings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 141 |
| ArrayOfTaskIds                        | Array of Task Ids.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 141 |
| ArrayOfText64                         | A set of names of max length 64 unicode characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 141 |
| ArrayOfTreatmentStruct                | Array of TreatmentStruct types.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 141 |
| ArrayOfUuid                           | Deprecated. Array of Unique Identifiers expressed as UUIDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 142 |
| ArrayOfWorldLocationStruct            | A polygonal chain (path) expressed as a sequence of geocentric points.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 142 |
| AttributeNamesType                    | Array of attribute names.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 142 |
| BinArrayType                          | Binary data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 142 |
| CommunicationNetworkArray             | References to a set of communication networks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 142 |
| ConnectionReceiverArray               | Connection characteristics for a number of receivers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 142 |
| Datetime18                            | A designation of a specified chronological point measured using Coordinated Universal Time (UTC) ISO 8601:2000 as a standard of reference, constrained to 'zero meridian' i.e. Zulu time zone only. This is expressed as a composite field using a compacted ISO notation YYYYMMDDHHMMSS.SSS where YYYY represents a year, MM represents a month in values from 00 to 12, and DD represents a day in values from 00 to 31, HH represents an hour, MM represents a minute, and SS.SSS represents the number of seconds and milliseconds. | 142 |
| GeocentricPolygon                     | A polygon defined as a closed polygonal chain of geocentric world locations where the last location is connected to the first.                                                                                                                                                                                                                                                                                                                                                                                                          | 143 |
| GeodeticPath                          | A sequence of geodetic locations defining a path where each segment is a great circle between locations.                                                                                                                                                                                                                                                                                                                                                                                                                                | 143 |
| GeodeticPolygon                       | A sequence of geodetic locations defining a geographical area bounded by a closed path where the first and last locations in the sequence are connected. Each point is a geodetic coordinate in WGS84 on the earth surface, and each segment is a great circle between locations.                                                                                                                                                                                                                                                       | 143 |
| IncomingConnectionArray               | A set of incoming connections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 143 |
| LinkStatusArray                       | The status of physical network links.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 143 |
| ManufacturerIdType                    | The manufacturer mnemonic code consisting of three 6 bit ASCII characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 143 |
| NETNArrayOfSupplyStruct               | A set of supply descriptions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 143 |
| NatoStockNumberArray13                | 13 digits (0-9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 144 |
| NetworkDeviceArray                    | A set of network devices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 144 |
| RequestedConnectionArray              | Characterisation of a set of requested connections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 144 |
| Text1                                 | Text of max length 1 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 144 |
| Text15                                | Text of max length 15 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 144 |
| Text2                                 | Text of max length 2 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 144 |
| Text20                                | Text of max length 20 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 144 |
| Text21                                | Text of max length 21 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 144 |
| Text24                                | Text of max length 24 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 145 |
| Text255                               | Text of max length 255 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 145 |
| Text3                                 | Text of max length 3 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 145 |
| Text5                                 | Text of max length 5 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 145 |
| Text64                                | Text of max length 64 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 145 |
| Text8                                 | Text of max length 8 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 145 |
| Text9                                 | Text of max length 9 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 145 |
| TransactionId                         | Unique identifier for a transaction. Encoded according to RFC 4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs.                                                                                                                                                                                                                                                                                                                                                                              | 145 |
| UUID                                  | 4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs. For example, 00112233-4455-8877-6699-aabbccddeeff is encoded as the bytes 00 11 22 33 44 55 88 77 66 99 aa bb cc dd ee ff.                                                                                                                                                                                                                                                                                                                 | 146 |
| UuidArrayOfHLAbyte16                  | Deprecated. UUIDs are exchanged as a byte array and are encoded according to RFC 4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs. For example, 00112233-4455-8877-6699-aabbccddeeff is encoded as the bytes 00 11 22 33 44 55 88 77 66 99 aa bb cc dd ee ff.                                                                                                                                                                                                                                | 146 |

---

## ArrayOfAgentConcentrationStruct SEQUENCE

Array of agents and their concentrations.

```
sequence<AgentConcentrationStruct> ArrayOfAgentConcentrationStruct;
```

Named by [CBRNDetector](#), [CBRNSensorUpdate](#).

---

## ArrayOfAgentMassStruct SEQUENCE

Array of agents and their masses.

```
sequence<AgentMassStruct> ArrayOfAgentMassStruct;
```

Named by [CBRNAircraft](#), [CBRNAmphibiousVehicle](#), [CBRNGroundVehicle](#), [CBRNMultiDomainPlatform](#), [CBRNPlatformUpdate](#), [CBRN-Spacecraft](#), [CBRNSubmersibleVessel](#), [CBRNSurfaceVessel](#).

---

## ArrayOfAgentTypeEnum SEQUENCE

Array of agents.

```
sequence<AgentTypeEnum16> ArrayOfAgentTypeEnum;
```

Named by [CBRNSensor](#), [TreatmentStruct](#).

---

## ArrayOfCBRNExposureStruct SEQUENCE

Array of agents and their dosages.

```
sequence<CBRNExposureStruct> ArrayOfCBRNExposureStruct;
```

Named by [CBRNCasualty](#), [CBRNHuman](#).

---

## ArrayOfCBRNSensorReadingStruct SEQUENCE

Array of sensor readings.

```
sequence<CBRNSensorReadingStruct> ArrayOfCBRNSensorReadingStruct;
```

Named by [CBRNSensor](#).

---

## ArrayOfCommunicationNetworks SEQUENCE

An array of Communication Networks.

```
sequence<CommunicationNetworkStruct> ArrayOfCommunicationNetworks;
```

Named by [EquipmentItem](#), [Unit](#).

---

## ArrayOfEntityStruct SEQUENCE

Data for one or more entities that comprise an entity list.

```
sequence<EntityStruct> ArrayOfEntityStruct;
```

Named by [NETNAggregate](#).

---

## ArrayOfHoldings SEQUENCE

Sequence of holdings.

```
sequence<Holding> ArrayOfHoldings;
```

Named by [Unit](#).

---

---

## ArrayOfNames SEQUENCE

Array of Names.

```
sequence<string> ArrayOfNames;
```

Named by [CapabilitiesSupportedETR](#).

---

## ArrayOfProbabilityHazardContourStruct SEQUENCE

Array of Probability Hazard Contours.

```
sequence<ProbabilityHazardContourStruct> ArrayOfProbabilityHazardContourStruct;
```

Named by [ProbabilityHazardContourGroup](#).

---

## ArrayOfProtectionEffectivenessStruct SEQUENCE

Array of the protection's effectiveness.

```
sequence<ProtectionEffectivenessStruct> ArrayOfProtectionEffectivenessStruct;
```

Named by [COLPRO](#).

---

## ArrayOfRawDataHazardContourStruct SEQUENCE

Array of Raw Data Hazard Contours.

```
sequence<RawDataHazardContourStruct> ArrayOfRawDataHazardContourStruct;
```

Named by [RawDataHazardContourGroup](#).

---

## ArrayOfRelations SEQUENCE

Sequence of relations.

```
sequence<Relation> ArrayOfRelations;
```

Named by [Force](#).

---

## ArrayOfRepairStruct SEQUENCE

List of repair descriptions (equipment and type of repairs).

```
sequence<RepairStruct> ArrayOfRepairStruct;
```

Named by [OfferRepair](#), [RepairComplete](#), [RequestRepair](#).

---

## ArrayOfRepairTypeEnum SEQUENCE

List of repair types

```
sequence<rpr.RepairTypeEnum16> ArrayOfRepairTypeEnum;
```

Named by [RepairStruct](#).

---

## ArrayOfResourceStatus SEQUENCE

The array of health states for a named resource.

```
sequence<ResourceStatusNumberStruct> ArrayOfResourceStatus;
```

Named by [Divide](#), [NETNAggregate](#).

---



## ArrayOfSensorStruct SEQUENCE

Array with definitio0ns of sensors, 1+ cardinality

```
sequence<SensorStruct> ArrayOfSensorStruct;
```

Named by [ElectronicSignatureStruct](#).

## ArrayOfSigmas6 SEQUENCE

Size of the initial Gaussian puff sigma values.

```
sequence<rpr.LengthMeterFloat32, 6> ArrayOfSigmas6;
```

Named by [ReleaseSizeStruct](#).

## ArrayOfSpottedEntities SEQUENCE

The spotted entities at a specific time

```
sequence<SpottedEntity> ArrayOfSpottedEntities;
```

Named by [SensorReport](#), [SpotReport](#).

## ArrayOfSpottedEquipment SEQUENCE

An array with spotted equipment at the spotted entity.

```
sequence<SpottedEquipment> ArrayOfSpottedEquipment;
```

Named by [SpottedEntity](#).

## ArrayOfStringType SEQUENCE

A generic representation of a set of strings.

```
sequence<string> ArrayOfStringType;
```

Named by [CapabilitiesSupportedMRM](#).

## ArrayOfTaskIds SEQUENCE

Array of Task Ids.

```
sequence<TransactionId> ArrayOfTaskIds;
```

Named by [CancelSpecifiedTasks](#).

## ArrayOfText64 SEQUENCE

A set of names of max length 64 unicode characters.

```
sequence<Text64> ArrayOfText64;
```

Named by [AddPassage](#), [CancelAllTasks](#), [CancelSpecifiedTasks](#), [ClearObstacle](#), [CreateMinefield](#), [CreateObstacle](#), [DamageStatusReport](#), [Dismount](#), [DisruptCommunication](#), [DisruptionEffect](#), [ETRReport](#), [ETRTask](#), [ETRTaskManagement](#), [EstablishCheckPoint](#), [FireAtEntity](#), [FireAtEntityWM](#), [FireAtLocation](#), [FireAtLocationWM](#), [FollowEntity](#), [InWeaponRangeReport](#), [Mount](#), [Move](#), [MoveIntoFormation](#), [MoveToEntity](#), [MoveToLocation](#), [Observe](#), [OperateCheckPoint](#), [Patrol](#), [PatrolRepeating](#), [PositionStatusReport](#), [RemoveCheckPoint](#), [RemovePassage](#), [ResourceStatusReport](#), [SendSafetyRelatedBroadcastMessage](#), [SendSafetyRelatedMessage](#), [SensorReport](#), [SetOrderedAltitude](#), [SetOrderedSpeed](#), [SetRulesOfEngagement](#), [SetTransmitterStatus](#), [SpotReport](#), [StatusReport](#), [StopAtSideOfRoad](#), [TaskStatusReport](#), [TurnToHeading](#), [TurnToOrientation](#), [UnderAttackStatusReport](#), [Wait](#).

## ArrayOfTreatmentStruct SEQUENCE

Array of TreatmentStruct types.

```
sequence<TreatmentStruct> ArrayOfTreatmentStruct;
```

Named by [CBRNHuman](#), [DecontaminationStation](#).

## ArrayOfUuid SEQUENCE

Deprecated. Array of Unique Identifiers expressed as UUIDs.

```
sequence<UuidArrayOfHLAByte16> ArrayOfUuid;
```

Named by [CBRNFacilityUpdate](#), [CBRNTreatmentCommand](#), [COLPROUpdate](#), [DecontaminationStationUpdate](#), [FederateApplication](#), [IPECommand](#), [InWeaponRangeReport](#), [InitiateTransfer](#), [Merge](#), [NETNAggregate](#), [NETNAircraft](#), [NETNAmphibiousVehicle](#), [NETNCulturalFeature](#), [NETNGroundVehicle](#), [NETNMultiDomainPlatform](#), [NETNSpacecraft](#), [NETNSubmersibleVessel](#), [NETNSurfaceVessel](#), [OfferTransport](#), [RequestTransfer](#), [RequestTransport](#), [RequestedConnection](#), [TransportDestroyedEntities](#), [TransportDisembarkmentStatus](#), [TransportEmbarkmentStatus](#).

## ArrayOfWorldLocationStruct SEQUENCE

A polygonal chain (path) expressed as a sequence of geocentric points.

```
sequence<rpr.WorldLocationStruct> ArrayOfWorldLocationStruct;
```

Named by [MoveToEntity](#), [MoveToLocation](#), [PathVariantStruct](#), [Patrol](#), [PatrolRepeating](#).

## AttributeNamesType SEQUENCE

Array of attribute names.

```
sequence<string> AttributeNamesType;
```

Named by [InitiateTransfer](#), [RequestTransfer](#).

## BinArrayType SEQUENCE

Binary data.

```
sequence<bool> BinArrayType;
```

Named by [AisBinaryMessageData](#), [AisMessage6](#), [AisMessage8](#).

## CommunicationNetworkArray SEQUENCE

References to a set of communication networks.

```
sequence<Text64> CommunicationNetworkArray;
```

Named by [NetworkDeviceStruct](#).

## ConnectionReceiverArray SEQUENCE

Connection characteristics for a number of receivers.

```
sequence<ConnectionReceiverStruct> ConnectionReceiverArray;
```

Named by [Connection](#).

## Datetime18 SEQUENCE

A designation of a specified chronological point measured using Coordinated Universal Time (UTC) ISO 8601:2000 as a standard of reference, constrained to 'zero meridian' i.e.  Zulu  time zone only. This is expressed as a composite field using a compacted ISO notation YYYYMMDDHHMMSS.SSS where YYYY represents a year, MM represents a month in values from 00 to 12, and DD represents a day in values from 00 to 31, HH represents an hour, MM represents a minute, and SS.SSS represents the number of seconds and milliseconds.

```
sequence<u16, 18> Datetime18;
```

Named by [AddPassage](#), [ClearObstacle](#), [CreateMinefield](#), [CreateObstacle](#), [DamageStatusReport](#), [Dismount](#), [DisruptCommunication](#), [ETRReport](#), [ETRTask](#), [EstablishCheckPoint](#), [FireAtEntity](#), [FireAtEntityWM](#), [FireAtLocation](#), [FireAtLocationWM](#), [FollowEntity](#), [InWeaponRangeReport](#), [Mount](#), [Move](#), [MoveIntoFormation](#), [MoveToEntity](#), [MoveToLocation](#), [Observe](#), [OperateCheckPoint](#), [Patrol](#), [PatrolRepeating](#), [PositionStatusReport](#), [RemoveCheckPoint](#), [RemovePassage](#), [ResourceStatusReport](#), [SendSafetyRelatedBroadcastMessage](#), [SendSafetyRelatedMessage](#), [SensorReport](#), [SetOrderedAltitude](#), [SetOrderedSpeed](#), [SetRulesOfEngagement](#), [SetTransmitterStatus](#), [SpotReport](#), [StatusReport](#), [StopAtSideOfRoad](#), [TaskStatusReport](#), [TurnToHeading](#), [TurnToOrientation](#), [UnderAttackStatusReport](#), [Wait](#).

## GeocentricPolygon SEQUENCE

A polygon defined as a closed polygonal chain of geocentric world locations where the last location is connected to the first.

```
sequence<rpr.WorldLocationStruct> GeocentricPolygon;
```

Named by [ATP45HazardArea](#), [ProbabilityHazardContourStruct](#), [RawDataHazardContourStruct](#).

## GeodeticPath SEQUENCE

A sequence of geodetic locations defining a path where each segment is a great circle between locations.

```
sequence<GeodeticLocation> GeodeticPath;
```

Named by [Path](#).

## GeodeticPolygon SEQUENCE

A sequence of geodetic locations defining a geographical area bounded by a closed path where the first and last locations in the sequence are connected. Each point is a geodetic coordinate in WGS84 on the earth surface, and each segment is a great circle between locations.

```
sequence<GeodeticLocation> GeodeticPolygon;
```

Named by [AreaVariantStruct](#), [GeoLocationReferenceVariant](#), [GeoReferenceVariant](#).

## IncomingConnectionArray SEQUENCE

A set of incoming connections.

```
sequence<IncomingConnectionStruct> IncomingConnectionArray;
```

Named by [CommunicationNode](#).

## LinkStatusArray SEQUENCE

The status of physical network links.

```
sequence<LinkStatusStruct> LinkStatusArray;
```

Named by [LinkStates](#).

## ManufacturerIdType SEQUENCE

The manufacturer mnemonic code consisting of three 6 bit ASCII characters.

```
sequence<u8, 3> ManufacturerIdType;
```

Named by [AisMessage24](#).

## NETNArrayOfSupplyStruct SEQUENCE

A set of supply descriptions.

```
sequence<NETNSupplyStruct> NETNArrayOfSupplyStruct;
```

Named by [Divide](#), [NETNAggregate](#), [OfferSupply](#), [RequestSupply](#), [SupplyComplete](#).

---

## NatoStockNumberArray13 SEQUENCE

13 digits (0-9)

```
sequence<u8, 13> NatoStockNumberArray13;
```

Named by [EquipmentItem](#), [Holding](#).

---

## NetworkDeviceArray SEQUENCE

A set of network devices.

```
sequence<NetworkDeviceStruct> NetworkDeviceArray;
```

Named by [CommunicationNode](#).

---

## RequestedConnectionArray SEQUENCE

Characterisation of a set of requested connections.

```
sequence<RequestedConnection> RequestedConnectionArray;
```

Named by [CommunicationNode](#).

---

## Text1 SEQUENCE

Text of max length 1 characters.

```
sequence<u16, 1> Text1;
```

Named by [EquipmentSymbolAmplificationStruct](#).

---

## Text15 SEQUENCE

Text of max length 15 characters.

```
sequence<u16, 15> Text15;
```

Named by [EquipmentSymbolAmplificationStruct](#), [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

## Text2 SEQUENCE

Text of max length 2 characters.

```
sequence<u16, 2> Text2;
```

Named by [EquipmentSymbolAmplificationStruct](#), [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

## Text20 SEQUENCE

Text of max length 20 characters.

```
sequence<u16, 20> Text20;
```

Named by [EquipmentSymbolAmplificationStruct](#), [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

## Text21 SEQUENCE

Text of max length 21 characters.

```
sequence<u16, 21> Text21;
```

Named by [EquipmentSymbolAmplificationStruct](#), [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

---

**Text24** SEQUENCE

Text of max length 24 characters.

```
sequence<u16, 24> Text24;
```

Named by [EquipmentSymbolAmplificationStruct](#).

---

**Text255** SEQUENCE

Text of max length 255 characters.

```
sequence<u16, 255> Text255;
```

Named by [EquipmentItem](#), [FederateApplication](#), [Force](#), [Installation](#), [Unit](#).

---

**Text3** SEQUENCE

Text of max length 3 characters.

```
sequence<u16, 3> Text3;
```

Named by [EquipmentSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

**Text5** SEQUENCE

Text of max length 5 characters.

```
sequence<u16, 5> Text5;
```

Named by [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

**Text64** SEQUENCE

Text of max length 64 characters.

```
sequence<u16, 64> Text64;
```

Named by [ArrayOfText64](#), [CommunicationNetwork](#), [CommunicationNetworkArray](#), [CommunicationNetworkStruct](#), [Connection](#), [DisruptCommunication](#), [IncomingConnectionStruct](#), [LinkStates](#), [NetworkDeviceStruct](#), [PhysicalNetwork](#), [RequestedConnection](#).

---

**Text8** SEQUENCE

Text of max length 8 characters.

```
sequence<u16, 8> Text8;
```

Named by [EquipmentSymbolAmplificationStruct](#), [InstallationSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

**Text9** SEQUENCE

Text of max length 9 characters.

```
sequence<u16, 9> Text9;
```

Named by [EquipmentSymbolAmplificationStruct](#), [UnitSymbolAmplificationStruct](#).

---

**TransactionId** SEQUENCE

Unique identifier for a transaction. Encoded according to RFC 4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs.

```
sequence<u8, 16> TransactionId;
```

Named by [AcceptOffer](#), [Activate](#), [AddPassage](#), [Aggregate](#), [ArrayOfTaskIds](#), [CancelAllTasks](#), [CancelOffer](#), [CancelRequestLOG](#), [CancelRequestTMR](#), [CancelSpecifiedTasks](#), [CapabilitiesSupportedETR](#), [CapabilitiesSupportedMRM](#), [ClearObstacle](#), [CreateMinefield](#), [CreateObstacle](#), [DamageStatusReport](#), [Deactivate](#), [Disaggregate](#), [Dismount](#), [DisruptCommunication](#), [Divide](#), [ETRReport](#), [ETRSimCon](#), [ETR-](#)

Task, ETRTaskManagement, EstablishCheckPoint, FireAtEntity, FireAtEntityWM, FireAtLocation, FireAtLocationWM, FollowEntity, InWeaponRangeReport, InitiateTransfer, LOGInteraction, LandSurfaceCondition, METOCInteraction, MRMInteraction, MagicMove, MagicResource, Merge, Mount, Move, MoveIntoFormation, MoveToEntity, MoveToLocation, Observe, OfferRepair, OfferService, OfferSupply, OfferTransfer, OfferTransport, OperateCheckPoint, Patrol, PatrolRepeating, PositionStatusReport, QuerySupportedCapabilitiesETR, QuerySupportedCapabilitiesMRM, ReadyToReceiveService, RejectOffer, RemoveCheckPoint, RemovePassage, RepairComplete, RequestLandSurfaceCondition, RequestMETOC, RequestMRM, RequestRepair, RequestService, RequestSubsurfaceLayerCondition, RequestSupply, RequestTransfer, RequestTransport, RequestTroposphereLayerCondition, RequestWaterSurfaceCondition, RequestWeatherCondition, ResourceStatusReport, ResponseMETOC, ResponseMRM, SendSafetyRelatedBroadcastMessage, SendSafetyRelatedMessage, SensorReport, ServiceComplete, ServiceReceived, ServiceStarted, SetOrderedAltitude, SetOrderedSpeed, SetRulesOfEngagement, SetTransmitterStatus, SpotReport, StatusReport, StopAtSideOfRoad, SubsurfaceLayerCondition, SupplyComplete, TMRInteraction, TaskStatusReport, TransferResult, TransportDestroyedEntities, TransportDisembarkmentStatus, TransportEmbarkmentStatus, TroposphereLayerCondition, TurnToHeading, TurnToOrientation, UnderAttackStatusReport, Wait, WaterSurfaceCondition, WeatherCondition.

## UUID SEQUENCE

4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs. For example, 00112233-4455-8877-6699-aabbccddeeff is encoded as the bytes 00 11 22 33 44 55 88 77 66 99 aa bb cc dd ee ff.

**sequence**<u8, 16> UUID;

Named by AddPassage, COMRoot, CancelAllTasks, CancelSpecifiedTasks, CapabilitiesSupportedETR, ClearObstacle, CommunicationNetworkStruct, CommunicationNode, Connection, ConnectionReceiverStruct, CreateMinefield, CreateObstacle, DamageStatusReport, Dismount, DisruptCommunication, ETRSimCon, ETRTask, ETRTaskManagement, EnvironmentCondition, EquipmentItem, EstablishCheckPoint, FireAtEntity, FireAtEntityWM, FireAtLocation, FireAtLocationWM, FollowEntity, GeoLocationReferenceVariant, GeoReferenceVariant, InWeaponRangeReport, IncomingConnectionStruct, Installation, LandSurfaceCondition, LinkStatusStruct, METOCRoot, MagicMove, MagicResource, Mount, Move, MoveIntoFormation, MoveToEntity, MoveToLocation, NetworkDeviceStruct, ORGRoot, Observe, OperateCheckPoint, Patrol, PatrolRepeating, PositionStatusReport, QuerySupportedCapabilitiesETR, Relation, RemoveCheckPoint, RemovePassage, RequestLandSurfaceCondition, RequestMETOC, RequestSubsurfaceLayerCondition, RequestTroposphereLayerCondition, RequestWaterSurfaceCondition, RequestWeatherCondition, RequestedConnection, ResourceStatusReport, ResponseMETOC, SendSafetyRelatedBroadcastMessage, SendSafetyRelatedMessage, SensorReport, SetOrderedAltitude, SetOrderedSpeed, SetRulesOfEngagement, SetTransmitterStatus, SpotReport, SpottedEntity, StatusReport, StopAtSideOfRoad, SubsurfaceLayerCondition, TaskStatusReport, TroposphereLayerCondition, TurnToHeading, TurnToOrientation, UnderAttackStatusReport, Unit, Wait, WaterSurfaceCondition, WeatherCondition.

## UuidArrayOfHLAByte16 SEQUENCE

Deprecated. UUIDs are exchanged as a byte array and are encoded according to RFC 4122, section 4.1.2 using 16 bytes. Also referred to as Variant 1 or RFC 4122/DCE 1.1 UUIDs. For example, 00112233-4455-8877-6699-aabbccddeeff is encoded as the bytes 00 11 22 33 44 55 88 77 66 99 aa bb cc dd ee ff.

**sequence**<u8, 16> UuidArrayOfHLAByte16;

Named by ATP45HazardArea, Activate, Aggregate, ArrayOfUuid, CBRNCasualty, CBRNDetector, CBRNDetectorAlarm, CBRNFacilityUpdate, CBRNPlatformUpdate, CBRNRelease, CBRNSensor, CBRNSensorUpdate, COLPROUpdate, Deactivate, DecontaminationStationUpdate, Disaggregate, Divide, EntityStruct, Merge, NETNAggregate, NETNAircraft, NETNAmphibiousVehicle, NETNCulturalFeature, NETNGroundVehicle, NETNHuman, NETNMultiDomainPlatform, NETNMunition, NETNNonHuman, NETNSpacecraft, NETNSubmersibleVessel, NETNSurfaceVessel, OfferRepair, OfferService, OfferSupply, OfferTransport, PathVariantStruct, PointVariantStruct, ProbabilityHazardContourGroup, QuerySupportedCapabilitiesMRM, RawDataHazardContourGroup, RepairStruct, RequestMRM, RequestRepair, RequestService, RequestSupply, RequestTransport, SEFacility, SEGeoObject, SendSafetyRelatedMessage, TransportDisembarkmentStatus, TransportEmbarkmentStatus, TreatmentStruct.

## Aliases

|                                    |                                                                                                                                                                                                           |     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| AltitudeMeterFloat64               | Generic representation of altitude defined by the context of use, i.e. height Above Mean Sea Level, height Above Ground Level.                                                                            | 147 |
| AtmosphericPressureFloat32         | Pressure measured in Millibar or Hecto Pascal. 1 mbar = 1 hPa                                                                                                                                             | 147 |
| BitsPerSecond                      | Data transfer rate.                                                                                                                                                                                       | 147 |
| DegreesPerSecondFloat32            | The turn rate in degrees per second, where: (a) zero value: not turning; (b) positive value: turning right; (c) negative value: turning left.                                                             | 148 |
| DensityRatioFloat32                | Ratio of density of two materials in range [0, 1].                                                                                                                                                        | 148 |
| DesignatedAreaCodeType             | A 10 bits value. Designated area code (DAC).                                                                                                                                                              | 148 |
| DirectionDegreesFloat32            | Compass direction measured clockwise relative to true north. Calculate values outside the range [0, 360) as modulo 360.                                                                                   | 148 |
| DosageKgSecondPerMeterCubedFloat32 | Dosage in SI units.                                                                                                                                                                                       | 148 |
| DraughtMeterFloat32                | The vertical distance between the waterline and the bottom of the hull (keel), with the thickness of the hull included; Draught determines the minimum depth of water a ship or boat can safely navigate. | 148 |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">EpochTimeSecInt64</a>             | The number of seconds since 1 Jan 1970 (wallclock time) or since the start of the simulation (logical time).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 148 |
| <a href="#">ExposureFloat32</a>               | Data type for exposure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 148 |
| <a href="#">FormationInt32</a>                | Defines the formation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 149 |
| <a href="#">FunctionIdType</a>                | A 6 bits value. Functional ID (FID).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 149 |
| <a href="#">IMOType</a>                       | A 30 bits value. The International Maritime Organization (IMO) number is a unique identifier for vessels. See <a href="https://www.itu.int/en/ITU-R">https://www.itu.int/en/ITU-R</a> . The IMO number is made of the three letters 'IMO' followed by a seven-digit number. This number consists of a six-digit sequential unique number followed by a check digit. The integrity of an IMO number can be verified using its check digit. This is done by multiplying each of the first six digits by a factor of 2 to 7 corresponding to their position from right to left. The rightmost digit of this sum is the check digit. For example, for IMO 9074729: $(9 \times 7) + (0 \times 6) + (7 \times 5) + (4 \times 4) + (7 \times 3) + (2 \times 2) = 139$ . This attribute represents the 7 digits value of the IMO number. The value shall be zero for not available (default). The value shall also be zero for inland vessels. | 149 |
| <a href="#">LatLongDegreesFloat64</a>         | Represents a measure of either latitude or longitude in decimal degrees of arc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 149 |
| <a href="#">MMSIType</a>                      | A 30 bits value. The MMSI number (Maritime Mobile Service Identity) is a unique nine-digit number for identifying an AIS station.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 149 |
| <a href="#">MassConcentrationFloat32</a>      | Concentration of a substance measured as kg/m <sup>3</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 149 |
| <a href="#">MassDensityFloat32</a>            | Density of substance measured as kg per cubic meter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 149 |
| <a href="#">MeanMetersFloat32</a>             | Mean of a Gaussian distribution, based on SI unit meter, unit symbol m.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 150 |
| <a href="#">PercentFloat64</a>                | A generic measure of percentage (0-100).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 150 |
| <a href="#">PrecipitationIntensityFloat32</a> | Light rain — when the precipitation rate is < 2.5 mm (0.098 in) per hour. Moderate rain — when the precipitation rate is between 2.5 mm (0.098 in) - 7.6 mm (0.30 in) or 10 mm (0.39 in) per hour. Heavy rain — when the precipitation rate is > 7.6 mm (0.30 in) per hour, or between 10 mm (0.39 in) and 50 mm (2.0 in) per hour. Violent rain — when the precipitation rate is > 50 mm (2.0 in) per hour.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 150 |
| <a href="#">QuantityFloat32</a>               | A generic floating-point quantity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 150 |
| <a href="#">QuantityFloat64</a>               | A generic floating-point quantity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 150 |
| <a href="#">QuantityInt32</a>                 | A generic discrete quantity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 150 |
| <a href="#">QuantityUInt32</a>                | Quantity in range [0, 2 <sup>32</sup> -1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 150 |
| <a href="#">RangeFloat32</a>                  | Range of sensor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 150 |
| <a href="#">SalinityFloat32</a>               | Practical Salinity Unit (PSU) measured in g/kg.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 151 |
| <a href="#">SerialNumberType</a>              | A 20 bits value. (Part B) Serial Number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 151 |
| <a href="#">ShipTypeType</a>                  | A 8 bits value. The type of ship and cargo. See <a href="https://www.itu.int/en/ITU-R">https://www.itu.int/en/ITU-R</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 151 |
| <a href="#">TimeSecInt32</a>                  | A generic time interval in seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 151 |
| <a href="#">UnitModelType</a>                 | A 4 bits value. (Part B) Unit Model Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 151 |
| <a href="#">VarianceMetersSquaredFloat32</a>  | Variance of a Gaussian distribution, based on SI unit meter squared, unit symbol m <sup>2</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 151 |

## AltitudeMeterFloat64 ALIAS

Generic representation of altitude defined by the context of use, i.e. height Above Mean Sea Level, height Above Ground Level.

```
alias AltitudeMeterFloat64 f64;
```

Named by [GeodeticPoint](#), [SetOrderedAltitude](#).

## AtmosphericPressureFloat32 ALIAS

Pressure measured in Millibar or Hecto Pascal. 1 mbar = 1 hPa

```
alias AtmosphericPressureFloat32 f32;
```

## BitsPerSecond ALIAS

Data transfer rate.

```
alias BitsPerSecond i64;
```

Named by [ConnectionReceiverStruct](#), [IncomingConnectionStruct](#), [LinkStatusStruct](#), [PhysicalGenericNetworkStruct](#).

## DegreesPerSecondFloat32 ALIAS

The turn rate in degrees per second, where: (a) zero value: not turning; (b) positive value: turning right; (c) negative value: turning left.

```
alias DegreesPerSecondFloat32 f32;
```

## DensityRatioFloat32 ALIAS

Ratio of density of two materials in range [0, 1].

```
alias DensityRatioFloat32 f32;
```

Named by [ReleaseDynamicsStruct](#).

## DesignatedAreaCodeType ALIAS

A 10 bits value. Designated area code (DAC).

```
alias DesignatedAreaCodeType i32;
```

Named by [AisBinaryMessageData](#), [AisMessage6](#), [AisMessage8](#).

## DirectionDegreesFloat32 ALIAS

Compass direction measured clockwise relative to true north. Calculate values outside the range [0, 360) as modulo 360.

```
alias DirectionDegreesFloat32 f32;
```

Named by [CurrentStruct](#), [DispositionStruct](#), [EntityStruct](#), [FormationPositionStruct](#), [FormationStruct](#), [WaveStruct](#), [WindStruct](#).

## DosageKgSecondPerMeterCubedFloat32 ALIAS

Dosage in SI units.

```
alias DosageKgSecondPerMeterCubedFloat32 f32;
```

Named by [CBRNEExposureStruct](#).

## DraughtMeterFloat32 ALIAS

The vertical distance between the waterline and the bottom of the hull (keel), with the thickness of the hull included; Draught determines the minimum depth of water a ship or boat can safely navigate.

```
alias DraughtMeterFloat32 f32;
```

## EpochTimeSecInt64 ALIAS

The number of seconds since 1 Jan 1970 (wallclock time) or since the start of the simulation (logical time).

```
alias EpochTimeSecInt64 i64;
```

Named by [ATP45HazardArea](#), [AppointmentStruct](#), [CBRNAlarmStruct](#), [CBRNSensorReadingStruct](#), [CBRNSensorUpdate](#), [MissionStruct](#), [ProbabilityHazardContourGroup](#), [RawDataHazardContourGroup](#).

## ExposureFloat32 ALIAS

Data type for exposure.

```
alias ExposureFloat32 f32;
```

Named by [RawDataHazardContourStruct](#).



## FormationInt32 ALIAS

Defines the formation.

```
alias FormationInt32 i32;
```

Named by [FormationPositionStruct](#).

## FunctionIdType ALIAS

A 6 bits value. Functional ID (FID).

```
alias FunctionIdType i32;
```

Named by [AisBinaryMessageData](#), [AisMessage6](#), [AisMessage8](#).

## IMOType ALIAS

A 30 bits value. The International Maritime Organization (IMO) number is a unique identifier for vessels. See <https://www.itu.int/en/ITU-R>. The IMO number is made of the three letters 'IMO' followed by a seven-digit number. This number consists of a six-digit sequential unique number followed by a check digit. The integrity of an IMO number can be verified using its check digit. This is done by multiplying each of the first six digits by a factor of 2 to 7 corresponding to their position from right to left. The rightmost digit of this sum is the check digit. For example, for IMO 9074729:  $(9 \times 7) + (0 \times 6) + (7 \times 5) + (4 \times 4) + (7 \times 3) + (2 \times 2) = 139$ . This attribute represents the 7 digits value of the IMO number. The value shall be zero for not available (default). The value shall also be zero for inland vessels.

```
alias IMOType i32;
```

## LatLongDegreesFloat64 ALIAS

Represents a measure of either latitude or longitude in decimal degrees of arc.

```
alias LatLongDegreesFloat64 f64;
```

Named by [GeodeticLocation](#), [GeodeticPoint](#).

## MMSIType ALIAS

A 30 bits value. The MMSI number (Maritime Mobile Service Identity) is a unique nine-digit number for identifying an AIS station.

```
alias MMSIType i32;
```

Named by [AcknowledgeData](#), [AisBaseStationReportAndUTCDateResponseData](#), [AisBinaryMessageData](#), [AisMessage](#), [AisMessage1](#), [AisMessage10](#), [AisMessage11](#), [AisMessage12](#), [AisMessage13](#), [AisMessage14](#), [AisMessage17](#), [AisMessage18](#), [AisMessage19](#), [AisMessage2](#), [AisMessage21](#), [AisMessage24](#), [AisMessage27](#), [AisMessage3](#), [AisMessage4](#), [AisMessage5](#), [AisMessage6](#), [AisMessage7](#), [AisMessage8](#), [AisMessage9](#), [AisNavigationData](#), [AisStaticAndVoyageData](#).

## MassConcentrationFloat32 ALIAS

Concentration of a substance measured as kg/m3.

```
alias MassConcentrationFloat32 f32;
```

Named by [AgentConcentrationStruct](#).

## MassDensityFloat32 ALIAS

Density of substance measured as kg per cubic meter.

```
alias MassDensityFloat32 f32;
```

Named by [CloudStruct](#), [HazeStruct](#), [SnowStruct](#).

## MeanMetersFloat32 ALIAS

Mean of a Gaussian distribution, based on SI unit meter, unit symbol m.

```
alias MeanMetersFloat32 f32;
```

Named by [ReleaseDistributionStruct](#).

## PercentFloat64 ALIAS

A generic measure of percentage (0-100).

```
alias PercentFloat64 f64;
```

Named by [ConnectionReceiverStruct](#), [DisruptionEffect](#), [IncomingConnectionStruct](#), [LinkStatusStruct](#), [PhysicalGenericNetworkStruct](#).

## PrecipitationIntensityFloat32 ALIAS

Light rain — when the precipitation rate is < 2.5 mm (0.098 in) per hour. Moderate rain — when the precipitation rate is between 2.5 mm (0.098 in) - 7.6 mm (0.30 in) or 10 mm (0.39 in) per hour. Heavy rain — when the precipitation rate is > 7.6 mm (0.30 in) per hour, or between 10 mm (0.39 in) and 50 mm (2.0 in) per hour. Violent rain — when the precipitation rate is > 50 mm (2.0 in) per hour.

```
alias PrecipitationIntensityFloat32 f32;
```

Named by [PrecipitationStruct](#).

## QuantityFloat32 ALIAS

A generic floating-point quantity.

```
alias QuantityFloat32 f32;
```

Named by [NETNSupplyStruct](#).

## QuantityFloat64 ALIAS

A generic floating-point quantity.

```
alias QuantityFloat64 f64;
```

Named by [Holding](#), [ResourceStatusNumberStruct](#).

## QuantityInt32 ALIAS

A generic discrete quantity.

```
alias QuantityInt32 i32;
```

Named by [SpottedEquipment](#).

## QuantityUInt32 ALIAS

Quantity in range [0, 2<sup>32</sup>-1]

```
alias QuantityUInt32 u32;
```

Named by [COLPRO](#), [DecontaminationStation](#).

## RangeFloat32 ALIAS

Range of sensor

```
alias RangeFloat32 f32;
```

Named by [SensorStruct](#).

**SalinityFloat32** ALIAS

Practical Salinity Unit (PSU) measured in g/kg.

```
alias SalinityFloat32 f32;
```

**SerialNumberType** ALIAS

A 20 bits value. (Part B) Serial Number.

```
alias SerialNumberType i32;
```

Named by [AisMessage24](#).

**ShipTypeType** ALIAS

A 8 bits value. The type of ship and cargo. See <https://www.itu.int/en/ITU-R>.

```
alias ShipTypeType u8;
```

**TimeSecInt32** ALIAS

A generic time interval in seconds.

```
alias TimeSecInt32 i32;
```

Named by [EstablishCheckPoint](#), [FireAtLocation](#), [FireAtLocationWM](#), [FollowEntity](#), [Move](#), [OperateCheckPoint](#), [PatrolRepeating](#), [Wait](#).

**UnitModelType** ALIAS

A 4 bits value. (Part B) Unit Model Code.

```
alias UnitModelType u8;
```

Named by [AisMessage24](#).

**VarianceMetersSquaredFloat32** ALIAS

Variance of a Gaussian distribution, based on SI unit meter squared, unit symbol m2.

```
alias VarianceMetersSquaredFloat32 f32;
```

Named by [ReleaseDistributionStruct](#).

**Optionals**

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| <a href="#">MaybeAggregateMissionEnum16</a>             | 152 |
| <a href="#">MaybeAidTypeEnumType</a>                    | 152 |
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| <a href="#">MaybeArrayOfCommunicationNetworks</a>       | 152 |
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| <a href="#">MaybeArrayOfResourceStatus</a>              | 153 |
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| <a href="#">MaybeArrayOfWorldLocationStruct</a>         | 153 |
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| <a href="#">MaybeCommunicationServiceTypeEnum</a>       | 153 |
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| MaybeGeoReferenceVariant                   | 154 |
| MaybeGeodeticLocation                      | 154 |
| MaybeGeodeticQuadrangle                    | 154 |
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| MaybeIceStruct                             | 154 |
| MaybeIncomingConnectionArray               | 154 |
| MaybeInstallationSymbolAmplificationStruct | 154 |
| MaybeLayerStruct                           | 154 |
| MaybeManeuverIndicatorEnumType             | 154 |
| MaybeModelResolutionTypeEnum32             | 154 |
| MaybeNETNArrayOfSupplyStruct               | 155 |
| MaybeNatoStockNumberArray13                | 155 |
| MaybeNavigationStatusEnumType              | 155 |
| MaybeNoOfferReasonEnumType                 | 155 |
| MaybeOfferTypeEnum32                       | 155 |
| MaybePathVariantStruct                     | 155 |
| MaybePointVariantStruct                    | 155 |
| MaybePrecipitationStruct                   | 155 |
| MaybeRoadIceConditionEnum16                | 155 |
| MaybeSeaStateEnum16                        | 155 |
| MaybeSedimentTypeEnum32                    | 155 |
| MaybeSnowStruct                            | 155 |
| MaybeSurfaceMoistureEnum16                 | 156 |
| MaybeUUID                                  | 156 |
| MaybeUnitSymbolAmplificationStruct         | 156 |
| MaybeUuidArrayOfHLAByte16                  | 156 |
| MaybeWaveStruct                            | 156 |
| MaybeWeaponControlOrderEnum8               | 156 |
| MaybeWindStruct                            | 156 |

## MaybeActiveStatusEnum8 OPTIONAL

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optional<ActiveStatusEnum8> MaybeActiveStatusEnum8;
```

## MaybeAggregateMissionEnum16 OPTIONAL

```
optional<AggregateMissionEnum16> MaybeAggregateMissionEnum16;
```

## MaybeAidTypeEnumType OPTIONAL

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## MaybeAppointmentStruct OPTIONAL

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```

## MaybeArrayOfCommunicationNetworks OPTIONAL

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## MaybeArrayOfEntityStruct OPTIONAL

```
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```

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**MaybeArrayOfHoldings** OPTIONAL

```
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```

---

**MaybeArrayOfRelations** OPTIONAL

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---

**MaybeArrayOfResourceStatus** OPTIONAL

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---

**MaybeArrayOfText64** OPTIONAL

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---

**MaybeArrayOfUuid** OPTIONAL

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---

**MaybeArrayOfWorldLocationStruct** OPTIONAL

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---

**MaybeCancellationReasonEnum32** OPTIONAL

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**MaybeCloudStruct** OPTIONAL

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---

**MaybeCommunicationNetworkTypeEnum** OPTIONAL

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---

**MaybeCommunicationServiceTypeEnum** OPTIONAL

```
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```

---

**MaybeCurrentStruct** OPTIONAL

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```

---

**MaybeEchelonEnum32** OPTIONAL

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---

---

**MaybeEquipmentSymbolAmplificationStruct** OPTIONAL

```
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```

---

**MaybeFormationStruct** OPTIONAL

```
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```

---

**MaybeGeoReferenceVariant** OPTIONAL

```
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```

---

**MaybeGeodeticLocation** OPTIONAL

```
optional<GeodeticLocation> MaybeGeodeticLocation;
```

---

**MaybeGeodeticQuadrangle** OPTIONAL

```
optional<GeodeticQuadrangle> MaybeGeodeticQuadrangle;
```

---

**MaybeHazeStruct** OPTIONAL

```
optional<HazeStruct> MaybeHazeStruct;
```

---

**MaybeIceStruct** OPTIONAL

```
optional<IceStruct> MaybeIceStruct;
```

---

**MaybeIncomingConnectionArray** OPTIONAL

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```

---

**MaybeInstallationSymbolAmplificationStruct** OPTIONAL

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```

---

**MaybeLayerStruct** OPTIONAL

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---

**MaybeManeuverIndicatorEnumType** OPTIONAL

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---

**MaybeModelResolutionTypeEnum32** OPTIONAL

```
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```

---

---

**MaybeNETNArrayOfSupplyStruct** OPTIONAL

```
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```

---

**MaybeNatoStockNumberArray13** OPTIONAL

```
optional<NatoStockNumberArray13> MaybeNatoStockNumberArray13;
```

---

**MaybeNavigationStatusEnumType** OPTIONAL

```
optional<NavigationStatusEnumType> MaybeNavigationStatusEnumType;
```

---

**MaybeNoOfferReasonEnumType** OPTIONAL

```
optional<NoOfferReasonEnumType> MaybeNoOfferReasonEnumType;
```

---

**MaybeOfferTypeEnum32** OPTIONAL

```
optional<OfferTypeEnum32> MaybeOfferTypeEnum32;
```

---

**MaybePathVariantStruct** OPTIONAL

```
optional<PathVariantStruct> MaybePathVariantStruct;
```

---

**MaybePointVariantStruct** OPTIONAL

```
optional<PointVariantStruct> MaybePointVariantStruct;
```

---

**MaybePrecipitationStruct** OPTIONAL

```
optional<PrecipitationStruct> MaybePrecipitationStruct;
```

---

**MaybeRoadIceConditionEnum16** OPTIONAL

```
optional<RoadIceConditionEnum16> MaybeRoadIceConditionEnum16;
```

---

**MaybeSeaStateEnum16** OPTIONAL

```
optional<SeaStateEnum16> MaybeSeaStateEnum16;
```

---

**MaybeSedimentTypeEnum32** OPTIONAL

```
optional<SedimentTypeEnum32> MaybeSedimentTypeEnum32;
```

---

**MaybeSnowStruct** OPTIONAL

```
optional<SnowStruct> MaybeSnowStruct;
```

---

---

**MaybeSurfaceMoistureEnum16** OPTIONAL

```
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```

---

**MaybeUUID** OPTIONAL

```
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```

---

**MaybeUnitSymbolAmplificationStruct** OPTIONAL

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```

---

**MaybeUuidArrayOfHLAByte16** OPTIONAL

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```

---

**MaybeWaveStruct** OPTIONAL

```
optional<WaveStruct> MaybeWaveStruct;
```

---

**MaybeWeaponControlOrderEnum8** OPTIONAL

```
optional<WeaponControlOrderEnum8> MaybeWeaponControlOrderEnum8;
```

---

**MaybeWindStruct** OPTIONAL

```
optional<WindStruct> MaybeWindStruct;
```

---



# rpr

453 types.

## Object classes

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ActiveSonar                 | Describes the state of an active sonar system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 158 |
| ActiveSonarBeam             | A sector of concentrated acoustic energy from an active sonar device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 158 |
| AdditionalPassiveActivities | Describes the steady state components of non-propulsion passive emissions such as those held in the Additional Narrowband Database (ANDB).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 159 |
| AggregateEntity             | A group of one or more separate objects that operate together as part of an organization. These objects may be discrete, may be other aggregate objects, or may be a mixture of both.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 159 |
| Aircraft                    | A platform entity that operates mainly in the air, such as aircraft, balloons, etc. This includes the entities when they are on the ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 160 |
| AmphibiousVehicle           | A platform entity that can operate both on the land and the sea.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 160 |
| ArealObject                 | A synthetic environment object that is geometrically anchored to the terrain with a set of three or more points, which comes to a closure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 160 |
| BaseEntity                  | A base class of aggregate and discrete scenario domain participants. The BaseEntity class is characterized by being located at a particular location in space and independently movable, if capable of movement at all. It specifically excludes elements normally considered to be a component of another element. The BaseEntity class is intended to be a container for common attributes for entities of this type. Since it lacks sufficient class specific attributes that are required for simulation purposes, federates cannot publish objects of this class. Certain simulation management federates, e.g. viewers, may subscribe to this class. Simulation federates will normally subscribe to one of the subclasses, to gain the extra information required to properly simulate the entity. | 160 |
| BreachObject                | A breach linear object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 161 |
| BreachableLinearObject      | A linear object that can be breached.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 161 |
| BreachablePointObject       | A point object that can be breached.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 161 |
| BurstPointObject            | A burst point object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 161 |
| CraterObject                | A pit, depression, or cavity formed in the surface of the earth by an explosion. The depression's shape can range from saucer to conical, depending largely of the depth of burst.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 162 |
| CulturalFeature             | Engineering and natural effects such as craters, bridges, vehicle tracks, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 162 |
| Designator                  | A system used to designate or mark a location or object, such as a laser designator which supports a laser-guided weapon engagement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 162 |
| EmbeddedSystem              | A base class used to associate components such as sensor and emitting systems with their parent entity object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 162 |
| EmitterBeam                 | A sector of concentrated energy from a device that radiates an electromagnetic signal. See also IEEE 1278.1-1995 Section 5.4.7.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 163 |
| EmitterSystem               | A device that radiates an electromagnetic signal, such as a radar or jammer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 163 |
| EnvironmentObject           | A base class of environment point, linear, or areal object classes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 164 |
| EnvironmentProcess          | An object class used to communicate information about environmental processes and effects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 164 |
| EnvironmentalEntity         | An object which has physical extent but not necessarily fixed size and shape, such as meteorological effects such as clouds or smokes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 164 |
| ExhaustSmokeObject          | An exhaust smoke linear object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 164 |
| Expendables                 | Countermeasures devices that are dispensed from another entity. The devices may be active emitters or passive reflectors of energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 165 |
| GriddedData                 | An object class used to depict global, spatially varying environmental effects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 165 |
| GroundVehicle               | A platform entity that operates wholly on the surface of the earth.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 165 |
| Human                       | A human lifeform.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 165 |
| IFF                         | Interrogator Friend or Foe (IFF) system, Air Traffic Control Beacon and Transponder system, collision avoidance and navigational aids systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 165 |
| JammerBeam                  | An emitter beam that is designed to jam or otherwise interfere or confuse another emitter system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 166 |
| Lifeform                    | A living military platform (human or not).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 166 |
| LinearObject                | A synthetic environment object that has size and an orientation and is geometrically anchored to the terrain with one point.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 167 |
| Minefield                   | An area of ground or water containing mines laid with or without a pattern.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 167 |
| MinefieldData               | The MinefieldData object class provides data about collections of mines within a minefield on an individual mine basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 167 |
| MinefieldLaneMarkerObject   | A minefield lane marker showing a cleared lane through a specific minefield, defined as a linear object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 168 |
| MinefieldObject             | A mine, mine weapon, mine row, mine strip, mine lane, mine marker or minefield, defined as an areal environment object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 169 |
| MultiDomainPlatform         | A platform entity that operates in more than one domain (excluding those combinations explicitly defined as subclasses of the superclass of this class).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 169 |
| Munition                    | A complete device charged with explosives, propellants, pyrotechnics, initiating composition, or nuclear, biological or chemical material for use in military operations, including demolitions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 169 |

|                             |                                                                                                                                                                                                                                                                                                                                  |     |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| NatoIFF                     | NATO Identification Friend or Foe (IFF) system that uses electromagnetic transmissions to which friendly forces' equipment automatically responds.                                                                                                                                                                               | 169 |
| NatoIFFInterrogator         | The part of an IFF system that first transmits electromagnetic signals.                                                                                                                                                                                                                                                          | 170 |
| NatoIFFTransponder          | The part of a specific IFF system that responds (for example by emitting pulses) to the electromagnetic signals.                                                                                                                                                                                                                 | 170 |
| NonHuman                    | An animal or other non-human lifeform.                                                                                                                                                                                                                                                                                           | 170 |
| OtherArealObject            | Areal objects other than Minefield objects.                                                                                                                                                                                                                                                                                      | 170 |
| OtherLinearObject           | Linear objects other than BreachableLinear, Breach, ExhaustSmoke, or MinefieldLaneMarker objects.                                                                                                                                                                                                                                | 170 |
| OtherPointObject            | Point objects other than Breachable, Burst, Crater, RibbonBridge, or Structure objects.                                                                                                                                                                                                                                          | 170 |
| PhysicalEntity              | A base class of all discrete platform scenario domain participants.                                                                                                                                                                                                                                                              | 171 |
| Platform                    | A physical object under the control of armed forces upon which sensor, communication, or weapon systems may be mounted.                                                                                                                                                                                                          | 172 |
| PointObject                 | A synthetic environment object that is geometrically anchored to the terrain with a single point.                                                                                                                                                                                                                                | 172 |
| PropulsionNoise             | Describes the steady state component of unintended passive emissions that are normally associated with the power plant                                                                                                                                                                                                           | 173 |
| RRB                         | RRB IFF transponder system                                                                                                                                                                                                                                                                                                       | 173 |
| RadarBeam                   | A sector of concentrated energy from a device that radiates an electromagnetic signal.                                                                                                                                                                                                                                           | 173 |
| Radio                       | Electronic devices for the communication of both audio and data, operated by entities belonging to armed forces.                                                                                                                                                                                                                 | 173 |
| RadioReceiver               | A device that converts incoming electromagnetic waves in the radio frequency range into information.                                                                                                                                                                                                                             | 173 |
| RadioTransmitter            | A device that sends out information encoded in electromagnetic waves in the radio frequency range.                                                                                                                                                                                                                               | 174 |
| RibbonBridgeObject          | A ribbon bridge object.                                                                                                                                                                                                                                                                                                          | 174 |
| Sensor                      | Sensors and emitters, such as stand-alone radars, jammers, and detection systems, that are not part of another platform or system described by another Physical Entity, and are operated by armed forces.                                                                                                                        | 174 |
| SovietIFF                   | Soviet type IFF system                                                                                                                                                                                                                                                                                                           | 175 |
| SovietIFFInterrogator       | Soviet IFF Interrogator                                                                                                                                                                                                                                                                                                          | 175 |
| SovietIFFTransponder        | Soviet IFF Transponder                                                                                                                                                                                                                                                                                                           | 176 |
| Spacecraft                  | A platform entity that operates mainly in space.                                                                                                                                                                                                                                                                                 | 176 |
| StructureObject             | A building structure, building rubble, stationary bridge, AVLb object.                                                                                                                                                                                                                                                           | 176 |
| SubmersibleVessel           | A platform entity that operates either on the surface of the sea, or beneath it.                                                                                                                                                                                                                                                 | 176 |
| Supplies                    | Supplies other than munitions, such as fuel, food and personnel.                                                                                                                                                                                                                                                                 | 176 |
| SurfaceVessel               | A platform entity that operates wholly on the surface of the sea.                                                                                                                                                                                                                                                                | 176 |
| UnderwaterAcousticsEmission | The underwater acoustic classes used to communicate underwater acoustic active, intentional emissions and Passive Signature or Unintentional Emissions information. These emissions are used during undersea warfare scenarios to detect, classify, and track hostile forces when electronic warfare mechanisms are unavailable. | 176 |

## ActiveSonar CLASS

Describes the state of an active sonar system.

| Name                | Type                             | Flags   | Description                                                                                  |
|---------------------|----------------------------------|---------|----------------------------------------------------------------------------------------------|
| acousticName        | ActiveSonarEnum16                | RO S BE | Defines the type of sonar being represented.                                                 |
| functionCode        | ActiveSonarFunction<br>CodeEnum8 | RO D BE | Declares the current primary use of the sonar. It may be used by simulators to infer intent. |
| acousticsIdentifier | Octet                            | RO S BE | Unique identifier for the sonar on an entity, Starts with 1                                  |

## ActiveSonarBeam CLASS

A sector of concentrated acoustic energy from an active sonar device.

| Name                             | Type               | Flags   | Description                                                                                                                 |
|----------------------------------|--------------------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| activeEmission<br>ParameterIndex | Integer16          | RO D BE | An index into the database of active (intentional) underwater acoustics emissions.                                          |
| activeSonarIdentifier            | string             | RO S BE | The RTI ID of the ActiveSonar emitting this beam                                                                            |
| azimuthBeamwidth                 | AngleRadianFloat32 | RO D BE | The horizontal width of the main beam (as opposed to any side lobes) measured from beam center to the 3 dB down point. Omni |

| Name               | Type                                         | Flags   | Description                                                                                                                                                                                                                                                     |
|--------------------|----------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| azimuthCenter      | <a href="#">AngleRadianFloat32</a>           | RO D BE | directional beams shall have a beam width of 0 radians.<br>The center azimuthal bearing of the main beam (as opposed to side lobes) in relation to the own ship heading, clockwise positive. Omnidirectional beams shall have an azimuthal center of 0 radians. |
| beamIdentifier     | <a href="#">Octet</a>                        | RO S BE | The identification of the active sonar beam, which must be unique on the active sonar system.                                                                                                                                                                   |
| elevationBeamwidth | <a href="#">AngleRadianFloat32</a>           | RO D BE | Vertical angle from beam center to 3db down point. 0 is omnidirectional                                                                                                                                                                                         |
| elevationCenter    | <a href="#">AngleRadianFloat32</a>           | RO D BE | The angle of axis of the beam center to the horizontal plane. Positive upward.                                                                                                                                                                                  |
| eventIdentifier    | <a href="#">EventIdentifierStruct</a>        | RO D BE | Used to associate changes to the beam with its ActiveSonar.                                                                                                                                                                                                     |
| scanPattern        | <a href="#">ActiveSonarScanPatternEnum16</a> | RO D BE | The pattern that describes the angular movement of the sonar beam during its sweep.                                                                                                                                                                             |

## AdditionalPassiveActivities CLASS

Describes the steady state components of non-propulsion passive emissions such as those held in the Additional Narrow-band Database (ANDB).

| Name              | Type                              | Flags   | Description                                                                    |
|-------------------|-----------------------------------|---------|--------------------------------------------------------------------------------|
| activityCode      | <a href="#">UnsignedInteger16</a> | RO D BE | Database index used to indicate the passive signature of the entity            |
| activityParameter | <a href="#">Integer16</a>         | RO D BE | The current state of this activity.                                            |
| isSilent          | <a href="#">bool</a>              | RO D BE | Used to silence an additional passive activity without destroying this object. |

## AggregateEntity CLASS

A group of one or more separate objects that operate together as part of an organization. These objects may be discrete, may be other aggregate objects, or may be a mixture of both.

| Name                   | Type                                                 | Flags   | Description                                                                                            |
|------------------------|------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------|
| aggregateMarking       | <a href="#">AggregateMarkingStruct</a>               | RO D BE | A unique marking or combination of characters used to distinguish the aggregate from other aggregates. |
| aggregateState         | <a href="#">AggregateStateEnum8</a>                  | RO D BE | An indicator of the extent of association of objects form an operating group.                          |
| dimensions             | <a href="#">DimensionStruct</a>                      | RO D BE | The size of the area covered by the units in the aggregate.                                            |
| entityIdentifiers      | <a href="#">RTObjectIdArray</a>                      | RO D BE | The identification of entities that are contained within the aggregate.                                |
| forceIdentifier        | <a href="#">ForceIdentifierEnum8</a>                 | RO D BE | The identification of the force that the aggregate belongs to.                                         |
| formation              | <a href="#">FormationEnum32</a>                      | RO D BE | The category of positional arrangement of the entities within the aggregate.                           |
| numberOfSilentEntities | <a href="#">Integer16</a>                            | RO D BE | The number of elements in the SilentEntities list.                                                     |
| numberOfVariableDatums | <a href="#">UnsignedInteger32</a>                    | RO D BE | The number of records in the VariableDatums structure.                                                 |
| silentAggregates       | <a href="#">SilentAggregateStructLengthlessArray</a> | RO D BE | The numbers and types, of silent aggregates contained in the aggregate. Silent aggregates              |

| Name                        | Type                                                    | Flags   | Description                                                                                                                                                                                                                                                                                      |
|-----------------------------|---------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| silentEntities              | <a href="#">SilentEntityStruct<br/>LengthlessArray</a>  | RO D BE | are sub-aggregates that are in the aggregate, but that are not separately represented in the virtual world.<br>The numbers and types, of silent entities in the aggregate. Silent entities are entities that are in the aggregate, but that are not separately represented in the virtual world. |
| subAggregate<br>Identifiers | <a href="#">RTObjectIdArray</a>                         | RO D BE | The identifications of aggregates represented in the virtual world that are contained in the aggregate.                                                                                                                                                                                          |
| variableDatums              | <a href="#">VariableDatumStruct<br/>LengthlessArray</a> | RO D BE | Extra data that describes the aggregate.                                                                                                                                                                                                                                                         |

## Aircraft CLASS

A platform entity that operates mainly in the air, such as aircraft, balloons, etc. This includes the entities when they are on the ground.

## AmphibiousVehicle CLASS

A platform entity that can operate both on the land and the sea.

## ArealObject CLASS

A synthetic environment object that is geometrically anchored to the terrain with a set of three or more points, which comes to a closure.

| Name                 | Type                                                    | Flags   | Description                                                                                                    |
|----------------------|---------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------|
| pointsData           | <a href="#">WorldLocationStruct<br/>LengthlessArray</a> | RO D BE | Specifies the physical location (a collection of points) of the object                                         |
| percentComplete      | <a href="#">PercentUnsigned<br/>Integer32</a>           | RO D BE | Specifies the percent completion of the object                                                                 |
| damagedAppearance    | <a href="#">DamageStatus<br/>Enum32</a>                 | RO D BE | Specifies the damaged appearance of the object                                                                 |
| objectPreDistributed | <a href="#">bool</a>                                    | RO D BE | Specifies whether or not the object was created before the start of the exercise                               |
| deactivated          | <a href="#">bool</a>                                    | RO D BE | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment) |
| smoking              | <a href="#">bool</a>                                    | RO D BE | Specifies whether or not the object is smoking (creating a smoke plume)                                        |
| flaming              | <a href="#">bool</a>                                    | RO D BE | Specifies whether or not the object is aflame                                                                  |

## BaseEntity CLASS

A base class of aggregate and discrete scenario domain participants. The BaseEntity class is characterized by being located at a particular location in space and independently movable, if capable of movement at all. It specifically excludes elements normally considered to be a component of another element. The BaseEntity class is intended to be a container for common attributes for entities of this type. Since it lacks sufficient class specific attributes that are required for simulation purposes, federates cannot publish objects of this class. Certain simulation management federates, e.g. viewers, may subscribe to this class. Simulation federates will normally subscribe to one of the subclasses, to gain the extra information required to properly simulate the entity.

| Name             | Type                                   | Flags   | Description                                    |
|------------------|----------------------------------------|---------|------------------------------------------------|
| entityType       | <a href="#">EntityTypeStruct</a>       | RO S BE | The category of the entity.                    |
| entityIdentifier | <a href="#">EntityIdentifierStruct</a> | RO S BE | The unique identifier for the entity instance. |

| Name            | Type                                 | Flags   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|--------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| isPartOf        | <a href="#">IsPartOfStruct</a>       | RO D BE | Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string. |
| spatial         | <a href="#">SpatialVariantStruct</a> | RO D BE | Spatial state stored in one variant record attribute.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| relativeSpatial | <a href="#">SpatialVariantStruct</a> | RO D BE | Relative spatial state stored in one variant record attribute.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## BreachObject CLASS

A breach linear object.

| Name           | Type                                                            | Flags   | Description                      |
|----------------|-----------------------------------------------------------------|---------|----------------------------------|
| segmentRecords | <a href="#">BreachStruct</a><br><a href="#">LengthlessArray</a> | RO D BE | Specifies a breach linear object |

## BreachableLinearObject CLASS

A linear object that can be breached.

| Name           | Type                                                                       | Flags   | Description                          |
|----------------|----------------------------------------------------------------------------|---------|--------------------------------------|
| segmentRecords | <a href="#">BreachableSegmentStruct</a><br><a href="#">LengthlessArray</a> | RO D BE | Specifies a breachable linear object |

## BreachablePointObject CLASS

A point object that can be breached.

| Name           | Type                                | Flags   | Description                                     |
|----------------|-------------------------------------|---------|-------------------------------------------------|
| breachedStatus | <a href="#">BreachedStatusEnum8</a> | RO D BE | Specifies the breached appearance of the object |

## BurstPointObject CLASS

A burst point object.

| Name           | Type                                     | Flags   | Description                                                                                                                                                                                                   |
|----------------|------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| percentOpacity | <a href="#">PercentUnsignedInteger32</a> | RO D BE | Specifies the opacity of the smoke                                                                                                                                                                            |
| cylinderSize   | <a href="#">UnsignedInteger32</a>        | RO D BE | Specifies the radius of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts |
| cylinderHeight | <a href="#">UnsignedInteger32</a>        | RO D BE | Specifies the height of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts |

| Name            | Type                                  | Flags   | Description                                                      |
|-----------------|---------------------------------------|---------|------------------------------------------------------------------|
| numberOfBursts  | <a href="#">UnsignedInteger32</a>     | RO D BE | Specifies the number of bursts in the instance of tactical smoke |
| chemicalContent | <a href="#">ChemicalContentEnum32</a> | RO D BE | Specifies the chemical content of the smoke                      |

## CraterObject CLASS

A pit, depression, or cavity formed in the surface of the earth by an explosion. The depression's shape can range from saucer to conical, depending largely of the depth of burst.

| Name       | Type                              | Flags   | Description                                                                                          |
|------------|-----------------------------------|---------|------------------------------------------------------------------------------------------------------|
| craterSize | <a href="#">UnsignedInteger32</a> | RO D BE | Specifies the diameter of the crater, where the center of the crater is at the point object location |

## CulturalFeature CLASS

Engineering and natural effects such as craters, bridges, vehicle tracks, etc.

| Name                 | Type                 | Flags   | Description                                                       |
|----------------------|----------------------|---------|-------------------------------------------------------------------|
| externalLightsOn     | <a href="#">bool</a> | RO D BE | Whether the cultural feature's external lights are on or not.     |
| internalHeatSourceOn | <a href="#">bool</a> | RO D BE | Whether the cultural feature's internal heat source is on or not. |
| internalLightsOn     | <a href="#">bool</a> | RO D BE | Whether the cultural feature's internal lights are on or not.     |

## Designator CLASS

A system used to designate or mark a location or object, such as a laser designator which supports a laser-guided weapon engagement.

| Name                         | Type                                        | Flags   | Description                                                                            |
|------------------------------|---------------------------------------------|---------|----------------------------------------------------------------------------------------|
| codeName                     | <a href="#">DesignatorCodeNameEnum16</a>    | RO S BE | The code name of the designator system.                                                |
| designatedObjectIdentifier   | <a href="#">string</a>                      | RO D BE | The object instance ID of the entity that is currently being designated (if any).      |
| designatorCode               | <a href="#">DesignatorCodeEnum16</a>        | RO D BE | The designator code being used by the designating entity.                              |
| designatorEmissionWavelength | <a href="#">WavelengthMicronFloat32</a>     | RO D BE | The wavelength of the designator system.                                               |
| designatorOutputPower        | <a href="#">PowerWattFloat32</a>            | RO D BE | The output power of the designator system.                                             |
| designatorSpotLocation       | <a href="#">WorldLocationStruct</a>         | RO D BE | The location, in the world coordinate system, of the designator spot.                  |
| deadReckoningAlgorithm       | <a href="#">DeadReckoningAlgorithmEnum8</a> | RO D BE | Dead reckoning algorithm used by the issuing object.                                   |
| relativeSpotLocation         | <a href="#">RelativePositionStruct</a>      | RO D BE | The location of the designator spot, relative to the object being designated (if any). |
| spotLinearAccelerationVector | <a href="#">AccelerationVectorStruct</a>    | RO D BE | The rate of change in linear velocity of the designator spot over time.                |

## EmbeddedSystem CLASS

A base class used to associate components such as sensor and emitting systems with their parent entity object.

| Name                 | Type                                   | Flags   | Description                                                                        |
|----------------------|----------------------------------------|---------|------------------------------------------------------------------------------------|
| entityIdentifier     | <a href="#">EntityIdentifierStruct</a> | RO S BE | The EntityIdentifier of the object which this embedded system is a part of.        |
| hostObjectIdentifier | <a href="#">string</a>                 | RO S BE | The RTI object instance ID of the object of which this embedded system is part of. |
| relativePosition     | <a href="#">RelativePositionStruct</a> | RO D BE | The position of the embedded system, relative to the host object's position.       |

## EmitterBeam CLASS

A sector of concentrated energy from a device that radiates an electromagnetic signal. See also IEEE 1278.1-1995 Section 5.4.7.1.

| Name                     | Type                                              | Flags   | Description                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| beamAzimuthCenter        | <a href="#">AngleRadianFloat32</a>                | RO D BE | The azimuth center of the emitter beam's scan volume relative to the emitter system.                                                                                                                |
| beamAzimuthSweep         | <a href="#">AngleRadianFloat32</a>                | RO D BE | The azimuth half-angle of the emitter beam's scan volume relative to the emitter system.                                                                                                            |
| beamElevationCenter      | <a href="#">AngleRadianFloat32</a>                | RO D BE | The elevation center of the emitter beam's scan volume relative to the emitter system.                                                                                                              |
| beamElevationSweep       | <a href="#">AngleRadianFloat32</a>                | RO D BE | The elevation half-angle of the emitter beam's scan volume relative to the emitter system.                                                                                                          |
| beamFunctionCode         | <a href="#">BeamFunctionCodeEnum8</a>             | RO D BE | The function of the emitter beam.                                                                                                                                                                   |
| beamIdentifier           | <a href="#">Octet</a>                             | RO S BE | The identification of the emitter beam (must be unique on the emitter system).                                                                                                                      |
| beamParameterIndex       | <a href="#">UnsignedInteger16</a>                 | RO D BE | The index, into the federation specific emissions database, of the current operating mode of the emitter beam.                                                                                      |
| effectiveRadiatedPower   | <a href="#">PowerRatioDecibelMilliwattFloat32</a> | RO D BE | The effective radiated power of the emitter beam.                                                                                                                                                   |
| emissionFrequency        | <a href="#">FrequencyHertzFloat32</a>             | RO D BE | The center frequency of the emitter beam.                                                                                                                                                           |
| emitterSystemIdentifier  | <a href="#">string</a>                            | RO S BE | The identification of the emitter system that is generating this emitter beam.                                                                                                                      |
| eventIdentifier          | <a href="#">EventIdentifierStruct</a>             | RO D BE | The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event. |
| frequencyRange           | <a href="#">FrequencyHertzFloat32</a>             | RO D BE | The bandwidth of the frequencies covered by the emitter beam.                                                                                                                                       |
| pulseRepetitionFrequency | <a href="#">FrequencyHertzFloat32</a>             | RO D BE | The Pulse Repetition Frequency of the emitter beam.                                                                                                                                                 |
| pulseWidth               | <a href="#">TimeMicrosecondFloat32</a>            | RO D BE | The pulse width of the emitter beam.                                                                                                                                                                |
| sweepSynch               | <a href="#">PercentFloat32</a>                    | RO D BE | The percentage of time a scan is through its pattern from its origin.                                                                                                                               |

## EmitterSystem CLASS

A device that radiates an electromagnetic signal, such as a radar or jammer.

| Name                | Type                                 | Flags   | Description                         |
|---------------------|--------------------------------------|---------|-------------------------------------|
| emitterFunctionCode | <a href="#">EmitterFunctionEnum8</a> | RO D BE | The function of the emitter system. |
| emitterType         | <a href="#">EmitterTypeEnum16</a>    | RO S BE | The type of the emitter system.     |



| Name            | Type                  | Flags   | Description                                                                                                                                                                                         |
|-----------------|-----------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| emitterIndex    | Octet                 | RO S BE | A unique number, which uniquely identifies the emitter system from other on the same host entity.                                                                                                   |
| eventIdentifier | EventIdentifierStruct | RO D BE | The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event. |

## EnvironmentObject CLASS

A base class of environment point, linear, or areal object classes.

| Name                        | Type                         | Flags   | Description                                                                                                 |
|-----------------------------|------------------------------|---------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | EntityIdentifierStruct       | RO S BE | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject Identifier | string                       | RO D BE | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier             | ForceIdentifierEnum8         | RO D BE | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                  | EnvironmentObject TypeStruct | RO S BE | Identifies the type of this EnvironmentObject instance                                                      |

## EnvironmentProcess CLASS

An object class used to communicate information about environmental processes and effects.

| Name                      | Type                       | Flags   | Description                                                                                                                                                             |
|---------------------------|----------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| processIdentifier         | EntityIdentifierStruct     | RO S BE | Identifies which process issued the environmental process update                                                                                                        |
| type                      | EnvironmentType Struct     | RO S BE | Identifies the environmental process type                                                                                                                               |
| modelType                 | EnvironmentModel TypeEnum8 | RO D BE | Identifies the model used for generating this environmental process update ; defined in 1278.1a as being exercise specific                                              |
| environmentProcess Active | bool                       | RO D BE | Specifies the status of the environmental process (active or inactive)                                                                                                  |
| sequenceNumber            | u16                        | RO D BE | Optional, used to support update sequencing ; if not used, set to EP_NO_SEQUENCE ; when used, set to zero for each exercise and incremented by one for each update sent |
| environmentRecData        | EnvironmentRecStruct Array | RO D BE | Specifies environment records                                                                                                                                           |

## EnvironmentalEntity CLASS

An object which has physical extent but not necessarily fixed size and shape, such as meteorological effects such as clouds or smokes.

| Name        | Type              | Flags   | Description                                      |
|-------------|-------------------|---------|--------------------------------------------------|
| opacityCode | OpacityCodeEnum32 | RO D BE | Specifies the density of an environmental entity |

## ExhaustSmokeObject CLASS

An exhaust smoke linear object.



| Name           | Type                                                                  | Flags   | Description                              |
|----------------|-----------------------------------------------------------------------|---------|------------------------------------------|
| segmentRecords | <a href="#">ExhaustSmokeStruct</a><br><a href="#">LengthlessArray</a> | RO D BE | Specifies an exhaust smoke linear object |

## Expendables CLASS

Countermeasures devices that are dispensed from another entity. The devices may be active emitters or passive reflectors of energy.

## GriddedData CLASS

An object class used to depict global, spatially varying environmental effects.

| Name             | Type                                                                                          | Flags   | Description                                                                                                                                                                                                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gridIdentifier   | <a href="#">EntityIdentifierStruct</a>                                                        | RO S BE | Identifies the environmental simulation application                                                                                                                                                                                                                          |
| coordinateSystem | <a href="#">EnvironmentData</a><br><a href="#">CoordinateSystem</a><br><a href="#">Enum16</a> | RO D BE | Specifies the coordinate system used to locate the data grid                                                                                                                                                                                                                 |
| numberOfGridAxes | <a href="#">Octet</a>                                                                         | RO D BE | Specifies the number of grid axes used to define the data grid (e.g. three grid axes for an x, y, z coordinate system)                                                                                                                                                       |
| constantGrid     | <a href="#">EnvironmentGridType</a><br><a href="#">Enum8</a>                                  | RO D BE | Specifies whether the grid axes remain constant for the life of the data grid                                                                                                                                                                                                |
| environmentType  | <a href="#">EnvironmentType</a><br><a href="#">Struct</a>                                     | RO S BE | Identifies the type of environmental entity being described                                                                                                                                                                                                                  |
| orientation      | <a href="#">OrientationStruct</a>                                                             | RO D BE | Specifies the orientation of the data grid, with Euler angles                                                                                                                                                                                                                |
| sampleTime       | <a href="#">UnsignedInteger64</a>                                                             | RO D BE | Specifies the valid time of the environmental data sample                                                                                                                                                                                                                    |
| totalValues      | <a href="#">UnsignedInteger32</a>                                                             | RO D BE | Specifies the number of data values that make up this grid                                                                                                                                                                                                                   |
| vectorDimension  | <a href="#">Octet</a>                                                                         | RO D BE | Specifies the number of data values at each grid point ; VectorDimension shall be one for scalar data, and shall be greater than one when multiple enumerated environmental data values are sent for each grid point (e.g. u, v, w wind components have VectorDimension = 3) |
| gridAxisInfo     | <a href="#">GridAxisStruct</a><br><a href="#">LengthlessArray</a>                             | RO D BE | Specifies information on grid axes                                                                                                                                                                                                                                           |
| gridDataInfo     | <a href="#">GridDataStruct</a><br><a href="#">LengthlessArray</a>                             | RO D BE | Specifies information on grid data representations                                                                                                                                                                                                                           |

## GroundVehicle CLASS

A platform entity that operates wholly on the surface of the earth.

## Human CLASS

A human lifeform.

## IFF CLASS

Interrogator Friend or Foe (IFF) system, Air Traffic Control Beacon and Transponder system, collision avoidance and navigational aids systems.

| Name                               | Type                                                                              | Flags   | Description                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|-----------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| beamAzimuthCenter                  | <a href="#">AngleRadianFloat32</a>                                                | RO D BE | The azimuth center of the IFF beam's scan volume relative to the IFF system.                                                                                                                                                                                                                                                       |
| beamAzimuthSweep                   | <a href="#">AngleRadianFloat32</a>                                                | RO D BE | The azimuth half-angle of the IFF beam's scan volume relative to the IFF system.                                                                                                                                                                                                                                                   |
| beamElevationCenter                | <a href="#">AngleRadianFloat32</a>                                                | RO D BE | The elevation center of the IFF beam's scan volume relative to the IFF system.                                                                                                                                                                                                                                                     |
| beamElevationSweep                 | <a href="#">AngleRadianFloat32</a>                                                | RO D BE | The elevation half-angle of the IFF beam's scan volume relative to the IFF system.                                                                                                                                                                                                                                                 |
| beamSweepSync                      | <a href="#">PercentFloat32</a>                                                    | RO D BE | The percentage of time a scan is through its pattern from its origin.                                                                                                                                                                                                                                                              |
| eventIdentifier                    | <a href="#">EventIdentifierStruct</a>                                             | RO D BE | Used to associate related events.                                                                                                                                                                                                                                                                                                  |
| fundamentalParameterData           | <a href="#">FundamentalParameterDataStruct</a><br><a href="#">LengthlessArray</a> | RO D BE | The fundamental energy radiation characteristics of the IFF/ATC/NAVAIDS system.                                                                                                                                                                                                                                                    |
| layer2DataAvailable                | <a href="#">bool</a>                                                              | RO D BE | Specifies if level 2 data is available for this IFF system. If level 2 data is available then the BeamAzimuthCenter, BeamAzimuthSweep, BeamElevationCenter, BeamElevationSweep, BeamSweepSync, FundamentalParameterData, SecondaryOperationalDataParameter1, and SecondaryOperationalDataParameter2 attributes shall be generated. |
| secondaryOperationalDataParameter1 | <a href="#">IffOperationalParameter1Enum8</a>                                     | RO D BE | Additional characteristics of the IFF/ATC/NAVAIDS emitting system.                                                                                                                                                                                                                                                                 |
| secondaryOperationalDataParameter2 | <a href="#">IffOperationalParameter2Enum8</a>                                     | RO D BE | Additional characteristics of the IFF/ATC/NAVAIDS emitting system.                                                                                                                                                                                                                                                                 |
| systemMode                         | <a href="#">IffSystemModeEnum8</a>                                                | RO D BE | Mode of operation.                                                                                                                                                                                                                                                                                                                 |
| systemName                         | <a href="#">IffSystemNameEnum16</a>                                               | RO S BE | Particular named type of the IFF system in use.                                                                                                                                                                                                                                                                                    |
| systemType                         | <a href="#">IffSystemTypeEnum16</a>                                               | RO S BE | General type of IFF system in use.                                                                                                                                                                                                                                                                                                 |
| systemIsOn                         | <a href="#">bool</a>                                                              | RO D BE | Whether or not the system is on.                                                                                                                                                                                                                                                                                                   |
| systemIsOperational                | <a href="#">bool</a>                                                              | RO D BE | Whether or not the system is operational.                                                                                                                                                                                                                                                                                          |

## JammerBeam CLASS

An emitter beam that is designed to jam or otherwise interfere or confuse another emitter system

| Name                    | Type                              | Flags   | Description                                                                                                                           |
|-------------------------|-----------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| jammingModeSequence     | <a href="#">UnsignedInteger32</a> | RO D BE | The jamming mode technique or series of techniques being applied.                                                                     |
| jammedObjectIdentifiers | <a href="#">RTIobjectIdArray</a>  | RO D BE | Identification of the emitter beams being jammed.                                                                                     |
| highDensityJam          | <a href="#">bool</a>              | RO D BE | When TRUE the receiving simulation can assume that all emitter beams that are in the scan pattern of the jammer beam are being jammed |

## Lifeform CLASS

A living military platform (human or not).

| Name          | Type                             | Flags   | Description                                        |
|---------------|----------------------------------|---------|----------------------------------------------------|
| flashLightsOn | <a href="#">bool</a>             | RO D BE | Whether the lifeform's flash lights are on or not. |
| stanceCode    | <a href="#">StanceCodeEnum32</a> | RO D BE | The stance of the lifeform.                        |

| Name                 | Type                                  | Flags   | Description                                         |
|----------------------|---------------------------------------|---------|-----------------------------------------------------|
| primaryWeaponState   | <a href="#">WeaponStateEnum32</a>     | RO D BE | The state of the soldier's primary weapon system.   |
| secondaryWeaponState | <a href="#">WeaponStateEnum32</a>     | RO D BE | The state of the soldier's secondary weapon system. |
| complianceState      | <a href="#">ComplianceStateEnum32</a> | RO D BE | The compliance of the lifeform.                     |

## LinearObject CLASS

A synthetic environment object that has size and an orientation and is geometrically anchored to the terrain with one point.

## Minefield CLASS

An area of ground or water containing mines laid with or without a pattern.

| Name                      | Type                                                                    | Flags  | Description                                                                                                  |
|---------------------------|-------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------|
| activeStatus              | <a href="#">MinefieldStatusEnum8</a>                                    | RO D C | Specifies the active status of the minefield                                                                 |
| forceIdentifier           | <a href="#">ForceIdentifierEnum8</a>                                    | RO D C | Identifies the force to which the minefield belongs                                                          |
| lane                      | <a href="#">MinefieldLaneEnum8</a>                                      | RO D C | Specifies whether or not the minefield has an active lane                                                    |
| minefieldAppearanceType   | <a href="#">MinefieldTypeEnum8</a>                                      | RO D C | Specifies the appearance information needed for displaying the symbology of the minefield                    |
| minefieldIdentifier       | <a href="#">EntityIdentifierStruct</a>                                  | RO S C | Uniquely identifies this minefield instance in association with the federate's site and application          |
| minefieldLocation         | <a href="#">WorldLocationStruct</a>                                     | RO D C | Specifies the location of the center of the minefield                                                        |
| minefieldOrientation      | <a href="#">OrientationStruct</a>                                       | RO D C | Specifies the orientation of the minefield, with Euler angles                                                |
| minefieldType             | <a href="#">EntityTypeStruct</a>                                        | RO D C | Specifies the minefield type                                                                                 |
| mineTypes                 | <a href="#">EntityTypeStruct</a><br><a href="#">LengthlessArray</a>     | RO D C | Specifies the type of each mine contained within the minefield                                               |
| perimeterPointCoordinates | <a href="#">PerimeterPointStruct</a><br><a href="#">LengthlessArray</a> | RO D C | Specifies the location of each perimeter point, relative to the minefield location                           |
| protocolMode              | <a href="#">MinefieldProtocolEnum8</a>                                  | RO D C | Specifies the mode (Heartbeat or Query Response Protocol) being used to communicate data about the minefield |
| state                     | <a href="#">bool</a>                                                    | RO D C | Specifies whether or not the minefield has been deactivated                                                  |

## MinefieldData CLASS

The MinefieldData object class provides data about collections of mines within a minefield on an individual mine basis.

| Name                    | Type                                                                 | Flags   | Description                                                                                              |
|-------------------------|----------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------|
| groundBurialDepthOffset | <a href="#">DepthMeter</a><br><a href="#">Float32LengthlessArray</a> | RO D BE | Specifies the offset of the origin of the mine coordinate system with respect to the ground surface      |
| fusing                  | <a href="#">MineFusingStruct</a><br><a href="#">LengthlessArray</a>  | RO D BE | Specifies the primary and secondary fuse and anti-handling device for each mine in a collection of mines |
| mineEmplacementTime     | <a href="#">ClockTimeStruct</a><br><a href="#">LengthlessArray</a>   | RO D BE | Specifies the real-world (UTC) emplacement time of each mine in a collection of mines                    |

| Name                           | Type                                                       | Flags   | Description                                                                                                                                                                                 |
|--------------------------------|------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mineEntityIdentifier           | MineIdentifier<br>LengthlessArray                          | RO D BE | Identifies the mine entity identifier; the MineEntityID in conjunction with the MinefieldID form the unique identifier for each mine                                                        |
| minefieldIdentifier            | string                                                     | RO D BE | Identifies the minefield to which the mines belong                                                                                                                                          |
| mineLocation                   | WorldLocationStruct<br>LengthlessArray                     | RO D BE | Specifies the location of the mine relative to the minefield location for each mine in a collection of mines                                                                                |
| mineOrientation                | OrientationStruct<br>LengthlessArray                       | RO D BE | Specifies the orientation of the center axis direction of fire of the mine, relative to the minefield Coordinate System for each mine in a collection of mines                              |
| mineType                       | EntityTypeStruct                                           | RO D BE | Specifies the type of mine for the collection of mines contained within the MinefieldData object                                                                                            |
| numberTripDetonation<br>Wires  | Unsigned<br>Integer8Lengthless<br>Array                    | RO D BE | Specifies the number of trip detonation wires that exist for each mine in a collection of mines                                                                                             |
| numberWireVertices             | Unsigned<br>Integer8Lengthless<br>Array                    | RO D BE | Specifies the number of vertices for each trip wire of each mine in a collection of mines                                                                                                   |
| paintScheme                    | MinefieldPaintScheme<br>LengthlessArray                    | RO D BE | Specifies the camouflage scheme/color of the mine                                                                                                                                           |
| reflectance                    | MineDielectric<br>DifferenceLengthless<br>Array            | RO D BE | Specifies the local dielectric difference between the mine and the surrounding soil                                                                                                         |
| scalarDetection<br>Coefficient | Unsigned<br>Integer8Lengthless<br>Array                    | RO D BE | Specifies the coefficient to be utilized to insure proper correlation between detectors located on different simulation platforms                                                           |
| sensorTypes                    | MinefieldSensorType<br>LengthlessArray                     | RO D BE | In QRP mode, specifies the requesting sensor types which were specified in the minefield query whereas in Heartbeat mode, specifies the sensor types that are being served by the minefield |
| snowBurialDepthOffset          | DepthMeter<br>Float32Lengthless<br>Array                   | RO D BE | Specifies the offset of the origin of the mine coordinate system with respect to the snow surface                                                                                           |
| thermalContrast                | TemperatureDegree<br>Celsius<br>Float32Lengthless<br>Array | RO D BE | Specifies the temperature difference between the mine and the surround soil in degrees Centigrade                                                                                           |
| waterBurialDepth<br>Offset     | DepthMeter<br>Float32Lengthless<br>Array                   | RO D BE | Specifies the offset of the origin of the mine coordinate system with respect to the water surface                                                                                          |
| wireVertices                   | WorldLocationStruct<br>LengthlessArray                     | RO D BE | Specifies the locations of vertices in a trip wire                                                                                                                                          |

## MinefieldLaneMarkerObject CLASS

A minefield lane marker showing a cleared lane through a specific minefield, defined as a linear object.

| Name           | Type                                         | Flags   | Description                                                                  |
|----------------|----------------------------------------------|---------|------------------------------------------------------------------------------|
| segmentRecords | MinefieldLaneMarker<br>StructLengthlessArray | RO D BE | Specifies a minefield lane marker showing a cleared lane through a minefield |

## MinefieldObject CLASS

A mine, mine weapon, mine row, mine strip, mine lane, mine marker or minefield, defined as an areal environment object.

| Name           | Type                | Flags   | Description                                               |
|----------------|---------------------|---------|-----------------------------------------------------------|
| breachedStatus | BreachedStatusEnum8 | RO D BE | Specifies the breached appearance of the minefield object |
| mineCount      | UnsignedInteger32   | RO D BE | Specifies the number of mines in the minefield            |

## MultiDomainPlatform CLASS

A platform entity that operates in more than one domain (excluding those combinations explicitly defined as subclasses of the superclass of this class).

## Munition CLASS

A complete device charged with explosives, propellants, pyrotechnics, initiating composition, or nuclear, biological or chemical material for use in military operations, including demolitions.

| Name                 | Type | Flags   | Description                                                         |
|----------------------|------|---------|---------------------------------------------------------------------|
| launcherFlashPresent | bool | RO D BE | Whether the flash of the munition being launched is present or not. |

## NatoIFF CLASS

NATO Identification Friend or Foe (IFF) system that uses electromagnetic transmissions to which friendly forces' equipment automatically responds.

| Name                       | Type                   | Flags   | Description                                                       |
|----------------------------|------------------------|---------|-------------------------------------------------------------------|
| alternateMode4             | IffAlternateMode4Enum8 | RO D BE | IFF Alternate Mode 4.                                             |
| modelEnabled               | bool                   | RO D BE | Specifies whether or not Mode 1 is enabled.                       |
| modelIsDamaged             | bool                   | RO D BE | Specifies whether or not Mode 1 is damaged.                       |
| modelIsMalfunctioning      | bool                   | RO D BE | Specifies whether or not Mode 1 is malfunctioning.                |
| modelIsOn                  | bool                   | RO D BE | Specifies whether or not Mode 1 is on.                            |
| mode2Enabled               | bool                   | RO D BE | Specifies whether or not Mode 2 is enabled.                       |
| mode2IsDamaged             | bool                   | RO D BE | Specifies whether or not Mode 2 is damaged.                       |
| mode2IsMalfunctioning      | bool                   | RO D BE | Specifies whether or not Mode 2 is malfunctioning.                |
| mode2IsOn                  | bool                   | RO D BE | Specifies whether or not Mode 2 is on.                            |
| mode3AEnabled              | bool                   | RO D BE | Specifies whether or not Mode 3 is enabled.                       |
| mode3AIsDamaged            | bool                   | RO D BE | Specifies whether or not Mode 3 is damaged.                       |
| mode3AIsMalfunctioning     | bool                   | RO D BE | Specifies whether or not Mode 3 is malfunctioning.                |
| mode3AIsOn                 | bool                   | RO D BE | Specifies whether or not Mode 3 is on.                            |
| mode4Enabled               | bool                   | RO D BE | Specifies whether or not Mode 4 is enabled.                       |
| mode4IsDamaged             | bool                   | RO D BE | Specifies whether or not Mode 4 is damaged.                       |
| mode4IsMalfunctioning      | bool                   | RO D BE | Specifies whether or not Mode 4 is malfunctioning.                |
| mode4IsOn                  | bool                   | RO D BE | Specifies whether or not Mode 4 is on.                            |
| mode4PseudoCrypto          | UnsignedInteger16      | RO D BE | Mode 4 Pseudo-Crypto value (0-4094).                              |
| mode4PseudoCryptoAvailable | bool                   | RO D BE | Specifies whether or not Mode 4 pseudo-crypto value is available. |
| mode5CEnabled              | bool                   | RO D BE | Specifies whether or not Parameter 5 - Mode C is enabled.         |

| Name                   | Type | Flags   | Description                                                                                               |
|------------------------|------|---------|-----------------------------------------------------------------------------------------------------------|
| mode5CIsDamaged        | bool | RO D BE | Specifies whether or not Parameter 5 - Mode C is damaged.                                                 |
| mode5CIsMalfunctioning | bool | RO D BE | Specifies whether or not Parameter 5 - Mode C is malfunctioning.                                          |
| mode5CIsOn             | bool | RO D BE | Specifies whether or not Parameter 5 - Mode C is on.                                                      |
| modeSEnabled           | bool | RO D BE | Specifies whether or not Parameter 6 - Mode S is enabled.                                                 |
| modeSIsDamaged         | bool | RO D BE | Specifies whether or not Parameter 6 - Mode S is damaged.                                                 |
| modeSIsMalfunctioning  | bool | RO D BE | Specifies whether or not Parameter 6 - Mode S is malfunctioning.                                          |
| modeSIsOn              | bool | RO D BE | Specifies whether or not Parameter 6 - Mode S is on.                                                      |
| modeSIsTcasI           | bool | RO D BE | Specifies which Traffic Alert/Collision Avoidance System is used. FALSE means TCAS I, TRUE means TCAS II. |

## NatoIFFInterrogator CLASS

The part of an IFF system that first transmits electromagnetic signals.

## NatoIFFTransponder CLASS

The part of a specific IFF system that responds (for example by emitting pulses) to the electromagnetic signals.

| Name                    | Type                                     | Flags   | Description                                                          |
|-------------------------|------------------------------------------|---------|----------------------------------------------------------------------|
| emergencyOn             | bool                                     | RO D BE | Specifies whether or not emergency is on.                            |
| identSquawkFlashOn      | bool                                     | RO D BE | Specifies whether or not the squawk ident/flash is on.               |
| mode1Code               | UnsignedInteger16                        | RO D BE | Mode 1 code, two octal values.                                       |
| mode2Code               | UnsignedInteger16                        | RO D BE | Mode 2 code, four octal values.                                      |
| mode3ACode              | UnsignedInteger16                        | RO D BE | Mode 3 code, four octal values.                                      |
| mode5CAltitude          | TransponderModeCAltitude100FootInteger16 | RO D BE | Parameter 5 - Mode C altitude.                                       |
| mode5CAltitudeAvailable | bool                                     | RO D BE | Specifies whether or not Parameter 5 - Mode C altitude is available. |
| stiOn                   | bool                                     | RO D BE | Specifies whether or not STI is on.                                  |

## NonHuman CLASS

An animal or other non-human lifeform.

## OtherArealObject CLASS

Areal objects other than Minefield objects.

## OtherLinearObject CLASS

Linear objects other than BreachableLinear, Breach, ExhaustSmoke, or MinefieldLaneMarker objects.

## OtherPointObject CLASS

Point objects other than Breachable, Burst, Crater, RibbonBridge, or Structure objects.

## PhysicalEntity CLASS

A base class of all discrete platform scenario domain participants.

| Name                            | Type                                           | Flags   | Description                                                                                                   |
|---------------------------------|------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------|
| acousticSignatureIndex          | Integer16                                      | RO D BE | Index used to obtain the acoustics (sound through air) signature state of the entity.                         |
| alternateEntityType             | EntityTypeStruct                               | RO S BE | The category of entity to be used when viewed by entities on the 'opposite' side.                             |
| articulatedParametersArray      | ArticulatedParameterStructLengthlessArray      | RO D BE | Identification of the visible parts, and their states, of the entity which are capable of independent motion. |
| camouflageType                  | CamouflageEnum32                               | RO D BE | The type of camouflage in use (if any).                                                                       |
| damageState                     | DamageStatusEnum32                             | RO D BE | The state of damage of the entity.                                                                            |
| engineSmokeOn                   | bool                                           | RO D BE | Whether the entity's engine is generating smoke or not.                                                       |
| firePowerDisabled               | bool                                           | RO D BE | Whether the entity's main weapon system has been disabled or not.                                             |
| flamesPresent                   | bool                                           | RO D BE | Whether the entity is on fire (with visible flames) or not.                                                   |
| forceIdentifier                 | ForceIdentifierEnum8                           | RO D BE | The identification of the force that the entity belongs to.                                                   |
| hasAmmunitionSupplyCap          | bool                                           | RO D BE | Whether the entity has the capability to supply other entities with ammunition.                               |
| hasFuelSupplyCap                | bool                                           | RO D BE | Whether the entity has the capability to supply other entities with fuel or not.                              |
| hasRecoveryCap                  | bool                                           | RO D BE | Whether the entity has the capability to recover other entities or not.                                       |
| hasRepairCap                    | bool                                           | RO D BE | Whether the entity has the capability to repair other entities or not.                                        |
| immobilized                     | bool                                           | RO D BE | Whether the entity is immobilized or not.                                                                     |
| infraredSignatureIndex          | Integer16                                      | RO D BE | Index used to obtain the infra-red signature state of the entity.                                             |
| isConcealed                     | bool                                           | RO D BE | Whether the entity is concealed or not.                                                                       |
| liveEntityMeasuredSpeed         | VelocityDecimeterPerSecondInteger16            | RO D BE | The entity's own measurement of speed (e.g. air speed for aircraft).                                          |
| marking                         | MarkingStruct                                  | RO D BE | A unique marking or combination of characters used to distinguish the entity from other entities.             |
| powerPlantOn                    | bool                                           | RO D BE | Whether the entity's power plant is on or not.                                                                |
| propulsionSystemsData           | PropulsionSystemDataStructLengthlessArray      | RO D BE | The basic operating data of the propulsion systems aboard the entity.                                         |
| radarCrossSectionSignatureIndex | Integer16                                      | RO D BE | Index used to obtain the radar cross section signature state of the entity.                                   |
| smokePlumePresent               | bool                                           | RO D BE | Whether the entity is generating smoke or not (intentional or unintentional).                                 |
| tentDeployed                    | bool                                           | RO D BE | Whether the entity has deployed tent or not.                                                                  |
| trailingEffectsCode             | TrailingEffectsCodeEnum32                      | RO D BE | The type and size of any trail that the entity is making.                                                     |
| vectoringNozzleSystemData       | VectoringNozzleSystemDataStructLengthlessArray | RO D BE | The basic operational data for the vectoring nozzle systems aboard the entity.                                |



## Platform CLASS

A physical object under the control of armed forces upon which sensor, communication, or weapon systems may be mounted.

| Name                  | Type             | Flags   | Description                                                     |
|-----------------------|------------------|---------|-----------------------------------------------------------------|
| afterburnerOn         | bool             | RO D BE | Whether the entity's afterburner is on or not.                  |
| antiCollisionLightsOn | bool             | RO D BE | Whether the entity's anti-collision lights are on or not.       |
| blackOutBrakeLightsOn | bool             | RO D BE | Whether the entity's black out brake lights are on or not.      |
| blackOutLightsOn      | bool             | RO D BE | Whether the entity's black out lights are on or not.            |
| brakeLightsOn         | bool             | RO D BE | Whether the entity's brake lights are on or not.                |
| formationLightsOn     | bool             | RO D BE | Whether the entity's formation lights are on or not.            |
| hatchState            | HatchStateEnum32 | RO D BE | The state of the entity's (main) hatch.                         |
| headLightsOn          | bool             | RO D BE | Whether the entity's headlights are on or not.                  |
| interiorLightsOn      | bool             | RO D BE | Whether the entity's internal lights are on or not.             |
| landingLightsOn       | bool             | RO D BE | Whether the entity's landing lights are on or not.              |
| launcherRaised        | bool             | RO D BE | Whether the entity's weapon launcher is in the raised position. |
| navigationLightsOn    | bool             | RO D BE | Whether the entity's navigation lights are on or not.           |
| rampDeployed          | bool             | RO D BE | Whether the entity has deployed a ramp or not.                  |
| runningLightsOn       | bool             | RO D BE | Whether the entity's running lights are on or not.              |
| spotLightsOn          | bool             | RO D BE | Whether the entity's spotlights are on or not.                  |
| tailLightsOn          | bool             | RO D BE | Whether the entity's tail lights are on or not.                 |

## PointObject CLASS

A synthetic environment object that is geometrically anchored to the terrain with a single point.

| Name                 | Type                     | Flags   | Description                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| location             | WorldLocationStruct      | RO D BE | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation          | OrientationStruct        | RO D BE | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete      | PercentUnsignedInteger32 | RO D BE | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance    | DamageStatusEnum32       | RO D BE | Specifies the damaged appearance of the object instance                                                                                                                                                                                                                                                            |
| objectPreDistributed | bool                     | RO D BE | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated          | bool                     | RO D BE | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking              | bool                     | RO D BE | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming              | bool                     | RO D BE | Specifies whether or not the object is aflame                                                                                                                                                                                                                                                                      |



## PropulsionNoise CLASS

Describes the steady state component of unintended passive emissions that are normally associated with the power plant

| Name                          | Type                                    | Flags   | Description                                                                                                                                                    |
|-------------------------------|-----------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hullMaskerOn                  | bool                                    | RO D BE | Whether the hull masker silencing system is on.                                                                                                                |
| passiveParameterIndex         | UnsignedInteger16                       | RO D BE | Database index to look up the entities acoustic signature                                                                                                      |
| propulsionPlant Configuration | PropulsionPlantEnum8                    | RO D BE | The operating mode of the propulsion plant                                                                                                                     |
| shaftRateData                 | ShaftDataStruct<br>LengthlessArray1Plus | RO D BE | Information about each of the propulsion shafts associated with the entity. Shafts are ordered from port to starboard, when looking from the stern to the bow. |

## RRB CLASS

RRB IFF transponder system

| Name             | Type  | Flags   | Description                                            |
|------------------|-------|---------|--------------------------------------------------------|
| code             | Octet | RO D BE | RRB Code (range 0-16)                                  |
| powerReduction   | bool  | RO D BE | Specifies whether or not power reduction is on.        |
| isDamaged        | bool  | RO D BE | Specifies whether or not the system is damaged.        |
| isMalfunctioning | bool  | RO D BE | Specifies whether or not the system is malfunctioning. |
| isOn             | bool  | RO D BE | Specifies whether or not the system is on.             |
| radarEnhancement | bool  | RO D BE | Specifies whether or not the radar enhancement is on.  |

## RadarBeam CLASS

A sector of concentrated energy from a device that radiates an electromagnetic signal.

| Name                    | Type            | Flags   | Description                                                                                                                     |
|-------------------------|-----------------|---------|---------------------------------------------------------------------------------------------------------------------------------|
| highDensityTrack        | bool            | RO D BE | When TRUE the receiving simulation can assume that all targets that are in the scan pattern of the radar beam are being tracked |
| trackObject Identifiers | RTObjectIdArray | RO D BE | Identification of the entities being tracked.                                                                                   |

## Radio CLASS

Electronic devices for the communication of both audio and data, operated by entities belonging to armed forces.

## RadioReceiver CLASS

A device that converts incoming electromagnetic waves in the radio frequency range into information.

| Name                           | Type                                  | Flags   | Description                                                                                    |
|--------------------------------|---------------------------------------|---------|------------------------------------------------------------------------------------------------|
| radioIndex                     | UnsignedInteger16                     | RO S BE | A number that uniquely identifies this radio receiver from other receivers on the host object. |
| receivedPower                  | PowerRatioDecibel<br>MilliwattFloat32 | RO D BE | The radio frequency power received, after applying any propagation loss and antenna gain.      |
| receivedTransmitter Identifier | string                                | RO D BE | The object instance ID of the transmitter that generated the received radio signal.            |
| receiverOperational Status     | ReceiverOperational<br>StatusEnum16   | RO D BE | The operational state of the radio receiver.                                                   |

## RadioTransmitter CLASS

A device that sends out information encoded in electromagnetic waves in the radio frequency range.

| Name                         | Type                                                       | Flags   | Description                                                                                                                                                                           |
|------------------------------|------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| antennaPatternData           | <a href="#">AntennaPatternVariantStructLengthlessArray</a> | RO D BE | The radiation pattern of the radio's antenna.                                                                                                                                         |
| cryptographicMode            | <a href="#">CryptographicModeEnum32</a>                    | RO D BE | The mode of the cryptographic system.                                                                                                                                                 |
| cryptoSystem                 | <a href="#">CryptographicSystemTypeEnum16</a>              | RO D BE | The type of cryptographic equipment in use.                                                                                                                                           |
| encryptionKeyIdentifier      | <a href="#">UnsignedInteger16</a>                          | RO D BE | The identification of the key used to encrypt the radio signals being transmitted. The transmitter and receiver should be considered to be using the same key if these numbers match. |
| frequency                    | <a href="#">FrequencyHertzUnsignedInteger64</a>            | RO D BE | The center frequency of the radio transmissions.                                                                                                                                      |
| frequencyBandwidth           | <a href="#">FrequencyHertzFloat32</a>                      | RO D BE | The bandpass of the radio transmissions.                                                                                                                                              |
| radioIndex                   | <a href="#">UnsignedInteger16</a>                          | RO S BE | A number that uniquely identifies this radio from others on the host.                                                                                                                 |
| radioInputSource             | <a href="#">RadioInputSourceEnum8</a>                      | RO D BE | Specifies which position or data port provided the input for the transmission.                                                                                                        |
| radioSystemType              | <a href="#">RadioTypeStruct</a>                            | RO S BE | The type of radio transmitter.                                                                                                                                                        |
| rFModulationSystemType       | <a href="#">RFModulationSystemTypeEnum16</a>               | RO D BE | The radio system type associated with this transmitter.                                                                                                                               |
| rFModulationType             | <a href="#">RFModulationTypeVariantStruct</a>              | RO D BE | Classification of the modulation type.                                                                                                                                                |
| spreadSpectrum               | <a href="#">SpreadSpectrumVariantStruct</a>                | RO D BE | Describes the spread spectrum characteristics of the transmission, such as frequency hopping or other spread spectrum transmission modes.                                             |
| streamTag                    | <a href="#">UnsignedInteger64</a>                          | RO D BE | A globally unique identifier for the associated audio stream                                                                                                                          |
| timeHopInUse                 | <a href="#">bool</a>                                       | RO D BE | Whether the radio is using time hopping or not.                                                                                                                                       |
| transmittedPower             | <a href="#">PowerRatioDecibelMilliwattFloat32</a>          | RO D BE | The average transmitted power.                                                                                                                                                        |
| transmitterOperationalStatus | <a href="#">TransmitterOperationalStatusEnum8</a>          | RO D BE | The current operational state of the radio transmitter.                                                                                                                               |
| worldLocation                | <a href="#">WorldLocationStruct</a>                        | RO D BE | The location of the antenna in the world coordinate system.                                                                                                                           |

## RibbonBridgeObject CLASS

A ribbon bridge object.

| Name             | Type                              | Flags   | Description                                                  |
|------------------|-----------------------------------|---------|--------------------------------------------------------------|
| numberOfSegments | <a href="#">UnsignedInteger32</a> | RO D BE | Specifies the number of segments composing the ribbon bridge |

## Sensor CLASS

Sensors and emitters, such as stand-alone radars, jammers, and detection systems, that are not part of another platform or system described by another Physical Entity, and are operated by armed forces.

| Name          | Type                 | Flags   | Description                                            |
|---------------|----------------------|---------|--------------------------------------------------------|
| antennaRaised | <a href="#">bool</a> | RO D BE | Whether the sensor/emitter's antenna is raised or not. |

| Name             | Type | Flags   | Description                                                                                                                  |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------|
| blackoutLightsOn | bool | RO D BE | Whether the sensor/emitter's blackout lights are on or not.                                                                  |
| lightsOn         | bool | RO D BE | Whether the sensor/emitter's lights are on or not.                                                                           |
| interiorLightsOn | bool | RO D BE | Whether the sensor/emitter's interior lights are on or not.                                                                  |
| missionKill      | bool | RO D BE | Whether the sensor/emitter has sustained damage that will prevent it carrying out its mission or not (e.g. damaged antenna). |

## SovietIFF CLASS

Soviet type IFF system

| Name                           | Type | Flags   | Description                                             |
|--------------------------------|------|---------|---------------------------------------------------------|
| parameter1Enabled              | bool | RO D BE | Specifies whether or not Parameter 1 is enabled.        |
| parameter1IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 1 is damaged.        |
| parameter1Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 1 is malfunctioning. |
| parameter1IsOn                 | bool | RO D BE | Specifies whether or not Parameter 1 is on.             |
| parameter2Enabled              | bool | RO D BE | Specifies whether or not Parameter 2 is enabled.        |
| parameter2IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 2 is damaged.        |
| parameter2Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 2 is malfunctioning. |
| parameter2IsOn                 | bool | RO D BE | Specifies whether or not Parameter 2 is on.             |
| parameter3Enabled              | bool | RO D BE | Specifies whether or not Parameter 3 is enabled.        |
| parameter3IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 3 is damaged.        |
| parameter3Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 3 is malfunctioning. |
| parameter3IsOn                 | bool | RO D BE | Specifies whether or not Parameter 3 is on.             |
| parameter4Enabled              | bool | RO D BE | Specifies whether or not Parameter 4 is enabled.        |
| parameter4IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 4 is damaged.        |
| parameter4Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 4 is malfunctioning. |
| parameter4IsOn                 | bool | RO D BE | Specifies whether or not Parameter 4 is on.             |
| parameter5Enabled              | bool | RO D BE | Specifies whether or not Parameter 5 is enabled.        |
| parameter5IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 5 is damaged.        |
| parameter5Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 5 is malfunctioning. |
| parameter5IsOn                 | bool | RO D BE | Specifies whether or not Parameter 5 is on.             |
| parameter6Enabled              | bool | RO D BE | Specifies whether or not Parameter 6 is enabled.        |
| parameter6IsDamaged            | bool | RO D BE | Specifies whether or not Parameter 6 is damaged.        |
| parameter6Is<br>Malfunctioning | bool | RO D BE | Specifies whether or not Parameter 6 is malfunctioning. |
| parameter6IsOn                 | bool | RO D BE | Specifies whether or not Parameter 6 is on.             |

## SovietIFFInterrogator CLASS

Soviet IFF Interrogator

## SovietIFFTransponder CLASS

Soviet IFF Transponder

## Spacecraft CLASS

A platform entity that operates mainly in space.

## StructureObject CLASS

A building structure, building rubble, stationary bridge, AVLB object.

## SubmersibleVessel CLASS

A platform entity that operates either on the surface of the sea, or beneath it.

## Supplies CLASS

Supplies other than munitions, such as fuel, food and personnel.

## SurfaceVessel CLASS

A platform entity that operates wholly on the surface of the sea.

## UnderwaterAcousticsEmission CLASS

The underwater acoustic classes used to communicate underwater acoustic active, intentional emissions and Passive Signature or Unintentional Emissions information. These emissions are used during undersea warfare scenarios to detect, classify, and track hostile forces when electronic warfare mechanisms are unavailable.

| Name            | Type                                  | Flags   | Description                                                                                                                                                                              |
|-----------------|---------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventIdentifier | <a href="#">EventIdentifierStruct</a> | RO D BE | The generating federate uses the Event Identifier to associate related events. The event number begins at one at the beginning of the exercise and is incremented by one for each event. |

### Fixed records

|                                          |                                                                                                                                                                                                                                                                           |     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AccelerationVectorStruct</a> | The magnitude of the change in linear velocity over time.                                                                                                                                                                                                                 | 181 |
| <a href="#">Acknowledge</a>              | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN PDU from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.                                             | 181 |
| <a href="#">AcknowledgeR</a>             | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN PDU, with a specified acknowledgement protocol, from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request. | 181 |
| <a href="#">AcousticTransient</a>        | Specifies the occurrence of a transient acoustic event such as torpedo tube floodings, hatch slams, and wrench drops.                                                                                                                                                     | 182 |
| <a href="#">ActionRequest</a>            | A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action.                                                                                                             | 182 |
| <a href="#">ActionRequestR</a>           | A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.                          | 182 |
| <a href="#">ActionRequestToObject</a>    | A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action.                                                                                             | 183 |
| <a href="#">ActionRequestToObjectR</a>   | A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.          | 183 |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                 |     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ActionResponse                    | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponse interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.                                                                                                                                                 | 183 |
| ActionResponseFromObject          | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToInteraction interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.                                                                                                                                      | 183 |
| ActionResponseFromObjectR         | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToInteractionR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.                                                                                                                                     | 183 |
| ActionResponseR                   | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.                                                                                                                                                | 184 |
| AggregateMarkingStruct            | Unique marking associated with an aggregate.                                                                                                                                                                                                                                                                                                                                                    | 184 |
| AngularVelocityVectorStruct       | The rate at which the orientation is changing over time, in body coordinates.                                                                                                                                                                                                                                                                                                                   | 184 |
| ApplicationSpecificRadioSignal    | A radio signal that can be used for an application specific encoding scheme.                                                                                                                                                                                                                                                                                                                    | 184 |
| ArealObjectTransaction            | An interaction for modifying instances of the ArealObject class.                                                                                                                                                                                                                                                                                                                                | 185 |
| ArticulatedParameterStruct        | Structure to specify a movable or attached part of an entity. Based on the Articulation Parameter record as specified in IEEE 1278.1-1995 section 5.2.5. Note that usage of this datatype for the PhyscialEntity object class attribute ArticulatedParametersArray and MunitionDetonation interaction class parameter ArticulatedPartData shall be in accordance with IEEE 1278.1-1995 Annex A. | 185 |
| ArticulatedPartsStruct            | Structure to represent the state of a movable part of an entity.                                                                                                                                                                                                                                                                                                                                | 186 |
| AttachedPartsStruct               | Structure to represent removable parts that may be attached to an entity.                                                                                                                                                                                                                                                                                                                       | 186 |
| AttributeChangeRequest            | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value.                                                                                                                                                                                                                                             | 186 |
| AttributeChangeRequestR           | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value. The Simulation Manager federate specifies the acknowledgement protocol to be used.                                                                                                                                                          | 186 |
| AttributeChangeResult             | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not.                                                                                                                             | 187 |
| AttributeChangeResultR            | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not. The Simulation Manager federate specifies the acknowledgement protocol to be used.                                          | 187 |
| AttributeValuePairStruct          | Pair of an AttributeHandle identifying an attribute and data for that attribute.                                                                                                                                                                                                                                                                                                                | 187 |
| AudioDataTypeStruct               | Specifies an encoded audio radio signal.                                                                                                                                                                                                                                                                                                                                                        | 187 |
| BeamAntennaStruct                 | Specifies the direction, pattern, and polarization of radiation from a radio transmitter's antenna.                                                                                                                                                                                                                                                                                             | 187 |
| BreachObjectTransaction           | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachObject class.                                                                                                                                                                                                                                                        | 188 |
| BreachStruct                      | Record specifying the characteristics of a breach linear object segment                                                                                                                                                                                                                                                                                                                         | 188 |
| BreachableLinearObjectTransaction | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachableLinearObject class.                                                                                                                                                                                                                                              | 188 |
| BreachablePointObjectTransaction  | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachablePointObject class.                                                                                                                                                                                                                                               | 189 |
| BreachableSegmentStruct           | Record specifying the characteristics of a breachable linear object segment                                                                                                                                                                                                                                                                                                                     | 189 |
| BurstPointObjectTransaction       | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BurstPointObject class.                                                                                                                                                                                                                                                    | 190 |
| COMBICStateRecStruct              | Record specifying COMBIC state record                                                                                                                                                                                                                                                                                                                                                           | 191 |
| ClockTimeStruct                   | Specification of the point in time of an occurrence. Based on the Clock Time record as specified in IEEE 1278.1-1995 section 5.2.8.                                                                                                                                                                                                                                                             | 191 |
| Collision                         | The act or instance of coming together with solid impact.                                                                                                                                                                                                                                                                                                                                       | 191 |
| CollisionElastic                  | The act or instance of coming together with solid impact in an elastic manner. An elastic collision allows a higher fidelity collision to be modeled, taking into account linear and rotational momentum transfer, variable elasticity, and momentum transfer that is dependent on surface orientation.                                                                                         | 192 |
| Comment                           | A Simulation Management (SIMAN) interaction used to insert messages and information into a log stream and closely matches the structures used by the Data interaction class. This information is usually unsolicited in nature. The information contained within the Interaction should be used for commenting purposes only.                                                                   | 192 |
| Cone1GeomRecStruct                | Record specifying Cone 1 geometry record                                                                                                                                                                                                                                                                                                                                                        | 193 |
| Cone2GeomRecStruct                | Record specifying Cone 2 geometry record                                                                                                                                                                                                                                                                                                                                                        | 193 |

|                                   |                                                                                                                                                                                                                                                                                                                             |     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ConstituentPartRelationshipStruct | The relationship of the constituent part object to its host object. Based on the Relationship record as specified in IEEE 1278.1a-1998 section 5.2.56.                                                                                                                                                                      | 193 |
| CraterObjectTransaction           | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the CraterObject class.                                                                                                                                                                                    | 193 |
| CreateEntity                      | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity. See DIS 4.5.5.4.1 and DIS 5.3.6 for details.                                                                                                                                                  | 194 |
| CreateEntityR                     | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity, using a specified acknowledgement service. See DIS 4.5.11.3.1 and DIS 5.3.12.1 for details.                                                                                                   | 194 |
| CreateObjectRequest               | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type.                                                                                                                                                 | 195 |
| CreateObjectRequestR              | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type. The Simulation Manager federate specifies the acknowledgement protocol to be used.                                                              | 195 |
| CreateObjectResult                | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not.                                                                                    | 195 |
| CreateObjectResultR               | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not. The Simulation Manager federate specifies the acknowledgement protocol to be used. | 195 |
| Data                              | A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQuery interaction (in which case the Data interaction contains the results of the query) or b) a SetData interaction (in which case the Data interaction contains the data that the federate was able to set).                          | 196 |
| DataQuery                         | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data.                                                                                                                                                                | 196 |
| DataQueryR                        | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data. The Simulation Manager federate specifies the acknowledgement protocol to be used.                                                                             | 196 |
| DataR                             | A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQueryR interaction (in which case the DataR interaction contains the results of the query) or b) a SetDataR interaction (in which case the DataR interaction contains the data that the federate was able to set).                      | 197 |
| DatabaseIndexRadioSignal          | A radio signal that represents transmission of a pre-determined message by using a pre-defined database index.                                                                                                                                                                                                              | 197 |
| DimensionRateStruct               | Record specifying length X, Y, Z rates                                                                                                                                                                                                                                                                                      | 197 |
| DimensionStruct                   | Bounding box in X,Y,Z axis.                                                                                                                                                                                                                                                                                                 | 198 |
| Ellipsoid1GeomRecStruct           | Record specifying Ellipsoid 1 geometry record                                                                                                                                                                                                                                                                               | 198 |
| Ellipsoid2GeomRecStruct           | Record specifying Ellipsoid 2 geometry record                                                                                                                                                                                                                                                                               | 198 |
| EncodedAudioRadioSignal           | A radio signal used to transmit voice/audio data encoded according to a standard encoding scheme.                                                                                                                                                                                                                           | 198 |
| EntityCoordinateVectorStruct      | Vector in the entity coordinate system. Based on the Entity Coordinate Vector record as specified in IEEE 1278.1-1995 section 5.2.33a.                                                                                                                                                                                      | 199 |
| EntityIdentifierStruct            | Unique, exercise-wide identification of the entity, or a symbolic group address referencing multiple entities or a simulation application. Based on the Entity Identifier record as specified in IEEE 1278.1-1995 section 5.2.14.                                                                                           | 199 |
| EntityTypeStruct                  | Type of entity. Based on the Entity Type record as specified in IEEE 1278.1-1995 section 5.2.16.                                                                                                                                                                                                                            | 199 |
| EnvironmentObjectTransaction      | A base interaction for modifying instances of point, linear and areal environment object classes.                                                                                                                                                                                                                           | 200 |
| EnvironmentObjectTypeStruct       | Record specifying the domain, the kind and the specific identification of the environment object                                                                                                                                                                                                                            | 200 |
| EnvironmentRecStruct              | Record specifying an environment (geometry or state) record                                                                                                                                                                                                                                                                 | 200 |
| EnvironmentTypeStruct             | Record specifying the kind of environment, the domain and any extra information necessary for describing the environmental entity                                                                                                                                                                                           | 200 |
| EventIdentifierStruct             | Identification of an event. Based on the Event Identifier record as specified in IEEE 1278.1-1995 section 5.2.18.                                                                                                                                                                                                           | 201 |
| EventReport                       | A Simulation Management (SIMAN) interaction designed to allow a federate to alert a Simulation Manager federate that a particular event has occurred.                                                                                                                                                                       | 201 |
| ExhaustSmokeObjectTransaction     | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the ExhaustSmokeObject class.                                                                                                                                                                              | 201 |
| ExhaustSmokeStruct                | Record specifying the characteristics of a linear object smoke segment                                                                                                                                                                                                                                                      | 202 |



|                                      |                                                                                                                                                                                                                                                                     |     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FederateIdentifierStruct             | Unique identification of the simulation application (federate) in an exercise, or a symbolic group address referencing multiple simulation applications. Based on the Simulation Address record as specified in IEEE 1278.1-1995 section 5.2.14.1.                  | 202 |
| FixedDatumStruct                     | Identifier and value for a fixed datum.                                                                                                                                                                                                                             | 202 |
| FlareStateRecStruct                  | Record specifying Flare state record                                                                                                                                                                                                                                | 203 |
| FundamentalParameterDataStruct       | Fundamental Parameter Data record.                                                                                                                                                                                                                                  | 203 |
| GaussPlumeGeomRecStruct              | Record specifying Gaussian Plume geometry record                                                                                                                                                                                                                    | 203 |
| GaussPuffGeomRecStruct               | Record specifying Gaussian Puff geometry record                                                                                                                                                                                                                     | 204 |
| GridAxisStruct                       | Record specifying grid Xi axis data                                                                                                                                                                                                                                 | 204 |
| GridDataStruct                       | Record specifying grid data representation                                                                                                                                                                                                                          | 204 |
| GridValueType0Struct                 | Record specifying type 0 data representation                                                                                                                                                                                                                        | 204 |
| GridValueType1Struct                 | Record specifying type 1 data representation                                                                                                                                                                                                                        | 205 |
| GridValueType2Struct                 | Record specifying type 2 data representation                                                                                                                                                                                                                        | 205 |
| IrregularGridAxisStruct              | Record specifying irregular (variable spacing) axis data                                                                                                                                                                                                            | 205 |
| IsPartOfStruct                       | Defines the spatial relationship between two objects.                                                                                                                                                                                                               | 205 |
| JTIDSTransmitterStruct               | Contains JTIDS specific information about the JTIDS transmitter system.                                                                                                                                                                                             | 206 |
| Line1GeomRecStruct                   | Record specifying Line 1 geometry record                                                                                                                                                                                                                            | 206 |
| Line2GeomRecStruct                   | Record specifying Line 2 geometry record                                                                                                                                                                                                                            | 206 |
| LinearObjectTransaction              | An interaction for modifying instances of the LinearObject class.                                                                                                                                                                                                   | 206 |
| LinearSegmentStruct                  | Specifies linear object segment characteristics.                                                                                                                                                                                                                    | 207 |
| Link11BTransmitterStruct             | Contains specific information about the Link 11B transmitter system.                                                                                                                                                                                                | 207 |
| Link11TransmitterStruct              | Contains specific information about the Link 11 transmitter system.                                                                                                                                                                                                 | 207 |
| MarkingStruct                        | Character set used in the marking and the string of characters to be interpreted for display.                                                                                                                                                                       | 208 |
| MineFusingStruct                     | Record specifying the type of primary fuse, the type of the secondary fuse and the anti-handling device status of a mine                                                                                                                                            | 208 |
| MinefieldLaneMarkerObjectTransaction | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldLaneMarkerObject class.                                                                                                               | 208 |
| MinefieldLaneMarkerStruct            | Record specifying a minefield lane marker segment                                                                                                                                                                                                                   | 209 |
| MinefieldObjectTransaction           | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldObject class.                                                                                                                         | 209 |
| MinefieldQuery                       | Provides the means by which a federate shall query a minefield simulation for information on the individual mines within the minefield operating in QRP mode.                                                                                                       | 209 |
| MinefieldResponseNACK                | A response to a MinefieldQuery providing information on individual mines within a minefield.                                                                                                                                                                        | 210 |
| MunitionDetonation                   | Communicates information associated with the impact or detonation of a munition                                                                                                                                                                                     | 210 |
| NamedLocationStruct                  | The discrete positional relationship of the constituent part object with respect to its host object. Based on the specifications in IEEE 1278.1a-1998 of the IsPartOf PDU 'Location of Part' (paragraph 5.3.9.4e) and 'Named Location' (paragraph 5.3.9.4f) fields. | 211 |
| OrientationStruct                    | The orientation of an object in the world coordinate system, as specified in IEEE Std 1278.1-1995 section 1.3.2.                                                                                                                                                    | 211 |
| OtherArealObjectTransaction          | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherArealObject class.                                                                                                                        | 211 |
| OtherLinearObjectTransaction         | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherLinearObject class.                                                                                                                       | 212 |
| OtherPointObjectTransaction          | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherPointObject class.                                                                                                                        | 212 |
| PerimeterPointStruct                 | Record specifying the location of a perimeter point                                                                                                                                                                                                                 | 213 |
| PlumeDimensionRateStruct             | Record specifying plume dimension rates                                                                                                                                                                                                                             | 213 |
| PlumeDimensionStruct                 | Record specifying plume dimensions                                                                                                                                                                                                                                  | 213 |
| Point2GeomRecStruct                  | Record specifying Point 2 geometry record                                                                                                                                                                                                                           | 213 |
| PointObjectTransaction               | An interaction for modifying instances of the PointObject class.                                                                                                                                                                                                    | 214 |
| PropulsionSystemDataStruct           | Information describing a propulsion system in terms of power settings and current RPM.                                                                                                                                                                              | 214 |
| RadioSignal                          | The wireless transmission and reception of audio or digital data by means of electromagnetic waves.                                                                                                                                                                 | 215 |
| RadioTypeStruct                      | Specifies the type of a radio.                                                                                                                                                                                                                                      | 215 |
| RawBinaryRadioSignal                 | A radio signal used to transmit raw binary data.                                                                                                                                                                                                                    | 215 |
| RecordQueryR                         | A Simulation Management (SIMAN) interaction designed to allow a Simulation Manager federate to request data, in record format, from another federate.                                                                                                               | 215 |
| RecordR                              | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RecordQueryR or SetRecordR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.          | 216 |
| RecordSetStruct                      | A set of records and record details.                                                                                                                                                                                                                                | 216 |
| RecordStruct                         | Record data.                                                                                                                                                                                                                                                        | 216 |
| RectVol1GeomRecStruct                | Record specifying Rectangular Volume 1 geometry record                                                                                                                                                                                                              | 217 |

|                                |                                                                                                                                                                                                                                                                                           |     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| RectVol2GeomRecStruct          | Record specifying Rectangular Volume 2 geometry record                                                                                                                                                                                                                                    | 217 |
| RectVol3GeomRecStruct          | Record specifying Rectangular Volume 3 geometry record                                                                                                                                                                                                                                    | 217 |
| RelativePositionStruct         | Relative position in right-handed Cartesian coordinates.                                                                                                                                                                                                                                  | 217 |
| RelativeRangeBearingStruct     | Relative position in polar coordinates.                                                                                                                                                                                                                                                   | 217 |
| RemoveEntity                   | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.                                                                                                                                            | 218 |
| RemoveEntityR                  | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.                                                                                                                                            | 218 |
| RemoveObjectRequest            | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.                                                                                                                                 | 218 |
| RemoveObjectRequestR           | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.                                                                                                                                 | 218 |
| RemoveObjectResult             | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequest interaction, and to inform the Simulation Manager federate whether the removal was successful or not.                                                                                | 219 |
| RemoveObjectResultR            | A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequestR interaction, and to inform the Simulation Manager federate whether the removal was successful or not.                                                                               | 219 |
| RepairComplete                 | Notifies the requesting entity that the requested repair has been completed.                                                                                                                                                                                                              | 219 |
| RepairResponse                 | Acknowledges the notification of the completion of a repair.                                                                                                                                                                                                                              | 219 |
| ResupplyCancel                 | Communicates the canceling of a service function by either the receiving or the supplying entity.                                                                                                                                                                                         | 219 |
| ResupplyOffer                  | Communicates the offer of supplies from a supplying entity to a receiving entity.                                                                                                                                                                                                         | 219 |
| ResupplyReceived               | Acknowledge the receipt of supplies.                                                                                                                                                                                                                                                      | 220 |
| RibbonBridgeObjectTransaction  | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the RibbonBridgeObject class.                                                                                                                                            | 220 |
| SINCGARSModulationStruct       | Modulation parameters for SINCGARS radio system                                                                                                                                                                                                                                           | 221 |
| ServiceRequest                 | A request for logistics support. The requesting entity issues the interaction to the supplying entity asking for repair or specific supplies.                                                                                                                                             | 221 |
| SetData                        | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values.                                                                                                                    | 221 |
| SetDataR                       | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values. The Simulation Manager federate specifies the acknowledgement protocol to be used.                                 | 221 |
| SetRecordR                     | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values (provided in record format).                                                                                        | 222 |
| ShaftDataStruct                | Rotation state of a ship's propulsion shaft                                                                                                                                                                                                                                               | 222 |
| SilentAggregateStruct          | These fields contain information about subaggregates not registered in the federation.                                                                                                                                                                                                    | 223 |
| SilentEntityStruct             | These fields contain information about entities not registered in the federation.                                                                                                                                                                                                         | 223 |
| SpatialFPStruct                | Spatial structure for Dead Reckoning Algorithm FPW (2) and FPB (6).                                                                                                                                                                                                                       | 223 |
| SpatialFVStruct                | Spatial structure for Dead Reckoning Algorithm FVW (5) and RVB (9).                                                                                                                                                                                                                       | 223 |
| SpatialRPStruct                | Spatial structure for Dead Reckoning Algorithm RPW (3) and RPB (7).                                                                                                                                                                                                                       | 224 |
| SpatialRVStruct                | Spatial structure for Dead Reckoning Algorithm RVW (4) and RVB (8).                                                                                                                                                                                                                       | 224 |
| SpatialStaticStruct            | Spatial structure for Dead Reckoning Algorithm Static (1).                                                                                                                                                                                                                                | 224 |
| Sphere1GeomRecStruct           | Record specifying Bounding Sphere & Sphere 1 geometry record                                                                                                                                                                                                                              | 225 |
| Sphere2GeomRecStruct           | Record specifying Sphere 2 geometry record                                                                                                                                                                                                                                                | 225 |
| SphericalHarmonicAntennaStruct | Specifies the direction and radiation pattern from a radio transmitter's antenna.                                                                                                                                                                                                         | 225 |
| StartResume                    | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause.                                                                                    | 225 |
| StartResumeR                   | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause. The Simulation Manager federate specifies the acknowledgement protocol to be used. | 226 |
| StopFreeze                     | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) freeze (pause) their simulation or b) stop their simulation.                                                                                          | 226 |
| StopFreezeR                    | A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) pause their simulation or b) stop their simulation. The Simulation Manager federate specifies the acknowledgement protocol to be used.                | 227 |
| StructureObjectTransaction     | An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the StructureObject class.                                                                                                                                               | 227 |
| SupplyStruct                   | Represents a single supply type and the quantity being offered or requested.                                                                                                                                                                                                              | 228 |



|                                                 |                                                                                                               |     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">TransferControl</a>                 | A Simulation Management (SIMAN) interaction, sent to initiate the transfer of control of an entity.           | 228 |
| <a href="#">UniformGeomRecStruct</a>            | Record specifying Uniform geometry record                                                                     | 228 |
| <a href="#">VariableDatumStruct</a>             | These fields shall specify the types of variable datum, their length, and their value.                        | 229 |
| <a href="#">VectoringNozzleSystemDataStruct</a> | Operational data for describing the vectoring nozzle systems being simulated.                                 | 229 |
| <a href="#">VelocityVectorStruct</a>            | The rate at which the position is changing over time.                                                         | 229 |
| <a href="#">WeaponFire</a>                      | Communicates information associated with the firing or launch of a munition.                                  | 229 |
| <a href="#">WorldLocationStruct</a>             | The location of an object in the world coordinate system, as specified in IEEE Std 1278.1-1995 section 1.3.2. | 230 |

## AccelerationVectorStruct STRUCTURE

The magnitude of the change in linear velocity over time.

| Name          | Type                                                     | Description                              |
|---------------|----------------------------------------------------------|------------------------------------------|
| xAcceleration | <a href="#">AccelerationMeterPerSecondSquaredFloat32</a> | Acceleration component along the X axis. |
| yAcceleration | <a href="#">AccelerationMeterPerSecondSquaredFloat32</a> | Acceleration component along the Y axis. |
| zAcceleration | <a href="#">AccelerationMeterPerSecondSquaredFloat32</a> | Acceleration component along the Z axis. |

Named by [Designator](#), [SpatialFVStruct](#), [SpatialRVStruct](#).

## Acknowledge STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN PDU from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.

| Name              | Type                                   | Description                                                                                                                                          |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                              |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                                                   |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | This field matches this response with the specific StartResume, StopFreeze, CreateEntity or RemoveEntity interaction sent by the simulation manager. |
| acknowledgeFlag   | <a href="#">AcknowledgeFlagEnum16</a>  | The type of interaction being acknowledged.                                                                                                          |
| responseFlag      | <a href="#">ResponseFlagEnum16</a>     | The type of response made to the interaction by the recipient.                                                                                       |

## AcknowledgeR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN PDU, with a specified acknowledgement protocol, from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.

| Name              | Type                                   | Description                                                                                                                                          |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                              |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                                                   |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | This field matches this response with the specific StartResume, StopFreeze, CreateEntity or RemoveEntity interaction sent by the simulation manager. |
| acknowledgeFlag   | <a href="#">AcknowledgeFlagEnum16</a>  | The type of interaction being acknowledged.                                                                                                          |

| Name         | Type                               | Description                                                    |
|--------------|------------------------------------|----------------------------------------------------------------|
| responseFlag | <a href="#">ResponseFlagEnum16</a> | The type of response made to the interaction by the recipient. |

## AcousticTransient STRUCTURE

Specifies the occurrence of a transient acoustic event such as torpedo tube floodings, hatch slams, and wrench drops.

| Name                 | Type                                   | Description                                                       |
|----------------------|----------------------------------------|-------------------------------------------------------------------|
| activityCode         | <a href="#">UnsignedInteger16</a>      | Index into a data base of transient acoustic events               |
| activityParameter    | <a href="#">Integer16</a>              | The current state of the activity specified by the activity code  |
| hostObjectIdentifier | <a href="#">string</a>                 | The Object Instance ID of the object emitting the sound           |
| relativePosition     | <a href="#">RelativePositionStruct</a> | The location of the sound relative to the hosts coordinate system |

## ActionRequest STRUCTURE

A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action.

| Name              | Type                                                                | Description                                                                                                                                                                                                                                                                  |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                                   | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| actionRequestCode | <a href="#">ActionEnum32</a>                                        | The action that the recipient(s) are requested to perform.                                                                                                                                                                                                                   |
| fixedDatums       | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | Optional additional data items (types and values)                                                                                                                                                                                                                            |
| variableDatumSet  | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | Optional additional data items (types and values). These data items are not of fixed length.                                                                                                                                                                                 |

## ActionRequestR STRUCTURE

A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                    | Type                                                                | Description                                                                                                                                                                                                                                                                  |
|-------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity       | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity         | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier       | <a href="#">UnsignedInteger32</a>                                   | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| actionRequestCode       | <a href="#">ActionEnum32</a>                                        | The action that the recipient(s) are requested to perform.                                                                                                                                                                                                                   |
| fixedDatums             | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | Optional additional data items (types and values)                                                                                                                                                                                                                            |
| variableDatumSet        | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | Optional additional data items (types and values). These data items are not of fixed length.                                                                                                                                                                                 |
| acknowledgementProtocol | <a href="#">AcknowledgementProtocolEnum8</a>                        | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## ActionRequestToObject STRUCTURE

A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action.

| Name              | Type                            | Description                                                               |
|-------------------|---------------------------------|---------------------------------------------------------------------------|
| objectIdentifiers | <a href="#">RTObjectIdArray</a> | The list of objects that are the intended recipients of this interaction. |
| actionRequestCode | <a href="#">ActionEnum32</a>    | The action that the recipient(s) are intended to perform.                 |

## ActionRequestToObjectR STRUCTURE

A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                        | Type                                              | Description                                                               |
|-----------------------------|---------------------------------------------------|---------------------------------------------------------------------------|
| objectIdentifiers           | <a href="#">RTObjectIdArray</a>                   | The list of objects that are the intended recipients of this interaction. |
| actionRequestCode           | <a href="#">ActionEnum32</a>                      | The action that the recipient(s) are intended to perform.                 |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement<br/>ProtocolEnum8</a> | The acknowledgement protocol to be used for this transaction              |

## ActionResponse STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponse interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.

| Name              | Type                                                 | Description                                                                                                  |
|-------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>               | The DIS entity ID of the entity or application sending the interaction.                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>               | The DIS entity ID of the entity or application which is the intended recipient of the interaction.           |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                    | This field matches this response with the specific ActionRequest interaction sent by the simulation manager. |
| requestStatus     | <a href="#">RequestStatusEnum32</a>                  | The status of the request that the recipient has been asked to perform.                                      |
| fixedDatums       | <a href="#">FixedDatumStruct<br/>LengthlessArray</a> | Additional, fixed length data items (types and values).                                                      |
| variableDatumSet  | <a href="#">VariableDatumStruct<br/>Array</a>        | Additional, non fixed length, data items (types and values).                                                 |

## ActionResponseFromObject STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToObject interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.

| Name         | Type                               | Description                                                             |
|--------------|------------------------------------|-------------------------------------------------------------------------|
| actionResult | <a href="#">ActionResultEnum32</a> | The status of the request that the recipient has been asked to perform. |

## ActionResponseFromObjectR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToObjectR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.

| Name         | Type                               | Description                                                             |
|--------------|------------------------------------|-------------------------------------------------------------------------|
| actionResult | <a href="#">ActionResultEnum32</a> | The status of the request that the recipient has been asked to perform. |

## ActionResponseR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.

| Name              | Type                                                                | Description                                                                                                  |
|-------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient of the interaction.           |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                                   | This field matches this response with the specific ActionRequest interaction sent by the simulation manager. |
| requestStatus     | <a href="#">RequestStatusEnum32</a>                                 | The status of the request that the recipient has been asked to perform.                                      |
| fixedDatums       | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | Additional, fixed length data items (types and values).                                                      |
| variableDatumSet  | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | Additional, non fixed length, data items (types and values).                                                 |

## AggregateMarkingStruct STRUCTURE

Unique marking associated with an aggregate.

| Name                | Type                                 | Description          |
|---------------------|--------------------------------------|----------------------|
| markingEncodingType | <a href="#">MarkingEncodingEnum8</a> | The type of marking. |
| markingData         | <a href="#">MarkingArray31</a>       | The marking itself.  |

Named by [AggregateEntity](#).

## AngularVelocityVectorStruct STRUCTURE

The rate at which the orientation is changing over time, in body coordinates.

| Name             | Type                                                  | Description                              |
|------------------|-------------------------------------------------------|------------------------------------------|
| xAngularVelocity | <a href="#">AngularVelocityRadianPerSecondFloat32</a> | Acceleration component about the X axis. |
| yAngularVelocity | <a href="#">AngularVelocityRadianPerSecondFloat32</a> | Acceleration component about the Y axis. |
| zAngularVelocity | <a href="#">AngularVelocityRadianPerSecondFloat32</a> | Acceleration component about the Z axis. |

Named by [Cone2GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [GaussPuffGeomRecStruct](#), [RectVol2GeomRecStruct](#), [SpatialRPStruct](#), [SpatialRVStruct](#), [Sphere2GeomRecStruct](#).

## ApplicationSpecificRadioSignal STRUCTURE

A radio signal that can be used for an application specific encoding scheme.

| Name             | Type                                                                     | Description                                                   |
|------------------|--------------------------------------------------------------------------|---------------------------------------------------------------|
| hostRadioIndex   | <a href="#">string</a>                                                   | The object instance ID of the radio transmitting this signal. |
| dataRate         | <a href="#">BitRateBitPerSecond</a><br><a href="#">UnsignedInteger32</a> | The rate at which the data is being transmitted.              |
| signalDataLength | <a href="#">BitsUnsignedInteger16</a>                                    | The length of the signal data.                                |

| Name                 | Type                                           | Description                                                              |
|----------------------|------------------------------------------------|--------------------------------------------------------------------------|
| signalData           | <a href="#">SignalDataLengthlessArray1Plus</a> | The signal data.                                                         |
| tacticalDataLinkType | <a href="#">TacticalDataLinkTypeEnum16</a>     | The type of tactical data link used to transmitted this signal (if any). |
| tDLMessageCount      | <a href="#">UnsignedInteger16</a>              | The number of tactical data link messages contained in this signal.      |
| userProtocolID       | <a href="#">UserProtocolEnum32</a>             | The ID of the user protocol in use.                                      |

## ArealObjectTransaction STRUCTURE

An interaction for modifying instances of the ArealObject class.

| Name                       | Type                                               | Description                                                                                                    |
|----------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>             | Identifies this EnvironmentObject instance (point, linear or areal)                                            |
| referencedObjectIdentifier | <a href="#">string</a>                             | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated    |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>               | Identifies the force that created or modified this EnvironmentObject instance                                  |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a>        | Identifies the type of this EnvironmentObject instance                                                         |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>           | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                     |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>           | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                   |
| pointsData                 | <a href="#">WorldLocationStructLengthlessArray</a> | Specifies the physical location (a collection of points) that defines the object                               |
| percentComplete            | <a href="#">PercentUnsignedInteger32</a>           | Specifies the percent completion of the object                                                                 |
| damagedAppearance          | <a href="#">DamageStatusEnum32</a>                 | Specifies the damaged appearance of the object                                                                 |
| objectPreDistributed       | <a href="#">bool</a>                               | Specifies whether or not the object was created before the start of the exercise                               |
| deactivated                | <a href="#">bool</a>                               | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment) |
| smoking                    | <a href="#">bool</a>                               | Specifies whether or not the object is smoking (creating a smoke plume)                                        |
| flaming                    | <a href="#">bool</a>                               | Specifies whether or not the object is flaming                                                                 |

## ArticulatedParameterStruct STRUCTURE

Structure to specify a movable or attached part of an entity. Based on the Articulation Parameter record as specified in IEEE 1278.1-1995 section 5.2.5. Note that usage of this datatype for the PhyscialEntity object class attribute ArticulatedParametersArray and MunitionDetonation interaction class parameter ArticulatedPartData shall be in accordance with IEEE 1278.1-1995 Annex A.

| Name                       | Type                              | Description                                                                                                                                                                                                                                                              |
|----------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| articulatedParameterChange | <a href="#">Octet</a>             | Indicator of a change to the part. This field shall be set to zero for each exercise and sequentially incremented by one for each change in articulation parameters. In the case where all possible values are exhausted, the numbers shall be reused beginning at zero. |
| partAttachedTo             | <a href="#">UnsignedInteger16</a> | Identification of the articulated part to which this articulation parameter is attached. This field shall contain the value zero if the articulated part is attached directly to the entity.                                                                             |

| Name           | Type                                        | Description                                                                                         |
|----------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------|
| parameterValue | <a href="#">ParameterValueVariantStruct</a> | Details of the parameter: whether it is an articulated or an attached part, and its type and value. |

Named by [ArticulatedParameterStructLengthlessArray](#).

## ArticulatedPartsStruct STRUCTURE

Structure to represent the state of a movable part of an entity.

| Name       | Type                                        | Description                                                                                                                                                                                                                                |
|------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| classVal   | <a href="#">ArticulatedPartsTypeEnum32</a>  | The type class uniquely identifies a particular articulated part on a given entity type. Guidance for uniquely assigning type classes to an entity's articulated parts is given in section 4.8 of the enumeration document (SISO-REF-010). |
| typeMetric | <a href="#">ArticulatedTypeMetricEnum32</a> | The type metric uniquely identifies the transformation to be applied to the articulated part.                                                                                                                                              |
| value      | <a href="#">Float32</a>                     | Value of the transformation to be applied to the articulated part.                                                                                                                                                                         |

Named by [ParameterValueVariantStruct](#).

## AttachedPartsStruct STRUCTURE

Structure to represent removable parts that may be attached to an entity.

| Name      | Type                             | Description                                                                |
|-----------|----------------------------------|----------------------------------------------------------------------------|
| station   | <a href="#">StationEnum32</a>    | Identification of the location (or station) to which the part is attached. |
| storeType | <a href="#">EntityTypeStruct</a> | The entity type of the attached part.                                      |

Named by [ParameterValueVariantStruct](#).

## AttributeChangeRequest STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value.

| Name              | Type                                               | Description                                                                      |
|-------------------|----------------------------------------------------|----------------------------------------------------------------------------------|
| objectIdentifiers | <a href="#">RTObjectIdArray</a>                    | The list of objects that are the intended recipients of this interaction.        |
| attributeValueSet | <a href="#">AttributeValuePairStructArray1Plus</a> | The set of attributes and their values, that the recipients are asked to update. |

## AttributeChangeRequestR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                    | Type                                               | Description                                                                      |
|-------------------------|----------------------------------------------------|----------------------------------------------------------------------------------|
| objectIdentifiers       | <a href="#">RTObjectIdArray</a>                    | The list of objects that are the intended recipients of this interaction.        |
| attributeValueSet       | <a href="#">AttributeValuePairStructArray1Plus</a> | The set of attributes and their values, that the recipients are asked to update. |
| acknowledgementProtocol | <a href="#">AcknowledgementProtocolEnum8</a>       | The acknowledgement protocol to be used for a transaction                        |

## AttributeChangeResult STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not.

| Name                  | Type                                                                   | Description                                                                        |
|-----------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| objectIdentifier      | <a href="#">string</a>                                                 | The ID of the object replying to the AttributeChangeRequest interaction.           |
| attributeChangeResult | <a href="#">ResponseFlagEnum16</a>                                     | Indicates ability to comply.                                                       |
| attributeValueSet     | <a href="#">AttributeValuePair</a><br><a href="#">StructArray1Plus</a> | The set of attributes and their values that the recipient has been able to update. |

## AttributeChangeResultR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                        | Type                                                                   | Description                                                                        |
|-----------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">string</a>                                                 | The ID of the object replying to the AttributeChangeRequest interaction.           |
| attributeChangeResult       | <a href="#">ResponseFlagEnum16</a>                                     | Indicates ability to comply.                                                       |
| attributeValueSet           | <a href="#">AttributeValuePair</a><br><a href="#">StructArray1Plus</a> | The set of attributes and their values that the recipient has been able to update. |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement</a><br><a href="#">ProtocolEnum8</a>       | The acknowledgement protocol to be used for this transaction                       |

## AttributeValuePairStruct STRUCTURE

Pair of an AttributeHandle identifying an attribute and data for that attribute.

| Name                | Type                                | Description                                   |
|---------------------|-------------------------------------|-----------------------------------------------|
| attributeHandle     | <a href="#">UnsignedInteger32</a>   | AttributeHandle identifying attribute.        |
| numberOfBytesAValue | <a href="#">OctetArray</a>          | Value for the specified attribute.            |
| paddingTo32         | <a href="#">OctetPadding32Array</a> | Brings the record length to a 32-bit boundary |

Named by [AttributeValuePairStructArray1Plus](#).

## AudioDataTypeStruct STRUCTURE

Specifies an encoded audio radio signal.

| Name         | Type                                                                     | Description                                                    |
|--------------|--------------------------------------------------------------------------|----------------------------------------------------------------|
| streamTag    | <a href="#">UnsignedInteger64</a>                                        | Identifier for the audio stream. This must be globally unique. |
| encodingType | <a href="#">EncodingTypeEnum32</a>                                       | The type of encoding used for the audio data                   |
| sampleRate   | <a href="#">BitRateBitPerSecond</a><br><a href="#">UnsignedInteger32</a> | The number of samples per second in the audio data.            |
| dataLength   | <a href="#">BitsUnsignedInteger16</a>                                    | The length of the signal data.                                 |
| sampleCount  | <a href="#">UnsignedInteger32</a>                                        | The number of samples in this transmission.                    |
| data         | <a href="#">SignalDataLengthless</a><br><a href="#">Array1Plus</a>       | The signal data.                                               |

Named by [EncodedAudioRadioSignal](#).

## BeamAntennaStruct STRUCTURE

Specifies the direction, pattern, and polarization of radiation from a radio transmitter's antenna.



| Name                       | Type                                      | Description                                                                                                                                                        |
|----------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| beamOrientation            | <a href="#">OrientationStruct</a>         | The rotation that transforms the reference coordinate system into the beam coordinate system.                                                                      |
| beamAzimuthBeamwidth       | <a href="#">AngleRadianFloat32</a>        | The full width of the beam to the -3 dB power density points in the x-y plane of the beam coordinate system.                                                       |
| beamElevation<br>Beamwidth | <a href="#">AngleRadianFloat32</a>        | The full width of the beam to the -3 dB power density points in the x-z plane of the beam coordinate system.                                                       |
| referenceSystem            | <a href="#">ReferenceSystem<br/>Enum8</a> | The reference coordinate system with respect to which beam direction is specified                                                                                  |
| ez                         | <a href="#">Float32</a>                   | The magnitude of the Z-component (in beam coordinates) of the Electrical field at some arbitrary single point in the mainbeam and in the far field of the antenna. |
| ex                         | <a href="#">Float32</a>                   | The magnitude of the X-component (in beam coordinates) of the Electrical field at some arbitrary single point in the mainbeam and in the far field of the antenna  |
| beamPhaseAngle             | <a href="#">Float32</a>                   | The phase angle between Ez and Ex in radians.                                                                                                                      |

Named by [AntennaPatternVariantStruct](#).

## BreachObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachObject class.

| Name                           | Type                                             | Description                                                                                                 |
|--------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier               | <a href="#">EntityIdentifierStruct</a>           | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject<br>Identifier | <a href="#">string</a>                           | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier                | <a href="#">ForceIdentifierEnum8</a>             | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                     | <a href="#">EnvironmentObject<br/>TypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier           | <a href="#">FederateIdentifier<br/>Struct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier            | <a href="#">FederateIdentifier<br/>Struct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |
| segmentRecords                 | <a href="#">BreachStruct<br/>LengthlessArray</a> | Specifies a breach linear object                                                                            |

## BreachStruct STRUCTURE

Record specifying the characteristics of a breach linear object segment

| Name              | Type                                | Description                                                |
|-------------------|-------------------------------------|------------------------------------------------------------|
| segmentParameters | <a href="#">LinearSegmentStruct</a> | Specifies the breach linear object segment characteristics |
| padding           | <a href="#">OctetArray4</a>         | Padding to 64 bits                                         |

Named by [BreachStructLengthlessArray](#).

## BreachableLinearObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachableLinearObject class.

| Name             | Type                                   | Description                                                         |
|------------------|----------------------------------------|---------------------------------------------------------------------|
| objectIdentifier | <a href="#">EntityIdentifierStruct</a> | Identifies this EnvironmentObject instance (point, linear or areal) |



| Name                       | Type                                                   | Description                                                                                                 |
|----------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| referencedObjectIdentifier | <a href="#">string</a>                                 | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>                   | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a>            | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>               | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>               | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |
| segmentRecords             | <a href="#">BreachableSegmentStructLengthlessArray</a> | Specifies a breachable linear object                                                                        |

## BreachablePointObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachablePointObject class.

| Name                       | Type                                        | Description                                                                                                                                                                                                                                                                                                        |
|----------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                                                                                                                                                                                                                                |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated                                                                                                                                                                                                        |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                                                                                                                                                                                                                                      |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                                                                                                                                                                                                                             |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                         |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                       |
| location                   | <a href="#">WorldLocationStruct</a>         | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation                | <a href="#">OrientationStruct</a>           | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete            | <a href="#">PercentUnsignedInteger32</a>    | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance          | <a href="#">DamageStatusEnum32</a>          | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed       | <a href="#">bool</a>                        | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated                | <a href="#">bool</a>                        | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking                    | <a href="#">bool</a>                        | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming                    | <a href="#">bool</a>                        | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |
| breachedStatus             | <a href="#">BreachableStatusEnum8</a>       | Specifies the breached appearance of the object                                                                                                                                                                                                                                                                    |

## BreachableSegmentStruct STRUCTURE

Record specifying the characteristics of a breachable linear object segment

| Name              | Type                                 | Description                                                                                                                                   |
|-------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| segmentParameters | <a href="#">LinearSegmentStruct</a>  | Specifies the breachable linear object segment characteristics                                                                                |
| breachLength      | <a href="#">UnsignedInteger32</a>    | Specifies the breachable linear object segment as 8 portions of length = BreachLength                                                         |
| breachedState     | <a href="#">BreachedStatusEnum8</a>  | Specifies the breached appearance of the breachable linear object segment                                                                     |
| segmentBreached   | <a href="#">BreachedStatusArray8</a> | Specifies whether the segment portion beginning at the segment origin + (i*BreachLength) and extending BreachLength meters is breached or not |
| padding           | <a href="#">OctetArray7</a>          | Padding to 64 bits                                                                                                                            |

Named by [BreachableSegmentStructLengthlessArray](#).

## BurstPointObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BurstPointObject class.

| Name                        | Type                                        | Description                                                                                                                                                                                                                                                                                                        |
|-----------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                                                                                                                                                                                                                                |
| referencedObject Identifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated                                                                                                                                                                                                        |
| forceIdentifier             | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                                                                                                                                                                                                                                      |
| objectType                  | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                                                                                                                                                                                                                             |
| requestingIdentifier        | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                         |
| receivingIdentifier         | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                       |
| location                    | <a href="#">WorldLocationStruct</a>         | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation                 | <a href="#">OrientationStruct</a>           | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete             | <a href="#">PercentUnsignedInteger32</a>    | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance           | <a href="#">DamageStatusEnum32</a>          | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed        | <a href="#">bool</a>                        | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated                 | <a href="#">bool</a>                        | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking                     | <a href="#">bool</a>                        | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming                     | <a href="#">bool</a>                        | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |
| percentOpacity              | <a href="#">PercentUnsignedInteger32</a>    | Specifies the opacity of the smoke                                                                                                                                                                                                                                                                                 |
| cylinderSize                | <a href="#">UnsignedInteger32</a>           | Specifies the radius of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts                                                                                                      |

| Name            | Type                                  | Description                                                                                                                                                                                                   |
|-----------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cylinderHeight  | <a href="#">UnsignedInteger32</a>     | Specifies the height of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts |
| numberOfBursts  | <a href="#">UnsignedInteger32</a>     | Specifies the number of bursts in the instance of tactical smoke                                                                                                                                              |
| chemicalContent | <a href="#">ChemicalContentEnum32</a> | Specifies the chemical content of the smoke                                                                                                                                                                   |

## COMBICStateRecStruct STRUCTURE

Record specifying COMBIC state record

| Name                   | Type                                 | Description              |
|------------------------|--------------------------------------|--------------------------|
| timeSinceCreation      | <a href="#">UnsignedInteger32</a>    | Time since creation      |
| munitionSource         | <a href="#">EntityTypeStruct</a>     | Munition source          |
| numberOfSources        | <a href="#">Integer32</a>            | Number of sources        |
| geometryIndex          | <a href="#">UnsignedInteger16</a>    | Geometry index           |
| sourceType             | <a href="#">UnsignedInteger32</a>    | Source type              |
| barrageRate            | <a href="#">Float32</a>              | Barrage rate             |
| barrageDuration        | <a href="#">Float32</a>              | Barrage duration         |
| barrageCrosswindLength | <a href="#">Float32</a>              | Barrage crosswind length |
| barrageDownwindLength  | <a href="#">Float32</a>              | Barrage downwind length  |
| detonationVelocity     | <a href="#">VelocityVectorStruct</a> | Detonation velocity      |
| padding                | <a href="#">OctetArray4</a>          | Padding field            |

Named by [EnvironmentRecVariantStruct](#).

## ClockTimeStruct STRUCTURE

Specification of the point in time of an occurrence. Based on the Clock Time record as specified in IEEE 1278.1-1995 section 5.2.8.

| Name            | Type                                       | Description                                                |
|-----------------|--------------------------------------------|------------------------------------------------------------|
| hours           | <a href="#">ClockTimeHourInteger32</a>     | The number of hours since 0000 hours, January 1, 1970 UTC. |
| timePastTheHour | <a href="#">TimestampUnsignedInteger32</a> | The time past the hour indicated in the Hours field.       |

Named by [ClockTimeStructLengthlessArray](#), [StartResume](#), [StartResumeR](#), [StopFreeze](#), [StopFreezeR](#).

## Collision STRUCTURE

The act or instance of coming together with solid impact.

| Name                        | Type                                   | Description                                                                                     |
|-----------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------|
| collidingObjectIdentifier   | <a href="#">string</a>                 | The object instance ID of the object that the issuing object has collided with.                 |
| issuingObjectMass           | <a href="#">MassKilogramFloat32</a>    | The mass of the issuing object.                                                                 |
| issuingObjectVelocityVector | <a href="#">VelocityVectorStruct</a>   | The velocity vector of the issuing object at the moment of impact.                              |
| collisionType               | <a href="#">CollisionTypeEnum8</a>     | The type of collision.                                                                          |
| collisionLocation           | <a href="#">RelativePositionStruct</a> | The location of the collision relative to the object that the issuing object has collided with. |

| Name                    | Type                                  | Description                                                                                                |
|-------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------|
| eventIdentifier         | <a href="#">EventIdentifierStruct</a> | An ID assigned by the issuing object to associate related collision events.                                |
| issuingObjectIdentifier | <a href="#">string</a>                | The object instance ID of the object that has detected the collision and issued the collision interaction. |

## CollisionElastic STRUCTURE

The act or instance of coming together with solid impact in an elastic manner. An elastic collision allows a higher fidelity collision to be modeled, taking into account linear and rotational momentum transfer, variable elasticity, and momentum transfer that is dependent on surface orientation.

| Name                        | Type                                         | Description                                                                                                |
|-----------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| collidingObjectIdentifier   | <a href="#">string</a>                       | The object instance ID of the object that the issuing object has collided with.                            |
| issuingObjectMass           | <a href="#">MassKilogramFloat32</a>          | The mass of the issuing object.                                                                            |
| issuingObjectVelocityVector | <a href="#">VelocityVectorStruct</a>         | The velocity vector of the issuing object at the moment of impact.                                         |
| collisionType               | <a href="#">CollisionTypeEnum8</a>           | The type of collision.                                                                                     |
| collisionLocation           | <a href="#">RelativePositionStruct</a>       | The location of the collision relative to the object that the issuing object has collided with.            |
| eventIdentifier             | <a href="#">EventIdentifierStruct</a>        | An ID assigned by the issuing object to associate related collision events.                                |
| issuingObjectIdentifier     | <a href="#">string</a>                       | The object instance ID of the object that has detected the collision and issued the collision interaction. |
| coefficientOfRestitution    | <a href="#">Float32</a>                      | The degree that energy is conserved in a collision.                                                        |
| intermediateResultXX        | <a href="#">Float32</a>                      | X-X Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| intermediateResultXY        | <a href="#">Float32</a>                      | X-Y Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| intermediateResultXZ        | <a href="#">Float32</a>                      | X-Z Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| intermediateResultYY        | <a href="#">Float32</a>                      | Y-Y Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| intermediateResultYZ        | <a href="#">Float32</a>                      | Y-Z Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| intermediateResultZZ        | <a href="#">Float32</a>                      | Z-Z Component of the positive semi-definite Collision Intermediate Result matrix.                          |
| unitSurfaceNormal           | <a href="#">EntityCoordinateVectorStruct</a> | The normal vector to the surface at the point of collision detection.                                      |

## Comment STRUCTURE

A Simulation Management (SIMAN) interaction used to insert messages and information into a log stream and closely matches the structures used by the Data interaction class. This information is usually unsolicited in nature. The information contained within the Interaction should be used for commenting purposes only.

| Name              | Type                                     | Description                                                                                           |
|-------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>   | The DIS entity ID of the entity or application sending the interaction.                               |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>   | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction. |
| variableDatumSet  | <a href="#">VariableDatumStructArray</a> | The set of data items (types and values) associated with the interaction.                             |

## Cone1GeomRecStruct STRUCTURE

Record specifying Cone 1 geometry record

| Name           | Type                                | Description                            |
|----------------|-------------------------------------|----------------------------------------|
| vertexLocation | <a href="#">WorldLocationStruct</a> | Vertex location X, Y, Z                |
| orientation    | <a href="#">OrientationStruct</a>   | Orientation, specified by Euler angles |
| height         | <a href="#">Float32</a>             | Height                                 |
| peakAngle      | <a href="#">Float32</a>             | Peak angle                             |
| padding        | <a href="#">OctetArray4</a>         | Padding field                          |

Named by [EnvironmentRecVariantStruct](#).

## Cone2GeomRecStruct STRUCTURE

Record specifying Cone 2 geometry record

| Name            | Type                                        | Description                            |
|-----------------|---------------------------------------------|----------------------------------------|
| vertexLocation  | <a href="#">WorldLocationStruct</a>         | Vertex location X, Y, Z                |
| orientation     | <a href="#">OrientationStruct</a>           | Orientation, specified by Euler angles |
| velocity        | <a href="#">VelocityVectorStruct</a>        | Velocity Vx, Vy, Vz                    |
| angularVelocity | <a href="#">AngularVelocityVectorStruct</a> | Angular velocity Vx, Vy, Vz            |
| height          | <a href="#">Float32</a>                     | Height                                 |
| heightRate      | <a href="#">Float32</a>                     | Variation of height                    |
| peakAngle       | <a href="#">Float32</a>                     | Peak angle                             |
| peakAngleRate   | <a href="#">Float32</a>                     | Variation of peak angle                |
| padding         | <a href="#">OctetArray4</a>                 | Padding field                          |

Named by [EnvironmentRecVariantStruct](#).

## ConstituentPartRelationshipStruct STRUCTURE

The relationship of the constituent part object to its host object. Based on the Relationship record as specified in IEEE 1278.1a-1998 section 5.2.56.

| Name     | Type                                          | Description                                                                              |
|----------|-----------------------------------------------|------------------------------------------------------------------------------------------|
| nature   | <a href="#">ConstituentPartNatureEnum16</a>   | The nature or purpose for the joining of the constituent part object to the host object. |
| position | <a href="#">ConstituentPartPositionEnum16</a> | The position of the constituent part object with respect to the host object.             |

Named by [IsPartOfStruct](#).

## CraterObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the CraterObject class.

| Name                       | Type                                        | Description                                                                                                 |
|----------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |

| Name                 | Type                                     | Description                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestingIdentifier | <a href="#">FederateIdentifierStruct</a> | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                         |
| receivingIdentifier  | <a href="#">FederateIdentifierStruct</a> | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                       |
| location             | <a href="#">WorldLocationStruct</a>      | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation          | <a href="#">OrientationStruct</a>        | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete      | <a href="#">PercentUnsignedInteger32</a> | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance    | <a href="#">DamageStatusEnum32</a>       | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed | <a href="#">bool</a>                     | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated          | <a href="#">bool</a>                     | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking              | <a href="#">bool</a>                     | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming              | <a href="#">bool</a>                     | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |
| craterSize           | <a href="#">UnsignedInteger32</a>        | Specifies the diameter of the crater, where the center of the crater is at the point object location                                                                                                                                                                                                               |

## CreateEntity STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity. See DIS 4.5.5.4.1 and DIS 5.3.6 for details.

| Name              | Type                                   | Description                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The FederateIdentifier field identifies the federate that is responsible for creating the entity (if possible). The complete parameter defines the entity ID of the new entity (see also note [18]).                                                                         |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |

## CreateEntityR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity, using a specified acknowledgement service. See DIS 4.5.11.3.1 and DIS 5.3.12.1 for details.

| Name              | Type                                   | Description                                                                                                                                                                                          |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                                                                              |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The FederateIdentifier field identifies the federate that is responsible for creating the entity (if possible). The complete parameter defines the entity ID of the new entity (see also note [18]). |

| Name                        | Type                                              | Description                                                                                                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestIdentifier           | <a href="#">UnsignedInteger32</a>                 | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement<br/>ProtocolEnum8</a> | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## CreateObjectRequest STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type.

| Name              | Type                                                    | Description                                                                                                                                                                                                                                                                  |
|-------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectClass       | <a href="#">UnsignedInteger32</a>                       | The class of object to be created.                                                                                                                                                                                                                                           |
| attributeValueSet | <a href="#">AttributeValuePair<br/>StructArray1Plus</a> | The set of attributes, and associated values, to be used to initialize the object.                                                                                                                                                                                           |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                       | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |

## CreateObjectRequestR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                        | Type                                                    | Description                                                                                                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectClass                 | <a href="#">UnsignedInteger32</a>                       | The class of object to be created.                                                                                                                                                                                                                                           |
| attributeValueSet           | <a href="#">AttributeValuePair<br/>StructArray1Plus</a> | The set of attributes, and associated values, to be used to initialize the object.                                                                                                                                                                                           |
| requestIdentifier           | <a href="#">UnsignedInteger32</a>                       | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement<br/>ProtocolEnum8</a>       | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## CreateObjectResult STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not.

| Name               | Type                               | Description                                                                                                 |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| createObjectResult | <a href="#">ResponseFlagEnum16</a> | Indicates ability to comply.                                                                                |
| requestIdentifier  | <a href="#">UnsignedInteger32</a>  | This field matches this response with the specific CreateObject interaction sent by the simulation manager. |

## CreateObjectResultR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not. The Simulation Manager federate specifies the acknowledgement protocol to be used.



| Name               | Type                               | Description                                                                                                 |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| createObjectResult | <a href="#">ResponseFlagEnum16</a> | Indicates ability to comply.                                                                                |
| requestIdentifier  | <a href="#">UnsignedInteger32</a>  | This field matches this response with the specific CreateObject interaction sent by the simulation manager. |

## Data STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQuery interaction (in which case the Data interaction contains the results of the query) or b) a SetData interaction (in which case the Data interaction contains the data that the federate was able to set).

| Name              | Type                                                                | Description                                                                                                         |
|-------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                             |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                  |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                                   | This field matches this response with the specific SetData or DataQuery interaction sent by the simulation manager. |
| fixedDatums       | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | The set of data items (types and values), of fixed length, that the recipient can return for this interaction.      |
| variableDatumSet  | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | The set of data items (types and values), of variable length, that the recipient can return for this interaction.   |

## DataQuery STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data.

| Name                      | Type                                                               | Description                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity         | <a href="#">EntityIdentifierStruct</a>                             | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity           | <a href="#">EntityIdentifierStruct</a>                             | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                                                                                                                                                                           |
| requestIdentifier         | <a href="#">UnsignedInteger32</a>                                  | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| timeInterval              | <a href="#">UnsignedInteger32</a>                                  | The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single Data interaction in response to this interaction.                                                                                            |
| fixedDatumIdentifiers     | <a href="#">DatumIdentifier</a><br><a href="#">LengthlessArray</a> | The set of fixed length data items (types) that the recipient is requested to supply data for.                                                                                                                                                                               |
| variableDatum Identifiers | <a href="#">DatumIdentifier</a><br><a href="#">LengthlessArray</a> | The set of variable length data items (types) that the recipient is requested to supply data for.                                                                                                                                                                            |

## DataQueryR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name              | Type                                   | Description                                                                                        |
|-------------------|----------------------------------------|----------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                            |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient of the interaction. |



| Name                         | Type                                                               | Description                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestIdentifier            | <a href="#">UnsignedInteger32</a>                                  | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| timeInterval                 | <a href="#">UnsignedInteger32</a>                                  | The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single Data interaction in response to this interaction.                                                                                            |
| fixedDatumIdentifiers        | <a href="#">DatumIdentifier</a><br><a href="#">LengthlessArray</a> | The set of fixed length data items (types) that the recipient is requested to supply data for.                                                                                                                                                                               |
| variableDatum<br>Identifiers | <a href="#">DatumIdentifier</a><br><a href="#">LengthlessArray</a> | The set of variable length data items (types) that the recipient is requested to supply data for.                                                                                                                                                                            |
| acknowledgement<br>Protocol  | <a href="#">Acknowledgement</a><br><a href="#">ProtocolEnum8</a>   | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## DataR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQueryR interaction (in which case the DataR interaction contains the results of the query) or b) a SetDataR interaction (in which case the DataR interaction contains the data that the federate was able to set).

| Name                        | Type                                                                | Description                                                                                                         |
|-----------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| originatingEntity           | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                             |
| receivingEntity             | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                  |
| requestIdentifier           | <a href="#">UnsignedInteger32</a>                                   | This field matches this response with the specific SetData or DataQuery interaction sent by the simulation manager. |
| fixedDatums                 | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | The set of data items (types and values), of fixed length, that the recipient can return for this interaction.      |
| variableDatumSet            | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | The set of data items (types and values), of variable length, that the recipient can return for this interaction.   |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement</a><br><a href="#">ProtocolEnum8</a>    | The acknowledgement protocol to be used for a transaction                                                           |

## DatabaseIndexRadioSignal STRUCTURE

A radio signal that represents transmission of a pre-determined message by using a pre-defined database index.

| Name                 | Type                                                                 | Description                                                                        |
|----------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------|
| hostRadioIndex       | <a href="#">string</a>                                               | The object instance ID of the radio transmitting this signal.                      |
| databaseIndex        | <a href="#">UnsignedInteger32</a>                                    | The index into the federation specific database of stored signals.                 |
| duration             | <a href="#">TimeMillisecond</a><br><a href="#">UnsignedInteger32</a> | The duration of the stored signal to be replayed.                                  |
| startOffset          | <a href="#">TimeMillisecond</a><br><a href="#">UnsignedInteger32</a> | The offset, from the start of the stored signal, that the signal is replayed from. |
| tacticalDataLinkType | <a href="#">TacticalDataLinkType</a><br><a href="#">Enum16</a>       | The type of tactical data link used to transmitted this signal (if any).           |
| tDLMessageCount      | <a href="#">UnsignedInteger16</a>                                    | The number of tactical data link messages contained in this signal.                |

## DimensionRateStruct STRUCTURE

Record specifying length X, Y, Z rates

| Name  | Type                                          | Description                |
|-------|-----------------------------------------------|----------------------------|
| xRate | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of X axis length |
| yRate | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of Y axis length |
| zRate | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of Z axis length |

Named by [GaussPuffGeomRecStruct](#), [RectVol2GeomRecStruct](#).

## DimensionStruct STRUCTURE

Bounding box in X,Y,Z axis.

| Name        | Type                         | Description                    |
|-------------|------------------------------|--------------------------------|
| xAxisLength | <a href="#">MeterFloat32</a> | Length in meters along X axis. |
| yAxisLength | <a href="#">MeterFloat32</a> | Length in meters along Y axis. |
| zAxisLength | <a href="#">MeterFloat32</a> | Length in meters along Z axis. |

Named by [AggregateEntity](#), [Ellipsoid1GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [GaussPuffGeomRecStruct](#), [RectVol1GeomRecStruct](#), [RectVol2GeomRecStruct](#), [RectVol3GeomRecStruct](#).

## Ellipsoid1GeomRecStruct STRUCTURE

Record specifying Ellipsoid 1 geometry record

| Name             | Type                                | Description                            |
|------------------|-------------------------------------|----------------------------------------|
| centroidLocation | <a href="#">WorldLocationStruct</a> | Centroid location X, Y, Z              |
| sigmaValue       | <a href="#">DimensionStruct</a>     | Sigma dimensions                       |
| orientation      | <a href="#">OrientationStruct</a>   | Orientation, specified by Euler angles |

Named by [EnvironmentRecVariantStruct](#).

## Ellipsoid2GeomRecStruct STRUCTURE

Record specifying Ellipsoid 2 geometry record

| Name             | Type                                        | Description                            |
|------------------|---------------------------------------------|----------------------------------------|
| centroidLocation | <a href="#">WorldLocationStruct</a>         | Centroid location X, Y, Z              |
| sigmaValue       | <a href="#">DimensionStruct</a>             | Sigma dimensions                       |
| sigmaRate        | <a href="#">VelocityVectorStruct</a>        | Variation of sigma dimensions          |
| orientation      | <a href="#">OrientationStruct</a>           | Orientation, specified by Euler angles |
| velocity         | <a href="#">VelocityVectorStruct</a>        | Velocity Vx, Vy, Vz                    |
| angularVelocity  | <a href="#">AngularVelocityVectorStruct</a> | Angular velocity Vx, Vy, Vz            |
| padding          | <a href="#">OctetArray4</a>                 | Padding field                          |

Named by [EnvironmentRecVariantStruct](#).

## EncodedAudioRadioSignal STRUCTURE

A radio signal used to transmit voice/audio data encoded according to a standard encoding scheme.

| Name      | Type                                | Description                                             |
|-----------|-------------------------------------|---------------------------------------------------------|
| audioData | <a href="#">AudioDataTypeStruct</a> | The data associated with the encoded audio radio signal |

## EntityCoordinateVectorStruct STRUCTURE

Vector in the entity coordinate system. Based on the Entity Coordinate Vector record as specified in IEEE 1278.1-1995 section 5.2.33a.

| Name       | Type                         | Description                |
|------------|------------------------------|----------------------------|
| xComponent | <a href="#">MeterFloat32</a> | Component along the X axis |
| yComponent | <a href="#">MeterFloat32</a> | Component along the Y axis |
| zComponent | <a href="#">MeterFloat32</a> | Component along the Z axis |

Named by [CollisionElastic](#).

## EntityIdentifierStruct STRUCTURE

Unique, exercise-wide identification of the entity, or a symbolic group address referencing multiple entities or a simulation application. Based on the Entity Identifier record as specified in IEEE 1278.1-1995 section 5.2.14.

| Name               | Type                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| federateIdentifier | <a href="#">FederateIdentifierStruct</a> | Simulation application (federate) identifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| entityNumber       | <a href="#">UnsignedInteger16</a>        | Each entity in a given simulation application shall be given an entity identifier number unique to all other entities in that application. This identifier number is valid for the duration of the exercise; however, entity identifier numbers may be reused when all possible numbers have been exhausted. No entity shall have an entity identifier number of NO_ENTITY (0), ALL_ENTITIES (0xFFFF), or RQST_ASSIGN_ID (0xFFFE). The entity identifier number need not be registered or retained for future exercises. An entity identifier number equal to zero with valid site and application identification shall address a simulation application. An entity identifier number equal to ALL_ENTITIES shall mean all entities within the specified site and application. An entity identifier number equal to RQST_ASSIGN_ID allows the receiver of the CreateEntity interaction to define the entity identifier number of the new entity. The new entity will specify its entity identifier number in the Acknowledge interaction. |

Named by [netn.GeoLocationReferenceVariant](#), [netn.GeoReferenceVariant](#), [Acknowledge](#), [AcknowledgeR](#), [ActionRequest](#), [ActionRequestR](#), [ActionResponse](#), [ActionResponseR](#), [ArealObjectTransaction](#), [BaseEntity](#), [BreachObjectTransaction](#), [BreachableLinearObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [Comment](#), [CraterObjectTransaction](#), [CreateEntity](#), [CreateEntityR](#), [Data](#), [DataQuery](#), [DataQueryR](#), [DataR](#), [EmbeddedSystem](#), [EnvironmentObject](#), [EnvironmentObjectTransaction](#), [EnvironmentProcess](#), [EventReport](#), [ExhaustSmokeObjectTransaction](#), [GriddedData](#), [IsPartOfStruct](#), [LinearObjectTransaction](#), [Minefield](#), [MinefieldLaneMarkerObjectTransaction](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherLinearObjectTransaction](#), [OtherPointObjectTransaction](#), [PointObjectTransaction](#), [RecordQueryR](#), [RecordR](#), [RemoveEntity](#), [RemoveEntityR](#), [RibbonBridgeObjectTransaction](#), [SetData](#), [SetDataR](#), [SetRecordR](#), [StartResume](#), [StartResumeR](#), [StopFreeze](#), [StopFreezeR](#), [StructureObjectTransaction](#), [TransferControl](#).

## EntityTypeStruct STRUCTURE

Type of entity. Based on the Entity Type record as specified in IEEE 1278.1-1995 section 5.2.16.

| Name        | Type                              | Description                                                          |
|-------------|-----------------------------------|----------------------------------------------------------------------|
| entityKind  | <a href="#">Octet</a>             | Kind of entity.                                                      |
| domain      | <a href="#">Octet</a>             | Domain in which the entity operates.                                 |
| countryCode | <a href="#">UnsignedInteger16</a> | Country to which the design of the entity is attributed.             |
| category    | <a href="#">Octet</a>             | Main category that describes the entity.                             |
| subcategory | <a href="#">Octet</a>             | Subcategory to which an entity belongs based on the Category field.  |
| specific    | <a href="#">Octet</a>             | Specific information about an entity based on the Subcategory field. |
| extra       | <a href="#">Octet</a>             | Extra information required to describe a particular entity.          |

Named by [netn.AisEquipmentItem](#), [netn.CreateMinefield](#), [netn.CreateObstacle](#), [netn.EquipmentItem](#), [netn.FireAtEntityWM](#), [netn.FireAtLocationWM](#), [netn.Holding](#), [netn.InWeaponRangeReport](#), [netn.Installation](#), [netn.NETNSupplyStruct](#), [netn.ResourceStatusNumberStruct](#), [netn.SensorReport](#), [netn.SpottedEquipment](#), [netn.Unit](#), [AttachedPartsStruct](#), [BaseEntity](#), [COMBICStateRecStruct](#), [EntityTypeStruct](#), [LengthlessArray](#), [FlareStateRecStruct](#), [Minefield](#), [MinefieldData](#), [MinefieldQuery](#), [MunitionDetonation](#), [PhysicalEntity](#), [SilentAggregateStruct](#), [SilentEntityStruct](#), [SupplyStruct](#), [WeaponFire](#).

## EnvironmentObjectTransaction STRUCTURE

A base interaction for modifying instances of point, linear and areal environment object classes.

| Name                       | Type                                        | Description                                                                                                 |
|----------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |

## EnvironmentObjectTypeStruct STRUCTURE

Record specifying the domain, the kind and the specific identification of the environment object

| Name        | Type                  | Description                                                                               |
|-------------|-----------------------|-------------------------------------------------------------------------------------------|
| domain      | <a href="#">Octet</a> | Specifies the domain in which the object exists                                           |
| kind        | <a href="#">Octet</a> | Identifies the kind of object                                                             |
| category    | <a href="#">Octet</a> | Specifies the main category that describes the object                                     |
| subcategory | <a href="#">Octet</a> | Specifies a particular subcategory to which an object belongs based on the Category field |

Named by [netn.SEFacility](#), [ArealObjectTransaction](#), [BreachObjectTransaction](#), [BreachableLinearObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [CraterObjectTransaction](#), [EnvironmentObject](#), [EnvironmentObjectTransaction](#), [ExhaustSmokeObjectTransaction](#), [LinearObjectTransaction](#), [MinefieldLaneMarkerObjectTransaction](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherLinearObjectTransaction](#), [OtherPointObjectTransaction](#), [PointObjectTransaction](#), [RibbonBridgeObjectTransaction](#), [StructureObjectTransaction](#).

## EnvironmentRecStruct STRUCTURE

Record specifying an environment (geometry or state) record

| Name        | Type                                        | Description                                       |
|-------------|---------------------------------------------|---------------------------------------------------|
| index       | <a href="#">UnsignedInteger32</a>           | Identifies the sequentially numbered record index |
| dataVariant | <a href="#">EnvironmentRecVariantStruct</a> | Specifies geometry and state record alternatives  |

Named by [EnvironmentRecStructArray](#).

## EnvironmentTypeStruct STRUCTURE

Record specifying the kind of environment, the domain and any extra information necessary for describing the environmental entity

| Name       | Type                  | Description                   |
|------------|-----------------------|-------------------------------|
| entityKind | <a href="#">Octet</a> | Identifies the kind of entity |

| Name        | Type                              | Description                                                                                             |
|-------------|-----------------------------------|---------------------------------------------------------------------------------------------------------|
| domain      | <a href="#">Octet</a>             | Specifies a single primary domain in which the environmental condition exists                           |
| classVal    | <a href="#">UnsignedInteger16</a> | Identifies the type of environmental entity                                                             |
| category    | <a href="#">Octet</a>             | Specifies the main category that describes the environmental entity                                     |
| subcategory | <a href="#">Octet</a>             | Specifies a particular subcategory to which an environmental entity belongs based on the Category field |
| specific    | <a href="#">Octet</a>             | Identifies specific information about an environmental entity based on the Subcategory field            |
| extra       | <a href="#">Octet</a>             | Specifies extra information required to describe a particular environmental entity                      |

Named by [EnvironmentProcess](#), [GriddedData](#).

## EventIdentifierStruct STRUCTURE

Identification of an event. Based on the Event Identifier record as specified in IEEE 1278.1-1995 section 5.2.18.

| Name                     | Type                              | Description                                                                                                                                                                                                                                                                                                |
|--------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eventCount               | <a href="#">UnsignedInteger16</a> | The event number. Uniquely assigned by the simulation application (federate) that initiates the sequence of events. It shall be set to one for each exercise and incremented by one for each event. In the case where all possible values are exhausted, the numbers may be reused beginning again at one. |
| issuingObject Identifier | <a href="#">string</a>            | Identification of the object issuing the event.                                                                                                                                                                                                                                                            |

Named by [ActiveSonarBeam](#), [Collision](#), [CollisionElastic](#), [EmitterBeam](#), [EmitterSystem](#), [IFF](#), [MunitionDetonation](#), [UnderwaterAcoustics Emission](#), [WeaponFire](#).

## EventReport STRUCTURE

A Simulation Management (SIMAN) interaction designed to allow a federate to alert a Simulation Manager federate that a particular event has occurred.

| Name              | Type                                                                | Description                                                                                           |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                               |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction. |
| eventType         | <a href="#">EventTypeEnum32</a>                                     | The type of event being reported.                                                                     |
| fixedDatums       | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | The set of fixed size data items (types and values) associated with this event.                       |
| variableDatumSet  | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | The set of variable size data items (types and values) associated with this event.                    |

## ExhaustSmokeObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the ExhaustSmokeObject class.

| Name                        | Type                                   | Description                                                                                                 |
|-----------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">EntityIdentifierStruct</a> | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject Identifier | <a href="#">string</a>                 | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |

| Name                 | Type                                              | Description                                                                                  |
|----------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------|
| forceIdentifier      | <a href="#">ForceIdentifierEnum8</a>              | Identifies the force that created or modified this EnvironmentObject instance                |
| objectType           | <a href="#">EnvironmentObjectTypeStruct</a>       | Identifies the type of this EnvironmentObject instance                                       |
| requestingIdentifier | <a href="#">FederateIdentifierStruct</a>          | Identifies the simulation application sending the EnvironmentObjectTransaction interaction   |
| receivingIdentifier  | <a href="#">FederateIdentifierStruct</a>          | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction |
| segmentRecords       | <a href="#">ExhaustSmokeStructLengthlessArray</a> | Specifies an exhaust smoke linear object                                                     |

## ExhaustSmokeStruct STRUCTURE

Record specifying the characteristics of a linear object smoke segment

| Name              | Type                                     | Description                                                                  |
|-------------------|------------------------------------------|------------------------------------------------------------------------------|
| segmentParameters | <a href="#">LinearSegmentStruct</a>      | Specifies the linear object smoke segment characteristics                    |
| percentOpacity    | <a href="#">PercentUnsignedInteger32</a> | Specifies the opacity of the linear object smoke segment                     |
| attached          | <a href="#">bool</a>                     | Specifies whether the linear object smoke segment is attached to the vehicle |
| chemicalContent   | <a href="#">ChemicalContentEnum32</a>    | Specifies the chemical content of the linear object smoke segment            |

Named by [ExhaustSmokeStructLengthlessArray](#).

## FederateIdentifierStruct STRUCTURE

Unique identification of the simulation application (federate) in an exercise, or a symbolic group address referencing multiple simulation applications. Based on the Simulation Address record as specified in IEEE 1278.1-1995 section 5.2.14.1.

| Name          | Type                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| siteID        | <a href="#">UnsignedInteger16</a> | Each site shall be assigned a unique site identification. No individual site shall be assigned an identification number containing NO_SITE (0) or ALL_SITES (0xFFFF). An identification number equal to ALL_SITES (0xFFFF) shall mean all sites; this may be used to initialize or start all sites. The mechanism by which Site Identification numbers are assigned is part of federation agreements.                                                                                                                                                                                  |
| applicationID | <a href="#">UnsignedInteger16</a> | Each simulation application (federate) at a site shall be assigned an identification number unique within that site. No simulation application shall be assigned an identification number containing NO_APPLIC (0) or ALL_APPLIC (0xFFFF). An application identification number equal to ALL_APPLIC (0xFFFF) shall mean all applications; this may be used to start all applications within a site. One or more simulation applications may reside in a single host computer. The mechanism by which application identification numbers are assigned is part of federation agreements. |

Named by [ArealObjectTransaction](#), [BreachObjectTransaction](#), [BreachableLinearObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [CraterObjectTransaction](#), [EntityIdentifierStruct](#), [EnvironmentObjectTransaction](#), [ExhaustSmokeObjectTransaction](#), [LinearObjectTransaction](#), [MinefieldLaneMarkerObjectTransaction](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherLinearObjectTransaction](#), [OtherPointObjectTransaction](#), [PointObjectTransaction](#), [RibbonBridgeObjectTransaction](#), [StructureObjectTransaction](#).

## FixedDatumStruct STRUCTURE

Identifier and value for a fixed datum.

| Name                 | Type                                  | Description                          |
|----------------------|---------------------------------------|--------------------------------------|
| fixedDatumIdentifier | <a href="#">DatumIdentifierEnum32</a> | The identifier for this fixed datum. |
| fixedDatumValue      | <a href="#">UnsignedInteger32</a>     | The value for this fixed datum.      |

Named by [FixedDatumStructLengthlessArray](#).

## FlareStateRecStruct STRUCTURE

Record specifying Flare state record

| Name              | Type                              | Description         |
|-------------------|-----------------------------------|---------------------|
| timeSinceCreation | <a href="#">UnsignedInteger32</a> | Time since creation |
| source            | <a href="#">EntityTypeStruct</a>  | Source              |
| numberIntensity   | <a href="#">Integer32</a>         | Number intensity    |
| numberOfSources   | <a href="#">Integer32</a>         | Number of sources   |
| geometryIndex     | <a href="#">UnsignedInteger16</a> | Geometry index      |
| padding           | <a href="#">OctetArray2</a>       | Padding field       |

Named by [EnvironmentRecVariantStruct](#).

## FundamentalParameterDataStruct STRUCTURE

Fundamental Parameter Data record.

| Name            | Type                                              | Description                                                                                                                                                                                    |
|-----------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| erp             | <a href="#">PowerRatioDecibelMilliwattFloat32</a> | The average peak radiated power of the emission.                                                                                                                                               |
| frequency       | <a href="#">FrequencyHertzFloat32</a>             | The center frequency of the emission.                                                                                                                                                          |
| pgRF            | <a href="#">InterrogationsPerSecondFloat32</a>    | When applied to originators, this field shall specify the number of interrogations per second emitted. This field shall be set to zero when applied to a responder (i.e. transponder) systems. |
| pulseWidth      | <a href="#">TimeMicrosecondFloat32</a>            | The duration of the fundamental pulse of which the interrogation or reply is composed.                                                                                                         |
| burstLength     | <a href="#">Integer32</a>                         | The number of emissions generated in a single burst. This field shall contain zero for continuously emitting systems and shall contain the value 1 for responders.                             |
| applicableModes | <a href="#">IffApplicableModesEnum8</a>           | Specifies the modes to which the fundamental parameter data applies.                                                                                                                           |
| padding         | <a href="#">OctetArray3</a>                       | Padding to 32 bits                                                                                                                                                                             |

Named by [FundamentalParameterDataStructLengthlessArray](#).

## GaussPlumeGeomRecStruct STRUCTURE

Record specifying Gaussian Plume geometry record

| Name                | Type                                     | Description                            |
|---------------------|------------------------------------------|----------------------------------------|
| sourceLocation      | <a href="#">WorldLocationStruct</a>      | Source location X, Y, Z                |
| orientation         | <a href="#">OrientationStruct</a>        | Orientation, specified by Euler angles |
| plumeDimension      | <a href="#">PlumeDimensionStruct</a>     | Plume dimensions                       |
| plumeDimensionRate  | <a href="#">PlumeDimensionRateStruct</a> | Variation of plume dimensions          |
| leadingEdge         | <a href="#">Float32</a>                  | Leading edge                           |
| leadingEdgeVelocity | <a href="#">VelocityVectorStruct</a>     | Leading edge velocity                  |
| padding             | <a href="#">OctetArray4</a>              | Padding field                          |

Named by [EnvironmentRecVariantStruct](#).

## GaussPuffGeomRecStruct STRUCTURE

Record specifying Gaussian Puff geometry record

| Name                | Type                                        | Description                            |
|---------------------|---------------------------------------------|----------------------------------------|
| puffLocation        | <a href="#">WorldLocationStruct</a>         | Puff location X, Y, Z                  |
| originationLocation | <a href="#">WorldLocationStruct</a>         | Origination location X, Y, Z           |
| sigmaValue          | <a href="#">DimensionStruct</a>             | Sigma dimensions                       |
| sigmaRate           | <a href="#">DimensionRateStruct</a>         | Variation of sigma dimensions          |
| orientation         | <a href="#">OrientationStruct</a>           | Orientation, specified by Euler angles |
| velocity            | <a href="#">VelocityVectorStruct</a>        | Velocity Vx, Vy, Vz                    |
| angularVelocity     | <a href="#">AngularVelocityVectorStruct</a> | Angular velocity Vx, Vy, Vz            |
| centroidHeight      | <a href="#">Float32</a>                     | Centroid height                        |

Named by [EnvironmentRecVariantStruct](#).

## GridAxisStruct STRUCTURE

Record specifying grid Xi axis data

| Name                  | Type                                      | Description                                                                                                                                                                                                                           |
|-----------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| initialValue          | <a href="#">Float64</a>                   | Specifies the coordinate of the origin (or initial value) for the Xi axis                                                                                                                                                             |
| finalValue            | <a href="#">Float64</a>                   | Specifies the coordinate of the endpoint (or final value) for the Xi axis                                                                                                                                                             |
| totalNumberOfPoints   | <a href="#">UnsignedInteger16</a>         | Specifies the total number of grid points along the Xi domain axis                                                                                                                                                                    |
| interleafFactor       | <a href="#">Octet</a>                     | Specifies the integer valued interleaf factor along a domain (grid) Xi axis ; a value of one shall indicate no sub-sampling (interleaving), while integer values greater than one shall indicate the sampling frequency along an axis |
| numberOfPoints        | <a href="#">UnsignedInteger16</a>         | Specifies the number of grid locations along the Xi axis                                                                                                                                                                              |
| initialIndex          | <a href="#">UnsignedInteger16</a>         | Specifies the index of the initial grid point along the Xi domain axis                                                                                                                                                                |
| axisTypeAAlternatives | <a href="#">GridAxisTypeVariantStruct</a> | Specifies axis data alternatives                                                                                                                                                                                                      |

Named by [GridAxisStructLengthlessArray](#).

## GridDataStruct STRUCTURE

Record specifying grid data representation

| Name                            | Type                                                | Description                                     |
|---------------------------------|-----------------------------------------------------|-------------------------------------------------|
| sampleType                      | <a href="#">EnvironmentDataSampleTypeEnum16</a>     | Specifies the environmental data sample         |
| dataRepresentationAAlternatives | <a href="#">GridDataRepresentationVariantStruct</a> | Specifies grid data representation alternatives |

Named by [GridDataStructLengthlessArray](#).

## GridValueType0Struct STRUCTURE

Record specifying type 0 data representation



| Name                 | Type                                | Description                                   |
|----------------------|-------------------------------------|-----------------------------------------------|
| numberOfBytesAValues | <a href="#">OctetArray1Plus</a>     | Specifies the number of bytes                 |
| paddingTo32          | <a href="#">OctetPadding32Array</a> | Brings the record length to a 32-bit boundary |

Named by [GridDataRepresentationVariantStruct](#).

## GridValueType1Struct STRUCTURE

Record specifying type 1 data representation

| Name                  | Type                                | Description                                                                                    |
|-----------------------|-------------------------------------|------------------------------------------------------------------------------------------------|
| scale                 | <a href="#">Float32</a>             | Specifies the constant scale factor used to scale the environmental state variable data values |
| offset                | <a href="#">Float32</a>             | Specifies the constant offset used to scale the environmental state variable data values       |
| numberOfValuesAValues | <a href="#">Integer16Array1Plus</a> | Specifies the number of environmental state variable data values                               |
| paddingTo32           | <a href="#">OctetPadding32Array</a> | Brings the record length to a 32-bit boundary                                                  |

Named by [GridDataRepresentationVariantStruct](#).

## GridValueType2Struct STRUCTURE

Record specifying type 2 data representation

| Name                  | Type                              | Description                                                      |
|-----------------------|-----------------------------------|------------------------------------------------------------------|
| numberOfValuesAValues | <a href="#">Float32Array1Plus</a> | Specifies the number of environmental state variable data values |

Named by [GridDataRepresentationVariantStruct](#).

## IrregularGridAxisStruct STRUCTURE

Record specifying irregular (variable spacing) axis data

| Name                                    | Type                                             | Description                                                                                              |
|-----------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| coordinateScale                         | <a href="#">Float64</a>                          | Specifies the value that linearly scales the coordinates of the grid locations for the Xi axis           |
| coordinateOffset                        | <a href="#">Float64</a>                          | Specifies the constant offset value that shall be applied to the grid locations for the Xi axis          |
| numberOfGridLocations<br>AGridLocations | <a href="#">Unsigned<br/>Integer16Array1Plus</a> | Specifies the coordinate values for the Ni grid locations along the irregular (variable spacing) Xi axis |
| paddingTo64                             | <a href="#">OctetPadding64Array</a>              | Brings the record length to a 64-bit boundary                                                            |

Named by [GridAxisTypeVariantStruct](#).

## IsPartOfStruct STRUCTURE

Defines the spatial relationship between two objects.

| Name                        | Type                                                   | Description                                                                                          |
|-----------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| hostEntityIdentifier        | <a href="#">EntityIdentifierStruct</a>                 | The identifier of the entity of which the object is a constituent part.                              |
| hostRTIObject<br>Identifier | <a href="#">string</a>                                 | The RTI instance identifier of the object of which this object is a constituent part.                |
| relationship                | <a href="#">ConstituentPart<br/>RelationshipStruct</a> | The relationship of the constituent part object to its host object.                                  |
| namedLocation               | <a href="#">NamedLocationStruct</a>                    | The discrete positional relationship of the constituent part object with respect to its host object. |

Named by [BaseEntity](#).

## JTIDSTransmitterStruct STRUCTURE

Contains JTIDS specific information about the JTIDS transmitter system.

| Name                              | Type                                           | Description                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timeSlotAllocationLevel           | <a href="#">TimeSlotAllocationLevelEnum8</a>   | TSA level of fidelity.                                                                                                                                                                                                                                                                                          |
| transmittingTerminalPrimaryMode   | <a href="#">JTIDSPrimaryModeEnum8</a>          | The primary mode of a JTIDS system.                                                                                                                                                                                                                                                                             |
| transmittingTerminalSecondaryMode | <a href="#">JTIDSSecondaryModeEnum8</a>        | The JTIDS secondary mode of operation.                                                                                                                                                                                                                                                                          |
| synchronizationState              | <a href="#">JTIDSSynchronizationStateEnum8</a> | The state of synchronization that the JTIDS system has achieved.                                                                                                                                                                                                                                                |
| networkSynchronizationID          | <a href="#">UnsignedInteger32</a>              | TSA Levels 0-2: may be set to 0 (wildcard) or the unique 32-bit value of a simulated net. TSA Levels 3 and 4: set to the unique 32-bit value of a simulated net. Only an NTR can generate a Network Synchronization ID; all other participants use the ID obtained from the NTR to which they are synchronized. |

Named by [SpreadSpectrumVariantStruct](#).

## Line1GeomRecStruct STRUCTURE

Record specifying Line 1 geometry record

| Name               | Type                                | Description          |
|--------------------|-------------------------------------|----------------------|
| startPointLocation | <a href="#">WorldLocationStruct</a> | Start point location |
| endPointLocation   | <a href="#">WorldLocationStruct</a> | End point location   |

Named by [EnvironmentRecVariantStruct](#).

## Line2GeomRecStruct STRUCTURE

Record specifying Line 2 geometry record

| Name               | Type                                 | Description          |
|--------------------|--------------------------------------|----------------------|
| startPointLocation | <a href="#">WorldLocationStruct</a>  | Start point location |
| endPointLocation   | <a href="#">WorldLocationStruct</a>  | End point location   |
| startPointVelocity | <a href="#">VelocityVectorStruct</a> | Start point velocity |
| endPointVelocity   | <a href="#">VelocityVectorStruct</a> | End point velocity   |

Named by [EnvironmentRecVariantStruct](#).

## LinearObjectTransaction STRUCTURE

An interaction for modifying instances of the LinearObject class.

| Name                       | Type                                        | Description                                                                                                 |
|----------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |

| Name                 | Type                                      | Description                                                                                  |
|----------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|
| requestingIdentifier | <a href="#">FederateIdentifier Struct</a> | Identifies the simulation application sending the EnvironmentObjectTransaction interaction   |
| receivingIdentifier  | <a href="#">FederateIdentifier Struct</a> | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction |

## LinearSegmentStruct STRUCTURE

Specifies linear object segment characteristics.

| Name                 | Type                                      | Description                                                                                                                                                                              |
|----------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| segmentNumber        | <a href="#">UnsignedInteger32</a>         | Identifies the individual segment.                                                                                                                                                       |
| percentComplete      | <a href="#">PercentUnsigned Integer32</a> | Specifies the percent completion of the segment.                                                                                                                                         |
| location             | <a href="#">WorldLocationStruct</a>       | Specifies the location of the segment.                                                                                                                                                   |
| orientation          | <a href="#">OrientationStruct</a>         | Specifies the orientation of the segment.                                                                                                                                                |
| length               | <a href="#">UnsignedInteger16</a>         | Specifies the length of the segment, in meters, extending in the positive X direction.                                                                                                   |
| width                | <a href="#">UnsignedInteger16</a>         | Specifies the total width of the segment, in meters; one-half of the width shall extend in the positive Y direction, and one-half of the width shall extend in the negative Y direction. |
| height               | <a href="#">UnsignedInteger16</a>         | Specifies the height of the segment, in meters, above ground.                                                                                                                            |
| depth                | <a href="#">UnsignedInteger16</a>         | Specifies the depth of the segment, in meters, below ground level.                                                                                                                       |
| damagedState         | <a href="#">DamageStatus Enum32</a>       | Specifies the damaged appearance of the segment.                                                                                                                                         |
| deactivated          | <a href="#">bool</a>                      | Specifies whether or not the segment has been deactivated (it has ceased to exist in the synthetic environment).                                                                         |
| flaming              | <a href="#">bool</a>                      | Specifies whether or not the segment is aflame.                                                                                                                                          |
| objectPreDistributed | <a href="#">bool</a>                      | Specifies whether or not the segment was created before the start of the exercise.                                                                                                       |
| smoking              | <a href="#">bool</a>                      | Specifies whether or not the segment is smoking (creating a smoke plume).                                                                                                                |

Named by [BreachStruct](#), [BreachableSegmentStruct](#), [ExhaustSmokeStruct](#), [MinefieldLaneMarkerStruct](#).

## Link11BTransmitterStruct STRUCTURE

Contains specific information about the Link 11B transmitter system.

| Name            | Type                                           | Description                                      |
|-----------------|------------------------------------------------|--------------------------------------------------|
| reportingUnit   | <a href="#">ParticipatingUnit SourceNumber</a> | Reporting Unit (RU) number.                      |
| fidelityLevel   | <a href="#">Link1111BFidelityLevel Enum8</a>   | Fidelity level of the Link 11B simulation.       |
| linkState       | <a href="#">Link11BLinkState Enum8</a>         | The state the Link 11B terminal is currently in. |
| modeOf0peration | <a href="#">Link11BModeOf OperationEnum16</a>  | Link 11B terminal operation mode.                |

Named by [SpreadSpectrumVariantStruct](#).

## Link11TransmitterStruct STRUCTURE

Contains specific information about the Link 11 transmitter system.

| Name              | Type                                          | Description                                                                                                        |
|-------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| participatingUnit | <a href="#">ParticipatingUnitSourceNumber</a> | Participating Unit (PU) number.                                                                                    |
| fidelityLevel     | <a href="#">Link111BFidelityLevelEnum8</a>    | Fidelity Level of the Link 11 simulation.                                                                          |
| terminalMode      | <a href="#">Link11TerminalModeEnum8</a>       | Mode of the Link 11 terminal.                                                                                      |
| modeOfOperation   | <a href="#">Link11ModeOfOperationEnum16</a>   | Link 11 terminal operation mode.                                                                                   |
| netCycleTime      | <a href="#">TimeSecondUnsignedInteger16</a>   | The time in seconds required for the NCS to complete a polling sequence of all PUs. Not applicable to non-NCS PUs. |

Named by [SpreadSpectrumVariantStruct](#).

## MarkingStruct STRUCTURE

Character set used in the marking and the string of characters to be interpreted for display.

| Name                | Type                                 | Description                   |
|---------------------|--------------------------------------|-------------------------------|
| markingEncodingType | <a href="#">MarkingEncodingEnum8</a> | Character set representation. |
| markingData         | <a href="#">MarkingArray11</a>       | 11 element character string   |

Named by [PhysicalEntity](#).

## MineFusingStruct STRUCTURE

Record specifying the type of primary fuse, the type of the secondary fuse and the anti-handling device status of a mine

| Name               | Type                                  | Description                                         |
|--------------------|---------------------------------------|-----------------------------------------------------|
| primary            | <a href="#">MinefieldFusingEnum32</a> | Specifies the type of the primary fuse of a mine    |
| secondary          | <a href="#">MinefieldFusingEnum32</a> | Specifies the type of the secondary fuse of a mine  |
| antiHandlingDevice | <a href="#">bool</a>                  | Specifies the anti-handling device status of a mine |
| padding            | <a href="#">OctetArray3</a>           | Padding to 32 bits                                  |

Named by [MineFusingStructLengthlessArray](#).

## MinefieldLaneMarkerObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldLaneMarkerObject class.

| Name                       | Type                                                     | Description                                                                                                 |
|----------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>                   | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObjectIdentifier | <a href="#">string</a>                                   | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>                     | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a>              | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>                 | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>                 | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |
| segmentRecords             | <a href="#">MinefieldLaneMarkerStructLengthlessArray</a> | Specifies a minefield lane marker linear object                                                             |

## MinefieldLaneMarkerStruct STRUCTURE

Record specifying a minefield lane marker segment

| Name                | Type                                      | Description                                                        |
|---------------------|-------------------------------------------|--------------------------------------------------------------------|
| segmentParameters   | <a href="#">LinearSegmentStruct</a>       | Specifies the minefield lane marker segment characteristics        |
| visibleSideLocation | <a href="#">VisibleSideLocationEnum32</a> | Specifies the visible side(s) of the minefield lane marker segment |

Named by [MinefieldLaneMarkerStructLengthlessArray](#).

## MinefieldObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldObject class.

| Name                       | Type                                               | Description                                                                                                    |
|----------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>             | Identifies this EnvironmentObject instance (point, linear or areal)                                            |
| referencedObjectIdentifier | <a href="#">string</a>                             | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated    |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>               | Identifies the force that created or modified this EnvironmentObject instance                                  |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a>        | Identifies the type of this EnvironmentObject instance                                                         |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>           | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                     |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>           | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                   |
| pointsData                 | <a href="#">WorldLocationStructLengthlessArray</a> | Specifies the physical location (a collection of points) that defines the object                               |
| percentComplete            | <a href="#">PercentUnsignedInteger32</a>           | Specifies the percent completion of the object                                                                 |
| damagedAppearance          | <a href="#">DamageStatusEnum32</a>                 | Specifies the damaged appearance of the object                                                                 |
| objectPreDistributed       | <a href="#">bool</a>                               | Specifies whether or not the object was created before the start of the exercise                               |
| deactivated                | <a href="#">bool</a>                               | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment) |
| smoking                    | <a href="#">bool</a>                               | Specifies whether or not the object is smoking (creating a smoke plume)                                        |
| flaming                    | <a href="#">bool</a>                               | Specifies whether or not the object is flaming                                                                 |
| breachedStatus             | <a href="#">BreachedStatusEnum8</a>                | Specifies the breached appearance of the minefield object                                                      |
| mineCount                  | <a href="#">UnsignedInteger32</a>                  | Specifies the number of mines in the minefield                                                                 |

## MinefieldQuery STRUCTURE

Provides the means by which a federate shall query a minefield simulation for information on the individual mines within the minefield operating in QRP mode.

| Name                 | Type                                                | Description                                                                                             |
|----------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| minefieldIdentifier  | <a href="#">string</a>                              | Identifies the minefield to which this query is addressed                                               |
| perimeterPoints      | <a href="#">PerimeterPointStructLengthlessArray</a> | Specifies the location of each perimeter point in the requested area relative to the minefield location |
| queryFusing          | <a href="#">bool</a>                                | Specifies whether or not fusing is requested                                                            |
| queryMineOrientation | <a href="#">bool</a>                                | Specifies whether or not orientation is requested                                                       |

| Name                            | Type                               | Description                                                                       |
|---------------------------------|------------------------------------|-----------------------------------------------------------------------------------|
| queryGroundBurialDepthOffset    | bool                               | Specifies whether or not ground burial depth offset is requested                  |
| queryMineEmplacementAge         | bool                               | Specifies whether or not emplacement age is requested                             |
| queryPaintScheme                | bool                               | Specifies whether or not paint scheme is requested                                |
| queryReflectance                | bool                               | Specifies whether or not reflectance is requested                                 |
| queryScalarDetectionCoefficient | bool                               | Specifies whether or not scalar detection coefficient is requested                |
| querySnowBurialDepthOffset      | bool                               | Specifies whether or not snow burial depth offset is requested                    |
| queryThermalContrast            | bool                               | Specifies whether or not thermal contrast is requested                            |
| queryTripDetonationWire         | bool                               | Specifies whether or not trip detonation wire is requested                        |
| queryWaterBurialDepthOffset     | bool                               | Specifies whether or not water burial depth offset is requested                   |
| requestingEntityIdentifier      | string                             | Identifies the entity that requests the information from the minefield simulation |
| requestIdentifier               | UnsignedInteger8                   | Identifies the minefield query request                                            |
| requestedMineType               | EntityTypeStruct                   | Identifies the type of mine being queried by the requesting federate              |
| sensorTypes                     | MinefieldSensorTypeLengthlessArray | Specifies the types of sensors requesting the data                                |

## MinefieldResponseNACK STRUCTURE

A response to a MinefieldQuery providing information on individual mines within a minefield.

| Name                       | Type                                     | Description                                                                            |
|----------------------------|------------------------------------------|----------------------------------------------------------------------------------------|
| minefieldIdentifier        | string                                   | Identifies the minefield to which this query is addressed                              |
| missingRecordNumbers       | MissingRecordNumbersLengthlessArray1Plus | Specifies the record numbers that were not received in a sequence of minefield records |
| requestIdentifier          | UnsignedInteger8                         | Identifies the minefield query request                                                 |
| requestingEntityIdentifier | string                                   | Identifies the entity that requested the information from the minefield simulation     |

## MunitionDetonation STRUCTURE

Communicates information associated with the impact or detonation of a munition

| Name                   | Type                                      | Description                                                                                                                    |
|------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| articulatedPartData    | ArticulatedParameterStructLengthlessArray | The set of articulated parts affected by the detonation (including where on the articulated part the detonation has affected). |
| detonationLocation     | WorldLocationStruct                       | The location, in world coordinates, at which the munition detonated.                                                           |
| detonationResultCode   | DetonationResultCodeEnum8                 | The type of detonation (including no detonation).                                                                              |
| eventIdentifier        | EventIdentifierStruct                     | An ID, generated by the issuing federate, used to associate related fire and detonation events.                                |
| firingObjectIdentifier | string                                    | The object instance ID of the entity that fired the munition.                                                                  |
| finalVelocityVector    | VelocityVectorStruct                      | The velocity vector of the munition at the moment of the detonation.                                                           |
| fuseType               | FuseTypeEnum16                            | The type of fuse on the munition.                                                                                              |

| Name                       | Type                                   | Description                                                                           |
|----------------------------|----------------------------------------|---------------------------------------------------------------------------------------|
| munitionObjectIdentifier   | <a href="#">string</a>                 | The Object instance ID of the fired munition, if an object instance is registered.    |
| munitionType               | <a href="#">EntityTypeStruct</a>       | The type of munition that is detonating.                                              |
| quantityFired              | <a href="#">UnsignedInteger16</a>      | The number of rounds fired in the burst.                                              |
| rateOfFire                 | <a href="#">UnsignedInteger16</a>      | The rate of fire in rounds per minute when quantity > 1. Zero otherwise.              |
| relativeDetonationLocation | <a href="#">RelativePositionStruct</a> | The location of the detonation in coordinates relative to the target entity (if any). |
| targetObjectIdentifier     | <a href="#">string</a>                 | The object instance ID of the entity that the munition has detonated on (if any).     |
| warheadType                | <a href="#">WarheadTypeEnum16</a>      | The type of warhead on the munition.                                                  |

## NamedLocationStruct STRUCTURE

The discrete positional relationship of the constituent part object with respect to its host object. Based on the specifications in IEEE 1278.1a-1998 of the IsPartOf PDU 'Location of Part' (paragraph 5.3.9.4e) and 'Named Location' (paragraph 5.3.9.4f) fields.

| Name          | Type                                             | Description                                                                     |
|---------------|--------------------------------------------------|---------------------------------------------------------------------------------|
| stationNumber | <a href="#">Integer16</a>                        | The number of the particular station at which the constituent part is attached. |
| stationName   | <a href="#">StationNameLocationVariantStruct</a> | The name of the station where the constituent part is located.                  |

Named by [IsPartOfStruct](#).

## OrientationStruct STRUCTURE

The orientation of an object in the world coordinate system, as specified in IEEE Std 1278.1-1995 section 1.3.2.

| Name  | Type                               | Description                |
|-------|------------------------------------|----------------------------|
| psi   | <a href="#">AngleRadianFloat32</a> | Rotation about the Z axis. |
| theta | <a href="#">AngleRadianFloat32</a> | Rotation about the Y axis. |
| phi   | <a href="#">AngleRadianFloat32</a> | Rotation about the X axis. |

Named by [netn.SpottedEntity](#), [BeamAntennaStruct](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [Cone1GeomRecStruct](#), [Cone2GeomRecStruct](#), [CraterObjectTransaction](#), [Ellipsoid1GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [GaussPlumeGeomRecStruct](#), [GaussPuffGeomRecStruct](#), [GriddedData](#), [LinearSegmentStruct](#), [Minefield](#), [OrientationStructLengthlessArray](#), [OtherPointObjectTransaction](#), [PointObject](#), [PointObjectTransaction](#), [RectVol1GeomRecStruct](#), [RectVol2GeomRecStruct](#), [RectVol3GeomRecStruct](#), [RibbonBridgeObjectTransaction](#), [SpatialFPStruct](#), [SpatialFVStruct](#), [SpatialRPStruct](#), [SpatialRVStruct](#), [SpatialStaticStruct](#), [StructureObjectTransaction](#).

## OtherArealObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherArealObject class.

| Name                       | Type                                        | Description                                                                                                 |
|----------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |



| Name                 | Type                                                | Description                                                                                                    |
|----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| receivingIdentifier  | <a href="#">FederateIdentifier Struct</a>           | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                   |
| pointsData           | <a href="#">WorldLocationStruct LengthlessArray</a> | Specifies the physical location (a collection of points) that defines the object                               |
| percentComplete      | <a href="#">PercentUnsigned Integer32</a>           | Specifies the percent completion of the object                                                                 |
| damagedAppearance    | <a href="#">DamageStatus Enum32</a>                 | Specifies the damaged appearance of the object                                                                 |
| objectPreDistributed | <a href="#">bool</a>                                | Specifies whether or not the object was created before the start of the exercise                               |
| deactivated          | <a href="#">bool</a>                                | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment) |
| smoking              | <a href="#">bool</a>                                | Specifies whether or not the object is smoking (creating a smoke plume)                                        |
| flaming              | <a href="#">bool</a>                                | Specifies whether or not the object is flaming                                                                 |

## OtherLinearObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherLinearObject class.

| Name                        | Type                                         | Description                                                                                                 |
|-----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">EntityIdentifierStruct</a>       | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject Identifier | <a href="#">string</a>                       | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier             | <a href="#">ForceIdentifierEnum8</a>         | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                  | <a href="#">EnvironmentObject TypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier        | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier         | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |

## OtherPointObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherPointObject class.

| Name                        | Type                                         | Description                                                                                                 |
|-----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">EntityIdentifierStruct</a>       | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject Identifier | <a href="#">string</a>                       | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier             | <a href="#">ForceIdentifierEnum8</a>         | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                  | <a href="#">EnvironmentObject TypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier        | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier         | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |
| location                    | <a href="#">WorldLocationStruct</a>          | Specifies the location of the object based on x, y and z coordinates                                        |



| Name                 | Type                                     | Description                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orientation          | <a href="#">OrientationStruct</a>        | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete      | <a href="#">PercentUnsignedInteger32</a> | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance    | <a href="#">DamageStatusEnum32</a>       | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed | <a href="#">bool</a>                     | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated          | <a href="#">bool</a>                     | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking              | <a href="#">bool</a>                     | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming              | <a href="#">bool</a>                     | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |

## PerimeterPointStruct STRUCTURE

Record specifying the location of a perimeter point

| Name | Type                         | Description                |
|------|------------------------------|----------------------------|
| x    | <a href="#">MeterFloat32</a> | Specifies the X coordinate |
| y    | <a href="#">MeterFloat32</a> | Specifies the Y coordinate |

Named by [PerimeterPointStructLengthlessArray](#).

## PlumeDimensionRateStruct STRUCTURE

Record specifying plume dimension rates

| Name       | Type                                          | Description               |
|------------|-----------------------------------------------|---------------------------|
| lengthRate | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of plume length |
| widthRate  | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of plume width  |
| heightRate | <a href="#">VelocityMeterPerSecondFloat32</a> | Variation of plume height |

Named by [GaussPlumeGeomRecStruct](#).

## PlumeDimensionStruct STRUCTURE

Record specifying plume dimensions

| Name   | Type                         | Description  |
|--------|------------------------------|--------------|
| length | <a href="#">MeterFloat32</a> | Plume length |
| width  | <a href="#">MeterFloat32</a> | Plume width  |
| height | <a href="#">MeterFloat32</a> | Plume height |

Named by [GaussPlumeGeomRecStruct](#).

## Point2GeomRecStruct STRUCTURE

Record specifying Point 2 geometry record

| Name     | Type                                 | Description         |
|----------|--------------------------------------|---------------------|
| location | <a href="#">WorldLocationStruct</a>  | Location X, Y, Z    |
| velocity | <a href="#">VelocityVectorStruct</a> | Velocity Vx, Vy, Vz |
| padding  | <a href="#">OctetArray4</a>          | Padding field       |

Named by [EnvironmentRecVariantStruct](#).

## PointObjectTransaction STRUCTURE

An interaction for modifying instances of the PointObject class.

| Name                       | Type                                        | Description                                                                                                                                                                                                                                                                                                        |
|----------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifier           | <a href="#">EntityIdentifierStruct</a>      | Identifies this EnvironmentObject instance (point, linear or areal)                                                                                                                                                                                                                                                |
| referencedObjectIdentifier | <a href="#">string</a>                      | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated                                                                                                                                                                                                        |
| forceIdentifier            | <a href="#">ForceIdentifierEnum8</a>        | Identifies the force that created or modified this EnvironmentObject instance                                                                                                                                                                                                                                      |
| objectType                 | <a href="#">EnvironmentObjectTypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                                                                                                                                                                                                                             |
| requestingIdentifier       | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                         |
| receivingIdentifier        | <a href="#">FederateIdentifierStruct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                       |
| location                   | <a href="#">WorldLocationStruct</a>         | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation                | <a href="#">OrientationStruct</a>           | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete            | <a href="#">PercentUnsignedInteger32</a>    | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance          | <a href="#">DamageStatusEnum32</a>          | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed       | <a href="#">bool</a>                        | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated                | <a href="#">bool</a>                        | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking                    | <a href="#">bool</a>                        | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming                    | <a href="#">bool</a>                        | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |

## PropulsionSystemDataStruct STRUCTURE

Information describing a propulsion system in terms of power settings and current RPM.

| Name         | Type                                        | Description                                                                                 |
|--------------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| powerSetting | <a href="#">Float32</a>                     | The power setting of the propulsion system, after any response lags have been incorporated. |
| engineRPM    | <a href="#">RevolutionsPerMinuteFloat32</a> | The current engine speed.                                                                   |

Named by [PropulsionSystemDataStructLengthlessArray](#).

## RadioSignal STRUCTURE

The wireless transmission and reception of audio or digital data by means of electromagnetic waves.

## RadioTypeStruct STRUCTURE

Specifies the type of a radio.

| Name                | Type                                     | Description                                                                                |
|---------------------|------------------------------------------|--------------------------------------------------------------------------------------------|
| entityKind          | <a href="#">Octet</a>                    | The kind of the entity described in this struct.                                           |
| domain              | <a href="#">Octet</a>                    | The domain in which the radio entity operates.                                             |
| countryCode         | <a href="#">UnsignedInteger16</a>        | The country to which the design of the radio entity is attributed.                         |
| category            | <a href="#">Octet</a>                    | The main category that describes the radio entity.                                         |
| nomenclatureVersion | <a href="#">NomenclatureVersionEnum8</a> | The specific modification or individual unit type for a series and/or family of equipment. |
| nomenclature        | <a href="#">NomenclatureEnum16</a>       | The nomenclature for a particular communications device.                                   |

Named by [RadioTransmitter](#).

## RawBinaryRadioSignal STRUCTURE

A radio signal used to transmit raw binary data.

| Name                 | Type                                                 | Description                                                              |
|----------------------|------------------------------------------------------|--------------------------------------------------------------------------|
| hostRadioIndex       | <a href="#">string</a>                               | The object instance ID of the radio transmitting this signal.            |
| dataRate             | <a href="#">BitRateBitPerSecondUnsignedInteger32</a> | The rate at which the data is being transmitted.                         |
| signalDataLength     | <a href="#">BitsUnsignedInteger16</a>                | The length of the signal data.                                           |
| signalData           | <a href="#">SignalDataLengthlessArray1Plus</a>       | The signal data.                                                         |
| tacticalDataLinkType | <a href="#">TacticalDataLinkTypeEnum16</a>           | The type of tactical data link used to transmitted this signal (if any). |
| tDLMessageCount      | <a href="#">UnsignedInteger16</a>                    | The number of tactical data link messages contained in this signal.      |

## RecordQueryR STRUCTURE

A Simulation Management (SIMAN) interaction designed to allow a Simulation Manager federate to request data, in record format, from another federate.

| Name              | Type                                   | Description                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| timeInterval      | <a href="#">UnsignedInteger32</a>      | The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single RecordR interaction in response to this interaction.                                                                                         |
| eventType         | <a href="#">EventTypeEnum32</a>        | Specifies the type of event that the receiving entity or application should use to trigger the issue of a RecordR interaction in response to this query. If this is zero, then                                                                                               |

| Name                     | Type                                            | Description                                                                                                                                    |
|--------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| acknowledgement Protocol | <a href="#">Acknowledgement ProtocolEnum8</a>   | reporting shall be periodic based upon the value of the TimeInterval parameter.<br>The acknowledgement protocol to be used for the transaction |
| recordIdentifiers        | <a href="#">DatumIdentifier LengthlessArray</a> | Identifies the records for which information is requested                                                                                      |

## RecordR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RecordQueryR or SetRecordR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.

| Name                     | Type                                          | Description                                                                                                                   |
|--------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity        | <a href="#">EntityIdentifierStruct</a>        | The DIS entity ID of the entity or application sending the interaction.                                                       |
| receivingEntity          | <a href="#">EntityIdentifierStruct</a>        | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                            |
| requestIdentifier        | <a href="#">UnsignedInteger32</a>             | This field matches this response with the specific RecordQueryR or SetRecordR interaction sent by the simulation manager.     |
| eventType                | <a href="#">EventTypeEnum32</a>               | The type of event that caused the RecordR interaction to be issued.                                                           |
| acknowledgement Protocol | <a href="#">Acknowledgement ProtocolEnum8</a> | The acknowledgement protocol to be used for the transaction                                                                   |
| responseSerialNumber     | <a href="#">UnsignedInteger32</a>             | Used to identify the serial number of the RecordR interaction when more than one interaction is used to report record values. |
| recordSetData            | <a href="#">RecordSetStruct Array1Plus</a>    | The set of records containing the information requested                                                                       |

## RecordSetStruct STRUCTURE

A set of records and record details.

| Name                             | Type                                   | Description                                                                                                    |
|----------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| recordSetIdentifier              | <a href="#">DatumIdentifier Enum32</a> | This field shall specify the data structure used to convey the parameter values of the record for each record. |
| recordSetSerialNumber            | <a href="#">UnsignedInteger32</a>      | This field shall specify the serial number of the first record in the array.                                   |
| numberOfRecords<br>ARecordValues | <a href="#">RecordStructArray</a>      | This field contains the records of the format specified by the Record ID field.                                |

Named by [RecordSetStructArray1Plus](#).

## RecordStruct STRUCTURE

Record data.

| Name                         | Type                                | Description                                   |
|------------------------------|-------------------------------------|-----------------------------------------------|
| numberOfBits                 | <a href="#">UnsignedInteger32</a>   | Number of bits of data.                       |
| numberOfBytesARecord<br>Data | <a href="#">OctetArray</a>          | Data.                                         |
| paddingTo32                  | <a href="#">OctetPadding32Array</a> | Brings the record length to a 32-bit boundary |

Named by [RecordStructArray](#).

## RectVol1GeomRecStruct STRUCTURE

Record specifying Rectangular Volume 1 geometry record

| Name           | Type                                | Description                            |
|----------------|-------------------------------------|----------------------------------------|
| cornerLocation | <a href="#">WorldLocationStruct</a> | Corner location X, Y, Z                |
| dimensions     | <a href="#">DimensionStruct</a>     | Dimensions                             |
| orientation    | <a href="#">OrientationStruct</a>   | Orientation, specified by Euler angles |

Named by [EnvironmentRecVariantStruct](#).

## RectVol2GeomRecStruct STRUCTURE

Record specifying Rectangular Volume 2 geometry record

| Name            | Type                                        | Description                            |
|-----------------|---------------------------------------------|----------------------------------------|
| cornerLocation  | <a href="#">WorldLocationStruct</a>         | Corner location X, Y, Z                |
| dimensions      | <a href="#">DimensionStruct</a>             | Dimensions                             |
| dimensionsRate  | <a href="#">DimensionRateStruct</a>         | Variation of dimensions                |
| orientation     | <a href="#">OrientationStruct</a>           | Orientation, specified by Euler angles |
| velocity        | <a href="#">VelocityVectorStruct</a>        | Velocity Vx, Vy, Vz                    |
| angularVelocity | <a href="#">AngularVelocityVectorStruct</a> | Angular velocity Vx, Vy, Vz            |
| padding         | <a href="#">OctetArray4</a>                 | Padding field                          |

Named by [EnvironmentRecVariantStruct](#).

## RectVol3GeomRecStruct STRUCTURE

Record specifying Rectangular Volume 3 geometry record

| Name           | Type                                | Description                            |
|----------------|-------------------------------------|----------------------------------------|
| centerLocation | <a href="#">WorldLocationStruct</a> | Center location X, Y, Z                |
| dimensions     | <a href="#">DimensionStruct</a>     | Dimensions                             |
| orientation    | <a href="#">OrientationStruct</a>   | Orientation, specified by Euler angles |

Named by [EnvironmentRecVariantStruct](#).

## RelativePositionStruct STRUCTURE

Relative position in right-handed Cartesian coordinates.

| Name          | Type                         | Description                                                |
|---------------|------------------------------|------------------------------------------------------------|
| bodyXDistance | <a href="#">MeterFloat32</a> | The distance from the reference location along the X axis. |
| bodyYDistance | <a href="#">MeterFloat32</a> | The distance from the reference location along the Y axis. |
| bodyZDistance | <a href="#">MeterFloat32</a> | The distance from the reference location along the Z axis. |

Named by [netn.EntityStruct](#), [AcousticTransient](#), [Collision](#), [CollisionElastic](#), [Designator](#), [EmbeddedSystem](#), [MunitionDetonation](#), [StationNameLocationVariantStruct](#).

## RelativeRangeBearingStruct STRUCTURE

Relative position in polar coordinates.

| Name    | Type                               | Description                              |
|---------|------------------------------------|------------------------------------------|
| range   | <a href="#">LengthMeterFloat32</a> | The range from the reference location.   |
| bearing | <a href="#">AngleRadianFloat32</a> | The bearing from the reference location. |

Named by [StationNameLocationVariantStruct](#).

## RemoveEntity STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.

| Name              | Type                                   | Description                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |

## RemoveEntityR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.

| Name                        | Type                                              | Description                                                                                                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity           | <a href="#">EntityIdentifierStruct</a>            | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity             | <a href="#">EntityIdentifierStruct</a>            | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier           | <a href="#">UnsignedInteger32</a>                 | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| acknowledgement<br>Protocol | <a href="#">Acknowledgement<br/>ProtocolEnum8</a> | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## RemoveObjectRequest STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.

| Name              | Type                              | Description                                                                                                                                                                                                                                                                  |
|-------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifiers | <a href="#">RTObjectIdArray</a>   | The IDs of the objects to be removed.                                                                                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a> | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |

## RemoveObjectRequestR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.

| Name              | Type                              | Description                                                                                                                                                                                                                                                                  |
|-------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifiers | <a href="#">RTObjectIdArray</a>   | The IDs of the objects to be removed.                                                                                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a> | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |

| Name                        | Type                                              | Description                                               |
|-----------------------------|---------------------------------------------------|-----------------------------------------------------------|
| acknowledgement<br>Protocol | <a href="#">Acknowledgement<br/>ProtocolEnum8</a> | The acknowledgement protocol to be used for a transaction |

## RemoveObjectResult STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequest interaction, and to inform the Simulation Manager federate whether the removal was successful or not.

| Name               | Type                               | Description                                                                                                 |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| removeObjectResult | <a href="#">ResponseFlagEnum16</a> | Indicates ability to comply.                                                                                |
| requestIdentifier  | <a href="#">UnsignedInteger32</a>  | This field matches this response with the specific RemoveObject interaction sent by the simulation manager. |

## RemoveObjectResultR STRUCTURE

A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequestR interaction, and to inform the Simulation Manager federate whether the removal was successful or not.

| Name               | Type                               | Description                                                                                                 |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| removeObjectResult | <a href="#">ResponseFlagEnum16</a> | Indicates ability to comply.                                                                                |
| requestIdentifier  | <a href="#">UnsignedInteger32</a>  | This field matches this response with the specific RemoveObject interaction sent by the simulation manager. |

## RepairComplete STRUCTURE

Notifies the requesting entity that the requested repair has been completed.

| Name            | Type                             | Description                                         |
|-----------------|----------------------------------|-----------------------------------------------------|
| receivingObject | <a href="#">string</a>           | Object requesting repairs                           |
| repairingObject | <a href="#">string</a>           | Object that is able to perform the requested repair |
| repairType      | <a href="#">RepairTypeEnum16</a> | Type of repair performed                            |

## RepairResponse STRUCTURE

Acknowledges the notification of the completion of a repair.

| Name             | Type                              | Description                                         |
|------------------|-----------------------------------|-----------------------------------------------------|
| receivingObject  | <a href="#">string</a>            | Object requesting repairs                           |
| repairingObject  | <a href="#">string</a>            | Object that is able to perform the requested repair |
| repairResultCode | <a href="#">RepairResultEnum8</a> | Result of repair                                    |

## ResupplyCancel STRUCTURE

Communicates the canceling of a service function by either the receiving or the supplying entity.

| Name            | Type                   | Description                       |
|-----------------|------------------------|-----------------------------------|
| receivingObject | <a href="#">string</a> | Object that is receiving supplies |
| supplyingObject | <a href="#">string</a> | Object that has offered supplies  |

## ResupplyOffer STRUCTURE

Communicates the offer of supplies from a supplying entity to a receiving entity.

| Name            | Type                   | Description                                    |
|-----------------|------------------------|------------------------------------------------|
| receivingObject | <a href="#">string</a> | Object that the supplies are being offered to. |
| supplyingObject | <a href="#">string</a> | Object that is offering the supplies           |

| Name         | Type                                                            | Description                        |
|--------------|-----------------------------------------------------------------|------------------------------------|
| suppliesData | <a href="#">SupplyStruct</a><br><a href="#">LengthlessArray</a> | List of supplies that are offered. |

## ResupplyReceived STRUCTURE

Acknowledge the receipt of supplies.

| Name            | Type                                                            | Description                                 |
|-----------------|-----------------------------------------------------------------|---------------------------------------------|
| receivingObject | <a href="#">string</a>                                          | Object that is receiving the supplies       |
| supplyingObject | <a href="#">string</a>                                          | Object that is providing the supplies.      |
| suppliesData    | <a href="#">SupplyStruct</a><br><a href="#">LengthlessArray</a> | List of supplies taken by receiving object. |

## RibbonBridgeObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the RibbonBridgeObject class.

| Name                        | Type                                         | Description                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| objectIdentifier            | <a href="#">EntityIdentifierStruct</a>       | Identifies this EnvironmentObject instance (point, linear or areal)                                                                                                                                                                                                                                                |
| referencedObject Identifier | <a href="#">string</a>                       | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated                                                                                                                                                                                                        |
| forceIdentifier             | <a href="#">ForceIdentifierEnum8</a>         | Identifies the force that created or modified this EnvironmentObject instance                                                                                                                                                                                                                                      |
| objectType                  | <a href="#">EnvironmentObject TypeStruct</a> | Identifies the type of this EnvironmentObject instance                                                                                                                                                                                                                                                             |
| requestingIdentifier        | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                         |
| receivingIdentifier         | <a href="#">FederateIdentifier Struct</a>    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                                                                                                                                                                                                                       |
| location                    | <a href="#">WorldLocationStruct</a>          | Specifies the location of the object based on x, y and z coordinates                                                                                                                                                                                                                                               |
| orientation                 | <a href="#">OrientationStruct</a>            | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete             | <a href="#">PercentUnsigned Integer32</a>    | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance           | <a href="#">DamageStatus Enum32</a>          | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed        | <a href="#">bool</a>                         | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated                 | <a href="#">bool</a>                         | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking                     | <a href="#">bool</a>                         | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming                     | <a href="#">bool</a>                         | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |
| numberOfSegments            | <a href="#">UnsignedInteger32</a>            | Specifies the number of segments composing the ribbon bridge                                                                                                                                                                                                                                                       |



## SINCGARSModulationStruct STRUCTURE

Modulation parameters for SINCGARS radio system

| Name                            | Type                                | Description                                                                                |
|---------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|
| fHNetID                         | <a href="#">Integer16</a>           | Frequency hopping network identifier.                                                      |
| hopSetID                        | <a href="#">Integer16</a>           | Identification for the set of frequencies used when creating a hopping pattern.            |
| lockoutSetID                    | <a href="#">Integer16</a>           | Identification for the set of frequencies that are excluded from the hopping pattern.      |
| transmissionSecurity<br>Key     | <a href="#">Integer16</a>           | Identifies the transmission security key that is used when generating the hopping pattern. |
| fHSynchronizationTime<br>Offset | <a href="#">TimeSecondInteger32</a> | The offset to exercise time in seconds for the clock in the radio.                         |

Named by [SpreadSpectrumVariantStruct](#).

## ServiceRequest STRUCTURE

A request for logistics support. The requesting entity issues the interaction to the supplying entity asking for repair or specific supplies.

| Name             | Type                                                            | Description                                                                                                                                      |
|------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| requestingObject | <a href="#">string</a>                                          | Object requesting service                                                                                                                        |
| servicingObject  | <a href="#">string</a>                                          | Object that is able to provide the requested service                                                                                             |
| serviceType      | <a href="#">ServiceTypeEnum8</a>                                | Type of requested service                                                                                                                        |
| suppliesData     | <a href="#">SupplyStruct</a><br><a href="#">LengthlessArray</a> | For a service of resupply, the list of supplies to be exchanged. If the service requested is not resupply, then this parameter shall be omitted. |

## SetData STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values.

| Name              | Type                                                                | Description                                                                                                                                                                                                                                                                  |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                                                                                                                                                                           |
| requestIdentifier | <a href="#">UnsignedInteger32</a>                                   | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| fixedDatums       | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | The set of fixed length data items (types and values) that the recipient is requested to set.                                                                                                                                                                                |
| variableDatumSet  | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | The set of variables length data items (types and values) that the recipient is requested to set.                                                                                                                                                                            |

## SetDataR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                     | Type                                                                | Description                                                                                                                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity        | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity          | <a href="#">EntityIdentifierStruct</a>                              | The DIS entity ID of the entity or application which is the intended recipient of the interaction.                                                                                                                                                                           |
| requestIdentifier        | <a href="#">UnsignedInteger32</a>                                   | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| fixedDatums              | <a href="#">FixedDatumStruct</a><br><a href="#">LengthlessArray</a> | The set of fixed length data items (types and values) that the recipient is requested to set.                                                                                                                                                                                |
| variableDatumSet         | <a href="#">VariableDatumStruct</a><br><a href="#">Array</a>        | The set of variables length data items (types and values) that the recipient is requested to set.                                                                                                                                                                            |
| acknowledgement Protocol | <a href="#">Acknowledgement ProtocolEnum8</a>                       | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## SetRecordR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values (provided in record format).

| Name                     | Type                                                          | Description                                                                                                                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity        | <a href="#">EntityIdentifierStruct</a>                        | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity          | <a href="#">EntityIdentifierStruct</a>                        | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier        | <a href="#">UnsignedInteger32</a>                             | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| acknowledgement Protocol | <a href="#">Acknowledgement ProtocolEnum8</a>                 | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |
| recordSetData            | <a href="#">RecordSetStruct</a><br><a href="#">Array1Plus</a> | Specifies the information, in record format, to be set by the receiving entity.                                                                                                                                                                                              |

## ShaftDataStruct STRUCTURE

Rotation state of a ship's propulsion shaft

| Name              | Type                                                   | Description                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| currentShaftRate  | <a href="#">RevolutionsPerMinute Integer16</a>         | The current shaft speed. A positive shaft speed indicates that the shaft is rotating in a clockwise direction (when viewed from the stern to bow), and a negative shaft speed indicates an anti-clockwise direction.                                                                                                                                                |
| orderedShaftRate  | <a href="#">RevolutionsPerMinute Integer16</a>         | The ordered shaft speed. If the ordered shaft speed is different from the current shaft speed then the shaft will accelerate or decelerate at the shaft rate of change. A positive shaft speed indicates that the shaft is rotating in a clockwise direction (when viewed from the stern to bow), and a negative shaft speed indicates an anti-clockwise direction. |
| shaftRateOfChange | <a href="#">SpeedChangeRate RPMPerSecond Integer16</a> | The absolute value of the shaft's rotational acceleration. The shaft will change its rotational speed at this rate if the current shaft speed differs from the ordered shaft rate.                                                                                                                                                                                  |

Named by [ShaftDataStructLengthlessArray1Plus](#).

## SilentAggregateStruct STRUCTURE

These fields contain information about subaggregates not registered in the federation.

| Name                             | Type                              | Description                                                                                            |
|----------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| aggregateType                    | <a href="#">EntityTypeStruct</a>  | This field shall specify the aggregates common to this system list.                                    |
| numberOfAggregatesOf<br>ThisType | <a href="#">UnsignedInteger16</a> | This field shall specify the number of aggregates that have the type specified in AggregateType field. |

Named by [SilentAggregateStructLengthlessArray](#).

## SilentEntityStruct STRUCTURE

These fields contain information about entities not registered in the federation.

| Name                           | Type                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| numberOfEntitiesOf<br>ThisType | <a href="#">UnsignedInteger16</a>                          | This field shall specify the number of entities that have the type specified in the field EntityType.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| numberOfAppearance<br>Records  | <a href="#">UnsignedInteger16</a>                          | This field shall specify the number of Entity Appearance records that follow. This number shall be between zero and the number of entities of this type. Simulation applications representing the aggregate that do not model entity appearances shall set this field to zero. Simulation applications representing the aggregate that model entity appearances shall set this field to the number of entity appearances that deviate from the default appearance. Other simulations can safely assume that any entity appearances not specified are default appearances. |
| entityType                     | <a href="#">EntityTypeStruct</a>                           | This field shall specify the entity types common to the entities in this system list.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| entityAppearance               | <a href="#">Unsigned<br/>Integer32Lengthless<br/>Array</a> | This field shall specify the entity appearances of entities in the aggregate that deviate from the default. The length of the array is defined in the NumberOfAppearanceRecords field.                                                                                                                                                                                                                                                                                                                                                                                    |

Named by [SilentEntityStructLengthlessArray](#).

## SpatialFPStruct STRUCTURE

Spatial structure for Dead Reckoning Algorithm FPW (2) and FPB (6).

| Name           | Type                                 | Description                                                                                                                                                                                                                                                                                    |
|----------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| worldLocation  | <a href="#">WorldLocationStruct</a>  | Location of the object.                                                                                                                                                                                                                                                                        |
| isFrozen       | <a href="#">bool</a>                 | Whether the object is frozen or not.                                                                                                                                                                                                                                                           |
| orientation    | <a href="#">OrientationStruct</a>    | The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the world coordinate system to the entity coordinate system). |
| velocityVector | <a href="#">VelocityVectorStruct</a> | The rate at which an object's position is changing over time.                                                                                                                                                                                                                                  |

Named by [SpatialVariantStruct](#).

## SpatialFVStruct STRUCTURE

Spatial structure for Dead Reckoning Algorithm FVW (5) and RVB (9).

| Name          | Type                                | Description                          |
|---------------|-------------------------------------|--------------------------------------|
| worldLocation | <a href="#">WorldLocationStruct</a> | Location of the object.              |
| isFrozen      | <a href="#">bool</a>                | Whether the object is frozen or not. |

| Name               | Type                                     | Description                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orientation        | <a href="#">OrientationStruct</a>        | The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the world coordinate system to the entity coordinate system). |
| velocityVector     | <a href="#">VelocityVectorStruct</a>     | The rate at which an object's position is changing over time.                                                                                                                                                                                                                                  |
| accelerationVector | <a href="#">AccelerationVectorStruct</a> | The magnitude of the change in linear velocity of an object over time.                                                                                                                                                                                                                         |

Named by [SpatialVariantStruct](#).

## SpatialRPStruct STRUCTURE

Spatial structure for Dead Reckoning Algorithm RPW (3) and RPB (7).

| Name            | Type                                        | Description                                                                                                                                                                                                                                                                                    |
|-----------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| worldLocation   | <a href="#">WorldLocationStruct</a>         | Location of the object.                                                                                                                                                                                                                                                                        |
| isFrozen        | <a href="#">bool</a>                        | Whether the object is frozen or not.                                                                                                                                                                                                                                                           |
| orientation     | <a href="#">OrientationStruct</a>           | The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the world coordinate system to the entity coordinate system). |
| velocityVector  | <a href="#">VelocityVectorStruct</a>        | The rate at which an object's position is changing over time.                                                                                                                                                                                                                                  |
| angularVelocity | <a href="#">AngularVelocityVectorStruct</a> | The rate at which an object's orientation is changing over time.                                                                                                                                                                                                                               |

Named by [SpatialVariantStruct](#).

## SpatialRVStruct STRUCTURE

Spatial structure for Dead Reckoning Algorithm RVW (4) and RVB (8).

| Name               | Type                                        | Description                                                                                                                                                                                                                                                                                    |
|--------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| worldLocation      | <a href="#">WorldLocationStruct</a>         | Location of the object.                                                                                                                                                                                                                                                                        |
| isFrozen           | <a href="#">bool</a>                        | Whether the object is frozen or not.                                                                                                                                                                                                                                                           |
| orientation        | <a href="#">OrientationStruct</a>           | The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the world coordinate system to the entity coordinate system). |
| velocityVector     | <a href="#">VelocityVectorStruct</a>        | The rate at which an object's position is changing over time.                                                                                                                                                                                                                                  |
| accelerationVector | <a href="#">AccelerationVectorStruct</a>    | The magnitude of the change in linear velocity of an object over time.                                                                                                                                                                                                                         |
| angularVelocity    | <a href="#">AngularVelocityVectorStruct</a> | The rate at which an object's orientation is changing over time.                                                                                                                                                                                                                               |

Named by [SpatialVariantStruct](#).

## SpatialStaticStruct STRUCTURE

Spatial structure for Dead Reckoning Algorithm Static (1).

| Name          | Type                                | Description                                                                                                              |
|---------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| worldLocation | <a href="#">WorldLocationStruct</a> | Location of the object.                                                                                                  |
| isFrozen      | <a href="#">bool</a>                | Whether the object is frozen or not.                                                                                     |
| orientation   | <a href="#">OrientationStruct</a>   | The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes |

| Name | Type | Description                                                                                                                                                           |
|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |      | (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the world coordinate system to the entity coordinate system). |

Named by [SpatialVariantStruct](#).

## Sphere1GeomRecStruct STRUCTURE

Record specifying Bounding Sphere & Sphere 1 geometry record

| Name             | Type                                | Description               |
|------------------|-------------------------------------|---------------------------|
| centroidLocation | <a href="#">WorldLocationStruct</a> | Centroid location X, Y, Z |
| radius           | <a href="#">Float32</a>             | Radius                    |
| padding          | <a href="#">OctetArray4</a>         | Padding field             |

Named by [EnvironmentRecVariantStruct](#).

## Sphere2GeomRecStruct STRUCTURE

Record specifying Sphere 2 geometry record

| Name             | Type                                        | Description                 |
|------------------|---------------------------------------------|-----------------------------|
| centroidLocation | <a href="#">WorldLocationStruct</a>         | Centroid location X, Y, Z   |
| radius           | <a href="#">Float32</a>                     | Radius                      |
| radiusRate       | <a href="#">Float32</a>                     | Variation of radius         |
| velocity         | <a href="#">VelocityVectorStruct</a>        | Velocity Vx, Vy, Vz         |
| angularVelocity  | <a href="#">AngularVelocityVectorStruct</a> | Angular velocity Vx, Vy, Vz |

Named by [EnvironmentRecVariantStruct](#).

## SphericalHarmonicAntennaStruct STRUCTURE

Specifies the direction and radiation pattern from a radio transmitter's antenna.

| Name            | Type                                             | Description                                                                                                                                                                                                |
|-----------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| order           | <a href="#">UnsignedInteger32</a>                | The highest order of the expansion in spherical harmonics, counting from zero.                                                                                                                             |
| coefficients    | <a href="#">CoefficientsLengthlessArray1Plus</a> | Represents the power distribution from the antenna as the coefficients of a spherical harmonic expansion to the order given in the Order field. The length of the array is $N^2+2N+1$ (N being the Order). |
| referenceSystem | <a href="#">ReferenceSystemEnum8</a>             | This field shall specify the reference coordinate system with respect to which beam direction is specified.                                                                                                |
| padding         | <a href="#">OctetArray3</a>                      | Padding to 32 bits                                                                                                                                                                                         |

Named by [AntennaPatternVariantStruct](#).

## StartResume STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause.

| Name              | Type                                   | Description                                                                                           |
|-------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                               |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction. |

| Name              | Type                              | Description                                                                                                                                                                                                                                                                  |
|-------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| realWorldTime     | <a href="#">ClockTimeStruct</a>   | The real world time (GMT) that the entity or entities should start/resume at.                                                                                                                                                                                                |
| requestIdentifier | <a href="#">UnsignedInteger32</a> | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| simulationTime    | <a href="#">ClockTimeStruct</a>   | The simulation time that the entity or entities should use when they start/resume.                                                                                                                                                                                           |

## StartResumeR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                     | Type                                          | Description                                                                                                                                                                                                                                                                  |
|--------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity        | <a href="#">EntityIdentifierStruct</a>        | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity          | <a href="#">EntityIdentifierStruct</a>        | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| realWorldTime            | <a href="#">ClockTimeStruct</a>               | The real world time (GMT) that the entity or entities should start/resume at.                                                                                                                                                                                                |
| requestIdentifier        | <a href="#">UnsignedInteger32</a>             | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| simulationTime           | <a href="#">ClockTimeStruct</a>               | The simulation time that the entity or entities should use when they start/resume.                                                                                                                                                                                           |
| acknowledgement Protocol | <a href="#">Acknowledgement ProtocolEnum8</a> | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## StopFreeze STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) freeze (pause) their simulation or b) stop their simulation.

| Name              | Type                                   | Description                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a> | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a>      | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| realWorldTime     | <a href="#">ClockTimeStruct</a>        | The real world time (GMT) that the entity or entities should stop/freeze at.                                                                                                                                                                                                 |
| reason            | <a href="#">StopFreezeReason Enum8</a> | The reason for the stop or freeze.                                                                                                                                                                                                                                           |
| reflectValues     | <a href="#">bool</a>                   | Whether the entity or entities being stopped/frozen should continue to reflect values when stopped/frozen.                                                                                                                                                                   |

| Name                           | Type | Description                                                                                                                     |
|--------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------|
| runInternalSimulation<br>Clock | bool | Whether the entity or entities being stopped/frozen should continue to run their internal simulation clock when stopped/frozen. |
| updateAttributes               | bool | Whether the entity or entities being stopped/frozen should continue to update attributes when stopped/frozen.                   |

## StopFreezeR STRUCTURE

A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) pause their simulation or b) stop their simulation. The Simulation Manager federate specifies the acknowledgement protocol to be used.

| Name                           | Type                             | Description                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity              | EntityIdentifierStruct           | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity                | EntityIdentifierStruct           | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier              | UnsignedInteger32                | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| realWorldTime                  | ClockTimeStruct                  | The real world time (GMT) that the entity or entities should stop/freeze at.                                                                                                                                                                                                 |
| reason                         | StopFreezeReason<br>Enum8        | The reason for the stop or freeze.                                                                                                                                                                                                                                           |
| reflectValues                  | bool                             | Whether the entity or entities being stopped/frozen should continue to reflect values when stopped/frozen.                                                                                                                                                                   |
| runInternalSimulation<br>Clock | bool                             | Whether the entity or entities being stopped/frozen should continue to run their internal simulation clock when stopped/frozen.                                                                                                                                              |
| updateAttributes               | bool                             | Whether the entity or entities being stopped/frozen should continue to update attributes when stopped/frozen.                                                                                                                                                                |
| acknowledgement<br>Protocol    | Acknowledgement<br>ProtocolEnum8 | The acknowledgement protocol to be used for a transaction                                                                                                                                                                                                                    |

## StructureObjectTransaction STRUCTURE

An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the StructureObject class.

| Name                           | Type                            | Description                                                                                                 |
|--------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------|
| objectIdentifier               | EntityIdentifierStruct          | Identifies this EnvironmentObject instance (point, linear or areal)                                         |
| referencedObject<br>Identifier | string                          | Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated |
| forceIdentifier                | ForceIdentifierEnum8            | Identifies the force that created or modified this EnvironmentObject instance                               |
| objectType                     | EnvironmentObject<br>TypeStruct | Identifies the type of this EnvironmentObject instance                                                      |
| requestingIdentifier           | FederateIdentifier<br>Struct    | Identifies the simulation application sending the EnvironmentObjectTransaction interaction                  |
| receivingIdentifier            | FederateIdentifier<br>Struct    | Identifies the simulation application receiving the EnvironmentObjectTransaction interaction                |
| location                       | WorldLocationStruct             | Specifies the location of the object based on x, y and z coordinates                                        |



| Name                 | Type                                     | Description                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orientation          | <a href="#">OrientationStruct</a>        | Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system |
| percentComplete      | <a href="#">PercentUnsignedInteger32</a> | Specifies the percent completion of the object                                                                                                                                                                                                                                                                     |
| damagedAppearance    | <a href="#">DamageStatusEnum32</a>       | Specifies the damaged appearance of the object                                                                                                                                                                                                                                                                     |
| objectPreDistributed | <a href="#">bool</a>                     | Specifies whether or not the object was created before the start of the exercise                                                                                                                                                                                                                                   |
| deactivated          | <a href="#">bool</a>                     | Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment)                                                                                                                                                                                                     |
| smoking              | <a href="#">bool</a>                     | Specifies whether or not the object is smoking (creating a smoke plume)                                                                                                                                                                                                                                            |
| flaming              | <a href="#">bool</a>                     | Specifies whether or not the object is flaming                                                                                                                                                                                                                                                                     |

## SupplyStruct STRUCTURE

Represents a single supply type and the quantity being offered or requested.

| Name       | Type                             | Description                                                                                                                                           |
|------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| supplyType | <a href="#">EntityTypeStruct</a> | The type of supply being offered or requested.                                                                                                        |
| quantity   | <a href="#">Float32</a>          | The number of units of the supply type. The unit measure depends on the supply type and shall use the SI units of measurement used for such supplies. |

Named by [SupplyStructLengthlessArray](#).

## TransferControl STRUCTURE

A Simulation Management (SIMAN) interaction, sent to initiate the transfer of control of an entity.

| Name              | Type                                      | Description                                                                                                                                                                                                                                                                  |
|-------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originatingEntity | <a href="#">EntityIdentifierStruct</a>    | The DIS entity ID of the entity or application sending the interaction.                                                                                                                                                                                                      |
| receivingEntity   | <a href="#">EntityIdentifierStruct</a>    | The DIS entity ID of the entity or application which is the intended recipient(s) of the interaction.                                                                                                                                                                        |
| requestIdentifier | <a href="#">UnsignedInteger32</a>         | The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements. |
| transferType      | <a href="#">TransferTypeEnum8</a>         | The type of transfer to be performed.                                                                                                                                                                                                                                        |
| transferEntity    | <a href="#">string</a>                    | The ID of the object to be transferred.                                                                                                                                                                                                                                      |
| recordSetData     | <a href="#">RecordSetStructArray1Plus</a> | Specifies the information, in record format, to be set by the receiving entity.                                                                                                                                                                                              |

## UniformGeomRecStruct STRUCTURE

Record specifying Uniform geometry record

| Name    | Type                        | Description   |
|---------|-----------------------------|---------------|
| padding | <a href="#">OctetArray8</a> | Padding field |

Named by [EnvironmentRecVariantStruct](#).



## VariableDatumStruct STRUCTURE

These fields shall specify the types of variable datum, their length, and their value.

| Name        | Type                                        | Description                                                                                                                                                             |
|-------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| datumID     | <a href="#">DatumIdentifierEnum32</a>       | The fixed datum id represented by a 32-bit enumeration                                                                                                                  |
| datumLength | <a href="#">UnsignedInteger32</a>           | This field shall specify the length of the variable datum in bits.                                                                                                      |
| datumValue  | <a href="#">UnsignedInteger64Array1Plus</a> | Value of the variable datum defined by the Variable Datum ID and Variable Datum length. This field shall be padded at the end to make the length a multiple of 64-bits. |

Named by [VariableDatumStructArray](#), [VariableDatumStructLengthlessArray](#).

## VectoringNozzleSystemDataStruct STRUCTURE

Operational data for describing the vectoring nozzle systems being simulated.

| Name                      | Type                               | Description                                                            |
|---------------------------|------------------------------------|------------------------------------------------------------------------|
| horizontalDeflectionAngle | <a href="#">AngleDegreeFloat32</a> | The nozzle deflection angle in the horizontal body axis of the entity. |
| verticalDeflectionAngle   | <a href="#">AngleDegreeFloat32</a> | The nozzle deflection angle in the vertical body axis of the entity.   |

Named by [VectoringNozzleSystemDataStructLengthlessArray](#).

## VelocityVectorStruct STRUCTURE

The rate at which the position is changing over time.

| Name      | Type                                          | Description                          |
|-----------|-----------------------------------------------|--------------------------------------|
| xVelocity | <a href="#">VelocityMeterPerSecondFloat32</a> | Velocity component along the X axis. |
| yVelocity | <a href="#">VelocityMeterPerSecondFloat32</a> | Velocity component along the Y axis. |
| zVelocity | <a href="#">VelocityMeterPerSecondFloat32</a> | Velocity component along the Z axis. |

Named by [netn.CBRNRelease](#), [COMBICStateRecStruct](#), [Collision](#), [CollisionElastic](#), [Cone2GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [GaussPlumeGeomRecStruct](#), [GaussPuffGeomRecStruct](#), [Line2GeomRecStruct](#), [MunitionDetonation](#), [Point2GeomRecStruct](#), [RectVol2GeomRecStruct](#), [SpatialFPStruct](#), [SpatialFVStruct](#), [SpatialRPStruct](#), [SpatialRVStruct](#), [Sphere2GeomRecStruct](#), [WeaponFire](#).

## WeaponFire STRUCTURE

Communicates information associated with the firing or launch of a munition.

| Name                     | Type                                  | Description                                                                                                        |
|--------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| eventIdentifier          | <a href="#">EventIdentifierStruct</a> | An ID, generated by the issuing federate, used to associate related fire and detonation events.                    |
| fireControlSolutionRange | <a href="#">LengthMeterFloat32</a>    | The range used in the fire control solution. Zero if the range is unknown or inapplicable.                         |
| fireMissionIndex         | <a href="#">UnsignedInteger32</a>     | A unique index to identify the fire mission (used to associate weapon fire interactions in a single fire mission). |
| firingLocation           | <a href="#">WorldLocationStruct</a>   | The location, in world coordinates, from which the munition was launched.                                          |
| firingObjectIdentifier   | <a href="#">string</a>                | The object instance ID of the entity firing the munition.                                                          |
| fuseType                 | <a href="#">FuseTypeEnum16</a>        | The type of fuse on the munition.                                                                                  |
| initialVelocityVector    | <a href="#">VelocityVectorStruct</a>  | The velocity vector of the munition when fired.                                                                    |

| Name                     | Type                              | Description                                                                        |
|--------------------------|-----------------------------------|------------------------------------------------------------------------------------|
| munitionObjectIdentifier | <a href="#">string</a>            | The object instance ID of the fired munition, if an object instance is registered. |
| munitionType             | <a href="#">EntityTypeStruct</a>  | The type of munition being fired.                                                  |
| quantityFired            | <a href="#">UnsignedInteger16</a> | The number of rounds fired in the burst.                                           |
| rateOfFire               | <a href="#">UnsignedInteger16</a> | The rate of fire in rounds per minute when quantity > 1. Zero otherwise.           |
| targetObjectIdentifier   | <a href="#">string</a>            | The object instance ID of the intended target (if any).                            |
| warheadType              | <a href="#">WarheadTypeEnum16</a> | The type of warhead fitted to the munition being fired.                            |

## WorldLocationStruct STRUCTURE

The location of an object in the world coordinate system, as specified in IEEE Std 1278.1-1995 section 1.3.2.

| Name | Type                         | Description                                |
|------|------------------------------|--------------------------------------------|
| x    | <a href="#">MeterFloat64</a> | Distance from the origin along the X axis. |
| y    | <a href="#">MeterFloat64</a> | Distance from the origin along the Y axis. |
| z    | <a href="#">MeterFloat64</a> | Distance from the origin along the Z axis. |

Named by [netn.AddPassage](#), [netn.AppointmentStruct](#), [netn.ArrayOfWorldLocationStruct](#), [netn.CBRNAlarmStruct](#), [netn.CBRNRelease](#), [netn.CommunicationNode](#), [netn.EstablishCheckPoint](#), [netn.FireAtLocation](#), [netn.FireAtLocationWM](#), [netn.GeocentricPolygon](#), [netn.MagicMove](#), [netn.MoveIntoFormation](#), [netn.MoveToLocation](#), [netn.PointVariantStruct](#), [netn.PositionStatusReport](#), [netn.SpottedEntity](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [Cone1GeomRecStruct](#), [Cone2GeomRecStruct](#), [CraterObjectTransaction](#), [Designator](#), [Ellipsoid1GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [EnvironmentRecVariantStruct](#), [GaussPlumeGeomRecStruct](#), [GaussPuffGeomRecStruct](#), [Line1GeomRecStruct](#), [Line2GeomRecStruct](#), [LinearSegmentStruct](#), [Minefield](#), [MunitionDetonation](#), [OtherPointObjectTransaction](#), [Point2GeomRecStruct](#), [PointObject](#), [PointObjectTransaction](#), [RadioTransmitter](#), [RectVol1GeomRecStruct](#), [RectVol2GeomRecStruct](#), [RectVol3GeomRecStruct](#), [RibbonBridgeObjectTransaction](#), [SpatialFPStruct](#), [SpatialFVStruct](#), [SpatialRPStruct](#), [SpatialRVStruct](#), [SpatialStaticStruct](#), [Sphere1GeomRecStruct](#), [Sphere2GeomRecStruct](#), [StructureObjectTransaction](#), [WeaponFire](#), [WorldLocationStructLengthlessArray](#).

## Enumerations

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| <a href="#">AcknowledgementProtocolEnum8</a>       | Required reliability service                                 | 232 |
| <a href="#">ActionEnum32</a>                       | Action ID                                                    | 232 |
| <a href="#">ActionResultEnum32</a>                 | Request status                                               | 233 |
| <a href="#">ActiveSonarEnum16</a>                  | Acoustic system name                                         | 233 |
| <a href="#">ActiveSonarFunctionCodeEnum8</a>       | The current function being performed by the sonar            | 234 |
| <a href="#">ActiveSonarScanPatternEnum16</a>       | Acoustic scan pattern                                        | 234 |
| <a href="#">AggregateStateEnum8</a>                | Aggregate state                                              | 234 |
| <a href="#">AmplitudeAngleModulationTypeEnum16</a> | Detailed modulation types for Amplitude and Angle Modulation | 234 |
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| <a href="#">AngleModulationTypeEnum16</a>          | Detailed modulation types for Angle Modulation               | 235 |
| <a href="#">AntennaPatternTypeEnum32</a>           | The radiation pattern from an antenna.                       | 235 |
| <a href="#">ArticulatedPartsTypeEnum32</a>         | Articulated part type class                                  | 235 |
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|                                       |                                                                                               |     |
|---------------------------------------|-----------------------------------------------------------------------------------------------|-----|
| DetonationResultCodeEnum8             | Detonation result                                                                             | 246 |
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| EnvironmentDataCoordinateSystemEnum16 | Environment data coordinate system                                                            | 255 |
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| EnvironmentGridTypeEnum8              | Environment data grid type                                                                    | 256 |
| EnvironmentModelTypeEnum8             | Environment process model type                                                                | 256 |
| EnvironmentRecordTypeEnum32           | Environment record type                                                                       | 256 |
| EventTypeEnum32                       | Event type                                                                                    | 257 |
| ForceIdentifierEnum8                  | Force ID                                                                                      | 257 |
| FormationEnum32                       | Formation                                                                                     | 258 |
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| IffAlternateMode4Enum8                | IFF alternate mode 4                                                                          | 259 |
| IffApplicableModesEnum8               | IFF applicable modes                                                                          | 259 |
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| IffOperationalParameter2Enum8         | IFF operational parameter 2                                                                   | 259 |
| IffSystemModeEnum8                    | IFF system mode                                                                               | 259 |
| IffSystemNameEnum16                   | IFF system name                                                                               | 260 |
| IffSystemTypeEnum16                   | IFF system type                                                                               | 260 |
| JTIDSPrimaryModeEnum8                 | The primary mode of a JTIDS system [UID 173].                                                 | 260 |
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| Link11SignalWaveformEnum8             | Link 11 Signal Waveform [UID 734].                                                            | 262 |
| Link11TerminalModeEnum8               | Link 11 Terminal Mode [UID 468].                                                              | 263 |
| Link16VersionEnum8                    | Link 16 version [UID 800].                                                                    | 263 |
| MajorRFModulationTypeEnum16           | Major classification of the modulation type.                                                  | 263 |
| MarkingEncodingEnum8                  | Marking character set                                                                         | 263 |
| MinefieldFusingEnum32                 | Minefield fuse type                                                                           | 263 |
| MinefieldLaneEnum8                    | Minefield lane status                                                                         | 264 |
| MinefieldPaintSchemeEnum32            | Minefield paint scheme                                                                        | 264 |
| MinefieldProtocolEnum8                | Minefield communication protocol mode                                                         | 264 |
| MinefieldSensorTypeEnum32             | Minefield sensor type                                                                         | 264 |
| MinefieldStatusEnum8                  | Minefield status                                                                              | 265 |
| MinefieldTypeEnum8                    | Minefield type                                                                                | 265 |
| NomenclatureEnum16                    | The nomenclature for a specific radio system.                                                 | 265 |
| NomenclatureVersionEnum8              | The naming convention applied to the radio system by the manufacturer/<br>controlling agency. | 265 |
| OpacityCodeEnum32                     | Density of environmentals                                                                     | 266 |
| ParameterTypeEnum32                   | Parameter type                                                                                | 267 |
| PropulsionPlantEnum8                  | Propulsion plant configuration                                                                | 267 |
| PulseModulationTypeEnum16             | Detailed modulation types for Pulse Modulation                                                | 267 |
| RFModulationSystemTypeEnum16          | The radio system type associated with this transmitter.                                       | 267 |
| RadioInputSourceEnum8                 | Radio input source                                                                            | 268 |
| ReceiverOperationalStatusEnum16       | The operational state of a radio receiver.                                                    | 268 |
| ReferenceSystemEnum8                  | The reference coordinate system used                                                          | 268 |
| RepairResultEnum8                     | Result of repair                                                                              | 268 |
| RepairTypeEnum16                      | System repaired                                                                               | 268 |
| RequestStatusEnum32                   | Request status                                                                                | 270 |
| ResponseFlagEnum16                    | Response flag                                                                                 | 270 |
| SISOSTD002VersionEnum8                | SISO-STD-002 version [UID 736].                                                               | 270 |
| ServiceTypeEnum8                      | Service type                                                                                  | 270 |
| SpreadSpectrumEnum16                  | The type of spread spectrum characteristics employed by a transmitter.                        | 270 |

|                                                   |                                                                                                                      |     |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">StanceCodeEnum32</a>                  | Life form state                                                                                                      | 271 |
| <a href="#">StationEnum32</a>                     | Attached part station                                                                                                | 271 |
| <a href="#">StopFreezeReasonEnum8</a>             | Reason to stop                                                                                                       | 274 |
| <a href="#">TacticalDataLinkTypeEnum16</a>        | The type of tactical data link used to transmit a signal                                                             | 274 |
| <a href="#">TimeSlotAllocationLevelEnum8</a>      | Time Slot Allocation (TSA) level of fidelity. See SISO-STD-002 for detailed descriptions and requirements [UID 172]. | 275 |
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| <a href="#">TransferTypeEnum8</a>                 | Transfer type                                                                                                        | 275 |
| <a href="#">TransmitterOperationalStatusEnum8</a> | The current operational state of a radio transmitter.                                                                | 276 |
| <a href="#">UnmodulatedTypeEnum16</a>             | Detailed modulation types for an unmodulated carrier                                                                 | 276 |
| <a href="#">UserProtocolEnum32</a>                | The ID of the user protocol used to transmit a radio signal.                                                         | 276 |
| <a href="#">VisibleSideLocationEnum32</a>         | Minefield lane marker visible side                                                                                   | 277 |
| <a href="#">WarheadTypeEnum16</a>                 | Warhead                                                                                                              | 277 |
| <a href="#">WeaponStateEnum32</a>                 | Weapon position                                                                                                      | 278 |

## AcknowledgeFlagEnum16 ENUMERATION

Acknowledgment flags

Held as u16.

|                                   |   |
|-----------------------------------|---|
| <a href="#">createEntity</a>      | 1 |
| <a href="#">removeEntity</a>      | 2 |
| <a href="#">startResume</a>       | 3 |
| <a href="#">stopFreeze</a>        | 4 |
| <a href="#">transferOwnership</a> | 5 |

Named by [Acknowledge](#), [AcknowledgeR](#).

## AcknowledgementProtocolEnum8 ENUMERATION

Required reliability service

Held as u8.

|                                |   |
|--------------------------------|---|
| <a href="#">acknowledged</a>   | 0 |
| <a href="#">unacknowledged</a> | 1 |

Named by [ActionRequestR](#), [ActionRequestToObjectR](#), [AttributeChangeRequestR](#), [AttributeChangeResultR](#), [CreateEntityR](#), [CreateObjectRequestR](#), [DataQueryR](#), [DataR](#), [RecordQueryR](#), [RecordR](#), [RemoveEntityR](#), [RemoveObjectRequestR](#), [SetDataR](#), [SetRecordR](#), [StartResumeR](#), [StopFreezeR](#).

## ActionEnum32 ENUMERATION

Action ID

Held as u32.

|                                                                  |    |
|------------------------------------------------------------------|----|
| <a href="#">other</a>                                            | 0  |
| <a href="#">localStorageOfTheRequestedInformation</a>            | 1  |
| <a href="#">informSimulationManagerOfRanOutOfAmmunitionEvent</a> | 2  |
| <a href="#">informSimulationManagerOfKilledInActionEvent</a>     | 3  |
| <a href="#">informSimulationManagerOfDamageEvent</a>             | 4  |
| <a href="#">informSimulationManagerOfMobilityDisabledEvent</a>   | 5  |
| <a href="#">informSimulationManagerOfFireDisabledEvent</a>       | 6  |
| <a href="#">informSimulationManagerOfRanOutOfFuelEvent</a>       | 7  |
| <a href="#">recallCheckpointData</a>                             | 8  |
| <a href="#">recallInitialParameters</a>                          | 9  |
| <a href="#">initiateTetherLead</a>                               | 10 |
| <a href="#">initiateTetherFollow</a>                             | 11 |
| <a href="#">untether</a>                                         | 12 |
| <a href="#">initiateServiceStationResupply</a>                   | 13 |
| <a href="#">initiateTailgateResupply</a>                         | 14 |
| <a href="#">initiateHitchLead</a>                                | 15 |
| <a href="#">initiateHitchFollow</a>                              | 16 |
| <a href="#">unhitch</a>                                          | 17 |
| <a href="#">mount</a>                                            | 18 |
| <a href="#">dismount</a>                                         | 19 |
| <a href="#">startDailyReadinessCheck</a>                         | 20 |
| <a href="#">stopDailyReadinessCheck</a>                          | 21 |
| <a href="#">dataQuery</a>                                        | 22 |

|                                 |     |
|---------------------------------|-----|
| statusRequest                   | 23  |
| sendObjectStateData             | 24  |
| reconstitute                    | 25  |
| lockSiteConfiguration           | 26  |
| unlockSiteConfiguration         | 27  |
| updateSiteConfiguration         | 28  |
| querySiteConfiguration          | 29  |
| tetheringInformation            | 30  |
| mountIntent                     | 31  |
| acceptSubscription              | 33  |
| unsubscribe                     | 34  |
| teleportEntity                  | 35  |
| changeAggregateState            | 36  |
| requestStartPDU                 | 37  |
| wakeupGetReadyForInitialization | 38  |
| initializeInternalParameters    | 39  |
| sendPlanData                    | 40  |
| synchronizeInternalClocks       | 41  |
| run                             | 42  |
| saveInternalParameters          | 43  |
| simulateMalfunction             | 44  |
| joinExercise                    | 45  |
| resignExercise                  | 46  |
| timeAdvance                     | 47  |
| taccsfLosRequestType1           | 100 |
| taccsfLosRequestType2           | 101 |

Named by [ActionRequest](#), [ActionRequestR](#), [ActionRequestToObject](#), [ActionRequestToObjectR](#).

## ActionResultEnum32 ENUMERATION

Request status

Held as u32.

|                             |     |
|-----------------------------|-----|
| other                       | 0   |
| pending                     | 1   |
| executing                   | 2   |
| partiallyComplete           | 3   |
| complete                    | 4   |
| requestRejected             | 5   |
| retransmitRequestNow        | 6   |
| retransmitRequestLater      | 7   |
| invalidTimeParameters       | 8   |
| simulationTimeExceeded      | 9   |
| requestDone                 | 10  |
| taccsfLosReplyType1         | 100 |
| taccsfLosReplyType2         | 101 |
| joinExerciseRequestRejected | 201 |

Named by [ActionResponseFromObject](#), [ActionResponseFromObjectR](#).

## ActiveSonarEnum16 ENUMERATION

Acoustic system name

Held as u16.

|          |   |
|----------|---|
| other    | 0 |
| anBqq5   | 1 |
| anSsq62  | 2 |
| anSqs23  | 3 |
| anSqs26  | 4 |
| anSqs53  | 5 |
| alfs     | 6 |
| lfa      | 7 |
| anAqs901 | 8 |
| anAqs902 | 9 |

Named by [ActiveSonar](#).

## ActiveSonarFunctionCodeEnum8 ENUMERATION

The current function being performed by the sonar

Held as u8.

|                               |   |
|-------------------------------|---|
| other                         | 0 |
| platformSearchDetectTrack     | 1 |
| navigation                    | 2 |
| mineHunting                   | 3 |
| weaponSearchDetectTrackDetect | 4 |

Named by [ActiveSonar](#).

## ActiveSonarScanPatternEnum16 ENUMERATION

Acoustic scan pattern

Held as u16.

|                    |   |
|--------------------|---|
| scanPatternNotUsed | 0 |
| conical            | 1 |
| helical            | 2 |
| raster             | 3 |
| sectorSearch       | 4 |
| continuousSearch   | 5 |

Named by [ActiveSonarBeam](#).

## AggregateStateEnum8 ENUMERATION

Aggregate state

Held as u8.

|                        |   |
|------------------------|---|
| other                  | 0 |
| aggregated             | 1 |
| disaggregated          | 2 |
| fullyDisaggregated     | 3 |
| pseudoDisaggregated    | 4 |
| partiallyDisaggregated | 5 |

Named by [AggregateEntity](#).

## AmplitudeAngleModulationTypeEnum16 ENUMERATION

Detailed modulation types for Amplitude and Angle Modulation

Held as u16.

|                   |   |
|-------------------|---|
| other             | 0 |
| amplitudeAndAngle | 1 |

Named by [RFModulationTypeVariantStruct](#).

## AmplitudeModulationTypeEnum16 ENUMERATION

Detailed modulation types for Amplitude Modulation

Held as u16.

|                           |    |
|---------------------------|----|
| other                     | 0  |
| audioFrequencyShiftKeying | 1  |
| amplitudeModulation       | 2  |
| continuousWaveModulation  | 3  |
| doubleSideband            | 4  |
| independentSideband       | 5  |
| ssbLowerSideband          | 6  |
| ssbFullCarrier            | 7  |
| ssbReducedCarrier         | 8  |
| ssbUpperSideband          | 9  |
| vestigialSideband         | 10 |

Named by [RFModulationTypeVariantStruct](#).

## AngleModulationTypeEnum16 ENUMERATION

Detailed modulation types for Angle Modulation

Held as u16.

|                      |   |
|----------------------|---|
| other                | 0 |
| frequencyModulation  | 1 |
| frequencyShiftKeying | 2 |
| phaseModulation      | 3 |

Named by [RFModulationTypeVariantStruct](#).

## AntennaPatternTypeEnum32 ENUMERATION

The radiation pattern from an antenna.

Held as u32.

|                   |   |
|-------------------|---|
| omniDirectional   | 0 |
| beam              | 1 |
| sphericalHarmonic | 2 |

## ArticulatedPartsTypeEnum32 ENUMERATION

Articulated part type class

Held as u32.

|                              |      |                          |      |                         |      |
|------------------------------|------|--------------------------|------|-------------------------|------|
| other                        | 0    | primaryBoom              | 3488 | primaryDefenseSystems1  | 5056 |
| rudder                       | 1024 | primaryLauncherArm       | 3520 | primaryDefenseSystems2  | 5088 |
| leftFlap                     | 1056 | otherFixedPositionParts  | 3552 | primaryDefenseSystems3  | 5120 |
| rightFlap                    | 1088 | landingGearNose          | 3584 | primaryDefenseSystems4  | 5152 |
| leftAileron                  | 1120 | landingGearLeftMain      | 3616 | primaryDefenseSystems5  | 5184 |
| rightAileron                 | 1152 | landingGearRightMain     | 3648 | primaryDefenseSystems6  | 5216 |
| helicopterMainRotor          | 1184 | leftSideSecondaryWeapon  | 3680 | primaryDefenseSystems7  | 5248 |
| helicopterTailRotor          | 1216 | BayDoors                 |      | primaryDefenseSystems8  | 5280 |
| otherAircraftControlSurfaces | 1248 | rightSideSecondaryWeapon | 3712 | primaryDefenseSystems9  | 5312 |
| propellerNumber1             | 1280 | BayDoors                 |      | primaryDefenseSystems10 | 5344 |
| propellerNumber2             | 1312 | primaryTurretNumber1     | 4096 | primaryRadar1           | 5376 |
| propellerNumber3             | 1344 | primaryTurretNumber2     | 4128 | primaryRadar2           | 5408 |
| propellerNumber4             | 1376 | primaryTurretNumber3     | 4160 | primaryRadar3           | 5440 |
| leftStabilatorStabilator     | 1408 | primaryTurretNumber4     | 4192 | primaryRadar4           | 5472 |
| Number1                      |      | primaryTurretNumber5     | 4224 | primaryRadar5           | 5504 |
| rightStabilatorStabilator    | 1440 | primaryTurretNumber6     | 4256 | primaryRadar6           | 5536 |
| Number2                      |      | primaryTurretNumber7     | 4288 | primaryRadar7           | 5568 |
| leftRuddervatorRuddervator   | 1472 | primaryTurretNumber8     | 4320 | primaryRadar8           | 5600 |
| Number1                      |      | primaryTurretNumber9     | 4352 | primaryRadar9           | 5632 |
| rightRuddervator             | 1504 | primaryTurretNumber10    | 4384 | primaryRadar10          | 5664 |
| RuddervatorNumber2           |      | primaryGunNumber1        | 4416 | secondaryTurretNumber1  | 5696 |
| leftLeadingEdgeFlapSlat      | 1536 | primaryGunNumber2        | 4448 | secondaryTurretNumber2  | 5728 |
| rightLeadingEdgeFlapSlat     | 1568 | primaryGunNumber3        | 4480 | secondaryTurretNumber3  | 5760 |
| periscope                    | 2048 | primaryGunNumber4        | 4512 | secondaryTurretNumber4  | 5792 |
| genericAntenna               | 2080 | primaryGunNumber5        | 4544 | secondaryTurretNumber5  | 5824 |
| snorkel                      | 2112 | primaryGunNumber6        | 4576 | secondaryTurretNumber6  | 5856 |
| otherExtendableParts         | 2144 | primaryGunNumber7        | 4608 | secondaryTurretNumber7  | 5888 |
| landingGear                  | 3072 | primaryGunNumber8        | 4640 | secondaryTurretNumber8  | 5920 |
| tailHook                     | 3104 | primaryGunNumber9        | 4672 | secondaryTurretNumber9  | 5952 |
| speedBrake                   | 3136 | primaryGunNumber10       | 4704 | secondaryTurretNumber10 | 5984 |
| leftWeaponBayDoors           | 3168 | primaryLauncher1         | 4736 | secondaryGunNumber1     | 6016 |
| rightWeaponBayDoors          | 3200 | primaryLauncher2         | 4768 | secondaryGunNumber2     | 6048 |
| tankOrAPChatch               | 3232 | primaryLauncher3         | 4800 | secondaryGunNumber3     | 6080 |
| wingsweep                    | 3264 | primaryLauncher4         | 4832 | secondaryGunNumber4     | 6112 |
| bridgeLauncher               | 3296 | primaryLauncher5         | 4864 | secondaryGunNumber5     | 6144 |
| bridgeSection1               | 3328 | primaryLauncher6         | 4896 | secondaryGunNumber6     | 6176 |
| bridgeSection2               | 3360 | primaryLauncher7         | 4928 | secondaryGunNumber7     | 6208 |
| bridgeSection3               | 3392 | primaryLauncher8         | 4960 | secondaryGunNumber8     | 6240 |
| primaryBlade1                | 3424 | primaryLauncher9         | 4992 | secondaryGunNumber9     | 6272 |
| primaryBlade2                | 3456 | primaryLauncher10        | 5024 | secondaryGunNumber10    | 6304 |



|                           |      |
|---------------------------|------|
| secondaryLauncher1        | 6336 |
| secondaryLauncher2        | 6368 |
| secondaryLauncher3        | 6400 |
| secondaryLauncher4        | 6432 |
| secondaryLauncher5        | 6464 |
| secondaryLauncher6        | 6496 |
| secondaryLauncher7        | 6528 |
| secondaryLauncher8        | 6560 |
| secondaryLauncher9        | 6592 |
| secondaryLauncher10       | 6624 |
| secondaryDefenseSystems1  | 6656 |
| secondaryDefenseSystems2  | 6688 |
| secondaryDefenseSystems3  | 6720 |
| secondaryDefenseSystems4  | 6752 |
| secondaryDefenseSystems5  | 6784 |
| secondaryDefenseSystems6  | 6816 |
| secondaryDefenseSystems7  | 6848 |
| secondaryDefenseSystems8  | 6880 |
| secondaryDefenseSystems9  | 6912 |
| secondaryDefenseSystems10 | 6944 |
| secondaryRadar1           | 6976 |
| secondaryRadar2           | 7008 |
| secondaryRadar3           | 7040 |
| secondaryRadar4           | 7072 |
| secondaryRadar5           | 7104 |
| secondaryRadar6           | 7136 |
| secondaryRadar7           | 7168 |
| secondaryRadar8           | 7200 |
| secondaryRadar9           | 7232 |
| secondaryRadar10          | 7264 |
| deckElevator1             | 7296 |
| deckElevator2             | 7328 |
| catapult1                 | 7360 |
| catapult2                 | 7392 |
| jetBlastDeflector1        | 7424 |
| jetBlastDeflector2        | 7456 |
| arrestorWires1            | 7488 |
| arrestorWires2            | 7520 |
| arrestorWires3            | 7552 |
| wingOrRotorFold           | 7584 |
| fuselageFold              | 7616 |
| cargoDoor                 | 7648 |
| cargoRamp                 | 7680 |
| airToAirRefuelingBoom     | 7712 |

Named by [ArticulatedPartsStruct](#).

## ArticulatedTypeMetricEnum32 ENUMERATION

Articulated part type metric

Held as u32.

|               |    |
|---------------|----|
| position      | 1  |
| positionRate  | 2  |
| extension     | 3  |
| extensionRate | 4  |
| x             | 5  |
| xRate         | 6  |
| y             | 7  |
| yRate         | 8  |
| z             | 9  |
| zRate         | 10 |
| azimuth       | 11 |
| azimuthRate   | 12 |
| elevation     | 13 |
| elevationRate | 14 |
| rotation      | 15 |
| rotationRate  | 16 |



Named by [ArticulatedPartsStruct](#).

BeamFunctionCodeEnum8

ENUMERATION

Beam function

Held as u8.

|                               |    |
|-------------------------------|----|
| other                         | 0  |
| search                        | 1  |
| heightFinder                  | 2  |
| acquisition                   | 3  |
| tracking                      | 4  |
| acquisitionAndTracking        | 5  |
| commandGuidance               | 6  |
| illumination                  | 7  |
| rangeOnlyRadar                | 8  |
| missileBeacon                 | 9  |
| missileFuze                   | 10 |
| activeRadarMissileSeeker      | 11 |
| jammer                        | 12 |
| iff                           | 13 |
| navigationalOrWeather         | 14 |
| meteorological                | 15 |
| dataTransmission              | 16 |
| navigationalDirectionalBeacon | 17 |
| timeSharedSearch              | 20 |
| timeSharedAcquisition         | 21 |
| timeSharedTrack               | 22 |
| timeSharedCommand             | 23 |
| Guidance                      |    |
| timeSharedIllumination        | 24 |
| timeSharedJamming             | 25 |

Named by [EmitterBeam](#).

BreachedStatusEnum8

ENUMERATION

Breached appearance

Held as u8.

|                   |   |
|-------------------|---|
| noBreaching       | 0 |
| slightBreaching   | 1 |
| moderateBreaching | 2 |
| cleared           | 3 |

Named by [BreachablePointObject](#), [BreachablePointObjectTransaction](#), [BreachableSegmentStruct](#), [BreachedStatusArray8](#), [MinefieldObject](#), [MinefieldObjectTransaction](#).

CamouflageEnum32

ENUMERATION

Camouflage type

Held as u32.

|                    |   |
|--------------------|---|
| uniformPaintScheme | 0 |
| desertCamouflage   | 1 |
| winterCamouflage   | 2 |
| forestCamouflage   | 3 |
| genericCamouflage  | 4 |

Named by [PhysicalEntity](#).

ChemicalContentEnum32

ENUMERATION

Smoke chemical content

Held as u32.

|              |   |                  |   |
|--------------|---|------------------|---|
| other        | 0 | whitePhosphorous | 2 |
| hydrochloric | 1 | redPhosphorous   | 3 |

Named by [BurstPointObject](#), [BurstPointObjectTransaction](#), [ExhaustSmokeStruct](#).

## CollisionTypeEnum8 ENUMERATION

Collision type

Held as u8.

|           |   |
|-----------|---|
| inelastic | 0 |
| elastic   | 1 |

Named by [Collision](#), [CollisionElastic](#).

## CombinationModulationTypeEnum16 ENUMERATION

Detailed modulation types for Combination Modulation

Held as u16.

|                     |   |
|---------------------|---|
| other               | 0 |
| amplitudeAnglePulse | 1 |

Named by [RFModulationTypeVariantStruct](#).

## ComplianceStateEnum32 ENUMERATION

Life form compliance

Held as u32.

|                    |    |
|--------------------|----|
| other              | 0  |
| detained           | 1  |
| surrender          | 2  |
| usingFists         | 3  |
| verbalAbuse1       | 4  |
| verbalAbuse2       | 5  |
| verbalAbuse3       | 6  |
| passiveResistance1 | 7  |
| passiveResistance2 | 8  |
| passiveResistance3 | 9  |
| nonLethalWeapon1   | 10 |
| nonLethalWeapon2   | 11 |
| nonLethalWeapon3   | 12 |
| nonLethalWeapon4   | 13 |
| nonLethalWeapon5   | 14 |
| nonLethalWeapon6   | 15 |

Named by [Lifeform](#).

## ConstituentPartNatureEnum16 ENUMERATION

Relationship nature

Held as u16.

|                                                |    |
|------------------------------------------------|----|
| other                                          | 0  |
| hostFireableMunition                           | 1  |
| munitionCarriedAsCargo                         | 2  |
| fuelCarriedAsCargo                             | 3  |
| gunmountAttachedToHost                         | 4  |
| computerGeneratedForcesCarriedAsCargo          | 5  |
| vehicleCarriedAsCargo                          | 6  |
| emitterMountedOnHost                           | 7  |
| mobileCommandAndControlEntityCarriedAboardHost | 8  |
| entityStationedWithRespectToHost               | 9  |
| teamMemberInFormationWith                      | 10 |

Named by [ConstituentPartRelationshipStruct](#).

## ConstituentPartPositionEnum16 ENUMERATION

Relationship position

Held as u16.

|         |   |
|---------|---|
| other   | 0 |
| onTopOf | 1 |
| inside  | 2 |

Named by [ConstituentPartRelationshipStruct](#).

---

## ConstituentPartStationNameEnum16 ENUMERATION

Station name

Held as u16.

|                                |    |
|--------------------------------|----|
| other                          | 0  |
| aircraftWingstation            | 1  |
| shipsForwardGunmountStarboard  | 2  |
| shipsForwardGunmountPort       | 3  |
| shipsForwardGunmountCenterline | 4  |
| shipsAftGunmountStarboard      | 5  |
| shipsAftGunmountPort           | 6  |
| shipsAftGunmountCenterline     | 7  |
| forwardTorpedoTube             | 8  |
| aftTorpedoTube                 | 9  |
| bombBay                        | 10 |
| cargoBay                       | 11 |
| truckBed                       | 12 |
| trailerBed                     | 13 |
| wellDeck                       | 14 |
| onStationRangeBearing          | 15 |
| onStationXYZ                   | 16 |

---

## CryptographicModeEnum32 ENUMERATION

Represents the encryption mode of a cryptographic system.

Held as u32.

|                    |   |
|--------------------|---|
| basebandEncryption | 0 |
| diphaseEncryption  | 1 |

Named by [RadioTransmitter](#).

---

## CryptographicSystemTypeEnum16 ENUMERATION

Identifies the type of cryptographic equipment

Held as u16.

|                               |    |
|-------------------------------|----|
| other                         | 0  |
| ky28                          | 1  |
| ky58                          | 2  |
| narrowSpectrumSecureVoiceNsve | 3  |
| wideSpectrumSecureVoiceWsve   | 4  |
| sincgarslcom                  | 5  |
| ky75                          | 6  |
| ky100                         | 7  |
| ky57                          | 8  |
| kyv5                          | 9  |
| link11KG40aPNtds              | 10 |
| link11BKG40aS                 | 11 |
| link11KG40ar                  | 12 |

Named by [RadioTransmitter](#).

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## DamageStatusEnum32 ENUMERATION

Damaged appearance

Held as u32.

|          |   |              |   |                |   |
|----------|---|--------------|---|----------------|---|
| noDamage | 0 | slightDamage | 1 | moderateDamage | 2 |
|----------|---|--------------|---|----------------|---|

destroyed 3

Named by [netn.SEFacility](#), [netn.SensorStruct](#), [ArealObject](#), [ArealObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [CraterObjectTransaction](#), [LinearSegmentStruct](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherPointObjectTransaction](#), [PhysicalEntity](#), [PointObject](#), [PointObjectTransaction](#), [RibbonBridgeObjectTransaction](#), [StructureObjectTransaction](#).

## DatumIdentifierEnum32 ENUMERATION

Datum ID

Held as u32.

|                                       |       |                              |       |
|---------------------------------------|-------|------------------------------|-------|
| highFidelityHAVEQUICKRadio            | 3000  | disSiteId                    | 15100 |
| blankingSectorAttributeRecord         | 3500  | disHostId                    | 15200 |
| angleDeceptionAttributeRecord         | 3501  | disEntityId                  | 15300 |
| falseTargetsAttributeRecord           | 3502  | mountIntent                  | 15400 |
| dEPrecisionAimpointRecord             | 4000  | tetherUnthetherCommandId     | 15500 |
| dEAreaAimpointRecord                  | 4001  | teleportEntityDataRecord     | 15510 |
| directedEnergyDamageDescriptionRecord | 4500  | disAggregateId               | 15600 |
| cryptoControl                         | 5000  | ownershipStatus              | 15800 |
| mode5STransponderLocation             | 5001  | loads                        | 20000 |
| mode5STransponderLocationError        | 5002  | crewMembers                  | 21000 |
| squitterAirbornePositionReport        | 5003  | crewMemberId                 | 21100 |
| squitterAirborneVelocityReport        | 5004  | health                       | 21200 |
| squitterSurfacePositionReport         | 5005  | jobAssignment                | 21300 |
| squitterIdentificationReport          | 5006  | fuel                         | 23000 |
| gicb                                  | 5007  | fuelQuantityLiters           | 23100 |
| squitterEventDrivenReport             | 5008  | fuelQuantityGallons          | 23105 |
| antennaLocation                       | 5009  | ammunition                   | 24000 |
| basicInteractive                      | 5010  | ammunitionQuantity120mmHeat  | 24001 |
| interactiveMode4Reply                 | 5011  | ammunitionQuantity120mmSabot | 24002 |
| interactiveMode5Reply                 | 5012  | ammunitionQuantity127mmM8    | 24003 |
| interactiveBasicMode5                 | 5013  | ammunitionQuantity127mmM20   | 24004 |
| interactiveBasicModeS                 | 5014  | ammunitionQuantity762mmM62   | 24005 |
| iOEffect                              | 5500  | ammunitionQuantityM250Ukl8a1 | 24006 |
| iOCommunicationsNode                  | 5501  | ammunitionQuantityM250Ukl8a3 | 24007 |
| entityIdentification                  | 10000 | ammunitionQuantity762mmM80   | 24008 |
| entityType                            | 11000 | ammunitionQuantity127mm      | 24009 |
| concatenated                          | 11100 | ammunitionQuantity762mm      | 24010 |
| entityTypeKind                        | 11110 | minesQuantity                | 24060 |
| entityTypeDomain                      | 11120 | minesType                    | 24100 |
| entityTypeCountry                     | 11130 | minesKind                    | 24110 |
| entityTypeCategory                    | 11140 | minesDomain                  | 24120 |
| entityTypeSubcategory                 | 11150 | minesCountry                 | 24130 |
| entityTypeSpecific                    | 11160 | minesCategory                | 24140 |
| entityTypeExtra                       | 11170 | minesSubcategory             | 24150 |
| forcedId                              | 11200 | minesExtra                   | 24160 |
| description                           | 11300 | minesDescription             | 24300 |
| alternativeEntityType                 | 12000 | cargo                        | 25000 |
| alternativeEntityTypeKind             | 12110 | vehicleMass                  | 26000 |
| alternativeEntityTypeDomain           | 12120 | supplyQuantity               | 27000 |
| alternativeEntityTypeCountry          | 12130 | armament                     | 28000 |
| alternativeEntityTypeCategory         | 12140 | status                       | 30000 |
| alternativeEntityTypeSubcategory      | 12150 | activateEntity               | 30010 |
| alternativeEntityTypeSpecific         | 12160 | subscriptionState            | 30100 |
| alternativeEntityTypeExtra            | 12170 | roundTripTimeDelay           | 30300 |
| alternativeEntityTypeDescription      | 12300 | tadiJMessageCountLabel0      | 30400 |
| entityMarking                         | 13000 | tadiJMessageCountLabel1      | 30401 |
| entityMarkingCharacters               | 13100 | tadiJMessageCountLabel2      | 30402 |
| crewId                                | 13200 | tadiJMessageCountLabel3      | 30403 |
| taskOrganization                      | 14000 | tadiJMessageCountLabel4      | 30404 |
| regimentName                          | 14200 | tadiJMessageCountLabel5      | 30405 |
| battalionName                         | 14300 | tadiJMessageCountLabel6      | 30406 |
| companyName                           | 14400 | tadiJMessageCountLabel7      | 30407 |
| platoonName                           | 14500 | tadiJMessageCountLabel8      | 30408 |
| squadName                             | 14520 | tadiJMessageCountLabel9      | 30409 |
| teamName                              | 14540 | tadiJMessageCountLabel10     | 30410 |
| bumperNumber                          | 14600 | tadiJMessageCountLabel11     | 30411 |
| vehicleNumber                         | 14700 | tadiJMessageCountLabel12     | 30412 |
| unitNumber                            | 14800 | tadiJMessageCountLabel13     | 30413 |
| disIdentity                           | 15000 | tadiJMessageCountLabel14     | 30414 |

|                           |       |                                          |       |
|---------------------------|-------|------------------------------------------|-------|
| tadilJMessageCountLabel15 | 30415 | primaryTargetLine                        | 39000 |
| tadilJMessageCountLabel16 | 30416 | exercise                                 | 40000 |
| tadilJMessageCountLabel17 | 30417 | exerciseState                            | 40010 |
| tadilJMessageCountLabel18 | 30418 | restartRefresh                           | 40015 |
| tadilJMessageCountLabel19 | 30419 | afatdsFileName                           | 40020 |
| tadilJMessageCountLabel20 | 30420 | terrainDatabase                          | 41000 |
| tadilJMessageCountLabel21 | 30421 | missions                                 | 42000 |
| tadilJMessageCountLabel22 | 30422 | missionId                                | 42100 |
| tadilJMessageCountLabel23 | 30423 | missionType                              | 42200 |
| tadilJMessageCountLabel24 | 30424 | missionRequestTimeStamp                  | 42300 |
| tadilJMessageCountLabel25 | 30425 | exerciseDescription                      | 43000 |
| tadilJMessageCountLabel26 | 30426 | name                                     | 43100 |
| tadilJMessageCountLabel27 | 30427 | entities                                 | 43200 |
| tadilJMessageCountLabel28 | 30428 | version                                  | 43300 |
| tadilJMessageCountLabel29 | 30429 | guiseMode                                | 43410 |
| tadilJMessageCountLabel30 | 30430 | simulationApplicationActiveStatus        | 43420 |
| tadilJMessageCountLabel31 | 30431 | simulationApplicationRoleRecord          | 43430 |
| position                  | 31000 | simulationApplicationState               | 43440 |
| routeWaypointType         | 31010 | visualOutputMode                         | 44000 |
| milGrid10                 | 31100 | simulationManagerRole                    | 44100 |
| geocentricCoordinates     | 31200 | simulationManagerSiteId                  | 44110 |
| geocentricCoordinateX     | 31210 | simulationManagerApplicationId           | 44120 |
| geocentricCoordinateY     | 31220 | simulationManagerEntityId                | 44130 |
| geocentricCoordinateZ     | 31230 | simulationManagerActiveStatus            | 44140 |
| latitude                  | 31300 | afterActiveReviewRole                    | 44200 |
| longitude                 | 31400 | afterActiveReviewSiteId                  | 44210 |
| lineOfSight               | 31500 | afterActiveApplicationId                 | 44220 |
| lineOfSightX              | 31510 | afterActiveReviewEntityId                | 44230 |
| lineOfSightY              | 31520 | afterActiveReviewActiveStatus            | 44240 |
| lineOfSightZ              | 31530 | exerciseLoggerRole                       | 44300 |
| altitude                  | 31600 | exerciseLoggerSiteId                     | 44310 |
| destinationLatitude       | 31700 | exerciseLoggerApplicationId              | 44320 |
| destinationLongitude      | 31800 | exerciseEntityId                         | 44330 |
| destinationAltitude       | 31900 | exerciseLoggerActiveStatus               | 44340 |
| orientation               | 32000 | syntheticEnvironmentManagerRole          | 44400 |
| hullHeadingAngle          | 32100 | syntheticEnvironmentManagerSiteId        | 44410 |
| hullPitchAngle            | 32200 | syntheticEnvironmentManagerApplicationId | 44420 |
| rollAngle                 | 32300 | syntheticEnvironmentManagerEntityId      | 44430 |
| rollAngleX                | 32500 | syntheticEnvironmentManagerActiveStatus  | 44440 |
| rollAngleY                | 32600 | simnetDisTranslatorRole                  | 44500 |
| rollAngleZ                | 32700 | simnetDisTranslatorSiteId                | 44510 |
| appearance                | 33000 | simnetDisTranslatorApplicationId         | 44520 |
| ambientLighting           | 33100 | simnetDisTranslatorEntityId              | 44530 |
| lights                    | 33101 | simnetDisTranslatorActiveStatus          | 44540 |
| paintScheme               | 33200 | applicationRate                          | 45000 |
| smoke                     | 33300 | applicationTime                          | 45005 |
| trailingEffects           | 33400 | applicationTimestep                      | 45010 |
| flaming                   | 33500 | feedbackTime                             | 45020 |
| marking                   | 33600 | simulationRate                           | 45030 |
| minePlowsAttached         | 33710 | simulationTime                           | 45040 |
| mineRollersAttached       | 33720 | simulationTimestep                       | 45050 |
| tankTurretAzimuth         | 33730 | timeInterval                             | 45060 |
| failuresAndMalfunctions   | 34000 | timeLatency                              | 45070 |
| age                       | 34100 | timeScheme                               | 45080 |
| kilometers                | 34110 | exerciseElapsedTime                      | 46000 |
| damage                    | 35000 | elapsedTime                              | 46010 |
| cause                     | 35050 | environment                              | 50000 |
| mobilityKill              | 35100 | weather                                  | 51000 |
| firePowerKill             | 35200 | weatherCondition                         | 51010 |
| personnelCasualties       | 35300 | thermalCondition                         | 51100 |
| velocity                  | 36000 | thermalVisibilityFloatingPoint32         | 51110 |
| xVelocity                 | 36100 | thermalVisibilityUnsignedInteger32       | 51111 |
| yVelocity                 | 36200 | time                                     | 52000 |
| zVelocity                 | 36300 | timeString                               | 52001 |
| speed                     | 36400 | timeOfDayDiscrete                        | 52100 |
| acceleration              | 37000 | timeOfDayContinuous                      | 52200 |
| xAcceleration             | 37100 | timeMode                                 | 52300 |
| yAcceleration             | 37200 | timeScene                                | 52305 |
| zAcceleration             | 37300 | currentHour                              | 52310 |
| engineStatus              | 38100 | currentMinute                            | 52320 |

|                                   |       |                                               |       |
|-----------------------------------|-------|-----------------------------------------------|-------|
| currentSecond                     | 52330 | humidity                                      | 57200 |
| azimuth                           | 52340 | visibility                                    | 57300 |
| maximumElevation                  | 52350 | winds                                         | 57400 |
| timeZone                          | 52360 | speed2                                        | 57410 |
| timeRate                          | 52370 | windSpeedKnots                                | 57411 |
| simulationTime2                   | 52380 | windDirection                                 | 57420 |
| timeSunriseEnabled                | 52400 | windDirectionDegrees                          | 57421 |
| sunriseHour                       | 52410 | rainsoak                                      | 57500 |
| sunriseMinute                     | 52420 | tideSpeed                                     | 57610 |
| sunriseSecond                     | 52430 | tideSpeedKnots                                | 57611 |
| sunriseAzimuth                    | 52440 | tideDirection                                 | 57620 |
| timeSunsetEnabled                 | 52500 | tideDirectionDegrees                          | 57621 |
| sunsetHour                        | 52510 | haze                                          | 58000 |
| sunsetHour2                       | 52511 | hazeVisibilityMeters                          | 58100 |
| sunsetMinute                      | 52520 | hazeVisibilityMiles                           | 58105 |
| sunsetSecond                      | 52530 | hazeDensity                                   | 58200 |
| date                              | 52600 | hazeCeilingMeters                             | 58430 |
| dateEuropean                      | 52601 | hazeCeilingFeet                               | 58435 |
| dateUs                            | 52602 | contaminantsAndObscurants                     | 59000 |
| month                             | 52610 | contaminantObscurantType                      | 59100 |
| day                               | 52620 | persistence                                   | 59110 |
| year                              | 52630 | chemicalDosage                                | 59115 |
| clouds                            | 53000 | chemicalAirConcentration                      | 59120 |
| cloudLayerEnable                  | 53050 | chemicalGroundDeposition                      | 59125 |
| cloudLayerSelection               | 53060 | chemicalMaximumGroundDeposition               | 59130 |
| cloudVisibility                   | 53100 | chemicalDosageThreshold                       | 59135 |
| baseAltitudeMeters                | 53200 | biologicalDosage                              | 59140 |
| baseAltitudeFeet                  | 53250 | biologicalAirConcentration                    | 59145 |
| ceilingMeters                     | 53300 | biologicalDosageThreshold                     | 59150 |
| ceilingFeet                       | 53350 | biologicalBinnedParticleCount                 | 59155 |
| characteristics                   | 53400 | radiologicalDosage                            | 59160 |
| concentrationLength               | 53410 | communications                                | 60000 |
| transmittance                     | 53420 | channelType                                   | 61100 |
| radiance                          | 53430 | channelType2                                  | 61101 |
| precipitation                     | 54000 | channelIdentification                         | 61200 |
| rain                              | 54100 | alphaIdentification                           | 61300 |
| fog                               | 55000 | radiolIdentification                          | 61400 |
| visibilityMeters                  | 55100 | landLineIdentification                        | 61500 |
| visibilityMetersUnsignedInteger32 | 55101 | intercomIdentification                        | 61600 |
| visibilityMiles                   | 55105 | groupNetworkChannelNumber                     | 61700 |
| fogDensity                        | 55200 | radioCommunicationsStatus                     | 62100 |
| base                              | 55300 | stationaryRadioTransmittersDefaultTime        | 62200 |
| viewLayerFromAbove                | 55401 | movingRadioTransmittersDefaultTime            | 62300 |
| transitionRange                   | 55410 | stationaryRadioSignalsDefaultTime             | 62400 |
| bottomMeters                      | 55420 | movingRadioSignalDefaultTime                  | 62500 |
| bottomFeet                        | 55425 | radiolInitializationTransecSecurityKey        | 63101 |
| fogCeilingMeters                  | 55430 | radiolInitializationInternalNoiseLevel        | 63102 |
| fogCeilingFeet                    | 55435 | radiolInitializationSquelchThreshold          | 63103 |
| heavenlyBodies                    | 56000 | radiolInitializationAntennaLocation           | 63104 |
| sun                               | 56100 | radiolInitializationAntennaPatternType        | 63105 |
| sunVisible                        | 56105 | radiolInitializationAntennaPatternLength      | 63106 |
| sunPosition                       | 56110 | radiolInitializationBeamDefinition            | 63107 |
| sunPositionElevationDegrees       | 56111 | radiolInitializationTransmitHeartbeatTime     | 63108 |
| sunPositionAzimuth                | 56120 | radiolInitializationTransmitDistanceThreshold | 63109 |
| sunPositionAzimuthDegrees         | 56121 | radioChannelInitializationLockoutId           | 63110 |
| sunPositionElevation              | 56130 | radioChannelInitializationHopsetId            | 63111 |
| sunPositionIntensity              | 56140 | radioChannelInitializationPresetFrequency     | 63112 |
| moon                              | 56200 | radioChannelInitializationFrequencySyncTime   | 63113 |
| moonVisible                       | 56205 | radioChannelInitializationComsecKey           | 63114 |
| moonPosition                      | 56210 | radioChannelInitializationAlpha               | 63115 |
| moonPositionAzimuth               | 56220 | algorithmParameters                           | 70000 |
| moonPositionAzimuthDegrees        | 56221 | deadReckoningAlgorithmDra                     | 71000 |
| moonPositionElevation             | 56230 | draLocationThreshold                          | 71100 |
| moonPositionElevationDegrees      | 56231 | draOrientationThreshold                       | 71200 |
| moonPositionIntensity             | 56240 | draTimeThreshold                              | 71300 |
| horizon                           | 56310 | simulationManagementParameters                | 72000 |
| horizonAzimuth                    | 56320 | checkpointInterval                            | 72100 |
| horizonElevation                  | 56330 | transmitterTimeThreshold                      | 72600 |
| horizonHeading                    | 56340 | receiverTimeThreshold                         | 72700 |
| horizonIntensity                  | 56350 | interoperabilityMode                          | 73000 |

|                                        |        |                                        |        |
|----------------------------------------|--------|----------------------------------------|--------|
| simnetDataCollection                   | 74000  | areaOfInterestGroundImpactCircleCenter | 100632 |
| eventId                                | 75000  | Longitude                              |        |
| sourceSiteId                           | 75100  | areaOfInterestGroundImpactCircleRadius | 100633 |
| sourceHostId                           | 75200  | areaOfInterestType                     | 100634 |
| articulatedParts                       | 90000  | targetAggregateId                      | 100640 |
| articulatedPartsPartId                 | 90050  | gicIdentificationNumber                | 100650 |
| articulatedPartsIndex                  | 90070  | estimatedTimeOfFlightToTbmImpact       | 100660 |
| articulatedPartsPosition               | 90100  | estimatedInterceptTime                 | 100661 |
| articulatedPartsPositionRate           | 90200  | estimatedTimeOfFlightToNextWaypoint    | 100662 |
| articulatedPartsExtension              | 90300  | entityTrackEquipmentData               | 100700 |
| articulatedPartsExtensionRate          | 90400  | emissionEwData                         | 100800 |
| articulatedPartsX                      | 90500  | appearanceData                         | 100900 |
| articulatedPartsXRate                  | 90600  | commandOrderData                       | 101000 |
| articulatedPartsY                      | 90700  | environmentalData                      | 101100 |
| articulatedPartsYRate                  | 90800  | significantEventData                   | 101200 |
| articulatedPartsZ                      | 90900  | operatorActionData                     | 101300 |
| articulatedPartsZRate                  | 91000  | adaEngagementMode                      | 101310 |
| articulatedPartsAzimuth                | 91100  | adaShootingStatus                      | 101320 |
| articulatedPartsAzimuthRate            | 91200  | adaMode                                | 101321 |
| articulatedPartsElevation              | 91300  | adaRadarStatus                         | 101330 |
| articulatedPartsElevationRate          | 91400  | shootCommand                           | 101340 |
| articulatedPartsRotation               | 91500  | adaWeaponStatus                        | 101350 |
| articulatedPartsRotationRate           | 91600  | adaFiringDisciple                      | 101360 |
| draAngularXVelocity                    | 100001 | orderStatus                            | 101370 |
| draAngularYVelocity                    | 100002 | timeSynchronization                    | 101400 |
| draAngularZVelocity                    | 100003 | tomahawkData                           | 101500 |
| appearanceTrailingEffects              | 100004 | numberOfDetonations                    | 102100 |
| appearanceHatch                        | 100005 | numberOfIntercepts                     | 102200 |
| appearanceCharacterSet                 | 100008 | obtControlMt201                        | 200201 |
| capabilityAmmunitionSupplier           | 100010 | sensorDataMt202                        | 200202 |
| capabilityMiscellaneousSupplier        | 100011 | environmentalDataMt203                 | 200203 |
| capabilityRepairProvider               | 100012 | ownershipDataMt204                     | 200204 |
| articulationParameter                  | 100014 | acousticContactDataMt205               | 200205 |
| articulationParameterType              | 100047 | sonobuoyDataMt207                      | 200207 |
| articulationParameterValue             | 100048 | sonobuoyContactDataMt210               | 200210 |
| timeOfDayScene                         | 100058 | heloControlMt211                       | 200211 |
| latitudeNorthLocationOfWeatherCell     | 100061 | esmControlData                         | 200213 |
| longitudeEastLocationOfWeatherCell     | 100063 | esmContactDataMt214                    | 200214 |
| tacticalDriverStatus                   | 100068 | esmEmitterDataMt215                    | 200215 |
| sonarSystemStatus                      | 100100 | weaponDefinitionDataMt217              | 200216 |
| accomplishedAccept                     | 100160 | weaponPresetDataMt217                  | 200217 |
| upperLatitude                          | 100161 | obtControlMt301                        | 200301 |
| latitudeSouthLocationOfWeatherCell     | 100162 | sensorDataMt302                        | 200302 |
| westernLongitude                       | 100163 | environmentalDataMt303m                | 200303 |
| longitudeWestLocationOfWeatherCell     | 100164 | ownershipDataMt304                     | 200304 |
| cDROMNumberDiskIDForTerrain            | 100165 | acousticContactDataMt305               | 200305 |
| dTEDDiskID                             | 100166 | sonobuoyDataMt307                      | 200307 |
| altitude1                              | 100167 | sonobuoyContactDataMt310               | 200310 |
| tacticalSystemStatus                   | 100169 | heloScenarioEquipmentStatus            | 200311 |
| jtidsStatus                            | 100170 | esmControlDataMt313                    | 200313 |
| tadilJStatus                           | 100171 | esmContactDataMt314                    | 200314 |
| dsddStatus                             | 100172 | esmEmitterDataMt315                    | 200315 |
| weaponSystemStatus                     | 100200 | weaponDefinitionDataMt316              | 200316 |
| subsystemStatus                        | 100205 | weaponPresetDataMt317                  | 200317 |
| numberOfInterceptorsFired              | 100206 | pairingAssociationEMT56                | 200400 |
| numberOfInterceptorDetonations         | 100207 | pointerEMT57                           | 200401 |
| numberOfMessageBuffersDropped          | 100208 | reportingResponsibilityEMT58           | 200402 |
| satelliteSensorBackgroundYearDay       | 100213 | trackNumberEMT59                       | 200403 |
| satelliteSensorBackgroundHourMinute    | 100214 | idForLink11ReportingEMT60              | 200404 |
| scriptNumber                           | 100218 | remoteTrackEMT62                       | 200405 |
| entityTrackUpdateData                  | 100300 | link11ErrorRateEMT63                   | 200406 |
| localForceTraining                     | 100400 | trackQualityEMT64                      | 200407 |
| entityTrackIdentityData                | 100500 | gridlockEMT65                          | 200408 |
| entityForTrackEvent                    | 100510 | killEMT66                              | 200409 |
| iffFriendFoeStatus                     | 100520 | trackIdChangeResolutionEMT68           | 200410 |
| engagementData                         | 100600 | weaponsStatusEMT69                     | 200411 |
| targetLatitude                         | 100610 | link11OperatorEMT70                    | 200412 |
| targetLongitude                        | 100620 | forceTrainingTransmitEMT71             | 200413 |
| areaOfInterestGroundImpactCircleCenter | 100631 | forceTrainingReceiveEMT72              | 200414 |
| Latitude                               |        | interceptorAmplificationEMT75          | 200415 |

|                               |        |                                              |        |
|-------------------------------|--------|----------------------------------------------|--------|
| consumablesEMT78              | 200416 | alternativeEntityTypeKind1                   | 240010 |
| link11LocalTrackQualityEMT95  | 200417 | alternativeEntityTypeDomain1                 | 240011 |
| dlrpEMT19                     | 200418 | alternativeEntityTypeCountry1                | 240012 |
| forceOrderEMT52               | 200419 | alternativeEntityTypeCategory1               | 240013 |
| wilcoCantcoEMT53              | 200420 | alternativeEntityTypeSubCategory             | 240014 |
| emcBearingEMT54               | 200421 | alternativeEntityTypeSpecific1               | 240015 |
| changeTrackEligibilityEMT55   | 200422 | alternativeEntityTypeExtra1                  | 240016 |
| landMassReferencePoint        | 200423 | entityLocationX                              | 240017 |
| systemReferencePoint          | 200424 | entityLocationY                              | 240018 |
| puAmplification               | 200425 | entityLocationZ                              | 240019 |
| setDrift                      | 200426 | entityLinearVelocityX                        | 240020 |
| beginInitializationMt1        | 200427 | entityLinearVelocityY                        | 240021 |
| statusAndControlMt3           | 200428 | entityLinearVelocityZ                        | 240022 |
| scintillationChangeMt39       | 200429 | entityOrientationPsi                         | 240023 |
| link11IdControlMt61           | 200430 | entityOrientationTheta                       | 240024 |
| puGuardList                   | 200431 | entityOrientationPhi                         | 240025 |
| windsAloftMt14                | 200432 | deadReckoningAlgorithm                       | 240026 |
| surfaceWindsMt15              | 200433 | deadReckoningLinearAccelerationX             | 240027 |
| seaStateMt17                  | 200434 | deadReckoningLinearAccelerationY             | 240028 |
| magneticVariationMt37         | 200435 | deadReckoningLinearAccelerationZ             | 240029 |
| trackEligibilityMt29          | 200436 | deadReckoningAngularVelocityX                | 240030 |
| trainingTrackNotification     | 200437 | deadReckoningAngularVelocityY                | 240031 |
| tacanDataMt32                 | 200501 | deadReckoningAngularVelocityZ                | 240032 |
| interceptorAmplificationMt75  | 200502 | entityAppearance                             | 240033 |
| tacanAssignmentMt76           | 200503 | entityMarkingCharacterSet                    | 240034 |
| autopilotStatusMt77           | 200504 | entityMarking11Bytes                         | 240035 |
| consumablesMt78               | 200505 | capability                                   | 240036 |
| downlinkMt79                  | 200506 | numberArticulationParameters                 | 240037 |
| tinReportMt80                 | 200507 | articulationParameterId                      | 240038 |
| specialPointControlMt81       | 200508 | articulationParameterType2                   | 240039 |
| controlDiscretesMt82          | 200509 | articulationParameterValue2                  | 240040 |
| requestTargetDiscretesMt83    | 200510 | typeOfStores                                 | 240041 |
| targetDiscretesMt84           | 200511 | quantityOfStores                             | 240042 |
| replyDiscretesMt85            | 200512 | fuelQuantity                                 | 240043 |
| commandManeuversMt86          | 200513 | radarSystemStatus                            | 240044 |
| targetDataMt87                | 200514 | radioCommunicationSystemStatus               | 240045 |
| targetPointerMt88             | 200515 | defaultTimeForRadioTransmissionForStationary | 240046 |
| interceptDataMt89             | 200516 | Transmitters                                 |        |
| decrementMissileInventoryMt90 | 200517 | defaultTimeForRadioTransmissionForMoving     | 240047 |
| link4aAlertMt91               | 200518 | Transmitters                                 |        |
| strikeControlMt92             | 200519 | bodyPartDamagedRatio                         | 240048 |
| speedChangeMt25               | 200521 | nameOfTheTerrainDatabaseFile                 | 240049 |
| courseChangeMt26              | 200522 | nameOfLocalFile                              | 240050 |
| altitudeChangeMt27            | 200523 | aimpointBearing                              | 240051 |
| aclsAnSpn46Status             | 200524 | aimpointElevation                            | 240052 |
| aclsAircraftReport            | 200525 | aimpointRange                                | 240053 |
| sps67RadarOperatorFunctions   | 200600 | airSpeed                                     | 240054 |
| sps55RadarOperatorFunctions   | 200601 | altitude2                                    | 240055 |
| spq9aRadarOperatorFunctions   | 200602 | applicationStatus                            | 240056 |
| sps49RadarOperatorFunctions   | 200603 | autoIff                                      | 240057 |
| mk23RadarOperatorFunctions    | 200604 | beaconDelay                                  | 240058 |
| sps48RadarOperatorFunctions   | 200605 | bingoFuelSetting                             | 240059 |
| sps40RadarOperatorFunctions   | 200606 | cloudBottom                                  | 240060 |
| mk95RadarOperatorFunctions    | 200607 | cloudTop                                     | 240061 |
| killNoKill                    | 200608 | direction                                    | 240062 |
| cMTPc                         | 200609 | endAction                                    | 240063 |
| cMC4AirGlobalData             | 200610 | frequency                                    | 240064 |
| cMC4GlobalData                | 200611 | freeze                                       | 240065 |
| linksimCommentPdu             | 200612 | heading                                      | 240066 |
| nSSTownshipControl            | 200613 | identification                               | 240067 |
| other                         | 240000 | initialPointData                             | 240068 |
| massOfTheVehicle              | 240001 | latitude2                                    | 240069 |
| forceld2                      | 240002 | lights2                                      | 240070 |
| entityTypeKind1               | 240003 | linear                                       | 240071 |
| entityTypeDomain1             | 240004 | longitude2                                   | 240072 |
| entityTypeCountry1            | 240005 | lowAltitude                                  | 240073 |
| entityTypeCategory1           | 240006 | mfdFormats                                   | 240074 |
| entityTypeSubCategory         | 240007 | nctr                                         | 240075 |
| entityTypeSpecific1           | 240008 | numberProjectiles                            | 240076 |
| entityTypeExtra1              | 240009 | operationCode                                | 240077 |



|                        |        |                      |        |
|------------------------|--------|----------------------|--------|
| pitch                  | 240078 | refuelingBoomConnect | 240148 |
| profiles               | 240079 | altitudeAgl          | 240149 |
| quantity               | 240080 | calibratedAirspeed   | 240150 |
| radarModes             | 240081 | tacanChannel         | 240151 |
| radarSearchVolume      | 240082 | tacanBand            | 240152 |
| roll                   | 240083 | tacanMode            | 240153 |
| rotation               | 240084 | munition             | 500001 |
| scaleFactorX           | 240085 | engineFuel           | 500002 |
| scaleFactorY           | 240086 | storageFuel          | 500003 |
| shields                | 240087 | notUsed              | 500004 |
| steerpoint             | 240088 | expendable           | 500005 |
| spare1                 | 240089 | totalRecordSets      | 500006 |
| spare2                 | 240090 | launchedMunition     | 500007 |
| team                   | 240091 | association          | 500008 |
| text                   | 240092 | sensor               | 500009 |
| timeOfDay              | 240093 | munitionReload       | 500010 |
| trailFlag              | 240094 | engineFuelReload     | 500011 |
| trailSize              | 240095 | storageFuelReload    | 500012 |
| typeOfProjectile       | 240096 | expendableReload     | 500013 |
| typeOfTarget           | 240097 |                      |        |
| typeOfThreat           | 240098 |                      |        |
| uhfFrequency           | 240099 |                      |        |
| utmAltitude            | 240100 |                      |        |
| utmLatitude            | 240101 |                      |        |
| utmLongitude           | 240102 |                      |        |
| vhfFrequency           | 240103 |                      |        |
| visibilityRange        | 240104 |                      |        |
| voidAaaHit             | 240105 |                      |        |
| voidCollision          | 240106 |                      |        |
| voidEarthHit           | 240107 |                      |        |
| voidFriendly           | 240108 |                      |        |
| voidGunHit             | 240109 |                      |        |
| voidRocketHit          | 240110 |                      |        |
| voidSamHit             | 240111 |                      |        |
| weaponData             | 240112 |                      |        |
| weaponType             | 240113 |                      |        |
| weather2               | 240114 |                      |        |
| windDirection1         | 240115 |                      |        |
| windSpeed              | 240116 |                      |        |
| wingStation            | 240117 |                      |        |
| yaw                    | 240118 |                      |        |
| memoryOffset           | 240119 |                      |        |
| memoryData             | 240120 |                      |        |
| vasi                   | 240121 |                      |        |
| beacon                 | 240122 |                      |        |
| strobe                 | 240123 |                      |        |
| culture                | 240124 |                      |        |
| approach               | 240125 |                      |        |
| runwayEnd              | 240126 |                      |        |
| obstruction            | 240127 |                      |        |
| runwayEdge             | 240128 |                      |        |
| rampTaxiway            | 240129 |                      |        |
| laserBombCode          | 240130 |                      |        |
| rackType               | 240131 |                      |        |
| hud                    | 240132 |                      |        |
| roleFileName           | 240133 |                      |        |
| pilotName              | 240134 |                      |        |
| pilotDesignation       | 240135 |                      |        |
| modelType              | 240136 |                      |        |
| disType                | 240137 |                      |        |
| classVal               | 240138 |                      |        |
| channel                | 240139 |                      |        |
| entityType2            | 240140 |                      |        |
| alternativeEntityType2 | 240141 |                      |        |
| entityLocation         | 240142 |                      |        |
| entityLinearVelocity   | 240143 |                      |        |
| entityOrientation      | 240144 |                      |        |
| deadReckoning          | 240145 |                      |        |
| failureSymptom         | 240146 |                      |        |
| maxFuel                | 240147 |                      |        |

Named by [DatumIdentifierLengthlessArray](#), [FixedDatumStruct](#), [RecordSetStruct](#), [VariableDatumStruct](#).

## DeadReckoningAlgorithmEnum8 ENUMERATION

Dead-reckoning algorithm

Held as u8.

|           |   |
|-----------|---|
| other     | 0 |
| staticVal | 1 |
| drmFpw    | 2 |
| drmRpw    | 3 |
| drmRvw    | 4 |
| drmFvw    | 5 |
| drmFpb    | 6 |
| drmRpb    | 7 |
| drmRvb    | 8 |
| drmFvb    | 9 |

Named by [Designator](#).

## DesignatorCodeEnum16 ENUMERATION

Designator code

Held as u16.

|       |   |
|-------|---|
| other | 0 |
|-------|---|

Named by [Designator](#).

## DesignatorCodeNameEnum16 ENUMERATION

Designator code name

Held as u16.

|       |   |
|-------|---|
| other | 0 |
|-------|---|

Named by [Designator](#).

## DetonationResultCodeEnum8 ENUMERATION

Detonation result

Held as u8.

|                                      |    |                                         |    |
|--------------------------------------|----|-----------------------------------------|----|
| other                                | 0  | airBurst                                | 25 |
| entityImpact                         | 1  | killWithFragmentType1                   | 26 |
| entityProximateDetonation            | 2  | killWithFragmentType2                   | 27 |
| groundImpact                         | 3  | killWithFragmentType3                   | 28 |
| groundProximateDetonation            | 4  | killWithFragmentType1AfterFlyOutFailure | 29 |
| detonation                           | 5  | killWithFragmentType2AfterFlyOutFailure | 30 |
| none                                 | 6  | missDueToFlyOutFailure                  | 31 |
| heHitSmall                           | 7  | missDueToEndGameFailure                 | 32 |
| heHitMedium                          | 8  | missDueToFlyOutAndEndGameFailure        | 33 |
| heHitLarge                           | 9  |                                         |    |
| armorPiercingHit                     | 10 |                                         |    |
| dirtBlastSmall                       | 11 |                                         |    |
| dirtBlastMedium                      | 12 |                                         |    |
| dirtBlastLarge                       | 13 |                                         |    |
| waterBlastSmall                      | 14 |                                         |    |
| waterBlastMedium                     | 15 |                                         |    |
| waterBlastLarge                      | 16 |                                         |    |
| airHit                               | 17 |                                         |    |
| buildingHitSmall                     | 18 |                                         |    |
| buildingHitMedium                    | 19 |                                         |    |
| buildingHitLarge                     | 20 |                                         |    |
| mineClearingLineCharge               | 21 |                                         |    |
| environmentObjectImpact              | 22 |                                         |    |
| environmentObjectProximateDetonation | 23 |                                         |    |
| waterImpact                          | 24 |                                         |    |

Named by [MunitionDetonation](#).

## EmitterFunctionEnum8

ENUMERATION

Emitter system function

Held as u8.

|                                |    |                             |    |
|--------------------------------|----|-----------------------------|----|
| other                          | 0  | jammingNoise                | 64 |
| multiFunction                  | 1  | jammingDeception            | 65 |
| earlyWarningSurveillance       | 2  | decoy                       | 66 |
| heightFinding                  | 3  | navigationDistanceMeasuring | 71 |
| fireControl                    | 4  | Equipment                   |    |
| acquisitionDetection           | 5  | terrainFollowing            | 72 |
| tracking                       | 6  | weatherAvoidance            | 73 |
| guidanceIllumination           | 7  | proximityFuse               | 74 |
| firingPointLaunchPointLocation | 8  | instrumentationDeprecated   | 75 |
| ranging                        | 9  | radiosonde                  | 76 |
| radarAltimeter                 | 10 | sonobuoy                    | 77 |
| imaging                        | 11 | bathothermalSensor          | 78 |
| motionDetection                | 12 | towedCounterMeasure         | 79 |
| navigation                     | 13 | weaponNonLethal             | 96 |
| weatherMeterological           | 14 | weaponLethal                | 97 |
| instrumentation                | 15 |                             |    |
| identificationClassification   | 16 |                             |    |
| IncludingIff                   |    |                             |    |
| aaaAntiAircraftArtilleryFire   | 17 |                             |    |
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| airSearchBomb                  | 18 |                             |    |
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| groundControlIntercept         | 26 |                             |    |
| coastalSurveillance            | 27 |                             |    |
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Named by [EmitterSystem](#).

## EmitterTypeEnum16 ENUMERATION

Emitter name

Held as u16.

|                      |      |            |      |             |      |
|----------------------|------|------------|------|-------------|------|
| emitter1l250         | 5    | anAlq162   | 1638 | anApn195    | 2970 |
| emitter1rl138        | 10   | anAlq164   | 1639 | anApn198    | 3015 |
| emitter1226DeccaMil  | 45   | anAlq165   | 1640 | anApn200    | 3060 |
| emitter9b1348        | 46   | anAlq167   | 1642 | anApn202    | 3105 |
| emitter3km6          | 47   | anAlq172V1 | 1643 | anApn215    | 3106 |
| emitter9kr400        | 48   | anAlq172V2 | 1644 | anApn209    | 3120 |
| emitter9gr400        | 80   | anAlq172V3 | 1645 | anApn217    | 3150 |
| emitter9gr600        | 90   | anAlq176   | 1646 | anApn218    | 3152 |
| emitter9lv200Ta      | 135  | anAlq178   | 1647 | anApn224    | 3153 |
| emitter9lv200Tv      | 180  | anAlq184   | 1648 | anApn227    | 3154 |
| emitter9lv200tt      | 181  | anAlq184V9 | 1649 | anApn230    | 3155 |
| a310z                | 225  | anAlq188   | 1650 | anApn232    | 3156 |
| a325a                | 270  | anAlq214   | 1651 | anApn237a   | 3157 |
| a346z                | 315  | anAlr56    | 1652 | anApn234    | 3158 |
| a353b                | 360  | anAlq221   | 1653 | anApn235    | 3159 |
| a372a                | 405  | anAlr69    | 1654 | anApn238    | 3160 |
| a372b                | 450  | anAlq211V  | 1655 | anApn222    | 3161 |
| a372c                | 495  | anAlt16a   | 1656 | anApn239    | 3162 |
| a377a                | 540  | anAlt28    | 1658 | anApn241    | 3164 |
| a377b                | 585  | anAlt32a   | 1660 | anApn242    | 3166 |
| a380z                | 630  | anApd10    | 1665 | anApn243    | 3170 |
| a381z                | 675  | anApp50    | 1700 | anApn506    | 3195 |
| a398z                | 720  | anApp53    | 1710 | anApq72     | 3240 |
| a403z                | 765  | anApp59    | 1755 | anApq99     | 3285 |
| a409a                | 810  | anApp63ab  | 1800 | anApq100    | 3330 |
| a418a                | 855  | anApp63c   | 1805 | anApq102    | 3375 |
| a419z                | 900  | anApp63V2  | 1807 | anApq107    | 3376 |
| a429z                | 945  | anApp63V3  | 1809 | anApq109    | 3420 |
| a432z                | 990  | anApp65    | 1845 | anApq113    | 3465 |
| a434z                | 1035 | anApp66    | 1870 | anApq120    | 3510 |
| a401a                | 1080 | anApp67    | 1880 | anApq126    | 3555 |
| aa12Seeker           | 1095 | anApp68    | 1890 | anApq128    | 3600 |
| ad4a                 | 1096 | anApp70    | 1935 | anApq129    | 3645 |
| ades                 | 1097 | anApp73    | 1945 | anApq148    | 3690 |
| agave                | 1100 | anApp77    | 1960 | anApq153    | 3735 |
| agrion15             | 1125 | anApp78    | 1970 | anApq155    | 3770 |
| aiMk23               | 1170 | anApp79    | 1971 | anApq159    | 3780 |
| aidali               | 1215 | anApp80    | 1972 | anApq164    | 3785 |
| aim120a              | 1216 | anApp81    | 1974 | anApq166    | 3788 |
| albatrosMk2          | 1260 | anApp502   | 1980 | anApq180    | 3794 |
| wgu16B               | 1270 | anApn1     | 2025 | anApq181    | 3795 |
| boxSpring            | 1280 | anApn22    | 2070 | anAps31     | 3820 |
| boxSpringB           | 1282 | anApn59    | 2115 | anAps42     | 3825 |
| anaSps502            | 1305 | anApn69    | 2160 | anAps80     | 3870 |
| anritsuElectricAr30a | 1350 | anApn81    | 2205 | anAps88     | 3915 |
| antilopeV            | 1395 | anApn117   | 2250 | anAps88a    | 3916 |
| anAdm160             | 1398 | anApn118   | 2295 | anAps115    | 3960 |
| anAle50              | 1400 | anApn130   | 2340 | anAps116    | 4005 |
| anAlq76              | 1410 | anApn131   | 2385 | anAps120    | 4050 |
| anAlq99              | 1440 | anApn133   | 2430 | anAps121    | 4095 |
| anAlq100             | 1485 | anApn134   | 2475 | anAps124    | 4140 |
| anAlq101             | 1530 | anApn141V  | 2476 | anAps125    | 4185 |
| anAlq119             | 1575 | anApn147   | 2520 | anAps128    | 4230 |
| anAlq122             | 1585 | anApn150   | 2565 | anAps130    | 4275 |
| anAlq126a            | 1620 | anApn153   | 2610 | anAps133    | 4320 |
| anAlq128             | 1621 | anApn154   | 2655 | anAps134    | 4365 |
| anAlq126b            | 1622 | anApn155   | 2700 | anAps137    | 4410 |
| anAlq131             | 1626 | anApn159   | 2745 | anAps138    | 4455 |
| anAlq135cD           | 1628 | anApn177   | 2746 | anAps143V1  | 4465 |
| anAlq144aV3          | 1630 | anApn182   | 2790 | anAps143V3  | 4467 |
| anAlq153             | 1632 | anApn187   | 2835 | anAps143bV3 | 4468 |
| anAlq155             | 1634 | anApn190   | 2880 | anAps153    | 4475 |
| anAlq161A            | 1636 | anApn194   | 2925 | anAps150    | 4480 |

|                  |      |                 |      |                       |      |
|------------------|------|-----------------|------|-----------------------|------|
| anAps145         | 4482 | anSpg51Fc       | 5760 | anTpq37               | 7300 |
| anAps504         | 4490 | anSpg51cD       | 5761 | anTpq38V8             | 7301 |
| anApw22          | 4500 | anSpg52         | 5805 | anTpq47               | 7303 |
| anApw23          | 4545 | anSpg53         | 5850 | anTps43               | 7305 |
| anApx6           | 4590 | anSpg55b        | 5895 | anTps43e              | 7310 |
| anApx7           | 4635 | anSpg60         | 5940 | anTps59               | 7315 |
| anApx39          | 4680 | anSpg62         | 5985 | anTps63               | 7320 |
| anApx64V         | 4681 | anSpn11         | 6025 | anTps70V1             | 7322 |
| anApx72          | 4725 | anSpn35         | 6030 | anTps75               | 7325 |
| anApx76          | 4770 | anSpn41         | 6050 | anTpx46V7             | 7330 |
| anApx78          | 4815 | anSpn43         | 6075 | anTpy2                | 7333 |
| anApx100         | 4816 | anSpn43a        | 6076 | anUlk6a               | 7335 |
| anApx101         | 4860 | anSpn46         | 6085 | anUpn25               | 7380 |
| anApx113Aiff     | 4870 | anSpq2          | 6120 | anUps1                | 7425 |
| anApy1           | 4900 | anSpq9          | 6165 | anUps2                | 7426 |
| anApy2           | 4905 | anSpq9b         | 6166 | anUpx1                | 7470 |
| anApy3           | 4950 | anSpq34         | 6190 | anUpx5                | 7515 |
| anApy7           | 4952 | anSps4          | 6210 | anUpx11               | 7560 |
| anApy8           | 4953 | anSps5          | 6255 | anUpx12               | 7605 |
| anApy9           | 4954 | anSps5c         | 6300 | anUpx17               | 7650 |
| anApy10          | 4955 | anSps6          | 6345 | anUpx23               | 7695 |
| anArn21          | 4995 | anSps10         | 6390 | anVps2                | 7740 |
| anArn52          | 5040 | anSps21         | 6435 | apar                  | 7765 |
| anArn84          | 5085 | anSps28         | 6480 | aparna                | 7770 |
| anArn118         | 5130 | anSps37         | 6525 | apelcoAd77            | 7785 |
| anArn153V        | 5131 | anSps39a        | 6570 | apg71                 | 7830 |
| anArn153         | 5165 | anSps40         | 6615 | apn148                | 7875 |
| anArw73          | 5175 | anSps41         | 6660 | apn227                | 7920 |
| anAsb1           | 5220 | anSps48         | 6705 | apq113                | 7965 |
| anAsg21          | 5265 | anSps48c        | 6750 | apq120                | 8010 |
| anAsn137         | 5266 | anSps48e        | 6752 | apq148                | 8055 |
| anAsq108         | 5280 | anSps49         | 6795 | aps504V3              | 8100 |
| anAwg9           | 5310 | anSps49V1       | 6796 | ar3d                  | 8105 |
| anBps9           | 5355 | anSps49V2       | 6797 | plesseyAR5            | 8112 |
| anBps15          | 5400 | anSps49V3       | 6798 | ar320                 | 8115 |
| anBps15h         | 5401 | anSps49V4       | 6799 | ar327                 | 8120 |
| anBps16          | 5405 | anSps49V5       | 6800 | arM31                 | 8145 |
| anCrm30          | 5420 | anSps49V6       | 6801 | ari5954               | 8190 |
| anDpw23          | 5430 | anSps49V7       | 6802 | ari5955               | 8235 |
| anDsq26PhoenixMh | 5445 | anSps49V8       | 6803 | ari5979               | 8280 |
| anDsq28HarpoonMh | 5490 | anSps49aV1      | 6804 | argsn31               | 8281 |
| anFpn40          | 5495 | anSps52         | 6840 | arinc564BndxKingRdr1e | 8325 |
| anFpn62          | 5500 | anSps53         | 6885 | arinc700BndxKingRdr1e | 8370 |
| anFps16          | 5505 | anSps55         | 6930 | ark1                  | 8375 |
| anFps18          | 5507 | anSps52c        | 6945 | arsr3                 | 8380 |
| anFps89          | 5508 | anSps55cs       | 6970 | arsr18                | 8390 |
| anFps117         | 5510 | anSps55Ss       | 6975 | as2Kipper             | 8415 |
| anFps20r         | 5515 | anSps58         | 7020 | as2KipperMh           | 8460 |
| anFps77          | 5520 | anSps58c        | 7025 | as4Kitchen            | 8505 |
| anFps103         | 5525 | anSps59         | 7065 | as4KitchenMh          | 8550 |
| anGpn12          | 5527 | anSps64         | 7110 | as5KeltMh             | 8595 |
| anGpx6           | 5530 | anSps65         | 7155 | as6KingfishMh         | 8640 |
| anGpx8           | 5535 | anSps66         | 7175 | as7Kerry              | 8685 |
| anGrn12          | 5537 | anSps67         | 7200 | as7KerryMg            | 8730 |
| anMpn14          | 5539 | anSps73I        | 7201 | as15KentAltimeter     | 8735 |
| anMpq10          | 5540 | anSps69         | 7210 | aspideAamSamIII       | 8760 |
| anMpq46Hpill     | 5545 | anSps73         | 7215 | asr4                  | 8772 |
| anMpq4855Cwar    | 5550 | anSps88         | 7225 | asrO                  | 8775 |
| anMpq49          | 5551 | anSpy1          | 7245 | asr5                  | 8780 |
| anMpq50ParTa     | 5555 | anSpy1a         | 7250 | asr7                  | 8782 |
| anMpq51RorTt     | 5560 | anSpy1b         | 7252 | asr8                  | 8785 |
| anMpq53          | 5570 | anSpy1bV        | 7253 | asr9                  | 8790 |
| anMpq63          | 5571 | anSpy1d         | 7260 | raytheonASR10ss       | 8812 |
| anMpq64          | 5575 | anSpy1dV        | 7261 | at2SwatterMg          | 8820 |
| anSlq32          | 5576 | anSpy1f         | 7265 | atcr33                | 8840 |
| anSlq32V4        | 5578 | anTlq32ARMDecoy | 7269 | atcr33KM              | 8845 |
| anSpg34          | 5580 | anTpn17         | 7270 | atlasElektronkTrsN    | 8865 |
| anSpg50          | 5625 | anTpn24         | 7275 | atlas9600m            | 8867 |
| anSpg51          | 5670 | anTpq18         | 7280 | atlas9740vts          | 8870 |
| anSpg51CwiTi     | 5715 | anTpq36         | 7295 | avg65                 | 8910 |

|                        |       |                        |       |                   |       |
|------------------------|-------|------------------------|-------|-------------------|-------|
| avh7                   | 8955  | calypsoC61             | 10845 | decca2459         | 12735 |
| aviaconversia          | 8990  | calypsoli              | 10890 | deccaAws1         | 12780 |
| aviaconversialll       | 8995  | cardionCoastal         | 10895 | deccaAws2         | 12782 |
| avq20                  | 9000  | castorli               | 10935 | deccaAws4         | 12785 |
| avq21                  | 9005  | castor2jTtCrotaleNg    | 10940 | deccaAws42        | 12787 |
| avq30x                 | 9045  | catHouse               | 10980 | deccamar          | 12800 |
| avq50Rca               | 9075  | cdr431                 | 10985 | deccaRm326        | 12805 |
| avq70                  | 9090  | chSsN6                 | 10995 | deccaRm416        | 12825 |
| aws5                   | 9135  | chairBackTt            | 11000 | deccaRm914        | 12870 |
| aws6                   | 9180  | chairBacklll           | 11010 | deccaRm1690       | 12915 |
| aws6b300               | 9185  | lemz96l6               | 11020 | deccaSuper101Mk3  | 12960 |
| b597z                  | 9200  | cheeseBrick            | 11025 | diss1             | 13005 |
| b636z                  | 9205  | clamPipe               | 11070 | diss7             | 13006 |
| backBoard              | 9215  | clamshell              | 11115 | diss013           | 13007 |
| backNetAB              | 9225  | coastalGiraffe         | 11125 | rapierTtdn181     | 13050 |
| backTrap               | 9270  | colibri                | 11137 | rapier2000Tt      | 13055 |
| bAESystemsRT1805Apn    | 9280  | collinsWxr700x         | 11160 | dogEar            | 13095 |
| baltyk                 | 9310  | collinsDn101           | 11205 | dogHouse          | 13140 |
| ballEnd                | 9315  | contravesSeaHunterMk4  | 11250 | dm3               | 13141 |
| ballGun                | 9360  | cornCan                | 11260 | dm3b              | 13142 |
| bandStand              | 9405  | cr105Rmca              | 11270 | dm5               | 13143 |
| barLock                | 9450  | crossBird              | 11295 | don2              | 13185 |
| bassTilt               | 9495  | crossDome              | 11340 | donAB2Kay         | 13230 |
| badger                 | 9505  | crossLegs              | 11385 | donets            | 13275 |
| beacon                 | 9540  | crossOut               | 11430 | downBeat          | 13320 |
| beanSticks             | 9585  | crossSlot              | 11475 | draa2a            | 13365 |
| beeHind                | 9630  | crossSword             | 11520 | draa2b            | 13410 |
| bellCrownA             | 9640  | crossUp                | 11565 | drac39            | 13455 |
| bellCrownB             | 9642  | crossSwordFc           | 11610 | dragonEye         | 13477 |
| bellSquat              | 9643  | crotaleAcquisitionTa   | 11655 | drbc30b           | 13500 |
| bigBack                | 9645  | crotaleNgTa            | 11660 | drbc31a           | 13545 |
| bigBirdABC             | 9659  | crotaleTt              | 11665 | drbc32            | 13585 |
| bigBird                | 9660  | crotaleMGMissileSystem | 11700 | drbc32a           | 13590 |
| bigBirdDMod            | 9661  | cs10Ta                 | 11715 | drbc32d           | 13635 |
| bigBulge               | 9675  | csfVaran               | 11725 | drbc33a           | 13680 |
| bigBulgeA              | 9720  | cssN4mh                | 11735 | drbi10            | 13725 |
| bigBulgeB              | 9765  | cssC3cCas1m1M2Mh       | 11745 | drbi23            | 13770 |
| snar10                 | 9780  | cssC2bHy1aMh           | 11790 | drbj11b           | 13815 |
| bigMesh                | 9810  | cws2                   | 11835 | drbn30            | 13860 |
| bigNet                 | 9855  | cylinderHead           | 11880 | drbn32            | 13905 |
| billBoard              | 9885  | cymbeline              | 11902 | drbr51            | 13950 |
| billFold               | 9900  | cyranolli              | 11925 | drbv20b           | 13995 |
| blowpipeMg             | 9905  | cyranolv               | 11970 | drbv22            | 14040 |
| blueFox                | 9930  | cyranolvM              | 11975 | drbv26c           | 14085 |
| blueVixen              | 9935  | da0100                 | 12010 | drbv30            | 14130 |
| blueSilk               | 9945  | da0500                 | 12015 | drbv50            | 14175 |
| blueParrot             | 9990  | dawn                   | 12060 | drbv51            | 14220 |
| blueOrchid             | 10035 | deadDuck               | 12105 | drbv51a           | 14265 |
| bmDjg8715              | 10057 | decca20v909            | 12110 | drbv51b           | 14310 |
| boatSail               | 10080 | decca20v90s            | 12111 | drbv51c           | 14355 |
| boforsElectronic9lv331 | 10125 | decca45                | 12150 | dropKick          | 14400 |
| boforsEricssonSea      | 10170 | decca50                | 12195 | drua31            | 14445 |
| Giraffe50Hc            |       | decca71                | 12196 | drumTilt          | 14490 |
| bowlMesh               | 10215 | decca110               | 12240 | drumTiltA         | 14535 |
| boxBrick               | 10260 | decca170               | 12285 | drumTiltB         | 14545 |
| boxTail                | 10305 | deccahf2               | 12292 | dumbo             | 14580 |
| bmKg860186058606       | 10315 | decca202               | 12330 | ekcoe390          | 14590 |
| bps11a                 | 10350 | deccaD202              | 12375 | ecr90             | 14600 |
| bps14                  | 10395 | decca303               | 12420 | eggCupAB          | 14625 |
| bps15a                 | 10440 | decca535               | 12430 | ekcoe120          | 14660 |
| br15TokyoKeiki         | 10485 | decca626               | 12465 | ekco190           | 14670 |
| bridgemaste            | 10510 | decca629               | 12510 | elL8222           | 14710 |
| breadBin               | 10530 | decca914               | 12555 | elM2001b          | 14715 |
| bt271                  | 10575 | decca916               | 12600 | elM2022           | 14725 |
| bx732                  | 10620 | decca926               | 12610 | elM2200           | 14750 |
| buranD                 | 10642 | decca1070a             | 12615 | elM2207           | 14760 |
| buzzStand              | 10665 | decca1226Commercial    | 12645 | elM2216V          | 14770 |
| c5aMultiModeRadar      | 10710 | decca1290              | 12655 | elt361            | 14776 |
| caiman                 | 10755 | decca1626              | 12690 | elt572            | 14785 |
| cakeStand              | 10800 | decca2070              | 12691 | eltaEIM2221GmStgr | 14805 |

|                  |       |               |       |                      |       |
|------------------|-------|---------------|-------|----------------------|-------|
| eIM2228s3d       | 14806 | furuno1932    | 16590 | hotFlash             | 18315 |
| eIM2705          | 14807 | furuno701     | 16605 | ihS6                 | 18318 |
| elTaSis          | 14810 | furuno1940    | 16606 | hotShotTa            | 18320 |
| eIM2238          | 14811 | furuno7112    | 16650 | hotShotTt            | 18325 |
| emd2900          | 14850 | furuno240     | 16690 | hotShotMg            | 18330 |
| endTray          | 14895 | furuno2400    | 16695 | iffMkXiiAimsUpX29    | 18360 |
| esr1             | 14900 | furuno8051    | 16730 | iffMkXv              | 18405 |
| et316            | 14905 | g030aApd31    | 16735 | iffint               | 18406 |
| exocetType       | 14935 | ga0100        | 16740 | jackKnife            | 18407 |
| exocet1          | 14940 | gage          | 16785 | ifftrsp              | 18408 |
| exocet1Mh        | 14985 | gardenia      | 16800 | javelinMg            | 18410 |
| exocet2          | 15030 | garpin        | 16830 | jayBird              | 18450 |
| eyeBowl          | 15075 | gateGuard     | 16833 | jl7                  | 18454 |
| eyeShield        | 15120 | garpunBalE    | 16835 | jl10b                | 18455 |
| f332z            | 15140 | gemBx132      | 16875 | jlp40                | 18458 |
| falconClawTI     | 15155 | mpdr12        | 16880 | jrcNmd401            | 18460 |
| falconClawTT     | 15156 | gepardTt      | 16884 | jupiter              | 18495 |
| falcon           | 15160 | geranF        | 16888 | jupiterli            | 18540 |
| falconG          | 15161 | giraffe       | 16900 | jy8                  | 18550 |
| fanSongA         | 15165 | ginSlingTa    | 16915 | jy9                  | 18555 |
| fanSongBFTa      | 15200 | ginSling      | 16920 | jy9Modified          | 18556 |
| fanSongBFTt      | 15210 | ginSlingMg    | 16925 | jy11ew               | 18557 |
| fanSongCETa      | 15220 | goldenBar     | 16931 | jy14                 | 18560 |
| fanSongCETt      | 15230 | goldenBat     | 16932 | k376z                | 18585 |
| fanSongCEMg      | 15240 | goldenDome    | 16935 | kelvinHughes2a       | 18630 |
| fanSongBFfMg     | 15255 | goldenHeart   | 16940 | kelvinHughes149      | 18675 |
| fanTail          | 15300 | gpn22         | 16945 | kelvinHughesType1006 | 18720 |
| fb7Radar         | 15305 | grn9          | 16950 | kelvinHughesType1007 | 18765 |
| fcr1401          | 15310 | graveStone    | 16960 | kHFfamily            | 18780 |
| finCurve         | 15345 | greenStain    | 16965 | kh902m               | 18785 |
| fireCan          | 15390 | gridBow       | 17010 | khromK               | 18786 |
| fireDish         | 15435 | grillPanTt    | 17025 | kh1700               | 18795 |
| fireDomeTa       | 15470 | gt4           | 17031 | kingPin              | 18797 |
| fireDomeTt       | 15475 | guardsman     | 17055 | kg300                | 18805 |
| fireDomeTi       | 15480 | gunDishZsu234 | 17070 | kiteScreech          | 18810 |
| fireIron         | 15525 | hairNet       | 17100 | kiteScreechA         | 18855 |
| fireWheel        | 15570 | halfPlateA    | 17145 | kiteScreechB         | 18900 |
| fishBowl         | 15615 | halfPlateB    | 17190 | kivach               | 18945 |
| flapLid          | 15660 | hard          | 17220 | klj1                 | 18948 |
| flapTruck        | 15705 | harpoon       | 17225 | klj3Type1473         | 18950 |
| flapWheel        | 15750 | hawkScreech   | 17235 | knifeRest            | 18990 |
| flashDance       | 15795 | headLightA    | 17280 | knifeRestB           | 19035 |
| flatFaceABCD     | 15840 | headLights    | 17325 | knifeRestC           | 19037 |
| flatFaceE        | 15842 | headLightsC   | 17370 | kj2000               | 19040 |
| flatScreen       | 15885 | headLightsMgA | 17415 | kr75                 | 19050 |
| flatSpin         | 15930 | headLightsMgB | 17460 | ksaSrn               | 19080 |
| flatTwin         | 15975 | headLightsTt  | 17505 | ksaTsr               | 19125 |
| fl400            | 15980 | headNet       | 17550 | landFall             | 19170 |
| fledermaus       | 16020 | heartAcheB    | 17572 | landRollMg           | 19215 |
| flycatcher       | 16030 | henEgg        | 17595 | landRollTa           | 19260 |
| flyScreen        | 16065 | henHouse      | 17640 | landRollTt           | 19305 |
| flyScreenAB      | 16110 | henNest       | 17685 | lazur                | 19306 |
| flyTrapB         | 16155 | henRoost      | 17730 | lc150                | 19310 |
| fogLampMg        | 16200 | highBrick     | 17775 | leningraf            | 19350 |
| fogLampTt        | 16245 | highFix       | 17820 | lightBulb            | 19395 |
| foilTwo          | 16290 | highGuard     | 17842 | lmtNrAi6a            | 19400 |
| footBall         | 16300 | highLarkTi    | 17865 | ln55                 | 19440 |
| foxHunter        | 16335 | highLark1     | 17910 | ln66                 | 19485 |
| foxFIREFoxFireAI | 16380 | highLark2     | 17955 | longBow              | 19530 |
| foxFireIII       | 16390 | highLark4     | 18000 | longBrick            | 19575 |
| fr151a           | 16400 | highLune      | 18045 | longBull             | 19620 |
| fr1505Da         | 16410 | highPoleAB    | 18090 | longEye              | 19665 |
| fr2000           | 16420 | highScoop     | 18135 | longHead             | 19710 |
| furuno2855w      | 16421 | highScreen    | 18150 | longTalk             | 19755 |
| frontDome        | 16425 | highSieve     | 18180 | longTrack            | 19800 |
| frontDoor        | 16470 | hg9550        | 18190 | longTrough           | 19845 |
| frontPiece       | 16515 | hn503         | 18200 | lookTwo              | 19890 |
| furuno           | 16560 | homeTalk      | 18225 | loran                | 19935 |
| furuno1721       | 16561 | hornSpoon     | 18270 | lowBlowTa            | 19950 |
| furuno1730       | 16580 | hotBrick      | 18280 | lowBlowTt            | 19955 |

|                        |       |                       |       |                      |       |
|------------------------|-------|-----------------------|-------|----------------------|-------|
| lowBlowMg              | 19960 | nayada                | 21915 | plesseyAws4          | 23940 |
| lowSieve               | 19980 | neptun                | 21960 | plesseyAws6          | 23985 |
| lowTrough              | 20025 | nikeTt                | 21980 | plesseyRj            | 23990 |
| trs2050                | 20040 | nj81e                 | 21983 | plesseyType996       | 24030 |
| lw08                   | 20070 | nrj6a                 | 21985 | plinthNet            | 24075 |
| m1983Fcr               | 20090 | nutCan                | 21992 | pluto                | 24095 |
| m2240                  | 20115 | nrba50                | 22005 | pohjanpalo           | 24100 |
| m44                    | 20160 | nrba51                | 22050 | pollux               | 24120 |
| m401z                  | 20205 | nrbf20a               | 22095 | popGroup             | 24165 |
| m585z                  | 20250 | nrj5                  | 22110 | popGroupMg           | 24210 |
| m588z                  | 20295 | nysaB                 | 22140 | popGroupTa           | 24255 |
| ma1IffPortion          | 20340 | o524a                 | 22185 | popGroupTt           | 24300 |
| mareld                 | 20360 | o580b                 | 22230 | porkTrough           | 24345 |
| maType909              | 20385 | o625z                 | 22275 | positiveME           | 24385 |
| marcs152               | 20420 | o626z                 | 22320 | postBow              | 24390 |
| marconi1810            | 20430 | oceanMaster           | 22335 | postLamp             | 24435 |
| marconiCanadaHc75      | 20475 | oddGroup              | 22345 | potDrum              | 24480 |
| marconiS713            | 20495 | oddLot                | 22365 | potHead              | 24525 |
| marconiS1802           | 20520 | oddPair               | 22410 | potShot              | 24535 |
| marconiS247            | 20530 | oddRods               | 22411 | praetorian           | 24540 |
| marconiS810            | 20565 | oka                   | 22455 | CountermeasuresSuite |       |
| marconiSa10            | 20585 | okean                 | 22500 | primus40Wxd          | 24570 |
| marconiType967         | 20610 | okeana                | 22505 | primus300sl          | 24615 |
| marconiType968         | 20655 | okinxe12c             | 22545 | primus700            | 24618 |
| marconiType992         | 20700 | omega                 | 22590 | primus3000           | 24620 |
| marconiSignaalType1022 | 20745 | omeraOrb32            | 22635 | prora                | 24630 |
| marconiSignaalType910  | 20790 | omul                  | 22640 | prorapa1660          | 24635 |
| marconiSignaalType911  | 20835 | oneEye                | 22680 | ps05a                | 24650 |
| marconiSignaalType992r | 20880 | op28                  | 22690 | ps46A                | 24660 |
| melco3                 | 20915 | ops16b                | 22725 | ps70R                | 24705 |
| northropGrummanMESA    | 20920 | ops18                 | 22730 | ps860                | 24707 |
| meshBrick              | 20925 | ops28                 | 22740 | ps870                | 24709 |
| mirageIII              | 20950 | or2                   | 22770 | ps890                | 24710 |
| mk15Ciws               | 20970 | orb31s                | 22810 | puffBall             | 24750 |
| mk23                   | 21015 | orb32                 | 22815 | pvs200               | 24760 |
| mk23Tas                | 21060 | orionRtn10x           | 22860 | r76                  | 24770 |
| mk25                   | 21105 | otomatMK1             | 22900 | r41xxx               | 24775 |
| mk25Mod3               | 21110 | otomatMkliTeseo       | 22905 | rac30                | 24780 |
| mk35M2                 | 21150 | owlScreech            | 22950 | racal1229            | 24795 |
| mk92                   | 21195 | p360z                 | 22955 | racalAc2690Bt        | 24840 |
| mk92Cas                | 21240 | pa1660                | 22960 | racalDecca1216       | 24885 |
| mk92Stir               | 21285 | paintBox              | 22977 | racalDecca20v909     | 24890 |
| mk95                   | 21330 | palmFrond             | 22995 | racalDecca360        | 24930 |
| mks818                 | 21332 | modifiedPaintBox      | 22998 | racalDeccaAc1290     | 24975 |
| m1a1                   | 21340 | palmFrondAb           | 23040 | racalDeccaTm1229     | 25020 |
| mmAps705               | 21375 | patHandTt             | 23085 | racalDeccaTm1626     | 25065 |
| mmSpg74                | 21420 | patHandMg             | 23095 | racalDrbn34a         | 25110 |
| mmSpg75                | 21465 | pattyCake             | 23130 | radar24              | 25155 |
| mmSpn703               | 21490 | pawnCake              | 23175 | ran7s                | 25200 |
| mmSps702               | 21510 | pbr4Rubin             | 23220 | ran10s               | 25205 |
| mmSps768               | 21555 | peaSticks             | 23265 | ran11Lx              | 25245 |
| mmSps774               | 21600 | peelCone              | 23310 | rapierTa             | 25260 |
| model17c               | 21625 | peelGroup             | 23355 | rapier2000Ta         | 25265 |
| moon4                  | 21645 | peelGroupA            | 23400 | rapierMg             | 25270 |
| moonPie                | 21646 | peelGroupB            | 23445 | rashmi               | 25275 |
| mmrs                   | 21650 | peelGroupMG           | 23450 | rasit                | 25276 |
| model360               | 21655 | peelPair              | 23490 | rat31s               | 25280 |
| model378               | 21660 | phalanx               | 23525 | ratacLct             | 25285 |
| model970               | 21661 | philips9lv200         | 23535 | rattler              | 25287 |
| model974               | 21665 | philips9lv331         | 23580 | raws                 | 25288 |
| mpdr18X                | 21690 | philipsLv223          | 23625 | rawl02               | 25289 |
| mr1600                 | 21700 | philipsSeaGiraffe50Hc | 23670 | raytheon1220         | 25290 |
| mt305x                 | 21710 | pinJib                | 23690 | raytheon1302         | 25300 |
| muffCob                | 21735 | plankShad             | 23710 | raytheon1500         | 25335 |
| mushroom               | 21780 | plankShave            | 23715 | raytheon1645         | 25380 |
| mushroom1              | 21825 | plankShaveA           | 23760 | raytheon1650         | 25425 |
| mushroom2              | 21870 | plankShaveB           | 23805 | raytheon1900         | 25470 |
| n920z                  | 21880 | plateSteer            | 23850 | raytheon2502         | 25515 |
| nanjingB               | 21890 | plesseyAws1           | 23895 | raytheonTm16506x     | 25560 |
| nanjingC               | 21895 | plesseyAWS2           | 23925 | raytheonTm166012s    | 25605 |



|                        |       |                 |       |                   |       |
|------------------------|-------|-----------------|-------|-------------------|-------|
| ray1220xr              | 25630 | scoopPlate      | 27180 | slotBackVI        | 29435 |
| ray1401                | 25635 | scr584          | 27190 | slotRest          | 29440 |
| ray2900                | 25650 | seaArcher2      | 27225 | sma3Rm            | 29475 |
| raypath                | 25695 | seaHunter4Mg    | 27270 | sma3Rm20          | 29520 |
| rbe2                   | 25735 | seaHunter4Ta    | 27315 | sma3rm20aSmg      | 29565 |
| rct180                 | 25739 | seaHunter4Tt    | 27360 | smaBps704         | 29610 |
| rdm                    | 25740 | seaGull         | 27405 | smaSpin749V2      | 29655 |
| rdy                    | 25760 | seaNet          | 27450 | smaSpn703         | 29700 |
| rdn72                  | 25785 | seaSparrow      | 27451 | smaSpn751         | 29745 |
| rdr1a                  | 25830 | seaSpray        | 27495 | smaSpos748        | 29790 |
| rdr1e                  | 25835 | seaTiger        | 27540 | smaSpq2           | 29835 |
| rdr4a                  | 25840 | seaTigerM       | 27550 | smaSpq2d          | 29880 |
| rdr160xd               | 25850 | searchwater     | 27570 | smaSpq701         | 29925 |
| rdr1200                | 25875 | searchwater2000 | 27575 | smaSps702UpX      | 29970 |
| rdr1400                | 25885 | seleniaOrion7   | 27585 | smaSt2OtomatliMh  | 30015 |
| rdr1400C               | 25890 | seleniaType912  | 27630 | sr47a             | 30016 |
| rdr1500                | 25895 | selenniaRan12LX | 27675 | sma718Beacon      | 30060 |
| riceCake               | 25896 | selenniaRtn10x  | 27720 | smartL            | 30070 |
| remora                 | 25900 | seliniaArp1645  | 27765 | smogLamp          | 30075 |
| riceBowl               | 25910 | sg              | 27800 | snapShot          | 30080 |
| riceLamp               | 25920 | sgr10200        | 27810 | snoopDrift        | 30105 |
| ricePad                | 25965 | sgr10302        | 27855 | snoopHalf         | 30140 |
| riceScreen             | 26010 | sgr104          | 27870 | snoopHead         | 30150 |
| deccarm1070a           | 26011 | sheetBend       | 27900 | snoopPair         | 30195 |
| rm370bt                | 26015 | sheetCurve      | 27945 | snoopPlate        | 30240 |
| rockwellCollinsFMR200x | 26020 | shipGlobe       | 27990 | snoopPing         | 30255 |
| rolandBn               | 26055 | shipWheel       | 28035 | snoopSlab         | 30285 |
| rolandMg               | 26100 | sgr114          | 28080 | snoopTray         | 30330 |
| rolandTa               | 26145 | shoreWalkA      | 28125 | snoopTray1        | 30375 |
| rolandTt               | 26190 | shortHorn       | 28170 | snoopTray2        | 30420 |
| roundBall              | 26235 | shotDome        | 28215 | snoopWatch        | 30465 |
| roundHouse             | 26280 | sideGlobeJn     | 28260 | snowDrift         | 30470 |
| roundHouseB            | 26325 | sideNet         | 28280 | spb7              | 30475 |
| rs0250                 | 26327 | sideWalkA       | 28305 | so1               | 30510 |
| rt0250                 | 26330 | signaalDa02     | 28350 | so12              | 30520 |
| rtn1a                  | 26350 | signaalDa05     | 28395 | soACommunist      | 30555 |
| rumSling               | 26360 | signaalDa08     | 28440 | so69              | 30580 |
| rv2                    | 26370 | signaalLw08     | 28485 | sockEye           | 30600 |
| rv3                    | 26415 | signaalLwor     | 28530 | som64             | 30645 |
| rv5                    | 26460 | signaalM45      | 28575 | sorbsiya          | 30660 |
| rv10                   | 26505 | signaalMw08     | 28620 | spadaTt           | 30670 |
| rv15m                  | 26506 | signaalSmart    | 28665 | sparrowAIMRim7III | 30690 |
| rv17                   | 26550 | signaalSting    | 28710 | sperryrascar      | 30691 |
| rv18                   | 26595 | signaalStir     | 28755 | sperryM3          | 30700 |
| rv21                   | 26596 | signaalWm202    | 28800 | spg53f            | 30735 |
| rv377                  | 26610 | signaalWm25     | 28845 | spg70RTN10x       | 30780 |
| rvUm                   | 26640 | signaalWm27     | 28890 | spg74RTN20x       | 30825 |
| rxn260                 | 26660 | signaalWm28     | 28935 | spg75RTN30x       | 30870 |
| s1810cd                | 26670 | signaalZw01     | 28980 | spg76RTN30x       | 30915 |
| salamandre             | 26673 | signaalZw06     | 29025 | spinScanA         | 30960 |
| s1850m                 | 26675 | skiPole         | 29070 | spinScanB         | 31005 |
| sa2Guideline           | 26685 | skinHead        | 29115 | spinTrough        | 31050 |
| sa3Goa                 | 26730 | skipSpin        | 29160 | splashDrop        | 31095 |
| sa8GeckoDt             | 26775 | skyguardB       | 29180 | spn2              | 31096 |
| sa12TelarIII           | 26795 | skyguardTa      | 29185 | spn4              | 31097 |
| saN7GadflyTi           | 26820 | skyguardTt      | 29190 | spn30             | 31100 |
| saN11Cads1Un           | 26865 | skymaster       | 29200 | spn35a            | 31140 |
| saccadeMH              | 26900 | skyWatch        | 29205 | spn41             | 31185 |
| saltPotAB              | 26910 | skyRanger       | 29210 | spn42             | 31230 |
| sap14                  | 26920 | skyshadow       | 29215 | spn43a            | 31275 |
| sap518                 | 26925 | skyshieldTa     | 29220 | spn43b            | 31320 |
| sap518m                | 26926 | sl              | 29250 | spn44             | 31365 |
| saturneli              | 26955 | slAlq234        | 29270 | spn46             | 31410 |
| scanCan                | 27000 | slapShotE       | 29295 | spn703            | 31455 |
| scanFix                | 27045 | slimNet         | 29340 | spn728V1          | 31500 |
| scanOdd                | 27090 | slotBackA       | 29385 | spn748            | 31545 |
| scanThree              | 27135 | slotBackIII     | 29400 | spn750            | 31590 |
| scanterCsr             | 27140 | slotBackB       | 29430 | spo8              | 31592 |
| scorads                | 27141 | slotBackIV      | 29431 | spongeCake        | 31635 |
| scoreboard             | 27150 | slotBackBTopaz  | 29432 | spoonRest         | 31680 |

|                   |       |                        |       |                      |       |
|-------------------|-------|------------------------|-------|----------------------|-------|
| spoonRestA        | 31681 | t8408                  | 34290 | trisponde            | 36380 |
| spoonRestB        | 31682 | t8911                  | 34335 | tritonG              | 36390 |
| spoonRestD        | 31684 | t8937                  | 34380 | trs3033              | 36405 |
| spq712Ran12lx     | 31725 | t8944                  | 34425 | trs3405              | 36420 |
| sps6c             | 31770 | t8987                  | 34470 | trs3410              | 36425 |
| sps10f            | 31815 | tacanSurf              | 34505 | trs3415              | 36430 |
| sps12             | 31860 | tallKing               | 34515 | trsN                 | 36450 |
| sps58             | 31905 | tallMike               | 34560 | tse5000              | 36495 |
| sps64             | 31950 | tallPath               | 34605 | tsr333               | 36540 |
| sps161            | 31960 | teamWork               | 34625 | tubBrick             | 36563 |
| sps768RANEI       | 31995 | t1135                  | 34626 | tubeArm              | 36585 |
| sps774RAN10s      | 32040 | tancanSurf             | 34627 | twinEyes             | 36630 |
| spy790            | 32085 | thaadGbr               | 34640 | twinPill             | 36675 |
| squareHead        | 32130 | thd225                 | 34650 | twinScan             | 36720 |
| squarePair        | 32175 | thd1940                | 34670 | twinScanRo           | 36765 |
| squareSlot        | 32220 | thd1955Palmier         | 34680 | twoSpot              | 36810 |
| squareTie         | 32265 | thd5500                | 34695 | type222              | 36843 |
| squashDome        | 32310 | thinPath               | 34740 | type226              | 36846 |
| squatEye          | 32330 | thinSkin               | 34785 | type262              | 36855 |
| squintEye         | 32355 | thompsonCsfTa10        | 34795 | type275              | 36900 |
| sr47bG            | 32375 | thompsonCsfTh          | 34830 | type293              | 36945 |
| srn6              | 32400 | D1040Neptune           |       | type341              | 36946 |
| srn15             | 32445 | thompsonCsfCalypso     | 34875 | type343SunVisorB     | 36990 |
| srn745            | 32490 | thompsonCsfCastor      | 34920 | type347b             | 37035 |
| sro1              | 32535 | thompsonCsfCastorli    | 34965 | type347G             | 37038 |
| sro2              | 32580 | thompsonCsfDrbc32a     | 35010 | type352              | 37040 |
| ssC2bSamletMg     | 32625 | thompsonCsfDrbj11DE    | 35055 | type354              | 37045 |
| ssN2aBCssc        | 32670 | thompsonCsfDrbv15a     | 35100 | type363              | 37048 |
| ssN2aBCssc2a3a2Mh | 32715 | thompsonCsfDrbv15c     | 35145 | type404aCh           | 37050 |
| ssN2cSeeker       | 32760 | thompsonCsfDrbv22d     | 35190 | type405              | 37052 |
| ssN2cDStyx        | 32805 | thompsonCsfDrbv23b     | 35235 | type753              | 37075 |
| ssN2cDStyxCDMh    | 32850 | thompsonCsfDrua33      | 35280 | type756              | 37080 |
| ssN3SscSsC18Bn    | 32895 | thompsonCsfMarsDrbv21a | 35325 | type903              | 37125 |
| ssN3bSepalAl      | 32940 | thompsonCsfSeaTiger    | 35370 | type909Ti            | 37170 |
| ssN3bSepalMh      | 32985 | thompsonCsfTriton      | 35415 | type909Tt            | 37215 |
| ssN9Siren         | 33030 | thompsonCsfVegaWith    | 35460 | type910              | 37260 |
| ssN9SirenAl       | 33075 | Drbc32e                |       | type931Ch            | 37265 |
| ssN9SirenMh       | 33120 | trs2105                | 35480 | type965              | 37305 |
| ssN12SandboxAl    | 33165 | ht223                  | 35485 | type967              | 37350 |
| ssN12SandboxMh    | 33210 | trs2100                | 35490 | type968              | 37395 |
| ssN19Shipwreck    | 33255 | tieRods                | 35505 | type974              | 37440 |
| ssN19ShipwreckAl  | 33300 | tinShield              | 35550 | type975              | 37485 |
| ssN19ShipwreckMh  | 33345 | tinTrap                | 35570 | type978              | 37530 |
| ssN21Al           | 33390 | tirspander             | 35580 | type981              | 37534 |
| ssN22Sunburn      | 33435 | toadStool1             | 35595 | type992              | 37575 |
| ssN22SunburnMh    | 33480 | toadStool2             | 35640 | type993              | 37620 |
| ssN27SizzlerMH    | 33485 | toadStool3             | 35685 | type994              | 37665 |
| stoneCake         | 33525 | toadStool4             | 35730 | type10061            | 37710 |
| str41             | 33570 | toadStool5             | 35775 | type10062            | 37755 |
| straightFlushTa   | 33590 | tokenB                 | 35785 | type1022             | 37800 |
| straightFlushTt   | 33595 | tombStone              | 35800 | ukMk10               | 37845 |
| straightFlushlll  | 33600 | tonson                 | 35810 | ups220c              | 37850 |
| strikeOut         | 33615 | topBow                 | 35820 | upx110               | 37890 |
| strutCurve        | 33660 | topDome                | 35865 | upx27                | 37935 |
| strutPair         | 33705 | topKnot                | 35910 | urn20                | 37980 |
| strutPair1        | 33750 | topMesh                | 35955 | urn25                | 38025 |
| strutPair2        | 33795 | topPair                | 36000 | volexliilv           | 38045 |
| sunVisor          | 33840 | topPlate               | 36045 | w1028                | 38060 |
| superfledermaus   | 33860 | topPlateB              | 36046 | w8818                | 38070 |
| supersearcher     | 33870 | topSail                | 36090 | w8838                | 38115 |
| swiftRod1         | 33885 | type208                | 36120 | w8852                | 38120 |
| swiftRod2         | 33930 | topSteer               | 36135 | wallRust             | 38150 |
| t1166             | 33975 | topTrough              | 36180 | was74s               | 38160 |
| t1171             | 34020 | scrumHalfTa            | 36220 | waspHead             | 38205 |
| t1202             | 34040 | torScrumHalfTt         | 36225 | watchdog             | 38210 |
| t6004             | 34065 | scrumHalfMg            | 36230 | watchGuard           | 38250 |
| t6031             | 34110 | trackDish              | 36270 | watchman             | 38260 |
| t8067             | 34155 | torsoM                 | 36315 | westernElectricMk10  | 38295 |
| t8068             | 34200 | tqn2                   | 36320 | westinghouseADR4lrsr | 38320 |
| t8124             | 34245 | trapDoor               | 36360 |                      |       |

|                          |       |
|--------------------------|-------|
| westinghouseElectric     | 38340 |
| Spg50                    |       |
| westinghouseElectricW120 | 38385 |
| westinghouseSps29c       | 38430 |
| westinghouseSps37        | 38475 |
| wetEye                   | 38520 |
| wetEye2                  | 38525 |
| wetEyeMod                | 38565 |
| wgu41B                   | 38570 |
| wgu44B                   | 38572 |
| whiff                    | 38610 |
| whiffBrick               | 38655 |
| whiffFire                | 38700 |
| whiteHouse               | 38715 |
| wildCard                 | 38745 |
| witchEight               | 38790 |
| witchFive                | 38835 |
| wm2xSeries               | 38880 |
| wm2xSeriesCas            | 38925 |
| wsr74c                   | 38950 |
| wsr74s                   | 38955 |
| wxr700c                  | 38960 |
| woodGage                 | 38970 |
| yardRake                 | 39015 |
| yewLoop                  | 39060 |
| ylc4                     | 39073 |
| yoYo                     | 39105 |
| zooPark1                 | 39125 |
| zd12                     | 39131 |
| zw06                     | 39150 |
| anAlq136V1               | 39200 |
| anAlq136V2               | 39201 |
| anAlq136V3               | 39202 |
| anAlq136V4               | 39203 |
| anAlq136V5               | 39204 |
| anAlq162V2               | 39210 |
| anAlq162V3               | 39211 |
| anAlq162V4               | 39212 |
| zhukM                    | 45300 |

Named by [EmitterSystem](#).

## EncodingTypeEnum32 ENUMERATION

Radio signal encoding type

Held as u32.

|                        |     |
|------------------------|-----|
| encoding8BitMuLaw      | 1   |
| cvsdPerMilStd188113    | 2   |
| adpcmPerCcittG721      | 3   |
| encoding16BitLinearPcm | 4   |
| encoding8BitLinearPcm  | 5   |
| vqVectorQuantization   | 6   |
| unavailableForUse      | 7   |
| gSMFullRateEtsi0610    | 8   |
| gSMHalfRateEtsi0620    | 9   |
| speexNarrowBand        | 10  |
| encoding16BitLinear    | 100 |
| PCM2SComplementLittle  |     |
| Endian                 |     |
| unavailableForUse2     | 255 |

Named by [AudioDataTypeStruct](#).

## EnvironmentDataCoordinateSystemEnum16 ENUMERATION

Environment data coordinate system

Held as u16.

|                                                            |   |
|------------------------------------------------------------|---|
| rightHandedCartesianLocalTopographicProjectionEastNorthUp  | 0 |
| leftHandedCartesianLocalTopographicProjectionEastNorthDown | 1 |
| latitudeLongitudeHeight                                    | 2 |
| latitudeLongitudeDepth                                     | 3 |

Named by [GriddedData](#).

## EnvironmentDataRepresentationEnum16 ENUMERATION

Environment data representation type

Held as u16.

|                      |   |
|----------------------|---|
| environmentDataType0 | 0 |
| environmentDataType1 | 1 |
| environmentDataType2 | 2 |

## EnvironmentDataSampleTypeEnum16 ENUMERATION

Environment data sample type

Held as u16.

|                                  |   |
|----------------------------------|---|
| environmentDataSampleTypeUnknown | 0 |
|----------------------------------|---|

Named by [GridDataStruct](#).

## EnvironmentGridAxisTypeEnum8 ENUMERATION

Grid axis type

Held as u8.

|                       |   |
|-----------------------|---|
| regularGridAxisType   | 0 |
| irregularGridAxisType | 1 |

## EnvironmentGridTypeEnum8 ENUMERATION

Environment data grid type

Held as u8.

|              |   |
|--------------|---|
| constantGrid | 0 |
| updatedGrid  | 1 |

Named by [GriddedData](#).

## EnvironmentModelTypeEnum8 ENUMERATION

Environment process model type

Held as u8.

|                         |   |
|-------------------------|---|
| environmentModelUnknown | 0 |
|-------------------------|---|

Named by [EnvironmentProcess](#).

## EnvironmentRecordTypeEnum32 ENUMERATION

Environment record type

Held as u32.

|                           |         |                           |            |
|---------------------------|---------|---------------------------|------------|
| cOMBICStateRecordType     | 256     | rectangularVolRecord1Type | 5242880    |
| flareStateRecordType      | 259     | rectangularVolRecord3Type | 83886080   |
| boundingSphereRecordType  | 65536   | pointRecord2Type          | 167772160  |
| uniformGeometryRecordType | 327680  | lineRecord2Type           | 201326592  |
| pointRecord1Type          | 655360  | sphereRecord2Type         | 218103808  |
| lineRecord1Type           | 786432  | ellipsoidRecord2Type      | 268435456  |
| sphereRecord1Type         | 851968  | coneRecord2Type           | 805306368  |
| ellipsoidRecord1Type      | 1048576 | rectangularVolRecord2Type | 1342177280 |
| coneRecord1Type           | 3145728 | gaussianPlumeRecordType   | 1610612736 |

gaussianPuffRecordType

1879048192

**EventTypeEnum32** ENUMERATION

Event type

Held as u32.

|                                    |    |
|------------------------------------|----|
| other                              | 0  |
| ranOutOfAmmunition                 | 2  |
| killedInAction                     | 3  |
| damage                             | 4  |
| mobilityDisabled                   | 5  |
| fireDisabled                       | 6  |
| ranOutOfFuel                       | 7  |
| entityInitialization               | 8  |
| requestForIndirectFireOrCASMission | 9  |
| indirectFireOrCASMission           | 10 |
| minefieldEntry                     | 11 |
| minefieldDetonation                | 12 |
| vehicleMasterPowerOn               | 13 |
| vehicleMasterPowerOff              | 14 |
| aggregateStateChangeRequested      | 15 |
| preventCollisionDetonation         | 16 |
| ownershipReport                    | 17 |

Named by [EventReport](#), [RecordQueryR](#), [RecordR](#).**ForceIdentifierEnum8** ENUMERATION

Force ID

Held as u8.

|            |    |
|------------|----|
| other      | 0  |
| friendly   | 1  |
| opposing   | 2  |
| neutral    | 3  |
| friendly2  | 4  |
| opposing2  | 5  |
| neutral2   | 6  |
| friendly3  | 7  |
| opposing3  | 8  |
| neutral3   | 9  |
| friendly4  | 10 |
| opposing4  | 11 |
| neutral4   | 12 |
| friendly5  | 13 |
| opposing5  | 14 |
| neutral5   | 15 |
| friendly6  | 16 |
| opposing6  | 17 |
| neutral6   | 18 |
| friendly7  | 19 |
| opposing7  | 20 |
| neutral7   | 21 |
| friendly8  | 22 |
| opposing8  | 23 |
| neutral8   | 24 |
| friendly9  | 25 |
| opposing9  | 26 |
| neutral9   | 27 |
| friendly10 | 28 |
| opposing10 | 29 |
| neutral10  | 30 |

Named by [netn.SEFacility](#), [AggregateEntity](#), [ArealObjectTransaction](#), [BreachObjectTransaction](#), [BreachableLinearObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObjectTransaction](#), [CraterObjectTransaction](#), [EnvironmentObject](#), [EnvironmentObjectTransaction](#), [ExhaustSmokeObjectTransaction](#), [LinearObjectTransaction](#), [Minefield](#), [MinefieldLaneMarkerObjectTransaction](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherLinearObjectTransaction](#), [OtherPointObjectTransaction](#), [PhysicalEntity](#), [PointObjectTransaction](#), [RibbonBridgeObjectTransaction](#), [StructureObjectTransaction](#).

## FormationEnum32 ENUMERATION

Formation

Held as u32.

|          |   |
|----------|---|
| other    | 0 |
| assembly | 1 |
| vee      | 2 |
| wedge    | 3 |
| line     | 4 |
| column   | 5 |

Named by [netn.MoveIntoFormation](#), [AggregateEntity](#).

## FuseTypeEnum16 ENUMERATION

Fuse (detonator)

Held as u16.

|                                      |      |                                          |      |
|--------------------------------------|------|------------------------------------------|------|
| other                                | 0    | timedNoseMountedVariableDelay            | 2800 |
| intelligentInfluence                 | 10   | timedLongDelaySide                       | 2900 |
| sensor                               | 20   | timedSelectableDelay                     | 2910 |
| selfDestruct                         | 30   | timedImpact                              | 2920 |
| ultraQuick                           | 40   | timedSequence                            | 2930 |
| body                                 | 50   | proximity                                | 3000 |
| deepIntrusion                        | 60   | proximityActiveLaser                     | 3100 |
| multifunction                        | 100  | proximityMagneticMagpolarity             | 3200 |
| pointDetonationPd                    | 200  | proximityActiveDopplerRadar              | 3300 |
| baseDetonationBd                     | 300  | proximityRadioFrequencyRF                | 3400 |
| contact                              | 1000 | proximityProgrammable                    | 3500 |
| contactInstantImpact                 | 1100 | proximityProgrammablePrefragmented       | 3600 |
| contactDelayed                       | 1200 | proximityInfrared                        | 3700 |
| contact10msDelay                     | 1201 | command                                  | 4000 |
| contact20msDelay                     | 1202 | commandElectronicRemotelySet             | 4100 |
| contact50msDelay                     | 1205 | altitude                                 | 5000 |
| contact60msDelay                     | 1206 | altitudeRadioAltimeter                   | 5100 |
| contact100msDelay                    | 1210 | altitudeAirBurst                         | 5200 |
| contact125msDelay                    | 1212 | depth                                    | 6000 |
| contact250msDelay                    | 1225 | acoustic                                 | 7000 |
| contactElectronicObliqueContact      | 1300 | pressure                                 | 8000 |
| contactGraze                         | 1400 | pressureDelay                            | 8010 |
| contactCrush                         | 1500 | inert                                    | 8100 |
| contactHydrostatic                   | 1600 | dummy                                    | 8110 |
| contactMechanical                    | 1700 | practice                                 | 8120 |
| contactChemical                      | 1800 | plugRepresenting                         | 8130 |
| contactPiezoelectric                 | 1900 | training                                 | 8150 |
| contactPointInitiating               | 1910 | pyrotechnic                              | 9000 |
| contactPointInitiatingBaseDetonating | 1920 | pyrotechnicDelay                         | 9010 |
| contactBaseDetonating                | 1930 | electroOptical                           | 9100 |
| contactBallisticCapAndBase           | 1940 | electroMechanical                        | 9110 |
| contactBase                          | 1950 | electroMechanicalNose                    | 9120 |
| contactNose                          | 1960 | strikerless                              | 9200 |
| contactFittedInStandoffProbe         | 1970 | strikerlessNoseImpact                    | 9210 |
| contactNonAligned                    | 1980 | strikerlessCompressionIgnition           | 9220 |
| timed                                | 2000 | compressionIgnition                      | 9300 |
| timedProgrammable                    | 2100 | compressionIgnitionStrikerlessNoseImpact | 9310 |
| timedBurnout                         | 2200 | percussion                               | 9400 |
| timedPyrotechnic                     | 2300 | percussionInstantaneous                  | 9410 |
| timedElectronic                      | 2400 | electronic                               | 9500 |
| timedBaseDelay                       | 2500 | electronicInternallyMounted              | 9510 |
| timedReinforcedNoseImpactDelay       | 2600 | electronicRangeSetting                   | 9520 |
| timedShortDelayImpact                | 2700 | electronicProgrammed                     | 9530 |
| timed10msDelay                       | 2701 | mechanical                               | 9600 |
| timed20msDelay                       | 2702 | mechanicalNose                           | 9610 |
| timed50msDelay                       | 2705 | mechanicalTail                           | 9620 |
| timed60msDelay                       | 2706 |                                          |      |
| timed100msDelay                      | 2710 |                                          |      |
| timed125msDelay                      | 2712 |                                          |      |
| timed250msDelay                      | 2725 |                                          |      |

Named by [MunitionDetonation](#), [WeaponFire](#).

HatchStateEnum32

ENUMERATION

Hatch state

Held as u32.

|                                                  |   |
|--------------------------------------------------|---|
| notApplicable                                    | 0 |
| primaryHatchIsClosed                             | 1 |
| primaryHatchIsPopped                             | 2 |
| primaryHatchIsPoppedAndPersonIsVisibleUnderHatch | 3 |
| primaryHatchIsOpen                               | 4 |
| primaryHatchIsOpenAndPersonIsVisible             | 5 |

Named by [Platform](#).

IffAlternateMode4Enum8

ENUMERATION

IFF alternate mode 4

Held as u8.

|                |   |
|----------------|---|
| other          | 0 |
| valid          | 1 |
| invalid        | 2 |
| noResponse     | 3 |
| unableToVerify | 4 |

Named by [NatoIFF](#).

IffApplicableModesEnum8

ENUMERATION

IFF applicable modes

Held as u8.

|          |   |
|----------|---|
| other    | 0 |
| allModes | 1 |

Named by [FundamentalParameterDataStruct](#).

IffOperationalParameter1Enum8

ENUMERATION

IFF operational parameter 1

Held as u8.

|       |   |
|-------|---|
| other | 0 |
|-------|---|

Named by [IFF](#).

IffOperationalParameter2Enum8

ENUMERATION

IFF operational parameter 2

Held as u8.

|       |   |
|-------|---|
| other | 0 |
|-------|---|

Named by [IFF](#).

IffSystemModeEnum8

ENUMERATION

IFF system mode

Held as u8.

|             |   |                     |   |
|-------------|---|---------------------|---|
| noStatement | 0 | lowOrLowSensitivity | 5 |
| off         | 1 |                     |   |
| standby     | 2 |                     |   |
| normal      | 3 |                     |   |
| emergency   | 4 |                     |   |

Named by [IFF](#).

**IfSystemNameEnum16**

ENUMERATION

IFF system name

Held as u16.

|                                                   |    |
|---------------------------------------------------|----|
| other                                             | 0  |
| markX                                             | 1  |
| markXII                                           | 2  |
| atcrbs                                            | 3  |
| soviet                                            | 4  |
| modeS                                             | 5  |
| markXXiiAtcrbs                                    | 6  |
| markXXiiAtcrbsModeS                               | 7  |
| ari5954                                           | 8  |
| ari5983                                           | 9  |
| genericRRB                                        | 10 |
| genericMarkXIIA                                   | 11 |
| genericMode5                                      | 12 |
| genericMarkXIIACombinedInterrogatorTransponderCit | 13 |
| genericMarkXIICombinedInterrogatorTransponderCit  | 14 |
| genericTCASIACASITransceiver                      | 15 |
| genericTCASIIACASIITransceiver                    | 16 |
| genericMarkXA                                     | 17 |
| genericMarkXSif                                   | 18 |

Named by [IFF](#).

**IfSystemTypeEnum16**

ENUMERATION

IFF system type

Held as u16.

|                                            |    |
|--------------------------------------------|----|
| other                                      | 0  |
| markTransponder                            | 1  |
| markInterrogator                           | 2  |
| sovietTransponder                          | 3  |
| sovietInterrogator                         | 4  |
| rRBTransponder                             | 5  |
| markXIIAInterrogator                       | 6  |
| mode5Interrogator                          | 7  |
| modeSInterrogator                          | 8  |
| markXIIATransponder                        | 9  |
| mode5Transponder                           | 10 |
| modeSTransponder                           | 11 |
| markXIIACombinedInterrogatorTransponderCit | 12 |
| markXIICombinedInterrogatorTransponderCit  | 13 |
| tcasACASTransceiver                        | 14 |

Named by [IFF](#).

**JTIDSPrimaryModeEnum8**

ENUMERATION

The primary mode of a JTIDS system [UID 173].

Held as u8.

|                      |   |
|----------------------|---|
| ntr                  | 1 |
| jTIDSUnitParticipant | 2 |

Named by [JTIDSTransmitterStruct](#).

**JTIDSSecondaryModeEnum8**

ENUMERATION

The JTIDS secondary mode of operation [UID 174].

Held as u8.

|      |   |                      |   |
|------|---|----------------------|---|
| none | 0 | netPositionReference | 1 |
|------|---|----------------------|---|



|                               |   |
|-------------------------------|---|
| primaryNavigationController   | 2 |
| secondaryNavigationController | 3 |

Named by [JTIDSTransmitterStruct](#).

## JTIDSSynchronizationStateEnum8 ENUMERATION

Describes the state of synchronization that the JTIDS system has achieved [UID 175].

Held as u8.

|                       |   |
|-----------------------|---|
| noStatement           | 0 |
| initialNetEntry       | 1 |
| coarseSynchronization | 2 |
| fineSynchronization   | 3 |

Named by [JTIDSTransmitterStruct](#).

## Link1111BEncryptionFlagEnum8 ENUMERATION

Link 11/11B Encryption Flag [UID 735].

Held as u8.

|                  |   |
|------------------|---|
| noEncryptionUsed | 0 |
| encryptionUsed   | 1 |

## Link1111BFidelityLevelEnum8 ENUMERATION

Link 11/11B Fidelity Level [UID 467].

Held as u8.

|                |   |
|----------------|---|
| fidelityLevel0 | 0 |
| fidelityLevel1 | 1 |
| fidelityLevel2 | 2 |

Named by [Link11BTransmitterStruct](#), [Link11TransmitterStruct](#).

## Link11BDataSignalingRateEnum8 ENUMERATION

Link 11B Data Signaling Rate [UID 740].

Held as u8.

|             |   |
|-------------|---|
| noStatement | 0 |
| n1200Bps    | 3 |
| n2400Bps    | 4 |
| n600Bps     | 5 |

## Link11BLinkStateEnum8 ENUMERATION

Link 11B Link State [UID 737].

Held as u8.

|             |   |
|-------------|---|
| noStatement | 0 |
| inactive    | 1 |
| ready       | 2 |
| active      | 3 |
| operational | 4 |

Named by [Link11BTransmitterStruct](#).

## Link11BMessageSubTypeEnum8 ENUMERATION

Link 11B Message Sub Type [UID 739].

Held as u8.

|                   |   |               |   |
|-------------------|---|---------------|---|
| noStatement       | 0 | standbySignal | 2 |
| transmissionFrame | 1 |               |   |

---

## Link11BModeOfOperationEnum16 ENUMERATION

Link 11B Mode of Operation [UID 738].

Held as u16.

|                           |   |
|---------------------------|---|
| noStatement               | 0 |
| fullTransmissionOfData    | 1 |
| limitedTransmissionOfData | 2 |
| receiveOnly               | 3 |

Named by [Link11BTransmitterStruct](#).

---

## Link11BModulationStandardEnum8 ENUMERATION

Link 11B Modulation Standard [UID 741].

Held as u8.

|             |   |
|-------------|---|
| noStatement | 0 |
| ccittv23    | 1 |

---

## Link11DataSignalingRateEnum8 ENUMERATION

Link 11 Data Signaling Rate [UID 732].

Held as u8.

|             |   |
|-------------|---|
| noStatement | 0 |
| n1364Bps    | 1 |
| n2250Bps    | 2 |

---

## Link11MessageSubTypeEnum8 ENUMERATION

Link 11 Message Sub Type [UID 730].

Held as u8.

|               |   |
|---------------|---|
| noStatement   | 0 |
| interrogation | 1 |
| dataStart     | 2 |
| data          | 3 |
| dataStop      | 4 |

---

## Link11ModeOfOperationEnum16 ENUMERATION

Link 11 Mode of Operation [UID 470].

Held as u16.

|                |   |
|----------------|---|
| noStatement    | 0 |
| netSync        | 1 |
| netTest        | 2 |
| rollCall       | 3 |
| shortBroadcast | 4 |
| broadcast      | 5 |

Named by [Link11TransmitterStruct](#).

---

## Link11SignalWaveformEnum8 ENUMERATION

Link 11 Signal Waveform [UID 734].

Held as u8.

|                                    |   |
|------------------------------------|---|
| noStatementCLEWFormat              | 0 |
| conventionalLinkElevenWaveformClew | 1 |
| singleToneLinkElevenWaveformSlew   | 2 |

### Link11TerminalModeEnum8 ENUMERATION

Link 11 Terminal Mode [UID 468].

Held as u8.

|                       |   |
|-----------------------|---|
| noStatement           | 0 |
| networkControlStation | 1 |
| picket                | 2 |

Named by [Link11TransmitterStruct](#).

### Link16VersionEnum8 ENUMERATION

Link 16 version [UID 800].

Held as u8.

|               |     |
|---------------|-----|
| noStatement   | 0   |
| milStd6016c   | 1   |
| milStd6016d   | 2   |
| milStd6016e   | 3   |
| milStd6016f   | 4   |
| milStd6016fc1 | 5   |
| sTANAG5516Ed3 | 103 |
| sTANAG5516Ed4 | 104 |
| sTANAG5516Ed5 | 105 |
| sTANAG5516Ed6 | 106 |
| sTANAG5516Ed8 | 108 |

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMMessageRadioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.VMFRadioSignal](#).

### MajorRFModulationTypeEnum16 ENUMERATION

Major classification of the modulation type.

Held as u16.

|                                 |   |
|---------------------------------|---|
| other                           | 0 |
| amplitude                       | 1 |
| amplitudeAndAngle               | 2 |
| angle                           | 3 |
| combination                     | 4 |
| pulse                           | 5 |
| unmodulated                     | 6 |
| carrierPhaseShiftModulationCpsm | 7 |

### MarkingEncodingEnum8 ENUMERATION

Marking character set

Held as u8.

|                 |   |
|-----------------|---|
| other           | 0 |
| ascii           | 1 |
| armyMarkingCCTT | 2 |
| digitChevron    | 3 |

Named by [AggregateMarkingStruct](#), [MarkingStruct](#).

### MinefieldFusingEnum32 ENUMERATION

Minefield fuse type

Held as u32.

|          |   |          |   |
|----------|---|----------|---|
| noFuse   | 0 | tiltRod  | 4 |
| other    | 1 | command  | 5 |
| pressure | 2 | tripWire | 6 |
| magnetic | 3 |          |   |

Named by [MineFusingStruct](#).

### MinefieldLaneEnum8 ENUMERATION

Minefield lane status

Held as u8.

|                            |   |
|----------------------------|---|
| minefieldHasActiveLane     | 0 |
| minefieldHasAnInactiveLane | 1 |

Named by [Minefield](#).

### MinefieldPaintSchemeEnum32 ENUMERATION

Minefield paint scheme

Held as u32.

|                    |    |
|--------------------|----|
| other              | 0  |
| standard           | 1  |
| camouflageDesert   | 2  |
| camouflageJungle   | 3  |
| camouflageSnow     | 4  |
| camouflageGravel   | 5  |
| camouflagePavement | 6  |
| camouflageSand     | 7  |
| naturalWood        | 8  |
| clear              | 9  |
| red                | 10 |
| blue               | 11 |
| green              | 12 |
| olive              | 13 |
| white              | 14 |
| tan                | 15 |
| black              | 16 |
| yellow             | 17 |
| brown              | 18 |

Named by [MinefieldPaintSchemeLengthlessArray](#).

### MinefieldProtocolEnum8 ENUMERATION

Minefield communication protocol mode

Held as u8.

|               |   |
|---------------|---|
| heartbeatMode | 0 |
| qRPMode       | 1 |

Named by [Minefield](#).

### MinefieldSensorTypeEnum32 ENUMERATION

Minefield sensor type

Held as u32.

|                                                       |      |
|-------------------------------------------------------|------|
| other                                                 | 0    |
| unaidedEyeActivelySearching                           | 4096 |
| unaidedEyeNotActivelySearching                        | 4097 |
| binoculars                                            | 4098 |
| imageIntensifier                                      | 4099 |
| hMMWVOccupantActivelySearching                        | 4100 |
| hMMWVOccupantNotActivelySearching                     | 4101 |
| truckOccupantActivelySearching                        | 4102 |
| truckOccupantNotActivelySearching                     | 4103 |
| trackedVehicleOccupantClosedHatchActivelySearching    | 4104 |
| trackedVehicleOccupantClosedHatchNotActivelySearching | 4105 |
| trackedVehicleOccupantOpenHatchActivelySearching      | 4106 |
| trackedVehicleOccupantOpenHatchNotActivelySearching   | 4107 |
| flirGeneric35                                         | 8192 |

|                             |       |
|-----------------------------|-------|
| flirGeneric812              | 8193  |
| flirAstamidsI               | 8194  |
| flirAstamidsIi              | 8195  |
| flirGstamids35              | 8196  |
| flirGstamids812             | 8197  |
| flirHstamids35              | 8198  |
| flirHstamids812             | 8199  |
| flirCobra35                 | 8200  |
| flirCobra812                | 8201  |
| radarGeneric                | 12288 |
| radarGenericGpr             | 12289 |
| radarGstamidsI              | 12290 |
| radarGstamidsIi             | 12291 |
| radarHstamidsI              | 12292 |
| radarHstamidsIi             | 12293 |
| magneticGeneric             | 16384 |
| magneticAnpss11             | 16385 |
| magneticAnpss12             | 16386 |
| magneticGstamids            | 16389 |
| laserGeneric                | 20480 |
| laserAstamids               | 20481 |
| sonarGeneric                | 24576 |
| physicalGenericProbe        | 28672 |
| physicalProbeMetalContent   | 28673 |
| physicalProbeNoMetalContent | 28674 |
| multispectralGeneric        | 32768 |

Named by [MinefieldSensorTypeLengthlessArray](#).

## MinefieldStatusEnum8 ENUMERATION

Minefield status

Held as u8.

|          |   |
|----------|---|
| active   | 0 |
| inactive | 1 |

Named by [Minefield](#).

## MinefieldTypeEnum8 ENUMERATION

Minefield type

Held as u8.

|                            |   |
|----------------------------|---|
| mixedAntiPersonnelAntiTank | 0 |
| pureAntiPersonnel          | 1 |
| pureAntiTank               | 2 |

Named by [Minefield](#).

## NomenclatureEnum16 ENUMERATION

The nomenclature for a specific radio system.

Held as u16.

|                                |   |
|--------------------------------|---|
| other                          | 0 |
| anArn118                       | 1 |
| anArn139                       | 2 |
| genericGroundFixedTransmitter  | 3 |
| genericGroundMobileTransmitter | 4 |

Named by [RadioTypeStruct](#).

## NomenclatureVersionEnum8 ENUMERATION

The naming convention applied to the radio system by the manufacturer/controlling agency.

Held as u8.

|                                             |    |
|---------------------------------------------|----|
| other                                       | 0  |
| jointElectronicsType                        | 1  |
| DesignationSystemJetdsNon<br>SpecificSeries |    |
| manufacturerDesignation                     | 2  |
| nationalDesignation                         | 3  |
| jETDSARCSets1                               | 11 |
| jETDSARCSets2                               | 12 |
| jETDSARCSets3                               | 13 |
| jETDSARCSets4                               | 14 |
| jETDSBRCSets1                               | 15 |
| jETDSBRCSets2                               | 16 |
| jETDSBRCSets3                               | 17 |
| jETDSBRCSets4                               | 18 |
| jETDSCRCs1                                  | 19 |
| jETDSCRCs2                                  | 20 |
| jETDSCRCs3                                  | 21 |
| jETDSCRCs4                                  | 22 |
| jETDSDRCs1                                  | 23 |
| jETDSDRCs2                                  | 24 |
| jETDSDRCs3                                  | 25 |
| jETDSDRCs4                                  | 26 |
| jETDSFRCSets1                               | 27 |
| jETDSFRCSets2                               | 28 |
| jETDSFRCSets3                               | 29 |
| jETDSFRCSets4                               | 30 |
| jETDSGRCSets1                               | 31 |
| jETDSGRCSets2                               | 32 |
| jETDSGRCSets3                               | 33 |
| jETDSGRCSets4                               | 34 |
| jETDSKRCSets1                               | 35 |
| jETDSKRCSets2                               | 36 |
| jETDSKRCSets3                               | 37 |
| jETDSKRCSets4                               | 38 |
| jETDSMRCSets1                               | 39 |
| jETDSMRCSets2                               | 40 |
| jETDSMRCSets3                               | 41 |
| jETDSMRCSets4                               | 42 |
| jETDSPRCs1                                  | 43 |
| jETDSPRCs2                                  | 44 |
| jETDSPRCs3                                  | 45 |
| jETDSPRCs4                                  | 46 |
| jETDSSRCs1                                  | 47 |
| jETDSSRCs2                                  | 48 |
| jETDSSRCs3                                  | 49 |
| jETDSSRCs4                                  | 50 |
| jETDSTRCSets1                               | 51 |
| jETDSTRCSets2                               | 52 |
| jETDSTRCSets3                               | 53 |
| jETDSTRCSets4                               | 54 |
| jETDSVRCs1                                  | 55 |
| jETDSVRCs2                                  | 56 |
| jETDSVRCs3                                  | 57 |
| jETDSVRCs4                                  | 58 |
| jETDSWRCs1                                  | 59 |
| jETDSWRCs2                                  | 60 |
| jETDSWRCs3                                  | 61 |
| jETDSWRCs4                                  | 62 |
| jETDSZRCs1                                  | 63 |
| jETDSZRCs2                                  | 64 |
| jETDSZRCs3                                  | 65 |
| jETDSZRCs4                                  | 66 |

Named by [RadioTypeStruct](#).

## OpacityCodeEnum32 ENUMERATION

Density of environmentals

Held as u32.

|           |   |
|-----------|---|
| clear     | 0 |
| hazy      | 1 |
| dense     | 2 |
| veryDense | 3 |
| opaque    | 4 |

Named by [EnvironmentalEntity](#).

## ParameterTypeEnum32 ENUMERATION

Parameter type

Held as u32.

|                   |   |
|-------------------|---|
| articulatedPart   | 0 |
| attachedPart      | 1 |
| separation        | 2 |
| entityType        | 3 |
| entityAssociation | 4 |

## PropulsionPlantEnum8 ENUMERATION

Propulsion plant configuration

Held as u8.

|                  |   |
|------------------|---|
| other            | 0 |
| dieselElectric   | 1 |
| diesel           | 2 |
| battery          | 3 |
| turbineReduction | 4 |
| steam            | 6 |
| gasTurbine       | 7 |

Named by [PropulsionNoise](#).

## PulseModulationTypeEnum16 ENUMERATION

Detailed modulation types for Pulse Modulation

Held as u16.

|                 |   |
|-----------------|---|
| other           | 0 |
| pulse           | 1 |
| xBandTACANPulse | 2 |
| yBandTACANPulse | 3 |

Named by [RFModulationTypeVariantStruct](#).

## RFModulationSystemTypeEnum16 ENUMERATION

The radio system type associated with this transmitter.

Held as u16.

|                         |    |
|-------------------------|----|
| other                   | 0  |
| generic                 | 1  |
| hq                      | 2  |
| hqii                    | 3  |
| hqia                    | 4  |
| sincgars                | 5  |
| ccttSincgars            | 6  |
| epIrsEnhancedPosition   | 7  |
| LocationReportingSystem |    |
| jtidsMids               | 8  |
| link11                  | 9  |
| link11B                 | 10 |
| IBandSATCOM             | 11 |
| enhancedSINCGARS73      | 12 |
| navigationAid           | 13 |

Named by [RadioTransmitter](#).

## RadioInputSourceEnum8 ENUMERATION

Radio input source

Held as u8.

|                   |    |
|-------------------|----|
| other             | 0  |
| pilot             | 1  |
| copilot           | 2  |
| firstOfficer      | 3  |
| driver            | 4  |
| loader            | 5  |
| gunner            | 6  |
| commander         | 7  |
| digitalDataDevice | 8  |
| intercom          | 9  |
| audioJammer       | 10 |
| dataJammer        | 11 |
| gPSJammer         | 12 |
| gPSMeaconer       | 13 |

Named by [RadioTransmitter](#).

## ReceiverOperationalStatusEnum16 ENUMERATION

The operational state of a radio receiver.

Held as u16.

|                   |   |
|-------------------|---|
| off               | 0 |
| onButNotReceiving | 1 |
| onAndReceiving    | 2 |

Named by [RadioReceiver](#).

## ReferenceSystemEnum8 ENUMERATION

The reference coordinate system used

Held as u8.

|                   |   |
|-------------------|---|
| worldCoordinates  | 1 |
| entityCoordinates | 2 |

Named by [BeamAntennaStruct](#), [SphericalHarmonicAntennaStruct](#).

## RepairResultEnum8 ENUMERATION

Result of repair

Held as u8.

|                              |   |
|------------------------------|---|
| other                        | 0 |
| repairEnded                  | 1 |
| invalidRepair                | 2 |
| repairInterrupted            | 3 |
| serviceCanceledByTheSupplier | 4 |

Named by [RepairResponse](#).

## RepairTypeEnum16 ENUMERATION

System repaired

Held as u16.

|                              |    |                     |     |
|------------------------------|----|---------------------|-----|
| noRepairsPerformed           | 0  | engineCoolantLeak   | 60  |
| allRequestedRepairsPerformed | 1  | fuelFilter          | 70  |
| motorOrEngine                | 10 | transmissionOilLeak | 80  |
| starter                      | 20 | engineOilLeak       | 90  |
| alternator                   | 30 | pumps               | 100 |
| generator                    | 40 | filters             | 110 |
| battery                      | 50 | transmission        | 120 |



|                                             |       |
|---------------------------------------------|-------|
| brakes                                      | 130   |
| suspensionSystem                            | 140   |
| oilFilter                                   | 150   |
| hull                                        | 1000  |
| airframe                                    | 1010  |
| truckBody                                   | 1020  |
| tankBody                                    | 1030  |
| trailerBody                                 | 1040  |
| turret                                      | 1050  |
| propeller                                   | 1500  |
| environmentalFilters                        | 1520  |
| wheels                                      | 1540  |
| tire                                        | 1550  |
| track                                       | 1560  |
| gunElevationDrive                           | 2000  |
| gunStabilizationSystem                      | 2010  |
| gunnersPrimarySightGps                      | 2020  |
| commandersExtensionToTheGPS                 | 2030  |
| loadingMechanism                            | 2040  |
| gunnersAuxiliarySight                       | 2050  |
| gunnersControlPanel                         | 2060  |
| gunnersControlAssemblyHandleHandles         | 2070  |
| commandersControlHandlesAssembly            | 2090  |
| commandersWeaponStation                     | 2100  |
| commandersIndependentThermalViewerCitr      | 2110  |
| generalWeapons                              | 2120  |
| fuelTransferPump                            | 4000  |
| fuelLines                                   | 4010  |
| gauges                                      | 4020  |
| generalFuelSystem                           | 4030  |
| electronicWarfareSystems                    | 4500  |
| detectionSystems                            | 4600  |
| electronicWarfareRadioFrequency             | 4610  |
| electronicWarfareMicrowave                  | 4620  |
| electronicWarfareInfrared                   | 4630  |
| electronicWarfareLaser                      | 4640  |
| rangeFinders                                | 4700  |
| rangeOnlyRadar                              | 4710  |
| laserRangeFinder                            | 4720  |
| electronicSystems                           | 4800  |
| electronicSystemsRadioFrequency             | 4810  |
| electronicSystemsMicrowave                  | 4820  |
| electronicSystemsInfrared                   | 4830  |
| electronicSystemsLaser                      | 4840  |
| radios                                      | 5000  |
| communicationSystems                        | 5010  |
| intercoms                                   | 5100  |
| encoders                                    | 5200  |
| encryptionDevices                           | 5250  |
| decoders                                    | 5300  |
| decryptionDevices                           | 5350  |
| computers                                   | 5500  |
| navigationAndControlSystems                 | 6000  |
| fireControlSystems                          | 6500  |
| airSupply                                   | 8000  |
| lifeSupportFilters                          | 8010  |
| lifeSupportWaterSupply                      | 8020  |
| refrigerationSystem                         | 8030  |
| chemicalBiologicalAndRadiologicalProtection | 8040  |
| waterWashDownSystems                        | 8050  |
| decontaminationSystems                      | 8060  |
| hydraulicSystemWaterSupply                  | 9000  |
| coolingSystem                               | 9010  |
| winches                                     | 9020  |
| catapults                                   | 9030  |
| cranes                                      | 9040  |
| launchers                                   | 9050  |
| lifeBoats                                   | 10000 |
| landingCraft                                | 10010 |
| ejectionSeats                               | 10020 |

Named by [netn.ArrayOfRepairTypeEnum](#), [RepairComplete](#).

## RequestStatusEnum32 ENUMERATION

Request status

Held as u32.

|                             |     |
|-----------------------------|-----|
| other                       | 0   |
| pending                     | 1   |
| executing                   | 2   |
| partiallyComplete           | 3   |
| complete                    | 4   |
| requestRejected             | 5   |
| retransmitRequestNow        | 6   |
| retransmitRequestLater      | 7   |
| invalidTimeParameters       | 8   |
| simulationTimeExceeded      | 9   |
| requestDone                 | 10  |
| taccsfLosReplyType1         | 100 |
| taccsfLosReplyType2         | 101 |
| joinExerciseRequestRejected | 201 |

Named by [ActionResponse](#), [ActionResponseR](#).

## ResponseFlagEnum16 ENUMERATION

Response flag

Held as u16.

|                       |   |
|-----------------------|---|
| other                 | 0 |
| ableToComply          | 1 |
| unableToComply        | 2 |
| pendingOperatorAction | 3 |

Named by [Acknowledge](#), [AcknowledgeR](#), [AttributeChangeResult](#), [AttributeChangeResultR](#), [CreateObjectResult](#), [CreateObjectResultR](#), [RemoveObjectResult](#), [RemoveObjectResultR](#).

## SISOSTD002VersionEnum8 ENUMERATION

SISO-STD-002 version [UID 736].

Held as u8.

|                |   |
|----------------|---|
| sisoStd0022006 | 0 |
| sisoStd0022021 | 1 |

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMessageradioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.VMFRadioSignal](#).

## ServiceTypeEnum8 ENUMERATION

Service type

Held as u8.

|          |   |
|----------|---|
| other    | 0 |
| resupply | 1 |
| repair   | 2 |

Named by [ServiceRequest](#).

## SpreadSpectrumEnum16 ENUMERATION

The type of spread spectrum characteristics employed by a transmitter.

Held as u16.

|                       |   |                     |   |
|-----------------------|---|---------------------|---|
| none                  | 0 | link11SpectrumType  | 3 |
| sINCGARSFrequencyHop  | 1 | link11BSpectrumType | 4 |
| jtidsMidsSpectrumType | 2 |                     |   |

## StanceCodeEnum32 ENUMERATION

Life form state

Held as u32.

|                      |    |
|----------------------|----|
| notApplicable        | 0  |
| uprightStandingStill | 1  |
| uprightWalking       | 2  |
| uprightRunning       | 3  |
| kneeling             | 4  |
| prone                | 5  |
| crawling             | 6  |
| swimming             | 7  |
| parachuting          | 8  |
| jumping              | 9  |
| sitting              | 10 |
| squatting            | 11 |
| crouching            | 12 |
| wading               | 13 |
| surrender            | 14 |
| detained             | 15 |

Named by [Lifeform](#).

## StationEnum32 ENUMERATION

Attached part station

Held as u32.

|                   |     |                   |     |                    |     |
|-------------------|-----|-------------------|-----|--------------------|-----|
| nothingEmpty      | 0   | fuselageStation40 | 551 | fuselageStation80  | 591 |
| fuselageStation1  | 512 | fuselageStation41 | 552 | fuselageStation81  | 592 |
| fuselageStation2  | 513 | fuselageStation42 | 553 | fuselageStation82  | 593 |
| fuselageStation3  | 514 | fuselageStation43 | 554 | fuselageStation83  | 594 |
| fuselageStation4  | 515 | fuselageStation44 | 555 | fuselageStation84  | 595 |
| fuselageStation5  | 516 | fuselageStation45 | 556 | fuselageStation85  | 596 |
| fuselageStation6  | 517 | fuselageStation46 | 557 | fuselageStation86  | 597 |
| fuselageStation7  | 518 | fuselageStation47 | 558 | fuselageStation87  | 598 |
| fuselageStation8  | 519 | fuselageStation48 | 559 | fuselageStation88  | 599 |
| fuselageStation9  | 520 | fuselageStation49 | 560 | fuselageStation89  | 600 |
| fuselageStation10 | 521 | fuselageStation50 | 561 | fuselageStation90  | 601 |
| fuselageStation11 | 522 | fuselageStation51 | 562 | fuselageStation91  | 602 |
| fuselageStation12 | 523 | fuselageStation52 | 563 | fuselageStation92  | 603 |
| fuselageStation13 | 524 | fuselageStation53 | 564 | fuselageStation93  | 604 |
| fuselageStation14 | 525 | fuselageStation54 | 565 | fuselageStation94  | 605 |
| fuselageStation15 | 526 | fuselageStation55 | 566 | fuselageStation95  | 606 |
| fuselageStation16 | 527 | fuselageStation56 | 567 | fuselageStation96  | 607 |
| fuselageStation17 | 528 | fuselageStation57 | 568 | fuselageStation97  | 608 |
| fuselageStation18 | 529 | fuselageStation58 | 569 | fuselageStation98  | 609 |
| fuselageStation19 | 530 | fuselageStation59 | 570 | fuselageStation99  | 610 |
| fuselageStation20 | 531 | fuselageStation60 | 571 | fuselageStation100 | 611 |
| fuselageStation21 | 532 | fuselageStation61 | 572 | fuselageStation101 | 612 |
| fuselageStation22 | 533 | fuselageStation62 | 573 | fuselageStation102 | 613 |
| fuselageStation23 | 534 | fuselageStation63 | 574 | fuselageStation103 | 614 |
| fuselageStation24 | 535 | fuselageStation64 | 575 | fuselageStation104 | 615 |
| fuselageStation25 | 536 | fuselageStation65 | 576 | fuselageStation105 | 616 |
| fuselageStation26 | 537 | fuselageStation66 | 577 | fuselageStation106 | 617 |
| fuselageStation27 | 538 | fuselageStation67 | 578 | fuselageStation107 | 618 |
| fuselageStation28 | 539 | fuselageStation68 | 579 | fuselageStation108 | 619 |
| fuselageStation29 | 540 | fuselageStation69 | 580 | fuselageStation109 | 620 |
| fuselageStation30 | 541 | fuselageStation70 | 581 | fuselageStation110 | 621 |
| fuselageStation31 | 542 | fuselageStation71 | 582 | fuselageStation111 | 622 |
| fuselageStation32 | 543 | fuselageStation72 | 583 | fuselageStation112 | 623 |
| fuselageStation33 | 544 | fuselageStation73 | 584 | fuselageStation113 | 624 |
| fuselageStation34 | 545 | fuselageStation74 | 585 | fuselageStation114 | 625 |
| fuselageStation35 | 546 | fuselageStation75 | 586 | fuselageStation115 | 626 |
| fuselageStation36 | 547 | fuselageStation76 | 587 | fuselageStation116 | 627 |
| fuselageStation37 | 548 | fuselageStation77 | 588 | fuselageStation117 | 628 |
| fuselageStation38 | 549 | fuselageStation78 | 589 | fuselageStation118 | 629 |
| fuselageStation39 | 550 | fuselageStation79 | 590 | fuselageStation119 | 630 |

|                    |     |                    |     |                    |     |
|--------------------|-----|--------------------|-----|--------------------|-----|
| fuselageStation120 | 631 | leftWingStation62  | 701 | rightWingStation4  | 771 |
| fuselageStation121 | 632 | leftWingStation63  | 702 | rightWingStation5  | 772 |
| fuselageStation122 | 633 | leftWingStation64  | 703 | rightWingStation6  | 773 |
| fuselageStation123 | 634 | leftWingStation65  | 704 | rightWingStation7  | 774 |
| fuselageStation124 | 635 | leftWingStation66  | 705 | rightWingStation8  | 775 |
| fuselageStation125 | 636 | leftWingStation67  | 706 | rightWingStation9  | 776 |
| fuselageStation126 | 637 | leftWingStation68  | 707 | rightWingStation10 | 777 |
| fuselageStation127 | 638 | leftWingStation69  | 708 | rightWingStation11 | 778 |
| fuselageStation128 | 639 | leftWingStation70  | 709 | rightWingStation12 | 779 |
| leftWingStation1   | 640 | leftWingStation71  | 710 | rightWingStation13 | 780 |
| leftWingStation2   | 641 | leftWingStation72  | 711 | rightWingStation14 | 781 |
| leftWingStation3   | 642 | leftWingStation73  | 712 | rightWingStation15 | 782 |
| leftWingStation4   | 643 | leftWingStation74  | 713 | rightWingStation16 | 783 |
| leftWingStation5   | 644 | leftWingStation75  | 714 | rightWingStation17 | 784 |
| leftWingStation6   | 645 | leftWingStation76  | 715 | rightWingStation18 | 785 |
| leftWingStation7   | 646 | leftWingStation77  | 716 | rightWingStation19 | 786 |
| leftWingStation8   | 647 | leftWingStation78  | 717 | rightWingStation20 | 787 |
| leftWingStation9   | 648 | leftWingStation79  | 718 | rightWingStation21 | 788 |
| leftWingStation10  | 649 | leftWingStation80  | 719 | rightWingStation22 | 789 |
| leftWingStation11  | 650 | leftWingStation81  | 720 | rightWingStation23 | 790 |
| leftWingStation12  | 651 | leftWingStation82  | 721 | rightWingStation24 | 791 |
| leftWingStation13  | 652 | leftWingStation83  | 722 | rightWingStation25 | 792 |
| leftWingStation14  | 653 | leftWingStation84  | 723 | rightWingStation26 | 793 |
| leftWingStation15  | 654 | leftWingStation85  | 724 | rightWingStation27 | 794 |
| leftWingStation16  | 655 | leftWingStation86  | 725 | rightWingStation28 | 795 |
| leftWingStation17  | 656 | leftWingStation87  | 726 | rightWingStation29 | 796 |
| leftWingStation18  | 657 | leftWingStation88  | 727 | rightWingStation30 | 797 |
| leftWingStation19  | 658 | leftWingStation89  | 728 | rightWingStation31 | 798 |
| leftWingStation20  | 659 | leftWingStation90  | 729 | rightWingStation32 | 799 |
| leftWingStation21  | 660 | leftWingStation91  | 730 | rightWingStation33 | 800 |
| leftWingStation22  | 661 | leftWingStation92  | 731 | rightWingStation34 | 801 |
| leftWingStation23  | 662 | leftWingStation93  | 732 | rightWingStation35 | 802 |
| leftWingStation24  | 663 | leftWingStation94  | 733 | rightWingStation36 | 803 |
| leftWingStation25  | 664 | leftWingStation95  | 734 | rightWingStation37 | 804 |
| leftWingStation26  | 665 | leftWingStation96  | 735 | rightWingStation38 | 805 |
| leftWingStation27  | 666 | leftWingStation97  | 736 | rightWingStation39 | 806 |
| leftWingStation28  | 667 | leftWingStation98  | 737 | rightWingStation40 | 807 |
| leftWingStation29  | 668 | leftWingStation99  | 738 | rightWingStation41 | 808 |
| leftWingStation30  | 669 | leftWingStation100 | 739 | rightWingStation42 | 809 |
| leftWingStation31  | 670 | leftWingStation101 | 740 | rightWingStation43 | 810 |
| leftWingStation32  | 671 | leftWingStation102 | 741 | rightWingStation44 | 811 |
| leftWingStation33  | 672 | leftWingStation103 | 742 | rightWingStation45 | 812 |
| leftWingStation34  | 673 | leftWingStation104 | 743 | rightWingStation46 | 813 |
| leftWingStation35  | 674 | leftWingStation105 | 744 | rightWingStation47 | 814 |
| leftWingStation36  | 675 | leftWingStation106 | 745 | rightWingStation48 | 815 |
| leftWingStation37  | 676 | leftWingStation107 | 746 | rightWingStation49 | 816 |
| leftWingStation38  | 677 | leftWingStation108 | 747 | rightWingStation50 | 817 |
| leftWingStation39  | 678 | leftWingStation109 | 748 | rightWingStation51 | 818 |
| leftWingStation40  | 679 | leftWingStation110 | 749 | rightWingStation52 | 819 |
| leftWingStation41  | 680 | leftWingStation111 | 750 | rightWingStation53 | 820 |
| leftWingStation42  | 681 | leftWingStation112 | 751 | rightWingStation54 | 821 |
| leftWingStation43  | 682 | leftWingStation113 | 752 | rightWingStation55 | 822 |
| leftWingStation44  | 683 | leftWingStation114 | 753 | rightWingStation56 | 823 |
| leftWingStation45  | 684 | leftWingStation115 | 754 | rightWingStation57 | 824 |
| leftWingStation46  | 685 | leftWingStation116 | 755 | rightWingStation58 | 825 |
| leftWingStation47  | 686 | leftWingStation117 | 756 | rightWingStation59 | 826 |
| leftWingStation48  | 687 | leftWingStation118 | 757 | rightWingStation60 | 827 |
| leftWingStation49  | 688 | leftWingStation119 | 758 | rightWingStation61 | 828 |
| leftWingStation50  | 689 | leftWingStation120 | 759 | rightWingStation62 | 829 |
| leftWingStation51  | 690 | leftWingStation121 | 760 | rightWingStation63 | 830 |
| leftWingStation52  | 691 | leftWingStation122 | 761 | rightWingStation64 | 831 |
| leftWingStation53  | 692 | leftWingStation123 | 762 | rightWingStation65 | 832 |
| leftWingStation54  | 693 | leftWingStation124 | 763 | rightWingStation66 | 833 |
| leftWingStation55  | 694 | leftWingStation125 | 764 | rightWingStation67 | 834 |
| leftWingStation56  | 695 | leftWingStation126 | 765 | rightWingStation68 | 835 |
| leftWingStation57  | 696 | leftWingStation127 | 766 | rightWingStation69 | 836 |
| leftWingStation58  | 697 | leftWingStation128 | 767 | rightWingStation70 | 837 |
| leftWingStation59  | 698 | rightWingStation1  | 768 | rightWingStation71 | 838 |
| leftWingStation60  | 699 | rightWingStation2  | 769 | rightWingStation72 | 839 |
| leftWingStation61  | 700 | rightWingStation3  | 770 | rightWingStation73 | 840 |

|                     |     |
|---------------------|-----|
| rightWingStation74  | 841 |
| rightWingStation75  | 842 |
| rightWingStation76  | 843 |
| rightWingStation77  | 844 |
| rightWingStation78  | 845 |
| rightWingStation79  | 846 |
| rightWingStation80  | 847 |
| rightWingStation81  | 848 |
| rightWingStation82  | 849 |
| rightWingStation83  | 850 |
| rightWingStation84  | 851 |
| rightWingStation85  | 852 |
| rightWingStation86  | 853 |
| rightWingStation87  | 854 |
| rightWingStation88  | 855 |
| rightWingStation89  | 856 |
| rightWingStation90  | 857 |
| rightWingStation91  | 858 |
| rightWingStation92  | 859 |
| rightWingStation93  | 860 |
| rightWingStation94  | 861 |
| rightWingStation95  | 862 |
| rightWingStation96  | 863 |
| rightWingStation97  | 864 |
| rightWingStation98  | 865 |
| rightWingStation99  | 866 |
| rightWingStation100 | 867 |
| rightWingStation101 | 868 |
| rightWingStation102 | 869 |
| rightWingStation103 | 870 |
| rightWingStation104 | 871 |
| rightWingStation105 | 872 |
| rightWingStation106 | 873 |
| rightWingStation107 | 874 |
| rightWingStation108 | 875 |
| rightWingStation109 | 876 |
| rightWingStation110 | 877 |
| rightWingStation111 | 878 |
| rightWingStation112 | 879 |
| rightWingStation113 | 880 |
| rightWingStation114 | 881 |
| rightWingStation115 | 882 |
| rightWingStation116 | 883 |
| rightWingStation117 | 884 |
| rightWingStation118 | 885 |
| rightWingStation119 | 886 |
| rightWingStation120 | 887 |
| rightWingStation121 | 888 |
| rightWingStation122 | 889 |
| rightWingStation123 | 890 |
| rightWingStation124 | 891 |
| rightWingStation125 | 892 |
| rightWingStation126 | 893 |
| rightWingStation127 | 894 |
| rightWingStation128 | 895 |
| m16a42Rifle         | 896 |
| m249Saw             | 897 |
| m60MachineGun       | 898 |
| m203GrenadeLauncher | 899 |
| m136At4             | 900 |
| m47Dragon           | 901 |
| aawsMJavelin        | 902 |
| m18a1ClaymoreMine   | 903 |
| mk19GrenadeLauncher | 904 |
| m2MachineGun        | 905 |
| otherAttachedParts  | 906 |

Named by [AttachedPartsStruct](#).

## StopFreezeReasonEnum8 ENUMERATION

Reason to stop

Held as u8.

|                           |   |
|---------------------------|---|
| other                     | 0 |
| recess                    | 1 |
| termination               | 2 |
| systemFailure             | 3 |
| securityViolation         | 4 |
| entityReconstitution      | 5 |
| stopForReset              | 6 |
| stopForRestart            | 7 |
| abortTrainingResumeTacOps | 8 |

Named by [StopFreeze](#), [StopFreezeR](#).

## TacticalDataLinkTypeEnum16 ENUMERATION

The type of tactical data link used to transmit a signal

Held as u16.

|                                                           |    |
|-----------------------------------------------------------|----|
| other                                                     | 0  |
| padil                                                     | 1  |
| nATOLink1                                                 | 2  |
| atdl1                                                     | 3  |
| link11BTadilb                                             | 4  |
| situationalAwarenessDataLinkSadl                          | 5  |
| link16LegacyFormatJtidsTadilJ                             | 6  |
| link16LegacyFormatJtidsFdlTadilJ                          | 7  |
| link11ATadila                                             | 8  |
| ijms                                                      | 9  |
| link4ATadilc                                              | 10 |
| link4C                                                    | 11 |
| tibs                                                      | 12 |
| atl                                                       | 13 |
| constantSource                                            | 14 |
| abbreviatedCommandAndControl                              | 15 |
| milstar                                                   | 16 |
| aths                                                      | 17 |
| othgold                                                   | 18 |
| tacelint                                                  | 19 |
| weaponsDataLinkAww13                                      | 20 |
| abbreviatedCommandAndControl1                             | 21 |
| enhancedPositionLocationReportingSystemEplrs              | 22 |
| positionLocationReportingSystemPlrs                       | 23 |
| sincgars                                                  | 24 |
| haveQuickI                                                | 25 |
| haveQuickII                                               | 26 |
| haveQuickIIASaturn                                        | 27 |
| intraFlightDataLink1                                      | 28 |
| intraFlightDataLink2                                      | 29 |
| improvedDataModemIdm                                      | 30 |
| airForceApplicationProgramDevelopmentAfapd                | 31 |
| cooperativeEngagementCapabilityCec                        | 32 |
| forwardAreaAirDefenseFaadDataLinkFdl                      | 33 |
| groundBasedDataLinkGbdI                                   | 34 |
| intraVehicularInfoSystemIvis                              | 35 |
| marineTacticalSystemMts                                   | 36 |
| tacticalFireDirectionSystemTacfire                        | 37 |
| integratedBroadcastServiceIbs                             | 38 |
| airborneInformationTransferAbit                           | 39 |
| advancedTacticalAirborneReconnaissanceSystemAtarsDataLink | 40 |
| battleGroupPassiveHorizonExtensionSystemBgphesDataLink    | 41 |
| commonHighBandwidthDataLinkChbdl                          | 42 |
| guardrailInteroperableDataLinkIdl                         | 43 |
| guardrailCommonSensorSystemOneCss1DataLink                | 44 |
| guardrailCommonSensorSystemTwoCss2DataLink                | 45 |
| guardrailCSS2MultiRoleDataLinkMrdl                        | 46 |

|                                                    |     |
|----------------------------------------------------|-----|
| guardrailCSS2DirectAirToSatelliteRelayDasrDataLink | 47  |
| lineOfSightLosDataLinkImplementationLOSTether      | 48  |
| lightweightCDLLwcdl                                | 49  |
| I52mSr71                                           | 50  |
| rivetReachRivetOwlDataLink                         | 51  |
| seniorSpan                                         | 52  |
| seniorSpur                                         | 53  |
| seniorStretch                                      | 54  |
| seniorYearInteroperableDataLinkIdI                 | 55  |
| spaceCDL                                           | 56  |
| tr1ModeMISTAirborneDataLink                        | 57  |
| kuBandSATCOMDataLinkImplementationUav              | 58  |
| missionEquipmentControlDataLinkMecdl               | 59  |
| radarDataTransmittingSetDataLink                   | 60  |
| surveillanceAndControlDataLinkScdl                 | 61  |
| tacticalUAVVideo                                   | 62  |
| uHFSATCOMDataLinkImplementationUav                 | 63  |
| tacticalCommonDataLinkTcdl                         | 64  |
| lowLevelAirPictureInterfaceLlapi                   | 65  |
| weaponsDataLinkAgm130                              | 66  |
| automaticIdentificationSystemAis                   | 67  |
| weaponsDataLinkAim120                              | 68  |
| gc3                                                | 99  |
| link16StandardizedFormatJtidsMidsTadilJ            | 100 |
| link16EnhancedDataRateEdrjtidsMidsTadilJ           | 101 |
| jtidsMIDSNetDataLoadTimsToms                       | 102 |
| link22                                             | 103 |
| aFIWCIADSCommunicationsLinks                       | 104 |
| f22IntraFlightDataLinkIfdl                         | 105 |
| lBandSATCOM                                        | 106 |
| tSAFCommunicationsLink                             | 107 |
| enhancedSINCGARS73                                 | 108 |
| f35MultifunctionAdvancedDataLinkMadl               | 109 |
| cursorOnTarget                                     | 110 |

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMMessageRadioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.TDLBinaryRadioSignal](#), [link16.VMFRadioSignal](#), [ApplicationSpecificRadioSignal](#), [DatabaseIndexRadioSignal](#), [RawBinaryRadioSignal](#).

## TimeSlotAllocationLevelEnum8 ENUMERATION

Time Slot Allocation (TSA) level of fidelity. See SISO-STD-002 for detailed descriptions and requirements [UID 172].

Held as u8.

|                      |   |
|----------------------|---|
| lowFidelityLevel0    | 0 |
| lowFidelityLevel1    | 1 |
| mediumFidelityLevel2 | 2 |
| mediumFidelityLevel3 | 3 |
| highFidelityLevel4   | 4 |

Named by [JTIDSTransmitterStruct](#).

## TrailingEffectsCodeEnum32 ENUMERATION

Size of trailing effect

Held as u32.

|             |   |
|-------------|---|
| noTrail     | 0 |
| smallTrail  | 1 |
| mediumTrail | 2 |
| largeTrail  | 3 |

Named by [PhysicalEntity](#).

## TransferTypeEnum8 ENUMERATION

Transfer type

Held as u8.

|                          |    |
|--------------------------|----|
| other                    | 0  |
| entityPush               | 1  |
| entityPull               | 2  |
| entitySwap               | 3  |
| environmentalProcessPush | 4  |
| environmentalProcessPull | 5  |
| notUsed                  | 6  |
| cancelTransfer           | 7  |
| manualPullTransferEntity | 8  |
| manualPullTransfer       | 9  |
| EnvironmentalProcess     |    |
| removeEntity             | 10 |

Named by [TransferControl](#).

## TransmitterOperationalStatusEnum8 ENUMERATION

The current operational state of a radio transmitter.

Held as u8.

|                      |   |
|----------------------|---|
| off                  | 0 |
| onButNotTransmitting | 1 |
| onAndTransmitting    | 2 |

Named by [netn.AisEquipmentItem](#), [netn.SetTransmitterStatus](#), [RadioTransmitter](#).

## UnmodulatedTypeEnum16 ENUMERATION

Detailed modulation types for an unmodulated carrier

Held as u16.

|                        |   |
|------------------------|---|
| other                  | 0 |
| continuousWaveEmission | 1 |

Named by [RFModulationTypeVariantStruct](#).

## UserProtocolEnum32 ENUMERATION

The ID of the user protocol used to transmit a radio signal.

Held as u32.

|                                      |      |                               |      |
|--------------------------------------|------|-------------------------------|------|
| ccsil                                | 1    | longbowAirborneTacfireMessage | 6663 |
| a2atdSincgarsErf                     | 5    | longbowGroundTacfireMessage   | 6664 |
| a2atdCac2                            | 6    | longbowAFAPDMessage           | 6665 |
| battleCommand                        | 20   | longbowErfMessage             | 6666 |
| aFIWCIADSTrackReport                 | 30   |                               |      |
| aFIWCIADSCommC2Message               | 31   |                               |      |
| aFIWCIADSGroundControlInterceptorGci | 32   |                               |      |
| Command                              |      |                               |      |
| aFIWCVoiceTextMessage                | 35   |                               |      |
| modSAFTextRadio                      | 177  |                               |      |
| ccttSincgarsErfLockout               | 200  |                               |      |
| ccttSincgarsErfHopset                | 201  |                               |      |
| ccttSincgarsOtar                     | 202  |                               |      |
| ccttSincgarsData                     | 203  |                               |      |
| modSAFFwaForwardAirController        | 546  |                               |      |
| modSAFThreatAdaC3                    | 832  |                               |      |
| f16MtcAfapd                          | 1000 |                               |      |
| f16MtcIdl                            | 1100 |                               |      |
| automaticIdentificationSystemAis     | 1371 |                               |      |
| modSAFArtilleryFireControl           | 4570 |                               |      |
| agts                                 | 5361 |                               |      |
| gc3                                  | 6000 |                               |      |
| wncpData                             | 6010 |                               |      |
| spokenTextMessage                    | 6020 |                               |      |
| longbowIdmMessage                    | 6661 |                               |      |
| comancheldmMessage                   | 6662 |                               |      |



Named by [ApplicationSpecificRadioSignal](#).

## VisibleSideLocationEnum32 ENUMERATION

Minefield lane marker visible side

Held as u32.

|                  |   |
|------------------|---|
| leftSideVisible  | 0 |
| rightSideVisible | 1 |
| bothSideVisible  | 2 |

Named by [MinefieldLaneMarkerStruct](#).

## WarheadTypeEnum16 ENUMERATION

Warhead

Held as u16.

|                           |      |                        |      |
|---------------------------|------|------------------------|------|
| other                     | 0    | dustyVX                | 8320 |
| cargoVariableSubmunitions | 10   | gaTabun                | 8325 |
| fuelAirExplosive          | 20   | thickenedGATabun       | 8330 |
| glassBeads                | 30   | dustyGATabun           | 8335 |
| warhead1um                | 31   | gbSarin                | 8340 |
| warhead5um                | 32   | thickenedGBSarin       | 8345 |
| warhead10um               | 33   | dustyGBSarin           | 8350 |
| highExplosive             | 1000 | gdSoman                | 8355 |
| hePlastic                 | 1100 | thickenedGDSoman       | 8360 |
| heIncendiary              | 1200 | dustyGDSoman           | 8365 |
| heFragmentation           | 1300 | gf                     | 8370 |
| heAntitank                | 1400 | thickenedGF            | 8375 |
| heBomblets                | 1500 | dustyGF                | 8380 |
| heShapedCharge            | 1600 | biological             | 9000 |
| heContinuousRod           | 1610 | biologicalVirus        | 9100 |
| heTungstenBall            | 1615 | biologicalBacteria     | 9200 |
| heBlastFragmentation      | 1620 | biologicalRickettsia   | 9300 |
| heSteerableDartswithHE    | 1625 | biologicalGenetically  | 9400 |
| heDarts                   | 1630 | ModifiedMicroOrganisms |      |
| heFlechettes              | 1635 | biologicalToxin        | 9500 |
| heDirectedFragmentation   | 1640 |                        |      |
| heSemiArmorPiercing       | 1645 |                        |      |
| heShapedCharge            | 1650 |                        |      |
| Fragmentation             |      |                        |      |
| heSemiArmorPiercing       | 1655 |                        |      |
| Fragmentation             |      |                        |      |
| heHollowCharge            | 1660 |                        |      |
| heDoubleHollowCharge      | 1665 |                        |      |
| heGeneralPurpose          | 1670 |                        |      |
| heBlastPenetrator         | 1675 |                        |      |
| heRodPenetrator           | 1680 |                        |      |
| heAntipersonnel           | 1685 |                        |      |
| smoke                     | 2000 |                        |      |
| illumination              | 3000 |                        |      |
| practice                  | 4000 |                        |      |
| kinetic                   | 5000 |                        |      |
| mines                     | 6000 |                        |      |
| nuclear                   | 7000 |                        |      |
| nuclearIMT                | 7010 |                        |      |
| chemicalGeneral           | 8000 |                        |      |
| chemicalBlisterAgent      | 8100 |                        |      |
| hdMustard                 | 8110 |                        |      |
| thickenedHDMustard        | 8115 |                        |      |
| dustyHDMustard            | 8120 |                        |      |
| chemicalBloodAgent        | 8200 |                        |      |
| acHcn                     | 8210 |                        |      |
| ckCnci                    | 8215 |                        |      |
| cgPhosgene                | 8220 |                        |      |
| chemicalNerveAgent        | 8300 |                        |      |
| vx                        | 8310 |                        |      |
| thickenedVX               | 8315 |                        |      |

Named by [MunitionDetonation](#), [WeaponFire](#).

## WeaponStateEnum32 ENUMERATION

Weapon position

Held as u32.

|                |   |
|----------------|---|
| noWeapon       | 0 |
| stowed         | 1 |
| deployed       | 2 |
| firingPosition | 3 |

Named by [Lifeform](#).

### Variant records

|                                                     |                                                                                                                                                         |     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AntennaPatternVariantStruct</a>         | Specifies the radiation pattern from the antenna, its orientation in space, and the polarization of the radiation.                                      | 278 |
| <a href="#">EnvironmentRecVariantStruct</a>         | Record specifying either a geometry or a state record                                                                                                   | 278 |
| <a href="#">GridAxisTypeVariantStruct</a>           | Record specifying either regular (fixed spacing) or irregular (variable spacing) axis data                                                              | 279 |
| <a href="#">GridDataRepresentationVariantStruct</a> | Record specifying either type 0 or type 1 or type 2 data representation                                                                                 | 279 |
| <a href="#">ParameterValueVariantStruct</a>         | Variant record specifying the type of articulation parameter (articulated or attached part), and its type and value.                                    | 279 |
| <a href="#">RFModulationTypeVariantStruct</a>       | Specifies the major modulation type as well as certain detailed information specific to the type.                                                       | 279 |
| <a href="#">SpatialVariantStruct</a>                | Variant Record for a single spatial attribute.                                                                                                          | 280 |
| <a href="#">SpreadSpectrumVariantStruct</a>         | Identifies the actual spread spectrum technique in use.                                                                                                 | 280 |
| <a href="#">StationNameLocationVariantStruct</a>    | The station name at which the constituent part is located. In case of 'On Station', the alternative specifies its location relative to the host object. | 280 |

## AntennaPatternVariantStruct VARIANT

Specifies the radiation pattern from the antenna, its orientation in space, and the polarization of the radiation.

| Type                                           | Description                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">BeamAntennaStruct</a>              | Specifies the direction, pattern, and polarization of radiation from a radio transmitter's antenna, represented with respect to a beam coordinate system.              |
| <a href="#">SphericalHarmonicAntennaStruct</a> | Specifies the direction and radiation pattern from a radio transmitter's antenna, represented with respect to the world coordinate system or entity coordinate system. |

Named by [AntennaPatternVariantStructLengthlessArray](#).

## EnvironmentRecVariantStruct VARIANT

Record specifying either a geometry or a state record

| Type                                    | Description                                                |
|-----------------------------------------|------------------------------------------------------------|
| <a href="#">WorldLocationStruct</a>     | Specifies Point 1 geometry record alternative              |
| <a href="#">Point2GeomRecStruct</a>     | Specifies Point 2 geometry record alternative              |
| <a href="#">Line1GeomRecStruct</a>      | Specifies Line 1 geometry record alternative               |
| <a href="#">Line2GeomRecStruct</a>      | Specifies Line 2 geometry record alternative               |
| <a href="#">Sphere1GeomRecStruct</a>    | Specifies Bounding Sphere geometry record alternative      |
| <a href="#">Sphere1GeomRecStruct</a>    | Specifies Sphere 1 geometry record alternative             |
| <a href="#">Sphere2GeomRecStruct</a>    | Specifies Sphere 2 geometry record alternative             |
| <a href="#">Ellipsoid1GeomRecStruct</a> | Specifies Ellipsoid 1 geometry record alternative          |
| <a href="#">Ellipsoid2GeomRecStruct</a> | Specifies Ellipsoid 2 geometry record alternative          |
| <a href="#">Cone1GeomRecStruct</a>      | Specifies Cone 1 geometry record alternative               |
| <a href="#">Cone2GeomRecStruct</a>      | Specifies Cone 2 geometry record alternative               |
| <a href="#">RectVol1GeomRecStruct</a>   | Specifies Rectangular Volume 1 geometry record alternative |
| <a href="#">RectVol2GeomRecStruct</a>   | Specifies Rectangular Volume 2 geometry record alternative |
| <a href="#">GaussPlumeGeomRecStruct</a> | Specifies Gaussian Plume geometry record alternative       |

| Type                                   | Description                                                |
|----------------------------------------|------------------------------------------------------------|
| <a href="#">GaussPuffGeomRecStruct</a> | Specifies Gaussian Puff geometry record alternative        |
| <a href="#">UniformGeomRecStruct</a>   | Specifies Uniform geometry record alternative              |
| <a href="#">COMBICStateRecStruct</a>   | Specifies COMBIC state record alternative                  |
| <a href="#">FlareStateRecStruct</a>    | Specifies Flare state record alternative                   |
| <a href="#">RectVol3GeomRecStruct</a>  | Specifies Rectangular Volume 3 geometry record alternative |

Named by [EnvironmentRecStruct](#).

## GridAxisTypeVariantStruct VARIANT

Record specifying either regular (fixed spacing) or irregular (variable spacing) axis data

| Type                                    | Description                                                  |
|-----------------------------------------|--------------------------------------------------------------|
| <a href="#">IrregularGridAxisStruct</a> | Specifies irregular (variable spacing) axis data alternative |

Named by [GridAxisStruct](#).

## GridDataRepresentationVariantStruct VARIANT

Record specifying either type 0 or type 1 or type 2 data representation

| Type                                 | Description                                      |
|--------------------------------------|--------------------------------------------------|
| <a href="#">GridValueType0Struct</a> | Specifies type 0 data representation alternative |
| <a href="#">GridValueType1Struct</a> | Specifies type 1 data representation alternative |
| <a href="#">GridValueType2Struct</a> | Specifies type 2 data representation alternative |

Named by [GridDataStruct](#).

## ParameterValueVariantStruct VARIANT

Variant record specifying the type of articulation parameter (articulated or attached part), and its type and value.

| Type                                   | Description                          |
|----------------------------------------|--------------------------------------|
| <a href="#">ArticulatedPartsStruct</a> | Alternative for an articulated part. |
| <a href="#">AttachedPartsStruct</a>    | Alternative for an attached part.    |

Named by [ArticulatedParameterStruct](#).

## RFModulationTypeVariantStruct VARIANT

Specifies the major modulation type as well as certain detailed information specific to the type.

| Type                                               | Description                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------------------|
| <a href="#">AmplitudeModulationTypeEnum16</a>      | Detailed modulation type for the amplitude major modulation type.           |
| <a href="#">AmplitudeAngleModulationTypeEnum16</a> | Detailed modulation type for the amplitude and angle major modulation type. |
| <a href="#">AngleModulationTypeEnum16</a>          | Detailed modulation type for the angle major modulation type.               |
| <a href="#">CombinationModulationTypeEnum16</a>    | Detailed modulation type for the combination major modulation type.         |
| <a href="#">PulseModulationTypeEnum16</a>          | Detailed modulation type for the pulse major modulation type.               |
| <a href="#">UnmodulatedTypeEnum16</a>              | Detailed modulation type for the unmodulated major modulation type.         |

Named by [RadioTransmitter](#).

## SpatialVariantStruct VARIANT

Variant Record for a single spatial attribute.

| Type                                | Description                                                                                                                                                |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">SpatialStaticStruct</a> | Variant for representing a static object.                                                                                                                  |
| <a href="#">SpatialFPStruct</a>     | Variant for representing an object with a constant velocity (or low acceleration) linear motion in world coordinates.                                      |
| <a href="#">SpatialRPStruct</a>     | Variant for representing an object with a constant velocity (or low acceleration) linear motion, including rotation information, in world coordinates.     |
| <a href="#">SpatialRVStruct</a>     | Variant for representing an object with high speed or maneuvering at any speed, including rotation information, in world coordinates.                      |
| <a href="#">SpatialFVStruct</a>     | Variant for representing an object with high speed or maneuvering at any speed in world coordinates.                                                       |
| <a href="#">SpatialFPStruct</a>     | Variant for representing an object with a constant velocity (or low acceleration) linear motion in body axis coordinates.                                  |
| <a href="#">SpatialRPStruct</a>     | Variant for representing an object with a constant velocity (or low acceleration) linear motion, including rotation information, in body axis coordinates. |
| <a href="#">SpatialRVStruct</a>     | Variant for representing an object with high speed or maneuvering at any speed, including rotation information, in body axis coordinates.                  |
| <a href="#">SpatialFVStruct</a>     | Variant for representing an object with high speed or maneuvering at any speed in body axis coordinates.                                                   |

Named by [BaseEntity](#).

## SpreadSpectrumVariantStruct VARIANT

Identifies the actual spread spectrum technique in use.

| Type                                     | Description                                                                |
|------------------------------------------|----------------------------------------------------------------------------|
| <a href="#">SINCGARSModulationStruct</a> | Modulation parameters for SINCGARS radio system.                           |
| <a href="#">JTIDSTransmitterStruct</a>   | Modulation parameters for Link 16 radio system according to SISO-STD-002.  |
| <a href="#">Link11TransmitterStruct</a>  | Modulation parameters for Link 11 radio system according to SISO-STD-005.  |
| <a href="#">Link11BTransmitterStruct</a> | Modulation parameters for Link 11B radio system according to SISO-STD-005. |

Named by [RadioTransmitter](#).

## StationNameLocationVariantStruct VARIANT

The station name at which the constituent part is located. In case of 'On Station', the alternative specifies its location relative to the host object.

| Type                                       | Description                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------|
| <a href="#">RelativePositionStruct</a>     | The location of the constituent part object relative to the host object entity coordinate system. |
| <a href="#">RelativeRangeBearingStruct</a> | The location of the constituent part object relative to the host object in polar coordinates.     |

Named by [NamedLocationStruct](#).

### Arrays

|                                                            |                                                                                                                                                              |     |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AntennaPatternVariantStructLengthlessArray</a> | Represents an antenna's radiation pattern, its orientation in space, and the polarization of the radiation.                                                  | 282 |
| <a href="#">ArticulatedParameterStructLengthlessArray</a>  | Dynamic array of ArticulatedParameterStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements. | 282 |
| <a href="#">AttributeValuePairStructArray1Plus</a>         | Array of AttributeValuePairStruct.                                                                                                                           | 282 |
| <a href="#">BreachStructLengthlessArray</a>                | Specifies a breach linear object (a collection of segments)                                                                                                  | 282 |
| <a href="#">BreachableSegmentStructLengthlessArray</a>     | Specifies a breachable linear object (a collection of segments)                                                                                              | 282 |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| BreachedStatusArray8                          | Specifies the breached appearance for each individual segment portion of length = BreachLength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 283 |
| ClockTimeStructLengthlessArray                | Dynamic array of ClockTimeStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 283 |
| CoefficientsLengthlessArray1Plus              | Represents the power distribution from the antenna as the coefficients of a spherical harmonic expansion. The highest order of the expansion can be determined by the number of coefficients in the array.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 283 |
| DatumIdentifierLengthlessArray                | Array of DatumIdentifierEnum32.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 283 |
| DepthMeterFloat32LengthlessArray              | Specifies ground - snow - water burial depth offset for each mine in a collection of mines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 283 |
| EntityTypeStructLengthlessArray               | Dynamic array of EntityTypeStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 283 |
| EnvironmentRecStructArray                     | Specifies environment records as a collection of geometry and state records                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 283 |
| ExhaustSmokeStructLengthlessArray             | Specifies an exhaust smoke linear object (a collection of smoke segments)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 284 |
| FixedDatumStructLengthlessArray               | Array of FixedDatumStructs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 284 |
| Float32Array1Plus                             | Generic dynamic array of Float32 elements, containing at least one element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 284 |
| FundamentalParameterDataStructLengthlessArray | Array of Fundamental Parameter Data records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 284 |
| GridAxisStructLengthlessArray                 | Specifies detailed information for a collection of grid axes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 284 |
| GridDataStructLengthlessArray                 | Specifies detailed information for a collection of grid data representations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 284 |
| Integer16Array1Plus                           | Generic dynamic array of Integer16 elements, containing at least one element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 284 |
| MarkingArray11                                | String of characters represented by an 11 element character string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 284 |
| MarkingArray31                                | String of characters represented by a 31 element character string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 285 |
| MineDielectricDifferenceLengthlessArray       | Specifies local dielectric difference between the mine and the surrounding soil (reflectance) for each mine in a collection of mines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 285 |
| MineFusingStructLengthlessArray               | Specifies the type of primary fuse, the type of the secondary fuse and the anti-handling device status for a collection of mines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 285 |
| MineIdentifierLengthlessArray                 | Identifies the mine entity identifier for each mine in a collection of mines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 285 |
| MinefieldLaneMarkerStructLengthlessArray      | Specifies a minefield lane marker (a collection of segments)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 285 |
| MinefieldPaintSchemeLengthlessArray           | Specifies the camouflage scheme/color for each mine in a collection of mines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 285 |
| MinefieldSensorTypeLengthlessArray            | Specifies a collection of minefield sensor types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 285 |
| MissingRecordNumbersLengthlessArray1Plus      | Specifies missing record numbers as a collection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 285 |
| OctetArray                                    | Generic dynamic array of Octet elements, may also contain no elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 286 |
| OctetArray1Plus                               | Generic dynamic array of Octet elements, containing at least one element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 286 |
| OctetArray2                                   | Generic array of two Octet elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 286 |
| OctetArray3                                   | Generic array of three Octet elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 286 |
| OctetArray4                                   | Generic array of four Octet elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 286 |
| OctetArray7                                   | Generic array of seven Octet elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 286 |
| OctetArray8                                   | Generic array of eight Octet elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 286 |
| OctetPadding32Array                           | Generic dynamic array of meaningless Octet elements, to align the subsequent data structure to the next 32 bit octet boundary value (OBV). The array is encoded without array length, containing zero to three elements.                                                                                                                                                                                                                                                                                                                                                                                                                                              | 286 |
| OctetPadding64Array                           | Generic dynamic array of meaningless Octet elements, to align the subsequent data structure to the next 64 bit octet boundary value (OBV). The array is encoded without array length, containing zero to seven elements.                                                                                                                                                                                                                                                                                                                                                                                                                                              | 287 |
| OrientationStructLengthlessArray              | Dynamic array of OrientationStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 287 |
| PerimeterPointStructLengthlessArray           | Specifies the location of perimeter points (collection)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 287 |
| PropulsionSystemDataStructLengthlessArray     | A set of Propulsion System Data descriptions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 287 |
| RPRUserDefinedTag                             | The array shall be at least 8 bytes (octets) in size, which shall contain the time according to the DIS time stamp field format (IEEE 1278.1-1995 section 5.2.31) converted into hexadecimal American Standard Code for Information Interchange (ASCII) character representation (0-9 and A-F), with leading zeros included. The ordering of the characters shall be in accordance with section 5.1.1 of IEEE 1278.1-1995, that is most significant octet first, with the most significant bits first (i.e. the character for bits 4-7 precedes the character for bits 0-3). This encoding is equivalent to the result of the 'C'-statement 'sprintf(UserTag, "%08X", | 287 |

|                                                |                                                                                                                                                             |     |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|                                                | DIStimestamp),' where 'DIStimestamp' is represented in native format.                                                                                       |     |
|                                                | More user-supplied information may be included, starting from the 9th character, as specified in the federation agreements.                                 |     |
| RTObjectIdArray                                | Set of ID's of registered object instances.                                                                                                                 | 287 |
| RecordSetStructArray1Plus                      | Array of RecordSetStruct                                                                                                                                    | 287 |
| RecordStructArray                              | Array of RecordStruct                                                                                                                                       | 288 |
| ShaftDataStructLengthlessArray1Plus            | Array of propulsion shaft states, one per shaft                                                                                                             | 288 |
| SignalDataLengthlessArray1Plus                 | The audio or digital data conveyed in a radio transmission.                                                                                                 | 288 |
| SilentAggregateStructLengthlessArray           | Set of silent aggregates (aggregates not registered in the federation).                                                                                     | 288 |
| SilentEntityStructLengthlessArray              | A set of silent entities (entities not registered in the federation).                                                                                       | 288 |
| SupplyStructLengthlessArray                    | A list of supply types and the number of each being offered or requested.                                                                                   | 288 |
| TemperatureDegreeCelsiusFloat32LengthlessArray | Specifies thermal contrast for each mine in a collection of mines                                                                                           | 288 |
| UnsignedInteger16Array1Plus                    | Generic dynamic array of UnsignedInteger16 elements, containing at least one element.                                                                       | 288 |
| UnsignedInteger32LengthlessArray               | Generic dynamic array of UnsignedInteger32 elements, may also contain no elements. The array is encoded without array length, containing only the elements. | 289 |
| UnsignedInteger64Array1Plus                    | Generic dynamic array of UnsignedInteger64 elements, containing at least one element.                                                                       | 289 |
| UnsignedInteger8LengthlessArray                | Generic dynamic array of UnsignedInteger8 elements, may also contain no elements. The array is encoded without array length, containing only the elements.  | 289 |
| VariableDatumStructArray                       | Array of VariableDatumStruct                                                                                                                                | 289 |
| VariableDatumStructLengthlessArray             | Set of additional data associated with an aggregate.                                                                                                        | 289 |
| VectoringNozzleSystemDataStructLengthlessArray | A set of Vectoring Nozzle System Data description.                                                                                                          | 289 |
| WorldLocationStructLengthlessArray             | Dynamic array of WorldLocationStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.       | 289 |

## AntennaPatternVariantStructLengthlessArray SEQUENCE

Represents an antenna's radiation pattern, its orientation in space, and the polarization of the radiation.

**sequence**<AntennaPatternVariantStruct> AntennaPatternVariantStructLengthlessArray;

Named by [RadioTransmitter](#).

## ArticulatedParameterStructLengthlessArray SEQUENCE

Dynamic array of ArticulatedParameterStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.

**sequence**<ArticulatedParameterStruct> ArticulatedParameterStructLengthlessArray;

Named by [MunitionDetonation](#), [PhysicalEntity](#).

## AttributeValuePairStructArray1Plus SEQUENCE

Array of AttributeValuePairStruct.

**sequence**<AttributeValuePairStruct> AttributeValuePairStructArray1Plus;

Named by [AttributeChangeRequest](#), [AttributeChangeRequestR](#), [AttributeChangeResult](#), [AttributeChangeResultR](#), [CreateObjectRequest](#), [CreateObjectRequestR](#).

## BreachStructLengthlessArray SEQUENCE

Specifies a breach linear object (a collection of segments)

**sequence**<BreachStruct> BreachStructLengthlessArray;

Named by [BreachObject](#), [BreachObjectTransaction](#).

## BreachableSegmentStructLengthlessArray SEQUENCE

Specifies a breachable linear object (a collection of segments)

```
sequence<BreachableSegmentStruct> BreachableSegmentStructLengthlessArray;
```

Named by [BreachableLinearObject](#), [BreachableLinearObjectTransaction](#).

## BreachedStatusArray8 SEQUENCE

Specifies the breached appearance for each individual segment portion of length = BreachLength

```
sequence<BreachedStatusEnum8, 8> BreachedStatusArray8;
```

Named by [BreachableSegmentStruct](#).

## ClockTimeStructLengthlessArray SEQUENCE

Dynamic array of ClockTimeStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<ClockTimeStruct> ClockTimeStructLengthlessArray;
```

Named by [MinefieldData](#).

## CoefficientsLengthlessArray1Plus SEQUENCE

Represents the power distribution from the antenna as the coefficients of a spherical harmonic expansion. The highest order of the expansion can be determined by the number of coefficients in the array.

```
sequence<Float32> CoefficientsLengthlessArray1Plus;
```

Named by [SphericalHarmonicAntennaStruct](#).

## DatumIdentifierLengthlessArray SEQUENCE

Array of DatumIdentifierEnum32.

```
sequence<DatumIdentifierEnum32> DatumIdentifierLengthlessArray;
```

Named by [DataQuery](#), [DataQueryR](#), [RecordQueryR](#).

## DepthMeterFloat32LengthlessArray SEQUENCE

Specifies ground - snow - water burial depth offset for each mine in a collection of mines

```
sequence<DepthMeterFloat32> DepthMeterFloat32LengthlessArray;
```

Named by [MinefieldData](#).

## EntityTypeStructLengthlessArray SEQUENCE

Dynamic array of EntityTypeStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<EntityTypeStruct> EntityTypeStructLengthlessArray;
```

Named by [Minefield](#).

## EnvironmentRecStructArray SEQUENCE

Specifies environment records as a collection of geometry and state records

```
sequence<EnvironmentRecStruct> EnvironmentRecStructArray;
```

Named by [EnvironmentProcess](#).



---

## ExhaustSmokeStructLengthlessArray SEQUENCE

Specifies an exhaust smoke linear object (a collection of smoke segments)

```
sequence<ExhaustSmokeStruct> ExhaustSmokeStructLengthlessArray;
```

Named by [ExhaustSmokeObject](#), [ExhaustSmokeObjectTransaction](#).

---

## FixedDatumStructLengthlessArray SEQUENCE

Array of FixedDatumStructs.

```
sequence<FixedDatumStruct> FixedDatumStructLengthlessArray;
```

Named by [ActionRequest](#), [ActionRequestR](#), [ActionResponse](#), [ActionResponseR](#), [Data](#), [DataR](#), [EventReport](#), [SetData](#), [SetDataR](#).

---

## Float32Array1Plus SEQUENCE

Generic dynamic array of Float32 elements, containing at least one element.

```
sequence<Float32> Float32Array1Plus;
```

Named by [GridValueType2Struct](#).

---

## FundamentalParameterDataStructLengthlessArray SEQUENCE

Array of Fundamental Parameter Data records.

```
sequence<FundamentalParameterDataStruct> FundamentalParameterDataStructLengthlessArray;
```

Named by [IFF](#).

---

## GridAxisStructLengthlessArray SEQUENCE

Specifies detailed information for a collection of grid axes

```
sequence<GridAxisStruct> GridAxisStructLengthlessArray;
```

Named by [GriddedData](#).

---

## GridDataStructLengthlessArray SEQUENCE

Specifies detailed information for a collection of grid data representations

```
sequence<GridDataStruct> GridDataStructLengthlessArray;
```

Named by [GriddedData](#).

---

## Integer16Array1Plus SEQUENCE

Generic dynamic array of Integer16 elements, containing at least one element.

```
sequence<Integer16> Integer16Array1Plus;
```

Named by [GridValueType1Struct](#).

---

## MarkingArray11 SEQUENCE

String of characters represented by an 11 element character string.

```
sequence<Octet, 11> MarkingArray11;
```

Named by [MarkingStruct](#).

---



---

## MarkingArray31 SEQUENCE

String of characters represented by a 31 element character string.

```
sequence<Octet, 31> MarkingArray31;
```

Named by [AggregateMarkingStruct](#).

---

## MineDielectricDifferenceLengthlessArray SEQUENCE

Specifies local dielectric difference between the mine and the surrounding soil (reflectance) for each mine in a collection of mines

```
sequence<MineDielectricDifference> MineDielectricDifferenceLengthlessArray;
```

Named by [MinefieldData](#).

---

## MineFusingStructLengthlessArray SEQUENCE

Specifies the type of primary fuse, the type of the secondary fuse and the anti-handling device status for a collection of mines

```
sequence<MineFusingStruct> MineFusingStructLengthlessArray;
```

Named by [MinefieldData](#).

---

## MineIdentifierLengthlessArray SEQUENCE

Identifies the mine entity identifier for each mine in a collection of mines

```
sequence<MineIdentifier> MineIdentifierLengthlessArray;
```

Named by [MinefieldData](#).

---

## MinefieldLaneMarkerStructLengthlessArray SEQUENCE

Specifies a minefield lane marker (a collection of segments)

```
sequence<MinefieldLaneMarkerStruct> MinefieldLaneMarkerStructLengthlessArray;
```

Named by [MinefieldLaneMarkerObject](#), [MinefieldLaneMarkerObjectTransaction](#).

---

## MinefieldPaintSchemeLengthlessArray SEQUENCE

Specifies the camouflage scheme/color for each mine in a collection of mines

```
sequence<MinefieldPaintSchemeEnum32> MinefieldPaintSchemeLengthlessArray;
```

Named by [MinefieldData](#).

---

## MinefieldSensorTypeLengthlessArray SEQUENCE

Specifies a collection of minefield sensor types

```
sequence<MinefieldSensorTypeEnum32> MinefieldSensorTypeLengthlessArray;
```

Named by [MinefieldData](#), [MinefieldQuery](#).

---

## MissingRecordNumbersLengthlessArray1Plus SEQUENCE

Specifies missing record numbers as a collection

```
sequence<UnsignedInteger8> MissingRecordNumbersLengthlessArray1Plus;
```

Named by [MinefieldResponseNACK](#).

---

## OctetArray SEQUENCE

Generic dynamic array of Octet elements, may also contain no elements.

```
sequence<Octet> OctetArray;
```

Named by [AttributeValuePairStruct](#), [RecordStruct](#).

## OctetArray1Plus SEQUENCE

Generic dynamic array of Octet elements, containing at least one element.

```
sequence<Octet> OctetArray1Plus;
```

Named by [GridValueType0Struct](#).

## OctetArray2 SEQUENCE

Generic array of two Octet elements.

```
sequence<Octet, 2> OctetArray2;
```

Named by [FlareStateRecStruct](#).

## OctetArray3 SEQUENCE

Generic array of three Octet elements.

```
sequence<Octet, 3> OctetArray3;
```

Named by [FundamentalParameterDataStruct](#), [MineFusingStruct](#), [SphericalHarmonicAntennaStruct](#).

## OctetArray4 SEQUENCE

Generic array of four Octet elements.

```
sequence<Octet, 4> OctetArray4;
```

Named by [BreachStruct](#), [COMBICStateRecStruct](#), [Cone1GeomRecStruct](#), [Cone2GeomRecStruct](#), [Ellipsoid2GeomRecStruct](#), [GaussPlumeGeomRecStruct](#), [Point2GeomRecStruct](#), [RectVol2GeomRecStruct](#), [Sphere1GeomRecStruct](#).

## OctetArray7 SEQUENCE

Generic array of seven Octet elements.

```
sequence<Octet, 7> OctetArray7;
```

Named by [BreachableSegmentStruct](#).

## OctetArray8 SEQUENCE

Generic array of eight Octet elements.

```
sequence<Octet, 8> OctetArray8;
```

Named by [UniformGeomRecStruct](#).

## OctetPadding32Array SEQUENCE

Generic dynamic array of meaningless Octet elements, to align the subsequent data structure to the next 32 bit octet boundary value (OBV). The array is encoded without array length, containing zero to three elements.

```
sequence<Octet> OctetPadding32Array;
```

Named by [AttributeValuePairStruct](#), [GridValueType0Struct](#), [GridValueType1Struct](#), [RecordStruct](#).

## OctetPadding64Array SEQUENCE

Generic dynamic array of meaningless Octet elements, to align the subsequent data structure to the next 64 bit octet boundary value (OBV). The array is encoded without array length, containing zero to seven elements.

```
sequence<Octet> OctetPadding64Array;
```

Named by [IrregularGridAxisStruct](#).

## OrientationStructLengthlessArray SEQUENCE

Dynamic array of OrientationStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<OrientationStruct> OrientationStructLengthlessArray;
```

Named by [MinefieldData](#).

## PerimeterPointStructLengthlessArray SEQUENCE

Specifies the location of perimeter points (collection)

```
sequence<PerimeterPointStruct> PerimeterPointStructLengthlessArray;
```

Named by [Minefield](#), [MinefieldQuery](#).

## PropulsionSystemDataStructLengthlessArray SEQUENCE

A set of Propulsion System Data descriptions.

```
sequence<PropulsionSystemDataStruct> PropulsionSystemDataStructLengthlessArray;
```

Named by [PhysicalEntity](#).

## RPRUserDefinedTag SEQUENCE

The array shall be at least 8 bytes (octets) in size, which shall contain the time according to the DIS time stamp field format (IEEE 1278.1-1995 section 5.2.31) converted into hexadecimal American Standard Code for Information Interchange (ASCII) character representation (0-9 and A-F), with leading zeros included. The ordering of the characters shall be in accordance with section 5.1.1 of IEEE 1278.1-1995, that is most significant octet first, with the most significant bits first (i.e. the character for bits 4-7 precedes the character for bits 0-3). This encoding is equivalent to the result of the 'C'-statement 'sprintf(UserTag, '%08X', DIStimestamp),' where 'DIStimestamp' is represented in native format. More user-supplied information may be included, starting from the 9th character, as specified in the federation agreements.

```
sequence<u8> RPRUserDefinedTag;
```

## RTIobjectIdArray SEQUENCE

Set of ID's of registered object instances.

```
sequence<string> RTIobjectIdArray;
```

Named by [ActionRequestToObject](#), [ActionRequestToObjectR](#), [AggregateEntity](#), [AttributeChangeRequest](#), [AttributeChangeRequestR](#), [JammerBeam](#), [RadarBeam](#), [RemoveObjectRequest](#), [RemoveObjectRequestR](#).

## RecordSetStructArray1Plus SEQUENCE

Array of RecordSetStruct

```
sequence<RecordSetStruct> RecordSetStructArray1Plus;
```

Named by [RecordR](#), [SetRecordR](#), [TransferControl](#).

## RecordStructArray SEQUENCE

Array of RecordStruct

```
sequence<RecordStruct> RecordStructArray;
```

Named by [RecordSetStruct](#).

## ShaftDataStructLengthlessArray1Plus SEQUENCE

Array of propulsion shaft states, one per shaft

```
sequence<ShaftDataStruct> ShaftDataStructLengthlessArray1Plus;
```

Named by [PropulsionNoise](#).

## SignalDataLengthlessArray1Plus SEQUENCE

The audio or digital data conveyed in a radio transmission.

```
sequence<Octet> SignalDataLengthlessArray1Plus;
```

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMMessageRadioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.TDLBinaryRadioSignal](#), [link16.VMFRadioSignal](#), [ApplicationSpecificRadioSignal](#), [AudioDataTypeStruct](#), [RawBinaryRadioSignal](#).

## SilentAggregateStructLengthlessArray SEQUENCE

Set of silent aggregates (aggregates not registered in the federation).

```
sequence<SilentAggregateStruct> SilentAggregateStructLengthlessArray;
```

Named by [AggregateEntity](#).

## SilentEntityStructLengthlessArray SEQUENCE

A set of silent entities (entities not registered in the federation).

```
sequence<SilentEntityStruct> SilentEntityStructLengthlessArray;
```

Named by [AggregateEntity](#).

## SupplyStructLengthlessArray SEQUENCE

A list of supply types and the number of each being offered or requested.

```
sequence<SupplyStruct> SupplyStructLengthlessArray;
```

Named by [ResupplyOffer](#), [ResupplyReceived](#), [ServiceRequest](#).

## TemperatureDegreeCelsiusFloat32LengthlessArray SEQUENCE

Specifies thermal contrast for each mine in a collection of mines

```
sequence<TemperatureDegreeCelsiusFloat32> TemperatureDegreeCelsiusFloat32LengthlessArray;
```

Named by [MinefieldData](#).

## UnsignedInteger16Array1Plus SEQUENCE

Generic dynamic array of UnsignedInteger16 elements, containing at least one element.

```
sequence<UnsignedInteger16> UnsignedInteger16Array1Plus;
```

Named by [IrregularGridAxisStruct](#).

UnsignedInteger32LengthlessArray

SEQUENCE

Generic dynamic array of UnsignedInteger32 elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<UnsignedInteger32> UnsignedInteger32LengthlessArray;
```

Named by [SilentEntityStruct](#).

UnsignedInteger64Array1Plus

SEQUENCE

Generic dynamic array of UnsignedInteger64 elements, containing at least one element.

```
sequence<UnsignedInteger64> UnsignedInteger64Array1Plus;
```

Named by [VariableDatumStruct](#).

UnsignedInteger8LengthlessArray

SEQUENCE

Generic dynamic array of UnsignedInteger8 elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<UnsignedInteger8> UnsignedInteger8LengthlessArray;
```

Named by [MinefieldData](#).

VariableDatumStructArray

SEQUENCE

Array of VariableDatumStruct

```
sequence<VariableDatumStruct> VariableDatumStructArray;
```

Named by [ActionRequest](#), [ActionRequestR](#), [ActionResponse](#), [ActionResponseR](#), [Comment](#), [Data](#), [DataR](#), [EventReport](#), [SetData](#), [SetDataR](#).

VariableDatumStructLengthlessArray

SEQUENCE

Set of additional data associated with an aggregate.

```
sequence<VariableDatumStruct> VariableDatumStructLengthlessArray;
```

Named by [AggregateEntity](#).

VectoringNozzleSystemDataStructLengthlessArray

SEQUENCE

A set of Vectoring Nozzle System Data description.

```
sequence<VectoringNozzleSystemDataStruct> VectoringNozzleSystemDataStructLengthlessArray;
```

Named by [PhysicalEntity](#).

WorldLocationStructLengthlessArray

SEQUENCE

Dynamic array of WorldLocationStruct elements, may also contain no elements. The array is encoded without array length, containing only the elements.

```
sequence<WorldLocationStruct> WorldLocationStructLengthlessArray;
```

Named by [ArealObject](#), [ArealObjectTransaction](#), [MinefieldData](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#).

Quantities

|                                                          |                                                                                                                                                        |     |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AccelerationMeterPerSecondSquaredFloat32</a> | Linear acceleration vector composed of SI base units. Based on the Linear Acceleration Vector record as specified in IEEE 1278.1-1995 section 5.2.33b. | 290 |
| <a href="#">AngleDegreeFloat32</a>                       | Angle, based on unit degree (of arc), unit symbol °.                                                                                                   | 290 |
| <a href="#">AngleRadianFloat32</a>                       | Angle, based on SI derived unit radian, unit symbol rad.                                                                                               | 290 |

|                                                       |                                                                                                                                                |     |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">AngularVelocityRadianPerSecondFloat32</a> | Angular velocity vector composed of SI base units. Based on the Angular Velocity Vector record as specified in IEEE 1278.1-1995 section 5.2.2. | 290 |
| <a href="#">DepthMeterFloat32</a>                     | Depth, based on SI base unit meter, unit symbol m.                                                                                             | 290 |
| <a href="#">FrequencyHertzFloat32</a>                 | Frequency, based on SI derived unit hertz, unit symbol Hz.                                                                                     | 291 |
| <a href="#">FrequencyHertzUnsignedInteger64</a>       | Frequency of a radio transmission, in hertz.                                                                                                   | 291 |
| <a href="#">InterrogationsPerSecondFloat32</a>        | Number of interrogations per second.                                                                                                           | 291 |
| <a href="#">LengthMeterFloat32</a>                    | Length, based on SI base unit meter, unit symbol m.                                                                                            | 291 |
| <a href="#">MassKilogramFloat32</a>                   | Mass, based on SI base unit kilogram, unit symbol kg.                                                                                          | 291 |
| <a href="#">MeterFloat32</a>                          | Datatype based on SI base unit meter, unit symbol m.                                                                                           | 291 |
| <a href="#">MeterFloat64</a>                          | Datatype based on SI base unit meter, unit symbol m.                                                                                           | 291 |
| <a href="#">RevolutionsPerMinuteFloat32</a>           | Rotation speed expressed in revolutions per minute.                                                                                            | 291 |
| <a href="#">RevolutionsPerMinuteInteger16</a>         | Frequency of rotation, expressed in revolutions per minute.                                                                                    | 292 |
| <a href="#">TemperatureDegreeCelsiusFloat32</a>       | Temperature, based on SI derived unit degree Celsius, unit symbol °C.                                                                          | 292 |
| <a href="#">TimeMicrosecondFloat32</a>                | Time, based on SI base unit second, expressed in microsecond, unit symbol μs.                                                                  | 292 |
| <a href="#">TimeMillisecondUnsignedInteger32</a>      | Time, based on SI base unit second, expressed in millisecond, unit symbol ms.                                                                  | 292 |
| <a href="#">TimeSecondInteger32</a>                   | Time, based on SI base unit second, unit symbol s.                                                                                             | 292 |
| <a href="#">TimeSecondUnsignedInteger16</a>           | Time, based on SI base unit second, unit symbol s.                                                                                             | 292 |
| <a href="#">VelocityDecimeterPerSecondInteger16</a>   | Velocity/Speed measured in decimeter per second.                                                                                               | 292 |
| <a href="#">VelocityMeterPerSecondFloat32</a>         | Speed/Velocity in meter per second.                                                                                                            | 292 |
| <a href="#">WavelengthMicronFloat32</a>               | Wavelength expressed in micrometer.                                                                                                            | 293 |

## AccelerationMeterPerSecondSquaredFloat32 QUANTITY

Linear acceleration vector composed of SI base units. Based on the Linear Acceleration Vector record as specified in IEEE 1278.1-1995 section 5.2.33b.

```
quantity<f32, m/s²> AccelerationMeterPerSecondSquaredFloat32;
```

Named by [AccelerationVectorStruct](#).

## AngleDegreeFloat32 QUANTITY

Angle, based on unit degree (of arc), unit symbol °.

```
quantity<f32, °> AngleDegreeFloat32;
```

Named by [VectoringNozzleSystemDataStruct](#).

## AngleRadianFloat32 QUANTITY

Angle, based on SI derived unit radian, unit symbol rad.

```
quantity<f32, rad> AngleRadianFloat32;
```

Named by [netn.MagicMove](#), [netn.Move](#), [netn.MoveIntoFormation](#), [netn.PositionStatusReport](#), [netn.TurnToHeading](#), [netn.TurnToOrientation](#), [netn.UnderAttackStatusReport](#), [ActiveSonarBeam](#), [BeamAntennaStruct](#), [EmitterBeam](#), [IFF](#), [OrientationStruct](#), [RelativeRangeBearingStruct](#).

## AngularVelocityRadianPerSecondFloat32 QUANTITY

Angular velocity vector composed of SI base units. Based on the Angular Velocity Vector record as specified in IEEE 1278.1-1995 section 5.2.2.

```
quantity<f32, rad/s> AngularVelocityRadianPerSecondFloat32;
```

Named by [AngularVelocityVectorStruct](#).

## DepthMeterFloat32 QUANTITY

Depth, based on SI base unit meter, unit symbol m.

```
quantity<f32, m> DepthMeterFloat32;
```

Named by [DepthMeterFloat32LengthlessArray](#).

## FrequencyHertzFloat32 QUANTITY

Frequency, based on SI derived unit hertz, unit symbol Hz.

```
quantity<f32, hz> FrequencyHertzFloat32;
```

Named by [EmitterBeam](#), [FundamentalParameterDataStruct](#), [RadioTransmitter](#).

## FrequencyHertzUnsignedInteger64 QUANTITY

Frequency of a radio transmission, in hertz.

```
quantity<u64, hz> FrequencyHertzUnsignedInteger64;
```

Named by [RadioTransmitter](#).

## InterrogationsPerSecondFloat32 QUANTITY

Number of interrogations per second.

```
quantity<f32, hz> InterrogationsPerSecondFloat32;
```

Named by [FundamentalParameterDataStruct](#).

## LengthMeterFloat32 QUANTITY

Length, based on SI base unit meter, unit symbol m.

```
quantity<f32, m> LengthMeterFloat32;
```

Named by [netn.AisMessage19](#), [netn.AisMessage21](#), [netn.AisMessage24](#), [netn.AisMessage5](#), [netn.AisStaticAndVoyageData](#), [netn.ArrayOfSigmas6](#), [RelativeRangeBearingStruct](#), [WeaponFire](#).

## MassKilogramFloat32 QUANTITY

Mass, based on SI base unit kilogram, unit symbol kg.

```
quantity<f32, kg> MassKilogramFloat32;
```

Named by [netn.AgentMassStruct](#), [netn.CBRNRelease](#), [Collision](#), [CollisionElastic](#).

## MeterFloat32 QUANTITY

Datatype based on SI base unit meter, unit symbol m.

```
quantity<f32, m> MeterFloat32;
```

Named by [netn.FollowEntity](#), [netn.FormationStruct](#), [netn.GeodeticCircle](#), [netn.IceStruct](#), [netn.LandSurfaceCondition](#), [netn.LayerStruct](#), [netn.MoveIntoFormation](#), [netn.NetworkDeviceGenericTransmitterCharacteristicsStruct](#), [netn.PhysicalGenericNetworkStruct](#), [netn.SnowStruct](#), [netn.TroposphereLayerCondition](#), [netn.WaterSurface](#), [netn.WaterSurfaceCondition](#), [netn.WaveStruct](#), [netn.Weather](#), [netn.WeatherCondition](#), [DimensionStruct](#), [EntityCoordinateVectorStruct](#), [PerimeterPointStruct](#), [PlumeDimensionStruct](#), [RelativePositionStruct](#).

## MeterFloat64 QUANTITY

Datatype based on SI base unit meter, unit symbol m.

```
quantity<f64, m> MeterFloat64;
```

Named by [WorldLocationStruct](#).

## RevolutionsPerMinuteFloat32 QUANTITY

Rotation speed expressed in revolutions per minute.

```
quantity<f32, rpm> RevolutionsPerMinuteFloat32;
```

Named by [PropulsionSystemDataStruct](#).

## RevolutionsPerMinuteInteger16 QUANTITY

Frequency of rotation, expressed in revolutions per minute.

```
quantity<i16, rpm> RevolutionsPerMinuteInteger16;
```

Named by [ShaftDataStruct](#).

## TemperatureDegreeCelsiusFloat32 QUANTITY

Temperature, based on SI derived unit degree Celsius, unit symbol °C.

```
quantity<f32, °C> TemperatureDegreeCelsiusFloat32;
```

Named by [netn.LandSurfaceCondition](#), [netn.ReleaseDynamicsStruct](#), [netn.SubsurfaceLayer](#), [netn.SubsurfaceLayerCondition](#), [netn.TroposphereLayerCondition](#), [netn.WaterSurfaceCondition](#), [netn.Weather](#), [netn.WeatherCondition](#), [TemperatureDegreeCelsiusFloat32LengthlessArray](#).

## TimeMicrosecondFloat32 QUANTITY

Time, based on SI base unit second, expressed in microsecond, unit symbol µs.

```
quantity<f32, µs> TimeMicrosecondFloat32;
```

Named by [EmitterBeam](#), [FundamentalParameterDataStruct](#).

## TimeMillisecondUnsignedInteger32 QUANTITY

Time, based on SI base unit second, expressed in millisecond, unit symbol ms.

```
quantity<u32, ms> TimeMillisecondUnsignedInteger32;
```

Named by [netn.ConnectionReceiverStruct](#), [netn.IncomingConnectionStruct](#), [netn.LinkStatusStruct](#), [netn.PhysicalGenericNetworkStruct](#), [DatabaseIndexRadioSignal](#).

## TimeSecondInteger32 QUANTITY

Time, based on SI base unit second, unit symbol s.

```
quantity<i32, s> TimeSecondInteger32;
```

Named by [netn.ATP45HazardArea](#), [netn.CBRNDetector](#), [netn.CBRNRelease](#), [netn.CBRNSensor](#), [netn.CheckPoint](#), [netn.DecontaminationStation](#), [netn.TreatmentStruct](#), [netn.WaveStruct](#), [SINCGARSModulationStruct](#).

## TimeSecondUnsignedInteger16 QUANTITY

Time, based on SI base unit second, unit symbol s.

```
quantity<u16, s> TimeSecondUnsignedInteger16;
```

Named by [Link11TransmitterStruct](#).

## VelocityDecimeterPerSecondInteger16 QUANTITY

Velocity/Speed measured in decimeter per second.

```
quantity<u16, dm/s> VelocityDecimeterPerSecondInteger16;
```

Named by [PhysicalEntity](#).

## VelocityMeterPerSecondFloat32 QUANTITY

Speed/Velocity in meter per second.

```
quantity<f32, m/s> VelocityMeterPerSecondFloat32;
```



Named by [netn.AisMessage1](#), [netn.AisMessage18](#), [netn.AisMessage19](#), [netn.AisMessage2](#), [netn.AisMessage27](#), [netn.AisMessage3](#), [netn.AisMessage9](#), [netn.AisNavigationData](#), [netn.CurrentStruct](#), [netn.DispositionStruct](#), [netn.PositionStatusReport](#), [netn.SetOrderedSpeed](#), [netn.SpottedEntity](#), [netn.WindStruct](#), [DimensionRateStruct](#), [PlumeDimensionRateStruct](#), [VelocityVectorStruct](#).

## WavelengthMicronFloat32 QUANTITY

Wavelength expressed in micrometer.

```
quantity<f32, μm> WavelengthMicronFloat32;
```

Named by [Designator](#).

### Aliases

|                                                           |                                                                                                                                                                                                                                                                                               |     |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">BitRateBitPerSecondUnsignedInteger32</a>      | Rate of transmission, in bits per second.                                                                                                                                                                                                                                                     | 293 |
| <a href="#">BitsUnsignedInteger16</a>                     | Transmission size, in number of bits.                                                                                                                                                                                                                                                         | 293 |
| <a href="#">ClockTimeHourInteger32</a>                    | Time past on the clock in full hours since a specified point in time.                                                                                                                                                                                                                         | 293 |
| <a href="#">Float32</a>                                   | Single-precision floating point number.                                                                                                                                                                                                                                                       | 294 |
| <a href="#">Float64</a>                                   | Double-precision floating point number.                                                                                                                                                                                                                                                       | 294 |
| <a href="#">Integer16</a>                                 | Integer in the range $[-2^{15}, 2^{15}-1]$ .                                                                                                                                                                                                                                                  | 294 |
| <a href="#">Integer32</a>                                 | Integer in the range $[-2^{31}, 2^{31}-1]$ .                                                                                                                                                                                                                                                  | 294 |
| <a href="#">MineDielectricDifference</a>                  | Local dielectric difference between the mine and the surrounding soil (reflectance)                                                                                                                                                                                                           | 294 |
| <a href="#">MineIdentifier</a>                            | Specifies a mine entity identifier                                                                                                                                                                                                                                                            | 294 |
| <a href="#">Octet</a>                                     | Uninterpreted 8-bit value.                                                                                                                                                                                                                                                                    | 294 |
| <a href="#">ParticipatingUnitSourceNumber</a>             | PU number. Valid range [1,76] octal ([1,62] decimal).                                                                                                                                                                                                                                         | 295 |
| <a href="#">PercentFloat32</a>                            | Percentage                                                                                                                                                                                                                                                                                    | 295 |
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## BitRateBitPerSecondUnsignedInteger32 ALIAS

Rate of transmission, in bits per second.

```
alias BitRateBitPerSecondUnsignedInteger32 u32;
```

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMessageRadioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.TDLBinaryRadioSignal](#), [link16.VMFRadioSignal](#), [ApplicationSpecificRadioSignal](#), [AudioDataTypeStruct](#), [RawBinaryRadioSignal](#).

## BitsUnsignedInteger16 ALIAS

Transmission size, in number of bits.

```
alias BitsUnsignedInteger16 u16;
```

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMessageRadioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.TDLBinaryRadioSignal](#), [link16.VMFRadioSignal](#), [ApplicationSpecificRadioSignal](#), [AudioDataTypeStruct](#), [RawBinaryRadioSignal](#).

## ClockTimeHourInteger32 ALIAS

Time past on the clock in full hours since a specified point in time.

```
alias ClockTimeHourInteger32 i32;
```

Named by [ClockTimeStruct](#).

## Float32 ALIAS

Single-precision floating point number.

```
alias Float32 f32;
```

Named by [netn.FireAtEntityWM](#), [netn.FireAtLocationWM](#), [ArticulatedPartsStruct](#), [BeamAntennaStruct](#), [COMBICStateRecStruct](#), [CoefficientsLengthlessArray1Plus](#), [CollisionElastic](#), [Cone1GeomRecStruct](#), [Cone2GeomRecStruct](#), [Float32Array1Plus](#), [GaussPlumeGeomRecStruct](#), [GaussPuffGeomRecStruct](#), [GridValueType1Struct](#), [PropulsionSystemDataStruct](#), [Sphere1GeomRecStruct](#), [Sphere2GeomRecStruct](#), [SupplyStruct](#).

## Float64 ALIAS

Double-precision floating point number.

```
alias Float64 f64;
```

Named by [GridAxisStruct](#), [IrregularGridAxisStruct](#).

## Integer16 ALIAS

Integer in the range  $[-2^{15}, 2^{15}-1]$ .

```
alias Integer16 i16;
```

Named by [netn.ConnectionReceiverStruct](#), [netn.FireAtEntityWM](#), [netn.FireAtLocationWM](#), [netn.IncomingConnectionStruct](#), [AcousticTransient](#), [ActiveSonarBeam](#), [AdditionalPassiveActivities](#), [AggregateEntity](#), [Integer16Array1Plus](#), [NamedLocationStruct](#), [PhysicalEntity](#), [SINCGARSModulationStruct](#).

## Integer32 ALIAS

Integer in the range  $[-2^{31}, 2^{31}-1]$ .

```
alias Integer32 i32;
```

Named by [netn.MineCountVariantStruct](#), [netn.PhysicalGenericNetworkStruct](#), [COMBICStateRecStruct](#), [FlareStateRecStruct](#), [FundamentalParameterDataStruct](#).

## MineDielectricDifference ALIAS

Local dielectric difference between the mine and the surrounding soil (reflectance)

```
alias MineDielectricDifference f32;
```

Named by [MineDielectricDifferenceLengthlessArray](#).

## MineIdentifier ALIAS

Specifies a mine entity identifier

```
alias MineIdentifier u16;
```

Named by [MineIdentifierLengthlessArray](#).

## Octet ALIAS

Uninterpreted 8-bit value.

```
alias Octet u8;
```

Named by [link16.OctetArray10](#), [link16.OctetArray6](#), [link16.OctetLengthlessArray29Plus](#), [ActiveSonar](#), [ActiveSonarBeam](#), [ArticulatedParameterStruct](#), [EmitterBeam](#), [EmitterSystem](#), [EntityTypeStruct](#), [EnvironmentObjectTypeStruct](#), [EnvironmentTypeStruct](#), [GridAxisStruct](#),

[GriddedData](#), [MarkingArray11](#), [MarkingArray31](#), [OctetArray](#), [OctetArray1Plus](#), [OctetArray2](#), [OctetArray3](#), [OctetArray4](#), [OctetArray7](#), [OctetArray8](#), [OctetPadding32Array](#), [OctetPadding64Array](#), [RRB](#), [RadioTypeStruct](#), [SignalDataLengthlessArray1Plus](#).

## ParticipatingUnitSourceNumber ALIAS

PU number. Valid range [1,76] octal ([1,62] decimal).

```
alias ParticipatingUnitSourceNumber u8;
```

Named by [Link11BTransmitterStruct](#), [Link11TransmitterStruct](#).

## PercentFloat32 ALIAS

Percentage

```
alias PercentFloat32 f32;
```

Named by [netn.CloudStruct](#), [netn.IceStruct](#), [netn.MineCountVariantStruct](#), [netn.ProbabilityHazardContourStruct](#), [netn.ProtectionEffectivenessStruct](#), [netn.TreatmentStruct](#), [EmitterBeam](#), [IFF](#).

## PercentUnsignedInteger32 ALIAS

Percentage

```
alias PercentUnsignedInteger32 u32;
```

Named by [netn.ElectronicSignatureStruct](#), [netn.HUMINTSignatureStruct](#), [netn.VisualSignatureStruct](#), [ArealObject](#), [ArealObjectTransaction](#), [BreachablePointObjectTransaction](#), [BurstPointObject](#), [BurstPointObjectTransaction](#), [CraterObjectTransaction](#), [ExhaustSmokeStruct](#), [LinearSegmentStruct](#), [MinefieldObjectTransaction](#), [OtherArealObjectTransaction](#), [OtherPointObjectTransaction](#), [PointObject](#), [PointObjectTransaction](#), [RibbonBridgeObjectTransaction](#), [StructureObjectTransaction](#).

## PowerRatioDecibelMilliwattFloat32 ALIAS

Power ratio in decibels (dB) of a measured power referenced to 1 milliwatt (mW).

```
alias PowerRatioDecibelMilliwattFloat32 f32;
```

Named by [EmitterBeam](#), [FundamentalParameterDataStruct](#), [RadioReceiver](#), [RadioTransmitter](#).

## PowerWattFloat32 ALIAS

The unit of power is the watt (W), which is equal to one joule per second.

```
alias PowerWattFloat32 f32;
```

Named by [Designator](#).

## SpeedChangeRateRPMPerSecondInteger16 ALIAS

Angular acceleration

```
alias SpeedChangeRateRPMPerSecondInteger16 i16;
```

Named by [ShaftDataStruct](#).

## TimestampUnsignedInteger32 ALIAS

The time past the hour, scaled so that value 0 represents the start of the hour and value  $2^{31} - 1$  represents one time unit before the start of the next hour, thereby resulting in each time unit representing exactly  $3600/(2^{31})$  s, which is approximately 1.67638063 microsecond.

```
alias TimestampUnsignedInteger32 u32;
```

Named by [ClockTimeStruct](#).

## TransponderModeCAltitude100FootInteger16 ALIAS

Actual Mode C altitude in the range 0-126,000 feet in 100-foot increments.

```
alias TransponderModeCAltitude100FootInteger16 i16;
```

Named by [NatoIFFTransponder](#).

## UnsignedInteger16 ALIAS

Integer in the range  $[0, 2^{16}-1]$ .

```
alias UnsignedInteger16 u16;
```

Named by [link16.JTIDSLETRadioSignal](#), [link16.JTIDSMessageradioSignal](#), [link16.JTIDSVoiceCVSDRadioSignal](#), [link16.JTIDSVoiceLPC10RadioSignal](#), [link16.JTIDSVoiceLPC12RadioSignal](#), [link16.Link16RadioSignal](#), [link16.RTTABRadioSignal](#), [link16.RTTReplyRadioSignal](#), [link16.TDLBinaryRadioSignal](#), [link16.VMFRRadioSignal](#), [AcousticTransient](#), [AdditionalPassiveActivities](#), [ApplicationSpecificRadioSignal](#), [ArticulatedParameterStruct](#), [COMBICStateRecStruct](#), [DatabaseIndexRadioSignal](#), [EmitterBeam](#), [EntityIdentifierStruct](#), [EntityTypeStruct](#), [EnvironmentTypeStruct](#), [EventIdentifierStruct](#), [FederateIdentifierStruct](#), [FlareStateRecStruct](#), [GridAxisStruct](#), [LinearSegmentStruct](#), [MunitionDetonation](#), [NatoIFF](#), [NatoIFFTransponder](#), [PropulsionNoise](#), [RadioReceiver](#), [RadioTransmitter](#), [RadioTypeStruct](#), [RawBinaryRadioSignal](#), [SilentAggregateStruct](#), [SilentEntityStruct](#), [UnsignedInteger16Array1Plus](#), [WeaponFire](#).

## UnsignedInteger32 ALIAS

Integer in the range  $[0, 2^{32}-1]$ .

```
alias UnsignedInteger32 u32;
```

Named by [link16.NTPTimestampStruct](#), [Acknowledge](#), [AcknowledgeR](#), [ActionRequest](#), [ActionRequestR](#), [ActionResponse](#), [ActionResponseR](#), [AggregateEntity](#), [AttributeValuePairStruct](#), [AudioDataTypeStruct](#), [BreachableSegmentStruct](#), [BurstPointObject](#), [BurstPointObjectTransaction](#), [COMBICStateRecStruct](#), [CraterObject](#), [CraterObjectTransaction](#), [CreateEntity](#), [CreateEntityR](#), [CreateObjectRequest](#), [CreateObjectRequestR](#), [CreateObjectResult](#), [CreateObjectResultR](#), [Data](#), [DataQuery](#), [DataQueryR](#), [DataR](#), [DatabaseIndexRadioSignal](#), [EnvironmentRecStruct](#), [FixedDatumStruct](#), [FlareStateRecStruct](#), [GriddedData](#), [JTIDSTransmitterStruct](#), [JammerBeam](#), [LinearSegmentStruct](#), [MinefieldObject](#), [MinefieldObjectTransaction](#), [RecordQueryR](#), [RecordR](#), [RecordSetStruct](#), [RecordStruct](#), [RemoveEntity](#), [RemoveEntityR](#), [RemoveObjectRequest](#), [RemoveObjectRequestR](#), [RemoveObjectResult](#), [RemoveObjectResultR](#), [RibbonBridgeObject](#), [RibbonBridgeObjectTransaction](#), [SetData](#), [SetDataR](#), [SetRecordR](#), [SphericalHarmonicAntennaStruct](#), [StartResume](#), [StartResumeR](#), [StopFreeze](#), [StopFreezeR](#), [TransferControl](#), [UnsignedInteger32LengthlessArray](#), [VariableDatumStruct](#), [WeaponFire](#).

## UnsignedInteger64 ALIAS

Integer in the range  $[0, 2^{64}-1]$ .

```
alias UnsignedInteger64 u64;
```

Named by [AudioDataTypeStruct](#), [GriddedData](#), [RadioTransmitter](#), [UnsignedInteger64Array1Plus](#).

## UnsignedInteger8 ALIAS

Integer in the range  $[0, 2^8-1]$ .

```
alias UnsignedInteger8 u8;
```

Named by [MinefieldQuery](#), [MinefieldResponseNACK](#), [MissingRecordNumbersLengthlessArray1Plus](#), [UnsignedInteger8LengthlessArray](#).

### Optionals

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---

**MaybeDamageStatusEnum32** OPTIONAL

```
optional<DamageStatusEnum32> MaybeDamageStatusEnum32;
```

---

**MaybeEntityTypeStruct** OPTIONAL

```
optional<EntityTypeStruct> MaybeEntityTypeStruct;
```

---

**MaybeEnvironmentObjectTypeStruct** OPTIONAL

```
optional<EnvironmentObjectTypeStruct> MaybeEnvironmentObjectTypeStruct;
```

---

**MaybeForceIdentifierEnum8** OPTIONAL

```
optional<ForceIdentifierEnum8> MaybeForceIdentifierEnum8;
```

---

**MaybeLengthMeterFloat32** OPTIONAL

```
optional<LengthMeterFloat32> MaybeLengthMeterFloat32;
```

---

**MaybeMeterFloat32** OPTIONAL

```
optional<MeterFloat32> MaybeMeterFloat32;
```

---

**MaybeTemperatureDegreeCelsiusFloat32** OPTIONAL

```
optional<TemperatureDegreeCelsiusFloat32> MaybeTemperatureDegreeCelsiusFloat32;
```

---

**MaybeTransmitterOperationalStatusEnum8** OPTIONAL

```
optional<TransmitterOperationalStatusEnum8> MaybeTransmitterOperationalStatusEnum8;
```

---

**MaybeVelocityMeterPerSecondFloat32** OPTIONAL

```
optional<VelocityMeterPerSecondFloat32> MaybeVelocityMeterPerSecondFloat32;
```

---

**MaybeWorldLocationStruct** OPTIONAL

```
optional<WorldLocationStruct> MaybeWorldLocationStruct;
```

---

# Built-in types

The types the language provides. Every one is written the same way on the wire wherever it appears.

| Type   | Meaning                      |
|--------|------------------------------|
| bool   | classic boolean              |
| f32    | 32 bit floating point number |
| f64    | 64 bit floating point number |
| i16    | 16 bit signed integer        |
| i32    | 32 bit signed integer        |
| i64    | 64 bit signed integer        |
| string | unbounded string             |
| u16    | 16 bit unsigned integer      |
| u32    | 32 bit unsigned integer      |
| u64    | 64 bit unsigned integer      |
| u8     | 8 bit unsigned integer       |

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